



2025 STATE OF THE MARKET REPORT FOR THE NEW YORK ISO MARKETS

POTOMAC
ECONOMICS

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Guide to Acronyms

BPCG	Bid Production Cost Guarantee	MW	Megawatts
CAF	Capacity Accreditation Factor	MWh	Megawatt-hours
CARC	Capacity Accreditation Resource Class	NYSERDA	NYS Energy Research & Development Authority
CLCPA	Climate Leadership and Community Protection Act	NYSRC	NYS Reliability Council
CONE	Cost of New Entry	NYTO	New York Transmission Owner
CP	PJM's Capacity Performance rule	OOM	Out-of-merit or Out-of-market
CRI	Cost of Reliability Improvement	ORDC	Operating Reserve Demand Curve
CRIS	Capacity Resource Interconnect. Service	MRI	Marginal Reliability Impact
CRM	Constraint Reliability Margin	PFP	ISO-NE's Pay for Performance rule
CTS	Coordinated Transaction Scheduling	PPTN	Public Policy Transmission Need
DEC	Department of Environmental Conservation	PSC	NYS Public Service Commission
DPS	Department of Public Service	PTC	Federal Production Tax Credit
DMNC	Demonstrated Maximum Net Capability	REC	Renewable Energy Credit
FERC	Federal Energy Regulatory Commission	RGGI	Regional Greenhouse Gas Initiative
GFC	Going Forward Cost	RTC	Real-Time Commitment model
GTDC	Graduated Transmission Demand Curve	RTD	Real-Time Dispatch model
ICAP	Installed Capacity	SCR	Special Case Resource
IPR	Intermittent Power Resource	SDU	System Deliverability Upgrade
IRM	Installed Capacity Margin	TCC	Transmission Congestion Contract
ITC	Federal Investment Tax Credit	TSL	Transmission Security Limit
kV	Kilo-volt	TVR	Transient Voltage Recovery
LBMP	Location-Based Marginal Price	UCAP	Unforced Capacity
LCR	Locational Capacity Requirement	UOL	Upper Operating Limit
LIPA	Long Island Power Authority	VFT	Variable Frequency Transformer
LOLE	Loss of Load Expectation		
MMbtu	Millions of British Thermal Units		

EXECUTIVE SUMMARY

As NYISO’s Market Monitor Unit (MMU), we evaluate the competitive performance of NYISO’s wholesale electricity markets, identify market flaws, and recommend market design improvements. We also evaluate the market power mitigation rules designed to limit anticompetitive conduct that would erode the benefits of the competitive markets. This State of the Market Report presents this evaluation for 2025.

NYISO operates competitive wholesale markets to satisfy New York’s electricity needs. These markets establish prices that reflect the value of energy at each location on the network. They deliver significant benefits by coordinating resource commitment and dispatch to meet the system’s demand at the lowest cost. These markets also provide competitive incentives for resources to perform reliably in the short term and make efficient investment and retirement decisions in the long term. The installed capacity market supplements the energy and ancillary services markets to satisfy NYISO’s planning reliability requirements.

New York State law requires the Public Service Commission (“PSC”) to implement regulations so the electricity generation fleet eliminates carbon dioxide emissions by 2040. The PSC is also responsible for overseeing electric corporations to ensure reliable service at just and reasonable rates. As these mandates become harder to reconcile, the NYISO’s energy, ancillary services, and capacity markets can help channel investment toward projects that support significant progress toward environmental policy goals while maintaining reliability at the lowest possible cost.

Market Highlights in 2025

The NYISO markets performed competitively in 2025, and the mitigation measures were generally effective in limiting conduct that would raise energy and capacity prices above competitive levels. This subsection summarizes the market results and trends.

Natural Gas Prices

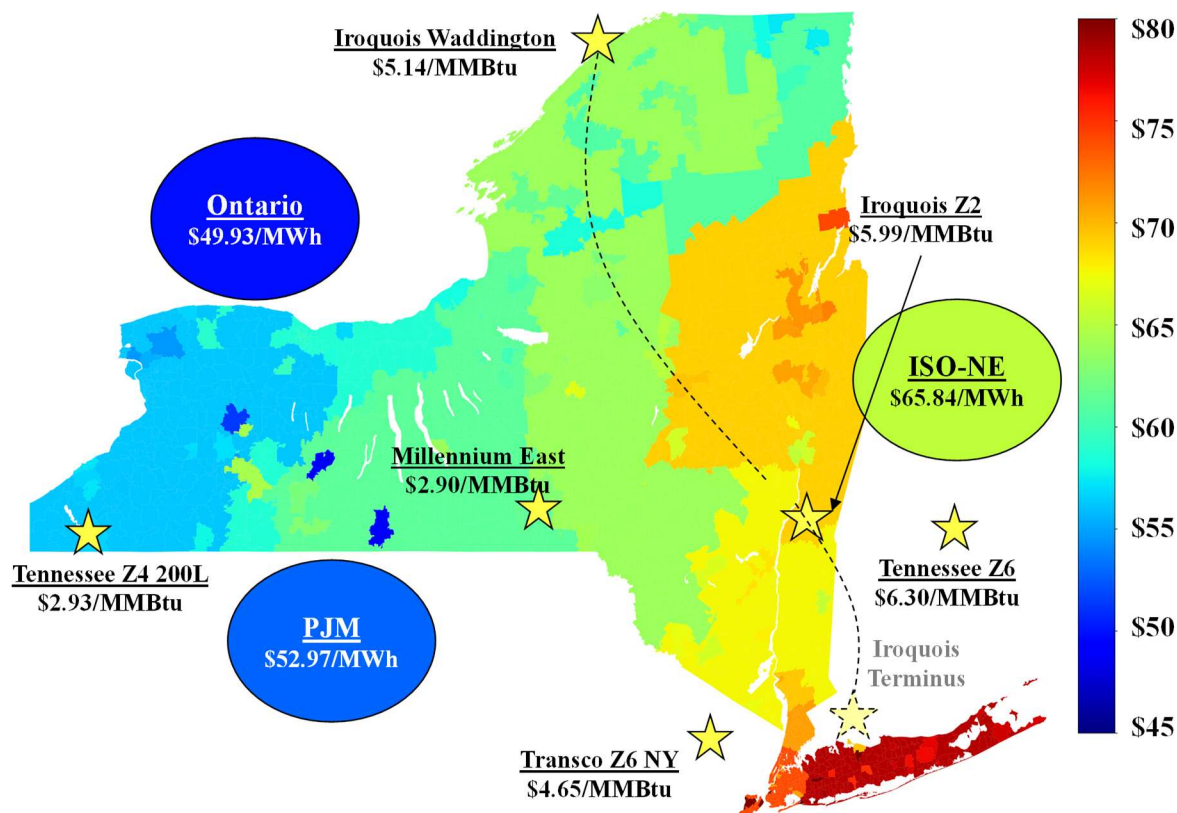
Electricity prices depend primarily on natural gas prices and load levels. Average gas prices rose by 60 to 112 percent across the State from 2024 to 2025. Gas prices averaged under \$3.25 per MMBtu in 2025 in western and central New York compared to \$5 to \$6.24 per MMBtu in eastern New York. Eastern New York experienced considerable price volatility in the winter because of west-to-east congestion on gas pipelines through the State. (See Section II.B).

Energy Prices and Transmission Congestion

Average energy prices rose from 2024 to 2025 by 66 to 79 percent primarily because of higher natural gas prices, increased exports to Quebec, and higher CO₂ emissions allowance prices.

Transmission congestion and losses in 2025 caused average real-time prices to range from \$54.40 per MWh in the West Zone to \$81 in Long Island. (See Section II.A)

Real-Time Energy Prices, Natural Gas Prices, and Congestion in 2025



CO₂ emissions allowance prices in the Regional Greenhouse Gas Initiative (“RGGI”) account for a growing share of electricity production costs and averaged nearly \$23 per ton in 2025. A typical gas-fired combined cycle generator in Central Zone likely incurred average RGGI compliance costs of more than \$10 per MWh (or ~28 percent of average production costs) in 2025. (See Section II.B)

Congestion has increased over the past two years on paths from central to eastern New York, within New York City, on exports to New England, and on imports from PJM, especially in cold weather when gas pipelines are congested. However, congestion from central to eastern New York became less frequent because of the recently built “AC Transmission” projects and dramatically reduced net imports from Quebec (1.1 GW of average imports in 2022 vs 550 MW of average exports in 2025).

Nationwide concern about the effect of rising electricity rates on the cost of living has been increasing. Although year-to-year NYISO market prices are volatile, they have a limited direct effect on residential retail rates because (1) residential rates are mainly driven by non-NYISO costs such as regulated transmission and distribution, and (2) New York utilities routinely hedge

wholesale costs on behalf of their customers. From 2021 to 2025, NYISO market prices equaled 34 percent of total residential retail rates in New York. During that period, statewide average residential rates rose by 3.2 cents per kWh (in real 2025 dollars) to 26.4 cents per kWh in 2025, while wholesale supply costs passed on to residential customers (including NYISO market purchases and hedged supply) rose by 1.4 cents to 10.2 cents per kWh.

Capacity Market

In the 2025/26 Capability Year, New York City capacity prices fell by 2 percent to \$132 per kW-year, while capacity prices in other regions increased by 38 percent to \$51 per kW-year. Prices outside New York City equaled 74 percent of the “Net Cost of New Entry (CONE)” for the rest-of-state region, although the Net CONE from the last Demand Curve Reset is far below the actual cost of new entry in rest-of-state. Higher Installed Reserve Margin (IRM) and lower imports drove the price increase outside New York City. The Locational Capacity Requirement (LCR) for New York City fell in 2025, increasing the capacity surplus relative to requirements, but a 41 percent increase in the UCAP Net CONE offset this effect.

Under-Investment in New and Existing Resources

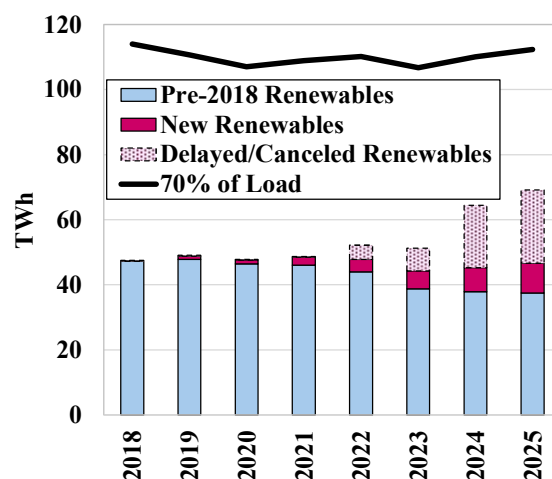
The NYISO markets are designed to provide efficient incentives for new investment, maintenance of existing generation, and retirement of resources that are no longer economic. Efficient markets compensate resources according to their value in serving load and satisfying reliability requirements. After the CLCPA was enacted in July 2019, many envisioned that State agencies would select all major new projects through procurements or direct investments, with NYISO market incentives playing a secondary role. However, nearly seven years later, State efforts to develop new clean resources are far behind schedule, and state policy has led to the retirement of 3.7 GW of conventional generation. In the past year, 1.2 GW of conventional generators submitted retirement notices, but NYISO retained them in service because of reliability concerns. Market reforms would help the NYISO provide adequate incentives for investment in new and existing resources to satisfy reliability needs. However, severe reliability issues are inevitable if New York cannot site and permit new investments in conventional generation.¹ This report evaluates progress in satisfying CLCPA requirements, the condition of the New York generation fleet, short-term capacity needs, and market incentives for capacity investment.

¹ Governor Hochul described the state’s current emission reduction targets by 2030 as “costly and unattainable” in a recent op-ed (See [Climate Action and Affordability Can and Must Go Hand-In-Hand](#)”, March 20, 2026, available [here](#)). State officials and agencies have increasingly acknowledged that the cost and difficulty of achieving the goals of the 2019 Climate Law have been underestimated, and the recently issued [2025 State Energy Plan](#) anticipates the need for “new or repowered” combustion unit capacity to maintain statewide system reliability before long-term decarbonization goals are achieved.

Investment in New Generation

New York has not permitted any new fossil-fuel generation since 2020. At the same time, new investment in renewables and storage has significantly lagged expectations. Figure ES-1 shows that the percentage of renewables in New York’s electricity mix has declined since the passage of CLCPA in 2019, which required 70 percent renewable generation by 2030. Reductions in hydro imports and generation drove this decline, which was only partly offset by new renewables (mainly behind-the-meter solar). Developers have canceled or delayed most utility-scale renewable projects the state contracted to enter service by 2025. Similarly, less than 100 MW of battery storage has been deployed since 2019, compared to the state’s goal of 3 GW in the NYISO markets by 2030 (Section IV.C).

Figure ES-1: Status of Renewable Projects



Renewable and storage development has been impeded by major cost increases, challenges in the permitting and interconnection processes, and potential revenues (including subsidies) being too low to support investment. We estimate that wind, solar, and battery storage resources in New York require state and federal subsidies equal to approximately half of their levelized costs to be financially viable. Recent changes in federal law will eliminate key subsidies for wind and solar projects not placed in service by the end of 2027 or under construction as of July 2026. Hence, while the completion of the CHPE project in 2026 will boost supply margins and energy imports, the outlook for new clean energy investment remains highly uncertain (Section IV.B).

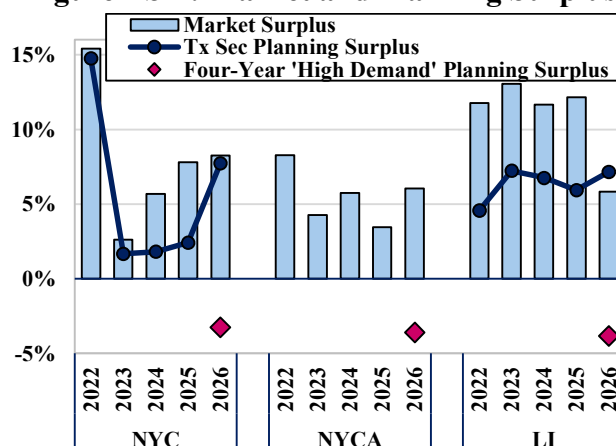
Declining Capacity Margins and the “Planning-Market Gap”

Figure ES-2 compares capacity surpluses above minimum summer requirements in the capacity market with those implied by NYISO’s short-term reliability planning studies.² Capacity margins declined from 2022 to 2025 as generator retirements outpaced new entry and as planning requirements became more conservative. The 1,250 MW Champlain Hudson Power Express (CHPE) facility will increase margins in New York City and statewide when it enters service in 2026. However, NYISO finds deficient (i.e., negative) margins in 2030 in all regions in its 2026 Q1 Short Term Assessment of Reliability that is used to determine whether generators can retire without triggering Reliability Needs.

² Reported Planning Surplus values reflect the Short Term Assessment of Reliability (STAR) from the same year, with the following adjustments: (a) we use the non-coincident peak load for transmission security margins in all years in line with NYISO’s current methodology, and (b) we assume CHPE in service and 2026 and that the Gowanus 2&3, Narrows, Danskammer, and Far Rockaway units remain in service in 2026.

Before 2026, capacity surpluses in New York City and Long Island were often much larger in the market than in planning studies. This divergence (i.e., planning-market gap) was caused by the use of higher load forecasts and generator outage rate assumptions in the planning studies for each year. NYISO recently addressed this divergence by updating the assumptions used to determine the market's LCRs for the 2026/27 Capability Year to align with the planning assessments, resulting in large increases in LCRs.

Figure ES-2: Market and Planning Surpluses



Despite apparently healthy capacity surpluses expected in 2026 following CHPE's entry, NYISO forecasts capacity deficiencies in 2030 primarily because it uses a "high demand" scenario that exceeds the expected baseline load forecast. Consequently, NYISO declared Reliability Needs in the downstate localities in response to each proposed retirement in the past year, retaining some affected units temporarily on a cost-of-service basis. This divergence between the expected market requirements and the planning requirement four years out under a conservative demand forecast will undermine the market's ability to attract new resources and retain existing resources needed for reliability. (Section V.B.4)

Market Incentives for Older Existing Generators

New York's fossil generation fleet is among the oldest in the country, with over 12 GW more than 40 years old. The policy-driven retirement of nuclear and other conventional generation and reduced imports from Quebec have greatly increased demand on New York's older fossil-fuel generators. Since 2019, the average capacity factor of fossil fuel generators over 40 years old has increased by 84 percent. This has increased the average time these units spend on maintenance by 29 percent, to more than 13 weeks per year. Without significant investment in new generating capacity, load growth in the coming years from data centers and electrification will need to be met primarily by increasing utilization of older units.

Recent market prices have been insufficient for many plants to recover their full going-forward costs (GFCs), which was the main cause of the 1.2 GW of retirement notices issued in the past year. As generators age, they require significant capital to remain in good condition. Generators earning revenues below their long-term GFCs often reduce costs by deferring maintenance and capex rather than retiring outright. This can gradually reduce performance and reliability in ways that may not be immediately apparent. We observe several gigawatts of capacity that performs relatively little maintenance and avoids generating when it would produce positive cash flow to reduce the risk of major outages or repairs (Section IV.A).

When the system has a large capacity surplus, it is efficient for older, costly units to retire. However, retirements and deferred maintenance are a major concern when market prices do not accurately signal reliability needs and reward reliable performance. It is therefore critical to align market outcomes with reliability planning requirements, as we discuss below.

Market Reforms to Support Reliability

Retirements of existing generation and weak new investment coinciding with the declaration of Reliability Needs suggests that NYISO market reforms are needed. We identify key shortcomings and recommended solutions in this report:

- The planning-market gap has widened due to the use of pessimistic assumptions in planning studies, including rapid load growth and indefinite delays of all planned generation and transmission projects. As a result, we estimate an effective target level of 750 MW in New York City in planning studies, compared to just 200 MW in the capacity market (Section V.B.4). NYISO’s recent proposal to assume large amounts of unplanned generator retirements will increase this target “level of excess” to ~1,000 MW in New York City and 1,800 MW statewide. We recommend that NYISO increase the “level of excess” in the capacity market to align with the planning process.³ (Rec #2025-1a)
- Some generators underreport forced outages and derates, which causes the generation fleet’s forced outage rates to be underestimated and exacerbates the planning-market gap. We recommend that NYISO strengthen compliance monitoring to ensure that generators correctly log forced outages and derates and support efficient accreditation (Sections III.A and IX.F and Rec #2025-1b)

New investment in generation will ultimately be needed to support future load growth. Storage and renewable projects are likely to enter service in the coming years, but the State increasingly recognizes that current targets are unattainable and/or excessively costly. The State can facilitate competitive, cost-effective, and reliable NYISO markets by allowing new or repowered gas-fired generation to be built alongside policy-supported clean energy sources.

Capacity Market Performance

While the overall design of the capacity market is sound, several reforms are needed to ensure that it provides efficient incentives to all resource types and locations and avoids costly out-of-market procurements. The previous section highlights market reforms that would align capacity prices with the cost of investment needed to satisfy the planning reliability needs. Because of widespread concerns about electricity affordability, it is also important to consider efficiency-

³ To avoid an excessive increase in consumer costs, it is also important to consider whether the assumptions underlying the reliability planning targets are reasonable. For additional discussion of the planning assumptions and level of excess, see our May 2026 memo on the Reliability Planning Process [here](#).

enhancing market reforms that would reduce consumer costs and soften the impact of the reforms discussed above. The market reforms recommended below would have mixed effects, but they are likely to reduce consumers' capacity costs overall.

Improving Locational Pricing Signals

Efficient capacity markets should provide accurate locational signals that reflect the value of capacity for meeting reliability needs in each region. Section IX.B finds that NYISO's current four-zone capacity market structure tends to overcompensate capacity in export-constrained sub-regions and undercompensate capacity at high-value locations. Because capacity prices do not reflect within-region transmission bottlenecks, NYISO relies on: (a) the Deliverability Study component of the interconnection process, which acts as a major barrier to new development (including \$2.2 billion in preliminary upgrade costs recently identified in the Cluster Study), and (b) higher IRM and LCRs to offset the impacts of resources that are not actually deliverable despite earning full compensation. This framework raises consumer costs, protects incumbent resources from competition, and inhibits efficient entry of new resources.

To address these shortcomings, we recommend strengthening locational incentives through: (1) locational Capacity Accreditation Factors that reflect both technology characteristics and deliverability across modeled transmission constraints in the short term (Rec. #2025-3); and (2) more granular and dynamically updated capacity zones in the longer term (Rec. #2022-4). Both approaches would facilitate granting Financial Capacity Transfer Rights that protect suppliers that fund deliverability upgrades from changes in locational capacity value. These reforms would reflect locational reliability needs directly in market outcomes, producing more efficient market signals and allowing NYISO to remove large parts of the Deliverability Study from the interconnection process (Section IX.B).

Efficient Compensation When LCRs Are Set by Transmission Security Limits

In recent years, LCRs of the downstate localities have been set at minimum values derived from Transmission Security Limits (TSLs). NYISO calculates the TSL floors in a deterministic assessment designed to protect against the locality's two largest contingencies. The TSL study also uses resource availability assumptions that differ from those in the probabilistic resource adequacy model used in the IRM and LCR study. As a result, several resource types are currently compensated inefficiently when LCRs are set at the TSL floors (Section IX.C):

- Some resources have lower availability in transmission security studies than in the capacity market accreditation. These include demand response (SCRs), offshore wind, and large units that form one of the locality's two largest contingencies. NYISO directly raises the TSL floors to offset the capacity these units sell. As a result, we estimate they will be overpaid by \$74 million in the 2026/27 capability year in NYC.

- Beginning in May 2026, gas-fired resources without firm winter fuel will face major reductions in capacity payments (42 percent in New York City) because of the introduction of non-firm Capacity Accreditation Factors (CAFs). Because the CAFs are based only on resource adequacy modeling, these units will be undercompensated for their contribution to TSL requirements, which may lead to inefficient retirements.
- The current demand curve shapes overstate the value of surplus capacity above the TSL floors, which are highly conservative by design.

To address these issues, we recommend: (1) Paying resources for capacity according to the requirements they help satisfy (Recommendation 2022-1); and (2) Developing sloped demand curves that reflect the marginal value of surplus capacity when a TSL determines an LCR (Recommendation 2023-4).

Improving Winter Capacity Market Incentives

The NYISO designed its capacity market primarily to satisfy a single reserve margin requirement based on summer peak load, assuming summer resources would also be available in winter. However, many resource types, such as capacity imports and gas and oil units affected by fuel constraints, now have winter availability that cannot be predicted from their summer capacity. Winter resource adequacy risk is growing and accounted for 14.8 percent of reliability risk in the 2026/27 IRM Study. Therefore, NYISO's markets must evolve to value and procure summer and winter capacity as separate products (Section IX.D).

NYISO has taken important steps to introduce seasonal elements into the capacity market, including establishing seasonal demand curve reference prices, accrediting firm fuel capability, and filing changes to set distinct summer and winter requirements. However, major shortcomings remain:

- Firm fuel and UDR import elections require commitments 16 months before the peak winter months, creating excessive risk for suppliers and discouraging efficient participation.
- Generators' firm fuel elections do not feed back into market parameters or affect prices and accreditation factors. Hence, market prices and incentives may not accurately reflect the amount of reliable winter capacity supplied relative to demand.

We continue to recommend establishing seasonal capacity requirements, CAFs, and demand curves (Rec. #2022-2). This recommendation would build on NYISO's recent improvements by setting independent summer and winter requirements that reflect the amount of capacity needed for reliability in each season. This approach would allow market participants make firm fuel and seasonal import decisions closer to the corresponding capability period and reflect those decisions appropriately in market outcomes.

Accommodating Large Load Growth

NYISO has experienced a rapid increase in large load interconnection requests in recent years, including requests from data centers and energy-intensive manufacturers. As of March 2026, over 12 GW of large load projects were in NYISO's queue, and more than 8 GW entered the queue in 2025 alone. This far exceeds the ~500 MW of large loads that entered service over the past decade and creates substantial uncertainty regarding future system demand. Most of the queued projects do not propose on-site generation or commit to flexible operation during peak periods, which creates a risk that load growth could outpace the development of new generation and storage resources needed to maintain reliability.

Proposed large load growth raises concerns about the adequacy of NYISO's load interconnection and system planning processes:

- NYISO's role in the large load interconnection process is largely informational and is limited to assessing transmission impacts through system impact studies. These studies do not impose binding upgrades, evaluate resource adequacy impacts, or ensure that future changes in a project's operations (e.g., flexible cryptocurrency loads converting to non-flexible computing workloads) are reassessed for reliability impacts.
- NYISO has limited visibility into which queued projects will ultimately be constructed, which complicates reliability planning. Excluding proposed load projects from planning studies risks understating future reliability needs, while including highly speculative and uncertain load growth will trigger inefficient out-of-market procurement actions that ratepayers bear.

Large loads enter into interconnection agreements with New York Transmission Owners ("NYTOs"), subject to State and local siting and permitting processes. While large loads are typically responsible for local upgrade and interconnection facility costs, State and local authorities may approve them for siting, permitting, and interconnection without any study of whether they will trigger resource adequacy violations. The NYPSC has recently opened a proceeding to consider reforms to the State's large load interconnection framework.

Accordingly, we recommend improved coordination between NYISO's quarterly short-term reliability planning process and State and local authorities to evaluate the amount of headroom available for load development in each region of the system. NYISO's evaluation of the BPS impacts of large loads would help State and local authorities make decisions in the public interest. At the same time, NYISO's planning processes would benefit from better information about each large load facility's interconnection agreement and status. We further recommend creating separate tracks for large loads seeking firm or non-firm service that can be reflected in the requirements interconnection agreements. (Section XII.D and Rec #2024-2b).

Deliverability Testing and Transmission Planning Processes

The large amount of proposed new renewable and storage projects in NYISO’s interconnection queue has focused attention on transmission planning and interconnection issues. It is efficient for the developer to bear the costs of upgrades needed for a new project to interconnect reliably. However, new projects should not bear a disproportionate share of the cost of upgrades that benefit others because doing so will deter efficient investment.

Concerns with the Deliverability Testing Process

To obtain rights to sell capacity (“CRIS”), new projects must complete the Deliverability Study portion of the interconnection process and pay for network upgrades (“SDUs”) needed to make their capacity deliverable to load. The current deliverability test has structural flaws that cause it to produce unreasonable estimates of the SDUs needed. It assumes vast amounts of proposed new projects enter service simultaneously and that no existing resources retire. These assumptions require new projects to be deliverable under highly unrealistic scenarios.. As a result, recent Deliverability Studies have assigned prohibitively costly SDUs to projects in certain areas (Section V.A).

NYISO recently proposed changes to the deliverability test that will identify fewer SDUs, particularly in the Highways portion of the test. While this change will allow new entrants to avoid SDUs in some areas, deliverability will continue to pose an obstacle for projects affected by Byway constraints and some Highway interfaces. Identifying fewer SDUs may also leave some resources with CRIS rights that are not actually deliverable. In such situations, NYISO’s process will raise IRM and/or LCR requirements to offset the impact of deliverability limitations, inefficiently raising consumer costs.

To address these issues, we recommend improving the granularity of price signals in the capacity market (Recs #2025-3 and #2022-4). Doing so would allow NYISO to address deliverability limitations through market prices as they arise and eliminate major portions of the Deliverability Test.

Improvements to Transmission Planning Process

The costs of regulated transmission projects recovered through NYISO rate schedules have risen from approximately \$0.50 per MWh of statewide load in 2021 to \$3.00 per MWh in 2025 and will continue to rise because of major projects approved or under evaluation. In Section V.B, we discuss NYISO’s centralized transmission planning process and suggest improvements so more efficient projects are selected.

The assumptions and techniques used in NYISO’s planning models (particularly the Outlook and PPTN studies) affect the transmission needs identified and solutions selected. NYISO has

improved its planning models in recent studies, including modeling more realistic siting and dispatch of renewables and storage in the ongoing 2026 Outlook study. Section V.B.5 discusses the remaining issues and recommends improvements to address them. (Recommendation 2022-3) Modeling improvements would help ensure that future transmission needs are assessed accurately and that solicitations select the most efficient candidate projects.

Out-of-Market Operator Actions and Out of Market Retention

When prices do not reflect the marginal value of energy and reserves at each location, NYISO must rely on out-of-market (“OOM”) commitments and dispatch to manage transmission constraints and reserve requirements that the market models do not fully represent. These actions tend to suppress prices in constrained areas. When flexible resources in constrained areas are undercompensated, capacity prices must increase to compensate, raising costs to consumers and weakening incentives for resources to locate where they are most needed.

Over time, the absence of efficient market signals undermines the profitability of resources that are essential for reliability. This can lead to proposed retirements of needed generators that are then retained through out-of-market mechanisms, such as Interim Service Provider (ISP) and Reliability Must Run (RMR) contracts at cost-of-service rates. OOM dispatch and retention mechanisms produce inefficient outcomes because units needed to manage critical constraints or maintain reliability do not earn adequate market revenues. This leads to poor maintenance, inefficient retirements, and higher long-term costs.

New York City

Supplemental commitments occurred on 121 days in 2025 across several load pockets in New York City. Because current market models do not fully represent these reliability needs, energy and reserve prices are often suppressed and fail to reflect the true cost of reliability in these areas. We recommend the NYISO model local reserve requirements that drive these supplemental commitments, which should provide more efficient price signals for flexible resources in these areas. (See Section VII.D and Recommendation 2024-1)

Supplemental commitments are made for areas of New York City that overlap with areas where NYISO and Con Edison have identified reliability needs. Energy and ancillary services pricing that better reflects locational value improves the market’s ability to retain needed units and attract investment (such as distributed storage) while avoiding OOM interventions.

Long Island

OOM commitments of oil-fired peaking units for Transient Voltage Recovery (TVR) requirements on the East End of Long Island were frequent, occurring in nearly 600 hours in 2025. These actions have led to inefficient price signals in the area, reducing average LBMPs by

~14 percent. To provide more efficient incentives for scheduling and new investment, we recommend NYISO model East End TVR needs in the market software using surrogate constraints. (See Section VIII.D.2 and Recommendation 2021-3)

The current Long Island reserve requirement was sometimes inadequate to satisfy multiple-contingency criteria. Modeling these reserve requirements in Long Island would improve efficiency and encourage flexible new resources to locate where they are most valuable. Therefore, we recommend that NYISO implement Long Island reserve requirements that will maintain reliability rather than rely on OOM actions. (See Appendix Section III.D and Recommendation 2024-1)

Peaking units were frequently OOM-committed to manage 69 kV transmission constraints in the Far Rockaway load pocket. OOM commitments occurred for ~370 hours in 2024 and increased to 695 hours in 2025. These units were scheduled for roughly 1,360 hours in 2024 and 1,300 hours in 2025, indicating that as OOM commitments increased, market-based commitments decreased. At the same time, the share of output from oil-firing (rather than gas-firing) increased from 2 percent in 2024 to 50 percent in 2025. BPCG uplift from these units increased from \$0.6 million in 2024 to nearly \$4.5 million in 2025 and was allocated statewide rather than locally. The units submitted a deactivation notice in 2025 but have been retained for reliability and are being temporarily operated under an ISP rate in 2026 at an additional cost of \$0.5 million per month. This example highlights inefficiencies that arise when a generator is frequently scheduled through OOM actions, since such generators (a) generally do not benefit from operating in the most cost-effective manner, and (b) the cost of OOM actions are often not allocated consistent with cost causation principles.⁴

In 2025, NYISO declared Reliability Needs on Long Island caused by the proposed retirements of the Far Rockaway (105 MW) and Pinelawn (74 MW) facilities. Both facilities were temporarily retained as Interim Service Providers at a cost of \$1.0 million per month above market payments allocated to Long Island customers.⁵ Capacity prices on Long Island have historically been lower than going-forward costs, and most units there are supported by long-term contracts with LIPA at above-market rates. Most of these contracts are set to expire before summer 2028, creating a risk that revenues will be inadequate to maintain needed capacity levels. The two proposed retirements previously had LIPA contracts that expired recently. The

⁴ When a generator is committed economically in some hours and through OOM actions in other hours, the generator does not receive a BPCG payment unless the financial losses from OOM commitments exceed the profits from economically scheduled hours. Hence, OOM actions can significantly reduce the profitability of a fuel-efficient generator, making it more profitable for the generator to operate on a more expensive fuel.

⁵ Far Rockaway entered into a new contract with LIPA and withdrew its retirement notice in April 2026, and NYISO concluded that Pinelawn may retire and concluded its ISP rate in May 2026.

capacity market demand curves for Long Island use a Net CONE that is lower than the true cost of new capacity, which may lead to further retirements or poor maintenance of capacity.

Upstate New York

OOM commitments in upstate regions have increased in recent years, primarily to satisfy multi-contingency criteria in the North Country load pockets. These commitments occurred on 205 days in 2024 and 119 days in 2025. We recommend incorporating reserve requirements in local load pockets into the market models to improve scheduling efficiency and establish more efficient market signals for new investment. (See Section VII.D and Recommendation 2024-1)

Energy and Ancillary Services Market Performance

We evaluate market performance based on how efficiently the market schedules resources and sets real-time prices, particularly during tight operating conditions. Efficient prices are important because they reward resources that perform flexibly and reliably during tight real-time conditions.

Dynamic Reserve Requirements

As fast-start peaking units retire and intermittent generation increases, congestion patterns and operating reserve constraints have become more variable. Consequently, NYISO does not schedule operating reserves efficiently when it could meet local reserve needs more cost-effectively by reducing imports and increasing internal generation rather than holding reserves on internal units. Accordingly, we have recommended that NYISO dynamically determine optimal reserve requirements for both local and systemwide needs. (Rec #2015-16 and #2016-1)

NYISO is working to deploy the “Dynamic Reserve” project in 2028. However, the proposed design has several key deficiencies that may limit its benefits. While it introduces more efficient reserve modeling for some requirements, it defers significant elements, particularly in the day-ahead market, where most reserve costs are incurred. The project also postpones key elements, such as intrazonal and N-1-1-0 constraints, which limits potential efficiency gains.

While Dynamic Reserve requirements should allow for efficient scheduling and pricing of reserves and energy at each location and appropriate cost allocation, the proposed design would: (a) uplift reserve costs statewide that should be allocated to resources that cause the NYISO to procure additional reserves; (b) treat non-firm and virtual imports as physical energy, which will lead to under-scheduling of physical resources and continued OOM actions; (c) over-procure local 30-minute reserves in the day-ahead market when load is over-scheduled relative to the forecast; (d) retain some static reserve requirements because of concerns related to potential deliverability limitations for reserves across constrained local transmission; and (e) allocate reserve costs to statewide load rather than to loads within relevant reserve areas.

Taken together, these issues reduce pricing accuracy, weaken investment signals, and diminish overall market efficiency, suggesting that further design improvements should be considered before implementation.

Preparing the System for Thunderstorm Alerts (TSAs)

TSAs require conservative transmission operations in the Con Edison service territory, effectively reducing transfer capability from upstate to downstate regions by 1 to 2 GW. Because these constraints are modeled in the real-time market but not in the day-ahead market, the market tends to under-commit downstate generation and over-commit upstate supply. This leads to inefficient commitment, increased use of high-cost peaking units in real time, and significant price divergence between day-ahead and real-time markets. Virtual traders have historically helped correct these divergences by purchasing load downstate and selling virtual energy upstate. However, the resulting balancing congestion shortfalls are often large, which are uplifted to load and difficult for market participants to hedge.

Although virtual trading helps correct inconsistent modeling between the day-ahead and real-time market, NYISO could more efficiently model TSA constraints in the day-ahead market when they are predictable. We evaluated the benefits of this solution by incorporating TSA constraints into the day-ahead market using a weather-based thunderstorm forecasting model. The model demonstrates strong predictive accuracy, particularly at higher probability thresholds and during high-load conditions. Our analysis shows that TSA-related congestion costs are highly concentrated in these periods, averaging \$300 to \$500 per MWh, and remain modest under lower load conditions. Notably, just 50 of the 1,377 high-risk hours analyzed in 2025 accounted for 99 percent of TSA-related congestion costs.

These findings indicate that targeted modeling of TSA constraints in high-risk hours could significantly improve market efficiency. Accordingly, we recommend that NYISO incorporate TSA constraints into the day-ahead market in high-risk hours based on forecasted thunderstorm probability and load. This would improve commitment efficiency, enhance price formation, and reduce uplift by shifting costs into market prices. (See Section VI.D and Rec #2025-5)

Market Performance under Reserve Shortage Conditions

Reserve shortages occurred in approximately 7 percent of real-time intervals in 2025 and increased average LBMPs by 10 to 15 percent across regions. Most shortages are transitory and arise from rapid or unforeseen changes in load, external interchange, and other system conditions. Accordingly, shortage pricing plays a critical role in rewarding flexible generation that is available and performing reliably, particularly as fast-start units retire and intermittent generation is added to the system.

Shortage pricing levels should be high enough to reflect the true value of reserves and avoid reliance on OOM actions. As discussed in Section VII.A, current operating reserve demand curves are set below: (a) the cost of OOM actions required to maintain reserves when neighboring control areas are also short; and (b) the marginal reliability value of reserves during deep shortages. This issue is particularly significant because shortage prices are much higher in neighboring markets. Resources in ISO-NE and PJM can receive \$4,000 to \$12,000 per MWh during relatively small shortages, compared to \$750 to \$3,000 per MWh in NYISO even during deep shortages. This disparity can incentivize exports from New York to neighboring regions despite more severe local shortages, especially during periods of natural gas scarcity. Hence, we recommend NYISO raise its reserve demand curves to address these issues. (See Section VII.A.1 and Recommendation 2017-2)

Market Performance under Transmission Shortages

Transmission shortages occur when power flows over a transmission facility exceed the applicable operating limit. This can arise either from a lack of available resources to relieve the constraint or a market design that allows small constraint violations when the cost of relief would exceed \$200 per MWh. In 2025, these localized shortages occurred in ~20 percent of real-time intervals, with frequencies ranging from less than 2 percent in the North Zone to more than 12 percent on Long Island. Transmission shortages had an average annual LBMP impact ranging from negative ~\$1 per MWh in the West Zone to nearly \$6.5 per MWh on Long Island.

“Offline GT pricing” continues to undermine pricing efficiency by preventing the market software from fully recognizing real-time transmission shortages. As a result, congestion prices do not accurately represent the severity of actual shortages. Currently, NYISO’s real-time pricing model assumes that offline GTs can respond to dispatch instructions within five minutes, even though they are not physically capable of doing so. Consequently, the market model may underestimate transmission scarcity, leading NYISO to over-constrain transmission in some areas that rely heavily on gas turbines to compensate for these differences. The largest impacts occur on Long Island, where correcting this issue would have increased average LBMPs by up to \$3.7 per MWh in 2025. To address this inefficiency, we recommend NYISO eliminate offline fast-start pricing from the real-time dispatch model. (See Section VII.B and Recommendation 2020-2)

Real-time Pricing during Gas Turbine Starts

NYISO was the first market to adopt fast-start pricing, which allows peaking units with quick-start capability and limited commitment constraints to “set price” when committed. Fast-start pricing is an essential element of an efficient wholesale market design. However, when NYISO’s real-time scheduling software economically commits gas turbines (“GTs”) with minimum run times longer than one hour, it treats them as having a one-hour minimum run time for scheduling

purposes, but the pricing algorithm does not let them set LBMP. As a result, real-time prices often fail to reflect the true costs of maintaining reliability when these GTs start.

In 2025, this issue depressed real-time LBMPs in approximately 9 percent of intervals in New York City and 4 percent of intervals on Long Island. Correcting this issue could increase LBMPs during peak-demand hours on most days and yield an estimated annual net revenue gain of \$2 to \$6 per kW-year across downstate load pockets.

Hence, we recommend that NYISO revise its real-time fast-start pricing criteria so fast-start pricing eligibility is based on the minimum run time used for scheduling, rather than the offer parameter value. This alignment would improve price formation and provide more appropriate investment signals for market participants. (See Section VII.C and Rec #2023-2)

Operations of PAR-Controlled Lines between New York City and Long Island

While most phase angle regulators (PARs) are operated to reduce production costs, several operate under bilateral contract terms regardless of economic efficiency. The most significant inefficiencies in this report are associated with two transmission lines that normally transfer up to 300 MW from Long Island to New York City under a wheeling agreement between Consolidated Edison (“ConEd”) and Long Island Power Authority (“LIPA”).

In 2025, operation of these lines under the wheeling agreement resulted in power flowing in the inefficient direction during approximately 77 percent of hours, which **increased**: (a) the production costs by an estimated \$12 million; (b) CO₂ emissions by an estimated 423 thousand tons; and (c) NO_x emissions by an estimated 498 tons.

As intermittent renewables are integrated into New York City and Long Island, the operational flexibility of these lines will become increasingly important, particularly if they can reduce renewable curtailment. Accordingly, this report recommends that NYISO work with the parties to the ConEd-LIPA wheeling agreement to explore potential changes that would enable more efficient use of these facilities (See Appendix Section III.I and Rec #2012-8.)

When the lines are scheduled more efficiently or in a way that reduces congestion, they produce congestion surpluses in the day-ahead market. In 2025, this resulted in approximately \$8 million in congestion surpluses. Under current market rules, such surpluses are socialized among all transmission owners in proportion to their TCC revenues, rather than being allocated to the facility owners responsible for scheduling them. This weakens incentives to work with the NYISO to schedule them efficiently. The following subsection discusses a recommendation to allocate day-ahead congestion surpluses and shortfalls to individual facilities using a flow-based approach. This would strengthen incentives to schedule the lines between New York City and Long Island efficiently.

Allocation of Day-Ahead Congestion Residuals

Day-ahead congestion shortfalls and surpluses, known as “residuals”, arise when day-ahead network capability differs from the capability modeled in the TCC auctions. Allocating these residuals on a “cost causation” basis is generally beneficial because it provides efficient financial incentives for Transmission Owners (TOs) to maintain equipment, configure the transmission system to minimize congestion, and schedule outages when they are least likely to increase congestion.

Currently, most residuals resulting from “Qualifying” changes in modeled transfer capability between TCC auctions and day-ahead markets are allocated to the responsible TOs. However, TOs receive any remaining shortfalls and surpluses in proportion to their TCC auction revenues. In 2025, this approach allocated \$62 million in surpluses and \$31 million in shortfalls in proportion to TCC auction revenues. We estimate that: (a) \$33 million in surpluses was associated with incremental transfer capability from recent Segment A and Segment B Public Policy Transmission Projects; (b) \$21 million in surpluses resulted from shifting generation and load patterns, primarily on transmission paths in Long Island and New York City and across external interfaces; and (c) \$8 million in surpluses was related to the use of 901/903 lines between Con Ed and LIPA (as discussed in the previous subsection). This report also provides a breakdown of shortfalls that associates the vast majority with specific operational causes.

The current socialized allocation methodology does not align with the underlying causes of these congestion residuals, which weakens incentives for TOs to operate their transmission equipment efficiently and, in some cases, encourages overselling transmission capability in the TCC auctions. Therefore, we recommend that NYISO revise the allocation of day-ahead congestion residuals to be based on changes in scheduled utilization of the transmission system between the TCC auctions and the day-ahead market. This adjustment would enable transmission owners to recover the value of transmission scheduled in the day-ahead market, even if the capacity was not fully sold in the TCC auctions. (See Section VIII.B.2 and Recommendation 2023-1).

Participation in the NYISO Market by Demand Side Resources

NYISO has sought to transition capacity-selling loads from the legacy emergency demand response (“SCR”) program to the new DER participation model, but the current rules impose significant burdens on DERs beyond those imposed on generators that sell capacity. DERs are treated as fully dispatchable resources able to follow 5-minute instructions, but they cannot reflect commitment parameters (e.g., curtailment start cost and minimum curtailment duration) in their energy bids. As a result, DERs lack bidding tools available to generators to limit their schedules so they align with the cost of curtailing and the flexibility of their load.

FERC Order 745 required demand response resources participating in wholesale electric markets to be paid LBMP for curtailment when the LBMP exceeds a “net-benefits test” threshold

(“NBT”). This gives demand response a double benefit for curtailing because it both avoids the cost of purchasing electricity and is paid the LBMP as if it produced energy. This also violates a core market principle that payments to sellers be funded by receipts from buyers. Consequently, NYISO customers must bear the cost of LBMP payments through uplift. In addition, demand reductions are measured against a baseline that often deviates significantly from the load’s actual consumption before curtailment. These inefficient market rules have led to 81 GWh of inefficient over-consumption (where loads have an incentive to consume when their value of consumption is lower than the LBMP), 6 percent of all demand reductions being phantom curtailment (where the baseline rules give them credit for curtailing more load than they were capable of consuming), and \$9 million of uplift costs from double-compensation. Eliminating LBMP payments to DERs for curtailing would require changes in Commission policy. (see Section XII).

Overview of Recommendations

Our analysis indicates that NYISO’s electricity markets performed well in 2025, although we recommend additional enhancements. Some recommendations address emerging issues that will become increasingly important as the system evolves and the State moves forward with its clean energy policies.

The first table below summarizes our high-priority recommendations. The majority of our recommendations were made prior reports, but we make five new recommendations in this report. In general, the recommendations that are designated as “high priority” are those that produce the largest economic efficiencies by lowering production costs of satisfying the system’s needs or improving incentives for participants to make efficient long-term decisions. These are often foundational changes with effects that are difficult to quantify because they fundamentally alter incentives in some aspect of the market.

The second table provides a list of issues that caused significant inefficiencies or misallocation of costs or revenues that we have sought to address with one of our recommendations. These are grouped into: “Misallocated Costs” which include situations where a specific category of wholesale market cost is allocated in a manner that is clearly inconsistent with cost-causation principles or beneficiary-pays principles, “Payment for Non-Services” which include when a resource is compensated for contributing to satisfying a particular reliability or operational requirement but does not actually make a contribution, and “Inefficient Operations Due to Incentives or Market Rules” which include situations where a valuable asset is operated in a manner that raises production costs due to an inefficient incentive or a market rule requiring unnecessary actions. This does not include a quantitative estimate for every recommendation.

Section XIII provides a complete list of recommendations and a detailed discussion of each. In total, we have 23 outstanding recommendations that are discussed in that section.

High Priority Recommendations in the 2025 SOM Report

Number	Section	Recommendation	NYISO Project Scope: (2026 / 2027)
<i>Capacity Market – Increasing Recognition of Reliability Needs in the Market</i>			
2025-1	V.B.4, III.A, IX.F	Address the “planning-market gap” by: a) Using transparent criteria to develop planning assumptions and calibrate the Level of Excess, and b) Improving GADS compliance monitoring	--
<i>Capacity Market – Avoiding Over-Payment to Resources with Lower Reliability Value</i>			
2025-2 & 2022-4	IX.B.1 & IX.B.2	Enhance locational pricing of capacity by: • Using the CAF framework to reflect intrazonal deliverability limitations (near-term) & • Implementing more granular capacity zones and a dynamic process for updating them (long-term)	--
2022-1	IX.C	Compensate capacity suppliers based on their contribution to transmission security needs when they determine the locational capacity requirements	<i>Reliability Attribute Based Capacity Pricing:</i> (-- / Market Design Complete) ⁶
2022-2	IX.D	Establish seasonal capacity requirements, CAFs, and demand curves and adjust derating factors to allow CARC/UDR elections close to the Capability Period	<i>Winter Reliability Capacity Enhancements:</i> (Development Complete / Deployment) ⁷
<i>Energy Market Enhancements – Pricing and Performance Incentives for Flexible Resources</i>			
2015-16 & 2016-1	XIII.B	Dynamically adjust operating reserve requirements to align with reliability needs & Develop rules for efficient pricing and settlement when operating reserve suppliers provide congestion relief	<i>Dynamic Reserves:</i> (Development Complete / Development Complete) ⁸
2024-1	VII.D, VIII.D.1	Use the reserve market to satisfy reserve needs driven by intrazonal constraints	--
2017-2	VII.A	Modify operating reserve demand curves to improve shortage pricing and ensure NYISO reliability	--

⁶ It is unclear whether the proposed project will address the issues driving this recommendation.

⁷ The Winter Reliability Capacity Enhancements will provide a partial solution to this recommendation. See Section B discussion of Recommendation 2022-2 for details.

⁸ See Section B discussion of Recommendation 2015-16 for a list of concerns with the current project scope.

Table of Misallocated Costs, Payment for Non-Services, and Inefficient Operations in 2025

Issue	Annual Amount	Recommend / SOM Section
Misallocated Costs:		Total: \$221M
(1) Local reserve charges are allocated statewide.	\$40M	2015-16
(2) Uplift from OOM dispatch & commitment for local reliability allocated statewide.	\$16M	2018-1
(3) LSEs of SCRs and DERs avoid ICAP obligations for qualified DR capacity, shifting obligations to other LSEs in the subzone.	\$34M	2025-3
(4) Largest two NYCA contingencies not paying for impact on reserve requirements, shifting costs statewide.	\$38M	2015-16
(5) Day-ahead congestion surpluses allocated based on TCC auction revenue rather than to the responsible party.	\$62M	2023-1
(6) Day-ahead congestion shortfalls allocated based on TCC auction revenue rather than to the responsible party.	\$31M	2023-1
Payment for Non-Services:		Total: \$78M
(1) Largest two contingencies in NYC capacity market are paid for capacity at the top of the unit when LCRs are based on TSL even though this capacity does not help satisfy the underlying reliability need.	\$12M	2022-1
(2) SCRs paid for capacity when LCRs based on TSL even though this capacity does not help satisfy the underlying reliability need.	\$14M	2022-1
(3) Long Island reserves providers are paid the SENY reserve prices instead of the marginal value of Long Island reserves, which is much lower.	\$18M	2019-1
(4) Uplift from paying DERs and SCRs as if they generate electricity.	\$34M	See comments in RM10-17
Inefficient Operations Due to Incentives or Market Rules:		Total: \$41M
(1) Lines between New York City and Long Island scheduled to flow in the high-cost area to the low-cost area.	\$12M	2012-8
(2) NYCA reserve requirements are inefficiently high because they exceed the schedules of the largest two contingencies	\$25M	2015-16
(3) NYC reserve requirements are inefficiently high given the schedule of the largest supply contingency	\$4M	2015-16

I. INTRODUCTION

This report assesses the efficiency and competitiveness of New York’s wholesale electricity markets in 2025.⁹ The NYISO operates competitive wholesale markets to satisfy the electricity needs of New York. These markets include:

- Day-ahead and real-time markets that simultaneously optimize scheduling and pricing of energy, operating reserves, and regulation;
- A capacity market that ensures the NYISO markets produce efficient long-term economic signals that guide decisions to invest in new and existing generation, transmission, and demand response resources (and/or retire uneconomic existing resources); and
- A market for transmission rights that allows participants to hedge the congestion costs associated with using the transmission network while helping to recoup the embedded cost and operations and maintenance costs of the transmission system.

The energy and ancillary services markets establish prices that reflect the value of energy at each location on the network. They deliver significant benefits by coordinating the commitment and dispatch of generation to ensure that resources are started and dispatched each day to reliably meet the system’s demands at the lowest cost. The coordination provided by the markets is essential because of the physical characteristics of electricity. This coordination affects not only the prices and production costs of electricity, but also the reliability with which it is delivered.

The NYISO markets have several key features that are designed to allow the power of markets to satisfy the needs of the system efficiently, including:

- Simultaneous optimization of energy, operating reserves, and regulation, which efficiently allocates resources to provide these products;
- Locational requirements in its operating reserve and capacity markets, which play a crucial role in signaling the need for resources in transmission-constrained areas;
- Capacity demand curves that reflect the value of incremental capacity to the system and provide for increased stability in market signals;
- Ancillary services demand curves, which contribute to efficient prices during shortages when resources are insufficient to satisfy all of needs of the system;
- A real-time commitment system (i.e., RTC) that commits quick-start units (that can start within 10 or 30 minutes) and schedules external transactions. RTC runs every 15 minutes, optimizing over a two-and-a-half hour period.

⁹ NYISO MST 30.10.1 states: “The Market Monitoring Unit shall prepare and submit to the Board an annual report on the competitive structure of, market trends in, and performance of, other competitive conditions in or affecting, and the economic efficiency of, the New York Electric Markets. Such report shall include recommendations for the improvement of the New York Electric Markets or of the monitoring, reporting and other functions undertaken pursuant to Attachment O and the Market Mitigation Measures.”

- A market scheduling system (i.e., Coordinated Transaction Scheduling) to coordinate an economic evaluation of interchange transactions between markets 15 to 30 minutes ahead of when the power flows in real-time.
- Mechanisms that allow gas turbines and demand-response resources to set energy prices when they are needed, which is essential for ensuring that price signals are efficient during peak demand conditions.
- A real-time dispatch system (i.e., RTD) that runs every five minutes and optimizes over a one-hour period, allowing the market to anticipate the upcoming needs and move resources to efficiently satisfy the needs.

These market designs provide substantial benefits to the region by:

- Ensuring that the lowest-cost supplies are used to meet demand in the short-term; and
- Establishing transparent price signals that facilitate efficient forward contracting and govern generation and transmission investment and retirement decisions in the long-term. Relying on private investment shifts the risks and costs of poor decisions from New York's consumers to investors.

As State policy-makers promote environmental goals by investing in electricity generation and transmission,¹⁰ the markets will adapt if the generation fleet shifts from being primarily fossil fuel-based, controllable, and centralized to having higher levels of intermittent renewables and distributed generation. The NYISO markets facilitate these investments by providing:

- Useful information regarding the value of electricity and cost of production throughout the State, enabling clean energy procurements to select more efficient proposals and transmission planning processes to identify needs appropriately and select the most efficient solutions; and
- Critical incentives not only for placing new resources where they are likely to be most economical and deliverable to consumers but also for keeping conventional resources that help integrate clean energy resources while maintain system reliability.

However, policies to promote renewable generation and other zero-emission resources are lagging far behind the State's ambitious goals. In the meantime, the State's existing generation fleet is becoming older and more costly to maintain, while load growth is increasing after more than a decade of decline. Given the slow pace of clean energy investment, it will be impossible to maintain reliability without some investments in new fossil fuel generating capacity.

It is critical for the NYISO markets to improve alignment between the market incentives and the reliability needs of the system to signal the need for adequate supply while minimizing costs to consumers. Section XIII of the report provides a number of recommendations that are intended to achieve these objectives.

¹⁰ For instance, see the New York's Climate Leadership and Community Protection Act ("CLCPA").

II. OVERVIEW OF MARKET TRENDS AND HIGHLIGHTS

This section discusses significant market trends and highlights in 2025. It evaluates energy and capacity costs, wholesale and retail electricity costs, fuel prices, generation patterns, demand patterns, and ancillary services prices.

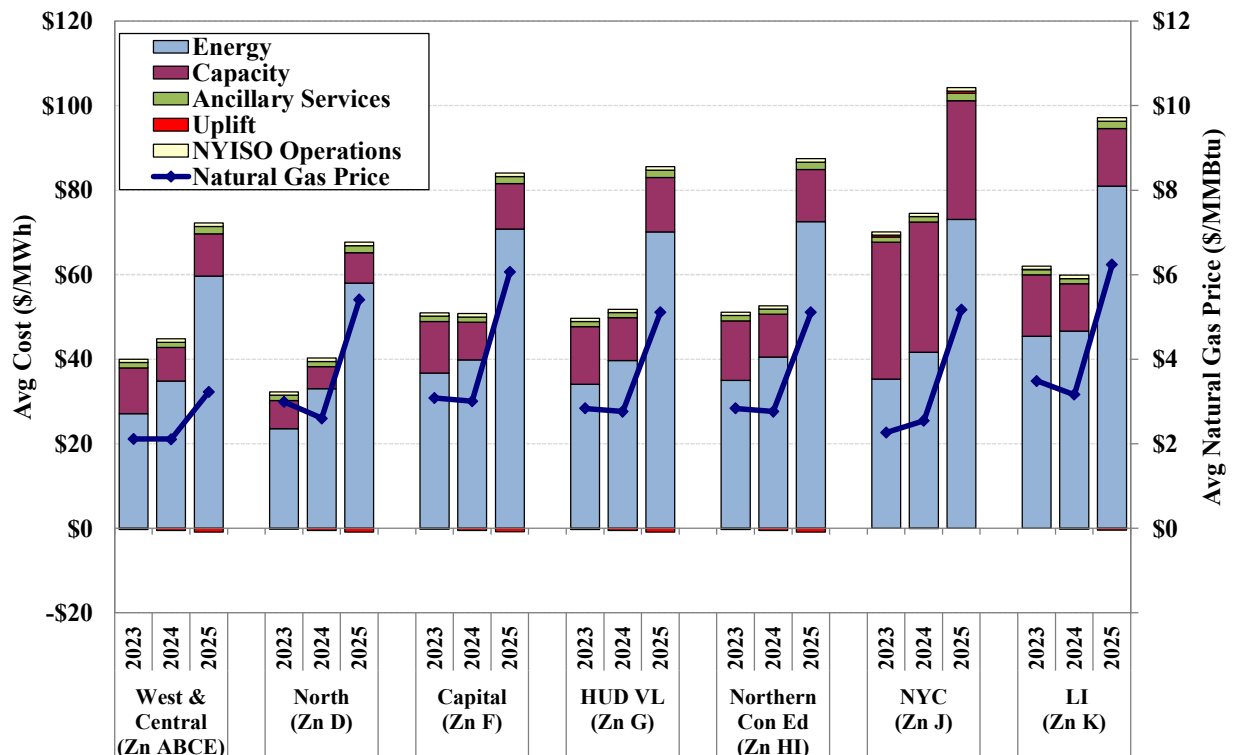
A. Wholesale and Retail Market Costs

Figure 1 summarizes wholesale market costs to consumers over the past three years by showing the all-in price for electricity, which reflects the average cost of serving load from the NYISO markets. The major components of this metric include:

- The energy component is the load-weighted average real-time energy price.
- The capacity component is based on monthly spot auction clearing prices and capacity procured in each area, allocated over energy consumption in that area.

All other components are the costs divided by the real-time load in the area.¹¹

Figure 1: Average All-In Price by Region
2023 - 2025



¹¹ Section I.A of the Appendix provides a detailed description of the all-in price calculation.

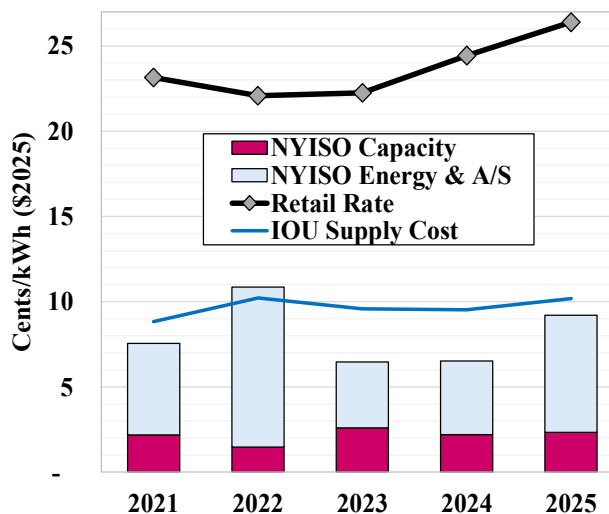
In 2025, average all-in prices ranged from \$66 per MWh in the West Zone to over \$104 per MWh in New York City. All-in prices rose in most regions primarily because of higher energy costs, which rose due to the following factors:

- Natural gas prices rose in all regions, especially in eastern New York. Gas prices rose by ~60 percent in western New York and 107-to-112 percent in eastern New York.
- The trend of increasing net exports to Quebec continued as the average rose by 400 MW from 2024 to 2025, while hydroelectric production in New York was down 330 MW year-over-year.
- CO₂ emissions allowance costs rose due to higher Regional Greenhouse Gas Initiative (“RGGI”) prices. RGGI prices rose by 8 percent from the prior year, averaging ~\$23 per ton in 2025, which translates to approximately \$10.20 per MWh in incremental costs for a typical gas-fired combined cycle generator.
- Weather conditions, both warmer in the summer and colder in the winter, contributed to an increase (1 percent) in average load and a 10 percent increase in peak load.
- Congestion price differences increased, especially across the Central East interface and in eastern New York load zones.

Capacity costs also rose by 21 to 36 percent across the state, except in New York City where capacity costs fell by 8 percent. The reasons for changes are discussed in Section IX.A, but they include a higher statewide Installed Reserve Margin (“IRM”) and a lower Locational Capacity Requirement (“LCR”) in New York City.

Figure 2 compares average residential retail electric rates in New York with average energy and capacity prices from the NYISO market from 2021 to 2025 in constant 2025 dollars. The figure also shows the average wholesale supply component of residential bills reported by the state’s investor-owned utilities (IOUs). NYISO combined energy and capacity prices averaged 34 percent of retail residential rates over the period, capacity prices alone average 9 percent. The majority of retail charges come from the combination of transmission and distribution cost of service rates, taxes, and other surcharges. The supply charges of IOUs are more stable but generally higher on average than wholesale prices due to the effects of distribution losses, short-term hedges and long term supply contracts entered into by IOUs. Statewide residential retail rates have risen by 3.2 cents per kWh from 2021 to 2025 in real 2025 dollars, while IOU supply costs rose by 1.4 cents over the same period.

Figure 2: Statewide Residential Retail and Wholesale Prices

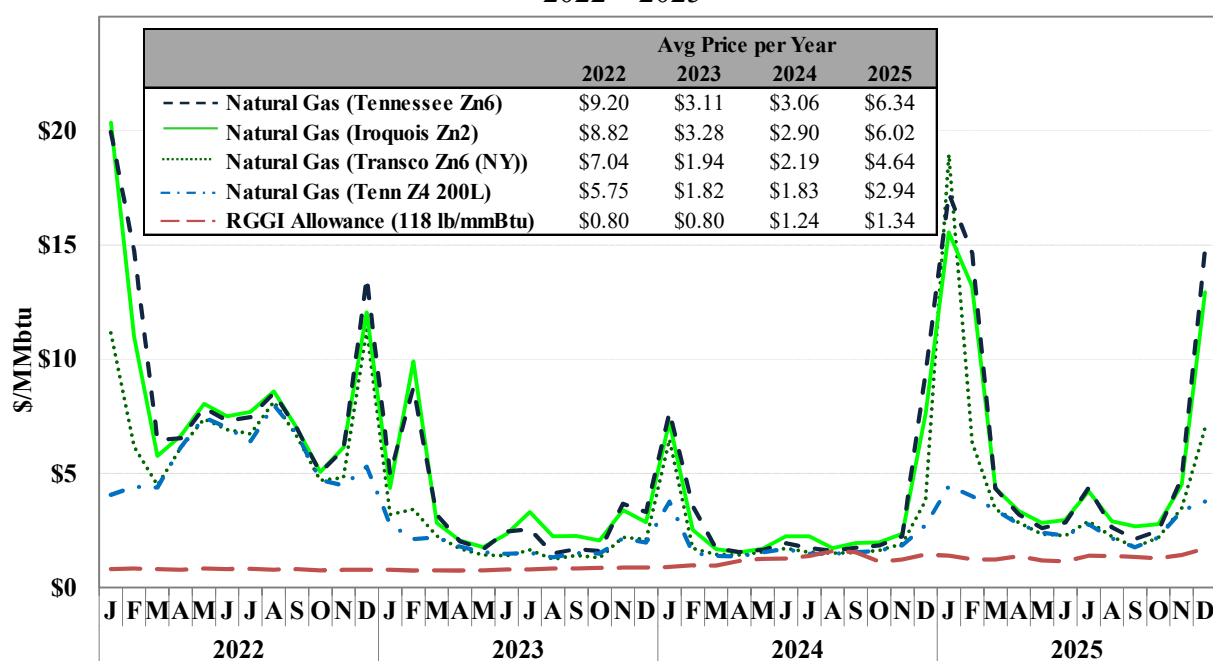


B. Prices for Natural Gas and Emission Allowances

Fossil fuel price fluctuations have been the primary driver of changes in wholesale energy prices historically. Figure 3 displays monthly and annual natural gas prices from 2022 to 2025 for several key indices and the average monthly RGGI allowance prices converted to units of \$ per MMBtu based on the emission rate of a gas-fired generators.¹²

Average annual natural gas prices for trading hubs around New York rose sharply in 2025 from the prior two years. Colder winter weather and hotter summer weather contributed to higher gas prices in 2025. Cold winter weather affected natural gas prices in eastern New York more severely than in western New York. Eastern New York gas prices rose 107 to 112 percent year-over-year.

Figure 3: Average Natural Gas Prices and RGGI Allowance Prices
2022 – 2025



Production Costs for Gas-fired Generators

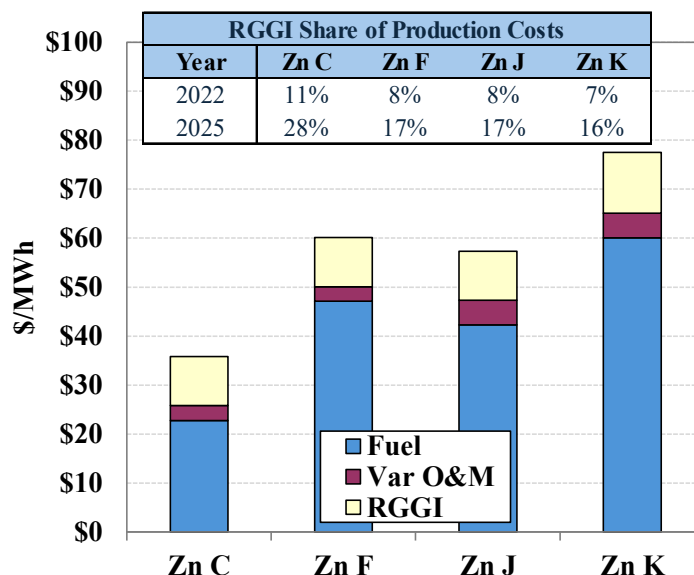
Natural gas costs typically account for the majority of production costs of gas-fired generators in New York. Congestion on the gas pipeline network translates into significant differences in production costs of gas-fired generators in different regions of the state. These cost differences contribute to congestion on the electric transmission system, especially between central and

¹² A RGGI allowance satisfies the compliance obligation for one short ton of CO₂ emissions. We convert to \$ per MMBtu by multiplying the RGGI allowance price by the 118 lbs/MMBtu emissions rate ÷ 2000 lbs/ton.

eastern New York. Figure 4 summarizes the production costs for gas-fired generators that are typically on the margin in most hours in four zones in 2025.¹³

Fuel costs accounted for an estimated 64 to 78 percent of the production costs on gas-fired generators in 2025. The next largest contributor RGGI allowance costs. Since RGGI prices are uniform across the state, allowance costs made up a larger share of total production costs outside eastern New York. In the Central Zone, RGGI allowances costs averaged 28 percent of production costs for a typical gas generator, up from 11 percent in 2022. Rising RGGI allowance prices have contributed to rising energy costs in New York over the past several years.

**Figure 4: Typical Production Costs
Gas-Fired Generators, 2025**



C. Demand Levels

Demand is a key driver of wholesale market outcomes. Higher demand levels drive high-cost peaking resources to set prices more frequently, and congestion into load centers generally increases as demand levels rise. Reserve shortages occur more frequently during periods of high demand, which can drive average prices up despite the relative brevity of those events. Annual peak demand forecasts are used to determine the MW-requirements in the capacity market.

Table 1 shows the following load statistics for the New York Control Area (NYCA) since 2016: (a) annual summer peak; (b) reconstituted annual summer peak; (c) annual winter peak; and (d) annual average load. The reconstituted summer peak incorporates any demand response that was activated during the peak load hour, either by utility deployment or by the NYISO. Therefore, a reconstituted peak load gives a truer sense of the supply resource requirements.

The average load across the system was 1 percent higher in 2025 than in 2024, but load is down over the past decade due to slow economic growth and the effects of energy efficiency and

¹³ Zones C, F, and J (i.e., Central, Capital, and New York City zones) assume a gas-fired combined cycle generator with 7,500 Btu/kWh heat rate. Zone K (i.e., Long Island) assumes an average heat rate of 8,750 Btu/kWh due to more frequent operation of higher heat rate units there. Variable Operating and Maintenance (“Var O&M”) costs are assumed to be \$3 per MWh in Zones C and F, and \$5 per MWh in Zones J and K. All regions assume a CO₂ emissions rate of 118 lb/MMBtu.

behind-the-meter generation. In 2025, peak demand rose significantly (by 10 percent) from 2024 because of a strong heat wave in late-June. The official peak demand was the highest observed since 2018. The reconstituted (for the effects of demand response) peak load in 2025 was the highest in over a decade. NYISO estimated that demand response resources shaved nearly 1 GW off the peak load on the hottest day (June 24, 2025). As of June 2025, only

**Table 1: Peak and Average Load Levels for NYCA
2016 to 2025**

Year	Load (GW)			
	Summer Peak (as Reported)	Summer Peak (Reconstituted)	Winter Peak	Annual Average
2016	32.1	32.5	24.2	18.3
2017	29.7	29.7	24.3	17.9
2018	31.9	32.5	25.1	18.4
2019	30.4	30.4	24.7	17.8
2020	30.7	31.2	22.5	17.1
2021	30.9	31.3	22.5	17.3
2022	30.5	31.2	23.2	17.4
2023	30.2	30.5	23.4	16.8
2024	28.9	29.8	23.1	17.2
2025	31.9	32.9	23.5	17.3

one demand response resource had transitioned from the NYISO Special Case Resource (“SCR”) program to the Distributed Energy Resource (“DER”). Unlike SCRs, demand response resources participating as DERs are often held in reserve (i.e., consume energy while being curtailable on short notice) rather than shut down under peak conditions. As more resources transition to the DER program, we may observe higher peak load levels.

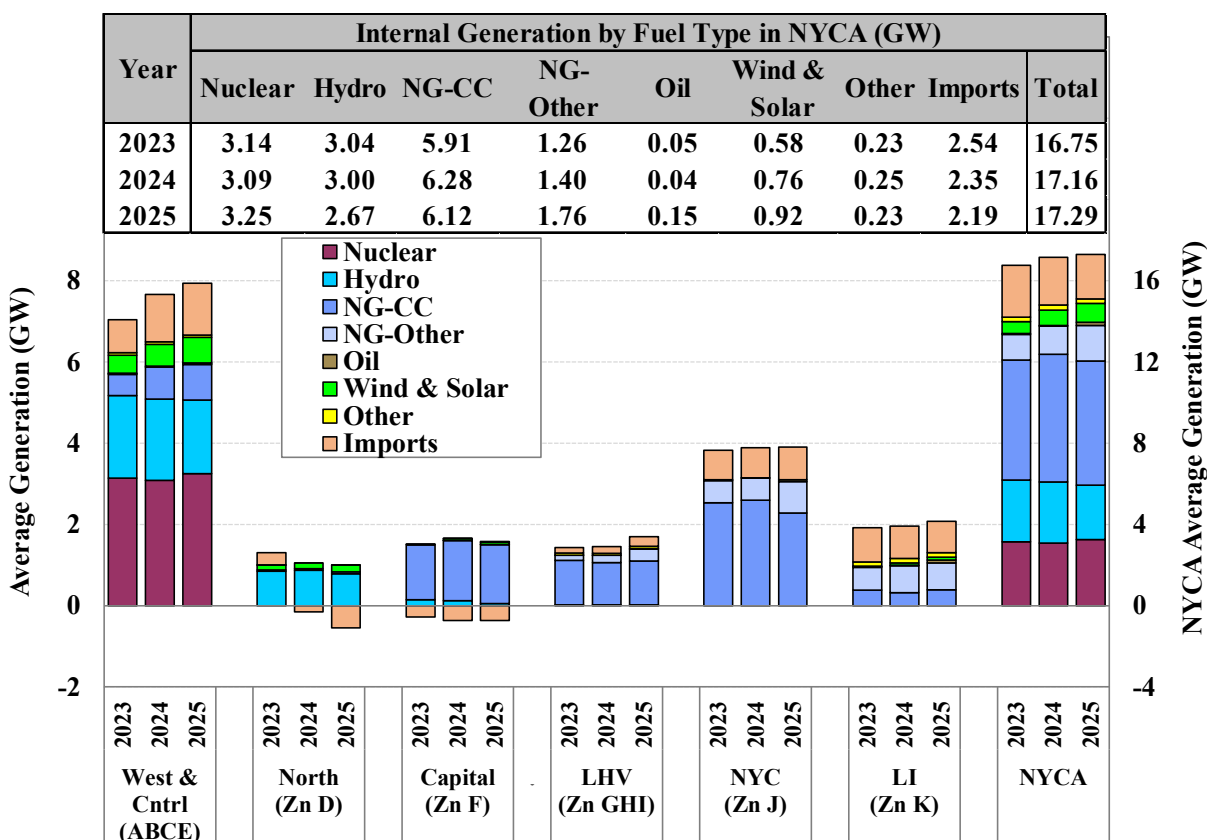
D. Generation by Fuel Type

Additions of new renewable generation and reduced net imports have led to changes in the mix of fuels used to generate electricity. Figure 5 displays annual generation by resource type from 2023 to 2025 (including net imports).¹⁴

More than half of all internal generation has come from gas-fired resources for the past three years, comprising 52 percent of the total in 2025. Output from Wind and Solar generation resources increased by 60 percent from 2023 to 2025 primarily due to the entry of several new resources since late 2023. However, total wind and solar utility-scale generation remained relatively low (accounting for 5 percent of the total). Net imports fell by 160 MW from 2024 because of continued increases (~390 MW) in net exports to Quebec. Flows from Quebec, which averaged over 1.3 GW in the import direction as recently as 2021, have shifted to an average of 550 MW in the export direction in 2025.

¹⁴ Net imports are assigned as follows: (a) Ontario imports to West & Central; (b) Quebec imports to North; (c) imports over the primary PJM interface are split 7, 47, and 46 percent to NYC, Lower Hudson Valley, and West & Central, respectively; (d) net imports over the primary ISO-NE interface are split 55 and 45 percent Capital and Lower Hudson Valley, respectively; and (e) Scheduled Lines where they inject (i.e., Cross-Sound and Neptune Cables, and 1385 Line in Long Island and the HTP and Linden VFT Lines in NYC).

**Figure 5: Generation by Type and Net Imports to New York
2023 - 2025**



E. Ancillary Services Markets

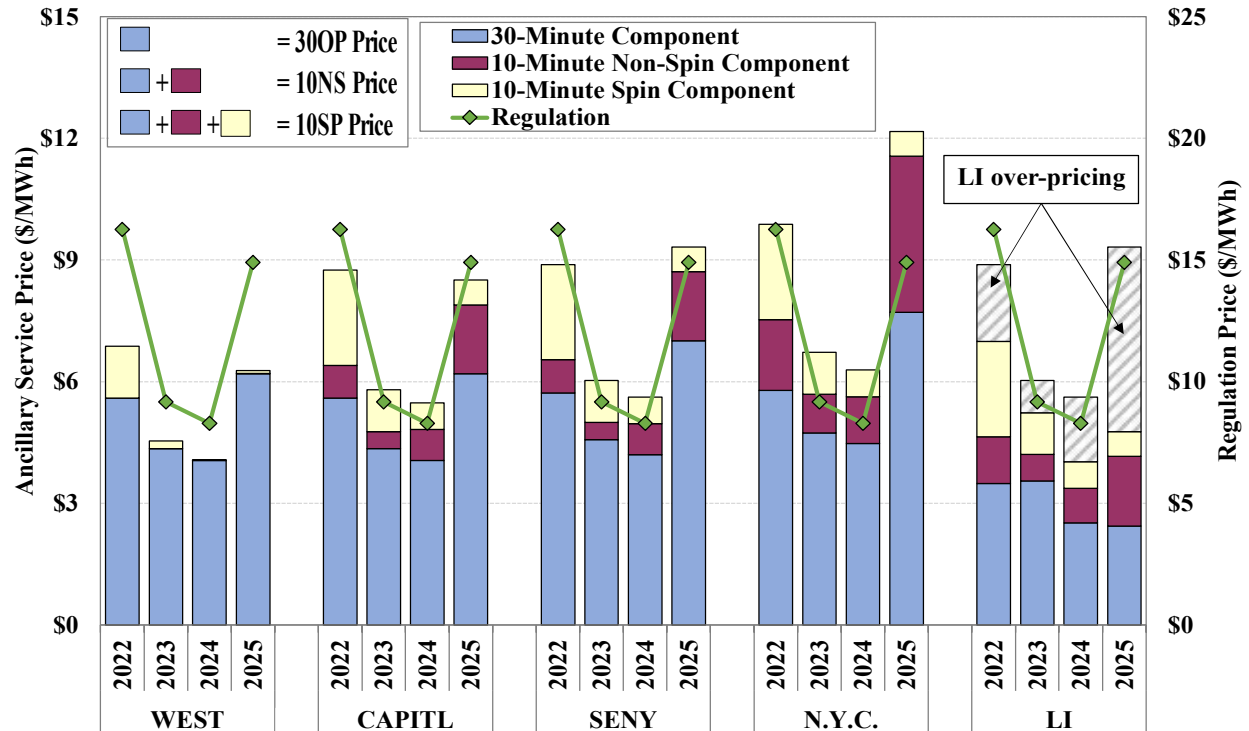
Scheduling of ancillary services is co-optimized with energy scheduling because much of the cost of providing ancillary services is the opportunity cost of not providing energy when it otherwise would be economic to produce. Co-optimization ensures that these opportunity costs are efficiently reflected in Location Based Marginal Prices (LBMPs) and reserve prices. Despite their small contribution to the overall system costs, the ancillary services markets provide additional energy and ancillary services revenues that reward resources that have high rates of availability, especially peaking units.

Figure 6 shows the average prices of the four ancillary services products by location in the day-ahead market in each of the past four years.¹⁵ There is a peculiar provision in NYISO's tariff which directs it to settle with Long Island generators at the reserve clearing prices for Zones GHI rather than the reserve clearing prices that would ordinarily result from the scheduling of Long Island resources. The figure shows the clearing prices for Long Island that would normally

¹⁵ See Appendix Section I.I for additional information regarding the ancillary services markets and detailed description of this chart.

result from the scheduling process, and it shows the amount of over-pricing that results from the requirement to settle at Zones GHI prices for the 10-minute spinning reserve product.

Figure 6: Average Day-Ahead Ancillary Services Prices
2022 - 2025



Reserve clearing prices rose in all regions from 2024 to 2025 due to higher loads, higher fuel costs (which raise opportunity costs of providing reserves), reductions in reliability commitments (which result in additional online reserves) in New York City, and transmission outages that reduced the availability of low-cost generation. In particular, the average price spread between Zones GHI and New York City for 10-minute reserves rose from \$0.67 per MWh in 2024 to \$2.85 per MWh in 2025.

The retirement of peaking generation in New York City has led to significant differences between the value of reserves on Long Island and other areas in Southeast New York. This has exacerbated differences between the reserve payments to Long Island generators and the value of those reserves. We estimate over-payments of \$17 million to Long Island generators for reserves in the day-ahead market in 2025 because of the rule to settle them at the prices for Zones GHI.

Local reserve requirements for eastern New York, southeast New York, New York City, and Long Island were binding more frequently in 2025, raising a long-standing concern with the allocation of reserve costs. Instead of allocating local reserve charges to the areas that benefit from additional reliability, local reserve charges are allocated statewide. In 2025, a total of \$39.7 million of local reserve charges were allocated statewide rather than to the loads that benefit.

III. CAPACITY PERFORMANCE DURING PEAK SUMMER CONDITIONS

Generator retirements and limited new supply investment in recent years has increased the strain on older existing generators. Increased load growth may exacerbate this issue in the coming years. NYISO has adopted multiple reforms to ensure that the capacity market is structured to appropriately compensate resources for their reliability contribution.

This section analyzes the availability of several resource classes during the hottest periods of the summer of 2025 and considers whether their contribution to reliability is commensurate with the qualification and accreditation of their capacity. This section also evaluates the performance of capacity sold from other control areas, which have been called more frequently in recent years, and analyzes the performance of large curtailable loads during these periods.

A. Fossil Fuel Generator Performance

NYISO experienced several days of high load in summer 2025 requiring nearly all generators to be scheduled for energy or reserves. These periods offer opportunities to assess the reliability of generation capacity that is otherwise used rarely or not at all (outside of scheduled tests). In this subsection, we find that less capacity was available in peak periods than NYISO's metrics for estimating generator availability would have predicted. These metrics affect generators' compensation and incentives (via their UCAP ratings) as well as NYISO's assessments of system reserve margin requirements and reliability.

It is challenging to assess whether generator reliability is overestimated in the market because: (a) there are few data points of very high load days, (b) many units remain unscheduled for energy on all but the highest load days, and (c) in some cases, 'planned' or 'maintenance' outages (which can in theory be rescheduled or recalled) mask situations where the unit really cannot be made available. Hence, we expect to refine the assessments found in this section as more data becomes available.

Background on Generator Performance Assessment

Fossil-fuel generators account for most output from internal resources in the NYISO. The vast majority of this production comes from natural gas-fired generation, but some oil-fired generation is necessary on the hottest summer days to meet peak loads. The last new fossil-fuel generator that entered the market did so in 2020. Since then, over 3.7 GW of existing fossil-fuel and nuclear generation has left the market through retirements and ICAP Ineligible Forced Outages (IIFO). The elimination of this capacity, which included both baseload and peaking resources, has increased the NYISO's reliance on many of the oldest generators that still exist in the market today. Indeed, the average capacity factor of generators at least 40 years-old in 2025 increased 84 percent from 2019 to 2025.

As capacity margins fall, it is important to evaluate the rules and procedures that determine:

- **Capacity Qualification:** The quantity of Installed Capacity (“ICAP”) that a generator is qualified to sell depends on its Dependable Maximum Net Capability (“DMNC”) value (provided that it has sufficient CRIS rights).
- **Capacity Accreditation:** The quantity of Unforced Capacity (“UCAP”) that a generator with no availability limitations can sell depends on its ICAP value times (1-minus-EFORd), where the EFORd is a measure of the generator’s likelihood of being forced out when needed.

The NYISO has procedures for testing generators to determine their DMNC values and uses NERC’s Generator Availability Data System (“GADS”) framework for estimating the EFORd of each generator. However, it is difficult to assess the results of these procedures and rules because peak conditions that threaten reliability occur very infrequently. Since the summer of 2025 had unusually hot weather, we examined the performance of each fossil fuel generator relative to its qualification and accreditation values.

Analysis of Generator Availability

We evaluated performance of existing fossil fuel generators during two heat waves in 2025: one from June 23-25 and one from July 28-30. We analyze the peak hours of each day (Hours 15 to 20) when ambient conditions are most severe, reporting performance in seven categories:

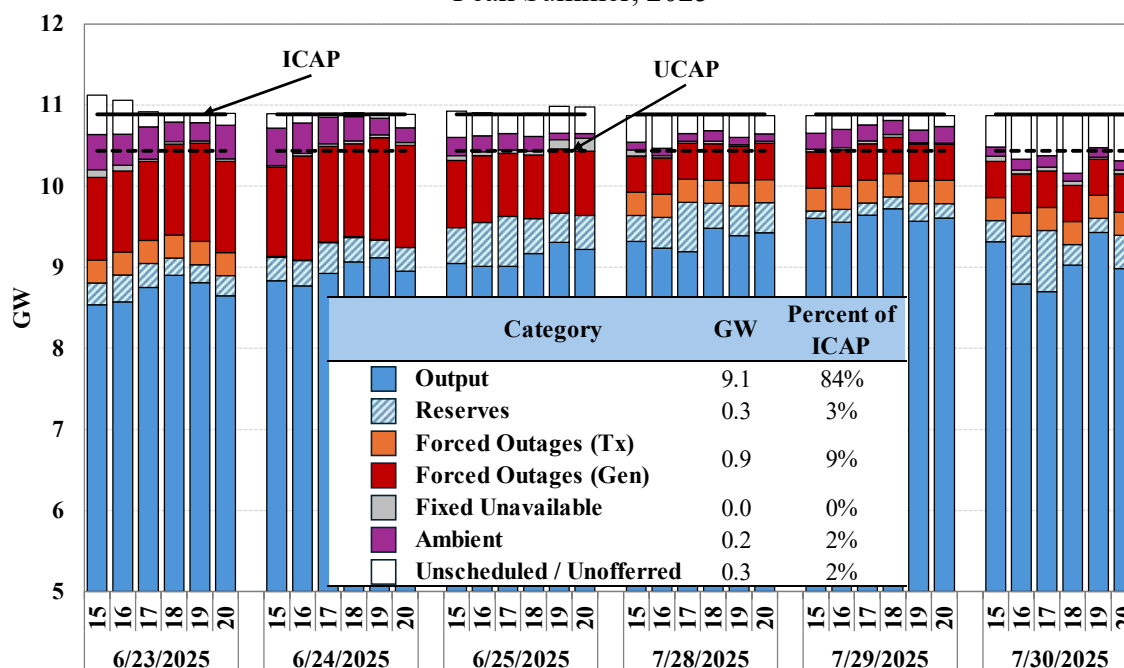
- **Output:** total hourly actual output produced by generators in each class.
- **Reserves:** total hourly operating reserves scheduled on generators in each class.
- **Ambient:** total hourly capacity unavailable to the market due to ambient conditions that were more severe than those assumed in the Dependable Maximum Net Capability (DMNC) tests used by generators to establish ICAP ratings.¹⁶
- **Outages (Tx):** total hourly capacity unavailable due to outages and derates on generators that occurred due to transmission limitations on the high side of the step-up transformer.
- **Outages (Gen):** total hourly capacity unavailable to the market due to outages and derates on generators that occurred due to equipment problems with the generators. This includes forced outages reported by generators through NYISO’s GADs system as well as capacity that was unavailable due to outages that we believe were incorrectly characterized as ‘planned’ rather than ‘forced’ (such as planned maintenance outages that extended substantially past their original schedules into the peak summer period, rendering the unit unavailable). We discuss the importance of improving outage logging and scheduling practices in Section IX.F.
- **Fixed Unavailable:** total hourly capacity unavailable to the market via economic dispatch due to the generators offering with a Self-Fixed bid type in real-time where the self-committed level was less than the Upper Operating Limit (UOL) of the generator.

¹⁶ See Section IX.E for details on ambient conditions and their impact on resource capabilities.

- Unscheduled / Unoffered:** total hourly capacity not scheduled for energy or reserves due to economics or because the capacity was not offered into the market without a known equipment failure or ambient condition to explain its unavailability. It is likely that much of this could have operated if needed, but some of these units likely failed to offer or offered at high prices to mask derated capacity or capacity with increased risk of outage.

We compared the availability of each class of resources against the expected availability from those units as derived from the Unforced Capacity (UCAP) ratings used to sell capacity into the NYISO ICAP Markets. The UCAP value is equal to the ICAP * (1 – EFORD) for a generator, where EFORD estimates the probability that the generator will be unavailable when otherwise in demand based on historic reporting of forced outages and forced derates. Ideally, the demonstrated capability of each resource class would closely align with the expected performance, i.e., the UCAP value. However, persistent, large discrepancies between the average performance of resources and the UCAP suggest that EFORD values do not adequately predict unavailability due to forced outages and derates during peak conditions.¹⁷ Figure 7 through Figure 9 show the results of our performance evaluation for three classes of fossil fuel generators: combined cycles, steam turbine generators, and fast-start peaking resources.

Figure 7: Performance Evaluation of Combined Cycles
Peak Summer, 2025



¹⁷ We discuss this further in Section IX.F.

Figure 8: Performance Evaluation of Steam Turbines
Peak Summer, 2025

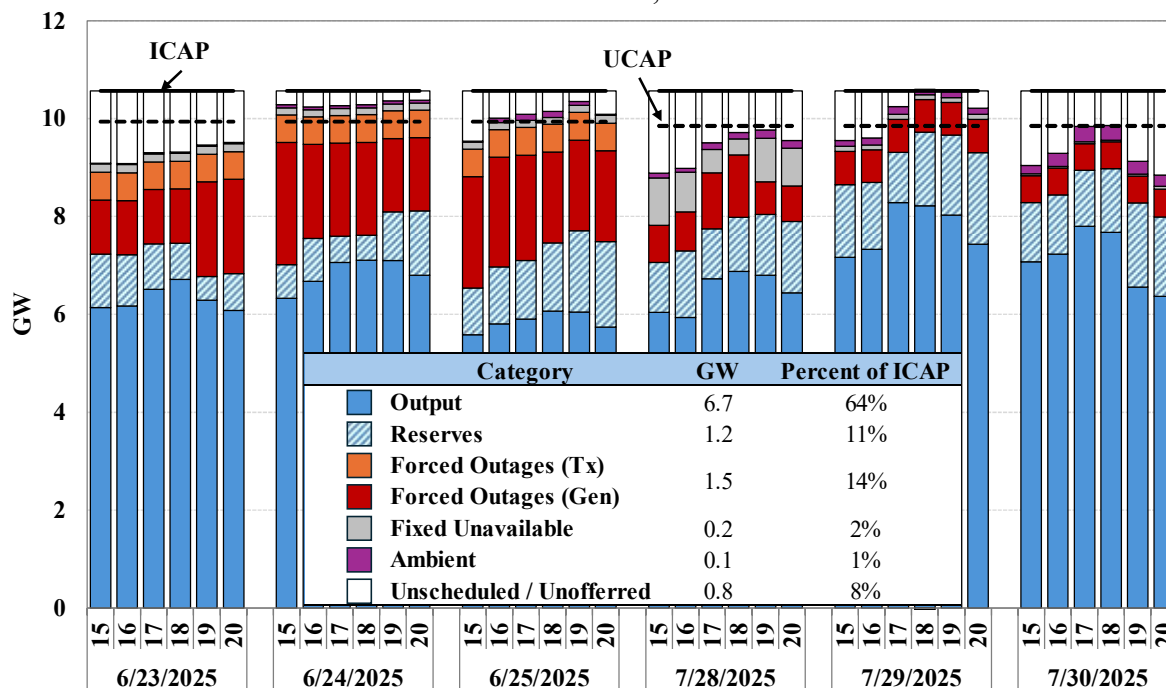
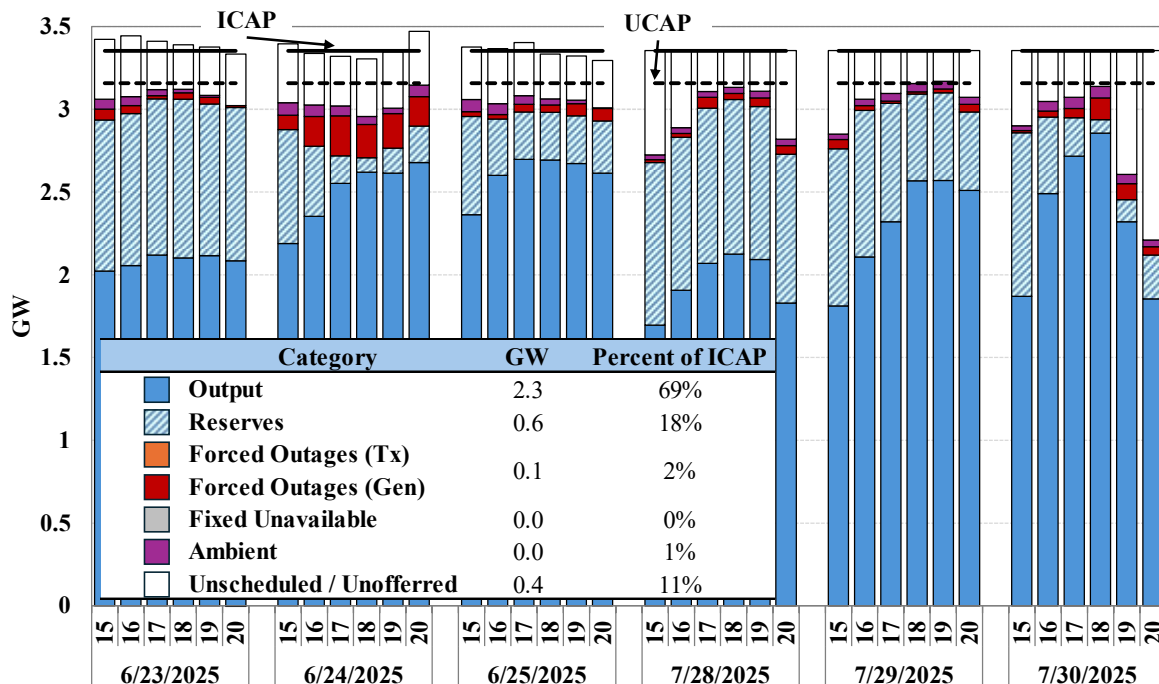


Figure 9: Performance Evaluation of Peaking Units
Peak Summer, 2025



The capacity evaluated (24.8 GW) across all three classes of generator had a weighted-average EFORd of 5.5 percent, but the unavailable rate due to forced outages and deratings averaged 10.1 percent during these heat waves. These results reinforce the importance of NYISO efforts to

improve the rules and procedures related to the qualification and accreditation of conventional generators. In particular:

- *Ambient water and air conditions* – These caused 2 percent of ICAP to be unavailable on average on these days. The NYISO has implemented changes to the DMNC determination process to help address this issue beginning in the 2026 Summer Capability Period.¹⁸ We will evaluate the adequacy of these changes in future reports.
- *Unscheduled Capacity* – This includes (a) slow-start capacity that was offered but not committed and (b) online capacity with latent reserves that were not scheduled.
- *Fixed Unavailable and Other Unoffered* – This includes capacity offered in the DAM but not scheduled and then not offered in real-time: (a) on the upper ranges of online units and (b) on peaking units.
- *Outages (Tx)* – This caused 2 percent of ICAP to be unavailable and includes outages caused by equipment on the high side of the generator step up transformer. The resource adequacy component of the IRM and LCR studies consider these outages, but the TSL calculation did not in the 2025/26 Capability Year. NYISO addressed this contributor to the planning-market gap when it set the TSLs for the 2026/27 Capability Year.
- *Outages (Gen)* – This caused 8 percent of ICAP to be unavailable and includes outages and deratings of generator equipment, which is higher than the weighted-average EFORD, of 5.5 percent. We reviewed submissions to GADS and found that 11 percent of this capacity was not reported as a forced outage or derate, suggesting that improper reporting to GADS likely contributes to underestimation of EFORD for some units.

Implications for Generator Qualification and Accreditation

Our analysis suggests that generators' current EFORD ratings overstate their availability during critical peak events. This leads to under-procurement of capacity to satisfy planning requirements and undermines confidence in the use of EFORD data from GADS for planning purposes. Under-estimated EFORD values are caused by a combination of outage reporting issues and calculation methodology:

Under-reporting of outages – generators sometimes do not accurately report or classify outages in GADS. This includes outages that are mischaracterized as 'planned' (so that they do not affect the unit's EFORD calculation), even when the outage would be more appropriately classified as 'forced' (e.g. in circumstances where the unit would not be able to operate if needed for reliability). Hence, we recommend NYISO help address the "planning-market gap" by improving GADS compliance monitoring. (see #2025-1b and Section IX.F)

Limited set of Demand Hours – a significant amount of capacity operates very infrequently or not at all outside of scheduled DMNC tests. This causes some units to have very low EFORD values based on a miniscule sample size of 'demand' hours, and creates incentives for such units to avoid operating in order to avoid revealing their actual reliability.

¹⁸ We discuss this further in Section IX.E.

Calculation of EFORD for Upper Output Ranges – Performance analysis during the heat waves highlighted that those classes of generator that are more likely to require the operations of ancillary equipment to operate in the highest operating ranges are more prone to forced outages and derates than the EFORD metric assumes. Common types of ancillary equipment or infrequently utilized operating practices that are required to access the upper ranges of many combined cycles and steam turbines include but are not limited to:

- Utilization of duct firing in a Heat Recovery Steam Generator (HRSG)
- Peak firing of combustion turbines
- Top heaters being taken out of service on steam turbines
- Overpressure tolerances of steam turbine control systems

The EFORD calculation does not make distinctions between a service hour at a low output level and those at high output levels. To the extent that higher operating levels of certain types of generators are less reliable but seldomly operated, EFORD may be insufficient to adequately translate ICAP to UCAP as currently utilized. We plan to analyze alternatives to the established EFORD methodology to more accurately calculate the marginal reliability value of generators based on operating ranges in upcoming quarterly State of the Market reports.

B. Non-Fossil Fuel Generator Performance

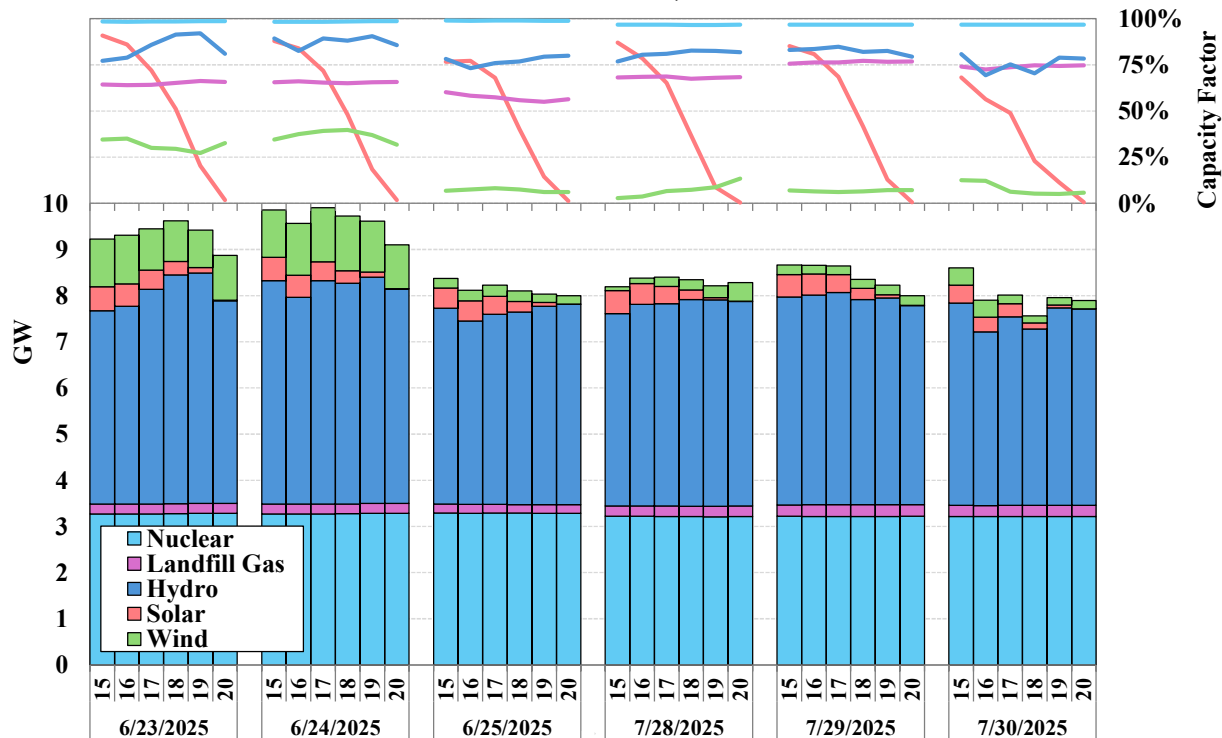
A significant minority of generation in the NYISO comes from non-fossil fuel generators. Although many of these resources are less controllable than traditional fossil units, the performance of this generation is critical to system reliability during peak summer conditions. Figure 10 displays the actual output of non-fossil fuel generators by fuel type during the same days and hours as covered in the analyses above. The upper portion displays the capacity factor of each class of resource to indicate how much of the installed capacity was utilized for energy on the peak summer days. Solar includes on front-of-the-meter (FTM) units in this analysis.

Nuclear and hydro resources make up 89 percent of the non-fossil fuel FTM generation during peak summer conditions. Capacity factors for both of those fuel types were high in summer 2025, with nuclear averaging 98 percent capacity factor and hydro averaging 81 percent. Nuclear output tends to be flat while most of the hydro capacity in the state is more flexible and frequently scheduled for reserves, leading it to have a lower capacity factor.

FTM solar resources had high capacity factors during the early-afternoon hours. However, solar generation fell from an average of 83 percent in Hour 15 to 1 percent in Hour 20. The increased penetration of solar resources, especially behind-the-meter units, in recent years has shifted the peak load hour later in the day. This highlights the importance of accrediting solar resources (and other technologies) based on the marginal reliability value of capacity.¹⁹

¹⁹ The CAFs for Solar generation ranged between 10.05 and 12.33 percent in the 2025/26 Capability Year.

Figure 10: Performance Evaluation of Non-Fossil Fuel Generators
Peak Summer, 2025



Wind generation performance was less predictable than the other fuel types mentioned. Capacity factors among the wind fleet reached highs of 40 percent in Hour 18 on June 24 and lows of 5 percent on July 30. Consequently, total wind output during peak summer hours ranged from a low of 153 MW to a high of 1.2 GW. The variability of wind output during critical peak events is captured in NYISO's resource adequacy modeling and the marginal Capacity Accreditation Factors of wind generators.²⁰

C. External Resource Capacity Performance

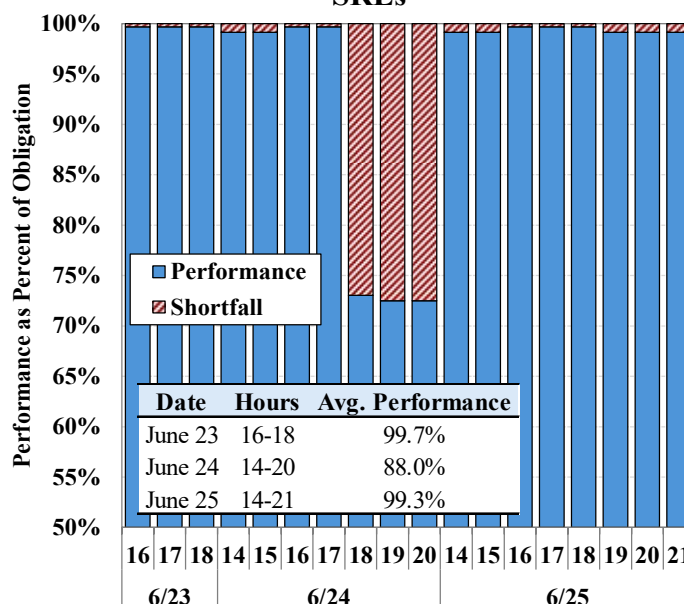
Capacity sales into the NYISO market from resources in external control areas were significant in 2025. In June 2025, roughly 7 percent of UCAP supply came from neighboring regions and was evenly split between imports to upstate and imports to downstate. As capacity margins in New York decline due to generator retirements and slow investment in new capacity, external capacity is being scheduled more frequently in the operating day. External ICAP resources are obligated to obtain sufficient transmission service to deliver when called upon in the regular Supplemental Resource Evaluation (SRE) process after the day-ahead market clears. Failure to transmit the energy to the NYISO can result in penalties.²¹

²⁰ The CAFs for Land-based Wind generation ranged between 16.61 and 18.20 percent in the 2025/26 Capability Year, while Offshore wind was 35.79 percent.

²¹ MST Section 5.12.12.2.

NYISO operators issued calls for SREs of external ICAP from resources outside the NYCA on each day of the June heat wave. Figure 11 summarizes our analysis of the performance of the suppliers of the roughly 2.5 GW of external capacity and found that each hour of SRE commitments saw at least a small amount of external capacity fail to flow as required (between 10 and 25 MW). Additionally, neighboring ISOs cut imports that failed to schedule the required firm transmission service during several peak hours of June 24. Performance was strong in most hours, but the combination of failures by suppliers that failed to schedule the imports and transactions that were cut for inadequate transmission service resulted in overall performance of only 70 percent during the highest load hours on June 24.

Figure 11: External Capacity Performance during SREs



External capacity resources that fail to deliver capacity during SRE events are subject to a deficiency charge equal to:²²

$$Deficiency\ Charge = 1.5 * PRICE * \left(\frac{1000\ kW}{1\ MW} \right) * \left(\frac{\sum_{n=1}^N (\max(ICAP_n^{MWh} - SRE_n^{MWh}, 0))}{N} \right)$$

where “N” is the number of hours in the month when the resource was committed by SRE instruction. For example, if the capacity price is \$5 per kW-month, and a generator is committed by SRE for two hours in a month, the deficiency charge will be \$3,750 per MWh that it fails to deliver, but if the generator is called for 30 hours, the deficiency charge will be only \$125 per MWh that it fails to deliver. Thus, the deficiency charge can be substantial and even exceed the capacity payment by 50 percent for the month (in the case where an external capacity resource fails to deliver every time it is called in a month), but the deficiency charge could still be insufficient to deter poor performance in some cases.

The marginal incentives from PJM and ISO New England during reserve shortages can exceed \$5,000 and \$10,000 per MWh because of the Capacity Performance and Pay-for-Performance rules. If an external SRE call coincides with a reserve shortage in a neighboring region, the resource may have a strong financial disincentive to deliver the obligated capacity to NYISO. It would be beneficial for NYISO to consider whether the deficiency charge is adequate.

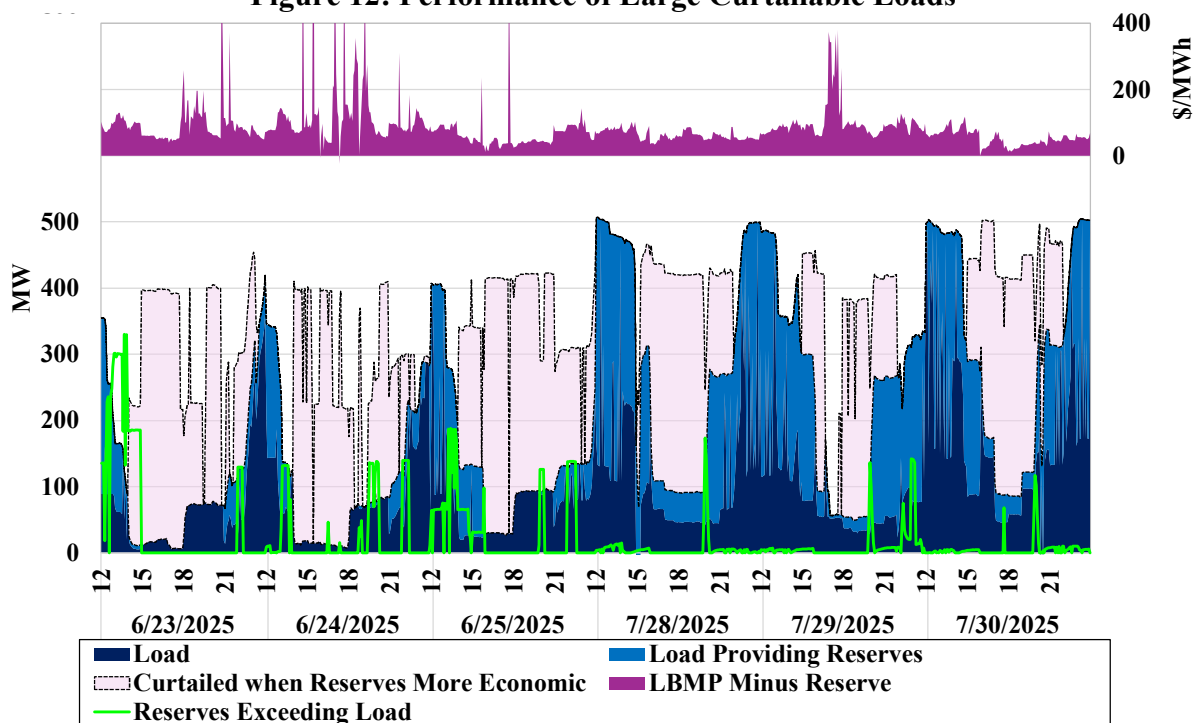
²² See MST 5.12.12.2 for detailed description of the Deficiency Charge and associated variables. See ICAP Manual 4.9.4 for description of applicable sanction period.

D. Performance of Large Curtailable Loads

NYISO's demand response programs facilitate voluntary curtailment by loads in exchange for energy and capacity payments. Various large consumers throughout New York, including industrial loads and data centers, participate in these programs. In Summer 2025, approximately 600 MW of large loads were enrolled in programs capable of participating in NYISO's real-time markets including DER Aggregations, behind-the-meter net generation (BTM:NG) resources, and loads that were enrolled in both the Special Case Resource (SCR) and Demand Side Ancillary Service Provider (DSASP) programs.²³ During high demand periods, these loads have incentives to curtail and/or provide real-time operating reserves while continuing to consume.

Figure 12 shows behavior of large loads enrolled in these programs in the hours beginning 12 through 23 of June 23-25 and July 28-30. The bottom portion shows loads' consumption and reserve schedules while the upper portion shows the real-time LBMP minus the spinning reserve price (based on Zone A prices). When LBMPs are much higher than reserve prices, it is generally economic for large loads to curtail. In the bottom, the light blue area indicates load that was online but providing real-time reserves. The pink area indicates load that curtailed when selling reserves would likely have been more economic. Finally, the green line shows the amount of reserve capacity sold by loads in excess of their actual consumption in the same interval (indicating that these reserves could not be converted to demand reduction if needed).

Figure 12: Performance of Large Curtailable Loads



²³ Additional loads sell capacity through the SCR program but are not capable of participating in the real-time market – see Section XII of this report for an overview of NYISO's demand response programs.

We draw the following observations about large loads' performance from Figure 12:

- *Good Overall Performance* – The vast majority of large load capacity participating as DER, BTM:NG and SCR+DSASP curtailed or provided reserves during the peak afternoon and evening hours on the highest load days of 2025.
- *Some Loads Did Not Curtail* – Approximately 50 to 100 MW did not curtail or sell reserves in late afternoon/evening hours. On June 23-25, some SCR+DSASP resources curtailed for four hours (satisfying their SCR performance obligation) and then increased load in hour beginning 18 while the event was still underway. On July 28-30, some loads simply did not curtail despite activation of the SCR program. In both cases, activation of the SCR program made these resources ineligible to provide reserves (even if it was possible for them to do so).
- *Market-Based Scheduling More Efficient than DR Curtailment Programs* – Up to 400 MW of large loads were curtailed when providing reserves would have been more economic. Large loads can be curtailed outside of the real-time market through the SCR program or utility demand response programs such as Commercial System Relief Programs (CSRP). Loads in these programs curtail to comply with performance requirements even if continuing to consume while providing reserves would be more economic. While some of this may be addressed by the recent sunset of the DSASP program (which resulted in inefficiencies when SCR+DSASP resources that could sell reserves were curtailed by SCR activations), curtailment of loads in DER aggregations through utilities' CSRP programs will continue to preclude economic reserve sales.
- *Inaccurate Baselines Cause Differences between Scheduled and Actual Outcomes* – Some resources sold reserves exceeding their actual load, although the frequency and amount fell between June (when there was no prior experience with DER Aggregation operations) and July. In July, some resources continued to sell reserves exceeding their load while ramping up after being curtailed. To avoid scheduling phantom reserve capacity, resources should update offers to reflect actual capability throughout the day.
- *BTM Loads Unable to Provide Reserves* – NYISO's BTM:NG loads generally performed well, fully curtailing during peak periods, although this program does not allow loads to provide reserves when efficient to do so.

Overall, large loads exhibited a strong ability to curtail under peak conditions. The recent transition of most large loads from the SCR and DSASP programs to the DER program will improve the efficiency of scheduled load curtailments.²⁴ However, the actual availability of DERs to curtail load is frequently inconsistent with the Economic Customer Baseline Load ("ECBL") that is used to limit their availability in the NYISO market systems. When the ECBL of a DER is lower than its curtailable load, the DER is unable to offer its full capability into the market. When the ECBL is higher than the DER's curtailable load, it is difficult to ensure that the market does not schedule the DER to provide more energy and reserves than it is capable of providing. It will be important to refine the DER model rules and procedures in the coming years to take advantage of the full potential of demand response in the NYISO markets.

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See Section XII for details.

IV. LONG-TERM INVESTMENT SIGNALS AND POLICY IMPLEMENTATION

A well-functioning wholesale market establishes transparent and efficient price signals to guide investment and retirement decisions. Several suppliers proposed to retire generating capacity in 2025, even as NYISO has raised reliability concerns driven by future load growth and retirements. New York State has not permitted new fossil generation to be built since 2020, so the only new investment has been in clean energy resources (including wind, solar, batteries and transmission) that earn a combination of NYISO market revenues and state and federal subsidies.

The NYISO markets play a crucial role in: (a) motivating investors to maintain existing generators in a reliability condition, and (b) attracting clean energy investors to the most valuable technologies and locations. This section evaluates:

- Incentives to maintain existing generation resources in good repair (subsection A)
- Incentives for investment in new generation and storage resources (subsection B), and
- Progress of clean energy investment in the NYISO markets since passage of New York’s 2019 Climate Leadership and Community Protection Act (subsection C).

A. Incentives to Maintain Existing Generators

The NYISO markets are the primary or sole revenue source for most existing generators in New York. NYISO is experiencing an increasing rate of proposed retirements, with over 1,200 MW of fossil capacity issuing deactivation notices in the past year.²⁵ This subsection examines the profitability of maintaining existing resources in service under recent market conditions.

Figure 13 shows estimated average net revenues and going-forward costs (GFCs) for several existing technology types from 2023 through 2025. We consider these generator types to be “at-risk” of retirements because they contain units that have issued deactivation notices in recent years and/or have net revenues near or below GFCs. The “Estimated GFC” is the long run cost of maintaining an existing unit in reliable condition. This includes fixed operating costs such as labor, maintenance (often performed by specialized contractors), property taxes, lease or real estate opportunity cost, administrative costs and the amortized cost of major periodic capex and repairs. Plant owners may defer certain costs (such as major maintenance and capex) at the risk of degrading performance and greater risk of catastrophic outage. Hence, we also show a “Short-Term GFC” reflecting only unavoidable operating costs. We estimate generic GFCs based on a variety of public and non-public sources.

²⁵ The following facilities notified NYISO of their intent to deactivate in the past year: Far Rockaway GT 1 & 2 (Zone K), Pinelawn Power (Zone K), Gowanus Barges 2 & 3 (Zone J), Narrows Barges 1 & 2 (Zone J), and Danskammer units 1-4 (Zone G). Generator deactivation notices are posted on NYISO’s Short Term Reliability Process page ([link](#)). NYISO’s Reliability Planning Process may retain generators proposing to retire at cost-of-service rates if the retirement would result in reliability violations. As of April 2026, Far Rockaway entered into a long-term contract with LIPA and withdrew its retirement notice, while NYISO determined that Pinelawn will cease receiving Interim Service Provider payments on May 15 and may retire.

Beginning in May 2026, fossil generators in zones F-K that lack a firm fuel supply in winter will be assigned the Non-Firm Capacity Accreditation Factor (CAF) and face major reductions in capacity revenues (up to 45 percent reduction in zones F and G). To demonstrate the impending impact on plant economics, we estimate 2023-2025 capacity revenues for non-firm gas-only generations as if the Non-Firm CAFs for 2026/27 had been in effect.

Figure 13: Net Revenue and Going-Forward Costs of Existing At-Risk Generation Types²⁶
2023 – 2025 Average

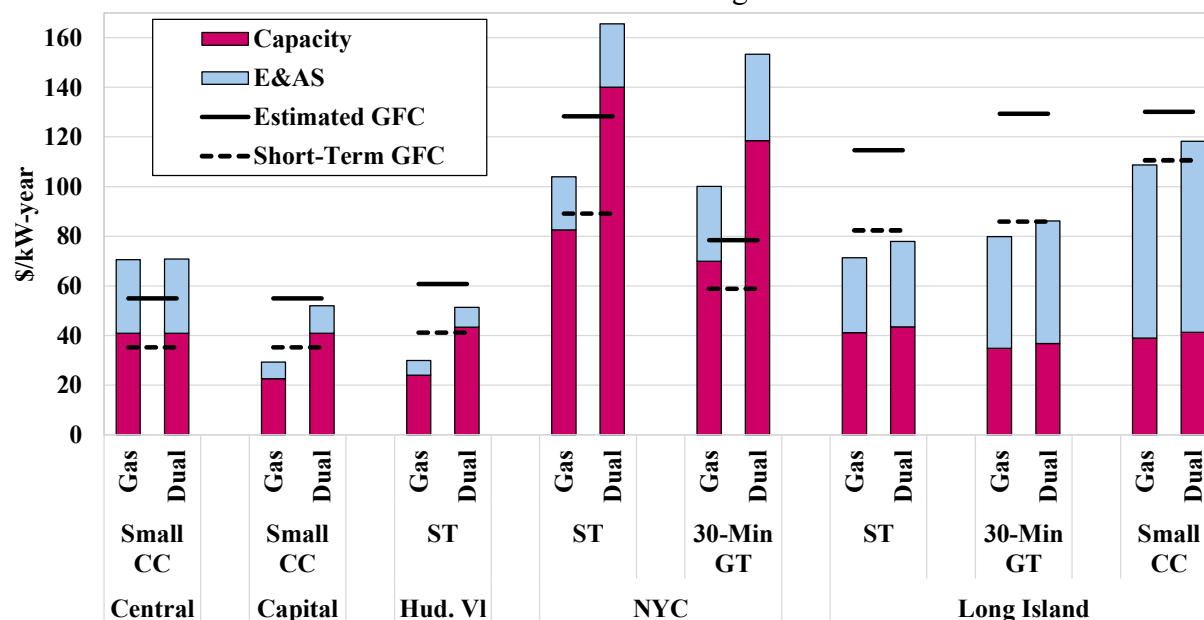


Figure 13 shows estimated net revenues using generic unit parameters, but there is major variation in the profitability of individual units within each type due to factors like age, state of repair, size, and location on the gas system. When revenues are low, plant owners in “harvest mode” may defer needed maintenance and capex and offer their units in a way that is designed to avoid being scheduled in the energy market (e.g. to avoid the risk of costly-to-repair outages associated with starting or generating). Hence, there may be many units that earn meaningfully lower revenues than those depicted in Figure 13.

Figure 13 shows that estimated net revenues are below GFCs for several of the generator types examined, especially for gas-only units that began being compensated based on Non-Firm capacity accreditation in May 2026. We observe the following trends in regional profitability:

- **Rest of State and Hudson Valley:** revenues have been below some units’ GFCs but sufficient to recover short-term fixed costs, reflecting a systemwide capacity surplus. If Non-Firm CAFs had been in effect, revenues would have been below short-term fixed

²⁶ See Appendix Section VII for additional detail on net revenue modeling assumptions and results. Generators in zones A-E currently cannot be assigned to the non-firm CAF, but NYISO has indicated this may be revisited in the coming years.

costs for some units. The 500 MW Danskammer steam turbine plant in Zone G (the majority of which lacks dual fuel capability according to NYISO's 2025 Gold Book) recently filed a deactivation notice proposing to retire in 2026. There are nearly 1.9 GW of small CCs (with plant size below 400 MW) in the rest-of-state region, with capacity factors averaging only 10 percent from 2023 to 2025 despite having heat rates that suggest more frequent operation would be profitable. Many of these units are likely in harvest mode and regularly submit offers designed to avoid operation.

- **New York City:** recent revenues have generally exceeded estimated GFCs due to high capacity prices. If Non-Firm CAFs had been in effect, revenues of gas-only steam turbines would have been below GFCs. There is over 1.0 GW of gas-only steam turbine capacity in New York City.
- **Long Island:** revenues were below short-term GFCs for many units. Generators on Long Island face higher GFCs than other parts of the state due to substantially higher property taxes and lease costs and generally higher operating costs. Nearly all capacity on Long Island has historically been under long-term contract with LIPA at rates substantially above market prices. However, most of these contracts are currently set to expire before summer 2028. Two plants whose contracts with LIPA expired recently – the Far Rockaway GT (105 MW) and Pinelawn CC (74 MW) and facilities – filed deactivation notices proposing to retire in 2025. Both were temporarily retained as Interim Service Providers until 2026 Q2, earning fixed avoidable cost rates of \$88 to \$113 per kW-year. LIPA recently negotiated a 10-year capacity contract with the owner of the Far Rockaway facility at a payment rate of approximately \$83 per kW-year.²⁷ Combined with our estimate for annual E&AS revenues for this technology, this implies a GFC over \$130 per kW-year. This includes the cost of repairing major equipment issues that reduced the plant's effective capacity since at least 2024, which could not be justified by market revenues alone.

1. Market Design Shortcomings Highlighted by Existing Unit Net Revenues

NYISO has recently highlighted potential reliability shortfalls in the coming decade if load grows and/or generators retire.²⁸ A well-functioning market would send long-term signals to maintain existing units in service when capacity shortages are expected. This may entail periods when revenues exceed GFCs, so that plant owners have incentives to remain in service and in good repair through surplus periods when revenues are below GFCs. In this report and elsewhere we have highlighted situations where improvements are needed to ensure that the NYISO markets properly incentivize retention and maintenance of existing generators:

- **Planning-Market Gap:** This is where the amount of capacity required to satisfy planning reliability needs exceeds the capacity market requirements. NYISO has recently retained units proposing to retire under cost-of-service rates through its Reliability Planning

²⁷ See November 12, 2025 memorandum to LIPA Board of Trustees seeking Authorization to Execute a Capacity Purchase Agreement with MPH Rockaway Peakers, LLC, available [here](#).

²⁸ See recent NYISO Reliability Planning Studies including the 2025 CRP, available [here](#).

Process even though there was a surplus in the capacity market for the corresponding region. Changes to NYISO's capacity market are needed to ensure it does not incentivize resources to retire or defer repairs in circumstances where they may be needed for reliability (see Section V.B).

- ***Demand Curves Don't Reflect Cost of New Entry for Reliability:*** NYISO recently began setting its capacity market demand curve reference points based on the Net Cost of New Entry (CONE) of a two-hour battery storage project, which is not reflective of the costs the system would need to incur to support long-term reliability (e.g. the Net CONE of a dual fuel gas turbine). As a result, the ICAP reference points are set inefficiently low in the rest-of-state region and Long Island and inefficiently high in New York City (see Section IX.A).
- ***Downstate Non-Firm Units Undercompensated:*** Gas-only units that are needed to satisfy summer transmission security requirements in New York City may earn revenues below GFCs after the non-firm CAFs take effect in May 2026. This is because the CAFs do not accurately indicate the unit's reliability value for transmission security requirements. In Section IX.C we recommend compensating capacity suppliers based on the value they provide for transmission security and resource adequacy requirements.
- ***Inadequate Locational Incentives:*** In some cases, favorably situated resources are undercompensated because market payments don't consider their ability to address locational needs of the system. This leads to inefficient retirement of resources needed to manage network constraints. For example, the retirement notices of the Far Rockaway and Pinelawn plants in Long Island in 2025 caused NYISO to identify Reliability Needs and retain both plants under cost-of-service rates despite market revenues below GFCs in their locations. This underscores the concern mentioned above that the capacity demand curves for some areas are set below the GFCs of many existing generators. In addition, to avoid premature retirement of units with particular locational value, NYISO should:
 - ✓ Expand the granularity of capacity pricing to better distinguish between high-value and low-value locations (see section IX.B), and
 - ✓ Consistently model transmission constraints (including lower-voltage constraints) in the day-ahead and real-time markets (see section VII.D).

2. Evidence on Economic Life of Fossil-Fuel Generators

When capacity margins are small, market revenues can significantly exceed GFCs of existing generators (as in New York City in recent years). This happens because the Net CONE used to set capacity prices is estimated considering a hypothetical new unit's ability to earn future market revenues when it is no longer 'new'.

Several major M&A transactions for portfolios of primarily gas-fired generators in the US wholesale power markets were completed or announced in 2025 and early 2026. Tight market conditions driven by load growth and recent generator retirements are commonly cited as drivers of these deals. While most of the assets were outside of New York, their valuations provide insight into the market value of older gas-fired generators under conditions of rising demand.

Table 2 summarizes recent transactions, including statistics on the portion of each portfolio consisting of gas CC and CT units, the age of the generators purchased, and the purchase price per kW of capacity. Each of the ten transactions shown consisted primarily of CC and CT assets, and in many cases all or most of the portfolio consisted of generators over 20 years old. These generators retained significant value after 20+ years of life, with older generator portfolio valuations ranging from 45 to 74 percent of the capital cost of a new dual fuel combustion turbine in zone C estimated in NYISO's 2024 Demand Curve Reset.

Table 2: Summary of Recent Gas-Fired Portfolio Transactions in US Wholesale Markets²⁹

Transaction Close	Buyer	Seller	Primary Markets	Total Capacity (GW)	Percent Gas CT	Percent Gas CC	Price per kW	Average Age of CC and CT (Years)	Percent of CC/CT over 20 Years Old
Jun-25	Capital Power	LS Power	PJM	2.1	46%	54%	1,060	15	46%
Jun-25	Partners Group	Avenue Capital	CAISO	1.9	35%	63%	1,128	22	80%
Sep-25	CPS	ProEnergy	ERCOT	1.6	100%	0%	865	3	0%
Oct-25	Vistra	Lotus	PJM, ISO-NE, NYISO	2.5	8%	92%	760	24	87%
Nov-25	Talen Energy	Caithness / BlackRock	PJM	2.9	0%	100%	1,343	4	0%
Jan-26	Constellation	Calpine	ERCOT, CAISO, PJM, ISO-NE	27.4	5%	87%	969	23	81%
Jan-26	NRG	LS Power	PJM, ERCOT, NYISO	12.7	48%	35%	1,024	24	84%
2026 (expected)	LS Power	Constellation	PJM	4.4	0%	84%	1,139	20	61%
2026 (expected)	Talen Energy	ECP	PJM	2.7	19%	81%	1,296	23	100%
2026 (expected)	Vistra	Cogentrix	PJM, ERCOT, ISO-NE	5.1	20%	80%	789	19	57%

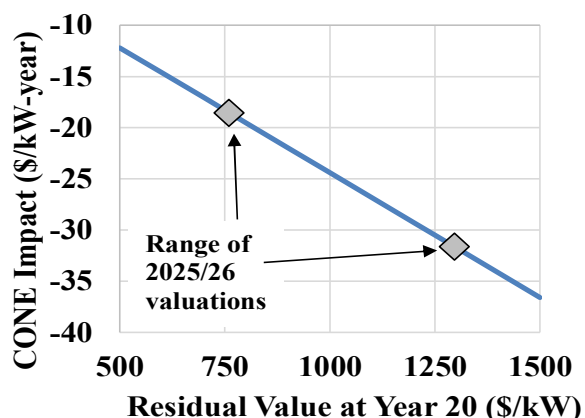
NYISO has historically assumed an economic life of 20 years or less for the proxy unit used to set the capacity market demand curves. This causes an upward bias in the estimated Net CONE, which would be lower if consideration were given to the residual value the unit may retain after 20 years.

Figure 14 estimates the potential range of impacts on Net CONE that would result from considering the residual value of capacity that was sold in recent deals. Table 2 reports seven deals in which the average age of capacity was between 19 and 24 years old and where the

²⁹ Purchase prices and lists of assets in each portfolio are from publicly reported details. Since capacity amounts are not reported in uniform units in the trade press (e.g. nameplate vs. net capacity), we show the average summer and winter net capacity for each portfolio based on EIA data and calculate the price per kW based on the publicly reported total purchase price. NRG's acquisition of assets from LS Power also included the CPOWER demand response platform in addition to the generation portfolio shown.

residual value ranged from \$760 to \$1,296 per kW. The figure shows how such substantial residual valuations would affect the annualized CONE of a generator built 20 years earlier. NYISO is currently exploring improvements to its Demand Curve Reset process and it would be appropriate to include consideration of residual value in any improvements. A more thorough assessment of residual value in a Demand Curve Reset study might consider valuation adjustments applicable to the proxy unit technology (many of the recent portfolios include CCs), location, and potential for future market expectations less favorable than those underlying recent transactions.

Figure 14: Residual Value Impact on 20-Year CONE Estimate



B. Incentives for Investment in New Generation

New York has not permitted new fossil generation projects to be built since 2020. Consequently, new investment in the NYISO markets currently takes the form of renewables, battery storage and transmission that receive a combination of wholesale market revenues and state and federal subsidies. The NYISO markets are important for these resources because they provide the revenues needed beyond subsidy payments, signal which clean energy projects provide the greatest system benefits, and provide transparency on the cost of state policies. If concerns about load growth, reliability, and affordability prompt New York to revisit its prohibition on new gas-fired projects, market prices will play a crucial role in attracting needed investment.

1. Incentives for Investment in Dispatchable Resources

Figure 15 compares average net revenues to the estimated cost of new entry (CONE) for battery storage and gas combustion turbines with dual fuel capability in 2023 to 2025. For battery storage, we include the levelized value of the federal Investment Tax Credit and New York's Bulk Storage Incentive as revenues. We show storage energy and ancillary services (E&AS) revenues under alternate energy market participation strategies: offering the full capacity of the battery as spinning reserves in the day-ahead market, or offering no day-ahead reserves (which creates additional flexibility for energy arbitrage and real-time market participation).³⁰

³⁰ Details on estimated net revenues by year can be found in Appendix Section VII. We estimate state incentives for storage using the levelized value of the \$75 per kWh from NYSEDA's Bulk Storage Incentive, which would have been available to projects during the study period. The PSC recently approved a new bulk incentive program design for energy storage.

The profitability of new generation varies by technology and zone, and it has been influenced by volatility in energy and capacity markets. In addition to market signals, federal incentives play a major role in the projects' profitability. We observe the following for specific technologies:

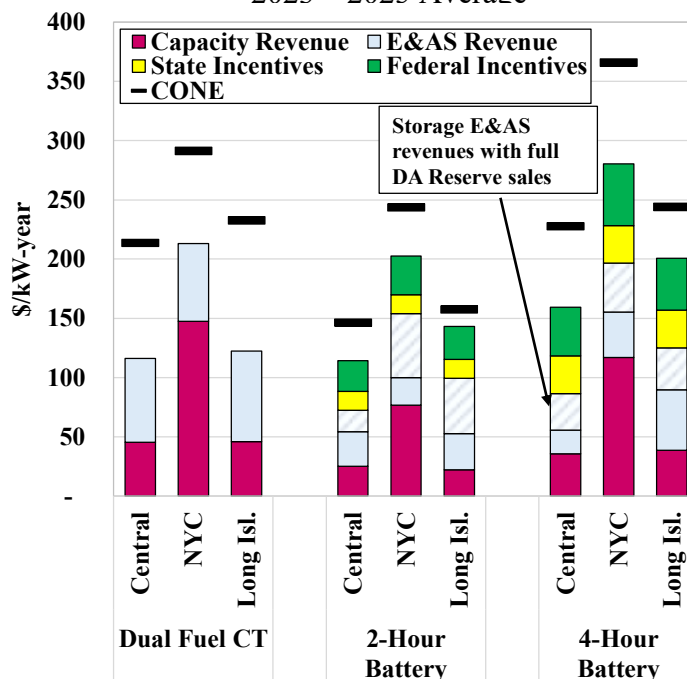
Gas-fired Combustion Turbines –

Estimated annual revenues for a new CT were below the CONE at all locations from 2023 to 2025. This is primarily because (a) all regions except New York City had large surpluses relative to the capacity market requirement during this period and (b) the estimated Net CONE underlying the ICAP Demand Curves prior to May 2025 did not adequately account for cost increases that occurred

since the 2020 Demand Curve Reset, resulting in prices far below the Net CONE of a new CT in New York City despite relatively tight conditions in 2023 and 2024. The New York DEC has declined to permit new combustion turbines in the past five years, citing their inconsistency with State climate law.³¹ However, investment in new peaking capacity may be needed for reliability in the coming years if significant demand growth due to large loads and electrification occurs.

Energy storage – Market revenues of battery storage have historically been below levels needed to justify investment, even after including federal and state incentives. Storage projects earn capacity revenues based on their Capacity Accreditation Factors (CAFs). In the 2025/26 capability year, the CAFs of storage with two-hour and four-hour durations were 65 and 79 percent in New York City and 53 and 87 percent in Long Island. We estimate that storage resources would have maximized net E&AS revenue by selling their full capacity as day-ahead reserves, particularly in New York City where this strategy exceeded that of selling no day-ahead reserves by over 300 percent for 2-hour storage and 200 percent for 4-hour storage. State plans to deploy large quantities of storage over the next several years are likely to have a downward impact on the CAFs and reserve revenues of storage.

Figure 15: Net Revenue and CONE for New Dispatchable Resources
2023 – 2025 Average



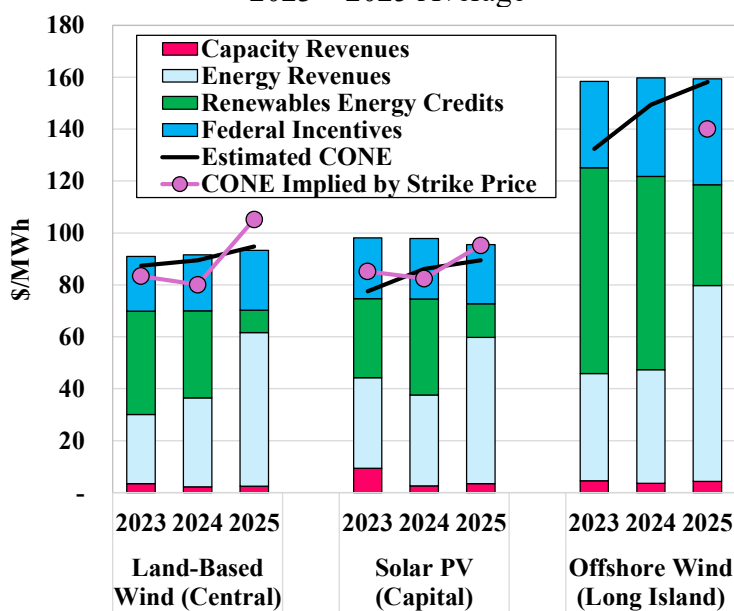
³¹ See permit denial letters from New York Department of Environmental Conservation for Astoria Replacement and Danskammer Generating Station projects available at [link](#), and [link](#), respectively.

The NYPSC recently approved a process to procure 3 GW of bulk battery storage by 2030 over three annual procurements, the first of which was issued in 2025. These procurements will replace the relatively modest Bulk Storage Incentive with a new Index Storage Credit structure comparable to the Index REC currently used in procurements of renewables (see below). We estimate that Index Storage Credit payments will need to provide 32 to 40 percent of total storage revenues (in addition to the 14 to 18 percent provided by federal subsidies) to incentivize investment in four-hour storage projects.

2. Incentives for Investment in Intermittent Renewables

Figure 16 shows estimated average revenues of intermittent renewable technologies from 2023 to 2025 compared to their estimated Cost of New Entry (CONE). REC revenues are estimated using reported Index REC strike prices for wind and solar projects that are currently in service and offshore wind projects that are under development.³² CONE values are estimated based on generic cost data from public sources. Figure 16 also shows the CONE value implied by the average Index REC strike price for each technology, based on the average strike price of projects deployed in 2023 and 2024 and the average strike price of projects that are currently under development for 2025.³³

Figure 16: Net Revenue and CONE for New Renewable Generation
2023 – 2025 Average



³² NYSERDA's Index REC contract structure is designed to provide developers a partial hedge against wholesale market prices. Index REC values are calculated on a monthly basis as the project's Strike Price minus a reference energy price and reference capacity price. For more details, see [here](#). For this analysis, we used Index REC strike prices of \$73.6/MWh for solar, \$70.0/MWh for land-based wind, \$133.9 for offshore wind in Zone J, and \$124.9 for offshore wind in Zone K.

³³ See Appendix VII.C for detailed assumptions used in this analysis. Average strike prices include projects with active new capacity projects (excluding repowerings) and Index REC prices reported by NYSERDA as of March 2025. NYSERDA strike prices are in fixed nominal dollars; we convert these reported prices to a real \$2025 price with equivalent present value over the lifetime of the project. We add estimated revenues from federal incentives to the strike price to derive the 'implied' CONE. For offshore wind, strike prices include the effects of contract provisions that reduce the strike price for projects receiving federal incentives.

Estimated revenues of renewables were generally sufficient to justify new investment. This is primarily due to state and federal subsidies, which accounted for approximately 54 percent of revenues for land-based wind, 51 percent of revenues for solar PV, and 64 percent of revenues for offshore wind. Energy market revenues were much higher in 2025 due to higher market prices, resulting in lower estimated REC payments under the Index REC structure. Recent changes in federal law will eliminate key subsidies for wind and solar projects that are not placed in service by the end of 2027 or under construction as of July 4, 2026. These subsidies make up approximately 25 percent of wind and solar revenues and their elimination will present a major challenge to project economics without a major increase in Index REC strike prices.

Zonal and Technology Discounts of Wind and Solar

New York's procurements for renewable energy have generally been designed to hedge against some market risks while retaining incentives for developers to maximize project value. Under the Index REC structure, the project's REC price in each month is equal to a fixed strike price minus 'index' energy and capacity prices derived from zonal average prices. Hence, the project is hedged against changes in overall energy and capacity prices, but faces two major market risks related to its location and generation pattern:

- *Nodal Discount* – Revenues to renewables will deviate from their Index REC strike price if the LBMP at the project's location differs from the load zone where it is located due to transmission constraints; and
- *Technology Discount* – Revenues of renewables will deviate from the Index REC strike price if prices during hours when the renewable resource generates are lower than the average price across all hours.

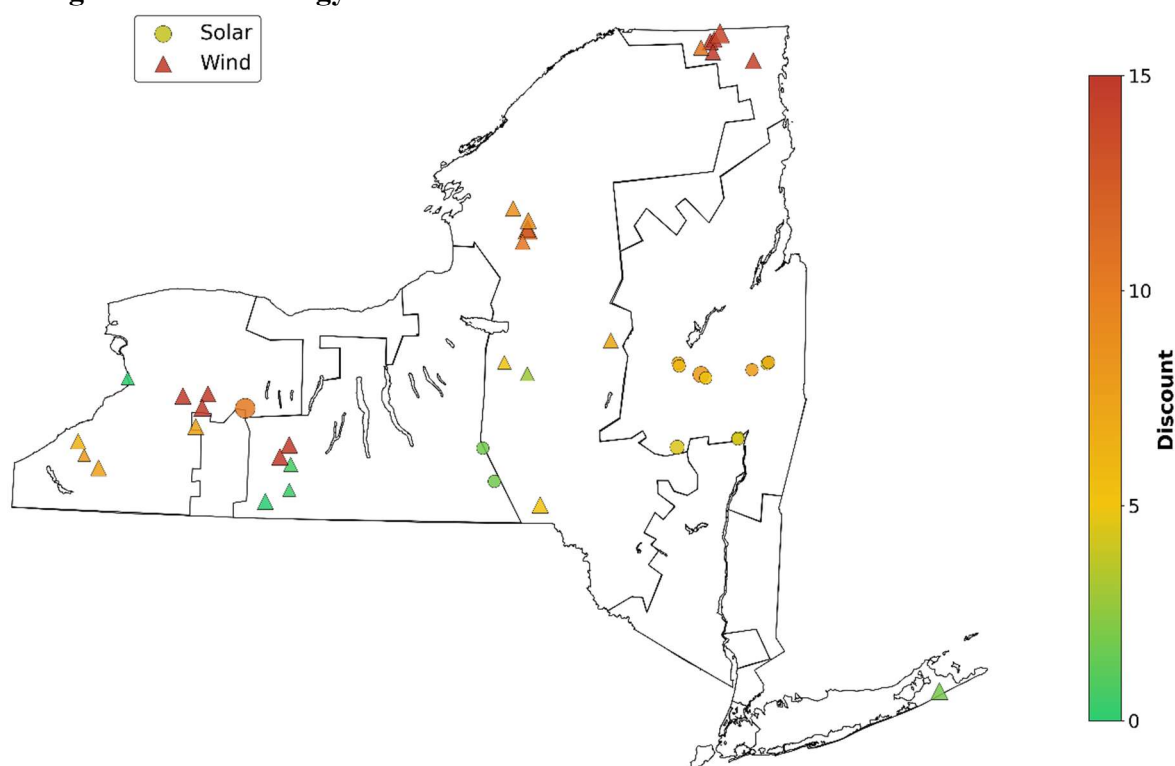
Figure 17 shows the discounts (including technology and nodal effects) at locations of existing wind and solar in 2025. The discount at each location is calculated as the difference between the generation-weighted nodal real-time LBMP and simple average zonal day-ahead LBMP.³⁴

Figure 17 indicates that many locations of renewable generators had substantial nodal and technology discounts, ranging from -\$2 to \$25 per MWh on an annual basis for wind and \$2 to \$10 per MWh for solar. The largest discounts occurred at locations where renewable generation exacerbates intra-zone transmission constraints, especially generators that increase loading on:

- 230 kV and 115 kV circuits flowing power from western to central New York, and
- 115 kV circuits flowing power from northern to central New York.

Nodal discounts reduced energy net revenues to intermittent renewable generation by as much as 42 percent below levels shown in Figure 16 in 2025.

³⁴ Discounts are calculated on a monthly basis and the annual average shown is weighted by monthly generation levels of the resource type. Generation weights are calculated using forecasted wind and solar generation levels to exclude effects of unit-specific outages and economic curtailment.

Figure 17: Technology and Nodal Discounts of Wind and Solar Resources in 2025

C. Status of Clean Energy Investment in NYISO

New York passed the Climate Leadership and Community Protection Act (CLCPA) in 2019, requiring major changes to the state's electricity sector. The CLCPA mandates an electric supply mix that is 70 percent renewable by 2030 and 100 percent zero emissions by 2040. It also includes resource specific targets including 9 GW of offshore wind by 2035 and 3 GW of energy storage deployed by 2030 (and the NY PSC later raised its storage deployment target to 6 GW by 2030). If realized, these goals would have dramatic impacts on the NYISO markets. This subsection discusses the current state of renewable energy deployments in the NYISO system.

Figure 18 compares planned and actual generation from renewable sources. The solid bars show generation by renewables in service as of 2025 and hydropower imports from Quebec. The patterned bars show planned generation from projects that were awarded contracts to sell renewable energy credits (RECs) by NYSERDA since 2018 that have not yet entered service. Contracts for RECs have been the primary mechanism used by the state to incentivize development of renewables. For projects that canceled their original NYSERDA contracts and were subsequently re-awarded new contracts, the original target in-service date is shown.

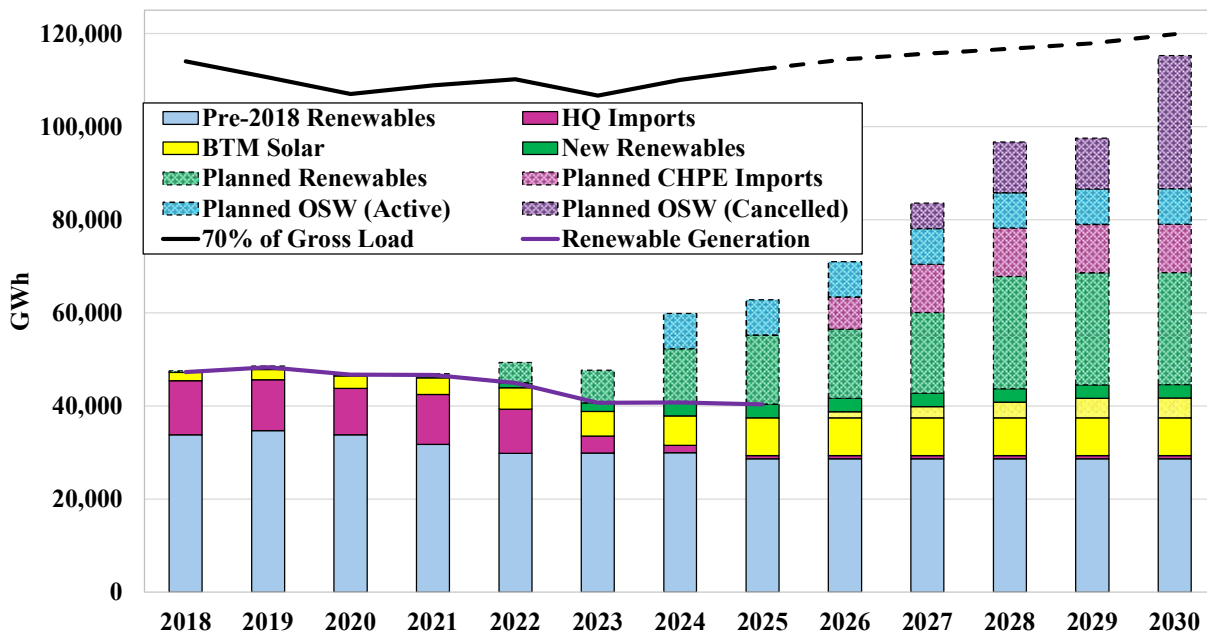
Figure 18: Planned and Actual Renewable Generation and Imports

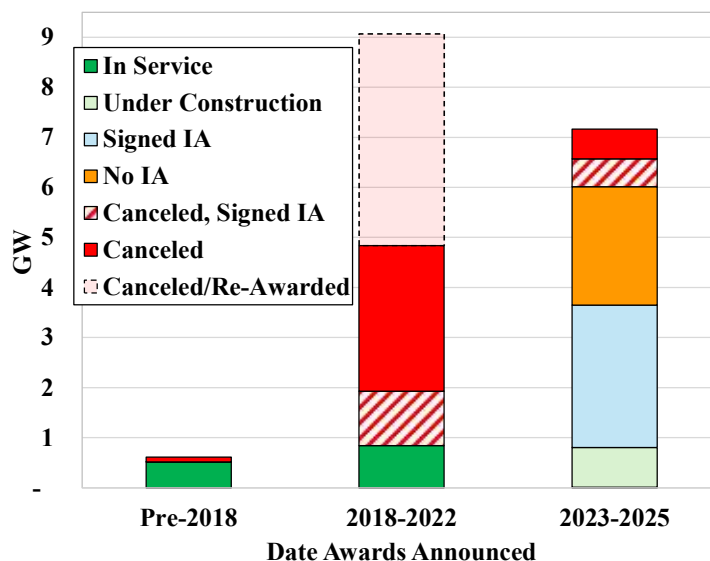
Figure 18 shows that renewable generation remains far behind the pace of development targeted by state solicitations and the 70 percent target. Renewables decreased from approximately 29 percent of New York’s supply in 2018 to 25 percent in 2025. New York added approximately 9,100 GWh of annual generation by new renewables and behind-the-meter solar during this time (6 percent of gross load), but generation from existing hydropower resources declined and imports from Hydro-Quebec collapsed between 2022 and 2025. Hydropower output is affected by climate conditions and recent low levels may change in future years. An increase in hydro imports from Quebec is expected following completion of the Champlain-Hudson Power Express (CHPE) transmission project, expected in 2026.

Renewable projects in New York have faced a variety of challenges since the passage of the CLCPA, including: major cost increases after initial contracts were negotiated, a challenging siting and permitting environment, interconnection process delays and uncertainty, changes in federal law and permitting actions, weak non-performance penalties in contracts, and a high degree of market risk (driven in part by uncertainty in the pace of future policy-driven renewable deployments). Below we summarize the status of land-based renewables, offshore wind and battery storage development:

Land-Based Renewables: NYSERDA has issued nine solicitations for new land-based renewables (primarily wind and solar) since 2017, awarding contracts to over 16 GW of projects. As Figure 18 shows, about 16 percent of the renewable generation targeted by 2025 has entered service. Figure 19 summarizes land-based renewable projects that have been awarded Tier 1 REC contracts by NYSERDA and their status in NYISO’s interconnection queue as of March

2026.³⁵ The vast majority of projects awarded contracts prior to 2023 have been canceled. Many of the canceled projects were subsequently awarded new contracts, typically at higher prices. For projects that were canceled and re-awarded, Index REC Strike Prices increased by an average of 40 percent. Of the 6.0 GW of currently active projects, 3.6 GW have completed NYISO's interconnection process and signed interconnection agreements (IAs) but only 800 MW are listed as 'Under Construction' in NYISO's interconnection queue as of March 2026. Of the currently active projects with signed IAs, 1.0 GW are land-based wind and 2.7 GW are solar PV.

Figure 19: Status of NYSERDA Land-Based Renewable Awards



Offshore Wind: NYSERDA awarded contracts to seven projects totaling 8.3 GW, but all seven initial awards have been canceled. Two of the canceled projects (1.7 GW) were re-awarded at higher Index REC strike prices in a 2023 solicitation. The two re-awarded projects (Empire Wind and Sunrise Wind) have signed interconnection agreements and are currently under construction.

Battery Storage: The NYPSC has adopted a target of 6 GW of energy storage deployed by 2030, including at least 3 GW participating in the NYISO markets. A large number of storage projects are in the interconnection queue, but few have been completed and 100 MW are currently listed as 'Under Construction' as of March 2026. Only 60 MW with an average duration of 1.5 hours have been deployed under the state's bulk storage program since 2019. Revenues of storage have generally been insufficient to justify investment even after considering state and federal subsidies (see Figure 15). The NYPSC recently approved a new three-year process to procure 3 GW of bulk storage projects and issued its first annual solicitation in 2025. These solicitations will use a new Index Storage Credit contract structure that may accelerate storage deployments by lowering market risk for project owners and likely offering much higher incentive levels than the current bulk storage program.

³⁵ Data taken from NYSERDA's renewable project database (see [here](#)) and NYISO's Interconnection Queue as of March 2026. Initial COD Targets are from NYSERDA announcements of solicitation results (see [here](#)). For projects that canceled original contracts and were subsequently re-awarded, the original award is included as 'Canceled' and the latest award is included based on its current status.

D. Conclusions on Investment Signals

The NYISO markets provide a critical revenue source for new and existing resources. A well-functioning wholesale market facilitates investment and reduces costs while retaining capacity needed to satisfy reliability criteria. In this section we have highlighted the following observations regarding generators' net revenues:

- A significant amount of existing generation capacity likely earns revenues below the going-forward costs (GFCs) required to remain in service and in good repair. While it is efficient for older generators to retire when there is surplus capacity, NYISO has recently highlighted reliability concerns and retained generators proposing to retire under cost-of-service arrangements. NYISO should implement changes to ensure that payments accurately reflect the value that each generator provides to the system.
- Investment in new renewable and storage resources has lagged expectations. Market prices are generally sufficient to support investment in wind and solar in conjunction with state and federal incentives, which account for over half of total revenues. However, investment in these resources has been modest due to cost increases, siting and permitting challenges, nodal and technology discounts, and other factors. Recent changes to federal tax incentives will create significant challenges for future renewable development. Battery storage remains uneconomic based on market prices, and large subsidies under the state's new Index Storage Credit framework may be required to motivate investment.

V. DELIVERABILITY TESTING AND TRANSMISSION PLANNING PROCESSES

New transmission investment occurs through centralized planning processes. These include the NYISO’s Comprehensive System Planning Process and the interconnection deliverability study procedures, which identifies upgrades funded by resource developers. We evaluate the performance of these processes and consider potential opportunities for improvement. Subsection A contains our evaluation of the deliverability study component of the interconnection process, while Subsection B discusses NYISO’s centralized planning processes.

A. Deliverability Study Process for New Resources

NYISO’s interconnection process is designed to ensure that new generation and storage projects can reliably interconnect to the network and deliver their capacity to load. This process often requires developers to pay for costly transmission upgrades in order to sell energy and capacity in the NYISO markets. In this subsection we discuss how the Deliverability Study component of the interconnection process creates inefficient barriers to new generation and storage investment.

Background on the Study Process

NYISO recently redesigned its interconnection study process, launching the first “Cluster Study” in 2024 (ongoing at time of writing). Previously, the most significant interconnection study was known as the Class Year process. Both the Cluster Study and Class Year processes were designed to jointly study the impacts of a group of proposed projects and allocate network upgrade costs required for them to reliably interconnect and participate in the NYISO markets. The Cluster Study is intended to improve upon the previous process with a streamlined set of studies, shorter timelines, more information provided to developers early in the process, and improved incentives for the progression of projects that are more likely to complete the process.

The Cluster Study identifies the following categories of upgrades (these same categories were identified under the Class Year process):³⁶

- Connecting Transmission Owner’ Attachment Facilities (CTOAF): dedicated facilities required to connect the project to the transmission owner’s network.
- System Upgrade Facilities (SUF): network upgrades needed for the group of studied projects to comply with NYISO’s Minimum Interconnection Standard (MIS). Projects must agree to pay the cost allocated to them for identified upgrades to receive the right to sell energy (ERIS). The MIS identifies adverse reliability impacts of interconnecting the project, but it considers normal operating actions that would avoid these impacts (e.g. reduction in output or curtailment of the resource as needed).

³⁶ For additional details on the interconnection process, see NYISO’s Transmission Expansion and Interconnection Manual, available [here](#).

- **System Deliverability Upgrades (SDU):** upgrades needed for resources to be deliverable under NYISO’s Deliverability Interconnection Standard (DIS). Projects must pay for identified upgrades to receive the right to sell capacity (CRIS). The DIS ensures that new projects receiving CRIS rights and existing resources can simultaneously deliver their output throughout the capacity zone without violating any transmission constraints.

Once required upgrades and cost allocations are identified, each developer must choose whether to pay for required upgrades to receive ERIS or CRIS rights or withdraw from the study through an iterative process. Recent Class Year studies have taken approximately two years to complete, while the Cluster Study is designed to last 590 days. During this time, new resources seeking to interconnect that are not part of the study must wait until a new study begins.

Recent Deliverability Cost Allocations

Table 3 summarizes initial cost allocations and outcomes for projects that were assigned SDUs in the two most recently completed Class Year studies. Class Year 2021 (completed in January 2023) studied projects requesting 10 GW of CRIS, of which 4.1 GW were found to be fully or partially undeliverable with total initial SDU costs of \$1.5 billion. Class Year 2023 (completed in January 2025) studied projects requesting 14 GW of CRIS, of which 0.9 GW were found to be fully or partially undeliverable with total initial SDU costs of \$400 million. Table 3 indicates the amount of CRIS that was ultimately accepted by developers who chose to pay for SDUs or accept only partial CRIS. In some cases, developers’ final cost allocations declined or their deliverable MWs increased as other projects withdrew from the study.

Table 3: Summary of Initial Deliverable MWs and SDUs in Class Year 2021 and 2023³⁷

Area	Requested CRIS (MW)	Deliverable CRIS (MW)	Initial SDU Cost (\$/kW UCAP)	Accepted CRIS (MW)
<i>Class Year 2021</i>				
ROS - Thousand Island	564	252	1,334	55 (10%)
NYC - Staten Island	650	121	795	121 (19%)
Long Island - East	1,351	0	506	1,111 (82%)
Long Island - Central	160	0	1,712	0 (0%)
Long Island - West	1,356	0	528	0 (0%)
<i>Class Year 2023</i>				
Long Island - East	714	242	864	162 (23%)
Long Island - Central	170	75	743	109 (64%)
Long Island - West	40	0	199	40 (100%)

The initial SDU cost allocations summarized in Table 3 are prohibitively costly for most projects and could not be recovered in many cases even if capacity prices were set at the Net Cost of New Entry (CONE). Consequently, most of the developers that were allocated costs for SDUs

³⁷ See Section IV.A of our 2023 NYISO State of the Market report (available [here](#)) and Section IV.A of our 2024 NYISO State of the Market Report (available [here](#)) for more detail on these cost allocations.

ultimately withdrew from their Class Year Study rather than accept CRIS. In addition to significant costs, SDUs may have development times that render the proposed generation or storage project non-viable. For example, a major SDU affecting most of the requested capacity on Long Island in Class Year 2023 was estimated to have a development lead time of 8 years.

The number of projects proposing to interconnect has grown in recent studies, and this is likely to lead to SDUs being identified in additional areas. While the SDUs summarized in Table 3 were driven by local “byway” constraints, preliminary results for the initial Cluster Study (underway at time of writing) identified \$2.2 billion in SDUs, including upgrades of the Volney East and Total East “highway” interfaces and various byway facilities in the New York City, Long Island and Rest of State regions.³⁸ These SDUs affected 8.3 GW of the 15.7 GW of requested CRIS examined, with cost allocations ranging from \$20 to \$2,800 per kW of UCAP. While NYISO has proposed process changes that would reduce the magnitude of these SDUs, it highlights how the deliverability test can have a large impact on the cost and complexity of new project development.

Issues with Deliverability Process

The deliverability test in its current form is an inefficient barrier to new entry that insulates existing resources from competition. While it is appropriate to ensure that capacity sold in the NYISO market is deliverable to load, the current deliverability process does so in a way that is overly burdensome to new project development and may often require costly upgrades that are not actually needed. The deliverability study suffers from key structural problems:

- **All proposed new projects are studied simultaneously.** This exacerbates the number of bottlenecks identified and the magnitude of SDUs needed to make all projects deliverable. This approach leads to identifying unrealistic SDUs because most of the proposed projects will never enter service. The size of final SDUs tends to decline as projects drop out through the study’s iterative decision process, but developers must commit to high initial SDUs before knowing how they may change, raising their risk.
- **All existing units are assumed to remain in service.** As a result, new projects must be deliverable in addition to all existing capacity. This is unrealistic because a large influx of new capacity would increase pressure on older generators to retire. Similarly, all projects that accepted SDUs in previous deliverability studies are assumed to be in service, even those whose development has paused and that may never actually be built. Hence, the process identifies upgrades needed to make deliverable a much larger amount of total capacity than will realistically exist at any single point in time.
- **Deterministic methods are poorly suited to new technologies.** Most projects seeking to interconnect are wind, solar and battery storage, whose output during critical hours is best assessed probabilistically. However, the deliverability test is based on a single snapshot of the system. This requires NYISO to choose a single deterministic value for

³⁸ See NYISO March 26, 2026 presentation to Transmission Planning Advisory Subcommittee “Transition Cluster Study 2024 Preliminary Deliverability Study”, available [here](#).

the output of all wind and solar resources. Doing so inevitably over-estimates their output in key hours (leading to excessive SDUs) or under-estimates it (allowing projects to obtain CRIS when they will not be deliverable in some critical conditions).

These features of the deliverability test cause several adverse outcomes: (a) new projects are deterred from entering the market, (b) incumbent generators that are affected by the same deliverability constraints as new entrants are shielded from competition, and (c) renewable and storage projects may require higher state contract payments, which are passed on to ratepayers.

Improving or Replacing the Deliverability Process

NYISO recently proposed changes to its deliverability test that will result in identifying fewer SDUs, particularly in the Highways portion of the test.³⁹ NYISO estimates that under its proposal, it would identify \$952 million in preliminary SDUs for the initial Cluster Study (as opposed to \$2.2 billion under the current methodology). While a reduction in SDUs will ease project development in many areas, NYISO's proposal does not resolve the basic structural issues discussed above, so deliverability will continue to pose an obstacle for projects affected by Byway constraints and some Highway interfaces. (Most of the remaining Cluster Study SDUs under NYISO's proposal affect projects in Staten Island and eastern Long Island.) However, identifying fewer SDUs may lead to situations where resources with CRIS rights are not actually deliverable. In such situations, NYISO's process will raise IRM and/or LCR requirements to offset the impact of the deliverability limitations, inefficiently raising consumer costs.

Market enhancements could reduce the need for most elements of the deliverability test. NYISO currently relies on the capacity market to manage deliverability on major interfaces (defining the boundaries of the G-J Locality, New York City and Long Island capacity zones), so new resources are not tested for deliverability across these interfaces. In principle, any interface modeled in NYISO's resource adequacy model (GE-MARS) can be used to define capacity zones in this way. For example, the UPNY-SENY interface between zones A-F and G-I was once studied in the deliverability highways test and caused SDUs to be identified, but NYISO ceased studying it in the deliverability test after the creation of the G-J locality in 2013.

Hence, in Section IX.B we propose two approaches to improving the number of locational price signals in the NYISO market.⁴⁰ Either of these approaches would allow NYISO to remove interfaces represented in the capacity market from the Deliverability Study process. This would allow deliverability constraints to be reflected in prices and capacity requirements as new projects enter service, rather than identified in advance based on unrealistic resource mix assumptions.

³⁹ See NYISO presentation Deliverability Test Methodology Changes at March 25, 2026 Management Committee meeting, available [here](#).

⁴⁰ See Recommendations 2022-4 and 2025-3 in Section XIII.

B. Market Impacts of System Planning Processes

NYISO’s centralized transmission planning processes are designed to identify and fund the transmission investments that are most cost-effective for satisfying reliability needs, achieving public policy goals, and/or reducing congestion. These planning processes are important because the markets generally do not provide efficient incentives for merchant transmission investment.

Projects selected in the planning processes are funded through regulated cost recovery rather than market revenues. Selecting inefficient projects can raise ratepayer costs and crowd out more economic merchant solutions (transmission and non-transmission). This subsection discusses market impacts of NYISO’s reliability and economic/policy planning processes.

3. Overview of NYISO’s System Planning Processes

NYISO performs centralized transmission planning through the Comprehensive System Planning process (CSPP).⁴¹ The CSPP consists of the following processes:

- The *Reliability Planning Process* identifies reliability needs in the short and long term and solicits solutions for needs for Bulk Power Transmission Facilities (BPTFs);
- The *Economic Planning Process* studies potential future congestion and includes a phase to evaluate projects proposed by developers to relieve congestion, but no transmission has ever been built through this process. The main product of this process is the System & Resource Outlook (“the Outlook”) that assesses future congestion and is used as the basis for evaluations of projects in other processes, such as the PPTPP.
- The *Public Policy Transmission Planning Process (PPTPP)* solicits, evaluates and selects projects designed to address policy-driven needs identified by the NY State Public Service Commission, such as integration of renewable energy.

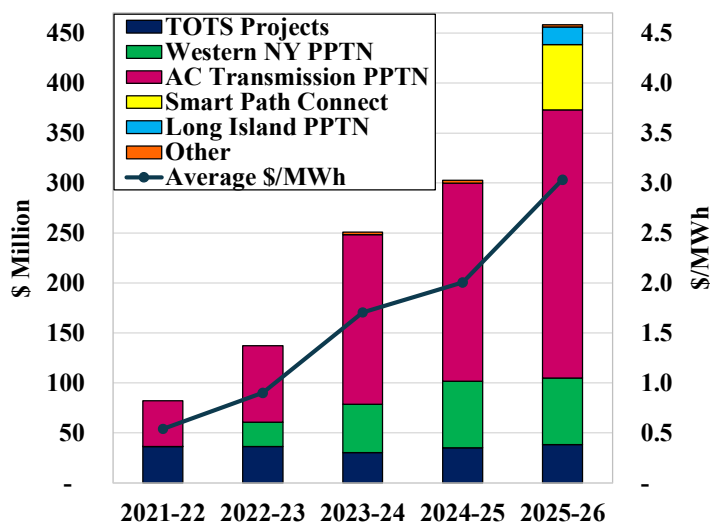
In addition to these NYISO planning mechanisms, transmission planning carried out by utilities and state agencies affects the wholesale markets. New York’s electric utilities plan investments in lower-voltage local transmission for local reliability and renewable integration through their PSC-approved rate cases and the state’s Coordinated Grid Planning Process. The state also has a process to identify “Priority Transmission Projects” regarded as needing rapid approval to comply with state policy mandates, to be built by the New York Power Authority.⁴² In recent years, bulk system transmission investment has primarily taken place through the PPTPP. NYISO has identified solutions for three Public Policy Transmission Needs (PPTN): the Western New York, the AC Transmission, and the Long Island Offshore Wind Export PPTNs. A fourth PPTN to integrate offshore wind in New York City was issued in 2023 but withdrawn in 2025.

⁴¹ For more information about the CSPP, see [here](#).

⁴² For example, see October 15, 2020 Order in NYPSC Case 20-E-0197, identifying the “Northern New York” bulk transmission project as a Priority Transmission Project.

The costs of transmission projects selected through NYISO’s planning processes are allocated to loads under NYISO’s OATT rate schedules. Figure 20 shows the total annual cost recovery quantities and approximate cost per MWh (averaged across all NYCA load) for transmission projects recovered through NYISO rate schedules in recent years. In practice, costs allocated to customers in each transmission owner area or load zone are not always proportional to load and vary by project. The costs of transmission projects recovered through NYISO rate schedules have increased in recent years and will continue to rise, driven by public policy projects. The embedded costs of transmission owners’ network that are not planned by NYISO are recovered through state-regulated retail rates, congestion revenues from the TCC auctions and day-ahead market, Transmission Service Charges (TSCs), and the NYPA Transmission Adjustment Charge (NTAC). In 2025, the TSC and NTAC averaged \$10.3 per MWh systemwide, with significant variation across transmission owner areas.

Figure 20: NYISO Regulated Transmission Costs



4. NYISO’s Reliability Planning Process and the “Planning-Market Gap”

NYISO’s Reliability Planning Process (RPP) is designed to identify potential violations of reliability criteria and solicit solutions. Key studies include the Short-Term Assessment of Reliability (STAR), which is performed quarterly and has a five-year horizon, and the Reliability Needs Assessment (RNA), which is performed every two years has a ten-year horizon. Proposed generator retirements are studied in the STAR, and the generator may be retained temporarily under regulated rates if its retirement would result in reliability violations within four years of the STAR study. NYISO issues solicitations through its short-term or long-term processes for permanent solutions to Reliability Needs, which may include market-based or regulated projects.

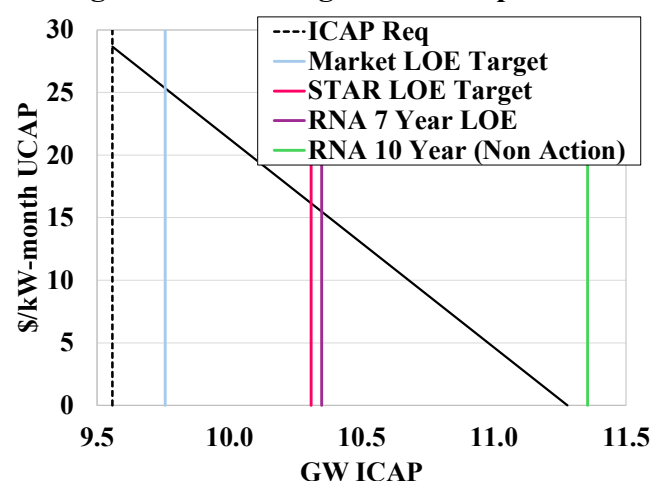
NYISO has identified Reliability Needs at an increasing pace in recent years. Five generating plants totaling 1.2 GW of capacity notified NYISO of their intent to retire in 2025, and NYISO identified Reliability Needs in its STAR process driven by each of them. Of these, 180 MW on Long Island intended to retire in 2025 and are currently being retained as Interim Service Providers (ISPs). The remainder intended to retire in the summer of 2026. At the time of writing, solicitations for permanent solutions to these Reliability Needs are underway and it is likely that some or all of the proposed retirements will continue to be retained under regulated rates until long-term solutions can be developed.

We have highlighted a “planning-market gap” where NYISO’s reliability planning studies implicitly require more capacity to satisfy reliability criteria than the capacity market is designed to procure.⁴³ This causes prices to indicate a capacity surplus in situations where the planning process identifies reliability violations. As a consequence, NYISO does not provide adequate incentives for generators to remain in service and invest in needed maintenance and capex. For example, NYISO compensated the Far Rockaway GT 1&2 and Pinelawn CC plants on Long Island for fixed costs \$7.3 and \$9.2 per kW-month of ICAP, respectively, under Interim Service Provider rates to address Reliability Needs in Q1 2026, which is significantly higher than expected market revenues for generators on Long Island.

NYISO recently modified its methodology for setting market LCRs to better align with the RPP, resulting in a major increase in the LCRs for 2026/27.⁴⁴ However, the Short Term Assessment of Reliability continues to target significantly larger capacity margins than the market. When assessing Reliability Needs caused by proposed generator retirements, the STRP uses a “High Demand” forecast and excludes any planned new generation or transmission additions. Consequently, NYISO’s 2026 Q1 STAR found a Lower Hudson Valley Reliability Need caused by the proposed retirement of the 500 MW Danskammer facility even though the capacity market surplus was over 1,100 MW in May 2026. NYISO recently proposed to assume that large amounts of ‘aging generation’ will retire as a base case assumption in years 6-10 of its Reliability Needs Assessment (RNA), which will further exacerbate the planning-market gap.

Figure 21 illustrates the planning-market gap in New York City. It shows the Summer 2026 ICAP demand curve, indicating the capacity market’s tariff-prescribed level of excess (LOE) and the target LOE implied by the STAR and the 2026 RNA. We estimate the STAR’s target LOE as the four-year impact of NYISO’s “High Demand” load forecast plus the size of the retiring facility (based on the 282 MW Gowanus 2&3). NYISO’s current methodology will identify a Reliability Need driven by a proposed generator retirement if there is not *currently* a capacity surplus at least as large as this LOE. We estimate the RNA’s target LOE based on assumed retirements of Aging Generation in the 2026 RNA assumptions by years 7 and 10.

Figure 21: Planning-Market Gap for NYC



⁴³ See our comments on the 2025-2034 Comprehensive Reliability Plan, available [here](#).

⁴⁴ NYISO modified its calculation of transmission security limit (TSL) floors in the LCR Study to use a non-coincident load forecast and generator outage rates consistent with the RPP. See November 20, 2025 ICAPWG presentation on 2026-2027 TSL Floor Value Methodology, available [here](#).

Reliability Needs driven by aging generation in year 7 of the study could trigger regulated solutions, while needs in year 10 are primarily informational but indicate how the planning LOE in subsequent RNAs might grow.

NYISO has recently proposed to identify long-term Reliability Needs using multiple scenarios, replacing its prior approach of using a single Base Case. To the extent that some scenarios include higher load growth or less generation availability than the Base Case, this proposal is likely to result in a larger planning-market gap than depicted in Figure 21. To ensure that the market continues to provide sufficient incentives for generators to remain in service and in good repair, we recommend that NYISO consider necessary changes to market parameters (such as the target level of excess) to eliminate planning-market gaps (Recommendation #2025-1a).

5. Assumptions Used in Economic Planning Models

Transmission planning has major impacts on prices and incentives in the NYISO markets. The location and magnitude of apparent transmission needs in the Public Policy and Economic planning processes are driven by modeling assumptions in NYISO's Outlook study. The Outlook models are also used in the PPTN process to evaluate proposed projects' benefits. Hence, the assumptions and techniques used in the Outlook models have major implications for the NYISO markets because they affect which transmission needs are identified and which solutions are selected.

In our reviews of recently completed planning studies, we have identified various modeling assumptions that caused some categories of benefits to be significantly overestimated and others to be underestimated.⁴⁵ In order to provide the most realistic and informative results to planners and market participants, we recommend the following improvements in future planning studies (Recommendation #2022-3). NYISO has proposed model improvements in its 2025 Outlook study (underway at time of writing) that would address several of these recommendations:

- Perform an 'optimized' production cost model sensitivity case in which renewable capacity in locations with high marginal rates of curtailment is relocated to locations with lower marginal rates of curtailment (addressed in 2025 Outlook);
- Model realistic local capacity requirements driven by changes in the resource mix and transmission network;
- Model procurement of ancillary services in production cost models, considering how future needs will be driven by resource mix changes;
- Improve the siting and dispatch pattern of storage investments in MAPS to more realistically reflect market incentives (addressed in 2025 Outlook);

⁴⁵ See MMU Review of 2021-2040 System & Resource Outlook, August 2022, available [here](#) and MMU Evaluation of the Long Island Offshore Wind Export PPTP Report, May 2023, available [here](#).

- Represent drivers of winter risk including fuel unavailability in the capacity expansion model; and
- Consider transmission outages and day-ahead net load forecast error when estimating production cost savings.

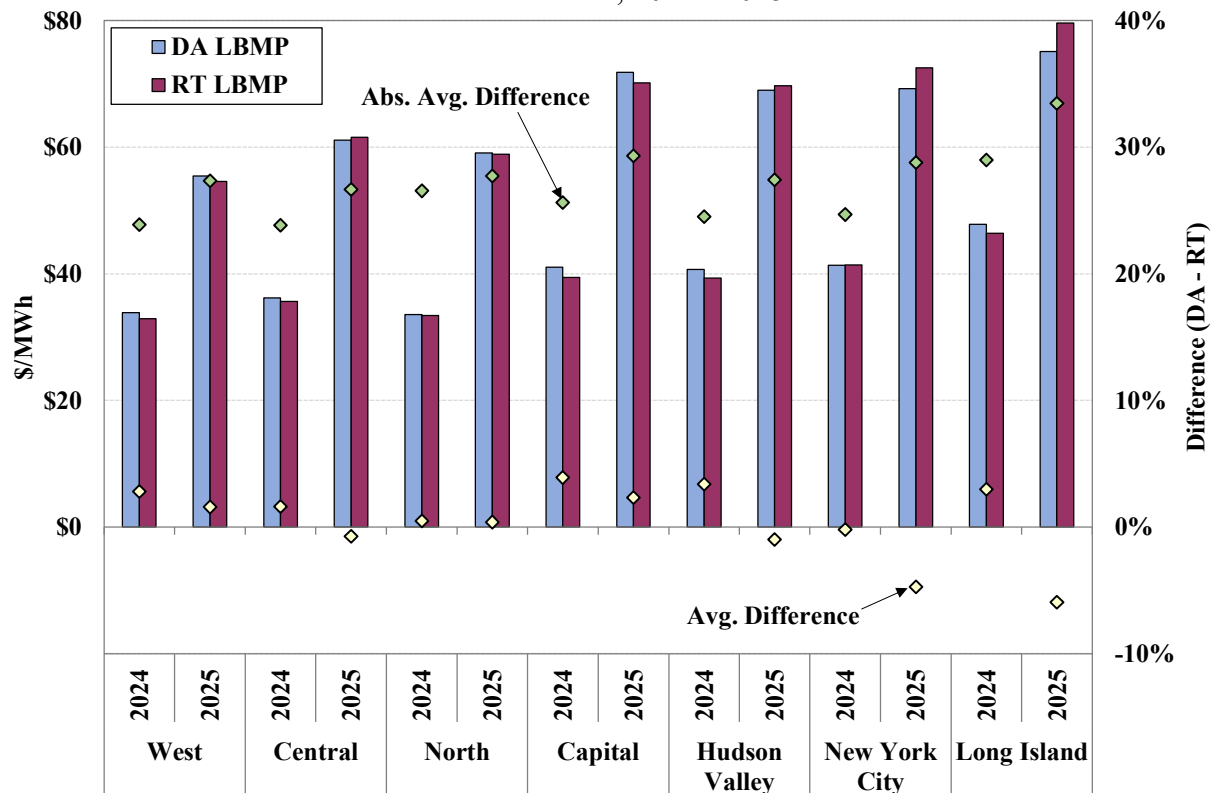
VI. DAY-AHEAD MARKET PERFORMANCE

The day-ahead market enables firms to make forward purchases and sales of power for delivery in real-time the next day. This allows participants to hedge their portfolios and manage real-time price volatility. In a well-functioning market, the day-ahead and real-time prices will not diverge systematically because participants will adjust their purchases and sales to arbitrage such differences. Price convergence is desirable also because it promotes the efficient commitment of generation, procurement of natural gas, and scheduling of external transactions. In this section, we evaluate the convergence of day-ahead and real-time energy prices (in subsection A), day-ahead scheduling patterns (in subsection B), virtual trading (in subsection C), and day-ahead scheduling when thunderstorm alerts are anticipated (in subsection D).

A. Day-Ahead to Real-Time Price Convergence

The following figure evaluates price convergence at the zonal level by reporting the percentage difference between the average day-ahead price and the average real-time price in select zones.

Figure 22: Price Convergence between Day-Ahead and Real-Time Markets
Select Zones, 2024 - 2025



The figure also reports the average absolute value of the difference between hourly day-ahead and real-time prices. These statistics are shown on an annual basis.⁴⁶

Day-ahead prices in 2025 were higher on average than real-time prices in most zones between West Zone and Capital Zone. Day-ahead prices in those regions were 0.4 percent to 2.3 percent higher than in real-time. Only the Central Zone experienced a small real-time price premium (0.7 percent). In a competitive market, day-ahead prices typically exhibit a small premium over real-time prices, while real-time prices are often higher during volatile periods.

All Southeast New York zones saw real-time prices exceed day-ahead on average. Real-time premiums were especially large in New York City (4.7 percent) and in Long Island (5.6 percent) due to real-time reserve shortages that were severe during periods of extended heat waves in the summer and real-time congestion patterns. Gas prices rose in 2025 and the weather during the hottest summer days and coldest winter days were more severe than in 2024. Real-time prices were higher in June and July in most regions when peak load conditions occurred and reserve shortages were most common. On the other hand, day-ahead price premiums were largest in January and December when gas prices and price volatility were highest.

Long Island remains more susceptible to real-time price premiums than other areas for several reasons. First, the generation fleet is older and non-quick start units are relatively slow-ramping steam units, contributing to more real-time price volatility in the morning and evening ramping hours. Second, the local gas distribution company provides less flexibility to generators that increase or decrease gas schedules intraday, and Long Island is more reliant on oil-fired generation. Third, Long Island is less connected to other areas through the high voltage transmission system.

New York City congestion increased in 2025 and volatility was higher in real-time largely because of tighter supply conditions. Transmission outages reduced the availability of combined cycle generation, while lower levels of supplemental commitment for local reliability reduced the amount of surplus operating reserves available to satisfy the local requirement for New York City and contributing to a significant increase in local 10-minute and 30-minute reserve prices.⁴⁷ In addition, summer peak load conditions were more severe, which contributed to real-time premiums in 2025.

B. Day-Ahead Load Scheduling

Under-scheduling load generally leads to lower day-ahead prices, while over-scheduling tends to raise day-ahead prices above those in real-time. Figure 23 shows the average day-ahead

⁴⁶ Section I.H in the Appendix evaluates the monthly variations of average day-ahead and real-time energy.

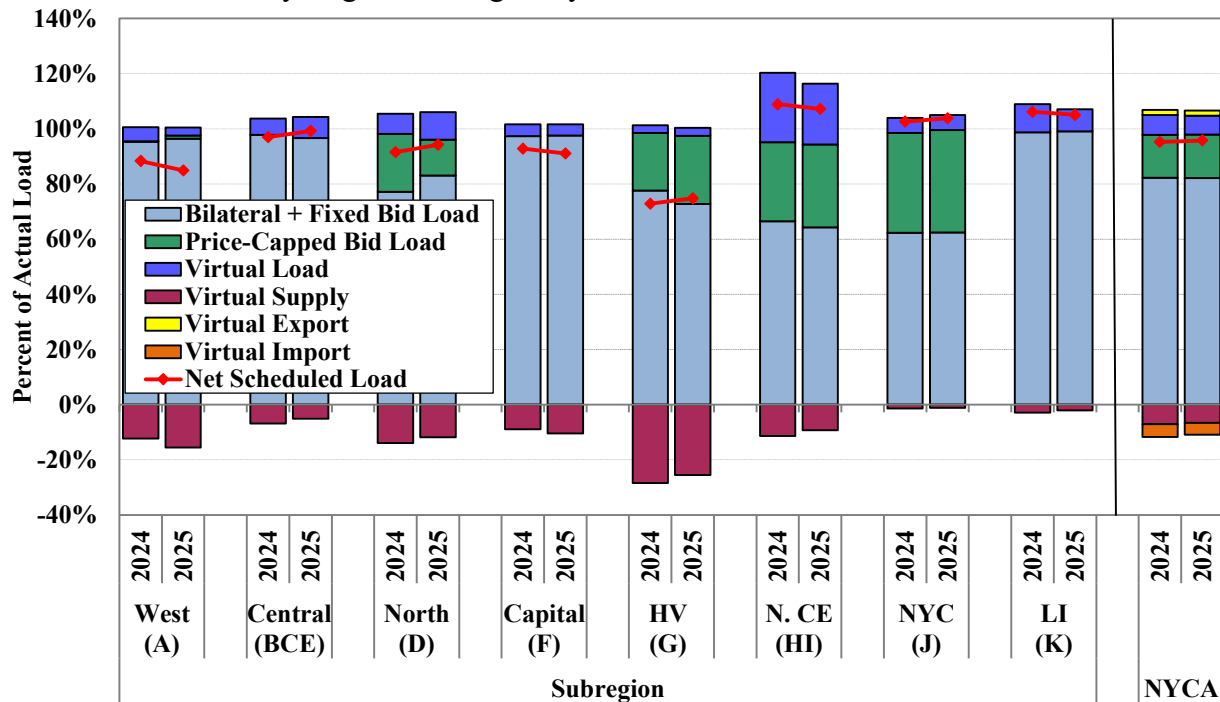
⁴⁷ Section II.E for details about increased operating reserve prices in 2025.

schedules of physical load, virtual trades, and virtual imports and exports as a percentage of real-time load in 2024 and 2025 for several regions.

Overall, net scheduled load in the day-ahead market was approximately 96 percent of actual NYCA load during daily peak load hours in 2025, slightly higher than in 2024. This pattern of net under-scheduling at the NYCA level is driven by several factors that reduce the incidence and severity of high real-time prices, including:

- The large quantity of available offline peaking generation and available import capability that can respond to unexpected real-time events,
- Out-of-market actions (i.e., SRE commitments and OOM dispatch) that bring online additional energy and reserves after the day-ahead market, and
- The tendency for renewable generators to under-schedule in the day-ahead market.

Figure 23: Day-Ahead Load Scheduling versus Actual Load
By Region, During Daily Peak Load Hours, 2024 – 2025



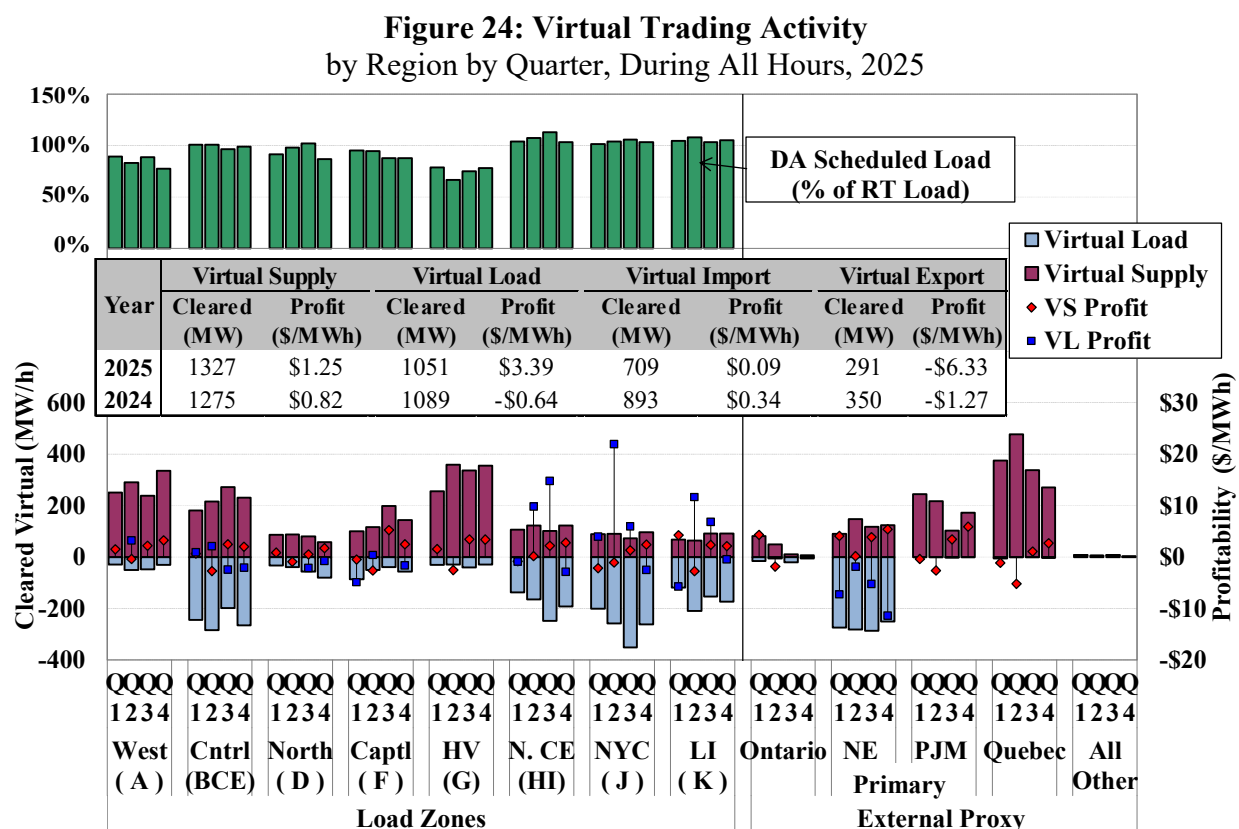
Average net load scheduling tends to be higher where volatile real-time congestion leads to very high (rather than low) real-time prices. Net scheduled load was highest in Northern ConEd (i.e., Zones H & I), New York City, and Long Island, which are prone to high real-time price spikes when unforeseen transmission and/or generation outages occur. Furthermore, these areas are affected by reductions in transmission capability that occur during thunderstorm alerts, which are evaluated in subsection D. Higher day-ahead net load scheduling generally helps increase commitment of resources in these areas, which ameliorates high real-time price events.

Localized congestion into these areas has increased in recent years following the exit from the

market of approximately 800 MW of peaking capacity between the summers of 2022 and 2023. Continued higher congestion levels likely contribute to higher day-ahead load scheduling.

C. Virtual Trading

Virtual trading helps align day-ahead prices with real-time prices, which is particularly beneficial when systematic inconsistencies between day-ahead and real-time markets would otherwise cause the prices to diverge. Such price divergence ultimately raises costs by undermining the efficiency of the resource commitments in the day-ahead market. Figure 24 summarizes virtual trading by location in 2025, including internal zones and external proxy buses.⁴⁸



The profits and losses of virtual load and supply have varied over time, reflecting the difficulty of predicting volatile real-time prices. Virtual traders netted a profit of approximately \$1.8 million in 2024 and \$30.3 million in 2025. Higher energy prices and increased price volatility provided additional opportunities for virtual traders in 2025. Virtual trading at internal zones in 2025 was very profitable, totaling \$45.8 million, while virtual trading at interfaces lost a total of

⁴⁸ See Figure A-37 in the Appendix for a detailed description of the chart. The method used in this report to calculate virtual quantities and profits has changed from past State of the Market reports. Specifically, DA scheduled transactions at the external interfaces that were cut or curtailed by system operators are excluded from the numbers in this report, resulting in higher net profitability numbers for virtual imports and exports.

\$15.5 million. The overall average rate of virtual profitability was positive at \$1.02 per MWh. In general, low virtual profitability indicates that the markets are relatively well-arbitraged, while virtual losses are unlikely to persist for a significant period.

The most consistent losses were incurred by virtual exports at the primary New England interface. We found that losing virtual trading positions at that interface were partly related to transactions associated with the sale of Renewable Energy Credits (RECs) to New England states that are typically profitable when factoring in the sale of RECs against these losses.

D. Day-Ahead Scheduling before Thunderstorm Alerts

When thunderstorms are forecasted in the Con Edison service territory, NYSRC rules require NYISO to operate the transmission system more conservatively by securing it as if the first contingency has already occurred. This results in a sudden reduction in transfer capability from upstate New York to downstate areas.⁴⁹ These events, known as Thunderstorm Alerts (TSAs), routinely reduce upstate-to-downstate transfer capability by approximately 1 to 2 GW relative to day-ahead scheduled levels, requiring increased reliance on higher-cost peaking resources in downstate areas in the real-time market.

Under the current market rules, TSA-related constraints are modeled only in the real-time market when storms conditions are present, and they are not incorporated into the day-ahead market. Consequently, the day-ahead market tends to under-commit generation in downstate areas and over-commit supply in upstate areas when thunderstorms are anticipated.

Virtual traders and other participants respond to these predictable inconsistencies by bidding-up day-ahead prices in downstate areas when TSA conditions are anticipated, which helps ensure that sufficient resources are committed. However, these load purchases are speculative and uncoordinated, since individual market participants cannot precisely predict the magnitude of transfer capability reductions or the actions of other firms. Consequently, significant discrepancies between day-ahead and real-time schedules often persist. These discrepancies can lead to extreme pricing and large balancing congestion shortfalls that are uplifted to Load Serving Entities, making such costs difficult to hedge financially.

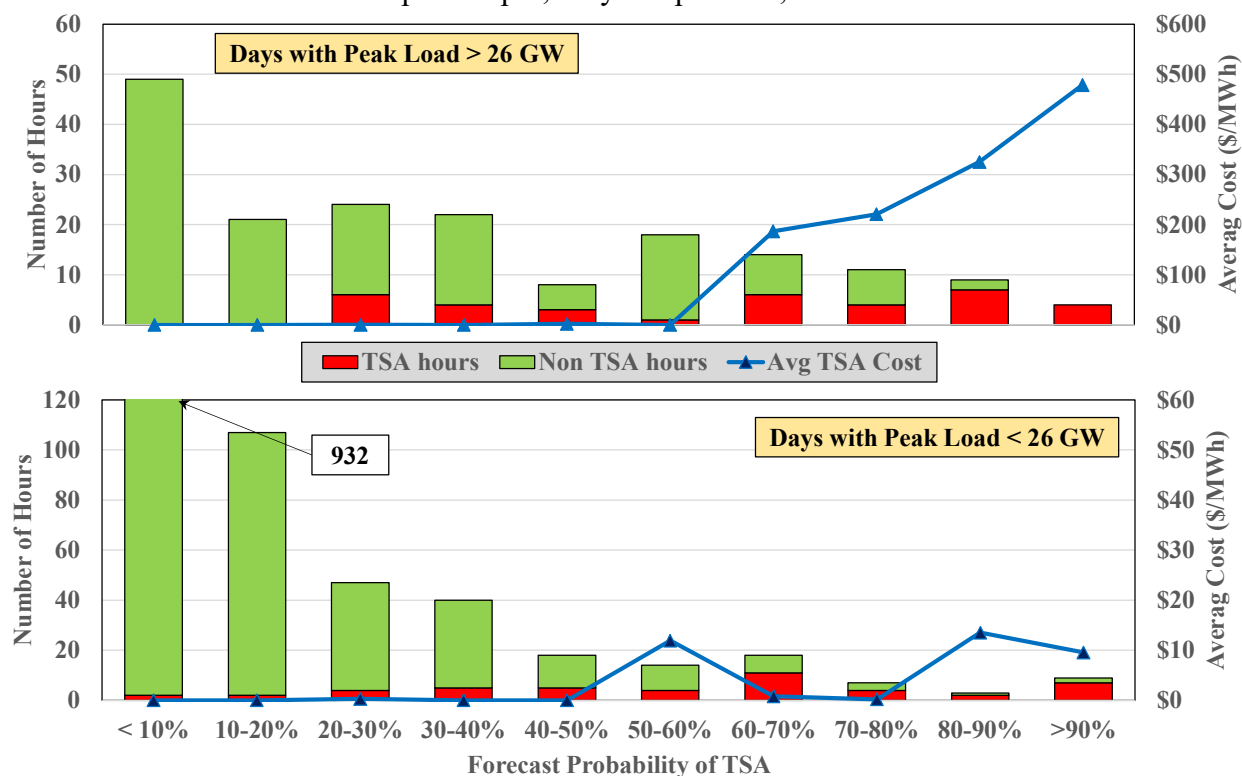
This subsection explores a potential approach to address these inefficiencies by incorporating TSA constraints into the day-ahead market. Specifically, we examine the use of weather-based TSA forecasts combined with modified GTDC values that reflect expected TSA-related congestion costs, enabling the day-ahead market to better anticipate and reflect reliability-driven transmission limitations during periods of elevated thunderstorm risk.

⁴⁹ See Reliability Rule G.1.B.R4 in *Reliability Rules & Compliance Manual for Planning and Operating the New York State Power System*, New York State Reliability Council.

We developed a weather-based TSA forecast model using publicly-available data.⁵⁰ Figure 25 presents the TSA probabilities produced by the forecasting model for 2025. The analysis focuses on the period when TSA events most frequently occur, specifically, the afternoon hours from 13 to 21 during the months of May through September. For each hour in this period, the model generates a probability of TSA occurrence. The x-axis categorizes each hour by the TSA probability. For each probability tranche, red bars show the number of hours in which TSA events actually occurred, while green bars represent hours without TSA events.

The figure also illustrates the relationship between TSA probability and associated congestion costs. These costs are measured as the difference in the congestion component of LBMPs between Zone G and Zone J attributable to TSA constraints. The cost is zero for non-TSA hours and TSA hours without binding TSA constraints. For each probability tranche, the reported cost reflects the average across all hours, both TSA and non-TSA, within that tranche. Because TSA impacts are typically most pronounced during high-load periods, the analysis further distinguishes between high-load days (when peak load exceeds 26 GW), shown in the top panel, and all other days, shown in the bottom panel.

Figure 25: Forecasted TSA Probability vs. Actual TSA Congestion Costs
1pm - 10pm, May - September, 2025



⁵⁰ See Section III.J in the Appendix for a more detailed description of the forecast model.

The TSA forecasting model demonstrates strong predictive accuracy. Across all hours in the study period, when the model forecasts a probability above 70 percent, TSA events occurred in approximately 65 percent of those hours. This accuracy increases with higher probability thresholds, reaching 80 percent of hours for forecasts above 80 percent. Model performance is stronger during high-load conditions. On days when peak load exceeded 26 GW, TSA events occurred in approximately 85 percent of hours when the forecast probability exceeded 80 percent.

The congestion costs from TSA events were modest on days with peak load levels below 26 GW, averaging approximately \$10 per MWh in hours when the forecast probability exceeded 80 percent. In contrast, on days with peak load above 26 GW, TSA congestion costs increased substantially, averaging \$300 to \$500 per MWh during hours with forecast probability above 80 percent. Since TSA constraints are not modeled in the day-ahead market, these costs are reflected in the form of large balancing congestion shortfalls uplifted to load. Section VIII.C discusses the calculation and allocation of these costs.

Since TSA events can be predicted with a high degree of precision, it would be efficient to incorporate them directly into the day-ahead market during hours of high risk of costly TSA conditions. Of the 1,377 hours evaluated, we found just 50 hours with a high risk of costly TSA conditions (defined as hours with TSA likelihood above 60 percent and load above 26 GW or TSA likelihood above 80 percent and load below 26 GW). Yet, these high-risk hours accounted for 99 percent of TSA congestion costs. Hence, we recommend that NYISO consider modeling TSA constraints in the day-ahead market during hours with high risk of costly TSA conditions.⁵¹ In addition, day-ahead GTDC values could be reduced for TSA constraints with lower forecasted TSA probabilities and load levels.

Incorporating TSA constraints into the day-ahead market would improve commitment decisions by ensuring that an appropriate level of generation is scheduled in downstate areas when TSA conditions are likely. This would reduce the need for costly real-time redispatch, narrow the gap between day-ahead and real-time outcomes, and improve overall scheduling and pricing efficiency. Moreover, it would allow a greater share of TSA-related costs to be reflected in market prices rather than uplift, thereby improving cost allocation, enhancing hedging opportunities, and strengthening incentives for market participants to respond efficiently to system conditions.

⁵¹ See Recommendation #2025-5.

VII. MARKET OPERATIONS

The purpose of the wholesale market is to coordinate resources efficiently to satisfy demand while maintaining system reliability. The day-ahead market should commit the lowest-cost resources necessary to meet expected conditions on the following day, and the real-time market should dispatch the available resources efficiently. Market prices should reflect the cost of satisfying demand while maintaining security and reliability.

Efficient real-time prices promote competitive supplier behavior, encourage demand response participation, and provide appropriate investment signals for new resources and transmission in the most valuable locations. During shortages of ancillary services and/or transmission capability, real-time prices should reflect the value of foregone supply and incentivize resources to support reliability at each location of the system. Effective system operations are also important, as they significantly affect wholesale market efficiency and costs.

In this section, we evaluate five aspects of market operations, focusing on the efficiency of incentives, scheduling, and pricing, particularly during tight operating conditions:

- Market Performance under Ancillary Services Shortage Conditions (sub-section A)
- Market Performance during Transmission Shortages (sub-section B)
- Real-Time Pricing of GTs Bidding Multi-Hour Minimum Run Times (sub-section C)
- Supplemental Commitment for Reliability (sub-section D)
- Uplift from Bid Production Cost Guarantee (BPCG) payments (sub-section E)

This section also discusses several recommendations to enhance pricing efficiency and strengthen performance incentives in both the day-ahead and real-time markets.

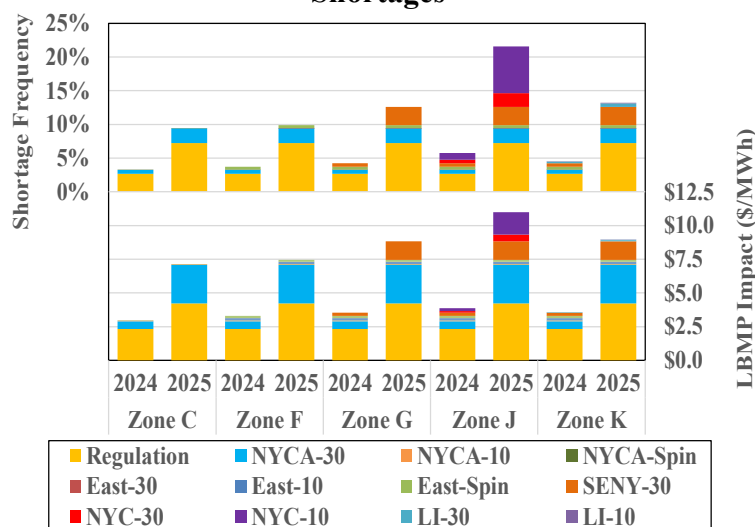
A. Market Performance under Ancillary Services Shortage Conditions

This subsection evaluates market operations and price formation during shortages of operating reserves and regulation service. Figure 26 presents the frequency of operating reserves and regulation shortages and their impact on LBMPs for five representative zones in 2024 and 2025. The stacked bars in the upper portion of the figure show the frequencies of shortages for each ancillary service product in the applicable zones, while the stacked bars in the lower portion show their impact on LBMPs in those zones.

Shortages occurred more frequently in 2025 than in 2024. This increase was most pronounced for reserve products whose demand curves include low-value (e.g., \$25 or \$40 per MWh) segments, such as regulation, system-wide and SENY 30-minute reserves, and reserves in New York City and Long Island. The NYISO has reserve requirements with low-value segments to help ensure that reserves are distributed across the system rather than being concentrated in

specific areas from which the reserves may not be fully deliverable following a contingency. The significant increase in shortages of these products reflected higher energy prices (resulting from higher natural gas prices, higher RGGI allowance prices, increased exports to Quebec, reduced hydroelectric output, and transmission outages limiting availability of low-cost generation in New York City) and the associated opportunity costs of providing reserves. Under these conditions, it can be more economic to allow shortages of these reserves than to procure them through more expensive redispatch options.

Figure 26: LBMP Impact of Ancillary Services Shortages



Regulation shortages remained the single most significant type of ancillary services shortage, occurring in 7 percent of intervals in 2025. At the regional level, system-wide and SENY 30-minute reserve shortages were most common, occurring in 2 to 3 percent of intervals, while other reserve shortages were relatively infrequent in 2025. At the zone level, New York City reserve shortages were the most common, occurring in 7 percent of intervals. Collectively, in 2025, these shortages increased average LBMPs by 10 to 15 percent across the system. Thus, ancillary services shortages have a significant impact on investment signals, shifting incentives toward flexible generation that is able to provide ancillary services at low cost.

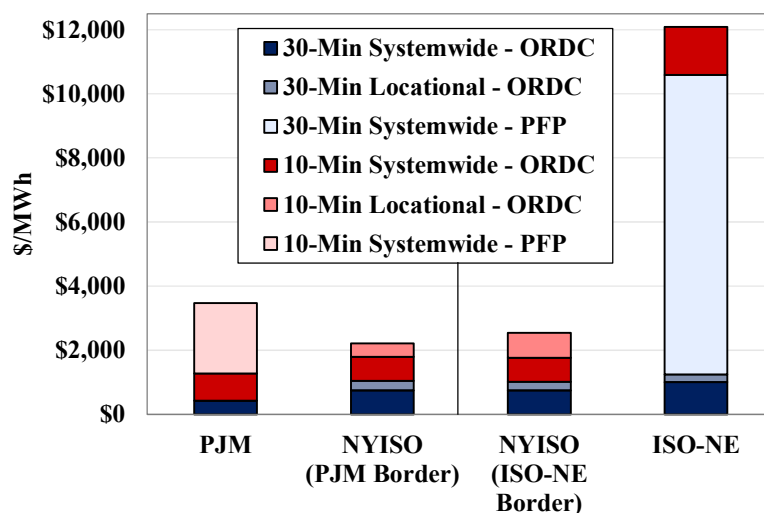
During operating reserve shortages, real-time prices should accurately reflect the foregone value of reserves, providing adequate incentives to attract resources needed to maintain reliability without resorting to out-of-market actions. The subsection below evaluates NYISO's shortage pricing, including the efficiency of the Operating Reserve Demand Curve (ORDC) in reflecting the reliability value of each class of reserves.

1. ORDC Price Levels in NYISO versus Neighboring Areas

While most of the LBMP impact of shortage pricing in the NYISO market comes from low-value segments, NYISO has high-value segments (e.g., \$500 to \$775 per MWh) for requirements that are essential for meeting mandatory transmission security and reliability criteria at the system-level, for Eastern New York, and for Southeast New York. However, NYISO's ORDCs

are low relative to shortage pricing levels in PJM and ISO New England.⁵² Figure 27 compares NYISO’s shortage pricing incentives with those provided by its neighbors, both of which have shortage pricing in their energy markets that is supplemented by additional “pay-for-performance” settlements in the capacity market.⁵³ The figure shows that shortage pricing is generally much lower in New York than in the neighboring markets:

Figure 27: Shortage Pricing in NYISO vs. Neighboring Markets



- During deep shortages of 30-minute reserves, NYISO shortage pricing, including locational adders, approaches roughly \$1,000 per MWh, significantly lower than ISO-NE’s current level of approximately \$10,600 per MWh.
- During deep shortages involving multiple 30-minute and 10-minute reserve requirements, NYISO invokes statewide shortage pricing levels that can exceed \$2,000 per MWh under severe conditions. However, this remains substantially lower than the current total shortage pricing in neighboring markets – nearly \$3,500 per MWh in PJM and over \$12,000 per MWh in ISO-NE.

Consequently, when NYISO is in a much less reliable state than PJM or ISO-NE, market participants will have strong incentives to export power from (or reduce imports to) NYISO. This pricing disparity will tend to undermine reliability in New York or necessitate out-of-market actions by NYISO operators to maintain system reliability.

2. ORDC Price Levels Compared to the Reliability Value of Reserves

In addition to comparing NYISO’s shortage pricing with its neighbors, we evaluate how effectively NYISO’s shortage pricing reflects the reliability risks from reserve shortages. Shortage pricing is determined by an RTO’s ORDC for each class of reserves. Ideally, the ORDC should indicate the marginal reliability value of reserves, based on the value of being able to serve the load, referred to as the Value of Lost Load (VOLL). As reserve levels fall, the risk of load shedding increases, implying the marginal value of reserves can be estimated as:

⁵² ISO New England has Pay For Performance (“PFP”) rules, and PJM has Capacity Performance (“CP”) rules. These rules provide incentives similar to shortage pricing through adjustments to capacity payments.

⁵³ See Figure A-82 in the Appendix for description of this chart.

$$\text{Expected Value of Lost Load (EVOLL)} = \text{VOLL} \times \text{Probability of losing load}$$

Based on studies examining the value of reliability, a reasonable estimate of NYISO's VOLL is approximately \$30,000 per MWh.⁵⁴ The slope of an efficient ORDC depends on how rapidly the probability of losing load increases as operating reserves decreases, which is estimated from the likelihoods of random contingencies and conditions that could arise during a shortage in the NYISO market. To quantify the relationship between available operating reserves and the probability of losing load, we used a Monte Carlo simulation tool that considered random forced generation outages, wind forecast errors, net load forecast errors, and import curtailments by neighboring areas (collectively referred to as "unexpected losses of net supply").⁵⁵

Figure 28: Comparison of MMU Economic ORDC to the Current 10-Minute ORDCs
NYCA Wide 10-Minute Reserves

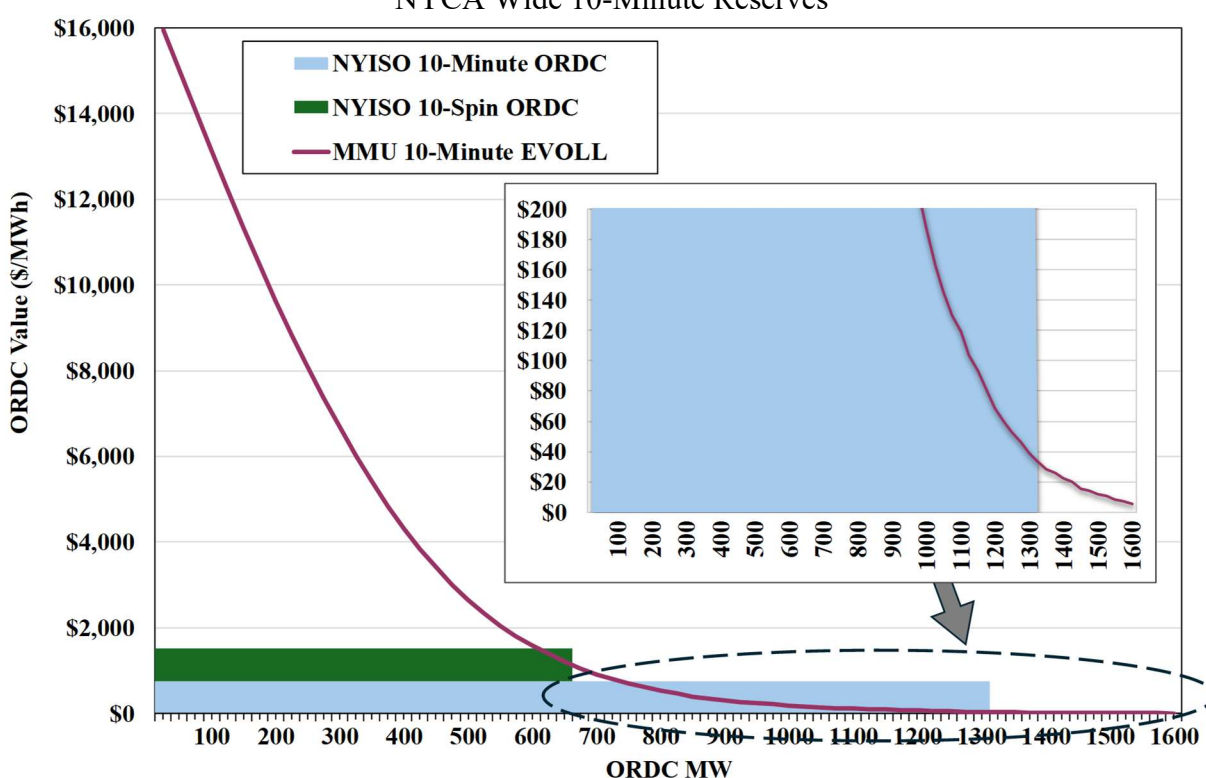


Figure 28 compares our estimates of the EVOLL curve for NYCA-wide 10-minute reserve requirements to the current ORDC utilized in the NYISO market, including both 10-minute spin and 10-minute total reserves that together reflect the total value of 10-minute reserves.

This analysis reveals that the current ORDC curves significantly undervalue the marginal reliability value of 10-minute reserves when available 10-minute reserve levels drop below 600

⁵⁴ See Section V.G in the Appendix for the mentioned studies.

⁵⁵ See Section V.G in the Appendix for the simulation methodology in more details.

MW. Specifically, the MMU-estimated EVOLL curve rises above \$16,000 per MWh when 10-minute reserves are nearly depleted, which is 10 times higher than the current combined ORDC value of \$1,525 per MWh for 10-Minute Spin and 10-Minute Total reserves. However, when reserve levels are above 800 MW up to 1,310 MW, the existing 10-minute total ORDC is significantly higher than the estimated marginal reliability value of these reserves. 10-minute reserves have some value beyond the minimum requirement of 1,310 MW.

A comparable analysis of the 30-minute ORDC is discussed in Appendix Section V.G. It shows that the existing 30-minute ORDC in the NYISO market undervalues 30-minute reserves when they are at or below approximately 725 MW. The estimated marginal reliability value at near-depletion is roughly \$2,800 per MWh, nearly four times of the \$750 per MWh set by the current ORDC.

Finally, the MMU-estimated EVOLL curves extend beyond the existing requirements of 1,310 MW for 10-minute reserves and 2,620 MW for 30-minute reserves. These estimates of the reliability value of holding additional reserves beyond the base requirements support the NYISO's plan to implement "uncertainty reserve" requirements as part of the *Balancing Intermittency* project in 2026.

3. ORDC Price Levels – Conclusions

We recommend NYISO revise its current ORDC curves to accomplish two primary objectives:

- **Schedule resources necessary to satisfy reliability criteria without resorting to OOM actions** – Current ORDCs may not provide sufficient incentives, especially since PJM and ISO New England have adopted unreasonably strong shortage pricing incentives.
- **Achieve better alignment with the estimated reliability value of reserves** – Existing statewide ORDCs undervalue reserves during deep shortages of 10-minute and 30-minute operating reserves when reliability risks are highest.

This recommendation should be a high priority because the demand for resources to respond to emergency conditions in real-time will become increasingly important.⁵⁶ The large-scale entry of intermittent renewables will significantly increase net load forecast uncertainty and, consequently, the estimated reliability value of reserves. Implementing this recommendation will improve incentives for generation and load flexibility. Any cost increase associated with higher ORDCs would be offset by corresponding reductions in capacity market demand curves.

B. Market Performance during Transmission Shortages

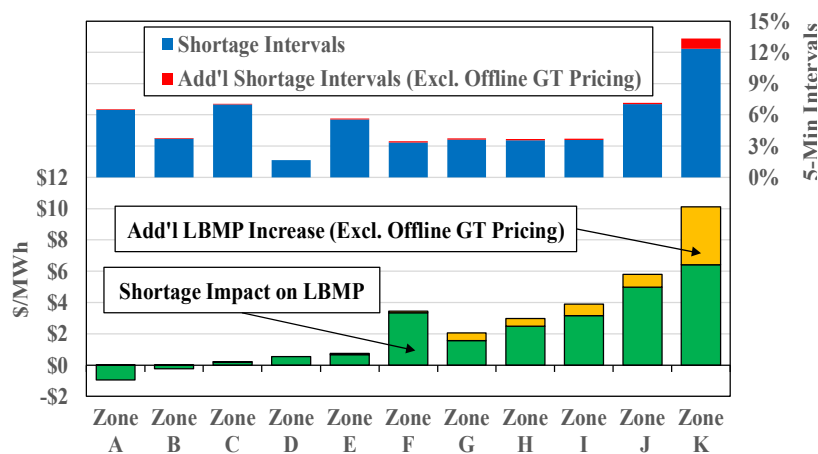
It is crucial for the market to establish efficient prices during transmission shortages that accurately reflect their severity and properly value the effects of transmission bottlenecks across

⁵⁶ See Recommendation 2017-2.

the system. Most transmission shortages are minor and short-lived, posing no significant threat to security or reliability. This is largely because NYISO uses a Constraint Reliability Margin (CRM) of 5 to 100 MW, which provides a buffer between modeled flows and transmission limits for each facility. This margin ensures that transient deviations between schedules and actual output do not lead to violations of transmission security criteria.⁵⁷ To reflect the severity of shortages, NYISO employs a Graduated Transmission Demand Curve (GTDC), which aligns MW shortage levels with each facility's CRM. This approach results in moderately high prices during slight shortages, escalating as the shortage grows more severe.⁵⁸

Figure 29 shows the frequency of transmission shortages and their impact on real-time LBMPs across eleven load zones in 2025. The blue bars in the upper portion of the figure show the share of five-minute intervals in which transmission shortages contributed to zonal LBMPs, while the green bars in the lower portion show the average annual impact of these shortages on LBMPs in each zone.

Figure 29: RT LBMP Impact of Transmission Shortages in 2025



In 2025, transmission shortages occurred in approximately 20 percent of all 5-minute market intervals, up from 9 percent in 2024. This increase was driven largely by higher generation redispatch costs associated with elevated energy prices. Most of these were small violations of the lower steps of the GTDC. Transmission shortages affected zonal LBMPs least frequently in the North Zone (1.6 percent of intervals) and most frequently in the Long Island Zone (12.3 percent of intervals). The average annual LBMP impact ranged from approximately -\$0.95 per MWh in the West Zone to nearly \$6.5 per MWh in the Long Island zone.

The figure also highlights an ongoing issue that undermines efficient price formation by preventing the market software from fully recognizing real-time transmission shortages. The

⁵⁷ A default CRM value of 10 MW is used for 69 kV and 115 kV constraints, while a default CRM value of 20 MW is used for most facilities at higher voltage levels. NYISO may adjust the CRM for an individual facility based on operating experience as necessary to maintain security.

⁵⁸ Since November 14, 2023, most transmission constraints use a GTDC with five steps of equal length at: \$200/MWh, \$350/MWh, \$600/MWh, \$1,500/MWh, and \$2,500/MWh. Each step is one-fifth of the CRM in length. Constraint violations larger than the CRM use a GTDC price level of \$4,000/MWh.

issue arises from the NYISO’s “offline GT pricing” logic, which assumes that offline GTs are dispatchable within five minutes, even though they cannot realistically start that quickly. As a result, constraint shadow prices and associated congestion costs are artificially suppressed when offline GTs are treated as if they can start up in five minutes.

The red bars in the upper portion of the figure show the additional intervals in which transmission outages would have affected each zone if the offline GT pricing logic was eliminated. Correspondingly, the orange bars in the lower portion show the estimated additional LBMP increases in each zone.⁵⁹ The largest effects are observed on Long Island, where 0.6 percent of intervals would have experienced more severe shortages and an additional 1.0 percent of intervals would have experienced transmission shortages and average LBMP would have increased by approximately \$3.7 per MWh in 2025. This result is consistent with Long Island’s heavy reliance on peaking resources that are treated as capable of starting in five minutes.

The offline GT pricing logic also contributes discrepancies between modeled and actual transmission flows, reducing the effectiveness of the real-time market models in maintaining transmission security, particularly in constrained areas such as Long Island. To address these discrepancies, the NYISO may apply larger CRMs on certain constraints, such as the 345 kV lines connecting upstate New York to Long Island.

Figure 30 shows our analysis of Long Island transmission facilities, comparing the magnitude of transmission constraint violations calculated by market software (represented by the blue points), which assumes offline GTs can generate output, with the actual constraint violations recognizing that offline GTs remain offline and produce no output (represented by red points).⁶⁰

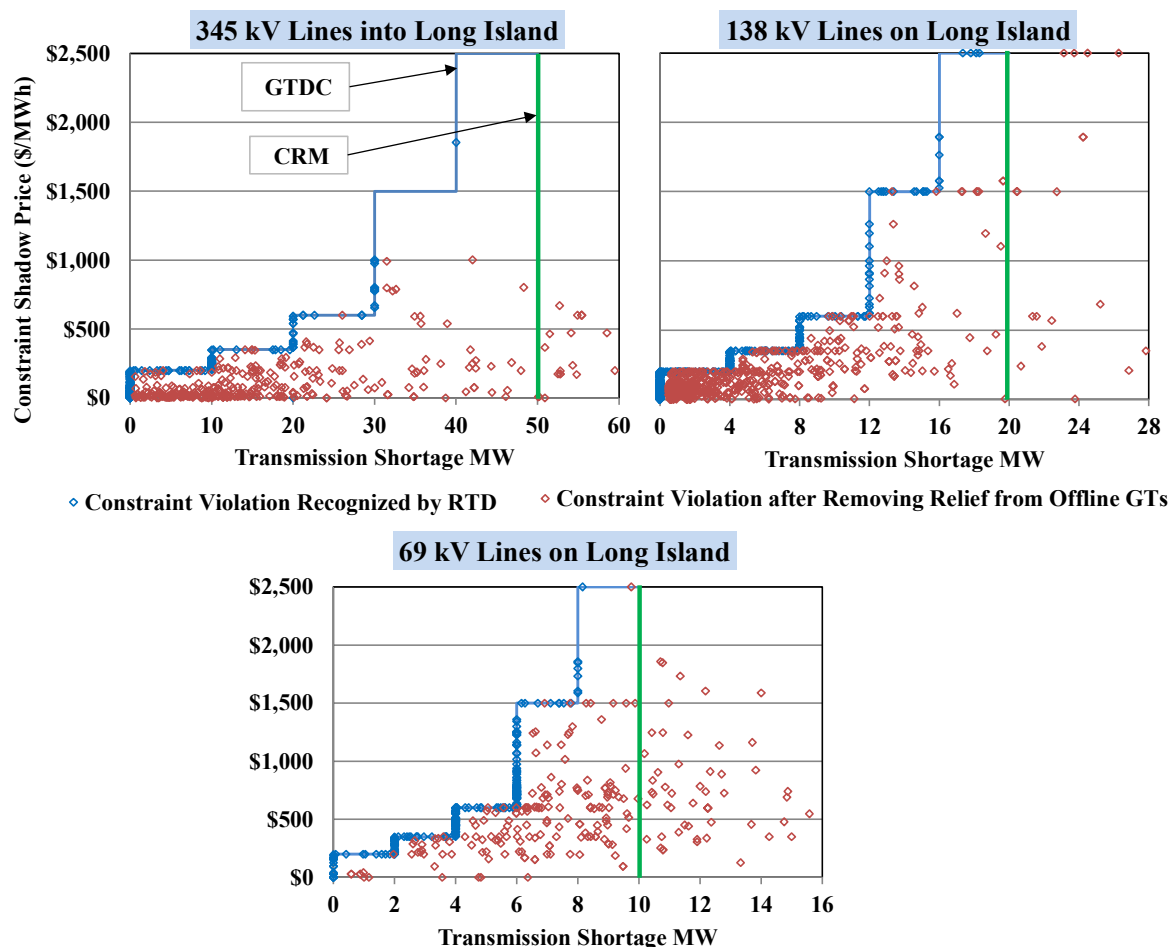
Most of the GTs that are treated by NYISO’s market software as capable of responding to a 5-minute dispatch instruction while offline are located in Long Island. This “offline GT pricing” practice may lead NYISO to reduce modeled transfer limits, thereby constraining transmission flows at artificially low levels in areas that rely more on peaking units such as Long Island. Consequently, this leads to unnecessary generation dispatch and inflated production costs during most periods. Given these inefficiencies, we recommend that NYISO eliminate offline fast-start pricing.⁶¹

⁵⁹ The estimated increase in LBMP is based on the corresponding increase in GTDC values after removing the impact of offline peaking resources.

⁶⁰ See Figure A-86 in the Appendix for description of the chart.

⁶¹ See Recommendation 2020-2.

Figure 30: Transmission Constraint Shadow Prices and Violations
With and Without “Relief” from Offline GTs, 2025



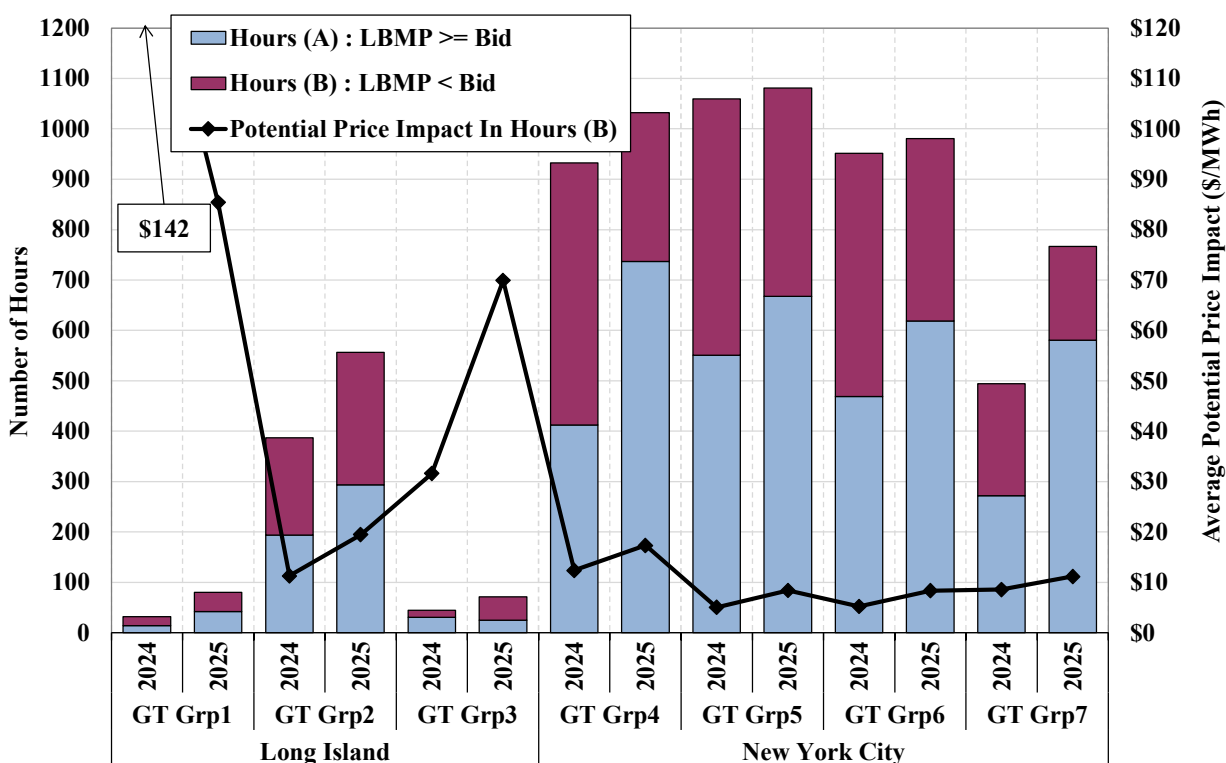
C. Real-Time Pricing of Gas Turbines Bidding Multi-Hour Minimum Run Time

Gas turbines run more frequently under tight conditions, making it particularly important to establish efficient real-time prices that incent resources to be flexible. Under existing fast-start pricing rules, real-time prices include the as-offered deployment costs of fast-start units, including start-up, incremental energy, and minimum generation costs.

However, fast-start units with minimum run time offers exceeding one hour are currently excluded from fast-start pricing eligibility. Specifically, the real-time market software (including RTC and RTD) and settlement rules ignore these units' actual minimum run time offers, deeming them to have a one-hour minimum run time whenever they submit economic real-time offers. Yet, despite being treated the same (for scheduling purposes) as units offering a one-hour minimum run time, these units are not eligible to set prices. This prevents LBMPs from accurately reflecting the true costs of maintaining reliability when these gas turbines are needed.

Figure 31 evaluates prices during economic commitments of gas turbines offering multi-hour minimum run times in the real-time market in 2024 and 2025.⁶² Gas turbines in New York City and Long Island are combined into seven groups based on their electric connection to the grid.

Figure 31: Prices During Commitments of GTs Offering Multi-Hour Min Run Times
2024 - 2025



Economic starts were more frequent among the four GT groups in New York City, ranging from approximately 500 to 1,100 hours each year over the past two years. However, LBMPs fell below GT costs in 41 percent of these hours. This disparity was largely due to these GTs not being eligible to set prices when their minimum run time offers exceed one hour. We estimate that if these GTs were allowed to set prices, the average LBMP during these hours would have been increased by up to \$5.17 per MWh at various locations in New York City.

On Long Island, although the three GT groups had fewer economic starts in each year, GT operating costs exceeded LBMPs in nearly 50 percent of these start hours. The potential price impact was considerably higher at these Long Island locations, with estimated LBMP increases ranging from \$11 to \$142 per MWh during the affected hours.

Overall, in 2025, the ineligibility of these GTs to set prices resulted in depressed real-time LBMPs in approximately 9 percent of intervals in New York City and 4 percent of intervals on

⁶² Economic GT commitments include GT start-ups made economically by RTC, RTD, and RTD-CAM, excluding self-schedules. See Figure A-73 in the Appendix for more details of this analysis.

Long Island. The potential increase in LBMPs would translate to an annual net revenue increase of \$2 to \$6 per kW-year across various load pockets in New York City and Long Island, although the price impacts could extend to broader areas depending on congestion patterns.

Given these inconsistencies between real-time prices and scheduling, we recommend NYISO revise its fast-start pricing criteria. Specifically, fast-start pricing eligibility should be based on the minimum run time used for scheduling, rather than on the submitted offer parameter, which is currently disregarded for real-time scheduling. By aligning these criteria, NYISO can enhance price efficiency and provide more appropriate investment signals for market participants.⁶³

D. Supplemental Commitment for Reliability

Supplemental commitments occur when a generating unit is committed to address local or systemwide reliability needs not reflected in the day-ahead or real-time markets. There are four primary types of supplemental commitments:

- **Day-Ahead Reliability Units (DARU)** commitment occurs before the day-ahead market at the request of transmission owners or NYISO for anticipated reliability needs;
- **Day-Ahead Local Reliability Rule (LRR)** commitment occurs within the day-ahead market's economic commitment process specifically to meet un-priced local reliability needs in New York City;
- **Supplemental Resource Evaluation (SRE)** commitment occurs after the day-ahead market at the request of transmission owners or NYISO for reliability; and
- **Forecast Pass Commitment (FCT)** occurs within the day-ahead market after the economic pass if it does not schedule enough physical resources to satisfy forecasted load and reserve requirements.

These out-of-market commitments highlight gaps in the market design, indicating a need for reliability services that the market is currently not procuring efficiently. OOM commitments depress energy and reserves prices, which undermines market incentives to maintain reliability and generates uplift costs. Hence, it is important to minimize supplemental commitments and look for ways to procure the underlying reliability services through the day-ahead and real-time market systems.

1. Supplemental Commitment in New York State

Figure 32 summarizes four types of reliability commitment by region in 2024 and 2025. In 2025, an average of approximately 376 MW of capacity was committed for reliability, marking a

⁶³ See Recommendation 2023-2.

6 percent decrease from 2024.⁶⁴ This decline was largely attributable to a 47 percent reduction in New York City, as discussed further below.

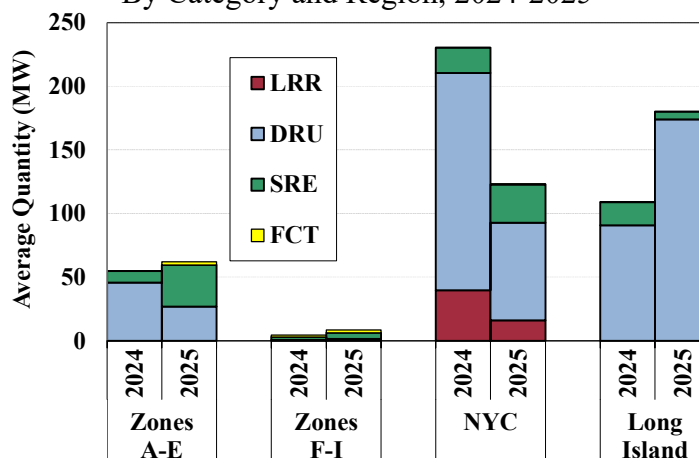
Conversely, reliability commitments on Long Island increased by 68 percent, partially offsetting the overall decrease. In recent years, large steam turbines have been supplementally committed with increasing frequency to manage high-voltage conditions during light-load periods, particularly in shoulder months. In 2025, such commitments occurred on 73 days and

accounted for the majority of reliability commitments on Long Island. Overall, Long Island accounted for 47 percent of all reliability commitments in 2025, the largest share ever observed.

Reliability commitments on Long Island to satisfy N-1-1-0 criteria (i.e., maintaining normal line loading following the two largest contingencies) occurred on 11 days in 2025. This indicates that the current Long Island 30-minute reserve requirement is inadequate to maintain reliability under certain conditions. The absence of these criteria in the market software forces system operators to resort to OOM commitments when needed, leading to understated prices and poor market incentives. Modeling reserve requirements in the Long Island load pockets where these OOM actions are used would improve efficiency and encourage new resources to locate where they are most valuable. Hence, we recommend that the NYISO implement local reserve requirements in Long Island that are adequate to maintain reliability rather than rely on out-of-market actions.⁶⁵

Although the day-ahead and real-time markets schedule resources to satisfy reserve requirements on Long Island, reserve providers are currently not paid reserve clearing prices corresponding to these requirements. Instead, they are paid based on the clearing prices for the larger Southeast New York region. Compensating reserve providers in accordance with the market scheduling decisions would improve market incentives, providing better signals to new investors over the long term. Hence, we recommend the NYISO set day-ahead and real-time reserve clearing prices considering reserve constraints for Long Island, which the NYISO plans to implement with the Dynamic Reserves project.⁶⁶

Figure 32: Supplemental Commitment for Reliability in New York
By Category and Region, 2024-2025



⁶⁴ See Section V.I in the Appendix for a description of the figure.

⁶⁵ See Recommendation 2024-1.

⁶⁶ See Recommendation 2019-1. See “Long Island Reserve Constraint Pricing”, MIWG, February 7, 2024.

In upstate regions, OOM commitments to satisfy N-1-1 criteria in the North Country load pockets occurred on 119 days in 2025, down from 205 days in 2024, reflecting fewer transmission outages that limited the import capability into the pocket. Despite this decline, such OOM commitments still accounted for nearly half of all reliability-committed capacity in upstate regions. As in New York City and Long Island, we recommend modeling these local reserve requirements to improve market efficiency and establish proper market signals for future investments.⁶⁷ An additional 45 percent of reliability commitments in upstate regions were made on high load days to address systemwide capacity deficiency concerns.

2. Reliability Commitment in New York City and North Country Load Pockets

Table 4 further examines reliability commitments made by the local transmission owner in New York City and by NYISO for North Country load pockets. Although these reliability commitments declined in 2025, they still represented more than 40 percent of all reliability commitments.⁶⁸ We evaluate these OOM commitments to assess whether they are necessary for reliability and cost-effective, and to identify potential gaps in NYISO's market design.⁶⁹

**Table 4: Reliability Commitment in New York City and North Country
2025**

Category of Reliability-Committed Capacity	New York City		North Country	
	Committed Capacity (GWh)	% of Total	Committed Capacity (GWh)	% of Total
Economic	500	33%	58	34%
Verified - Needed	93	6%	78	46%
Verified - Headroom	296	19%	22	13%
Unverified	644	42%	11	7%
Total GWh	1533		168	

In 2025, OOM commitments to satisfy reserve requirements occurred on 119 days in the North Country load pockets and 121 days in New York City load pockets. In New York City, 39 percent of these commitments were verified by the MMU as either economic or necessary to satisfy specific reliability requirements based on applicable system conditions, including forecasted load, generation and transmission availability, and contingencies. Conversely, 80 percent of North Country commitments were verified as economic or necessary for reliability.

⁶⁷ See Recommendation 2024-1.

⁶⁸ See Section V.I in the Appendix for more details for this analysis.

⁶⁹ NYISO's *Day Ahead Scheduling Manual*, Section 4.2.6 requires a TO requesting the commitment of unit for reliability to provide the reason and NYISO to review and validate the request.

Many “unverified” reliability commitments in New York City likely result from several factors: (a) DARU requests are routinely made three to five days in advance for consecutive days, when forecasts are often inherently less accurate and when some sources of supply have not yet been scheduled; (b) the local transmission owner may have operational requirements that are either not visible or not evaluated by the NYISO models; or (c) conservative assumptions are sometimes applied under special system conditions. While recent new transmission upgrades have reduced the frequency of reliability commitments, they have also changed the boundaries of load pockets for which the commitments were made. Future reports will seek to quantify: (a) the extent to which the current process leads to over-commitment of resources due to forecast errors, and (b) the specific local requirements that are not adequately modeled in the NYISO models.

A significant portion of the committed capacity, 19 percent in New York City and 13 percent in North Country, was categorized as surplus headroom on the units beyond the amount of capacity need we could verify, including hours committed to satisfy Minimum Run Time requirements. This was due to limited flexible generation options available to meet these reliability needs. Smaller flexible resources like batteries and DERs could offer more cost-effective reliability solutions in some areas. However, current market rules do not provide incentives for satisfying these local needs. Given the impact of OOM reliability commitments on resource scheduling and pricing, we recommend NYISO model the underlying N-1-1 and N-1-1-0 requirements explicitly as local reserve requirements. This would encourage investment in smaller dispatchable resources (e.g., batteries and DERs) to effectively satisfy these reliability needs.⁷⁰

3. Forecast Pass Commitment

Forecast pass commitments were infrequent in 2025, and the associated committed capacity was small. Nonetheless, we identified two issues in this process.

First, certain quick-start capacity was incorrectly categorized as slow-start capacity in the Forecast Pass. Consequently, most Forecast Pass commitments would not have been necessary if these quick-start units had been properly classified.⁷¹ Addressing this issue will require software changes.

Second, our evaluation showed that the physical energy and reserves scheduled in the day-ahead market were frequently below forecasted load and reserve requirements.⁷² As a result, the NYISO relies on substantial reserves from resources that are not compensated in the day-ahead market or obligated to be available the next day. As natural gas pipeline constraints limit fuel supplies, it becomes riskier to rely on resources that have no obligation or incentive to buy fuel.

⁷⁰ See Recommendation 2024-1.

⁷¹ See Section V. I in the Appendix for more information about this analysis.

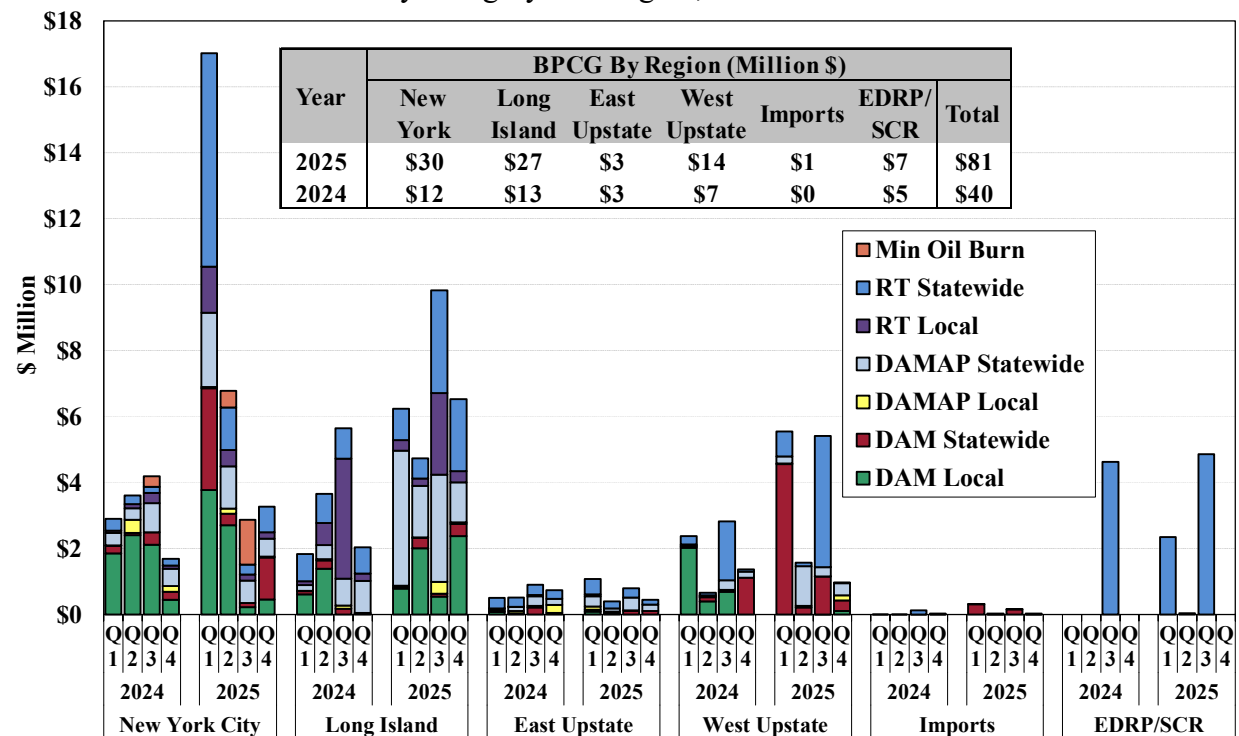
⁷² See Section V. I in the Appendix for more information about this analysis.

This issue could be efficiently addressed by explicitly modeling these needs in the market with appropriate dynamic reserve requirements.⁷³ In some cases, these reserve requirements could be satisfied by resources with longer lead times than those currently providing 10- and 30-minute reserves. However, rather than introduce new longer lead-time reserve products, it is likely more efficient to incorporate such requirements in the existing market framework by raising the 30-minute reserve requirement (at least in the short-term).

E. Guarantee Payment Uplift Charges

When suppliers scheduled by NYISO do not fully recover their as-offered costs from energy and ancillary services sales, they receive supplemental guarantee payments. NYISO recovers these payments through guarantee payment uplift charges. However, these uplift charges are difficult to hedge and do not provide transparent economic signals to market participants and potential investors. Therefore, it is important to minimize these charges. When the markets reflect reliability requirements and system conditions efficiently, uplift charges should be relatively low. Figure 33 shows guarantee payment uplift for four local reliability categories and three non-local reliability categories in 2024 and 2025 on a quarterly basis.⁷⁴

Figure 33: Uplift Costs from Guarantee Payments in New York
By Category and Region, 2024 – 2025



⁷³ See Recommendation 2015-16.

⁷⁴ See Section V.J in the Appendix for a more detailed description of this analysis.

Guarantee payment uplift totaled \$81 million in 2025, marking a 102 percent increase from 2024. The increase was driven largely by higher natural gas prices in 2025. BPCG payments to demand resources were relatively high in both years, driven by multiple SCR activations by NYISO on high-load days.

1. New York City

Local Reserve Needs – In New York City, nearly 30 percent of BPCG uplift (or \$8 million) was paid to generators committed to satisfy N-1-1-0 local reliability requirements, down from 60 percent in 2024. This decline reflects a corresponding reduction in such commitments. Since a portion of the costs of OOM commitments are recovered through guarantee payments, market incentives to satisfy these requirements remain very weak. Accordingly, we have recommended that NYISO explicitly model local reserve requirements to satisfy these N-1-1-0 needs, which would reduce associated BPCG uplift and provide more transparent and efficient price signals.⁷⁵

Transmission Constraints – An additional \$8 million in uplift was paid to peaking resources that were OOM committed and dispatched in both day-ahead and real-time markets to manage congestion into the Greenwood pocket. These OOM actions were driven by ring-bus-related contingencies that were not incorporated in the NYISO market model during outages of the Gowanus-Greenwood 138 kV line from mid-February through late April. Of this uplift, approximately \$1.7 million associated with day-ahead commitments was allocated to **local** load, while the remaining \$6.3 million associated with real-time operations was socialized across **system-wide** load. Given the localized nature of the need, the system-wide allocation is not consistent with cost-causation principles. In late April, NYISO implemented enhanced modeling of ring bus configurations in its market systems. With congestion now managed through the market, the associated costs are appropriately allocated to the **local** area benefiting from congestion relief.

Most of the remaining uplift was attributable to OOM actions by operators to manage fuel oil inventories in constrained pockets on cold days with gas supply constraints.

2. Long Island

Reserve Pricing Inefficiency – Of the \$11 million of DAMAP uplift incurred on Long Island in 2025, most resulted from an inconsistency between the scheduling and pricing of reserves whereby Long Island generators are paid at to the Southeast New York reserve price rather the marginal value of reserves on Long Island. This has inflated reserve clearing prices and DAMAP uplift. Our analysis indicates that incorporating the full costs of reserve constraints into reserve prices would have reduced DAMAP uplift on Long Island by more than 60 percent (approximately \$6.5 million). Accordingly, we have recommended NYISO set reserve clearing

⁷⁵ See Recommendation 2024-1.

prices for Long Island that reflect all binding reserve constraints in the market scheduling model and NYISO plans to implement the recommendation (as discussed in Subsection D.1).

Voltage Requirements – Approximately \$9 million of uplift was categorized as local BPCG uplift, with more than 60 percent (nearly \$6 million) paid to large steam turbines that were frequently OOM-committed to manage high voltage during light-load periods. Most of the remaining uplift in this category (approximately \$2.5 million) was paid to high-cost peaking resources frequently needed in the summer months to satisfy the Transient Voltage Recovery (TVR) needs on the East End of Long Island. We have recommended NYISO consider explicitly modeling local TVR requirements on Long Island in both the day-ahead and real-time markets.⁷⁶ Our analysis indicates that such improvements would have a significant impact on LBMPs in the Long Island load pockets, resulting in more efficient market signals that support investment needed to satisfy reliability criteria and relieve congestion.⁷⁷

Transmission Constraints – An additional \$4.5 million in uplift was paid to peaking resources in the Far Rockaway load pocket, where they were frequently needed to manage 69 kV transmission constraints associated with contingencies not currently modeled in the market system. Most of this uplift was socialized across system-wide load, which is not appropriate given the localized nature of the need, similar to the allocation issue associated with OOM actions in the Greenwood load pocket discussed earlier. Section VIII.D discusses the inefficient incentives that result from frequently managing transmission constraints with OOM actions rather than market-based congestion management.

3. Upstate Regions

Local Reserve Needs – Upstate regions accounted for ~20 percent of all BPCG uplift in 2025. More than \$5 million was paid to units supplementally committed to satisfy local reserve needs in the North Country load pockets on 119 days, although the uplift is allocated statewide. Incorporating more of these requirements into the day-ahead and real-time markets would enhance market efficiency and allocate the costs to the appropriate area.⁷⁸

System Capacity Needs on Peak Days – Over \$4 million was paid on high load days to high-cost generators that were supplementally committed for system-wide capacity deficiencies.

⁷⁶ See Recommendation 2021-3.

⁷⁷ See Section III.D in the Appendix for this analysis.

⁷⁸ See Recommendation 2024-1.

VIII. TRANSMISSION CONGESTION AND TCC CONTRACTS

Congestion arises when the transmission network lacks sufficient capacity to dispatch the least expensive generators to satisfy demand. When congestion occurs, the market software establishes Location-Based Marginal Prices (LBMPs) to reflect the cost of serving load at each location on the network. These LBMPs reflect that higher-cost generation is required at locations where transmission constraints limit the ability to deliver lower-cost power.

Congestion charges are applied to purchases and sales in the day-ahead and real-time markets based on the congestion components of LBMPs. Market participants can hedge day-ahead congestion charges by owning Transmission Congestion Contracts (TCCs), which entitle the holder to payments corresponding to the day-ahead congestion charges between two locations. However, no TCCs are sold for real-time congestion, as most power is scheduled through the day-ahead market.

Transmission owners recover part of the embedded cost of building and maintaining the transmission network through revenues from TCC sales and day-ahead congestion. When transmission capability is sold in the TCC auctions, day-ahead congestion revenues are used to compensate TCC holders. Any residual revenue is paid to transmission owners. The remaining embedded costs are recouped by Transmission Owners through demand charges and other charges to real-time withdrawals.

This section discusses five key aspects of congestion management in 2025:

- Day-ahead and real-time transmission congestion revenues (Subsection A),
- Day-ahead congestion shortfalls and surpluses (Subsection B),
- Balancing congestion shortfalls and surpluses (Subsection C),
- Transmission constraints managed using out-of-market actions (Subsection D),
- TCC prices and payments (Subsection E),

In addition, general congestion patterns are summarized in the Appendix Section III.B, while the Market Operations section and its corresponding appendix evaluate other elements of congestion management.⁷⁹

A. Day-ahead and Real-time Transmission Congestion Revenues

This subsection analyzes congestion that is managed by scheduling resources in the day-ahead and real-time markets to provide transmission relief.

⁷⁹ See evaluations of pricing during transmission shortages (VII.B) and coordinated congestion management with PJM (Appendix V.D).

Figure 34 evaluates overall congestion revenues and shortfalls over the past ten years and provides annual summaries for the following categories:

- **Day-ahead Congestion**

Revenues – These revenues are collected by NYISO when power is scheduled to flow across congested transmission facilities in the day-ahead market.

- **Day-ahead Congestion Shortfalls (and Surpluses)**

– Shortfalls occur when day-ahead congestion revenue

collections are less than payments to TCC holders. This typically happens when the quantity of TCCs sold exceeds the transmission capability modeled in the day-ahead market. Such shortfalls highlight the impact of outages and other factors that reduce transmission capability over constrained interfaces. Conversely, surpluses occur when day-ahead schedules utilize transmission capability not sold in the TCC auctions.

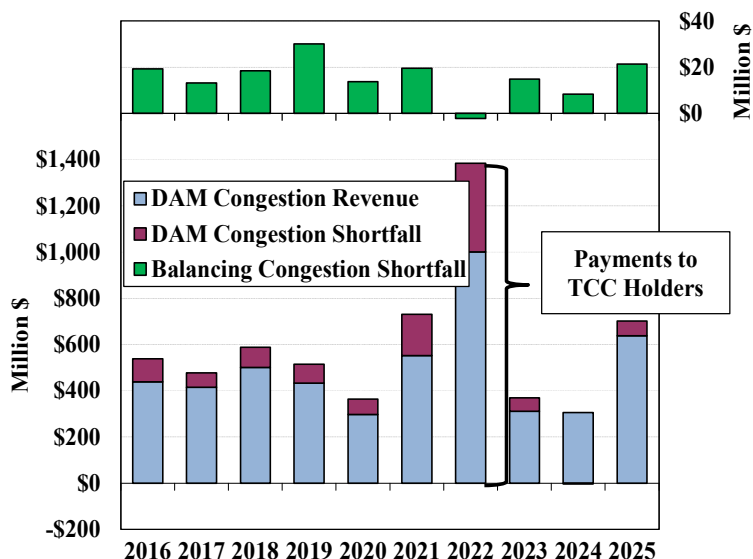
- **Balancing Congestion Shortfalls (and Surpluses)** – These arise when actual real-time flows over a constraint differ from the flows scheduled in the day-ahead market. Shortfalls occur when day-ahead scheduled flows exceed real-time flows, often due to outages, loop flows, modeling inefficiencies, or other operational limitations. Surpluses occur when real-time schedules utilize more transmission capability than is scheduled in the day-ahead market.

The figure shows that day-ahead congestion revenues, day-ahead congestion shortfalls, and balancing congestion shortfalls all increased from 2024.

Figure 35 compares day-ahead congestion revenues on the most congested transmission facilities in 2024 and 2025. Day-ahead congestion revenues totaled \$638 million in 2025, of which 90 percent occurred on the transmission paths shown in the figure, representing a 108 percent increase from 2024. The primary driver of this increase was higher natural gas prices and wider gas price spreads between regions, particularly in the winter months (January, February, December), where nearly 70 percent of 2025 congestion occurred.

Central to eastern NY – In the winter, these paths were frequently binding due to large gas spreads between central and eastern New York. These transmission paths accounted for 23

Figure 34: Congestion Revenues and Shortfalls
2016 – 2025



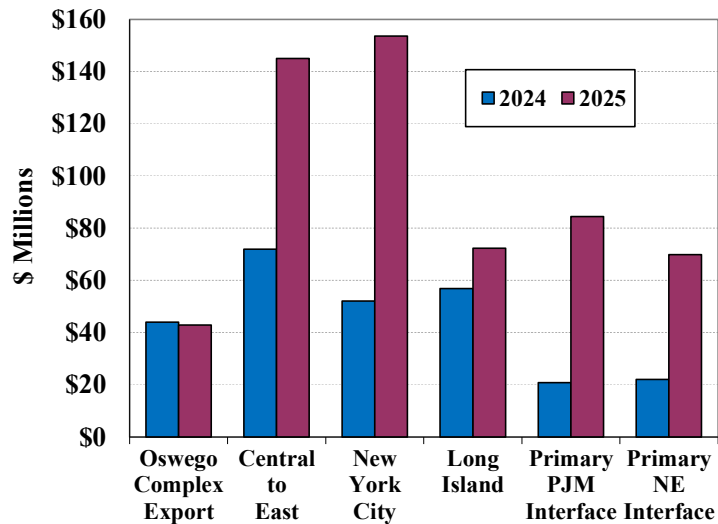
percent of total day-ahead congestion revenues in 2025, with more than 80 percent occurring during the winter months.

PJM to NY and NY to ISO-NE –

Similarly, elevated regional gas spreads drove substantial energy flows along the typical winter path between control areas, from PJM to New York and from New York to New England, leading to high congestion on these external interfaces. In addition, operators imposed several hundred megawatts of limit reductions on the primary

PJM interface in January to manage a large generator contingency on the Watercure-Oakdale 345 kV line, which further exacerbated congestion on the PJM interface.

**Figure 35: Day-Ahead Congestion Values by Path
2024 – 2025**



New York City – Transmission facilities in the city experienced the largest share of day-ahead congestion in 2025, with more than \$150 million in total congestion revenues, representing a 195 percent increase from 2024. This increase was largely attributable to two prolonged transmission outages. First, multiple Gowanus-Greenwood 138 kV facilities were out of service from late 2024 through early May 2025, which significantly increased congestion into the Greenwood load pocket during this period. Second, the Astoria Annex-E.13th St. 345 kV lines were forced out of service on most days from late April through the remainder of the year, limiting deliverable generation from the Astoria Annex and leading to export constraints from that area.

Oswego Complex Export – These constraints have been frequently binding in recent years, contributing over \$40 million in day-ahead congestion revenues in both 2024 and 2025. Nearly all of this congestion occurred on the Scriba-Volney 345 kV line. This facility typically became binding on high-load days when gas-fired and nuclear generation within the complex increased, or when transmission outages significantly reduced export capability, such as during the planned outage of the Clay-Independence 345 kV line from mid-September to early October. In addition, the constraint was frequently binding in the winter months when oil-fired generation from Oswego units was dispatched to address natural gas supply constraints.

Long Island – This congestion increased by 27 percent from the prior year despite fewer 345 kV transmission outages.⁸⁰ During the shoulder months, one or more of the Shore Road-to-Lake

⁸⁰ The Y50 line was out of service for roughly eight months in 2024, compared to only two months in 2025.

Success 138 kV lines and the Valley Stream-to-Stewart Ave 138 kV lines were frequently switched out of service to control high voltage under light load conditions, resulting in elevated congestion along their parallel paths. Additionally, the average flows from Long Island to New York City over the 901/903 lines returned to normal levels of ~200 MW, compared to just 70 MW in 2024, when these schedules were reduced on most days because of the extended outage of the Y50 line. The restoration of 901/903 flows contributed to higher congestion levels relative to 2024.

B. Day-Ahead Congestion Shortfalls and Surpluses

Day-ahead congestion revenue shortfalls and surpluses (known collectively as “residuals”) occur when day-ahead network capability differs from the modeled capability in the TCC auctions. Shortfalls arise when the day-ahead flows over a binding constraint are lower than the transfer capability used to sell TCCs, while surpluses occur when day-ahead flows exceed the transfer capability used in the TCC auctions.

In general, allocating surpluses and shortfalls based on cost causation principles is desirable because it provides efficient financial incentives for Transmission Owners (TOs) to maintain equipment, configure the transmission system appropriately, and schedule outages efficiently. This subsection evaluates various categories of residuals and assesses the extent to which their allocation aligns with these principles.

Shortfalls and surpluses are allocated to the responsible TO when they result from most changes in modeled transfer capability. These include qualifying transmission outages, return-to-service of transmission, facility uprates, and facility derates that can be attributed to a specific TO. This allocation is based on the flow impact of these factors on binding constraints in the day-ahead market, which is generally consistent with cost causation principles.⁸¹ However, the remaining shortfalls and surpluses are allocated in proportion to the TCC auction revenues received by each TO, which does not align with cost causation principles.⁸²

The following example illustrates how the allocation of congestion surpluses may deviate from cost causation principles. Consider a transmission constraint that binds in the day-ahead market with a scheduled flow of 300 MW, while TCCs were previously sold based on only 260 MW of transfer capability, implying that the constraint is undersold by 40 MW. If this additional 40 MW becomes available due to a facility uprate after the TCC auction, the responsible TO receives congestion surpluses associated with the additional transfer capability. Conversely, if the additional 40 MW arises simply because fewer TCCs were sold than the available transfer capability, the responsible TO receives only a small portion of the resulting congestion surpluses.

⁸¹ See OATT, Attachment N, Formula N-6 through N-14 for the calculation of these allocations.

⁸² See OATT, Attachment N, Formula N-15 for the calculation of this allocation.

This allocation method is inefficient because it unfairly penalizes TOs that own facilities on interfaces that bind under multiple transmission flow patterns.⁸³

Table 5 shows actual allocations of day-ahead congestion residuals over the past five years, categorized into two distinct groups. The first category includes allocations based on cost causation principles, following Formulas N-6 through N-14 in the OATT Attachment N. The second category consists of allocations based on the TCC revenue allocation, using Formula N-15. It is important to note that these numbers represent net annual allocations, which understate the allocation inefficiencies since surpluses on one path net out shortfalls on another path.

Table 5: Category of Day-Ahead Congestion Residual Allocations

Year	Cong. Residual Allocation (\$M)	
	Cost Causation	Formula N-15
2021	\$122.4	\$56.8
2022	\$326.5	\$57.4
2023	\$62.5	-\$4.1
2024	\$29.9	-\$32.8
2025	\$93.5	-\$31.4

The cost causation-based allocation consisted of mostly shortfalls, driven primarily by outages of transmission facilities. Total shortfalls in this category ranged from just \$30 million in 2024 to nearly \$330 million in 2022. These variations reflect the significance of outages and the associated congestion impacts in each year.

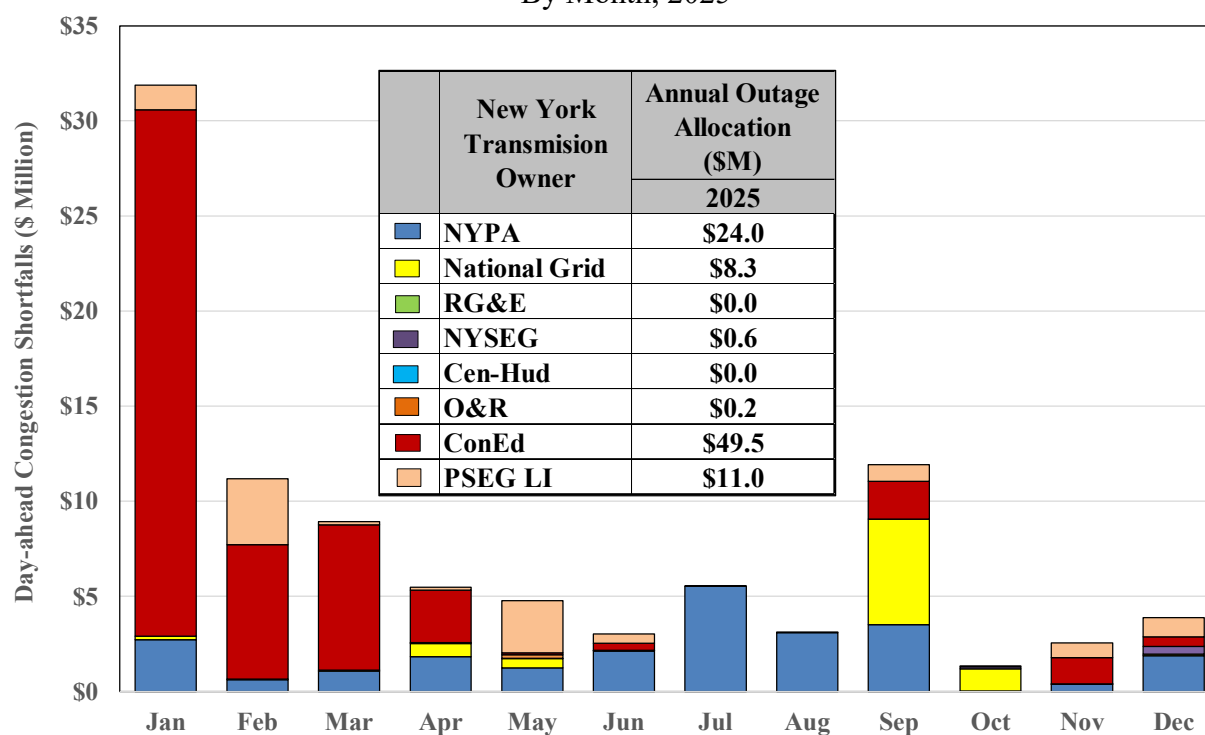
In contrast, Formula N-15 residuals were generally surpluses, except in 2021 and 2022. In those years, significant N-15 shortfalls occurred primarily due to reductions in the Central-East interface limit resulting from operational changes in nearby capacitors, static VAR compensators, and other transmission equipment where the responsible TO is not assigned the associated shortfalls. Consequently, when TCCs are oversold across the Central-East interface due to changes in the status of such equipment, one group of TOs receives excess TCC revenues, while the resulting day-ahead congestion revenue shortfalls are allocated to a different group of TOs. This allocation framework does not provide efficient incentives for the operation and maintenance of transmission equipment. In recent years, N-15 surpluses have become more pronounced, exceeding \$30 million in both 2024 and 2025.

1. Cost-Causation-Based Allocation of Shortfalls in 2025

⁸³ For example, suppose a 100 MW line between nodes A and B is constrained: (i) from A to B for 200 hours at a shadow price of \$5/MWh and (ii) from B to A for 150 hours at a shadow price of \$5/MWh. The line will provide \$17,500 of congestion revenue = 100 MW * 20 hours * \$5/MWh + 100 MW * 15 hours * \$5/MWh. However, the holder of a 100 MW TCC from node A to B (assuming a distribution factor of 100% for the TCC onto the line from A to B) will receive just \$2,500 = 100 MW * 20 hours * \$5/MWh *minus* 100 MW * 15 hours * \$5/MWh. This results in a \$15,000 revenue surplus, but the surplus is allocated to all TOs rather than just the owner of the line from node A to B.

Figure 36 further examines the net shortfalls allocated to each transmission owner on a cost-causation basis by month in 2025. Nearly all shortfalls in this category were concentrated among four of the eight transmission owners. Of these, two accounted for almost 80 percent of the total, with Con Edison accounting for the largest share at 53 percent (nearly \$50 million), followed by NYPA at 26 percent (approximately \$24 million).

Figure 36: Day-ahead Congestion Shortfalls Allocated to Responsible TOs
By Month, 2025



Con Edison Shortfalls – Most shortfalls allocated to Con Edison accrued from January through early May, driven by elevated congestion into the Greenwood load pocket in New York City. This congestion resulted from forced outages of multiple Gowanus-Greenwood 138 kV facilities, which significantly reduced import capability into the pocket.

NYPA Shortfalls – Shortfalls allocated to NYPA were distributed across most months of the year. Approximately 60 percent of these shortfalls resulted from forced outages of the Astoria Annex-E.13th St. 345 kV lines, which were out of service on most days from late April through the end of the year. These outages limited the deliverability of generation from the Astoria Annex. In addition, approximately \$5 million in shortfalls were associated with transmission outages in the North Zone to facilitate construction of the Smart Path Connect project.

PSEG LI Shortfalls – Approximately 45 percent of shortfalls (or \$5 million) allocated to PSEG Long Island were associated with the Shore Road-Lake Success 138 kV lines and the Valley Stream-Stewart Ave 138 kV lines, which were frequently switched out of service to manage high

voltage under light load conditions. An additional \$4 million accrued from late January through the end of March due to a forced outage of the Y50 345kV line.

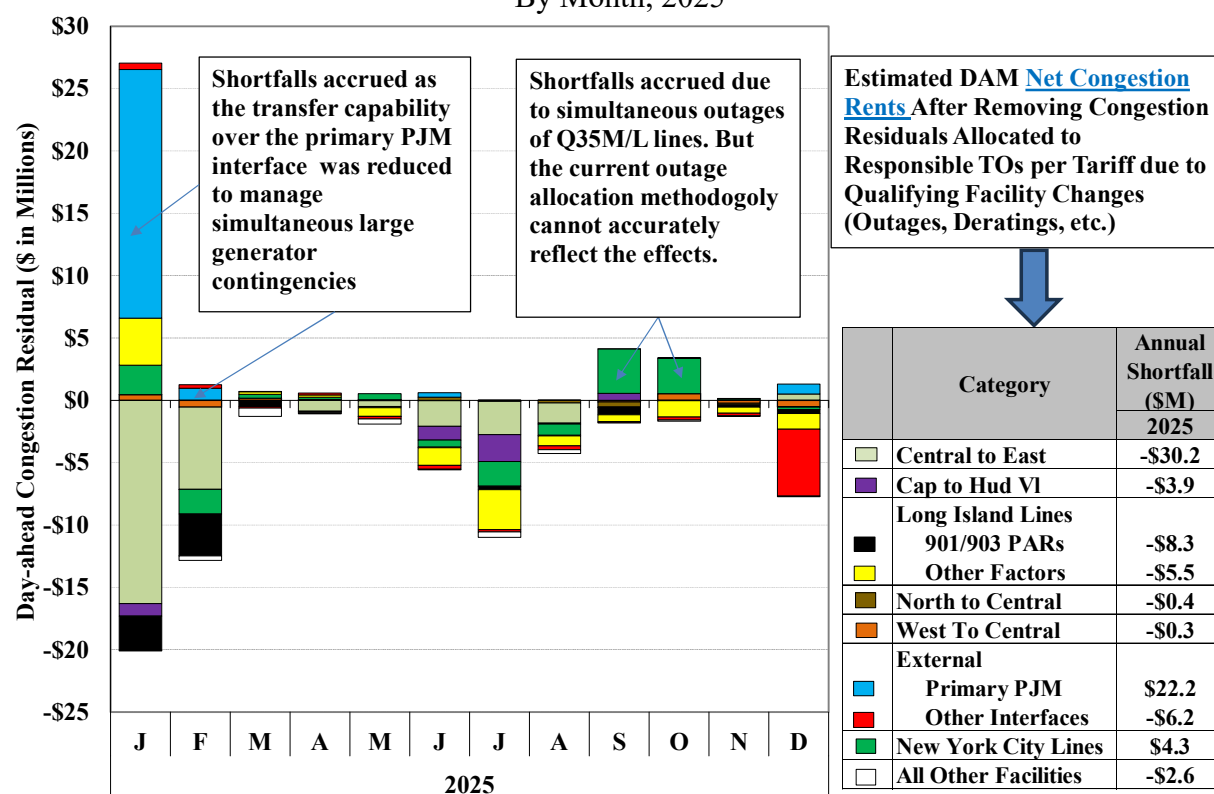
National Grid Shortfalls – Finally, nearly 80 percent of these shortfalls occurred during the planned outage of the Clay-Independence 345 kV line from mid-September to early October, which greatly reduced the export capability from the Oswego Complex generation.

Overall, most shortfalls in this category were driven by forced outages. While some shortfalls were attributable to planned outages, these were largely concentrated in the shoulder months. These results reflect that the cost-causation allocation mechanism has promoted efficient outage scheduling by providing transmission owners with incentives to minimize both the costs and duration of planned outages.

2. N-15 Allocation of Net Congestion Rents

The remaining congestion residuals, referred to as Net Congestion Rents, are socialized among all transmission owners in proportion to their TCC auction revenues. We estimated hourly net congestion rents associated with each binding transmission constraint and present them by month for major facility groups in the following figure.

Figure 37: Estimated N-15 Residuals by Facility Group
By Month, 2025



In 2025, net congestion rents totaled approximately \$31 million in surpluses, with the largest share accruing on new transmission facilities. Facilities associated with the AC Transmission Segment A and Segment B projects, as well as the Hurley Avenue Highway System Deliverability Upgrade project, were modeled as out-of-service in TCC auctions due to their eligibility for Incremental TCC awards. Under current NYISO practices, such facilities are not modeled as in-service in TCC auctions until the relevant Incremental TCC evaluation process concludes. However, these facilities were modeled as in-service in the day-ahead market, leading to more than \$33 million in congestion surpluses in 2025. These surpluses primarily accrued on transmission paths across the Central-East interface and from the Capital region to Hudson Valley. Despite being driven by specific transmission upgrades, these surpluses are allocated broadly across all transmission owners in proportion to their TCC auction revenues, which does not align with the allocation of transmission upgrade costs.

The second largest category of N-15 surpluses accrued on transmission paths that were constrained in the day-ahead market but undersold in the TCC auctions. Such congestion typically arises from hourly and daily changes in generation and load patterns, which shift congestion across the transmission network. These variations accounted for a significant share of N-15 surpluses on Long Island facilities from May through December, on New York City facilities in February and from June through October, and on the New England and Quebec interfaces in December. This pattern is becoming increasingly prevalent as intermittent renewable generation is added to the system. When TCCs are undersold on a given interface due to these shifting generation and load patterns, the resulting surpluses are currently allocated broadly across all transmission owners in proportion to TCC auction revenues, rather than to the TOs whose facilities enable additional transfers. As a result, TOs may not fully recover the value of their transmission assets when facilities are undersold in the TCC auctions, providing incentives to oversell transmission capability.⁸⁴

Another significant source of N-15 surpluses arose from differences in flow assumptions on PAR-controlled lines between the day-ahead market and TCC auctions. Specifically, the two PAR-controlled lines between New York City and Long Island (i.e., the 901/903 lines) generated approximately \$8 million in N-15 surpluses in 2025. These lines consistently caused congestion surpluses because their assumed flows from Long Island to New York City were typically 300 MW in the TCC auctions but significantly lower in the day-ahead market, particularly during the Y50 line outages from January to March. Since these flows are generally uneconomic and increase production costs, reducing the scheduled flows from TCC assumptions to the day-ahead market led to significant congestion surpluses. This also underscores that efficient scheduling the 901/903 lines would substantially reduce production costs. Even though Con Ed has contractual rights to schedule these facilities, if Con Ed reduces the schedule of these facilities,

⁸⁴ When a commodity is oversold in a forward market, it tends to depress forward prices relative to spot prices. Thus, the oversale of TCCs tends to reduce overall collections of revenue by transmission owners.

leading to production cost savings and congestion revenue surpluses, most of these surpluses are distributed to other TOs. Efficient allocation of these surpluses would provide much better scheduling incentives for the 901/903 lines.⁸⁵

However, these surpluses were partially offset by two primary sources of N-15 shortfalls in 2025. First, from late December 2024 through early February 2025, NYISO operators frequently imposed limit reductions of up to 700 MW on the primary PJM interface in both the day-ahead and real-time markets to secure the Watercure-Oakdale 345kV line for a contingency involving the simultaneous loss of over 2 GW of generation at the Nine Mile and Fitzpatrick nuclear generation facilities. These actions resulted in elevated congestion on the PJM interface and approximately \$26 million in day-ahead congestion shortfalls, including more than \$22 million in 2025. These shortfalls were socialized across all TOs in proportion to their TCC auction revenues, rather than allocated based on cost causation principles. If this constraint had been incorporated directly into the day-ahead and real-time market models, the LBMPs of the contingent generators would have been reduced to reflect their impact on congestion costs. This would have provided more efficient incentives for market participants to minimize the impact of the issue. To the extent that this led to day-ahead congestion shortfalls, these would have been assigned to the TOs that sell TCCs that source or sink at the relevant generator buses.

Second, certain N-15 shortfalls arose because the current cost-causation allocation methodology for outages may under-allocate shortfalls under certain simultaneous multiple-outage conditions. This is due to the methodology evaluating each outage individually, rather than capturing the combined impact of concurrent outages. For example, the simultaneous outages of the Q35M and Q35L 345 kV lines in New York City during September and October accounted for more than \$6 million in shortfalls. Similarly, the concurrent outages of the East Garden City-New Bridge Road 138 kV line and the East Garden City-Valley Stream 138 kV line on Long Island in January accounted for approximately \$2 million in shortfalls. Because these combined effects are not fully captured, the resulting shortfalls were socialized across all transmission owners, rather than allocated in accordance with cost causation principles.

In light of these inefficiencies, we recommend the NYISO revise the allocation of day-ahead congestion residuals that are currently socialized among TOs in proportion to TCC revenues. Instead, residuals should be allocated based on changes in the scheduled utilization of the transmission system between the TCC auctions and the day-ahead market.⁸⁶ This approach would allow transmission owners to recover the value of transmission usage in the day-ahead

⁸⁵ Recommendation 2012-8 would provide incentives to schedule these lines efficiently.

⁸⁶ Recommendation 2023-1 is to allocate day-ahead congestion rents consistent with cost causation principles. An approach for doing this was presented at the July 8, 2025 MIWG meeting in *Net Congestion Rent Assignment Simulation Examples*.

market, even when capacity was undersold in the TCC auctions, and would better align incentives with cost causation principles.

C. Balancing Congestion Shortfalls and Surpluses

Table 6 shows balancing congestion shortfalls by transmission facility group in 2025.^{87 88} Unlike day-ahead shortfalls, balancing congestion shortfalls are generally socialized to all NYCA load through Rate Schedule 1 charges.⁸⁹

1. Summary Balancing Congestion Residuals

Although net congestion shortfalls (excluding payments under the NY-PJM M2M agreement) were below \$29 million in 2025, gross shortfalls and surpluses were significantly larger, totaling approximately \$80 million and \$50 million, respectively. These large gross quantities were driven by a wide range of factors.

Table 6: Balancing Congestion Shortfalls in 2025

Facility Group	Annual Balancing Residuals (\$M)			
	M2M PARs	901/903 PARs	Surpluses	Shortfalls
TSA Constraints	\$3.3	\$0.0	-\$0.2	\$19.6
NYC Lines	-\$1.1	\$0.5	-\$5.3	\$11.1
Long Island Lines	\$0.0	\$5.3	-\$10.4	\$10.6
External - PJM	\$1.6	\$0.0	-\$4.9	\$8.5
External - NE	\$0.0	\$0.0	-\$3.9	\$3.4
External - OH	\$0.0	\$0.0	-\$3.1	\$2.1
External - HQ	\$0.0	\$0.0	-\$1.0	\$1.8
Central-to-East	-\$0.5	\$0.0	-\$4.5	\$2.7
Capital-to-HudVL	-\$0.1	\$0.0	-\$2.7	\$2.4
HudVL-to-Dunwod	\$4.4	\$0.0	-\$3.4	\$0.3
West-to-Central	-\$1.2	\$0.0	-\$5.9	\$1.3
North-to-Central	\$0.0	\$0.0	-\$2.1	\$1.0

The largest contributor to shortfalls was Thunderstorm Alert (TSA) events, which accounted for approximately \$23 million in net congestion shortfalls. These shortfalls were driven by substantially reduced transfer capability into downstate areas (i.e., Zones H-K) under TSA

⁸⁷ Appendix III. provides additional details on balancing congestion shortfalls.

⁸⁸ The balancing congestion shortfalls estimated in this table differ from actual balancing congestion shortfalls because the estimate: (a) is partly based on real-time schedules rather than metered injections and withdrawals; and (b) uses the original constraint shadow costs from the dispatch model therefore does not reflect the effect of price corrections.

⁸⁹ The only exception is that some balancing congestion shortfalls from TSA events are allocated to Zone J load.

conditions. More than 95 percent of these costs were concentrated on three days in July as discussed in greater detail later.

Another significant driver of shortfalls was the occurrence of transmission outages in real time that were not anticipated in the day-ahead market. In New York City, multiple Gowanus-Greenwood 138 kV facilities were out of service on most days from late 2024 through early May 2025. During this period, inconsistencies between day-ahead assumptions and the actual real-time status occurred on several occasions, leading to shortfalls, particularly in January and February. On Long Island, the Shore Road-Lake Success 138 kV lines and the Valley Stream-Stewart Ave 138 kV lines were frequently switched out of service to manage high voltage during light-load periods. While most of these actions were incorporated into the day-ahead market, line switching that was not scheduled until after the day-ahead market contributed to balancing shortfalls, particularly during the shoulder months.

The primary PJM interface was another notable source of shortfalls, accounting for more than \$8 million, primarily in January and December. In January, NYISO operators utilized OOM actions to secure the Watcure-Oakdale 345 kV line against a potential simultaneous outage of the Nine Mile 2 and Fitzpatrick nuclear units. These actions included curtailing scheduled transactions and reducing import limits across the primary PJM interface were used frequently until early February, when revised operating procedures were implemented. Managing this constraint outside the market resulted in balancing congestion shortfalls that were uplifted to load through Rate Schedule 1. If the constraint had instead been modeled directly in the market, the associated costs would have been reflected in LBMPs and allocated to the responsible generators. Similar OOM actions were taken in December to manage constraints on the E. Towanda-Hillside-North Waverly facilities, leading to additional shortfalls. While the Hillside-North Waverly 115 kV line was incorporated into both the day-ahead and real-time markets in February 2026, the E.Towanda-Hillside 230 kV line remains an NY-PJM M2M constraint modeled only in real time, often requiring costly redispatch or OOM actions that contribute to shortfalls.

Relatively small shortfalls accrued on other internal and external facilities, typically driven by modest adjustments to transfer limits to reflect differences between modeled and actual system conditions, short-duration forced outages, or operational changes affecting nearby capacitors and static VAR compensators that influence Central-East interface limits.

Real-time operation of PARs also contributed to balancing congestion residuals. PARs under the NY-PJM M2M process generally produced surpluses on internal constraints, but contributed to shortfalls during TSA events, on the NY-PJM intertie, and along transmission paths from Hudson Valley to Dunwoodie. Additionally, the 901/903 PARs contributed nearly \$6 million in shortfalls, primarily from April through July, as they delivered more power to New York City in real-time than was scheduled day-ahead.

At the same time, surpluses partially offset these shortfalls. A common driver of surpluses was the shift in generation and load patterns from day-ahead to real-time. For example, on west-to-central transmission paths, higher real-time wind generation increased flows relative to day-ahead schedules, producing surpluses on binding constraints. In addition, real-time market responses that adjusted flows to neighboring areas often contributed to surplus outcomes. PARs internal to Zones J and K were also a consistent source of surpluses because they are generally used to relieve congestion.

2. TSA-related Congestion Residuals in July 2025

Under normal operations, NYISO secures each transmission facility against a single contingency. However, during Thunderstorm Alert (TSA) conditions, the system must be operated more conservatively, requiring certain facilities to be secured against multiple contingencies in the real-time market.

Figure 38 provides details of the TSA constraint that accounted for 71 percent of TSA uplift in July 2025. As shown in the map, six transmission lines carry the majority of flows from Zone G to Zones H and I. Line #1, the Lovett-Buchanan 345 kV line, is the limiting facility for this TSA constraint. Lines #2, #3, and #4, the two Pleasant Valley-Wood St.-Millwood 345 kV lines and one Pleasant Valley-Wood St.-Pleasantville 345 kV line, are contingent elements. Lines #5 and #6 (the Ramapo-Buchanan 345 kV line and a second Pleasant Valley-Wood St.-Pleasantville 345 kV line) are not part of the TSA constraint itself but are key components of the UPNY-Con Ed interface. Notably, Line #6 (the Y87 line) was out of service in the real-time market on July 1 but was modeled as in-service in the day-ahead market. This disparity increased the divergence between day-ahead and real-time flows across the constraint and contributed to elevated TSA uplift.

Figure 38: Map of an Example TSA Constraint

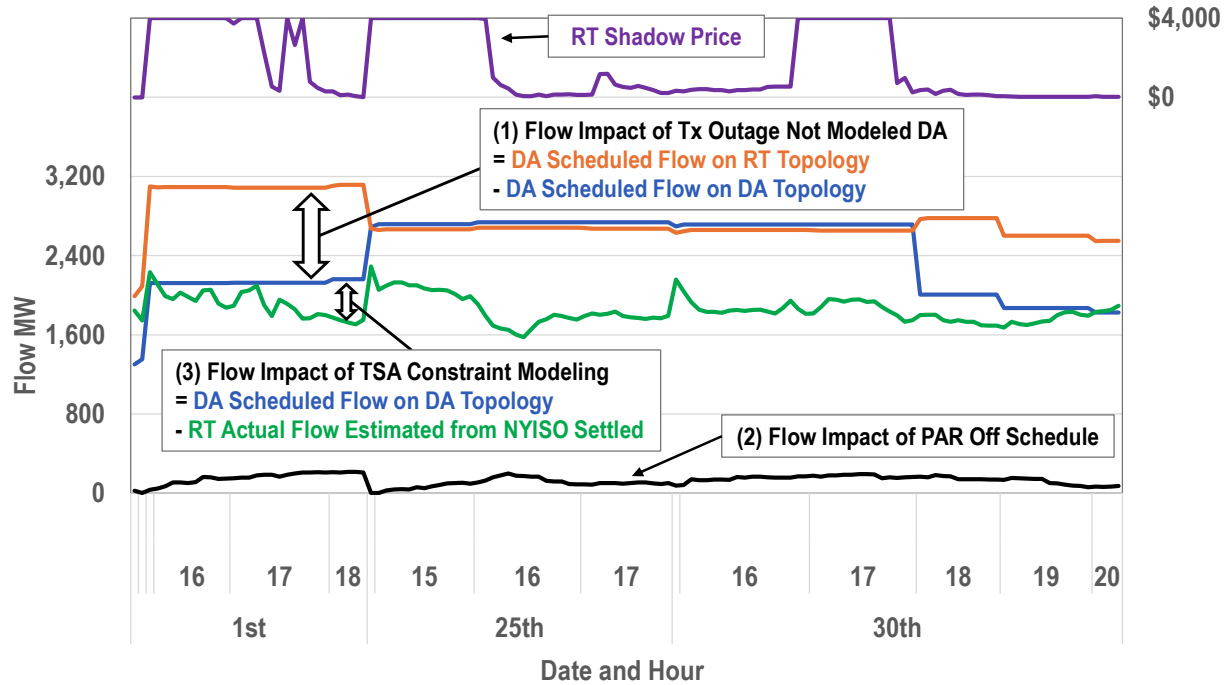


Figure 39 illustrates the methodology used to calculate TSA uplift for this TSA constraint. For each binding interval, the figure shows:

- Real-time shadow prices of the TSA constraint (purple line);
- Real-time flows based on real-time topology and schedules (green line);
- Day-ahead flows based on day-ahead topology and schedules (blue line);
- Day-ahead flows based on real-time topology and day-ahead schedules (orange line); and

- e) The impact of deviations in PAR flows from their scheduled levels based on real-time topology (black line).

Figure 39: Example Calculations of TSA Uplift Categories
Lovett-Buchanan Line for Contingency CE40 in July 2025



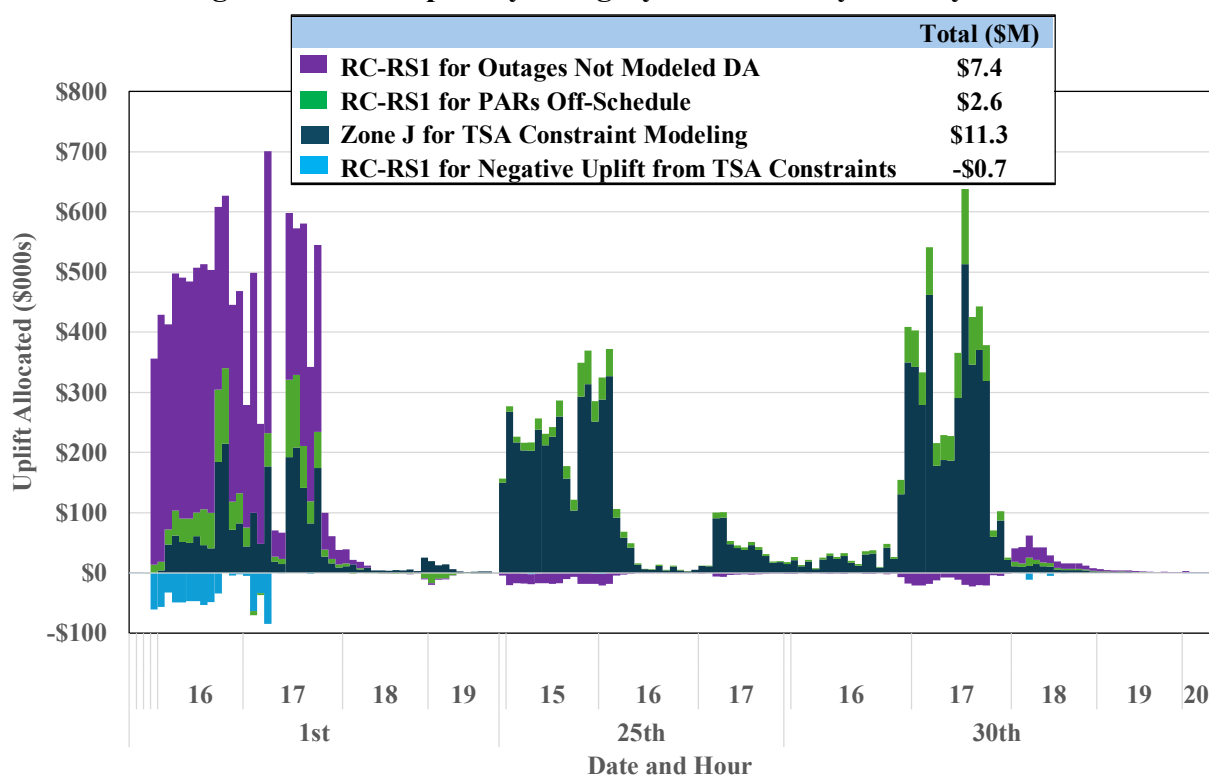
The total TSA uplift is calculated as $(a) \times [(d) - (b) + (e)]$, which can be decomposed into four components:

- 1) **Topology differences:** $(a) \times [(d) - (c)]$ captures uplift caused by differences between day-ahead and real-time network topology. For example, the Y87 outage on July 1 reduced available transfer capability in real time but was not reflected in the day-ahead model, leading to congestion shortfalls.
- 2) **PAR deviations:** $(a) \times (e)$ reflects uplift driven by PAR-controlled lines not maintaining scheduled flows during TSA conditions. The positive values of this term indicate that PAR operations reduced available transfer capacity, contributing to shortfalls.
- 3) **TSA-redispatch shortfalls:** $\text{Max}\{0, (a) \times [(c) - (b)]\}$ represents uplift directly attributable to TSA redispatch in real time, excluding topology and PAR effects. Under the tariff, only positive values (i.e., shortfalls) are allocated to Zone J load.
- 4) **TSA-redispatch surpluses:** $\text{Min}\{0, (a) \times [(c) - (b)]\}$ represents the surplus amounts calculated from TSA redispatch. Under the tariff, these surpluses are allocated through Rate Schedule 1.

Figure 40 summarizes our calculated TSA uplift by category for three days in July 2025 that accounted for nearly all TSA-related uplift in July. Total shortfalls across these days amounted to \$20.6 million. The Y87 outage was the primary driver on July 1, contributing more than \$7 million in shortfalls, while topology differences on the other two days had a smaller impact.

Deviations in PAR schedules contributed over \$2.5 million in shortfalls, indicating operational challenges with certain PAR-controlled facilities.

Figure 40: TSA Uplift by Category on Three Days in July 2025



Under current market rules, uplift from topology differences and PAR deviations, along with \$0.7 million in TSA-redispatch surpluses, are allocated through Rate Schedule 1. In contrast, approximately \$11 million in shortfalls were directly attributable to TSA redispatch in real time and are allocated to Zone J load because they benefit from the enhanced reliability provided by conservative operations during TSA events.

Initially, NYISO allocated TSA costs for July 2025 significantly different from the allocation summarized in Figure 40. In the initial invoices, NYISO allocated approximately \$55 million of uplift to Zone J. In the subsequent four-month settlement, NYISO reduced this amount to roughly \$3.5 million using its “Short-term Improvement Process,” resulting in rebates of about \$52 million to Zone J load, with the offsetting charges uplifted through Rate Schedule 1. NYISO has since decided to rebill July using an approach consistent with our estimates and to develop an automated “Long-term Improvement” process that is substantially better aligned with cost causation principles than either of the prior allocation methodologies.⁹⁰

⁹⁰ See April 8, 2026 MIWG presentation, *Storm Watch Redispatch (TSA) Costs*.

Looking back to the period from 2016 to 2024, NYISO allocated more than \$125 million in TSA-related shortfalls to Zone J load using the method initially in place for July 2025,⁹¹ more than double our estimate of approximately \$58 million for total shortfalls associated with TSA constraints.⁹² We strongly support NYISO’s efforts to implement the “Long-term Improvement” framework.

Overall, these results highlight two closely related issues. First, it would be beneficial to incorporate TSA-related transmission constraints into the day-ahead market when thunderstorms are forecasted. Doing so would allow the market to anticipate reduced transfer capability and reflect these conditions directly in day-ahead prices and schedules. This would reduce the divergence between day-ahead and real-time outcomes, thereby producing more efficient prices and lowering the need for real-time uplift payments.⁹³

Second, further improvements in the allocation of balancing congestion uplift (and surpluses) would be beneficial. Under current rules, only TSA-related shortfalls are allocated to local entities that require these additional reliability measures, while all other shortfalls and surpluses are socialized among system-wide load through Rate Schedule 1. The uplift allocation could more precisely distinguish between cost factors and aligning the allocation with underlying drivers would ensure that costs are borne by the entities that contribute to them, rather than being broadly socialized. This would strengthen incentives for market participants to respond efficiently to system conditions and promote more efficient market outcomes.

D. Management of Transmission Limitations using Out-of-Market Actions

Transmission constraints on facilities rated 100 kV and above are generally managed through the day-ahead and real-time markets. This approach provides several key benefits, including:

- **More efficient resource scheduling** that optimally balances the costs of satisfying demand, ancillary services, and transmission security requirements; and
- **More efficient price signals** that inform longer lead-time decisions, such as fuel procurement, generator commitment, external transaction scheduling, and investments in generation and transmission infrastructure.

However, certain transmission constraints, particularly those on lower voltage networks, are resolved primarily through Out-of-Market (“OOM”) actions, such as:

⁹¹ The TSA-related NYISO allocation is estimated using information published in NYISO CEO/COO Report, which reports the monthly average value of “TSA \$ per NYC MWh”.

⁹² The \$58 million represent our estimate of the total uplift accrued on TSA constraints. We did not conduct a detailed analysis to decompose this amount into the four categories of uplift discussed in this subsection.

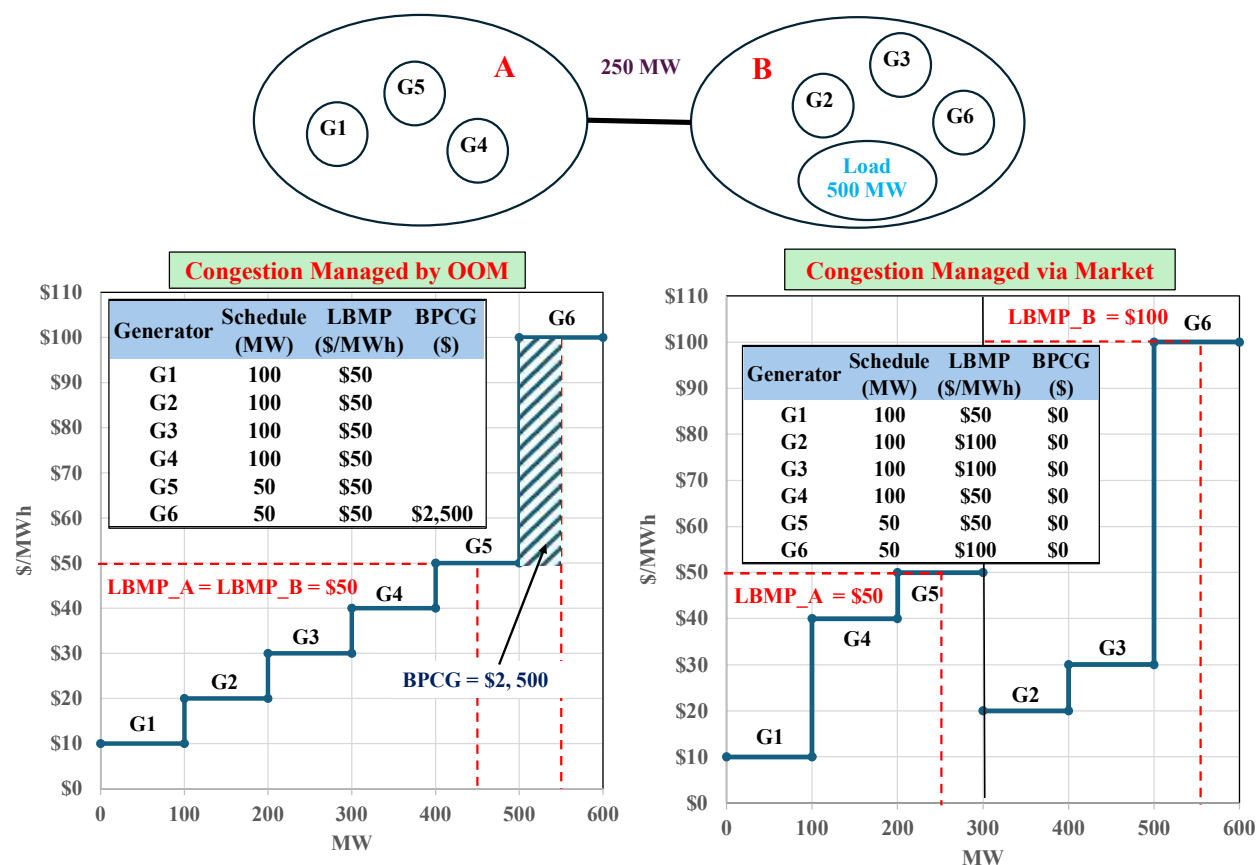
⁹³ Recommendation 2025-5 is to model TSA constraints in the day-ahead market. This is discussed further in Section VI.D.

- Out of merit dispatch and supplemental commitment of generation;
- Curtailment of external transactions and limitations on external interface transfer limits;
- Use of internal interface or constraint transfer limits that serve as proxies for limiting transmission facilities; and
- Adjusting PAR-controlled line flows on the higher voltage network.

In contrast to market-based scheduling, OOM actions can distort price signals and lead to inefficient compensation outcomes. When reliability is maintained through OOM actions rather than market-based pricing, real-time prices fail to reflect the true marginal cost of serving load. As a result, resources needed for reliability may not be adequately compensated through market revenues. Instead, high-cost units committed out of market are made whole through Bid Production Cost Guarantee (BPCG) payments, while lower-cost units dispatched economically may be under-compensated.

Figure 41 illustrates these differences in market efficiencies and incentives associated by comparing OOM actions with market-based scheduling in a simplified two-area example.

Figure 41: Illustration of Market Incentives for OOM Actions vs. Market Scheduling



In this example, area B is a load pocket with 500 MW of demand, served by six 100-MW generators (G1-G6) with costs ranging from \$10 per MWh to \$100 per MWh. Each area contains

three generators, with G6 in area B being the highest-cost unit. The tie line allows up to 250 MW of imports from area A to area B.

The bottom panels compare dispatch and compensation outcomes under the two approaches. In the OOM case (left panel), generator G6 is committed out of market for 50 MW to manage flows from area A to area B. Although a portion of G5's lower-cost output is displaced by G6's higher-cost output, G5 remains the marginal unit in the system-wide supply stack, setting the LBMP at \$50 per MWh in both areas. As a result, prices do not reflect the higher-cost resources required to maintain reliability. Because this price is below G6's cost, G6 receives \$2,500 in BPCG uplift to recover its cost, while all other generators are compensated at the lower LBMP.

In contrast, under the market-based scheduling approach (right panel), the same physical dispatch is achieved, but prices appropriately reflect the marginal cost of serving load given the transmission constraint. The import constraint between areas A and B separates the supply stacks, resulting in distinct prices in each area. In area A, G5 remains marginal and sets the price at \$50 per MWh, while in area B, G6 becomes the marginal unit and sets the price at \$100 per MWh. As a result, generators G2, G3, and G6 receive higher market revenues, and G6 fully recovers its costs through market prices without the need for BPCG uplift.

These differences highlight a key inefficiency of OOM actions. By relying on uplift payments to cover the costs of high-cost units, OOM dispatch systematically under-compensates low-cost resources. Over time, this distorts investment incentives in favor of older inefficient generating units. On the other hand, setting market clearing prices that reflect the cost of actions to maintain reliability leads to more efficient investment incentives.

OOM actions to manage constraints were common in several regions in 2024 and 2025. Table 7 summarizes the frequency of such actions by region, displaying the number of days in each year when OOM actions were used.⁹⁴

Long Island, New York City, and the North Zone and Mohawk Valley area experienced the most frequent OOM actions over the past two years. The remainder of this subsection discusses: (i) supplemental commitment for local reserve needs, (ii) OOM actions to manage voltage requirements, and (iii) OOM actions to manage transmission constraints.

Table 7: Summary of OOM Days for Managing Network Security and Reliability
2024-2025

Area	# of Days with OOM Actions	
	2024	2025
Central Zone	8	61
North & MHK VL	225	149
New York City	242	182
Long Island	182	254
All Other Regions	32	25

⁹⁴ See Section III.D in the Appendix for more details on the use of various resource types.

1. Supplemental Commitments for Local Reserve Requirements

Most OOM actions were supplemental commitments to satisfy local reserve requirements that are not fully represented in the day-ahead and real-time markets. In 2025, such commitments occurred on 121 days in the New York City load pockets and 119 days in the North Country load pockets. Notably, large OOM commitments in these local pockets often occurred even on days with relatively low reserve needs, leading to sizable surplus headroom on the OOM-committed units and substantial uplift costs.⁹⁵

On Long Island, supplemental commitments for this purpose were typically driven by tight system conditions, including severe weather, constrained gas supply, outages of interties, or generator trips. Although Long Island experienced relatively few of these OOM commitments over the past two years, it would be still beneficial to consider modeling the full reserve requirements in the day-ahead and real-time markets.

We have recommended incorporating N-1-1 and N-1-1-0 reserve requirements into the market software for key local areas, such as New York City, Long Island, and North Country load pockets, to improve scheduling efficiency, provide more efficient investment signals, and help integrate renewable and storage resources.⁹⁶

2. OOM Actions to Manage Voltage Requirements

OOM actions to address voltage issues increased significantly over the past year. On Long Island, large steam turbines were committed out of market on 168 days in 2025 to manage high voltage during light-load conditions. Additionally, oil-fired high-cost peaking resources were OOM-committed on 62 days to satisfy the Transient Voltage Recovery (TVR) requirements on the East End of Long Island during high-load conditions. These supplemental commitments suppressed market prices and resulted in approximately \$6 million in BPCG lift to steam turbines and \$2.5 million to peaking resources. Voltage-related OOM actions were less frequent outside Long Island, occurring on roughly 30 days in both New York City and the North Zone, and on fewer than 10 days in other regions.

The growing reliance on OOM actions for voltage control reflects evolving system conditions, including changes in the resource mix and shifting patterns of generation and load. In particular, increasing penetration of intermittent renewable resources, both before-the-meter and behind-the-meter, has contributed to greater variability in system voltage profiles. These resources can cause rapid fluctuations in output, leading to localized voltage swings that require active

⁹⁵ These OOM commitments are evaluated in more detail in Section VII.D.2.

⁹⁶ See Recommendation 2024-1.

management. In addition, the increasing use of power-electronics-based devices for control and switching further complicates voltage behavior, increasing the need for operational interventions.

Although incorporating voltage requirements into the market model would improve scheduling efficiency and price formation, doing so presents significant challenges due to the inherently highly nonlinear relationship between voltage and system conditions. However, the TVR requirement is an exception, as it can be approximated using surrogate constraints consistent with PSEG Long Island's operational guidelines. As discussed in Section III.E of the Appendix, such surrogate constraints could be developed to represent TVR requirements within the market models. Accordingly, we recommend that NYISO model East End TVR needs (using surrogate constraints) in the market software to improve operational efficiency and pricing outcomes.⁹⁷

3. OOM Actions to Manage Transmission Constraints

NYISO has a process to periodically evaluate and incorporate additional non-market-secured transmission constraints into the market models as needed. This process has reduced the needs for OOM actions to manage transmission bottlenecks in recent years. However, such OOM actions increased in several areas in 2025.

In western New York, beginning in late December 2024, NYISO identified a large generation contingency associated with the Watcure-Oakdale 345 kV line, which was not represented in the day-ahead and real-time market models. To manage this constraint, NYISO operators reduced limits on the primary PJM interface by up to 700 MW. These reductions remained in effect until February 1, 2025, when an enhanced operational procedure was implemented to mitigate the associated reliability risk. Over this 40-day period, these OOM actions resulted in approximately \$26 million in congestion shortfalls in the day-ahead and real-time markets. These costs were uplifted to ratepayers rather than being allocated to the responsible party.

In New York City, from mid-February through late April, both Gowanus-Greenwood 138 kV lines were out of service, creating a circuit breaker contingency within the Greenwood ring bus configuration that became the most limiting contingency for adjacent facilities. This contingency caused load to shift within the Greenwood load pocket in a manner not represented in the NYISO market model. To manage this constraint, NYISO OOM-committed peaking resources within the load pocket in both the day-ahead and real-time markets. Over this two-month period, these actions resulted in approximately \$1.7 million in day-ahead BPCG uplift allocated to local load and \$6.3 million in real-time BPCG uplift that was inappropriately socialized across system-wide load. NYISO implemented an enhanced modeling of ring bus configurations in its market models in late April, which should reduce the need for similar OOM actions going forward.

⁹⁷ See Recommendation 2021-3.

On Long Island, high-cost peaking resources were OOM-committed on 70 days to manage 69 kV transmission constraints in the Far Rockaway load pocket associated with an unmodeled bus contingency. As illustrated in Figure 41, OOM dispatch in these circumstances reward units for having higher variable as-bid costs, resulting in approximately \$4.5 million in BPCG uplift. While these units were previously both capable of burning natural gas, we observe that oil-fired operation at the facility has increased dramatically, accounting for 50 percent of total generation at the facility in 2025, up from less than 2 percent in 2024. When units are frequently scheduled through OOM actions, they are sometimes more profitable operating on a more expensive fuel. The affected units submitted deactivation notices but were retained under cost-of-service agreements.

Although these specific issues have been addressed through various operational measures, we strongly recommend that NYISO incorporate these constraints directly into the market models. Market-based congestion management produces more efficient price signals by reflecting the true marginal cost of serving load under system constraints, thereby providing stronger incentives for suppliers to maintain existing resources and invest in new capacity needed for reliability. In addition, any costs associated with congestion management are naturally allocated to responsible entities in accordance with cost causation principles. This improves economic efficiency by aligning financial responsibility with system impacts and strengthening incentives for market participants to respond efficiently to system conditions.

E. Transmission Congestion Contract Prices and Payments

We evaluate the performance of the TCC market by examining the consistency of TCC auction prices and congestion prices in the day-ahead market for the Winter 2024/25 and Summer 2025 Capability Periods (i.e., November 2024 to October 2025). Table 8 summarizes TCC cost and profit for the evaluation period separately for inter-zonal and intra-zonal TCCs.⁹⁸

- The *TCC Profit* measures the difference between the *TCC Payment* and the *TCC Cost*.
- The *TCC Cost* measures what market participants paid to obtain TCC rights from the TCC auctions. For a particular path, the *TCC Cost* is equal to the purchased TCC MW multiplied by the TCC price for that path.
- The *TCC Payment* is equal to the TCC MW between two points multiplied by the congestion cost difference in the day-ahead market between the two points.

Market participants purchasing TCCs in auctions covering the 12-month period from November 2024 to October 2025 realized net profits of \$234 million. Overall, TCC holders earned a positive return of 135 percent (measured as a weighted percentage of original TCC prices), compared to *negative* returns of 66 percent and 47 percent in the previous two 12-month periods.

⁹⁸ Appendix Section III.H describes how we break each TCC into inter-zonal and intra-zonal components.

Returns averaged 117 percent on inter-zonal transmission paths and 152 percent on intra-zonal paths.

Table 8: TCC Cost and Profit
Winter 2024/25 and Summer 2025 Capability Periods

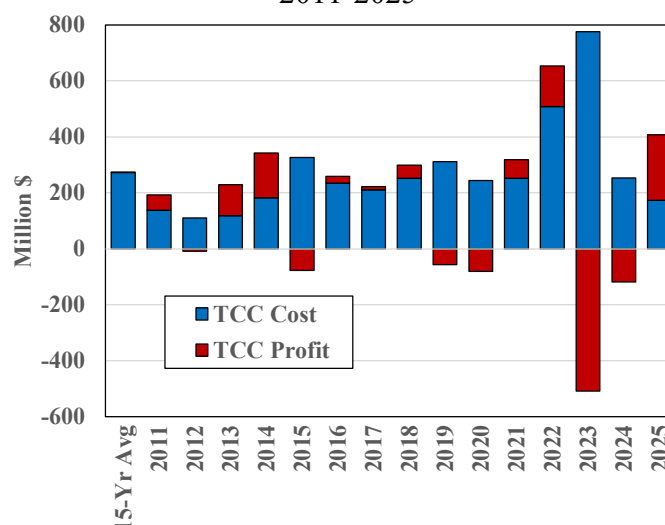
	TCC Cost (\$ Million)	TCC Profit (\$ Million)	Profit as a Percent of Cost
Intra-Zonal TCC			
Central Zone	\$26.2	\$4.2	16%
North Zone	\$3.7	\$10.3	278%
Mohawk VL	\$11.9	\$1.6	14%
Capital Zone	\$10.3	\$12.7	123%
New York City	\$17.7	\$96.9	547%
Long Island	\$14.9	\$8.9	60%
All Other	\$5.7	\$2.9	51%
Total	\$90.5	\$137.6	152%
Inter-Zonal TCC			
West NY to East NY	\$25.2	-\$3.8	-15%
New York to New England	\$27.6	\$7.9	29%
PJM to New York	\$30.2	\$64.6	214%
PJM to New England	\$10.8	\$15.9	147%
All Other	-\$11.5	\$11.9	-103%
Total	\$82.3	\$96.5	117%

These gains were driven by changes in congestion patterns that were not fully anticipated at the time of the auctions. TCC holders realized substantial gains of \$89 million on transmission paths between PJM and New York and between New York and New England, coinciding with a significant widening of regional natural gas price spreads from 2024 to 2025. In addition, gains of \$97 million were realized on transmission paths within New York City, corresponding to a 195 percent increase in day-ahead congestion revenues on these paths. This increase was driven primarily by higher natural gas prices and extended transmission outages in New York City during the winter months. Because these conditions were not fully anticipated at the time of the auctions, TCC holders realized gains on most intra-zonal and inter-zonal transmission paths in the affected regions.

More broadly, these results indicate that TCC prices generally align with congestion levels anticipated at the time of the auctions. Profits and losses of TCC bidders on most transmission paths are typically correlated with changes in day-ahead congestion patterns relative to prior years, emphasizing the importance of anticipated congestion levels in evaluating TCC profitability. However, unexpected congestion, often triggered by lengthy unplanned outages, remains a key source of variation in outcomes. TCC auction results also suggest that market expectations of congestion improve closer to real-time operations, consistent with the availability of more accurate information about the state of the transmission system and market conditions.

To place these results in a longer-term context, Figure 42 reports TCC costs and profits over the past fifteen Capability Periods.⁹⁹ Over this period, market participants realized net profits in nine years and net losses in the remaining six years, with most outcomes relatively modest. An important exception occurred in 2023, when participants experienced significant losses, primarily due to lower-than-expected congestion across the Central-East interface in the day-ahead market. These outcomes were driven by unexpected changes in natural gas prices, imports from Quebec, and transmission outages.¹⁰⁰

Figure 42: TCC Cost and Profit
2011-2025



Across the full 15-year period, the average profit was approximately \$0.2 million on average compared to costs of \$273 million per year. Over the long term, market participants seeking hedges are expected to pay a small premium on average in a competitive efficient TCC market, reflecting their willingness to pay in forward TCC auctions to hedge risks in the day-ahead market. The small average TCC profits may be partly attributable to the tendency for TCC auctions to oversell TCCs because the network model used in the TCC auctions typically has all-lines-in-service, while congestion in the day-ahead market often occurs when transmission outages lead to reduced transfer capability.

⁹⁹ Each Capability Period represents a 12-month period from November of the prior year through October of the current year.

¹⁰⁰ See Section VII.C in 2023 State of the Market Report for the New York ISO Markets (Potomac Economics) for more details.

IX. CAPACITY MARKET PERFORMANCE

The capacity market is designed to ensure that sufficient capacity is available to satisfy New York’s planning reserve requirements. This market provides economic signals that supplement the signals provided by the energy and ancillary services markets to facilitate new investment, retirement decisions, and participation by demand response.

The capacity auctions set clearing prices for four locations: New York City, Long Island, a Locality for Southeast New York (“the G-J Locality”), and NYCA. By setting a clearing price in each Locality, the capacity market facilitates investment where it is most valuable for satisfying NYISO’s planning needs. This section of the report discusses the following:

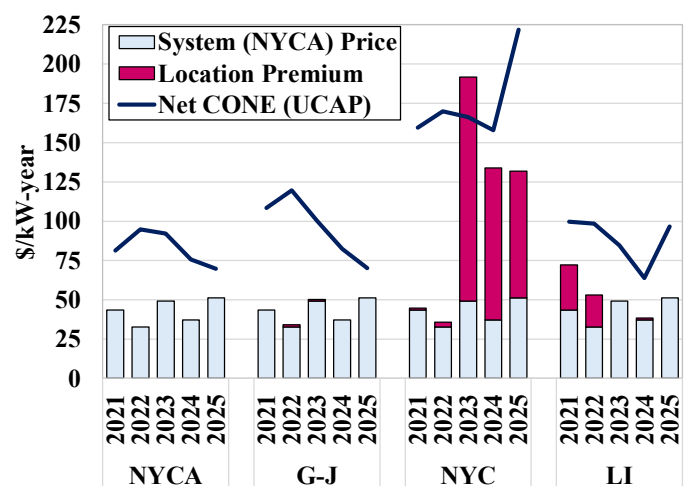
- A summary of recent capacity market results and notable developments (Subsection A);
- Recommended improvements to the design of the capacity market:
 - Improving locational price signals in the capacity market (Subsection B),
 - Compensating suppliers efficiently when locational capacity requirements are driven by transmission security limits (Subsection C),
 - Improving the market’s ability to support winter reliability (Subsection D) and,
 - Appropriately valuing conventional generating resources for their expected availability during critical peak conditions (Subsections E and F).

A. Recent Capacity Market Results

The ICAP Demand Curves determine how variations in the cleared supply of capacity affect clearing prices. Figure 43 summarizes total monthly spot auction prices in recent Capability Years, which run from May to the following April. The Net Cost of New Entry (Net CONE) is NYISO’s estimate of the capacity payments needed to attract new peaking capacity and is used to determine the Demand Curve reference prices.

Systemwide (NYCA) prices averaged \$42.8 per kW-year over the past five years and have generally been well below the Net CONE, reflecting capacity surplus conditions. Prices in the G-J Locality and Long Island have had little or no premium over NYCA in recent years. New York City prices averaged \$107.6 per

Figure 43: Capacity Spot Auction Prices
By Capability Year Starting In May of Year Shown



kW-year, with a large premium due to transmission limitations into the city. However, the NYC price was well below the Net CONE in most capability years.

Table 9 shows average spot auction prices for each locality for the 2025/26 Capability Year and year-over-year changes in key factors from the prior Capability Year.¹⁰¹

The systemwide capacity margin fell and prices rose, driven by an increase in the Installed Reserve Margin (IRM) and net reduction in UCAP supply. The IRM rose primarily due to: (i) a reduction in the NYC and G-J locational requirements, leading reliability criteria to be satisfied with a larger proportion of capacity outside of those areas, and (ii) updates to the IRM model including reduced support from emergency procedures.¹⁰² Total supply declined due to lower net imports, 110 MW (ICAP) of generator outages and retirements, and a net reduction in existing generators' Demonstrated Net Maximum Capability (DMNC). This was offset by the entry of 270 MW (ICAP) of renewables and storage since fall of 2024, but these resources provide a limited amount of UCAP under NYISO's capacity accreditation rules.

The New York City capacity surplus increased in 2025 due to a lower load forecast and Locational Capacity Requirement (LCR) compared to 2024. The LCR declined primarily because NYISO revised its Transmission Security Limit (TSL) floor calculation to use a coincident rather than non-coincident load forecast (NYISO reverted to a non-coincident forecast for 2026/27). These changes were partly offset by a reduction in supply caused by 48 MW of retirements and 58 MW of generator outages. Despite the increase in capacity margin, NYC

Table 9: Key Drivers of Capacity Spot Prices
2025/26 Capability Year

	NYCA	G-J	NYC	LI
UCAP Margin (Summer)				
2025 Margin (% of Requirement)	5.8%	16.4%	5.7%	11.7%
Net Change from Previous Yr	-2.3%	1.0%	2.1%	0.5%
Average Spot Price (Full Year)				
2025/26 Price (\$/kW-month)	\$4.28	\$4.28	\$10.98	\$4.28
Percent Change Yr-Yr	38%	38%	-2%	33%
Change in Demand				
Load Forecast (MW)	-72	-14	-163	36
IRM/LCR	2.4%	-2.2%	-1.9%	1.2%
ICAP Requirement (MW)	667	-346	-340	99
Summer UCAP Requirement (MW)	661	-270	-373	71
Change in Supply (Summer)				
Generator & UDR ICAP (MW)	154	-187	-109	228
Generator & UDR UCAP (MW)	17	-174	-194	110
Demand Response UCAP (MW)	-17	-31	-26	-7
Net Capacity Imports UCAP (MW)	-199			
Total UCAP Supply (MW)	-199	-204	-221	103
Change in Demand Curves				
UCAP Net CONE (\$/kW-yr)	-8%	-15%	41%	51%

¹⁰¹ For detailed capacity market results data, see Appendix VI.

¹⁰² The IRM is set annually by the New York State Reliability Council (NYSRC). The Final IRM Study for the 2024/25 capability year produced an IRM of 23.1 percent, but NYSRC adopted a final IRM of 22.0 percent considering the impact of NYISO's Transmission Security Limit (TSL) floors in the downstate localities. The IRM Study for the 2025/26 capability year produced an IRM of 24.4 percent concurrently with a reduction of the TSL floors. The IRM studies can be found on the NYSRC webpage [here](#).

prices declined by only 2 percent due to a 41 percent increase in the UCAP Net CONE following NYISO's 2024 Demand Curve Reset, which resulted in a major increase in the NYC demand curve reference points and higher prices at any given level of surplus.

Hedging in the NYISO Capacity Market in 2025

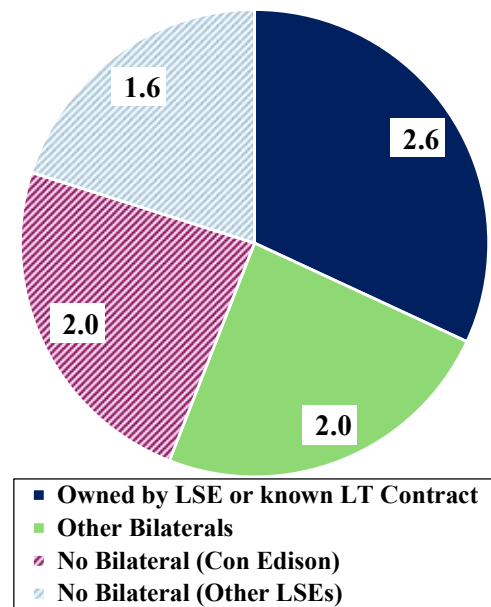
NYISO's spot capacity market serves as the basis for hedging and contracting by market participants seeking to manage price uncertainty. In 2025, 20 percent of systemwide capacity requirements were satisfied through NYISO's voluntary Strip and Monthly forward auctions, and 41 percent were satisfied through physical bilateral transactions. Participants also arrange financial capacity hedges, although the total amount of such transactions is unknown.

A significant amount of load engages in multi-year or long-term hedging of capacity price risk:

- New York's investor-owned utilities (IOUs) are required by the Public Service Commission to maintain supply portfolios that are partially, but not fully, hedged for retail and small commercial customers.¹⁰³ The IOUs routinely arrange capacity purchases several years in advance.¹⁰⁴
- The state's Public Power authorities (NYPA and LIPA) either directly own or enter into long term contracts with large amounts of generation capacity.
- Load-serving entities (LSEs) participating in New York's retail access market engage in a variety of hedging practices including bilateral contracting and ownership of generation by affiliates. Retail choice LSEs serve most of the state's commercial and industrial load and commonly offer fixed price supply contracts to customers.

Figure 44 summarizes the proportion of the total New York City UCAP Requirement that was covered by physical bilateral transactions in summer 2025. Approximately 32 percent was covered by bilaterals between an LSE and generation that is either owned or under long-term contract with that LSE or an affiliate. An additional 24 percent was covered by other bilaterals, including regular procurements made by Con Edison up to three years in advance and other private

Figure 44: Bilateral Sales Satisfying NYC UCAP



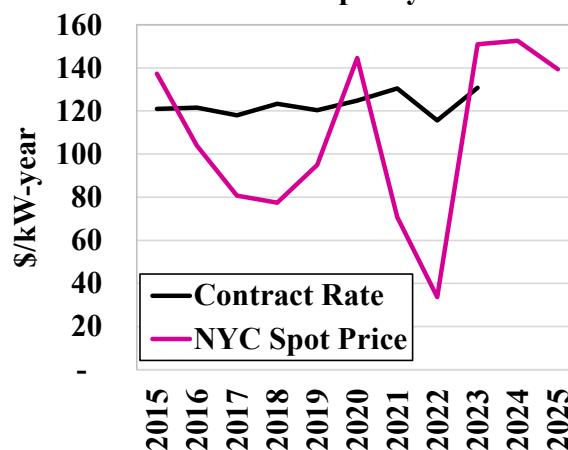
¹⁰³ See April 19, 2007 Order by NYPSC in Case 06-M-1017.

¹⁰⁴ For example, Con Edison indicated in its latest rate case that competitive RFPs for capacity products are issued one, two and three years in advance of the associated capability period. See Electric Supply Testimony in NYPSC Case 25-E-0072.

arrangements between LSEs and suppliers whose terms are not known. The remaining 44 percent of the UCAP Requirement not covered by bilaterals includes 2.0 GW allocated to Con Edison (49 percent of its UCAP Requirement) and 1.6 GW allocated to other LSEs including NYPA and other retail choice LSEs (40 percent of their UCAP Requirements). The extent of additional financial hedges by market participants is unknown.

Figure 45 compares historic NYC spot prices with the costs of capacity from firm contracts reported by Con Edison rate filings. Over the past decade, the costs of contracted capacity were higher on average but much more stable than the spot capacity price.

Figure 45: Con Edison Contracts vs NYC Capacity Price



Notable Developments in 2025

We highlight the following notable actions by NYISO impacting the capacity market in 2025:

- Demand Curve Reset starting in May 2025:** NYISO introduced demand curves that resulted in large reference price increases in New York City and Long Island and reductions in other areas (see Table 9). Reference price increases were driven by recent industry-wide cost increases, although these were partly offset by NYISO's decision to set demand curve reference prices based on the estimated Net CONE of a 2-hour battery system rather than that of a gas-fired combustion turbine (CT) as was done previously. The Net CONE of a 2-hour battery evaluated by NYISO did not accurately reflect the cost of capacity needed to satisfy reliability needs because NYISO did not properly consider that its capacity accreditation factor (CAF) is expected to fall dramatically as penetration increases and that it provides a limited contribution to transmission security.¹⁰⁵ In 2025, the 2-hour battery UCAP Net CONE was higher than the Net CONE of a dual fuel CT with a 20-year life by one percent in New York City, and lower than the CT Net CONE by 43 percent in Rest of State and 30 percent in Long Island.
- Retention of Capacity for Reliability:** generators totaling 1,247 MW submitted notices of intent to deactivate in 2025, including 571 MW in New York City, 178 MW in Long Island and 498 MW in the G-J Locality. NYISO identified Reliability Needs caused by each of these proposed retirements, triggering solicitations for solutions and possible retention under of the units proposing to retire at cost of service rates (see Section V.B.4). As of May 2025, all of these units remain in service, with Pinelawn eligible to deactivate on May 15. If any units are retained under contracts for reliability they will be required to offer as price takers in the capacity market, causing a downward market price impact.
- Updates to TSL Methodology:** Following the identification of Reliability Needs caused by proposed generator deactivations in 2025, NYISO revised its methodology to calculate

¹⁰⁵ See our December 2024 comments on NYISO's Demand Curve Reset filing, available [here](#).

Transmission Security Limit (TSL) floors to better align market requirements with the methodology of the Reliability Planning Process. Revisions include the use of non-coincident peak load in the TSL floor calculation and higher generator outage rate assumptions. These changes have a major upward impact on the LCRs for the 2026/27 Capability Year (4.5 percent in NYC and 6.9 percent in Long Island).¹⁰⁶

The potential for retention of capacity for reliability through out-of-market contracts highlights significant gaps between the reliability needs of the system and the representation of those needs in NYISO's capacity market. While NYISO's updates to the TSL Methodology for the 2026/27 Capability Year addressed key problems, significant gaps remain because: (a) locational reliability needs are not adequately represented in the capacity market, (b) the capacity demand curves are based on the cost of a 2-hour battery rather than a dual-fuel CT, and (c) NYISO's reliability planning process has adopted relatively conservative assumptions that require higher installed capacity margins than the capacity market was designed to maintain. These deficiencies are evaluated further in this section. Furthermore, the need to retain in service older inefficient existing capacity is also driven by the inability of investors to develop new cleaner fuel-efficient gas-fired generating capacity since 2019.

B. Assessment of Locational Pricing in the Capacity Market

An efficient capacity market would accurately signal the value of capacity at different locations in the system for meeting reliability requirements. In this subsection we discuss shortcomings in the zonal structure of NYISO's capacity market and recommend enhancements.

Overview of Zonal Structure

NYISO's capacity market consists of four pricing regions: New York City, Long Island, the G-J Locality, and Rest of State.¹⁰⁷ The boundaries between these regions roughly capture the locations of historical transmission bottlenecks that limit capacity deliverability during summer peak periods.¹⁰⁸ NYISO performs a New Capacity Zone study every four years to examine whether new capacity zones should be created. This process has created a new capacity zone only once, when the G-J Locality was created in 2013.

¹⁰⁶ See November 20, 2025 ICAPWG presentation on TSL floor methodology for 2026/27, available [here](#).

¹⁰⁷ In this subsection, a "capacity zone" refers to a pricing zone with a capacity market demand curve (such the NYCA and G-J Locality zones), a "region" refers to a part of a capacity zone that may have a distinct price (such as the Rest of State and GHI regions within the NYCA and G-J Locality), and an "area" refers to a part of the system that is separated from other areas by transmission bottlenecks. Areas are represented as "bubbles" in the GE-MARS topology and include (but are not limited to) the 11 historic load zones (A-K).

¹⁰⁸ Capacity deliverability broadly refers to the ability of generation to be delivered to load at times of peak system need. Assessments of deliverability examine whether the available generation in a region is simultaneously deliverable to load in a scenario where all generation is needed to avoid load shedding.

NYISO and the New York State Reliability Council (NYSRC) follow an annual process to establish the requirements and demand curves of each capacity zone:

1. NYISO and the Transmission Owners forecast peak load for the system and each locality;
2. NYSRC establishes a minimum Installed Reserve Margin (IRM) for the NYCA system;
3. NYISO sets separate demand curve reference prices for each pricing zone based on the estimated Net Cost of New Entry (Net CONE) in that zone;
4. NYISO establishes Locational Capacity Requirements (LCRs) for the three localities using its LCR Optimizer methodology.¹⁰⁹ This methodology uses minimum ‘floor’ values based on zonal Transmission Security Limits (TSLs). Currently, LCRs for all three localities are set based on the TSL floors.

During the monthly Spot Auctions, capacity in each locality is cleared against demand curves derived from the locality’s LCR and Net CONE. The localities have a ‘nested’ structure where prices are set at least as high as the ‘parent’ zone (the G-J Locality and Long Island are nested in NYCA, while NYC is nested in the G-J Locality).

Problems with Current Zonal Framework

The existing zonal framework and new zone creation process suffers from several deficiencies:

- Highway constraints not modeled – Generators in load zones that are separated by transmission constraints within an existing capacity region all receive the same price. For example, in recent LCR studies we have observed transmission bottlenecks within the Rest of State region (between zones A-B and zones C-F) and within the Lower Hudson Valley (between zone G and zones H-J). Binding constraints are likely to emerge between more highway interfaces as new entry changes historic congestion patterns.
- Byway constraints not modeled – Generators whose output is limited by transmission constraints within a load zone receive the full capacity price even when they are effectively not deliverable. For example, Staten Island resources are paid the premium New York City price even when intra-zone bottlenecks limit their ability to support New York City reliability. Similarly, recent deliverability studies have found binding constraints between eastern and western Long Island.
- Ineffective New Capacity Zone study – The New Capacity Zone study will not lead to creation of new zones in many situations where bottlenecks are present. It relies on a deterministic study process that considers only one set of system conditions. As a result, it fails to detect deliverability constraints that bind in NYISO’s probabilistic resource adequacy model. This inadequacy will grow as more intermittent and storage resources, whose output is not well represented by a deterministic snapshot, enter the market.
- Barriers to New Investment – NYISO’s Interconnection Process includes a Deliverability Study that examines whether the capacity of proposed new projects is deliverable throughout their capacity region. Projects attempting to enter potentially bottled areas

¹⁰⁹ See Appendix VI.E for discussion and critique of NYISO’s LCR Optimizer method.

may be assigned costs for System Deliverability Upgrades (SDUs). In recent years many applicants have been assigned prohibitively costly SDUs (see Section V.A), often based on highly unrealistic modeling scenarios where vast amounts of new capacity are assumed to enter simultaneously with no retirements of existing resources. The Deliverability Study process is necessitated by NYISO's relatively limited set of capacity zones, as it applies only to the project's deliverability *within* its capacity region.

The impact of these shortcomings is to over-compensate resources in bottled areas and under-compensate resources in high-value areas, which drives up capacity prices and retains excess capacity in service. This is because the IRM and LCR processes compensate for the presence of bottled capacity in a sub-regional area by inflating ICAP requirements instead of limiting procurement in the bottled area. Legacy resources in bottled areas have incentives to not retire and to retain their rights to sell capacity, preventing new entrants from entering those areas.

Overview of MMU Proposals

We recommend that NYISO implement changes to improve locational signals in the capacity market. Improved locational capacity pricing will: (a) provide stronger incentives to retain and invest in capacity where it is most valuable and (b) allow NYISO to significantly reduce the Deliverability Study process as an obstacle to new generator interconnection. Here we discuss two approaches:

- Set Capacity Accreditation Factors (CAFs) that reflect capacity deliverability, and/or
- Implement more granular capacity zones and a dynamic process for updating zones.

As part of either of these approaches, we recommend establishing Financial Capacity Transfer Rights (FCTRs), which would provide a hedge against location-driven changes in capacity payments for suppliers that fund deliverability upgrades or have done so in the past. When a transmission upgrade improves the capacity transfer limit between areas, the FCTR should provide compensation based on the difference in the value of capacity between those areas.

1. Locational Capacity Accreditation Factors (Rec. #2025-2)

Under this approach, NYISO would use the existing capacity accreditation process to discount payments to suppliers in bottled areas. Currently, NYISO assigns suppliers to a Capacity Accreditation Resource Class (CARC) by technology and capacity region. Instead, NYISO could establish CARCs for each combination of technology and area modeled in MARS. NYISO would then annually calculate a Capacity Accreditation Factor (CAF) for each CARC using the following process:

- 1) Calculate the Marginal Reliability Impact (MRI) for each CARC using NYISO's LCR Study model.¹¹⁰
- 2) For each CARC, calculate the CAF as the MRI of that CARC divided by the MRI of 'perfect' capacity at a reference location for the capacity region. The reference location would be selected as area with the highest MRI in the capacity region.

For example, this process would calculate the CAF of Wind resources in Zone A as the MRI of an additional 100 MW of wind in Zone A divided by the MRI of 100 MW of 'perfect' capacity in Zone F. This CAF therefore reflects the impacts of (1) the technology characteristics of the wind resource, and (2) any binding transmission bottlenecks between Zone A and the Rest of State region's reference area (Zone F).

Table 10 shows estimated locational CAFs for the Generator technology type using data from the 2026/27 LCR Study.¹¹¹

There were no significant deliverability limitations within regions, resulting in Generator CAFs of 100 percent at all modeled locations and suggesting that this proposal would not create significant market volatility in the short term.

Under this approach, locational value within a Capacity Region is reflected through changes to supplier UCAP ratings rather than prices. Potential reductions of supplier UCAP in bottled areas would not directly cause higher capacity prices. This is because NYISO's translation of the capacity requirements would cause a commensurate decrease in UCAP Requirements.

Table 10: Locational MRI and CAFs based on 2026/27 LCR Study

Capacity Region/Zone	Marginal Reliability Impact (Δ LOLE per 100MW)	Generator CAF
NYCA		
A - West	0.0068	100.0%
B - Genesee	0.0068	100.0%
C - Central	0.0068	100.0%
D - North	0.0068	100.0%
E - Mohawk Valley	0.0068	100.0%
F - Capital	0.0068	100.0%
G-J Locality		
G - Hudson Valley	0.0068	100.0%
H - Millwood	0.0068	100.0%
I - Dunwoodie	0.0068	100.0%
NYC		
J - N.Y.C	0.0068	100.0%
J3 - Staten Island	0.0068	100.0%
Long Island		
K - Long Island	0.0066	100.0%

¹¹⁰ The MRI is the change in system Loss of Load Expectation (LOLE) resulting from an addition of 100 MW of the technology/location combination represented by that CARC.

¹¹¹ We calculate the MRI of each location beginning with the 2026/27 LCR Case MARS model adjusted to the tariff-prescribed level of excess. For further detail, see V.I.E.

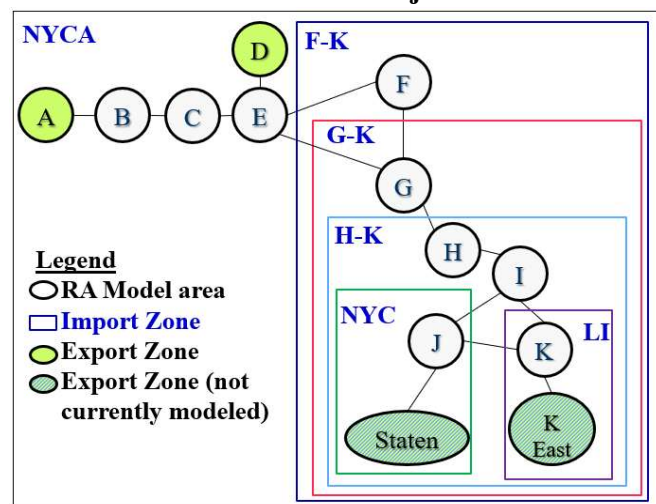
2. More Granular Capacity Zones (Rec. #2022-4)

This approach would expand the number of capacity zones and establish a more dynamic process for adding zones, while keeping the structure of the capacity market largely intact. We recommend the following process:¹¹²

1. Represent all major capacity deliverability bottlenecks in the resource adequacy model;
2. Designate capacity zones as import or export-constrained capacity zones based on the configuration of binding transmission constraints in the resource adequacy model;
3. Determine ICAP requirements for all import and export zones;
4. Establish import and export demand curves for use in the Spot Auction;
5. Apply a financial Capacity Constraint Pricing Credit or Charge to capacity payments of resources that positively or negatively impact aggregate deliverability between regions.

Figure 46 provides an example capacity zone topology under this alternative. It shows potential import and export capacity zones following completion of the Long Island PPTN transmission projects. Compared to today's capacity zones, the G-J Locality is expanded to include Zone K due to increased transfer limits within this area following the Long Island PPTN. New import zones are created downstream of the Central East (F-K) and UPNY-CONED (H-K) interfaces. Within the Rest of State region, export zones are created in western and northern New York. Finally, new areas not currently modeled directly in MARS are created in Staten Island and eastern Long Island, which lead to creation of export zones within the existing NYC and Long Island import zones. This arrangement of potential import and export zones is illustrative, and new or different zones could be created depending on the location of deliverability bottlenecks.

Figure 46: Illustration of Import and Export Zones After LI PPTN Projects In Service



Conclusions on Zonal Structure

The capacity market's current four zone structure causes inefficient market outcomes because (1) it does not recognize differences in the value of capacity at locations within existing regions, and (2) it obligates NYISO to manage within-region congestion through the interconnection Deliverability Study, which inefficiently inhibits new entry. We recommend increasing the

¹¹² For a detailed walkthrough of each of the steps in this recommendation, see VI.F.

locational granularity of capacity market incentives using one of the two concepts presented above. Improved locational market incentives would have the following principal effects:

- Allow NYISO to simplify the Deliverability Study by no longer requiring deliverability to be tested over interfaces that can be represented in the LCR Study model;¹¹³
- Allow for capacity bottlenecks to be reflected in the market as they emerge;
- Discount capacity payments to export-constrained areas that are overpriced;
- Protect asset owners who previously funded deliverability upgrades or elect to do so in the future through Financial Capacity Transfer Rights that provide a hedge against changes in locational prices and CAFs.

C. Impact of Transmission Security Limits on Efficient Capacity Payments

NYISO's process to set LCRs employs a minimum 'floor' value for each locality based on its Transmission Security Limit (TSL). In recent years, LCRs have increasingly been set at this 'TSL-floor'. This subsection discusses inefficient compensation that occurs when LCRs are set at the TSL floors and our recommendations for addressing this problem.

The purpose of the TSL-floors is to ensure that LCRs do not violate NYSRC/NPCC transmission security criteria. The TSL-floors are calculated using a deterministic study methodology that assesses the amount of capacity required in the locality under peak load conditions after applying a prescribed set of generation and transmission outage credible contingencies that vary by locality.¹¹⁴ The LCR is set at the higher of the TSL-floor or the level that NYISO's LCR Optimizer would otherwise determine based on probabilistic resource adequacy assessment.

In NYISO's planning studies, some resource types are assumed to contribute less towards transmission security requirements than resource adequacy requirements. In particular:

- Special Case Resources (SCRs) contribute 0 MW towards transmission security requirements because they are assumed to not be available under normal transfer criteria.
- Large resources (including generation and UDRs) can increase the transmission security requirement, which is intended to maintain reliability in the event that the largest two generation and/or transmission elements are lost.

Figure 47 shows the impact of these resource types on recent New York City LCRs. The stacked bars show the makeup of the TSL-floor, while the pink line shows the resource adequacy-based LCRs estimated as part of the annual IRM Study process. The NYC LCR has been set at the TSL-floor every year since 2023, with the TSL-floor exceeding the IRM Study's LCR estimate by 2.9 to 7.7 percent. SCRs and the largest generator contingency (the 985 MW Ravenswood Unit 3) raised the TSL-floors by 5.9 percent on average from 2023 through 2025. Beginning in

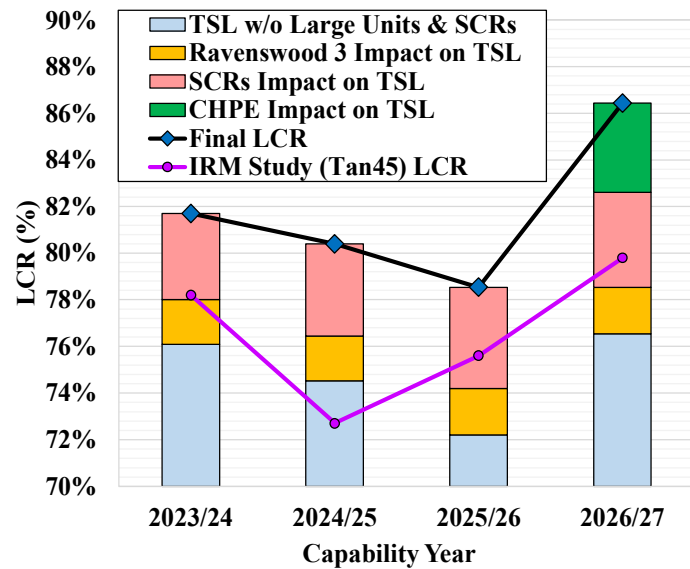
¹¹³ See our December 10, 2025 comments on NYISO's Deliverability Test, available [here](#).

¹¹⁴ See NYISO's 2026-2027 Locality Bulk Power Transmission Capability Report, available [here](#).

2026/27 with the expected entry of the 1,250 MW Champlain Hudson Power Express (CHPE) transmission project, large resources and SCRs will have a total impact of 9.9 percent on the TSL-floor.

The capacity market is designed to attract and retain capacity needed to satisfy planning reliability criteria. It is appropriate to set LCRs based on TSLs because, otherwise, a shortfall relative to the TSL-floor would trigger a regulated procurement of capacity through NYISO's Reliability Planning Process. However, LCRs based on TSL-floors result in inefficient capacity market compensation for two reasons:

Figure 47: New York City LCRs and TSLs



Resources that do not contribute to transmission security are overcompensated. The presence of very large resources, SCRs and offshore wind causes the TSL-floor to increase, so these resources provide less net supply towards meeting capacity requirements than they are paid for in the capacity market. This results in (1) higher consumer costs because these resources are paid more than the value they provide, and (2) inefficient investment incentives, such as for SCRs to convert to NYISO's DER resource type, which has different requirements and provides more value towards transmission security needs. In the 2025/26 Capability Year, we estimate that large resources and SCRs in New York City were overcompensated by up to \$26 million. We estimate this overpayment will grow to \$74 million in the 2026/27 capability year.

The ICAP Demand Curves overvalue surplus capacity beyond the transmission security requirement. Surplus capacity has less incremental benefit when requirements are based on transmission security as opposed to resource adequacy. This is because transmission security requirements are based on a deterministic and highly conservative scenario, regardless of its probability of occurring. Hence, the probability of load shedding due to insufficient capacity in a locality is vanishingly low when the locality satisfies transmission security requirements that significantly exceed resource adequacy-based requirements. Despite this, NYISO's capacity market uses demand curves that are of the same length for LCRs based on resource adequacy or transmission security. As a result, the demand curves overvalue surplus capacity beyond the TSL-based LCR and set prices that are inefficiently high when there is a surplus.

To address these inefficiencies: we recommend the following:

- Compensate capacity suppliers based on the requirements (e.g. transmission security or resource adequacy) that they contribute to meeting (Rec. 2022-1), and
- Develop sloped demand curves reflecting the marginal value of capacity when LCRs are set by TSL-floors. (Rec. 2023-4).

Under these recommendations, suppliers would receive a two-part capacity payment reflecting their resource adequacy and transmission security value, considering the resource's contribution to satisfying each requirement. NYISO would set prices for the resource adequacy and transmission security components using separate demand curves whose slopes appropriately reflect the value of surplus exceeding the requirement.¹¹⁵

D. Assessment of Seasonal Capacity Market Framework

The capacity market was designed to procure sufficient resources to reliably satisfy demand during summer peak conditions. It has been taken for granted that this design would also make sufficient resources available to satisfy demand at other times of the year. However, the evolving supply mix and demand patterns necessitate improvements to the market's ability to efficiently satisfy winter reliability needs. This subsection discusses these issues.

Recent Changes to Seasonal Market Structure

Resource adequacy risk in the NYISO system has historically been concentrated in summer because peak load is much higher in summer than winter. Winter resource adequacy risk is now growing relative to summer, comprising 14.8 percent of LOLE risk in the final IRM case for the 2026/27 capability year. The main drivers of growing winter risk include a large amount of gas-fired capacity that lacks firm fuel supplies in winter, rising winter load driven by heating electrification, and lower levels of capacity imports available from neighboring regions.¹¹⁶

NYISO's capacity market was historically not designed to distinguish the reliability value of capacity in summer and winter. While there are separate summer and winter capability periods and monthly spot auctions, they all use the same ICAP Requirements determined annually. NYISO's monthly prices are thus more akin to installment payments towards meeting an annual (summer-driven) requirement than signals of seasonal capacity value. In recent years, NYISO has made several market changes intended to better value seasonal capacity contributions:

¹¹⁵ For a more detailed discussion of the mechanics of this proposal, see our September 24, 2024 ICAPWG presentation (available [here](#)) and Section VIII.E of our 2024 State of the Market report (available [here](#)).

¹¹⁶ See Section VIII.G of our 2024 NYISO State of the Market Report (available [here](#)) and MMU Analysis of Gas Availability in Eastern New York, presentation to New York State Reliability Council (NYSRC) Installed Capacity Subcommittee, January 3, 2024, available [here](#).

- Seasonal Demand Curves – Beginning in the 2025/26 capability year, separate summer and winter demand curve reference prices are set so that the reference point is higher in the season that has more reliability risk in the IRM/LCR study.
- Firm Fuel Accreditation – Beginning in the 2026/27 capability year, gas and oil units in zones F through K must choose between the “firm” and “non-firm” capacity accreditation resource class (CARCs), which will have separate Capacity Accreditation Factors (CAFs). Generators can qualify for the firm CARC by committing to maintain firm gas pipeline transportation arrangements or sufficient stored inventory to operate for 56 hours across a seven-day period. The non-firm CAFs for 2026/27 are 55 percent in the Rest of State and GHI regions, 59 percent in New York City and 95 percent in Long Island, implying a major reduction in capacity revenues for non-firm generators.¹¹⁷
- Winter Reliability Capacity Enhancements project – NYISO recently filed a package of tariff revisions related to its seasonal capacity market design.¹¹⁸ These include: (1) separate summer and winter ICAP requirements based on the amounts of capacity modeled in the final IRM and LCR cases, (2) allocation of winter capacity obligations to Load-Serving Entities (LSEs) based on their winter peak load, and (3) requirements for UDR imports to make seasonal (rather than annual) elections of capacity sales.¹¹⁹ NYISO intends to implement these changes in the 2027/28 capability year.

Analysis of Current Seasonal Market Design Shortcomings

The changes discussed above significantly improve the market’s ability to support winter reliability. However, we have identified the following major shortcomings of the seasonal market design that are not resolved by the Winter Reliability Capacity Enhancements project:

Inflexible Firm Fuel and UDR Elections – Gas and oil generators are required to elect the firm or non-firm CARC during the IRM study process in August prior to the start of the May-April capability year, and they cannot move between CARCs after that date. This effectively requires a commitment to procure firm fuel 16 months before the associated peak winter period. This is poorly aligned with fuel procurement timelines and creates excessive financial risk for generators, who must commit to a CARC with limited information on expected capacity prices and the cost of acquiring firm fuel by acquiring primary firm gas pipeline transportation.

Similarly, NYISO’s recently proposed tariff revisions require UDR holders to elect the amount of winter capacity they will sell 16 months before the peak winter months. Since some UDR holders do not know how much capacity they will have available far in advance, they are likely to sell conservatively-low levels to avoid penalties. Gas units and UDRs will not be allowed to

¹¹⁷ See Final CARCS and CAFs on NYISO’s Capacity Accreditation webpage, available [here](#).

¹¹⁸ See NYISO filing in FERC Docket ER26-1431.

¹¹⁹ These changes address a deficiency in NYISO’s current process to set seasonal market parameters that causes extreme pricing when UDRs sell different amounts of capacity in summer and winter. See Section VIII.G of our 2023 NYISO State of the Market report and Recommendation #2023-5.

offer UCAP exceeding their elected quantities to the market if they are able to acquire it closer to the winter, which may lead to inefficient market outcomes and higher prices than necessary.

Prices Misaligned with Reliability Value – Under NYISO’s current approach, generators’ firm fuel elections have no impact on capacity prices or the winter risk level in the IRM Study (which in turn affects key market parameters such as the CAFs and reference points).¹²⁰ This is because the market parameters are set based on the assumptions modeled in the IRM Study, which currently does not consider the firm fuel elections.¹²¹

For example, the 2026/27 IRM study assumed 12.3 GW of generation from gas and dual fuel units in zones F through K would be available at extreme (90th percentile) winter peak load levels. Subsequently, 15.2 GW (UCAP) from these unit types elected the firm CARC for 2026/27.¹²² Modeling a higher quantity of firm fuel in the IRM Study would have resulted in lower ICAP requirements, lower prices and higher non-firm CAFs. Setting prices and CAFs that are misaligned from the system’s reliability needs could have major consequences including higher consumer costs, failure to attract efficient levels of firm fuel and capacity imports, and impacts on entry and exit decisions.

Seasonal Capacity Market Proposal

The current seasonal market framework is incomplete because it (a) places election requirements on firm fuel and UDR resources that impede efficient winter participation and (b) sets prices and CAFs that may be misaligned with reliability value. In a well-functioning market, prices and generator payments would adjust to changes in supply, providing efficient incentives to attract resources needed to maintain reliability in each season.

Hence, we recommend establishing seasonal capacity requirements, CAFs, and demand curves (Recommendation 2022-2). This recommendation builds on NYISO’s recently filed Winter Reliability Capacity Enhancements but more accurately signals the seasonal value of capacity

¹²⁰ The ICAP Requirements are derived from the levels of ICAP modeled in the final IRM and LCR studies. The ICAP requirements are converted to UCAP requirements using the average ICAP to UCAP derating factor for all resources in each locality. A larger amount of firm CARC elections increases the UCAP available for generators to sell but also results in higher UCAP requirements due to a lower average ICAP to UCAP derating factor, nullifying the impact on the spot market price.

¹²¹ The IRM Study process currently models a portion of the gas and dual fuel fleet as unavailable at high winter load levels based on estimates from historic data that do not incorporate fuel elections of generators for the capability year being studied. NYISO has indicated that it may increase reliance on generators’ elections in future IRM studies. This would improve the alignment of market outcomes and reliability, but it would also greatly increase risk faced by generators, who would be required to make fuel elections without information on the total amount of firm fuel to be modeled. For more detailed discussion, see March 17, 2025 ICAPWG presentation “Coordination of Firm Elections in the Capacity Market”, [here](#).

¹²² See NYISO presentation to February 4, 2026 NYSRC Installed Capacity Subcommittee, “Winter Fuel Availability Constraints Modeling Review: Comparison with 2026-2027 Firm Fuel Elections” available [here](#).

when the quantities of winter supply sources (such as firm gas, UDRs, imports and exports) are not known in advance, This would consist of the following:

- *Seasonal Requirements*: establish seasonal ICAP requirements that reflect the amount of capacity needed to satisfy the reliability criterion in each season. This could be done using the IRM/LCR modeling approach to determine separate requirements for satisfying summer and winter reliability targets.
- *Seasonal CAFs*: calculate separate summer and winter CAFs for each resource class. As a result, assumptions about relative summer and winter risk in the IRM/LCR study would not distort the value of the CAFs when there are changes in the supply mix.
- *Firm Elections Closer to Capability Period*: move the deadline for elections of firm fuel closer to the corresponding winter capability period (e.g., in the fall before the corresponding winter). This would better align firm fuel decisions with the timing of fuel contracting opportunities for suppliers, and create opportunities to suppliers to secure hedges through bilateral trades or the capacity strip auction to justify the additional costs of acquiring firm fuel.
- *Seasonal UCAP Demand Curves*: establish separate summer and winter capacity market demand curves in UCAP terms. The UCAP requirement would represent the amount of seasonally available capacity needed to comply with reliability criteria and would be derived from the amount of seasonally available UCAP modeled in the IRM/LCR study at criteria.¹²³ Reference prices of each season's demand curve would be set so that the price approaches the Net CONE of the reference technology when UCAP supply approaches the UCAP requirement in any season. It will be necessary to review the current demand curve shape and slopes to ensure that prices reflect the reliability value of capacity when risk is distributed across seasons.¹²⁴

Figure 48 illustrates how seasonal capacity requirements (determined based on the amount of capacity that satisfies the reliability criterion in each season) would translate into seasonal demand curves based on the UCAP-equivalent of the requirements.

Prices would result in the reference technology earning its Net CONE when reliability risk approaches the planning criterion (e.g. “1 day in 10 years” LOLE) in any one season or in aggregate across both seasons. Under this approach, changes in the amount of seasonal UCAP sold in the auction (for example, due to changes in the amount of firm fuel held by generators or

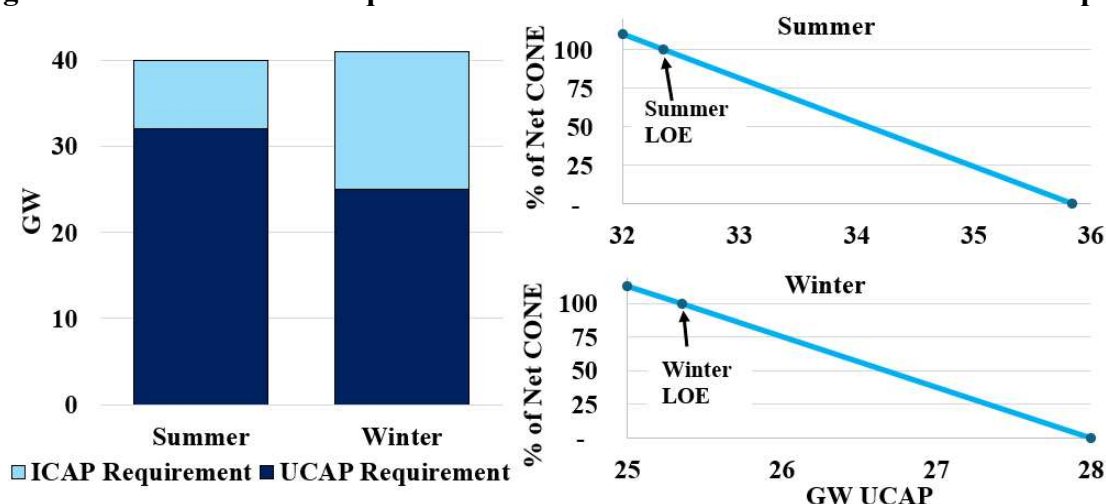
¹²³ The translation to a seasonal UCAP requirement should be done using the seasonal ICAP to UCAP ratio of the resource mix modeled in the IRM/LCR study (with adjustment for differences in generator EFORD values between the capacity market and IRM study). As a result, the importance of accurate assumptions regarding the quantity seasonally variable capacity (such as imports or firm gas) would be greatly reduced. For example, if the IRM/LCR study assumes no generators hold firm gas and a resource owner subsequently acquires firm gas, the UCAP requirement would remain fixed based on the ICAP-to-UCAP ratio of resources in the IRM/LCR study and the generator that acquired firm gas would sell additional UCAP, causing the market to clear further down the demand curve.

¹²⁴ For example, combined summer and winter capacity revenues should equal the reference unit's Net CONE when the risk level with the current capacity surplus is one half of the reliability criterion in both seasons.

net imports) would result in movement along the demand curve, treating seasonal UCAP from any source interchangeably. This has the following advantages:

- Summer and winter requirements are determined independently, so it is not necessary to predict relative levels of summer and winter supply accurately in the IRM/LCR study;
- Requirements and prices in summer and winter reflect the need for capacity that is reliably available (UCAP) in each season;
- Seasonal CAFs convert all resources' capacity to terms of equivalent marginal value, so that demand curves respond appropriately to changes in supply from any source; and
- Greater stability of seasonal prices and CAFs because they are not determined by sensitive estimates of relative winter and summer risk.

Figure 48: Illustration of Requirements and Demand Curves Under Seasonal Proposal



E. Excessive Qualification of Conventional Generators

NYISO tests the Dependable Net Maximum Capability (DMNC) of each generator on a seasonal basis. This test is used to estimate each generator's maximum output during weather conditions comparable to the expected peak load period of each season. The ICAP that a resource is qualified to sell in the capacity market is based on the lower of its DMNC and capacity interconnection rights (CRIS) quantity. NYISO has historically over-estimated the ICAP of fossil-fuel and nuclear resources affected by several issues which are listed below. Our recent State of the Market reports have recommended addressing these issues with improvements in the DMNC determination process.¹²⁵ For each issue, we summarize NYISO's recent efforts to improve the accuracy of DMNC quantities:

¹²⁵ See Recommendation #2021-4 in the 2024 State of the Market Report.

- *Emergency Capacity*: Capacity offered above a generator’s normal upper operating limit (UOLn) that is only activated under NYISO Emergency Operations.¹²⁶ Operators may not commit this capacity in practice because of concerns that the emergency capacity cannot operate in a reliable manner, thereby increasing the risk of outage to the normal range of the generator’s capacity.¹²⁷
 - NYISO sunset the Capacity Limited Resource (“CLR”) program and placed an obligation on most resources with emergency capacity ranges to offer that capacity in the normal operating range (or not sell it in the capacity market). This will partially address excessive qualification of emergency capacity.
- *Ambient Water Temperature Dependent*: Generators that have once-through water-cooled condensers experience diminished cooling capability as inlet water temperatures rise. Environmental restrictions also prohibit outlet water temperatures from exceeding defined thresholds. Therefore, many of these water-cooled units have reduced capability on hot summer days due to higher water temperatures.
 - NYISO altered the summer DMNC test window for cooling water dependent resources to the warmest summer months (i.e., July and August) during the hours between 10 AM and 10 PM. This will partially address water temperature dependent capacity.
- *Tidal Dependent*: Generators with once-through water-cooled condensers pulling water from tidal dependent sources (i.e., the southern regions of the Hudson River Estuary and coastal regions) see their capabilities rise and fall with changing tidal conditions due to variations in cooling water flow and pressure. We observe DMNC tests of these generators are usually conducted for one unit at a time during high tide conditions, leading to higher output levels than are achievable in peak summer conditions.
- *Relative Humidity Dependent*: Combustion turbines that are equipped with certain Inlet Cooling Systems are significantly impacted by increases in the relative humidity in the air. This impact increases as air temperatures rise, compounding this issue.
 - NYISO implemented new requirements on combustion turbines with certain evaporative cooling equipment to adjust to seasonal peak dry bulb and wet bulb temperature levels. The changes are expected to largely address excessive qualification for relative humidity effects.
- *Barometric Pressure Dependent*: Combustion turbine efficiency and capability is impacted by barometric pressure in a predictable manner. This relationship is positively correlated, i.e., turbine capability increases with higher pressure conditions (and decreases as pressure drops) because air density impacts mass flow through the turbine. This impact is predictably greater at summer peak load conditions since those correspond to the warmest weather days and hot air is less dense than colder air. Most affected

¹²⁶ See NYISO Emergency Operations [Manual](#).

¹²⁷ For example, if a 100 MW generator with 10 MW of emergency capacity has a 5 percent outage risk on the non-emergency range (i.e., the first 90 MW), then the effective UCAP of that capacity would be 85.5 MW (i.e., 95% of 90). If operating in the emergency range increases the outage risk of the facility to 15 percent, the true reliability value of the plant would be 85 MW, implying that the marginal value of the emergency capacity is *negative* 0.5 MW.

generators perform DMNC tests under the most favorable weather conditions possible, typically mild temperature, clear, sunny days in early or late summer. Barometric pressure is much higher on those types of days than on the typical peak load-type day at warmer air temperatures.

- *Cogeneration & Steam Demand:* Some units have reported derates from cogeneration units due to limitations associated with their host steam demand. Some resources in this category may sell capacity to NYISO without accounting for the full contractual obligations to their host steam demand.
 - NYISO enhanced its procedures to prevent cogeneration units from effectively double-selling capacity (i.e., selling capacity to NYISO that will not be available while satisfying obligations to their host steam demand).

We will evaluate the effects of the recent changes on capacity sales and system operations in a future quarterly SOM reports. Table 11 provides an overview of our recommendations related to each of these areas and the status of NYISO efforts to address them.

Table 11: Status of NYISO Projects to Address Excessive Qualification of Generators

Category	MMU Recommendation	NYISO Adaptation	Implementation Year
Emergency Capacity	Require emergency capacity to be accredited based on the marginal capacity value of the calculated EFORD of that operating range.	Sunset of the Capacity Limited Resource program.	2025/26
Relative Humidity	Require combustion turbines equipped with certain inlet cooling systems to adjust to make a relative humidity adjustment to DMNC test results.	Comparable to MMU's recommended approach.	2026/27
Barometric Pressure	Require combustion turbines to adjust DMNC test results to a barometric pressure consistent with expectations at peak conditions.	N/A	N/A
Water Temperature	Require once-through water cooled generators to adjust DMNC test results to expected water temperatures at peak summer conditions.	Require water-cooled generators to conduct DMNC tests during July or August between the hours of 10 AM and 10 PM.	2026/27
Tidal Conditions	Require coastal once-through cooled water dependent generators to adjust DMNC test results to water levels consistent with average tidal depth.	N/A	N/A
Cogeneration Steam Demand	Require cogeneration units to reduce ICAP sales based on actual host steam demand during peak load hours comparable to BTM:NG resources.	Strict enforcement of of NYISO Market Services Tariff 5.12.1.5.1	2025/26

While NYISO has already implemented or filed changes that will address much of the capacity affected by the issues above, the following additional changes to DMNC testing and ICAP qualification processes may further improve the accuracy of DMNC values:

- Calculate seasonal capacity ratings that are adjusted for ambient water temperatures and tidal conditions (in a similar procedure to what NYISO currently uses to adjust for ambient air conditions) for affected generators.
- Quantify the UCAP value of emergency capacity based on its marginal value of capacity determined by the Equivalent Forced Outage Rate of this range.
- Require cogeneration resources to be seasonally rated in a manner similar to Behind the Meter Net Generation (BTM:NG) resources, which takes into account host steam obligations during peak load conditions.
- Require stations with one or more combustion turbines to adjust for differences between barometric pressures during DMNC tests and expected conditions at the forecasted peak load.

F. Outage Scheduling and Forced Outage Rates

Generation owners invest significant capital in maintaining their facilities in a reliable condition to maximize their value over the long-term. Maintenance projects frequently last for weeks or months, so maintenance is generally conducted during low demand periods when reserve margins are largest. Maintenance costs are affected by the duration and timing of the outage, which affect the availability of plant staff and contractors. Generation owners sometimes propose scheduling outages that would leave a region deficient of capacity during a given period. NYTOs have local capacity needs and transmission outages that further limit the availability of outage time slots. Lastly, forced outages of generation and transmission may unexpectedly reduce reserve margins further and require rescheduling of certain outages. NYISO has the critical role of coordinating outages, which has become more challenging in recent years.

Capacity margins have fallen since 2019, which has reduced available outage time slots throughout the year. The emergence of winter reliability risks has increased the amount of time each year when lengthy outages cannot be accommodated. Since 2019, the average number of run hours of older (>40 years-old) fossil fuel generators has increased 81 percent, which has increased the time needed to perform maintenance on these units by 29 percent to an average of nearly 2,300 hours in 2025. NYISO's Outage Scheduling Manual outlines procedures for generation owners to propose outage schedules, for NYISO and NYTOs to approve them, and for dealing with forced outages and other unscheduled outages. The current process is a non-market-based queuing system that has worked reasonably well, but as outage slots become scarce, it may become necessary to use financial incentives to ration the available time slots.

Generators sometimes experience sudden operational issues that result in a forced outage or derate. In addition, generators may need to take generation offline temporarily to fix or replace equipment. When this happens, it is critical for generators to accurately report this information

to NERC and NYISO using the Generator Availability Data System (“GADS”) because NYISO uses this information to estimate forced outage rates for planning assessments and to discount capacity market revenues to each generator based on its performance. When outages are reported to GADS as planned or for maintenance, they do not count towards the generator’s forced outage rate, so there are strong financial incentives to underreport forced outages. NYISO has a process for reviewing compliance of individual generators with GADS reporting requirements.

This subsection evaluates outage scheduling with NYISO and reporting to GADS. The first part provides background on outage scheduling and reporting. The second part summarizes outage scheduling patterns in the peak summer and winter months. The third part analyzes generator performance and estimates potential underreporting of forced outages and derates to GADS.

1. Background on Outage Coordination with NYISO and Reporting to NERC

Generator owners must notify NYISO and the local NYTO when they propose to schedule an outage or when a unit is forced into an outage or derate.¹²⁸ Scheduled outages are reported to the Outage Management System (“OMS”) and must be approved by the ISO and NYTO prior to commencing. Forced outages and derates that occur unexpectedly are often notified to the ISO through the TO via an Out of Merit (“OOM”) designation. Generators can schedule outages up to 24 months in advance, and NYISO generally approves them when it forecasts sufficient reserve margin. Outages are generally not approved far in advance of the summer (from mid-June to early-September), but outages are frequently scheduled on short notice throughout the summer as reserve margins allow.

Even though winter reserve margins have shrunk, oil-fired and dual fuel units often schedule lengthy maintenance outages during peak demand periods with limited natural gas availability. Starting in the 2026/27 Capability Year, NYISO recognizes a higher capacity accreditation factor for generators with firm fuel in the winter (December to February). It is important for NYISO to adapt the outage scheduling procedures to prevent firm fuel generators from scheduling outages in periods when generators are paid a premium for being available on a firm fuel.

Market participants also report the status of generators after each month to GADS. This data reporting structure is used to estimate forced outage rates of generators and operates separately from NYISO’s outage scheduling processes. GADS and OMS use distinct criteria for classifying outages based on their differing purposes, so some outages that are *scheduled* in OMS may still be *forced* in GADS or vice versa. When outages and derates are not properly reported to GADS, it leads to inaccurate EFORd estimates at the resource-level and at the locality-level, which

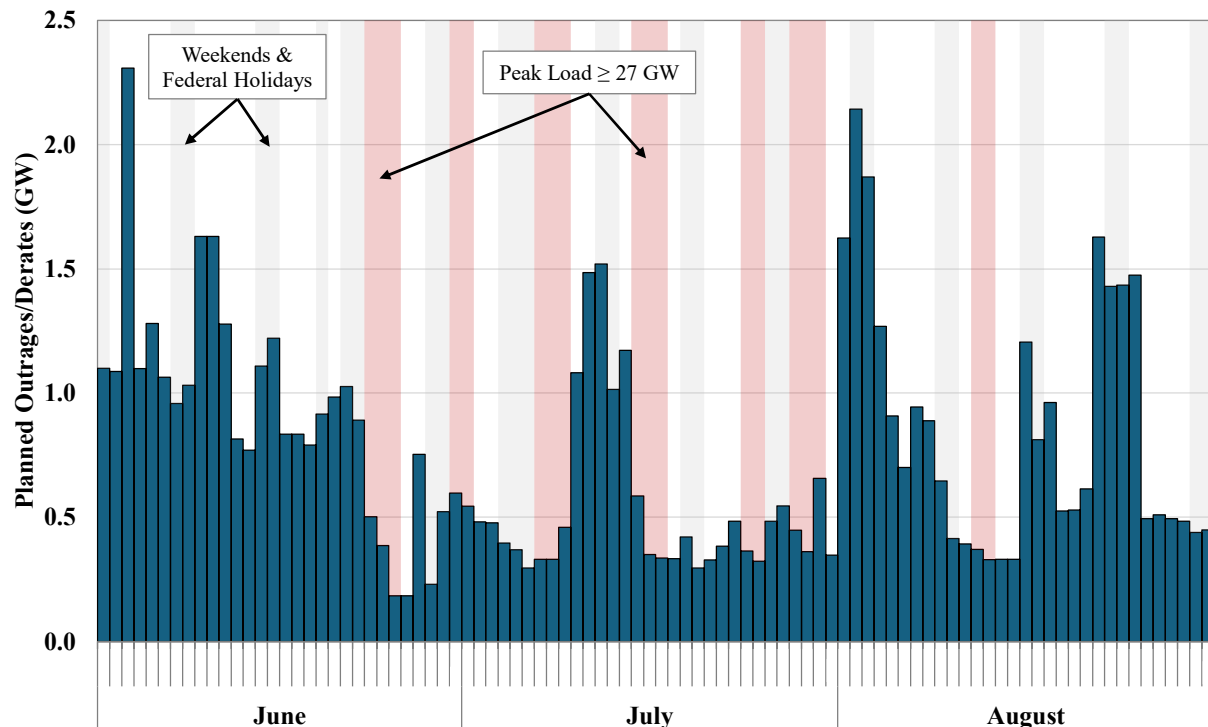
¹²⁸ See NYISO Outage Scheduling Manual Section 3.

affects planning studies, calculation of the IRM and LCRs, and other capacity market parameters.

2. Outage Scheduling during Peak Summer and Winter Conditions

Figure 49 and Figure 50 examine the planned outage schedules of fossil-fuel generators in the 2025 Summer and 2025/26 Winter Capability Periods. Figure 50 shows outages based on each resource's designated Firm, Non-Firm, or Partial Firm election in advance of the 2026/27 Capability Year.¹²⁹ Both figures show the average hourly capacity not offered into the DAM due to scheduled outages and derates. Figure 49 shows highlighted days when peak load exceeded 27 GW, indicating peak demand periods when relatively little surplus capacity is available in some areas. Figure 50 shows highlighted days when peak load exceeded 21 GW, which aligns with days when gas prices in some areas begin to reach parity with fuel oil prices.

Figure 49: Scheduled Outages and Derates during Peak Summer Months
Fossil-fuel Generators, 2025 Summer

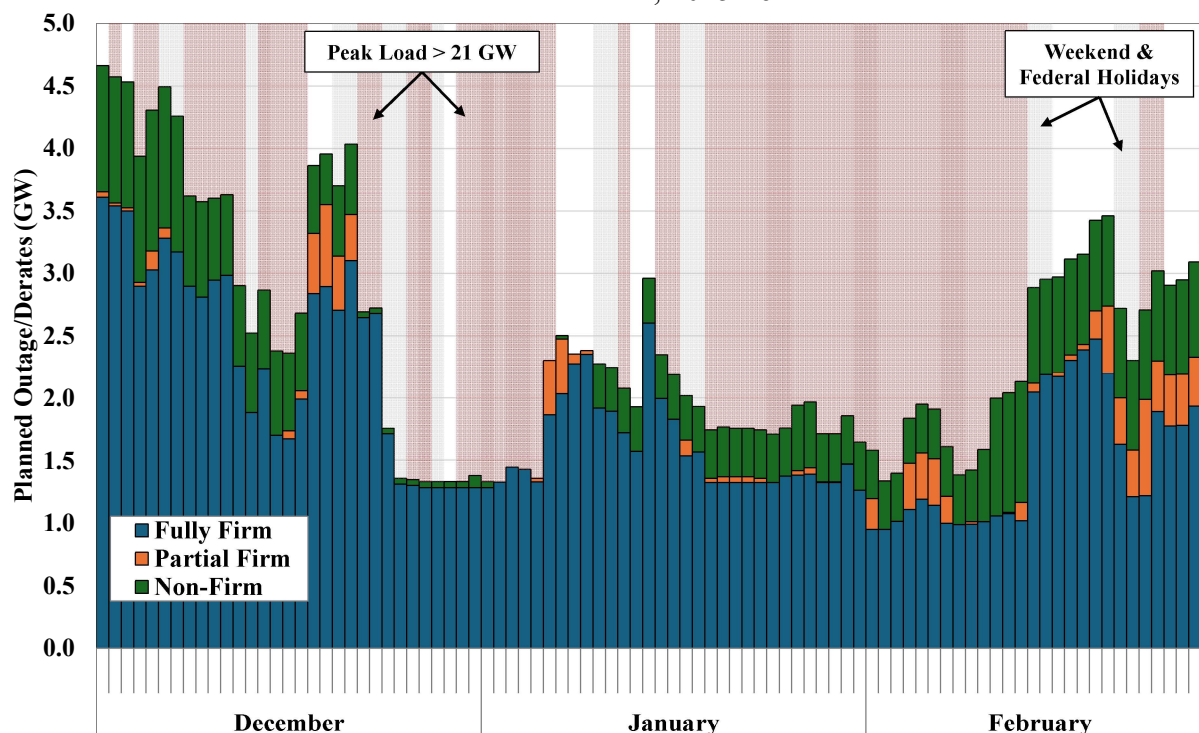


The figure shows outage schedule requests by fossil-fuel generators were approved on every day of the peak summer and peak winter periods. Even during the June 23-25 heat wave, one or more fossil-fuel generators were unavailable due to an ISO-approved outage when system peak

¹²⁹ Firm fuel requirements did not apply to any of the time periods covered in this report. The Firm/Non-Firm/Partial Firm designations are based on future elections and meant to illustrate how those resource classes schedule outages and derates in the current period, pre-firm fuel obligations.

loads exceeded 30 GW and grid operators utilized emergency operating procedures on each day. The duration of approved outages during the summer months was highly variable, but in one case, an outage that was originally scheduled to be completed in March was extended multiple times until July when the generator moved into a forced outage.

Figure 50: Scheduled Outages and Derates during Peak Winter Months
Fossil-fuel Generators, 2025/26 Winter



The winter months exhibited higher levels of scheduled outages than the summer. During the extended cold spell from late-January to early-February 2026, an average of 1.3 GW (~9 percent) of firm fuel capacity was unavailable due to approved, scheduled outages. All of this firm fuel capacity was located in Southeast New York and over 90 percent of the capacity was in outages scheduled for 50 days or more. NYISO will need to modify its outage scheduling practices before the winter of 2026/27 to ensure generators compensated for firm fuel are available during peak winter conditions.

As the NYISO generation fleet gets older, the average amount of time capacity resources are on outage is increasing, while the amount of capacity able to take outages each year has declined because the surplus capacity margin has fallen and because of the emergence of winter reliability risks in the months of December through February. Eventually, it will be necessary to ration the available maintenance slots in a more systematic way to provide generators with efficient scheduling incentives. Hence, we recommend (see #2025-4) NYISO consider whether the existing outage scheduling rules need to be revised to provide an incentive for generators to

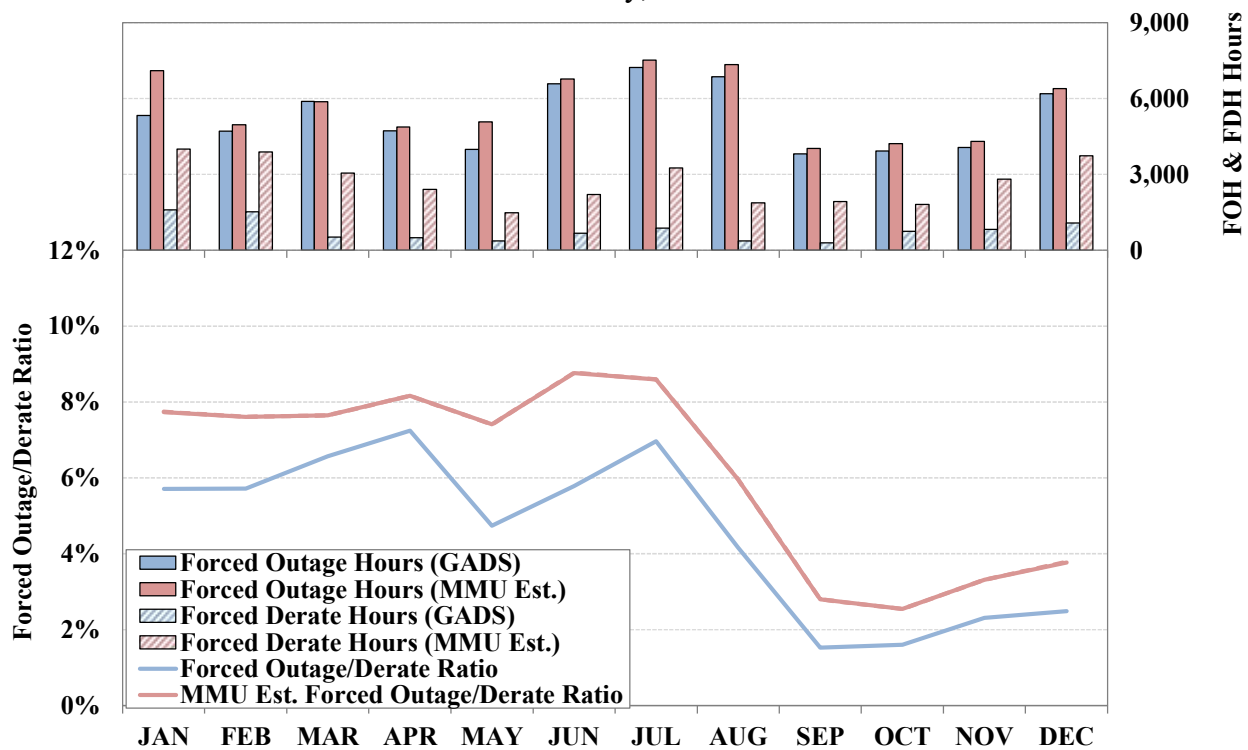
conduct planned and maintenance outages more quickly and at times that do not undermine reliability.

3. Monitoring for Unreported Forced Outages and Deratings

NYISO relies on GADS data to calculate EFORD values for applicable generators. Generators submit this data monthly in a distinct process from the outage scheduling process. We analyzed the extent to which reporting to GADS appears inconsistent with information submitted in the outage scheduling process and with the dispatch performance of those same generators. We are investigating cases of significant underreporting to assess the compliance of individual generation owners with their GADS reporting obligations. While some cases of potential underreporting may prove to be “false positives,” a significant share of the cases will likely prove to be cases of underreporting of forced outages or forced derates.

Figure 51 shows our analysis of outages and derates that may not be adequately reported to GADS as forced.¹³⁰ The top portion of the figure displays the contribution to outage MWh from forced outages and derates based on both the GADS data and the MMU estimates. The bottom portion of the figure displays the “forced outage/derate ratio” by month, one based on the GADS

Figure 51: Evaluation of GADS Data Submissions vs Market Results
Monthly, 2025



¹³⁰ The analysis evaluates fossil fuel generators in the NYISO.

data submitted by generators (blue line) and the other based on the MMU's internal screening of performance data (red line).¹³¹

This figure shows significant potential underreporting to GADS, especially of forced derate events. The average “forced outage/derate ratio” calculated from the GADS data was 4.5 percent for 2025, while our estimates based on generator performance data was significantly higher (7.4 percent).¹³² Our investigation has found several factors that contribute to this difference:

- ***Failure to Report Forced Outages and Deratings:*** In some cases, forced unavailability events (outages and deratings) show up in the OMS or OOM market data but do not appear in the GADS data. When forced outages and derates are not reported, a shortfall sanction should be levied by NYISO if the missing event is not identified before the next comparable Capability Period.¹³³
- ***Inconsistent Reporting of Forced Outage and Derating Durations:*** We observe forced outages and derates that are reported to GADS but with a shorter duration than appropriate from our review of the performance data. This is especially true with forced derates that are less apparent when a generator is not dispatched into its upper operating range.
- ***Outage Events Reported to OMS as Forced but to GADS as Planned:*** Some differences arise when outages are reported with comparable durations both to the NYISO and to GADS but the event types differ in the two systems. Sometimes, this occurs because NYISO approves short duration planned or maintenance outages where the minimum-advance notification requirement of these requests would not pass the NYISO procedure to screen generator outage requests that prevent planned outages during certain periods. In such cases, it is important to scrutinize information reported to GADS to ensure outages and derates are properly categorized.
- ***Failure to Continue Reporting Forced Derates Once Maintenance Outage Commences:*** Generators may experience issues that cause forced derates of the unit in real time, but not force the generator offline entirely. The NYISO and GADS often permits such resources to take Maintenance Outages to rectify the derate over the next weekend.¹³⁴ However, forced outages ought to be reported concurrently with the

¹³¹ For purposes of this analysis, we define the “forced outage/derate ratio” as the derating MWh divided by the total Energy equivalent of ICAP evaluated. For example, the Ratio for a 100 MW unit in January 2025 with 12 forced outage hours and 24 hours with a 10 percent forced derate would be calculated as $(12 \text{ hours} \times 100 \text{ MW} + 24 \text{ hours} \times 100 \text{ MW} \times 10 \text{ percent}) / (100 \text{ MW} \times 31 \text{ days} \times 24 \text{ hours per day}) = 1.9 \text{ percent}$.

¹³² These results exclude cases where the GADS data submitted to the NYISO was incomplete.

¹³³ For example, if a generator fails to report a forced outage in August 2025 but fixes the record in February 2026 prior to the 2026 Summer Capability Period, no ICAP penalties exist that could be levied on the resource since EFORs for a given Capability Period are based on status events from the prior to like-Capability Periods.

¹³⁴ The 2026 GADS DRI defines a Maintenance Outage as “An outage that can be deferred beyond the end of the next weekend (defined as Sunday at 2400 hours or as Sunday turns into Monday), but requires that the unit be removed from service, another outage state, or Reserve Shutdown state before the next Planned Outage (PO).”

Maintenance Outage until the equipment causing the derate is fixed.¹³⁵ GADS data submissions to the NYISO for situations of this nature rarely carry the forced derate into the Maintenance Outage causing an underreporting of forced derates.

- ***GADS Data Submitted Untimely:*** To qualify as Installed Capacity Suppliers in the NYISO, Generators must submit GADS data to the NYISO monthly by the 20th day of the month following the reporting month.¹³⁶ The tariff allows NYISO to sanction generators for late GADS submissions but never does so in practice.¹³⁷ Although most generators submit GADS data timely, some submit data several months late, hindering the processes used to evaluate compliance. As of January 21, 2026, the data for more than 1 GW worth of ICAP had not been updated since the end of June 2025.

Recommendation 2025-1b is to enhance GADS compliance monitoring to ensure generators log forced outages and derates comprehensively. Otherwise, the forced outage rates used in the capacity market will be underestimated, which may lead NYISO to implement more conservative assumptions in the reliability planning process. This should include clear guidance on the existing rules, stricter application of the existing sanction authority, more stringent criteria for initiating an investigation, and more unannounced UOL audits. In particular, NYISO should consider:

- ***Imposing sanctions for late GADS Data submissions:*** Efficient review of outages and derates requires timeliness that is hindered when GADS data submissions are late. The current procedures allow GADS data for the reporting month to be delayed up until the 20th day of the following month. Thus, even when data is submitted on time, 20 to 51 days may pass before the ISO and MMU have data necessary to flag discrepancies for review. We recommend that NYISO begin to apply the sanctions in MST 5.12.12.1 for late GADS data submissions.
- ***Increasing review of GADS data for accuracy based on generator performance:*** We observe that some derates are only reported to GADS when the generator is dispatched sufficiently high to make the derate apparent. However, the GADS Data Reporting Instructions (DRI) are clear that forced deratings last until the equipment causing the derate is repaired.¹³⁸
- ***Increasing review of GADS data for accuracy when outages are extended:*** We have observed instances when scheduled outages have been extended for weeks or months. Although these were not included in the analysis shown in Figure 51 (because they do not involve the performance data used in the analysis), we have found some instances when these outages should have been reported to GADS as forced outages rather than

¹³⁵ See footnote 138.

¹³⁶ See MST Section 5.12.5.1 and ICAP Manual Section 4.4.1. For example, January 26, 2026, GADS data would be due to the NYISO by February 20, 2026.

¹³⁷ See MST Section 5.12.12.1.

¹³⁸ “The derating ends when the equipment that caused the derating is returned to service, whether it is used at that time by the operators or not.” See GADS 2026 DRI, p. III-11.

maintenance outages or planned outage extensions.¹³⁹ Generation owners that fail to report these outages as forced benefit from increased capacity payments, but this failure leads to underestimates of forced outage rates in the planning processes.

¹³⁹ “For a planned outage, all of the specific individual maintenance and operational tasks to be performed are determined in advance and are referred to as the "original scope of work." The general task of repairing turbines, boilers, pumps, etc. is not considered a work scope because it does not define the individual tasks to be performed. For example, if a general task such as repair boiler is considered the work scope, it is impossible to conclude that any boiler work falls outside of the original scope of work. Discovery work and re-work which render the unit out of service beyond the estimated PO end date are not considered part of the original scope of work. A planned extension may be used only in instances where the original scope of work requires more time to complete than the estimated time. For example, if an inspection that is in the original scope of work for the planned outage takes longer than scheduled, the extra time should be coded as an extension (PE). However, if damage found during the inspection results in an extension of the outage, the extra time required to make repairs should be coded as a forced outage.” See GADS [2026 DRI](#), p. III-6.

X. EXTERNAL TRANSACTIONS

Wholesale markets facilitate the efficient use of both internal resources and transmission interfaces between control areas. The latter is beneficial because it allows:

- Low-cost resources in one area to compete to serve consumers in another area who might otherwise rely on higher-cost resources; and
- NYISO to access emergency power, reserves, and capacity from neighboring systems, thereby lowering the costs of meeting reliability standards across control areas.

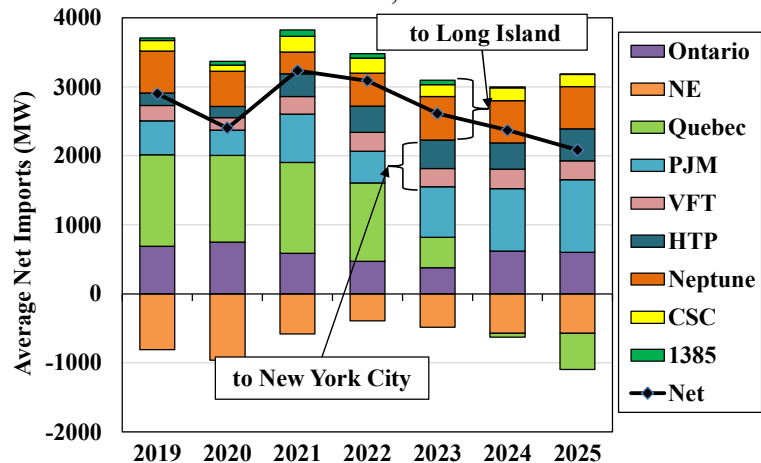
NYISO imports and exports substantial amounts of power from four adjacent control areas: New England, PJM, Ontario, and Quebec. In addition, Long Island and New York City are directly connected to PJM and New England via six controllable lines, which together can import up to roughly 2.7 GW directly to downstate areas.¹⁴⁰ Hence, NYISO's total import capability is large relative to its load, making efficient interface scheduling essential. In this section, we evaluate:

- Physical interchange between New York and neighboring control areas (sub-section A)
- Virtual import and export scheduling patterns (sub-section B)
- Performance of Coordinated Transaction Scheduling ("CTS") with ISO-NE and PJM, (sub-section C) including: (i) overall production cost savings from CTS; (ii) the impact of transaction fees imposed on CTS transactions; and (iii) drivers of forecast error in the models used to schedule CTS transactions.

A. Interchange between New York and Adjacent Areas

Figure 52 summarizes the net scheduled imports from neighboring control areas from 2019 through 2025 during peak hours (i.e., 6 am to 10 pm, Monday through Friday). In 2025, average total net imports from neighboring areas during peak hours were approximately 2,085 MW, marking a 12 percent decrease from 2024 and the lowest level since 2018. Net imports have declined steadily

Figure 52: Avg Net Imports from Neighboring Areas
Peak Hours, 2019-2025



¹⁴⁰ The controllable lines are: Cross Sound Cable, 1385 Line, Linden VFT Line, HTP Line, Neptune Cable, and A line, which is a PAR-controlled line that interconnects NYC to New Jersey and is scheduled as part of the primary PJM to NYISO interface and operated under the M2M JOA with PJM.

over the past three years, partly driven by reduced imports from Quebec. Despite this decline, imports continued to serve a significant portion of NYISO load, accounting for more than 11 percent of peak-hour demand in 2025.¹⁴¹

Controllable Interfaces to New York City and Long Island

On Long Island, net imports from neighboring control areas averaged 790 MW during peak hours in 2025, serving more than 32 percent of peak-hour demand. The Neptune Line was typically scheduled at its full available transfer capability, while flows across the Cross Sound Cable and the 1385 Line varied with gas price spreads between Long Island and New England.

In New York City, net imports over the HTP and Linden VFT lines averaged 735 MW during peak hours in 2025, an 11 percent increase from the prior year, and met approximately 12 percent of the City's peak-hour demand. Average scheduled flows over the HTP line have steadily increased from under 200 MW prior to 2020 to more than 450 MW in 2025. Lower natural gas prices in northern New Jersey and the limited geographic scope of the RGGI allowance market in PJM provide incentives for imports from PJM to New York City.

Primary Interfaces

Average net imports from neighboring areas across the four primary interfaces continued to decline in 2025, falling by 38 percent from 2024 to less than 560 MW during peak hours. This was the lowest level observed to date. This decline was driven primarily by increased exports to Quebec, although higher imports from PJM partially offset the overall decrease.

Among the primary interfaces, the Quebec interface historically accounted for the largest share of net imports to New York. However, this pattern reversed in 2024, when New York became a net exporter to Quebec for the first time. Net exports increased to an average of nearly 520 MW in 2025. The shift began in March 2023, when widespread wildfires across several Canadian provinces significantly disrupted generation and transmission capabilities in Quebec. Persistent drought conditions throughout 2024 and 2025 further reduced hydroelectric output, leading to sustained exports from New York to Quebec. In 2025, average net imports from Quebec occurred in two months, while the remaining ten months exhibited net exports, exceeding on average of 1 GW in December as water levels in the Saint Lawrence River reached record lows after an exceptionally warm and dry fall. This pronounced shift in interchange between Quebec and New York has had a material impact on prices and west-to-east congestion in New York.

Net imports across the primary interfaces with PJM and New England tracked changes in natural gas price spreads and emission allowance price differences between these regions. For example, during winter months, New York normally imported from PJM and exported to New England,

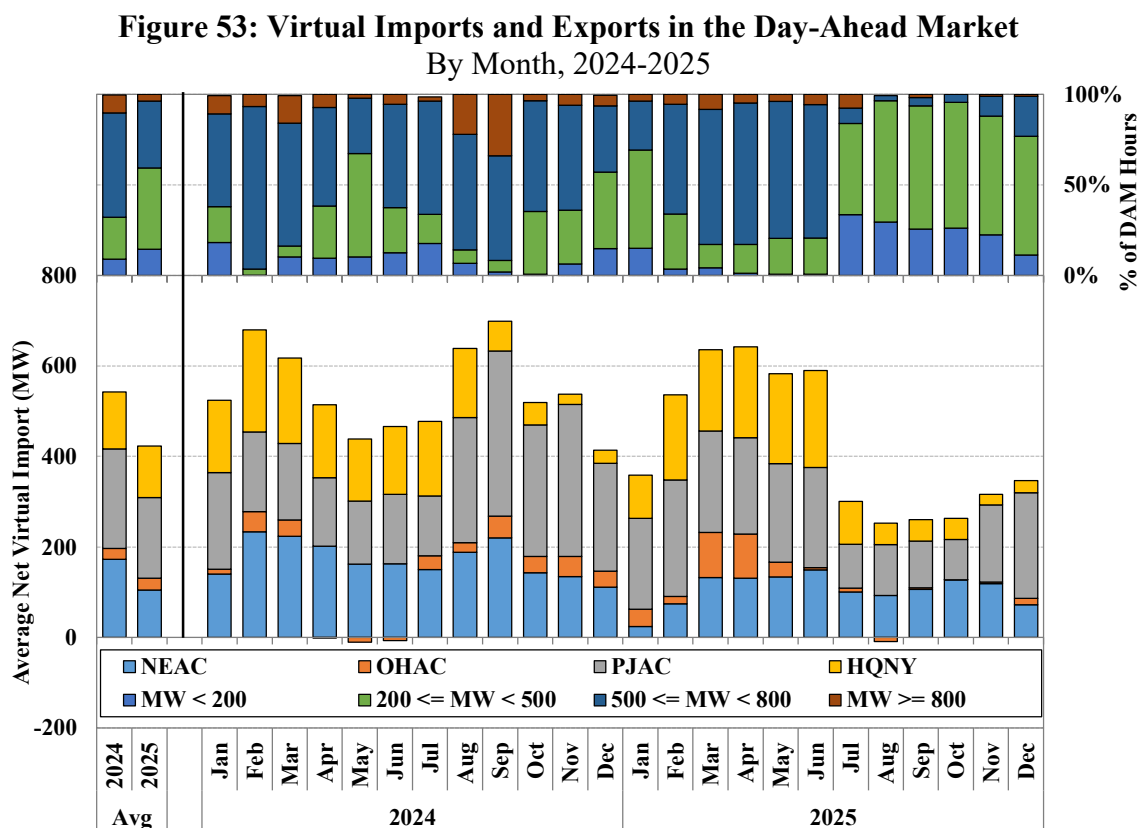
¹⁴¹ Figure A-54 to Figure A-57 in the Appendix provide additional details regarding interchange scheduling.

consistent with seasonal gas price spreads (i.e., New England > New York > PJM). These patterns highlight the strong interdependence between electricity and natural gas markets in the Northeast where natural gas is the primary fuel for power generation. In addition, imports from PJM have increased over the past three years as RGGI allowance prices have risen 68 percent from 2022 to 2025, adding around \$10.20 per MWh to the marginal cost of a typical combined cycle generator in NYISO relative to most areas of PJM.¹⁴²

B. Virtual Imports and Exports in the Day-Ahead Market

Traders frequently schedule transactions between NYISO markets and neighboring control areas in the day-ahead market but subsequently withdraw these transactions in the real-time market. We refer to these external transactions as “virtual” imports and exports, as they function similarly to ordinary virtual supply and load scheduled in the load zones.

Figure 53 examines the frequency and magnitude of scheduled net virtual imports in the day-ahead market in each month over the past two years.¹⁴³ The figure indicates that virtual external transactions between NYISO and neighboring control areas occurred in nearly every hour.



¹⁴² See Figure A-7 in the Appendix for more description of this analysis.

¹⁴³ See Figure A-38 in the Appendix for more description of this analysis.

In 2025, these transactions averaged more than 420 MW in the net import direction and exceeded 500 MW in approximately 40 percent of hours. Average net virtual imports declined by 22 percent from 2024 to 2025, driven largely by changes in the bidding patterns of one market participant beginning in July 2025. Despite the reduction, the relatively large and consistent volumes of net virtual transactions raise two concerns related to market performance.

First, virtual imports and exports are currently treated as physical energy in the day-ahead market but subsequently fail during post-DAM checkout with neighboring control areas. This can result in insufficient scheduling during the Forecast Pass of the day-ahead market, necessitating SRE commitments to address capacity deficiencies after the day-ahead market. This problem will continue under the current design for the Dynamic Reserves project, since virtual and non-firm imports will be counted toward satisfying operating reserve requirements in the day-ahead market, potentially leading to under-scheduling of physical energy and operating reserves.

Second, despite failing post-DAM checkout, virtual transactions are treated as available in RTC's *advisory* scheduling timeframe, but they are treated as unavailable in RTC's *binding* scheduling timeframe. This inconsistency can create unrealistic ramp constraints in the advisory timeframe, which subsequently distort real-time prices and schedules in the binding timeframe. Such issues have been observed from time to time at the Ontario interface and are likely to persist if left unaddressed.

Therefore, NYISO should consider measures to clearly distinguish virtual and non-firm imports in the day-ahead market and avoid treating them as equivalent to firm physical supply in the Forecast Pass and in the implementation of Dynamic Reserves.

C. Coordinated Transaction Scheduling with ISO-NE and PJM

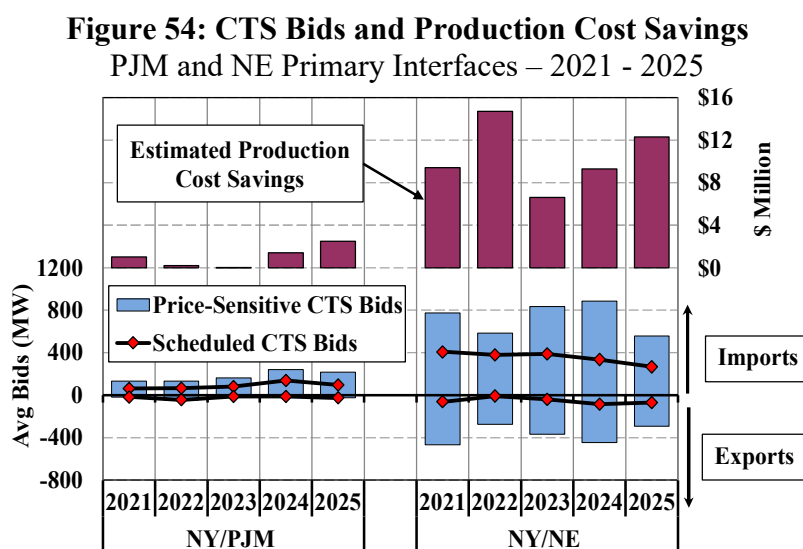
Coordinated Transaction Scheduling (CTS) allows two neighboring RTOs to exchange real-time market information to clear market participants' intra-hour external transactions more efficiently. CTS offers two key advantages over the traditional hourly LBMP-based scheduling system used at the interfaces with Ontario, Quebec, and between Long Island and Connecticut:

- **Flexibility:** Interface flows can be adjusted every 15 minutes instead of every 60 minutes, allowing for a more efficient response to changing real-time conditions.
- **Timeliness:** CTS schedules transactions much closer to the operating time. Hourly LBMP-based schedules are established up to 105 minutes in advance, while CTS schedules transactions less than 30 minutes ahead, benefiting from more accurate system information.

This subsection discusses several factors affecting CTS performance at the PJM and ISO New England interfaces, particularly the effects of transaction fees and short-term price forecasting errors. We also provide a detailed analysis of key factors contributing to inaccuracies in short-term price forecasts.

1. CTS Bids and Production Cost Savings

Under CTS, traders submit bids that are scheduled when the RTOs' forecasted price spread exceed the bid price. Therefore, it is critical for traders to submit a sufficient volume of price-sensitive bids. Figure 54 evaluates the price-sensitivity of bids at the primary PJM and ISO-NE interfaces and the associated market efficiency gains. The lower panel shows the average amount of price-sensitive bids and cleared schedules at each interface during peak hours (i.e., HB 7 to 22) over the past five years.¹⁴⁴ Only CTS bids are allowed at the ISO-NE interface, while CTS bids and LBMP-based bids are used at the PJM interface. The figure shows LBMP-based bids relative to the short-term forecast so the price-sensitivity of LBMP-based bids can be directly compared to that of CTS bids. The upper panel shows the market efficiency gains from CTS, measured by production cost savings.¹⁴⁵



The average amount of price-sensitive bids at the PJM interface was significantly lower than at the New England interface in each year from 2021 to 2025. On average, approximately 1,100 MW of bids (including both imports and exports) were offered in a price range of -\$10 to \$10 per MWh at the New England interface over this five-year period compared to just 200 MW at the PJM interface. Likewise, the average amount of cleared price-sensitive bids at the New England interface was more than three-times the average amount cleared at the PJM interface.

Over the past five years, interchange adjustments from the CTS process (relative to forecasted hourly schedules) occurred in approximately 85 percent of intervals at the New England interface, compared to 55 percent at the PJM interface. As a result, the estimated production cost savings from the NY/NE CTS process totaled \$52 million over this period, compared to just \$5

¹⁴⁴ CTS bids in the price range of -\$10 to \$10 per MWh are considered price-sensitive for this chart.

¹⁴⁵ For example, if the short-term price forecast in PJM is \$27, a \$5 CTS bid to import would be scheduled if the NYISO price forecast is greater than \$32. Likewise, a \$32 LBMP-based import offer would be scheduled under the same conditions. Thus, the LBMP-based offer would be shown in the figure as comparable to a \$5 CTS import bid. Section IV.C in the Appendix describes this figure in greater detail.

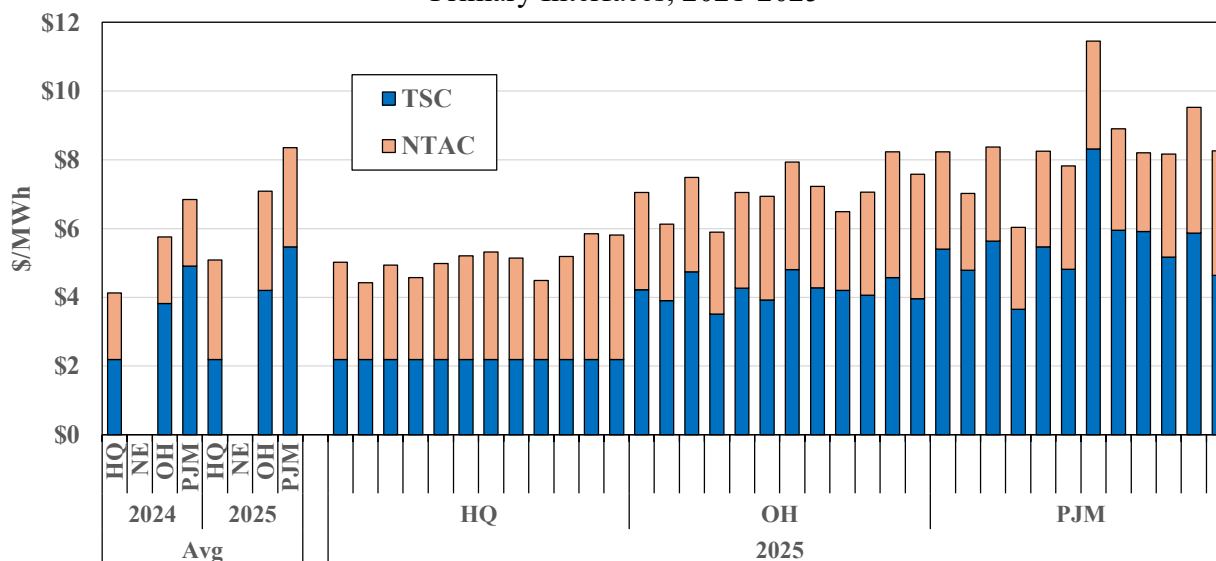
million at the primary PJM interface.¹⁴⁶ We find that production cost savings were significantly higher at the New England interface because of: (a) higher availability of price sensitive offers which allow more frequent intra-hour interchange adjustments, and (b) price forecast accuracy was better at the New England interface. Both factors are examined further in this subsection.

2. Impact of Fees Charged to CTS Transactions

The differences in performance between the two CTS processes are largely attributable to the large fees imposed at the PJM interface in contrast to the lack of substantial transmission charges or uplift charges on transactions between New York and New England.

Figure 55 shows the monthly transmission charges assessed by transmission owners on exports at the four primary interfaces with neighboring control areas in 2025. These charges include Transmission Service Charges (TSCs), assessed by the applicable transmission owners and represented by blue bars in the figure, as well as the NYPA Transmission Adjustment Charge (NTAC), represented by orange bars. There are no charges for the primary New England interface, since they are waived under the NY-NE CTS agreement. The figure also reports the annual average TSCs and NTACs at all interfaces in 2024 and 2025.

Figure 55: Transmission Service Charge Rates to Exports
Primary Interfaces, 2021-2025



Exports to PJM are typically subject to the highest transmission charges among all interfaces. In 2025, monthly charges ranged from approximately \$6 per MWh in April to nearly \$11.50 per

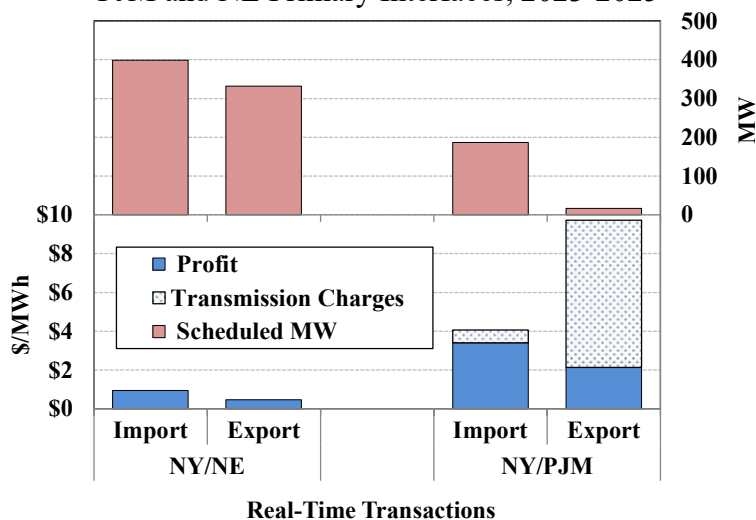
¹⁴⁶ Production cost savings are calculated relative to our estimates of scheduling that would have occurred under the previous hourly scheduling process, which we proxy based on the advisory schedules in NYISO's RTC model that are determined 30 minutes before each hour. Our evaluation tends to under-estimate the production cost savings, because the hourly schedules that we estimate would have occurred without CTS reflect some of the efficiencies that result from CTS.

MWh in July, averaging \$8.35 per MWh for the year. This increased ~23 percent from 2024, highlighting the volatility of these charges. Transmission charges were lower at other interfaces, averaging \$7.10 per MWh at the Ontario interface and \$5.10 per MWh at the HQ interface, while such charges at the New England interface are waived.

These charges are applied at the same rate as the average charges for firm network customers, even though exports under CTS are economically curtailable in real time and, therefore, don't contribute to the transmission planning needs of the NYISO or local TOs. These high charges discourage price-sensitive CTS export bidding, resulting in very few export bids submitted within the range of \$10 per MWh. In contrast, PJM charges a transmission rate on both physical imports and exports that use non-firm service at much lower levels, averaging less than \$1 per MWh.¹⁴⁷ This disparity has contributed to a higher volume of price-sensitive import bids relative to export bids. Overall, these transmission charges are a significant economic barrier to realizing the full potential benefits of the CTS process at the PJM interface.

Figure 56 examines the average profitability of scheduled real-time transactions and the average scheduled quantities at the two CTS interfaces from 2023 to 2025.¹⁴⁸ The pattern-filled bars in the lower portion of the figure show the average transmission charges over this period, which were \$7.60 per MWh for exports at the primary PJM interface and \$0.67 per MWh for imports, compared to \$0 per MWh at the New England interface.

Figure 56: Profit, Charges, and Quantities of Scheduled Real-Time External Transactions
PJM and NE Primary Interfaces, 2023-2025



At the New England interface, the average profitability of scheduled real-time transactions (including imports and exports) was ~\$0.70 per MWh over the three-year period. This relatively small margin indicates that CTS has been effective in promoting competition at the New England interface, as market participants have largely arbitrated away systematic price differences.

¹⁴⁷ Although PJM increased its Transmission Service Charge substantially to firm imports/exports to \$6.34/MWh in 2020, it kept the charge to non-firm transactions (including CTS transactions) at a low level of \$0.67/MWh. Also, PJM charges “real-time deviations” (which include imports and exports with a real-time schedule that is higher or lower than the day-ahead schedule) at a rate that averages less than \$1/MWh.

¹⁴⁸ Real-time external transactions here refer to external transactions that are only scheduled in the real-time market (excluding transactions scheduled in the day-ahead market and flow in real-time).

In contrast, at the PJM border, scheduled imports exhibited significantly higher average gross profitability of approximately \$4 per MWh, while scheduled exports showed even greater gross profitability of nearly \$10 per MWh. These outcomes reflect the fact that market participants schedule transactions only when expected inter-market price spreads are sufficient to offset the high and volatile transaction costs they incur. These costs include the transmission charges shown in the figure and additional uplift charges not depicted. Exports typically face combined uplift charges of \$2-\$2.5 per MWh across both markets, while imports incur uplift charges in PJM of around \$1 per MWh. As a result, participants schedule significantly lower transaction volumes, particularly for exports from NYISO to PJM, which are subject to the highest total fees.

Overall, these results demonstrate that imposing large transaction fees on low-margin trading activity strongly discourages transaction scheduling, which dramatically reduces trading volumes and liquidity, and may even reduce total collected revenues from fees.

We recommend eliminating or reducing these charges at the interfaces with PJM for several reasons.¹⁴⁹ First, as New York's resource mix evolves to include a greater share of intermittent renewable generation, it will become increasingly important to export surplus energy to neighboring regions during periods when in-state demand cannot absorb available supply. A more efficient CTS process would facilitate such transfers, supporting the successful integration of renewable resources. Conversely, high transaction costs will lead to more frequent curtailments of renewable generation that otherwise could be exported.

Second, it is unreasonable to apply transmission service charges to price-sensitive exports that are similar in magnitude to those assigned to network load customers. These charges are designed to recover the embedded cost of the transmission system, which is planned for the growth of network load. However, price-sensitive exports likely contribute nothing to the cost of the transmission system.

Third, we estimate that NYISO collected roughly \$2.2 million in export fees from real-time exports to PJM in 2025, while PJM collected \$4.1 million in export fees from real-time exports to NYISO. This suggests that lowering the export fee could actually lead to a higher overall collection of fees, because it would allow CTS transactions to be profitable under a wider range of conditions, encouraging greater participation and increasing total trading activity.

3. Evaluation of RTC Forecasting Error

The price forecasts produced by PJM were far less accurate than those produced by NYISO and ISO New England in recent years. Efficient CTS performance relies heavily on accurate price

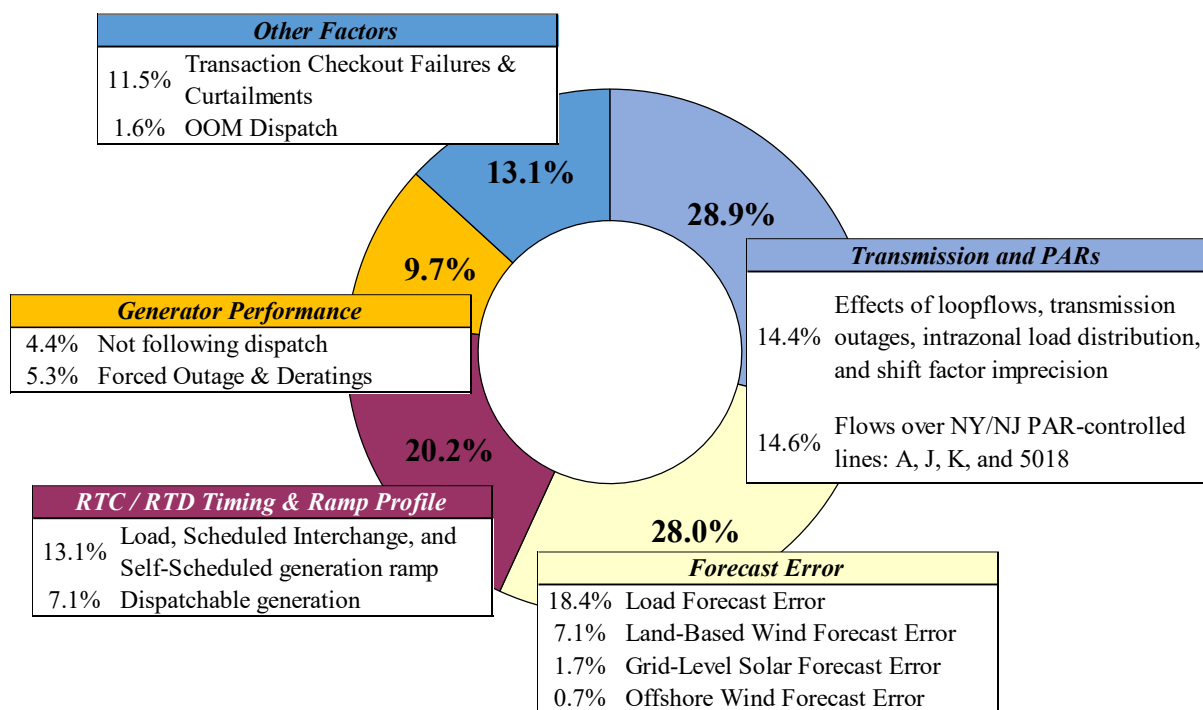
¹⁴⁹ See Recommendation 2015-9.

forecasting, so it is essential to evaluate market outcomes to identify drivers of forecast errors. This subsection summarizes our analysis of the factors contributing to NYISO’s forecast errors.

RTC schedules resources with commitment lead times ranging from 15 to 45 minutes, including external transactions and fast-start units. Inconsistencies between RTC and RTD prices may indicate that some RTC scheduling decisions and/or real-time prices are inefficient. We performed a systematic evaluation of the factors contributing to RTC/RTD price inconsistencies in 2025. This evaluation quantifies the impact of individual factors in each pricing interval, enabling a comparison of their relative significance over time. We expect this evaluation to be valuable as NYISO and stakeholders prioritize different projects to improve market performance.

Figure 57 summarizes the RTC/RTD divergence metric results for “detrimental” factors (i.e., factors that cause or contribute to differences between RTC and RTD) in 2025. Our findings show that the primary contributors to RTC price forecast errors in 2025 were largely consistent with prior years.¹⁵⁰

Figure 57: Detrimental Factors Causing Divergence Between RTC and RTD
2025



Transmission network modeling issues remained the most significant category in 2025, accounting for 29 percent of RTC/RTD divergences. This share declined modestly from the prior year, in part because the category of transaction curtailments increased significantly due to a

¹⁵⁰ See Appendix Section IV.D for details about this metric and “detrimental” and “beneficial” factors.

single event on June 24, when operator-initiated curtailments in response to reserve shortages across multiple markets caused large difference between RTC and RTD prices.

Key drivers within the transmission network category include:

- Variations in transfer capability available to NYISO-scheduled resources, primarily due to: (a) transmission outages; (b) changes in loop flows around Lake Erie and from New England; (c) inaccuracies in shift factor calculations for NYISO units, which result from the assumption that flows over PAR-controlled lines are not affected by generation re-dispatch; and (d) variations in intra-zone load distribution.
- Errors in the forecasted flows over PAR-controlled lines between the NYISO and PJM, specifically the 5018, A, and JK lines, which occur primarily because the RTC forecast: (a) lacks a module to predict loop flow variations from PJM across these lines, (b) assumes no PAR tap adjustments are made to control flows, and (c) incorrectly assumes that NYISO generation re-dispatch does not affect the flows across these lines.

The second-largest category was forecast errors in load (net of behind-the-meter solar) and production from grid-scale wind and solar resources, which accounted for 28 percent of the overall RTC/RTD divergence in 2025, up modestly from the prior year.¹⁵¹ Load forecast errors remained the dominant contributor within this category. During the hours between 10 and 18, the average RTC forecasted load was approximately 130 MW higher than average RTD load, contributing to a \$2.2 per MWh difference between average RTC LBMPs and RTD LBMPs over this period.¹⁵² Although operator adjustments to the RTC load forecast have declined modestly in recent years, they continue to be a significant driver of the load difference between RTC and RTD. The impact of forecasting errors related to intermittent wind and solar generation has been on the rise as renewable penetration has expanded in New York. Further, the accuracy of the load forecast is also increasingly affected by challenges in forecasting behind-the-meter solar production. Overall, the significance of this category is likely to increase as more intermittent generation enters the market in the coming years.

The third-largest category, which accounted for 20 percent of RTC/RTD divergence in 2025, was inconsistent assumptions related to the timing of the RTC and RTD evaluations. This includes inconsistent ramp profiles assumed for external interchange, load, self-scheduled generators, and dispatchable generators. For example, RTC assumes external transactions ramp to their schedule by the quarter-hour (i.e., at :00, :15, :30, and :45), while RTD assumes that external transactions start to ramp five minutes before the interval and reach their schedule five minutes after the interval, which is effectively five minutes later than the RTC assumption.¹⁵³

¹⁵¹ In this case, the forecast error is the difference between the forecast used by RTC and the forecast used by RTD, however, even the RTD forecast can differ from the actual real-time value.

¹⁵² See Figure A-70 in the Appendix for more details.

¹⁵³ Appendix Section IV.E shows the ramp profiles assumed by RTC and RTD for external transactions.

Addressing the sources of inconsistency between RTC and RTD is important for improving the performance of CTS with New England and PJM under present market conditions. Furthermore, New York's resource mix is transitioning away from traditional fossil-fuel generation towards: (a) intermittent renewable generation, which increases uncertainty of resource availability in real time, and (b) new types of peaking generators and energy storage resources, which must be deployed based on short-term forecasts of system conditions. A better-performing RTC will more efficiently schedule flexible resources in response to rapid changes in system conditions, which is critical for successful integration of renewable generation and for maintaining reliability in an evolving grid.

XI. COMPETITIVE PERFORMANCE OF THE MARKET

We regularly evaluate the competitive performance of the markets for energy, capacity, and other products. This section discusses our evaluation of 2025 market outcomes in three areas:

- Subsection A evaluates our screens for potential economic and physical withholding;
- Subsection B analyzes the application of market power mitigation measures in New York City and in other local areas when generation is committed for reliability; and
- Subsection C evaluates the use of the market power mitigation measures in the capacity market for New York City and the G-J Locality.

A. Potential Withholding in the Energy and Ancillary Services Market

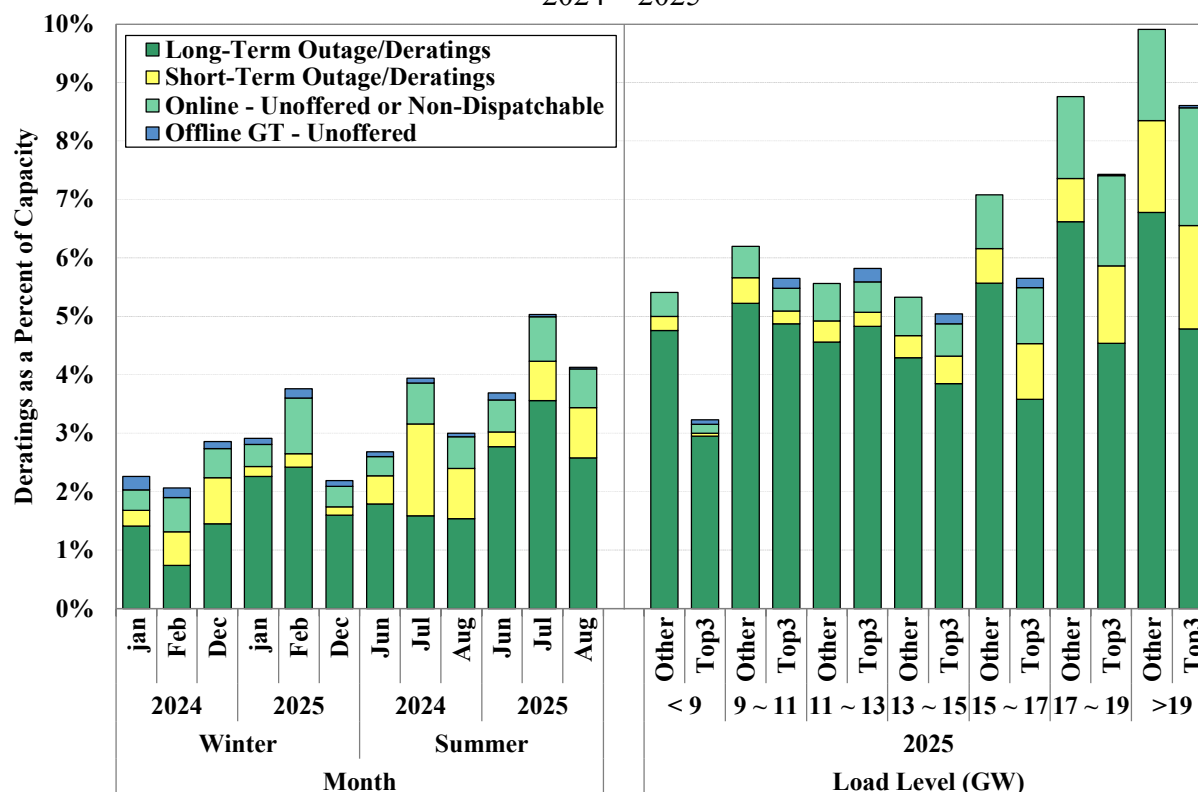
In a competitive market, suppliers have strong incentives to offer their supply at prices close to their short-run marginal production costs. Fuel costs account for the majority of short-run marginal costs for most generators, so the close correspondence of electricity prices and fuel prices is a positive indicator for the competitiveness of the NYISO's markets.

The “supply curve” for energy is relatively flat at low and moderate load levels and steeper at high load levels, which causes prices to be more sensitive to withholding and other anticompetitive conduct under high load conditions. Conditions arise when the supply cost curve becomes steep during periods of low and moderate demand, which could make such periods susceptible to potential anticompetitive conduct as well. Prices are also more sensitive to withholding in transmission-constrained areas where fewer suppliers compete to serve the load and manage the congestion into the area. Hence, our assessment focuses on potential withholding in Eastern New York because it contains the most import-constrained areas and is most susceptible to limitations on natural gas supply during peak winter periods.

In this competitive assessment, Figure 58 evaluates potential physical withholding by analyzing economic capacity that is not offered in real-time, either with or without a logged derating or outage. Deratings and outages are shown according to whether they are short-term (i.e., up to seven days) or long-term. Figure 59 evaluates potential economic withholding by estimating an “output gap” which is the amount of generation that is economic at the market clearing price but is not producing output because the supplier's offer parameters (economic or physical parameters) exceed the reference level by a given threshold. Both figures show quantities by season, load level, and the supplier's portfolio size.¹⁵⁴

¹⁵⁴ Both evaluations exclude capacity from hydro and other renewable generators. They also exclude nuclear units during maintenance outages, which cannot be scheduled when the generator is not economic. Mitigation Threshold refers to the threshold used for statewide mitigation, which is the lower of \$100 per MWh or 300 percent of the reference level. Threshold 1 is the 25 percent of the reference level, and Threshold 2 is 100 percent of the reference level. See Appendix Sections II.A and II.B for more details.

Figure 58: Unoffered Economic Capacity in Eastern New York
2024 – 2025



Overall levels of unoffered economic capacity were low but rose modestly in 2025. However, both long-and-short-term outages and deratings increased noticeably among generators in Eastern New York, averaging 6.5-percent in the largest portfolios during loads above 19 GW. Most of the capacity flagged for short-term outages and derates in peak hours was on combined cycle and conventional steam turbine units in Eastern New York. Combined cycle generators tend to be economic in all hours during peak periods so even short-duration outages can create a significant amount of flagged capacity. This summer witnessed more long-term outages as well, especially among combined cycle units. Supply chain issues continued to cause delays in return to service of combined cycles on lengthy outages.

Economic generation rendered unavailable due to planned outages has been an issue in the peak winter months, primarily when dual fuel generators take multi-month planned outages. Planned outage extensions also showed up during the June 2025 Heat Wave. One resource was identified as not offering due to a scheduled outage that was extended multiple times to a total of more than eight months. Our investigation determined that the generator was initially under a planned outage that changed to a forced outage due to issues that arose during the work.¹⁵⁵

¹⁵⁵ See Section IX.F for further discussion of this.

The amount of unoffered economic capacity increased when load levels in the east rose above 17 GW among all suppliers. Roughly 10 percent of economic supply was unavailable from the smaller portfolios during the highest load hours (greater than 19 GW) and nearly 9 percent of economic supply was unavailable from the Top 3 suppliers.

In the highest load hours 23-percent of the unoffered economic capacity of the Top 3 suppliers was unoffered or non-dispatchable capacity from online resources. Notable reasons for online resources not offering their upper output ranges included:

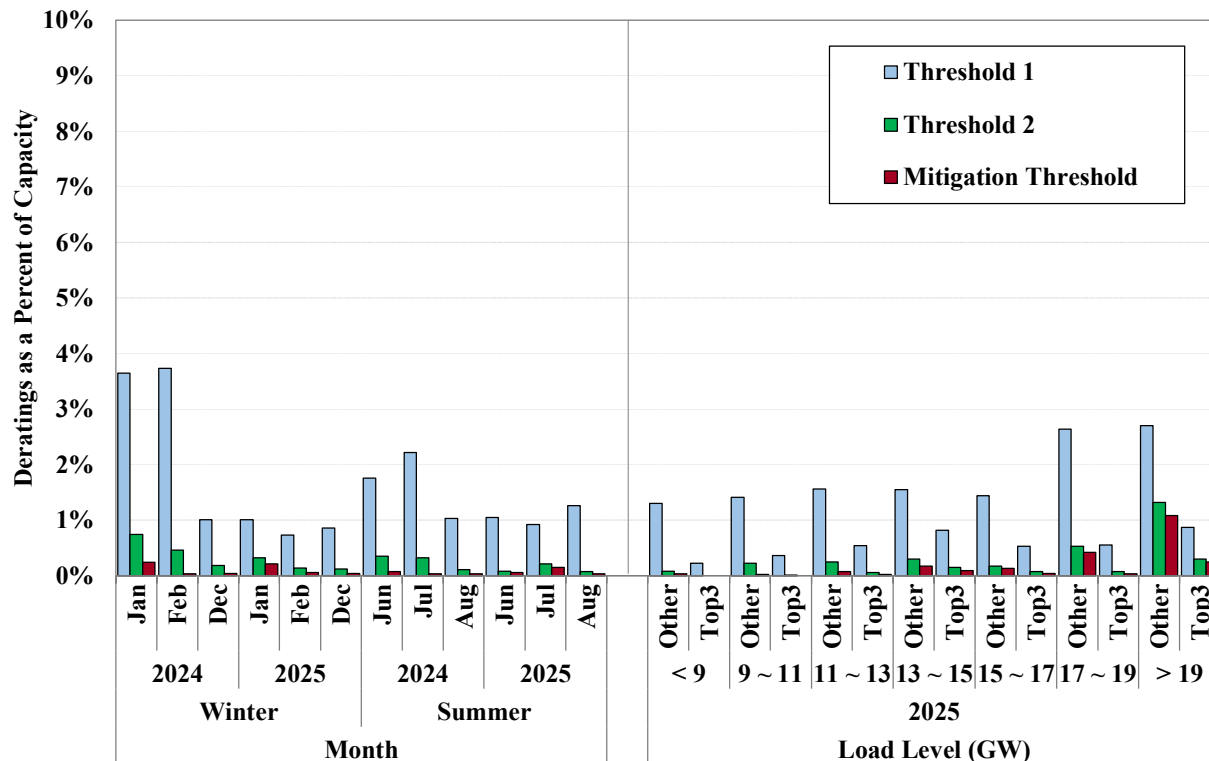
- *Inflexibility of duct-firing capacity* – Some combined cycles offer inflexibly in real-time to manage physical operating constraints on the duct-fired portion of the output range. Currently, the market models treat duct burners as capable of responding to AGC and eligible for 10-minute reserve products; however, duct-firing capacity is generally not flexible enough to provide either of these services.¹⁵⁶ NYISO's *Improve Duct-Firing Modeling* project will implement market changes in 2026 to address some of these issues.
- *Ambient conditions* – Capacity ratings for temperature-dependent resources change daily and hourly as ambient conditions vary. Relevant factors include air temperature, relative humidity, inlet water temperatures, and tidal levels. In 2025, NYISO was using unit-specific output factor curves that adjust for daily ambient air temperature changes, but NYISO will not use comparable information for humidity until summer 2026. Lack of information on inlet water temperatures and tidal levels, which affect the capability of generators with certain inlet cooling systems and water cooled condensers, make it difficult to quantify the precise amount of capacity unavailable due to ambient conditions.¹⁵⁷
- *Inability to follow basepoint* – Many older, smaller combined cycle generators upstate never installed the necessary equipment to follow 5-minute dispatch instructions automatically. Consequently, these units usually do not offer flexibly in real-time when committed. While some generators might make the necessary capital improvements to follow dispatch automatically, it is not cost-effective for others.
- *Cogeneration steam demand* – Several cogeneration resources in eastern New York sell steam under bilateral contracts and sell excess electric generation to the grid. Typically, host steam load takes priority over selling electricity to the grid and, therefore, electric capability from these resources is often derated when host steam demand rises.

The amount of output gap in Eastern New York remained very low in 2025, averaging 0.06 percent of total capacity at the statewide mitigation threshold and 1.1 percent at the lowest threshold evaluated (i.e., 25 percent above the Reference Level).

¹⁵⁶ Analysis of the affected capacity is provided in Section VII.C.

¹⁵⁷ NYISO efforts to address many of these ambient conditions are discussed further in Section IX.E.

Figure 59: Output Gap in Eastern New York
2024 – 2025



Much of the output gap in 2025 was attributable to units that typically have bid-based reference levels that are lower than the true marginal cost of generation. Thus, a significant portion of the capacity identified as output gap is due to low reference levels rather than inappropriately high energy offers.¹⁵⁸ To limit the potential for excessive mitigation in areas with strict mitigation measures (i.e., New York City), most NYC generators have cost-based Reference Levels.

Several instances of physical withholding were identified during the peak summer periods in 2025. A few isolated instances of withholding during the June Heat Wave were identified, investigated, and mitigated. All of these instances involved fossil fuel generators that were unoffered during the peak hours in real-time and not on a scheduled or legitimate forced outage. Other instances of withholding that involved market power were investigated regarding the Start-Up Notification Times of older vintage steam turbines in the state that prevented economic commitments in day-ahead and increased system prices. Although cases causing price impact that exceed state-wide mitigation or in-city thresholds are rare, instances have been identified in 2022, 2024, and 2025 that were referred to the NYISO with the recommendation of imposing

¹⁵⁸ NYISO Market Services Tariff Section 23 outlines three types of reference levels that a generator may have. The first type is the bid-based reference level, which is calculated as the average of accepted economic bids during unconstrained intervals over the past 90 days, adjusted for changes in day-ahead gas prices. This approach tends to under-state marginal costs for units that face fluctuating intraday fuel prices.

withholding sanctions. However, no sanctions for physical withholding have been imposed for these cases because of the slow progress of the penalty review and dispute resolution processes.

It is generally a positive indicator that the unoffered economic capacity and the output gap were comparable for top suppliers and other suppliers during high-load conditions when the market is most vulnerable to the exercise of market power. Overall, the patterns of unoffered capacity and output gap were consistent with competitive expectations and, outside of a few isolated cases, did not raise significant concern regarding the exercise of market power.

B. Automated Mitigation in the Energy Market

In New York City and other transmission-constrained areas, individual suppliers are sometimes needed to relieve congestion and may benefit from withholding supply (i.e., may have local market power). Likewise, when an individual supplier's units must be committed to maintain reliability, the supplier may benefit from raising its offer prices above competitive levels. In these cases, the market power mitigation measures effectively limit the ability of such suppliers to exercise market power. This section evaluates the use of three key mitigation measures:

- **Automated Mitigation Procedure (AMP) in New York City** – This is used in the day-ahead and real-time markets to mitigate offer prices of generators that are substantially above their reference levels (i.e., estimated marginal costs) when their offers would significantly raise the energy prices in transmission-constrained areas.¹⁵⁹
- **Reliability Mitigation in New York City** – When a generator is committed for local reliability, the start-up cost and minimum generation cost offers of the generator may be mitigated to its reference levels. A \$0 conduct threshold is used in the day-ahead market and the AMP conduct threshold is used in the real-time market.
- **Reliability Mitigation in Other Areas** – When a generator is committed for reliability and the generator is pivotal, the start-up cost and minimum generation cost offers of the generator may be mitigated to its reference levels. A conduct threshold of the higher of \$10 per MWh or 10 percent of the reference level is used.

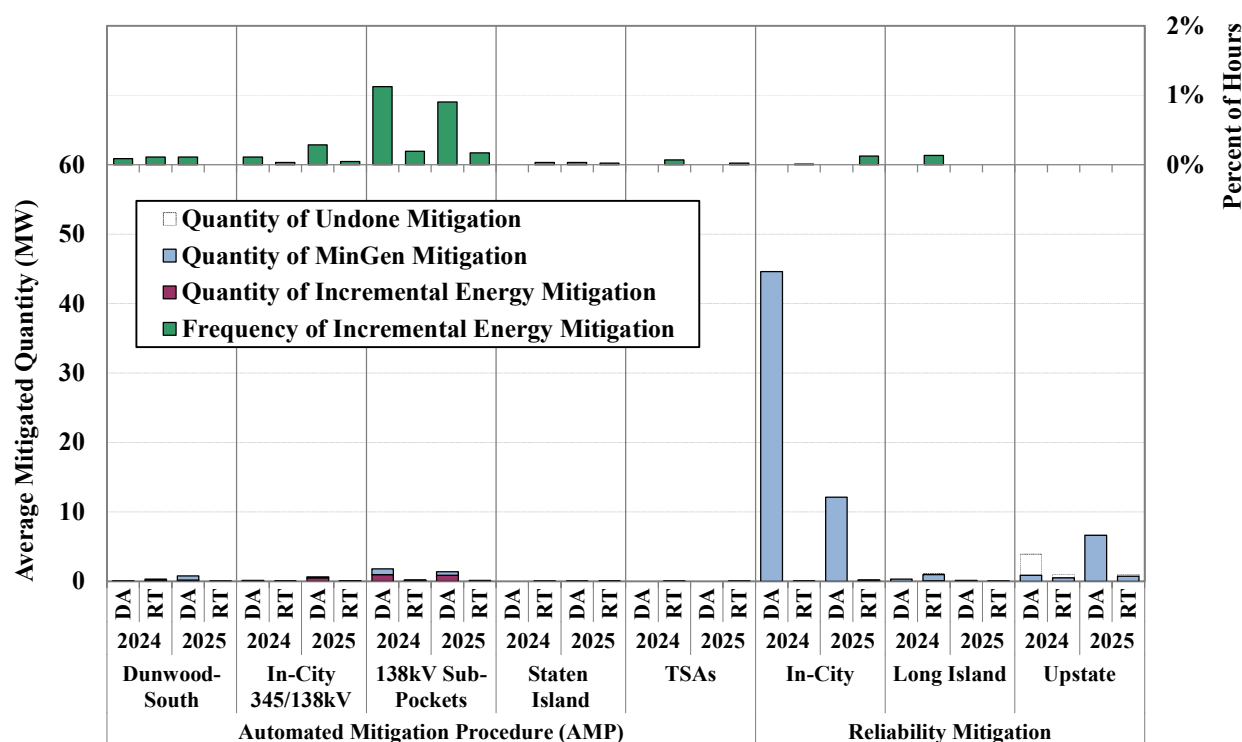
Figure 60 summarizes the market power mitigation (i.e., offer capping) that was imposed in the day-ahead and real-time markets in 2024 and in 2025. The figure shows that most mitigation occurs in the day-ahead market where most supply is scheduled. Reliability mitigation accounted for roughly 86 percent of all mitigation in 2024, nearly all of which occurred in the day-ahead market. In New York City, the amount of capacity committed for reliability and the frequency of mitigation decreased from 2024 to 2025 due to higher energy prices which made in-city steam units more economic than in prior years. The reliability mitigation is critical for

¹⁵⁹ The conduct and impact thresholds used by AMP are determined by the formula provided in the NYISO Market Services Tariff, Section 23.3.1.2.2.1.

ensuring that the market performs competitively because units that are needed for local reliability usually have market power.

AMP mitigation accounted for about 14 percent of total mitigation, up from 5 percent in 2024. This was not due to increased AMP mitigation, specifically; rather, the significant reduction in overall reliability mitigation was primarily responsible for AMP mitigation's net increase in overall share of mitigation in 2025. AMP mitigation only applies when there is an active transmission constraint.

Figure 60: Summary of Day-Ahead and Real-Time Mitigation
2024 - 2025



When natural gas prices become volatile, generators frequently use the Fuel Cost Adjustment (FCA) functionality to adjust their reference levels in the day-ahead and real-time markets. This has increased in frequency since the Indian Point units retired and eastern New York has become more reliant on gas-only and dual-fuel units. The FCA functionality is important because it allows a generator to reflect fuel cost variations closer to when the market clears. This helps the generator to avoid being mitigated and scheduled uneconomically.

C. Competition in the Capacity Market

The capacity market is designed to ensure that sufficient capacity is available to meet planning reserve margins by providing long-term signals for efficient investment in new and existing

generation, transmission, and demand response. NYISO has market power mitigation measures that are designed to ensure that the markets perform competitively.

Supply-side market power mitigation measures prevent or deter suppliers with market power from inflating prices above competitive levels by withholding economic capacity in these areas. The supply-side mitigation measures work by imposing an offer cap on pivotal suppliers in the spot auction and by imposing penalties on capacity otherwise withheld.¹⁶⁰

Buyer-side market power mitigation (BSM) measures are used in New York City and the G-J Locality to prevent entities from artificially depressing prices below competitive levels by subsidizing the entry of uneconomic capacity. The BSM measures work by imposing an offer floor on mitigated capacity, thereby preventing such capacity from depressing the clearing price. Beginning with NYISO's interconnection Class Year 2021, projects considered to contribute to New York state policy ("Excluded Facilities") are not subject to BSM evaluation. Projects that are not Excluded Facilities are exempted from an offer floor if they pass one of four evaluations.¹⁶¹

1. Application of the Supply-Side Mitigation Measures

Given the sensitivity of prices in the Mitigated Capacity Zones, the supply-side market power mitigation measures are important for ensuring that capacity prices in these zones are set at competitive levels. From time to time, the NYISO evaluates whether a proposal to remove capacity from a Mitigated Capacity Zone has a legitimate economic justification. We have found that the NYISO's evaluations in recent years have been in accordance with the tariff.

2. Application of the Buyer-Side Mitigation Measures

Class Year 2023, which concluded in January 2025, included intermittent renewables, energy storage, and an HVDC transmission project participating in the state's 'Tier 4' clean energy procurement in the mitigated capacity zones. All of the CY23 projects in the mitigated zones were considered to be Excluded Facilities and therefore not subject to BSM evaluation.

¹⁶⁰ See NYISO MST, Sections 23.4.5.2 to 23.4.5.6.

¹⁶¹ A new entrant can receive a BSM exemption under the provisions of: (a) Competitive Entry Exemption, (b) Part A Test Exemption, (c) Part B Test Exemption, and (d) Self-Supply Exemption. See MST Section 23.4.5.7.

XII. DEMAND RESPONSE PROGRAMS

Demand response participation enhances system reliability, reduces production costs, decreases price volatility, and reduces supplier market power. Even modest reductions in consumption by end users during high-price periods can significantly reduce the costs of committing and dispatching generation to satisfy system needs. These benefits underscore the value of wholesale markets that provide transparent economic signals for price-responsive loads and market processes that facilitate demand response.

NYISO facilitates demand response through several programs. The Special Case Resource (SCR) program and the Targeted Demand Response Program (TDRP) allow reliability demand response resources to be activated when NYISO or the local Transmission Owner forecasts a shortage. NYISO launched the DER and Aggregation Participation Model on April 16, 2024, which enables large individual consumers and consumer aggregations to participate more actively in the day-ahead and real-time markets. The Demand-Side Ancillary Services Program (DSASP) enabled economic demand response resources to participate in the ancillary services markets, although this program sunset on October 31, 2025, and many former DSASP resources transitioned to the Distributed Energy Resource (DER) and Aggregation Participation Model.

This section:

- Summarizes participation and deployment of demand response resources in 2025 (in subsection A),
- Evaluates the performance of the new DER model (in subsection B),
- Analyzes the extent to which demand response resources may respond to incentives to curtail load outside of NYISO and utility deployments (in subsection C), and
- Critiques the interconnection process for new large load facilities (in subsection D).

A. Participation in NYISO Demand Response Programs

Until 2025, SCRs accounted for nearly all of the capacity enrolled in the reliability-based programs, and the majority of revenues paid to them was for participating in the capacity market. Although the DER model was launched in April 2024, loads participating in the SCR program and/or DSASP program were reluctant to transition until NYISO ended the DSASP program in October 2025. Figure 61 summarizes enrollment of demand response resources in the SCR, DSASP, and DER programs in each month of 2025 through early-2026. Capacity values are listed in terms of ICAP sold in the NYISO installed capacity market.

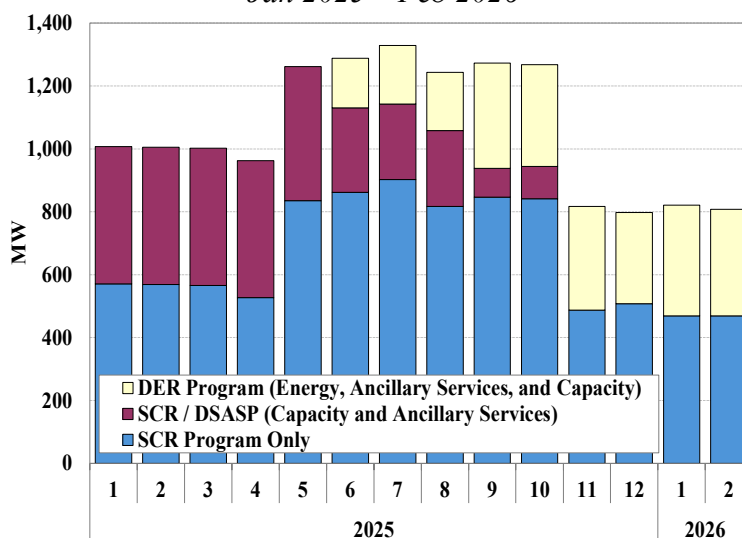
The figure shows a marked transition from the winter of 2024/25 to the winter of 2025/26. In January 2025, approximately 570 MW participated in the capacity market only (i.e., SCR Program only), while around 430 MW participated in the capacity and ancillary services

markets. By January 2026, SCR Program only participation had fallen to 470 MW, while most (80 percent of) loads that had previously participated in SCR and DSASP had moved to the DER program. Overall registration fell from 1,000 MW in January 2025 to 820 MW in January 2026 as some loads chose to stop or reduce participation in DR programs.

In the Summer 2025 Capability Period, registered capacity was higher (reflecting higher levels of participation and qualified capacity in the summer than in the winter). The summer results show DER participation began in June 2025 and by October 2025, most resources participating in the SCR and DSASP programs had already transitioned to the DER program. Ultimately, the retirement of the DSASP program drove some of the subsequent DER participation, but the DER program has also allowed these facilities to be scheduled through the NYISO's day-ahead and real-time markets.

Figure 61: Demand Response Registration

Jan 2025 – Feb 2026



Emergency Demand Response Deployments

NYISO activated SCR/EDRP resources more frequently in 2025 than in recent years:

- TDRPs were deployed in Zone J for local needs (J3, J4, and J8) on eleven days.
- SCR/EDRPs were deployed in all zones for NYCA-wide capacity needs on nine days.
- SCR/EDRPs were deployed in zones G through K on one day (July 17).
- SCR/EDRPs were deployed in zones J and K on one day (July 25)

Scarcity pricing was triggered on eight of the eleven days when SCR/EDRP activations occurred. Activations that coincide with no reserve shortages often arise due to the inefficiencies of the deployment model for SCRs. Under NYISO procedures, SCRs are called for a block of hours and all SCRs within the activation zones must be deployed. Since the operators cannot moderate the quantity and duration of SCR activations, it leads to some periods of low prices during SCR deployments. The DER model is designed to enhance flexibility in demand participation and alleviate some of these limitations of the SCR program.

Energy Payments to Demand Response Resources

When loads participating in NYISO demand response programs curtail consumption, they benefit in two ways. First, they avoid the cost of purchasing power at the applicable wholesale or retail rate. Second, under specific circumstances, they are also paid for the curtailed consumption as if they produced energy at the LBMP. This double-benefit results in significant uplift costs to other NYISO customers because it violates a basic principle of market exchange that payments to sellers must be funded by receipts from buyers. For example, if a load curtails 1 MW at the wholesale LBMP rate of \$100/MWh and a generator reduces production by 1 MW at \$100/MWh, the load reduces payments by \$100 and payments to the generator are reduced by the same amount. However, if the load is also *paid* \$100 after saving \$100, the \$100 payment is not balanced by revenue from other market participants. Therefore, the \$100 payment must be uplifted to NYISO customers.

NYISO's tariff makes energy payments to demand response resources for curtailments under specific circumstances:

- SCR & EDRP resources are paid the real-time zonal LBMP with a bid cost guarantee ("BCG") for a strike price of up to \$500/MWh during emergency demand response deployments events (which occurred on eleven days in 2025).
- DERs are paid for curtailments at the real-time nodal LBMP when it is at or above the monthly Net Benefits Threshold ("NBT"). NYISO developed the NBT to satisfy the requirements of Commission Order 745 to pay the LBMP to demand response resources for curtailing when it satisfies the "net-benefits test," which the Commission described as a "cost-effectiveness condition" that would "...ensure that the overall benefit of the reduced LMP that results from dispatching demand response resources exceeds the cost of dispatching and paying LMP to those resources."¹⁶²

Order 745 required ISOs to pay demand response resources for curtailing despite the objections of commenters who highlighted this was an inefficient "double-payment."¹⁶³ Order 745 stipulated ISOs implement a net-benefits test considering only short-term energy price elasticity and ignoring effects on prices in other hours, capacity prices, and investment. Hence, the net-benefits test does not ensure that ordinary electricity consumers actually benefit from this policy.

Figure 62 summarizes the energy settlements and BCG/BPCG payments for demand response resources participating in the SCR, EDRP, and DER programs in each month of 2025. It also displays the total demand reduction energy "supplied" by these resources in each month.

Energy settlements and BCG to demand response in 2025 totaled roughly \$33.8 million. SCR/EDRP resources accounted for 73 percent (\$24.8 million), while the remaining \$9 million

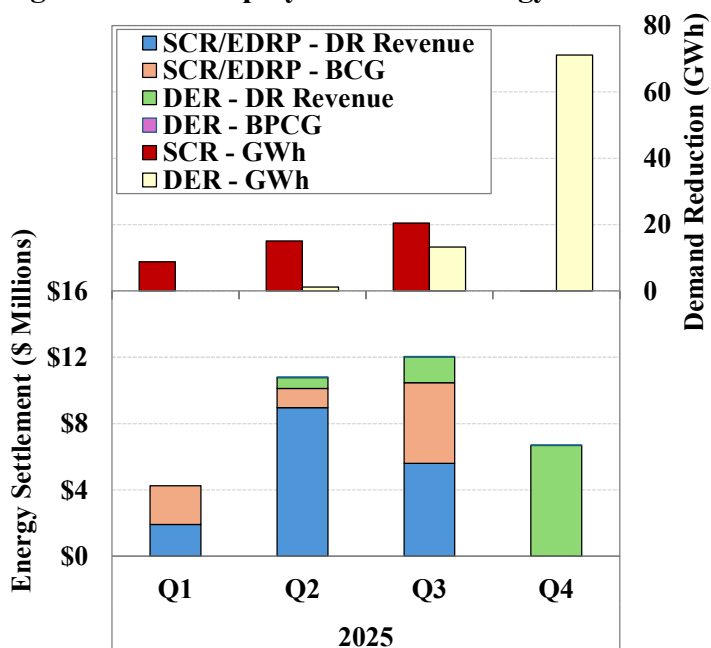
¹⁶² See Commission Order 745, P. 3.

¹⁶³ See Commission Order 745, dissent of Commissioner Moeller, page 5, citing Professor William W. Hogan.

was paid to DERs, which accounted for the majority (64 percent) of demand reduction quantities. Consequently, the weighted-average payments to DERs was \$105/MWh compared to \$560/MWh for SCR/EDRP. The average payment (per MWh) to DERs was significantly lower than the average to SCR and EDRP resources because:

- Most DERs are cryptocurrency mining loads, which have a much lower value of consumption than most other curtailable loads.
- SCR deployments are relatively infrequent because they only happen during peak operating conditions.

Figure 62: DR Deployments and Energy Settlements



B. Uneconomic Load and Phantom Curtailment Created by Order 745

NYISO implemented its DER and Aggregation participation model on April 16, 2024, allowing individual large consumers and aggregations of small consumers to participate more directly in the market. Registration in the DER program was slow until NYISO announced it would end the DSASP program. The first DER began participating in June 2025, and four others became active by the end of 2025. Registration has likely been slow due to several factors, with two standing out:

- All DER are considered “dispatch-only” and, therefore, unable to bid minimum notification time and commitment duration limits as most generators normally do.
- Managing compliance with energy and reserve offer obligations is difficult due to the requirements of the Economic Customer Baseline Load (“ECBL”) calculation.¹⁶⁴

Table 12 summarizes registration in the initial phase of the DER program. All resources participating in the DER program as of early

Table 12: DER Program Participation
July 2025 – March 2026

	2025		2026
	Q3	Q4	Q1
No. DER Resources	2	5	6
ICAP MW	235.3	314.2	334.8
Percent Demand Response	100%	100%	100%
Cryptocurrency Load Ratio	100%	100%	87%

¹⁶⁴ See NYISO Aggregation Manual Section 7.5.

2026 are demand response resources, and most are cryptocurrency mining loads. These loads are better able to manage the difficulties of being dispatch-only in the program due to their fast responsiveness, low value of consumption, and ability to cycle frequently.

FERC Order 745 requires that demand response resources participating in wholesale electric markets be paid LBMP for curtailment as long as the LBMP is above the “net-benefits test” threshold.¹⁶⁵ This created a double-benefit for DR for curtailing under specific conditions because they both avoid the cost of consuming energy and are paid the LBMP for curtailing. This double-benefit violates the principle that payments to sellers be funded by receipts from buyers. Hence, the LBMP payment must be uplifted to NYISO customers. In addition, we identify two inefficiencies that result from the requirements of Order 745: uneconomic load and phantom curtailment, which are discussed below.

Uneconomic Load

Figure 63 summarizes at the daily level the load reductions, uneconomic consumption, and energy “supplied” from loads participating in the DER program from November 2025 through January 2026. Uneconomic load includes consumption when the load’s estimated value of consumption is lower than the applicable retail or wholesale rate. The top portion of the chart displays the monthly NBT, the estimated value of consumption (i.e., opportunity cost) of cryptocurrency mining, and the capacity-weighted median incremental energy offer from participating cryptocurrency mining loads.¹⁶⁶

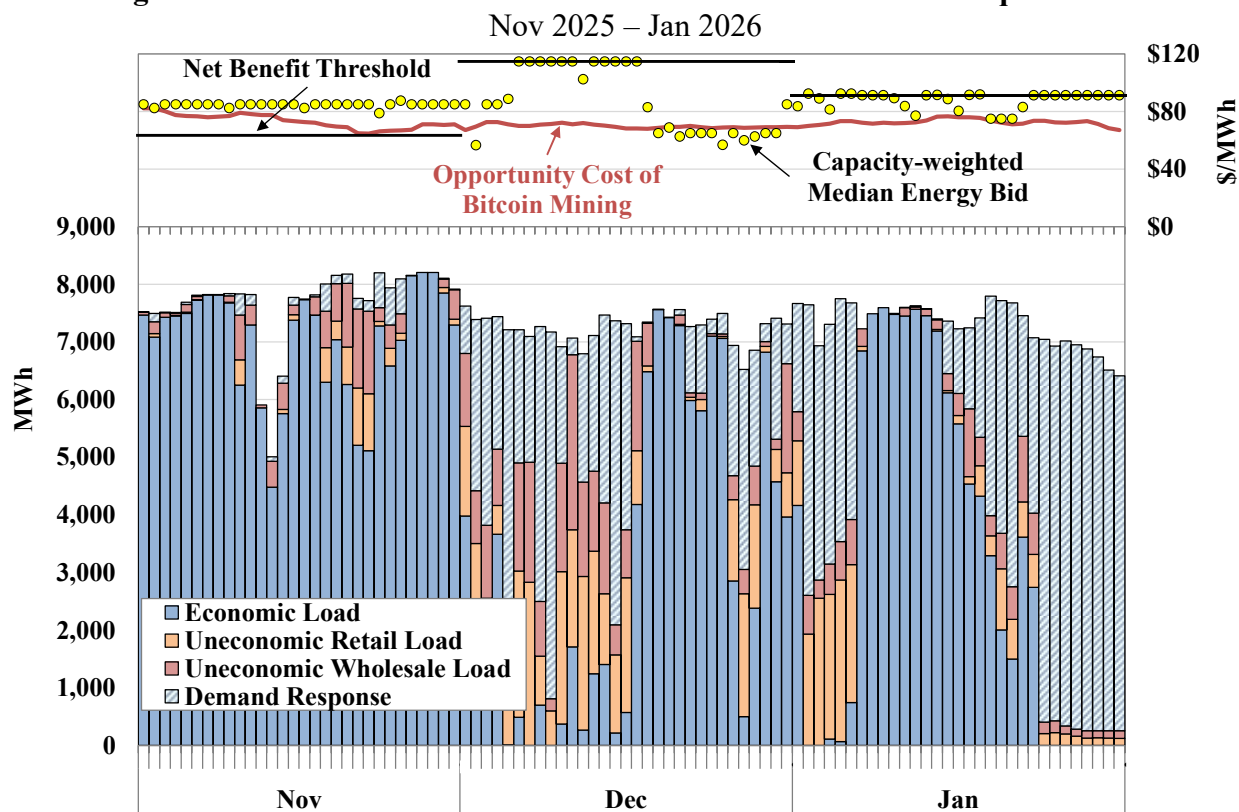
Figure 63 highlights some of the consequences of the Order 745 mandate to double-compensate demand response for curtailments. The figure shows the NBT exceeded the opportunity cost of cryptocurrency mining during most of the winter. By December 1, the estimated opportunity cost of mining Bitcoin had fallen to roughly \$70/MWh, while the NBT rose to \$114.75/MWh due primarily to higher expected winter gas prices. In an efficient market, cryptocurrency miners would curtail load at around \$70/MWh, but on many days in December and January, some loads raised their bid prices to the NBT level. Cryptocurrency miners have an incentive to consume energy at a loss under some conditions because doing so increases their Economic Consumer

¹⁶⁵ The threshold is meant to correspond to the point on the supply curve where a small reduction in load produces an LBMP decrease large enough that the net savings to load outweigh the additional payments to DR. *Technical Bulletin 245* outlines the procedure that the ISO uses to calculate monthly NBT. NYISO publishes the monthly NBT values on its website at <https://www.nyiso.com/documents/20142/1395792/Monthly-Net-Benefit-Offer-Floor-Prices.pdf/8df1d4dc-c6b6-77ae-02ce-c184bd189ee6>.

¹⁶⁶ The opportunity cost of bitcoin mining was calculated using the S19k XP antminer as the default miner. The daily value is calculated as equal to [Bitcoin Network Hash Price] * [Hash per miner] / [Power Requirement] * 1,000,000 / 24 * Efficiency. The assumptions for the default miner are: Power Requirement: 3,010 W; Hash per Miner: 0.14 PH, Efficiency: 0.95. Bitcoin Network Hash Price sourced from: <https://www.coinglass.com/pro/i/hashprice-index>.

Baseline Load (ECBL) values in future intervals. The quantity of uneconomic consumption from these loads depended on how frequently LBMPs were lower than the NBT but higher than the loads' consumption value. We estimate that 81 GWh of cryptocurrency load consumption occurred uneconomically with nearly half at facilities taking service under wholesale rates.

Figure 63: Inefficient Curtailment and Loads from Order 745 Requirements



Phantom Curtailment

DER curtailment is measured against the ECBL.¹⁶⁷ ECBL is calculated at the five-minute level and determined based on metered load and demand reductions during recent days.¹⁶⁸ Actual Demand Reductions, as defined in the NYISO tariffs, are the delta between the ECBL and the metered load in any interval where the unit receives a non-zero basepoint.¹⁶⁹ Consequently, when the ECBL is greater than the metered load immediately preceding the dispatch, a portion of

¹⁶⁷ ECBL is meant to identify the load level that a DR would be expected to be consuming at in each interval if not dispatched economically. It is calculated at the interval level based on the DR loads during the same interval in recent periods and credits the resources for demand response that takes place when LBMPs exceed the NBT. Demand response that occurs when LBMPs are less than NBT are not added back to the ECBL, creating a disincentive for DR to curtail at lower prices.

¹⁶⁸ Only demand reductions that occur when LBMP exceeds NBT are eligible to be included in ECBL.

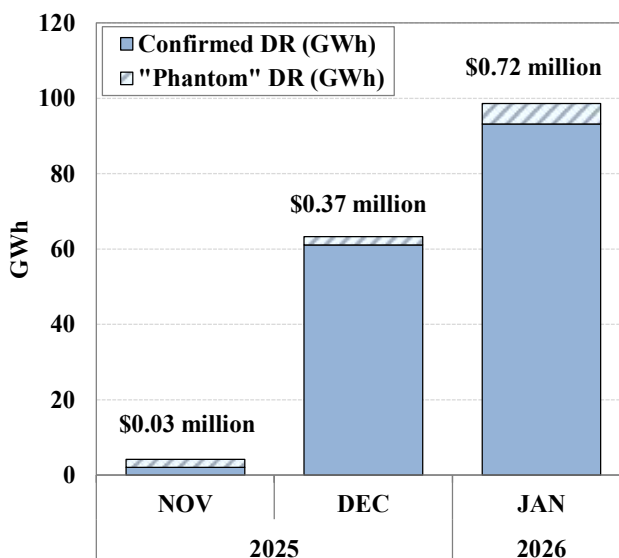
¹⁶⁹ OATT 24.2.

the reported demand reductions are “phantom” reductions derived by formula rather than actual reductions in load.¹⁷⁰

Figure 64 shows our analysis of DER curtailment from November 2025 through January 2026 which found that 6 percent of reported demand response energy was “phantom” demand response, i.e., demand response where metered load was less than ECBL but no actual demand reductions occurred. We estimate these phantom demand reductions received roughly \$1.1 million in demand response settlements over that period.

Our review of DER program performance in 2025 and early-2026 has reinforced the need to eliminate LBMP payments for DR curtailment. Eligibility for energy payments was required by FERC Order 745. This requirement has predictably led to inefficient over-consumption, phantom curtailment, and substantial uplift costs to other consumers. Eliminating these problems will require a change in Commission policy.

Figure 64: Analysis of “Phantom” Load
November 2025 – January 2026



C. Peak Load Management by End-Users

The NYISO markets provide indirect incentives for many end-use customers in New York to manage their contributions to system peak demand. This subsection discusses incentives for end users to manage their peak impacts outside of formal NYISO demand response programs.

NYISO Market Incentives for Peak Load Management by End Users

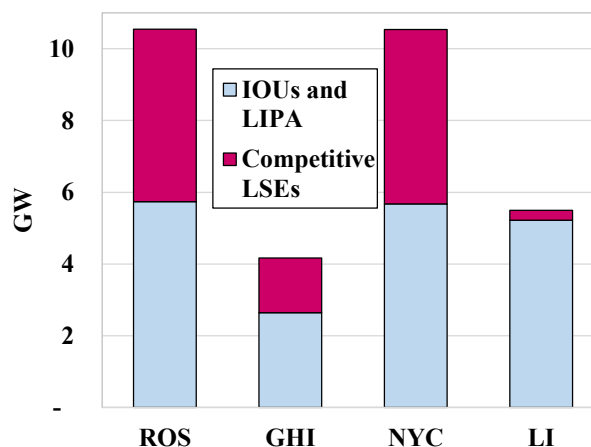
Load-serving entities (LSEs) act as buyers in NYISO’s capacity market and are required to purchase Unforced Capacity (UCAP) on behalf of their customers. Each LSE is assigned a share of the systemwide and locality UCAP Requirements based on its customers’ share of demand during the prior year’s system peak load hour.¹⁷¹ The LSE is then responsible for purchasing UCAP to cover that obligation through NYISO-administered auctions or bilateral transactions.

¹⁷⁰ Suppose a DR in interval n has 100 MW load, 0 MW basepoint, and 110 MW ECBL. In interval n+1 the basepoint increases to 10 MW, but the metered load and ECBL remain unchanged at 100 MW and 110 MW, respectively. By definition, the demand reductions in interval n+1 are 10 MW and the resource (assuming LBMP >= NBT) will settle on 10 MW of demand reduction despite no actual load shed.

¹⁷¹ See section 5.11 of NYISO’s Market Services Tariff (MST).

Figure 65 shows the amounts of load in each capacity region that are served by competitive LSEs versus the local utility. Loads can reduce their UCAP obligation by managing their demand during system peaks. NYISO’s tariff does not govern how LSEs pass the cost of capacity they purchase to individual end-use customers.¹⁷² Many customers (particularly in the residential sector) are charged flat or time-of-day volumetric prices that do not provide strong incentives for peak reduction. However, many customers have exposure to capacity prices and incentives for peak management:

Figure 65: Coincident Demand in Peak Load Hour in 2025 by LSE Category



ICAP Pass-Through by LSEs: Retail access in New York allows competitive LSEs to offer a variety of pricing terms tailored to their customers’ needs. In 2025, 40 percent of peak demand was served by competitive LSEs (including 9 percent by the New York Power Authority). The majority of commercial and industrial load in New York is served by competitive LSEs, especially for larger customer classes, while the investor-owned utilities and Long Island Power Authority act as the LSE for most residential load. Competitive LSEs commonly offer pricing arrangements that pass capacity savings to customers that reduce their demand during system peaks and provide related services such as notification of potential system peak events.¹⁷³

Distributed Storage Incentives: NYISO markets provide incentives for distributed (behind-the-meter and distribution level) storage that reduces system net peak load. In 2024, the New York PSC authorized additional incentive funding for 1.7 GW of distributed commercial and residential storage by 2030.¹⁷⁴ As of April 2026, 229 MW of distributed storage has been deployed through NYSERDA programs, including 49 MW deployed in New York City in 2025.

¹⁷² Retail pricing falls under the jurisdiction of the New York State Public Service Commission and the state’s retail access rules, while NYISO sets requirements applicable to LSEs.

¹⁷³ For example, one competitive LSE that serves load in New York describes the following service: “Since annual capacity and transmission rates are based on consumption during peak-setting events, many businesses actively curtail (reduce) their demand at those times to reduce costs. At Constellation, our Peak Load Management program notifies end users on a day-ahead and/or day-of basis of a potential capacity peak setting event. Consumers who choose to reduce their consumption may lower their capacity obligation for the following year”. Constellation Energy4Business blog, “Guide to Understanding Capacity and Transmission Costs”, retrieved February 23, 2026 ([link](#)).

¹⁷⁴ See NYSERDA, “2024 Energy Storage Order Overview”, available [here](#).

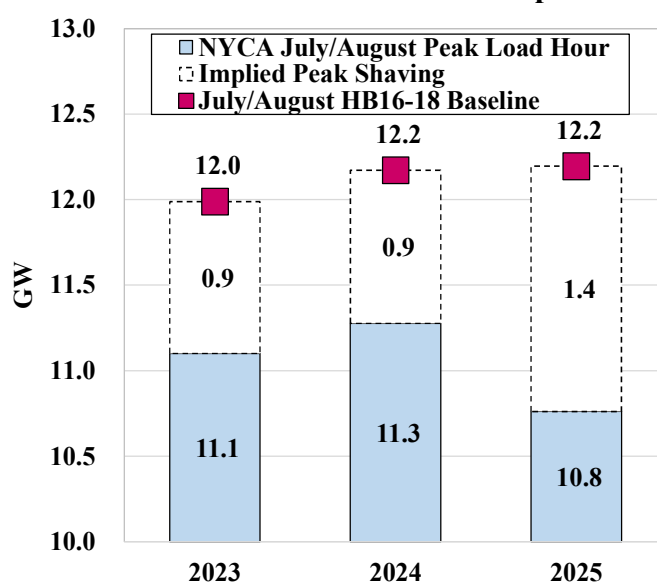
Over 700 MW of additional distributed storage projects have been approved for incentive funding since September 2024, of which 95 percent is in New York City.¹⁷⁵

Distributed storage projects that operate during system peak hours can earn payments based on NYISO capacity prices through New York’s Value Stack tariff or reduce capacity charges for load behind the meter. NYSERDA’s release of distributed storage incentives in 2025 resulted in vastly more subscriptions in New York City than in other areas of the state (even at much lower incentive rates), which is likely due in part to higher capacity prices in New York City.¹⁷⁶

Evidence of Peak Load Management by End Users

Consumption patterns under peak conditions show that electricity consumers respond rationally to incentives. Figure 66 compares the total demand of competitive LSEs during the NYCA coincident peak hour to their ‘baseline’ demand on July and August afternoon peaks.¹⁷⁷ Total demand of competitive LSEs was below baseline levels during the coincident peak by 0.9 GW in 2023 and 2024 and 1.4 GW in 2025. NYISO did not activate SCRs during the peak hours of 2023 and 2024 and no loads were enrolled in the DER participation model in those years. NYISO deployed both SCRs and DERs in the peak hour of 2025. Hence, Figure 66 suggests that large amounts of peak load management occur outside of NYISO demand response programs, with larger effects when demand response programs are activated.

Figure 66: Baseline and NYCA Coincident Peak Demand of Competitive



End users manage peak demand to reduce NYISO capacity costs and earn payments from programs operated by the investor-owned utilities, such as Commercial System Relief Programs (“CSRPs”). Figure 67 shows curtailment by loads in NYISO’s SCR program relative to their Average Coincident Load (ACL) values in the NYCA coincident peak hour of 2024, when the

¹⁷⁵ See “Retail and Bulk Energy Storage Incentive Programs Reported by NYSERDA: Beginning 2019” at data.ny.gov, accessed February 2026, available [here](#).

¹⁷⁶ See NYSERDA Retail Incentive Dashboard, available [here](#).

¹⁷⁷ We estimate each LSE’s baseline as the average of that LSE’s top ten highest demand hours during hours beginning 16-18 in July and August. This definition is intended to align with the timing of NYCA coincident peak demand hours in recent years. We calculated a baseline for total load served by each LSE in each transmission district and NYISO capacity region.

SCR program was not activated. SCRs curtailed by 782 MW, equivalent to 81 percent of the UCAP sold by SCRs in July 2024. Of this, 314 MW was instructed to curtail through an IOU demand response program, and the remainder curtailed without being instructed by NYISO or an IOU.

Improving Incentives for Peak Management

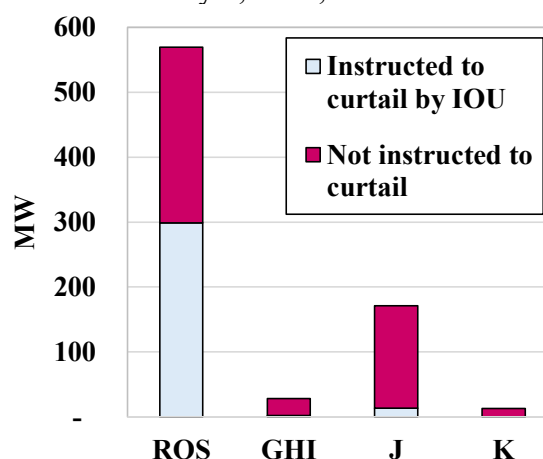
Peak demand management by end-users helps to maintain reliability and reduce capacity costs. It is important that NYISO's tariff rules governing the allocation of capacity costs to LSEs provide appropriate incentives for LSEs and their customers.

Currently, LSEs are responsible for a share of the system and locality UCAP requirements corresponding to their customers' share of peak load during the previous year's NYCA coincident peak load hour.¹⁷⁸ The summer peak load hour used for this purpose is limited to July and August non-holiday weekdays and the same NYCA-wide peak hour is used to determine LSEs' shares of NYCA and locality UCAP Requirements. This provides strong incentives to manage load during that peak hour, but does not provide incentives for:

- Peak load management outside of the single seasonal peak hour used to determine each LSE's share of UCAP Requirements;
- Peak load management during summer high load hours that occur outside of July and August; and
- Peak load management during locality peaks that occur at a different time than the NYCA coincident peak.

In addition, current rules inefficiently shift capacity market obligations from LSEs that serve loads enrolled in NYISO demand response programs to other LSEs. NYISO's tariff requires Demand Reductions by loads that sell capacity through NYISO programs to be reconstituted to their LSE's peak load forecast when determining the LSE's share of UCAP Requirements. This requirement is intended to prevent "double-counting" since these resources are credited as

Figure 67: Demand Reduction by SCRs without NYISO Instruction
July 8, 2024, HB17



¹⁷⁸ NYISO recently filed tariff revisions that would determine LSEs' UCAP obligations for the summer and winter capability periods based on their customers' share of demand during the previous year's summer and winter peak hours instead. See proposed revisions to Section 5.11.1 approved by NYISO's Management Committee on November 20, 2025.

suppliers in the capacity market.¹⁷⁹ However, no reconstitution occurs if the NYISO and/or TOs did not activate their demand response programs during the NYCA coincident peak hour (as was the case in 2023 and 2024). In this case, demand response providers that curtailed load during the system peak may simultaneously reduce their LSE's UCAP Obligation and earn capacity payments. We estimate that in the 2025/26 Capability Year, this issue resulted in shifting of \$34 million from LSEs that serve loads enrolled in NYISO programs to other LSEs.

Hence, we recommend making the following changes to how LSEs' UCAP Obligations are determined (Recommendation #2025-3):

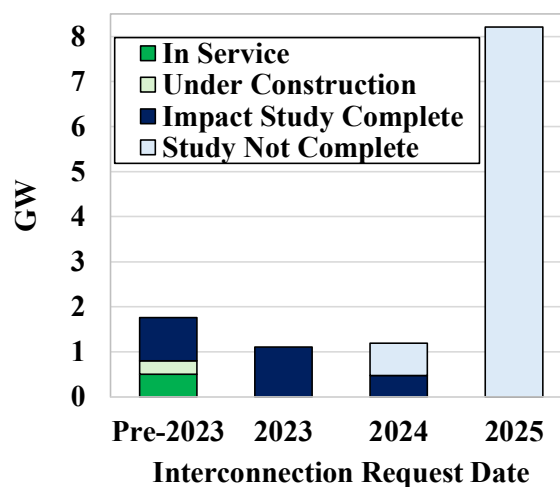
- Use a broader set of peak hours to determine LSEs' share of system and locality UCAP Requirements; and
- Include capacity participating in NYISO demand response programs in the demand forecasts of LSEs serving the associated customer accounts for purposes of determining LSEs' share of system and locality UCAP Requirements.

D. Critique of Large Load Interconnection Process

NYISO has recently received an unprecedented number of interconnection requests by proposed large load projects, including data centers and manufacturing facilities. As of March 2026 there were over 12 GW of large load projects in NYISO's interconnection queue, most of which entered the queue in 2025.¹⁸⁰

Figure 68 summarizes proposed large load projects in NYISO based on data from the Interconnection Queue as of March 2026. Approximately 500 MW of large loads (primarily cryptocurrency mining) have entered service in New York in the past decade. In recent years, a much larger number of loads have proposed to interconnect, including over 8 GW of applications received in 2025 alone. Most of the large loads currently in the queue have indicated that they do not intend to build new on-site generation or operate flexibly during system peak demand periods. NYISO monitors the regulatory and construction progress of large load projects in the queue, but NYISO has no way of predicting how many of the projects will actually be built in the long term.

Figure 68: Large Loads in NYISO Queue



¹⁷⁹ See NYISO Market Services Tariff section 5.11.1.

¹⁸⁰ Large loads are required to go through NYISO's interconnection process if they are over 10 MW (for bulk transmission-connected loads) over 80 MW (for local and distribution-connected loads).

NYISO's current role in New York's large load interconnection process is largely informational. Large loads are required to enter NYISO's queue and undergo a system impact study (SIS) to identify adverse reliability impacts on the bulk power transmission system, but the study does not produce any binding requirements for upgrades to maintain reliability, and it does not assess resource adequacy impacts of cumulative large load growth. After the project's SIS is approved by NYISO's Operating Committee, the load developer negotiates an Interconnection Agreement (IA) with the transmission owner (TO) in whose territory it will interconnect. NYISO is not party to these agreements.

We identify the following problems with the load interconnection process:

- ***Lack of information to State and local authorities about reliability impacts:*** some large load projects can be developed quickly and there is risk that their impact on system peak demand will outpace development of new generation and storage. While developers are sensitive to the cost of power, it is possible that load projects will continue to be built even at capacity price levels corresponding to resource adequacy violations (i.e., greater than the Net CONE). Currently, there is no formal process to inform State and local entities that regulate the siting and interconnection of large loads when new load development would undermine reliability.
- ***Lack of information to NYISO regarding future load interconnections:*** NYISO's Reliability Planning Process proactively identifies Reliability Needs driven by trends in load and resource availability. This process may retain generators proposing to retire and/or procure new generation or transmission under cost-of-service contracts. As NYISO highlighted in its 2025 Comprehensive Reliability Plan, this process has become increasingly fraught because NYISO has very limited visibility or control over the amount of large loads that will enter service. Excluding proposed load projects from planning studies used to identify actionable Reliability Needs raises the risk of future reliability violations, but including uncertain or speculative projects is likely to result in out-of-market procurement actions that are disruptive to the NYISO markets and whose costs are born by ratepayers.
- ***Lack of firm and non-firm load classifications:*** Once large load projects complete NYISO's study process and sign an IA, it is unclear if they face any restrictions on future changes in the facility's business model that may impact NYISO system reliability. For example, a project indicating that it plans to engage in cryptocurrency mining and flexibly curtail during system peaks may be found to have no adverse reliability impacts, but such a facility might later convert to a different and less flexible use case without being studied for the resulting reliability impact.

NYISO does not have the ability to address these problems unilaterally because load interconnection is primarily mediated through the TOs' interconnection processes and state and local approvals. The New York Public Service Commission (NYPSC) recently opened a proceeding to explore changes to the state's process for managing load interconnections.¹⁸¹

¹⁸¹ See NYPSC Docket 26-E-0045

Additionally, NYISO is currently pursuing a Market Project designed to identify improvements to its interconnection and planning processes.¹⁸² We make the following recommendations (see #2024-2):

- **NYISO’s planning process should study and report on large load impacts, not attempt to preemptively accommodate them.** Through its Short-Term Assessment of Reliability (STAR) and long-term Reliability Needs Assessment (RNA) reports, NYISO should quantify the amount of non-flexible large load that can be accommodated at various locations in the system before reliability violations (including resource adequacy, transmission security and others) arise. This information will serve as a critical input for utility and state planning processes to limit or impose conditions on interconnections that would cause reliability deficiencies. Construction of new generation to serve large load-driven demand should be coordinated through contracting by loads or merchant entry, not cost-of-service contracts designed to service potential future load interconnections.
- **Load interconnection studies and agreements should distinguish between firm and non-firm interconnections.** Non-firm (curtailable) loads should enjoy faster timelines and lower costs due to reduced system impact, but converting to a non-curtable (i.e., firm) business model should require them to undergo an additional reliability study.

¹⁸² See NYISO February 3, 2026 ESPWG/TPAS presentation “Load Interconnection Process: Challenges and Considerations”, available [here](#).

XIII. RECOMMENDATIONS

This report finds that the NYISO’s electricity markets performed well in 2025, although we recommend twenty-three additional enhancements to improve market performance or operations. We use a numbering system in which each recommendation is identified by the SOM report where it first appeared and the number used in that report. For example, Recommendation 2015-16 first appeared in the 2015 SOM Report as Recommendation #16. Most recommendations are carried over from last year, but Recommendations 2025-1 to 2025-5 are new. We designate some recommendations as “High Priority” or “Low Cost” based on our assessment of their effects on market efficiency relative to the cost or complexity of the proposed solution. The following table summarizes our current recommendations and NYISO market design projects that address them in the 2026 project plan and/or projected 2027 work based on the December 2025 Market Vision document (available [here](#)).

Table of Recommendations

Number	Section	Recommendation	Low Cost/ High Benefit	NYISO Project Scope: (2026 / 2027)
<i>Capacity Market – Increasing Recognition of Reliability Needs in the Market</i>				
2025-1	V.B.4, III.A, IX.F	Address the “planning-market gap” by: a) Using transparent criteria to develop planning assumptions and calibrate the Level of Excess, and b) Improving GADS compliance monitoring	✓	--
<i>Capacity Market – Avoiding Over-Payment to Resources with Lower Reliability Value</i>				
2025-2 & 2022-4	IX.B.1 & IX.B.2	Enhance locational pricing of capacity by: • Using the CAF framework to reflect intrazonal deliverability limitations (near-term) & • Implementing more granular capacity zones and a dynamic process for updating them (long-term)	✓	--
2025-3	XII.C	Align criteria for setting ICAP obligations with criteria for SCR and DER baseline values		--
2023-4	IX.C	Develop sloped demand curves reflecting the marginal value of surplus capacity for use when an LCR is determined by a Transmission Security Limit		--

Recommendations

Number	Section	Recommendation	Low Cost/ High Benefit	NYISO Project Scope: (2026 / 2027)
2022-1	IX.C	Compensate capacity suppliers based on their contribution to transmission security needs when they determine the locational capacity requirements	✓	<i>Reliability Attribute Based Capacity Pricing: (-- / Market Design Complete)</i> ¹⁸³
2022-2	IX.D	Establish seasonal capacity requirements, CAFs, and demand curves and adjust derating factors to allow CARC/UDR elections close to the Capability Period	✓	<i>Winter Reliability Capacity Enhancements: (Development Complete / Deployment)</i> ¹⁸⁴
Capacity Market – Other Enhancements				
2025-4	IX.F	Evaluate improvements to outage scheduling as the NYISO system shifts to have both summer and winter reliability risks		--
Energy Market Enhancements – Pricing and Performance Incentives for Flexible Resources				
2025-5	VI.D	Model TSA constraints in the DAM with a reduced GTDC based on forecasted weather and demand.		--
2024-1	VII.D, VIII.D.1	Use the reserve market to satisfy reserve needs driven by intrazonal constraints	✓	--
2023-1	VIII.B	Allocate congestion residuals to NYTOs based on incremental transfer capability scheduled in the DAM		--
2023-2	VII.C	Modify fast start pricing logic to base Minimum Run Time eligibility criteria on the treatment of the unit rather than the offer		--
2023-3	Appx. V.B	Revise tariff to provide disincentives for over-generation by generators with negative incremental costs		--
2021-3	VIII.D	Consider modeling transient voltage recovery constraints on Long Island in the energy market		--
2020-2	VII.B	Eliminate offline fast-start pricing from the real-time dispatch model		--

¹⁸³ It is unclear whether the proposed project will address the issues driving this recommendation.

¹⁸⁴ The Winter Reliability Capacity Enhancements will provide a partial solution to this recommendation. See discussion of Recommendation 2022-2 for details.

Number	Section	Recommendation	Low Cost/ High Benefit	NYISO Project Scope: (2026 / 2027)
2019-1	II.E	Set day-ahead and real-time reserve clearing prices considering reserve constraints for Long Island		<i>Dynamic Reserves:</i> (Development Complete / Development Complete)
2017-2	VII.A	Modify operating reserve demand curves to improve shortage pricing and ensure NYISO reliability	✓	--
2015-16 & 2016-1	XIII.B	Dynamically adjust operating reserve requirements to align with reliability needs & Develop rules for efficient pricing and settlement when operating reserve suppliers provide congestion relief	✓	<i>Dynamic Reserves:</i> (Development Complete / Development Complete) ¹⁸⁵
2015-9	X.C	Eliminate transaction fees for CTS transactions at the PJM-NYISO border		--
2012-8	VIII.B	Operate PAR-controlled lines between New York City and Long Island to minimize production costs and create financial rights that compensate affected TOs.		--
<i>Broad Market Enhancements</i>				
2024-2	XII	Evaluate reforms to enhance incentives for demand-side interconnection and participation in the NYISO markets. a) Allow DERs to submit offer parameters comparable to generators, and b) Coordinate with State and local entities regarding new load interconnections.		<i>DER Market Enhancements:</i> (Market Design Complete / Deployment) ¹⁸⁶
<i>Planning Process Enhancements</i>				
2022-3	V.B.5	Improve transmission planning assumptions and metrics to better identify and fund economically efficient transmission projects.		2026 Outlook Report enhances placement of new resources

¹⁸⁵ See Section XIII.B *Recommendation 2015-16* for a list of concerns with the current project scope.

¹⁸⁶ It is unclear whether the proposed project will address the issues driving this recommendation.

This section discusses each recommendation in more detail. The last subsection discusses several prior recommendations we did not include this year.

A. Criteria for High Priority Designation

As NYISO MMU, we recommend market rule changes to improve efficiency. In most cases, our recommendations are high-level changes and assume that NYISO will develop a detailed proposal vetted by stakeholders, and submit a Section 205 filing to FERC or make a procedural change. In some cases, we do not recommend a particular solution but may instead recommend NYISO evaluate the costs and benefits of addressing an issue through a rule change or software change.

In each report, we designate a few recommendations as “High Priority” or “Low Cost” based on our assessment of their effects on market efficiency or, in some cases, the magnitude of the market or pricing issue relative to the complexity of the proposed solution. When possible, we quantify benefits by estimating the production cost savings and/or investment cost savings a recommendation would produce because these are the most accurate measures of economic efficiency. We focus on maximizing economic efficiency because it minimizes the long-term costs of satisfying system needs. We generally do not rely on measures that reflect economic transfers from changing prices, such as consumer savings, because they do not measure economic efficiency.

We also consider the feasibility and implementation cost. Relatively quick or low-cost recommendations generally warrant higher priority because they yield higher benefit-to-cost ratios. Conversely, recommendations that are difficult to implement or have relatively uncertain benefits receive lower priority.

B. Discussion of Recommendations

Capacity Market – Increasing Recognition of Reliability Needs in the Market

2025-1: Address the “planning-market gap” by: (a) using transparent criteria to develop planning assumptions and calibrate the Level of Excess, and (b) improving GADS compliance monitoring. (High Priority)

NYISO has raised concerns about bulk power system reliability over the next ten years. Load growth is poised to exceed the previous decade; New York State policies to develop new zero-emission dispatchable generation have lagged expectations; and policies to shut down older conventional generation have driven the retirement of more than 3 GW of fuel-secure generating capacity in downstate areas (i.e., zones H-K). State policies inhibiting new conventional generation have also prevented new entry as reserve margins have dwindled. To address reliability concerns, the New York DEC delayed implementing air permit limit reductions that otherwise would have driven additional retirements of conventional generation in May 2025.

In recent years, NYISO has adopted increasingly conservative assumptions in its reliability planning process and is considering changes that would require higher capacity levels to avoid finding a reliability need. As NYISO's effective capacity requirements increase, NYISO should implement corresponding changes in the capacity market so the competitive wholesale market can efficiently incentivize maintenance of existing capacity and attract new supply where siting and permitting obstacles can be addressed. We recommend several initiatives to prevent a "planning-market gap" (a gap between reliability planning needs of the system and capacity market requirements that leads to out-of-market contracting for supply).

Recommendation 2025-1a is to develop reliability planning assumptions 2 to 10 years out based on objective criteria that can also calibrate the Level of Excess in the capacity market. The capacity market is designed to satisfy planning reliability needs through competitive market incentives. When planning studies use more conservative assumptions, the effective capacity planning requirement rises, so NYISO should make corresponding adjustments to the capacity market's targeted capacity level. This will require a disciplined approach to base-case and scenario development in reliability planning studies and corresponding adjustments to the target Level of Excess ("LOE") in the capacity market.

Recommendation 2025-1b to enhance GADS compliance monitoring would ensure generators fully log forced outages and derates. Otherwise, forced outage rates used in the capacity market will be understated, which may lead NYISO to adopt more conservative reliability planning assumptions. This should include clear guidance on existing rules, stricter criteria for initiating an investigation, and more unannounced UOL audits. Our analysis of generator operations found derating factors of conventional resources are understated because of underreporting.¹⁸⁷

This recommendation is high priority because operating a capacity market that provides investment incentives that reflect the system's planning reliability needs is essential. Both elements of this recommendation will align capacity market parameters with planning needs.

Capacity Market – Avoiding Over-Payment to Resources with Lower Reliability Value

2025-2: Enhance locational capacity settlements by using the CAF-determination framework to reflect intrazonal deliverability limitations. (High Priority)

NYISO's capacity market has four pricing zones in which all suppliers receive the same capacity price. However, the marginal value of capacity differs by location because of internal transmission constraints within each capacity zone. For example, bottlenecks can limit Staten Island capacity deliverability to the rest of New York City, but Staten Island suppliers receive the premium New York City price. This inflates consumer costs and weakens incentives to retain capacity in areas with reliability needs or retire capacity in oversupplied areas. The

¹⁸⁷ See Section IX.F.

deliverability planning process also imposes inefficiently high transmission upgrade costs on some new project developers, creating a barrier to entry in some areas. NYISO's tariff-defined New Capacity Zone creation process cannot create new capacity regions in a timely manner.

Hence, we have recommended (see #2022-4 below) implementing and dynamically updating an expanded set of capacity zones that reflect known bulk transmission bottlenecks on the NYISO system. If NYISO cannot implement recommendation #2022-4 in the near term, this recommendation (#2025-2) may achieve most underlying objectives and be easier to implement. This recommendation would keep the existing capacity zone configuration, but adjust the CAF of resources within each zone to reflect differences in locational value. For example, if the CAF of a Firm Fuel generator in New York City is 100 percent and the marginal reliability impact of a Firm Fuel generator in Staten Island is 80 percent of that in New York City, then the Staten Island generator would receive an 80 percent CAF for the Capability Year.

This recommendation would limit capacity supply in export-constrained zones more efficiently than the deliverability test, which sometimes creates excessive barriers to economic new entry. It would also discount capacity payments to resources in oversupplied areas, yielding significant consumer savings, prompting retirement of non-deliverable generation, and creating opportunities for new entry.

This recommendation is high priority because: (1) consumers overpay for capacity in export-constrained areas; (2) changing reliability needs (such as from the growing importance of winter reliability) make it critical for the capacity market to accurately signal the value of retaining and attracting capacity where needed; and (3) inefficient barriers to new entry exist in areas where generation is not fully deliverable within one of the existing four capacity zones.

2025-3: Align criteria for setting ICAP obligations with criteria for SCR and DER baseline values.

Demand response programs let load customers earn revenue by being partially or fully curtailable under peak load conditions, helping offset energy costs. LSE capacity obligations are based on consumption during the prior year's peak load hour, while loads in demand response programs can sell capacity based on a baseline that depends on consumption during a wider set of hours. In principle, this charges each load a capacity obligation it can sell back to the wholesale market. However, current rules allow loads to acquire a baseline that exceeds their obligation due to inconsistency between the hour that sets obligations and the hours that set the baseline.¹⁸⁸ For example, if a 10 MW SCR is not instructed to curtail in the peak load hour but reduces load to 2 MW, its LSE's UCAP obligation will be based on 2 MW but the resource will

¹⁸⁸ NYTOs are obligated to "add back" load that was curtailed in "Demand Reductions" called under NYISO and NYTO programs. However, even if NYTOs begin to "add back" load to reflect the amount of demand reductions from curtailments, the ICAP tags of these resources will be too low if (a) there are no program curtailments in the annual peak load hour and (b) loads curtail anyway.

still be able to sell 10 MW, resulting in a double payment for curtailment. This results in shifting of capacity costs to other LSEs.

We recommend aligning the criteria for setting LSE capacity obligations with the criteria for setting SCR and DER baseline values so demand-side resources cannot be net sellers of capacity. This could be done by adjusting the allocation of ICAP Requirements to LSEs based on the difference between the enrolled demand response capacity and metered peak load of their customers. In addition, LSE capacity obligations should be set based on load in a wider set of hours to better align with hours used for demand response baseline calculations and strengthen incentives for peak load management.

2023-4: Develop sloped demand curves reflecting the marginal value of surplus capacity for use when an LCR is determined by a Transmission Security Limit.

NYISO developed the sloped demand curves when the IRM and LCRs were typically based on probabilistic resource adequacy criteria. The slope should reflect that the marginal reliability value of capacity declines but remains positive as surplus capacity rises. In recent years, NYISO has often based LCRs on deterministic transmission security criteria, which do not explicitly quantify the reliability contribution of surplus supply under conditions more extreme than the planning criteria. NYISO uses the same sloped demand curves whether LCRs are based on resource adequacy or transmission. NYISO should develop sloped demand curves for TSL-based LCRs that reflect the value of surplus capacity given expected load forecast uncertainty and random variation in resource availability.

2022-1: Compensate capacity suppliers based on their contribution to transmission security needs when they determine locational capacity requirements. (High Priority, Current Effort)

NYISO determines Locational Capacity Requirements (LCRs) annually using the “LCR Optimizer” method, but applies a minimum floor in each locality designed to meet transmission security criteria. NYISO recently took steps to align its calculation of Transmission Security Limits (TSLs) used in the LCR process with the methodology used in its reliability planning studies. This has resulted in LCRs being set at the TSL-based floor in multiple localities, and the importance of TSLs is expected to grow.

Some resources, including large-contingency resources and Special Case Resources (SCRs), provide limited value for meeting transmission security planning requirements. NYISO does not count SCRs toward transmission security requirements, while supply resources that are among the two largest in an area tend to increase the area’s capacity requirement. For example, individual supply resources larger than 700 MW generally increase New York City’s capacity requirement. Consequently, when NYISO sets LCRs using the TSL methodology, the presence of these resources increases the LCR. This raises consumer costs and weakens investment

incentives because some suppliers receive payments based on requirements they do not help satisfy. We recommend adjusting capacity payments to reflect each resource's contribution to meeting underlying resource adequacy and transmission security requirements.¹⁸⁹ For large-contingency resources, this should not apply to the portion of the unit that does not increase the Transmission Security Limit.

Status: The 2025 Market Vision states that NYISO plans to design in 2027 a way to “enhance the Capacity Market’s ability to reflect and price transmission security” in the *Reliability Attribute Based Capacity Pricing* project.

2022-2: Establish seasonal capacity requirements, CAFs, and demand curves consistent and adjust derating factors to allow CARC and UDR elections close to the Capability Period. (High Priority, Current Effort)

NYISO's capacity market uses the same installed capacity requirement year-round. The level of surplus supply in each spot capacity auction depends on installed capacity relative to this annual requirement, which often bears little relationship to seasonal reliability risk. Seasonal risk depends on seasonally available supply (including deratings) and seasonal load. As a result, seasonal prices are set by installed capacity (regardless of actual availability) and may not provide incentives consistent with seasonal reliability risk. The annual requirement-setting process may also be poorly suited to the timing of winter fuel procurement decisions. We recommend separate seasonal capacity requirements and demand curves that reflect the supply needed to maintain reliability in each season, and Capacity Accreditation Factors that reflect the reliability value of each CARC in each season. We also recommend adjusting derating factors to account for differences between the CARC and UDR elections considered in the IRM study and the actual availability of resources in the capacity market.

Status: NYISO recently filed to deploy market design changes that would partially address this recommendation in 2027 under the *Winter Reliability Capacity Enhancement* project. These changes include: (1) separate summer and winter ICAP requirements based on the amounts of capacity modeled in the final IRM and LCR cases; (2) allocation of winter capacity obligations to Load-Serving Entities (LSEs) based on their winter peak load; and (3) required seasonal (rather than annual) elections of capacity sales for UDR imports. However, even with these changes, the rules impose election requirements on firm fuel and UDR resources that impede efficient winter participation and set prices and CAFs that may be misaligned with reliability value. Hence, we recommend moving to a capacity market with separate seasonal requirements, demand curves, and other parameters.

2022-4: Enhance locational capacity pricing by implementing more granular capacity zones and a dynamic process for updating them. (High Priority)

¹⁸⁹ See discussion in Section IX.C.

NYISO's capacity market has four pricing zones in which all suppliers receive the same capacity price. However, the marginal value of capacity differs by location because of internal transmission constraints within each zone. For example, bottlenecks limit Staten Island capacity deliverability to the rest of New York City, but Staten Island suppliers receive the premium New York City price. This inflates consumer payments and weakens incentives to retain capacity in areas with reliability needs or retire capacity in oversupplied areas. The deliverability planning process also imposes inefficiently high transmission upgrade costs on some new project developers, creating a barrier to entry in some areas. NYISO's tariff-defined zone creation process cannot create new capacity zones in a timely manner.

We recommend implementing and dynamically updating an expanded set of capacity zones that reflect known bulk transmission bottlenecks on the NYISO system. This process would set requirements for all load zones and designated sub-zone areas using the LCR Optimizer method. It would price capacity using demand curves for zones with binding transmission constraints in NYISO's resource adequacy model GE-MARS. NYISO would also need to define export demand curves for surplus zones constrained by transmission bottlenecks.

Because no zone configuration will perfectly reflect key planning constraints, the recommendation also includes a capacity constraint pricing (CCP) component applied in capacity settlement. This incremental locational price adder would ensure each resource's economic signal reflects its effects on key planning constraints.

This recommendation is high priority because: (1) consumers already significantly overpay because export-constrained areas are overpriced; (2) changing reliability needs (such as the growing importance of winter reliability) make it critical for the capacity market to accurately signal the value of retaining and attracting capacity where needed; and (3) inefficient barriers to new entry exist in areas where generation is not fully deliverable within one of the existing four capacity zones.

Status: The 2025 Market Vision states that NYISO plans to begin thinking about this issue in 2028 for possible deployment after 2030 under the *Capacity Zone Redesign* project.

Capacity Market – Other Enhancements

2025-4: Evaluate improvements to outage scheduling as the NYISO system shifts to have both summer and winter reliability risks.

We recommend evaluating whether NYISO should revise outage scheduling rules to incentivize generators to complete planned and maintenance outages more quickly and at times that do not undermine reliability. As the NYISO generation fleet ages, the average annual outage times are rising. At the same time, the ability to take outages has declined because surplus capacity margins have fallen and winter reliability risks have emerged in December through February.

Eventually, NYISO will need to ration available maintenance slots more systematically to provide efficient scheduling incentives.

Energy Market Enhancements – Pricing and Performance Incentives for Flexible Resources

2025-5: Model TSA constraints in the Day-Ahead Market with a reduced GTDC based on forecasted weather and demand.

When thunderstorms are forecasted around the ConEd service territory, NYSRC rules require NYISO to secure the transmission system as if the first transmission contingency has already occurred, causing sudden reductions in transfer capability from upstate NY to downstate areas.¹⁹⁰ These events (“thunderstorm alerts”) routinely curtail upstate-to-downstate transmission flows by 1 to 2 GW below day-ahead scheduled levels, forcing NYISO to rely more on expensive downstate peaking resources.

NYISO models these “TSA constraints” in the real-time market when storms are near, but not in the day-ahead market. As a result, the day-ahead market tends to under-commit downstate generation and over-commit upstate when storms are predicted. However, virtual traders and other market participants respond to predictable day-ahead/real-time price inconsistencies by bidding up day-ahead prices downstate when thunderstorms are forecasted. This helps commit adequate day-ahead resources, but day-ahead and real-time schedules often remain inconsistent because individual virtual traders cannot precisely predict how much to over-schedule downstate load in the day-ahead market. In addition, this typically produces large balancing congestion shortfalls that are uplifted to Load Serving Entities,¹⁹¹ making these charges difficult to hedge financially.

To address these issues, we recommend NYISO consider modeling TSA constraints in the day-ahead market in hours with significant thunderstorm risk, setting the GTDC level based on the thunderstorm forecast probability and load level. This would commit a more appropriate amount of downstate generation when thunderstorms are likely.

2024-1: Use the reserve market to satisfy reserve needs driven by intrazonal constraints. (High Priority)

NYISO must maintain sufficient energy and operating reserves to meet local (sub-zone) reliability needs under N-1-0, N-1-1, and N-1-1-0 criteria in New York City, Long Island, and

¹⁹⁰ See Reliability Rule G.1.B.R4 in *Reliability Rules & Compliance Manual for Planning and Operating the New York State Power System*, New York State Reliability Council.

¹⁹¹ See February 24, 2026 MIWG presentation, *Allocation of Congestion Uplift from Thunderstorm Watch Events in July 2025*.

other areas of New York State. Because the markets do not schedule and price these local needs, NYISO satisfies them with out-of-market commitments in most hours. NYISO recovers these costs through make-whole payments rather than energy and operating reserve clearing prices. Heavy reliance on make-whole payments distort short-term performance incentives and long-term investment incentives for resources that can meet local needs. They also weaken incentives for demand-side participation in the wholesale market. We recommend NYISO implement local reserve requirements in load pockets where NYISO otherwise relies on costly out-of-market actions.

We designate this recommendation as High Priority partly because significant transmission and resource mix changes have occurred recently and are planned in the coming years. New transmission and peaker retirements are shifting the boundaries of reserve-constrained areas, while intermittent renewable interconnections are producing larger and more variable operating reserve needs. Localized operating reserve requirements would help NYISO maintain reliability efficiently by strengthening investment incentives for new and existing low-cost reserve providers.

Status: NYISO has deferred studying this recommendation until 2028 when it tentatively plans a Market Concept Proposed project called “More Granular Operating Reserves”.

2023-1: Allocate congestion residuals to NYTOs based on incremental transfer capability scheduled in the Day-Ahead Market.

NYTOs recover a large share of high-voltage transmission costs through DAM Congestion Revenues and TCC auctions. TCC auction revenues are allocated to each NYTO in proportion to the value of its transmission facilities in the auctions, while NYTOs are charged when outages of their equipment reduce scheduled transfers in the DAM. However, when the DAM schedules additional transmission capability (above what was sold in the TCC auction), NYISO allocates the resulting revenues in proportion to the TCC revenue allocation, regardless of which NYTO’s facilities enabled the additional scheduling. As a result, NYTOs do not recover the value of transmission capability they provide when their assets are not sold in the TCC auctions, which discourages efficient day-ahead scheduling. For example, the 901 and 903 lines scheduled an average of 184 MW from Long Island to New York City in 2025 even though day-ahead prices on the Long Island side averaged \$6.15/MWh higher. Although more efficient scheduling of the 901 and 903 lines would produce significant day-ahead congestion surpluses, the NYTO that schedules the lines has little incentive to do so because most resulting revenues would flow to other transmission owners.¹⁹²

¹⁹² See discussion of Recommendation 2012-8 related to the operation of these facilities.

In 2025, after NYISO allocated day-ahead congestion shortfalls from outages attributed to a responsible NYTO, NYISO allocated a net day-ahead congestion revenue surplus of \$31 million to NYTOs in proportion to TCC revenue allocations. We decomposed this into \$62 million of congestion revenue surpluses from higher scheduled flows across constrained interfaces in the DAM and \$31 million of revenue shortfalls from transmission outages and equipment issues that current rules and practices did not allocate to the responsible party. Allocating these surpluses and shortfalls to the responsible party would be fairer and would improve incentives to operate the transmission system efficiently.

We recommend NYISO revise the allocation of DAM congestion residuals based on changes in scheduled transmission utilization between the TCC auctions and the day-ahead market.¹⁹³ This would strengthen incentives for efficient operations and adoption of grid-enhancing technologies by allowing each NYTO to recover the value of transmission scheduled in the DAM even if the capacity was not sold in the TCC auctions.

2023-2: Modify fast start pricing logic to base Minimum Run Time eligibility criteria on the treatment of the unit rather than the offer.

Fast-start pricing lets small quick-start resources with high minimum output levels (relative to maximum output) set the clearing price when their output displaces higher-cost resources in the DAM and RT markets. Resources can set price if they can start in 30 minutes or less and have a 1-hour minimum run time. Currently, by offering to start in 30 minutes or less in the real-time market, a generator agrees to be treated as having a 1-hour minimum run time for scheduling, even if it submitted an offer with a longer minimum run time.¹⁹⁴ However, quick-start resources that submit an offer with a Minimum Run Time longer than one hour are treated as ineligible to set price as Fast-Start Resources in the real-time market, even though the market software schedules them as if they have a 1-hour minimum run time.

We recommend NYISO revise fast-start pricing eligibility to use the minimum run time used for scheduling, not the offer parameter value that the real-time scheduling system ignores for scheduling. This would make real-time LBMPs more consistent with the cost of serving load at each location. In 2025, this would have affected LBMPs in New York City in 9 percent of intervals and on Long Island in 4 percent of intervals.¹⁹⁵ We believe NYISO can make this

¹⁹³ An approach for doing this was presented at the July 8, 2025 Market Issues Working Group meeting in *Net Congestion Rent Assignment Simulation Examples*.

¹⁹⁴ NYISO MST 4.4.1.4 states: “RTC will make all economic commitment/de-commitment decisions based upon available offers assuming Suppliers internal to the NYCA have a minimum run time...not longer than one hour;”

¹⁹⁵ See Section VII.C for details.

change without a tariff modification because the tariff bases eligibility on the minimum run time a unit “has,” not the Minimum Run Time offer submitted.¹⁹⁶

2023-3: Improve settlement rules to provide disincentives for over-generation by generators with negative incremental costs.

Control area operators maintain system security by redispatching generation to match load throughout the day. Good utility practice requires generators to make reasonable efforts to follow dispatch instructions within equipment limits. To support this practice, NYISO imposes over- and under-generation penalties to incentivize compliance. Units that over-generate by more than 3 percent of their upper operating limit are penalized by: (i) not receiving LBMP revenue for production above the 3 percent level when LBMP is positive and paying the LBMP when it is negative; and (ii) incurring a small share of regulation capacity costs in that interval. For generators with positive incremental costs, this penalty typically motivates compliance because they benefit financially from following instructions. For generators with negative incremental costs, the penalty is often insufficient because they may still profit from exceeding instructions beyond the 3 percent band, which can impact reliability. We recommend NYISO work with stakeholders to revise over-generation penalties so generators with negative incremental costs do not benefit from over-generating.

2021-3: Consider modeling transient voltage recovery constraints on Long Island in the energy market.

Market software does not represent transient voltage recovery (TVR) criteria for the East End of Long Island, so NYISO often satisfies these criteria by scheduling generation out of market during the summer. This can lead to inefficient scheduling and weak incentives for resources that can satisfy TVR criteria. We recommend NYISO represent these criteria in the day-ahead and real-time markets using surrogate constraints so generation scheduled to satisfy East End TVR criteria is compensated appropriately. Appendix III.E illustrates how surrogate constraints could satisfy TVR criteria in the market models. This would improve scheduling when units that satisfy TVR requirements affect thermal congestion and would make LBMPs more consistent with the cost of serving load on the East End of Long Island. In 2025, this would have increased average East End LBMPs by an estimated \$11 per MWh.¹⁹⁷

2020-2: Eliminate offline fast-start pricing from the real-time dispatch model.

NYISO’s real-time market runs a dispatch model that updates prices and schedules every five minutes. The model currently treats offline 10-minute gas turbines (units that can start in ten

¹⁹⁶ NYISO MST 2.6 defines “Fast-Start Resource: A Generator that...(3) has a minimum run time of one hour or less...”

¹⁹⁷ See Section VIII.D for details.

minutes) as if they can follow a 5-minute signal. Because these units cannot respond in five minutes, they routinely receive schedules they cannot follow. This causes under-generation, inconsistencies between scheduled and actual transmission flows, and inefficient prices that do not reflect supply-demand balance. We recommend NYISO eliminate offline fast-start pricing. This would make real-time LBMPs more consistent with the cost of serving load at each location. In 2025, this would have increased LBMPs on Long Island by an estimated \$3.70 per MWh and in other portions of Southeast New York by \$0.50 to \$0.80 per MWh.¹⁹⁸

2019-1: Set day-ahead and real-time reserve clearing prices considering reserve constraints for Long Island. (Current Effort)

The day-ahead and real-time markets schedule resources to meet reserve requirements, including 10-minute spinning reserves, 10-minute total reserves, and 30-minute total reserves on Long Island. However, Long Island reserve providers are not paid reserve prices corresponding to these requirements. Instead, NYISO pays them based on clearing prices for the larger Southeast New York region. This inflates Long Island reserve prices, increasing reserve charges to ratepayers by an estimated \$17 million and DAMAP uplift by an estimated \$6.5 million in 2025.¹⁹⁹ Paying reserve providers based on day-ahead and real-time scheduling decisions would improve incentives in both markets and provide better long-term signals to new investors.

Status: NYISO plans to address this recommendation through the planned *Long Island Reserve Constraint Pricing* project in 2028 along with the deployment of *Dynamic Reserves*, which will include the capability to dynamically procure reserves on Long Island.²⁰⁰

2017-2: Modify operating reserve demand curves to improve shortage pricing and ensure NYISO reliability. (High Priority)

Shortage prices during operating reserve deficiencies are too low to encourage efficient actions that preserve reliability. This is because the market's demand for operating reserves does not fully reflect their value in ensuring load is served – the value of lost load (VOLL). This weakens incentives to take actions that avoid load shedding in the short term and reduces longer-term incentives to replace inflexible, poor-performing resources with fast-ramping generation and storage. Shortage prices are also sometimes too low to schedule available resources needed to meet reliability requirements, forcing operators to rely on out-of-market actions.

PFP (“Pay For Performance”) rules in ISO New England and PJM exacerbate this problem by providing much higher incremental compensation for energy and reserves during reserve

¹⁹⁸ See Section VIII.B for details.

¹⁹⁹ See Sections II.E and VII.E for details.

²⁰⁰ See *Long Island Reserve Constraint Pricing*, presented to the Market Issues Working Group on February 7, 2024.

shortages than NYISO. Resources selling into ISO-NE receive over \$10,000/MWh and those selling into PJM receive more than \$4,000/MWh during even slight reserve shortages, while NYISO sets prices between \$750 and \$3,000/MWh during *deep* reserve shortages. This drives inefficient imports and exports during tight regional conditions and harms NYISO reliability as energy is drawn to neighboring markets even when NYISO shortages are much deeper.

We recommend NYISO modify its operating reserve demand curves to provide efficient incentives and ensure reliability during shortages. NYISO should target operating reserve demand curve step values so:

- Clearing prices rise to efficient levels given VOLL and the load-shedding risk at the depth of the reserve shortage;
- Incentive effects of neighbors' PFP rules are minimized;
- The real-time market schedules available resources so NYISO operators do not need out-of-market actions to maintain reliability; and
- NYISO real-time scheduling models prioritize appropriately when multiple reserve requirements and/or transmission constraints are simultaneously in shortage.

This recommendation is high priority because real-time responsiveness during emergency conditions will become increasingly important as capacity margins fall. Intermittent renewable resources entry and conventional generator retirements will likely increase net load forecast uncertainty and create new operational challenges. This recommendation strengthens incentives for generation and load flexibility and for efficient use of regional resources to preserve reliability. Higher operating reserve demand curves would increase consumer costs in the energy market, but a corresponding reduction in capacity market demand curves would offset them.

2016-1: Develop rules for efficient pricing and settlement when operating reserve providers provide congestion relief. (Current Effort, High Priority)

NYISO must maintain flows so that no transmission facility would be loaded above its Long-Term Emergency (LTE) rating after a contingency occurs. In some cases, NYISO can use operating reserves and other post-contingency operating actions to meet this requirement. This lets NYISO increase transmission utilization into load centers, reducing production costs and load-center pollution. Because the market does not compensate reserve providers for congestion relief, it does not provide efficient investment signals for flexible resources. We recommend NYISO compensate operating reserves that help manage congestion.

New York City has lost much of its peaking generation over the last four years and may lose more than half of its remaining peaking generators by 2030 under New York State environmental policies. Some retiring peaking resources are currently utilized for thousands of hours per year to manage congestion by providing offline reserves, reducing production costs and enabling higher import levels without adding load-center pollution. If NYISO does not compensate reserve

providers consistent with their value, investors are unlikely to site new resources to relieve congestion or to develop flexible operating characteristics. This will become more important as NYISO interconnects additional intermittent generation to the New York City transmission system because this will increase the variability in congestion patterns.

Status: NYISO plans to address this recommendation with the deployment of *Dynamic Reserves* in 2028.²⁰¹

2015-9: Eliminate transaction fees for CTS transactions at the PJM-NYISO border.

CTS with PJM has delivered efficiency benefits well below expectations since NYISO implemented it in 2014. Since ISO-NE implemented CTS in 2015, we have observed much greater CTS bidding at the ISO-NE interface. Lower CTS utilization with PJM is partly driven by the relatively large fees charged on CTS transactions with PJM, while NYISO and ISO-NE eliminated fees at their interface. We estimate NYISO collected just \$2.2 million in CTS export fees in 2025 because export fees were usually higher than expected profits from exporting to PJM. A lower export fee could increase total fee collections by making exports profitable under a wider range of conditions. CTS with PJM is unlikely to function effectively while transaction fees and uplift charges are large relative to expected inter-market spreads. In addition, when upstate renewable surpluses cannot be delivered downstate because of transmission constraints, CTS export fees impose significant costs on renewable generators exporting surplus power, which will tend to increase long-run REC costs for ratepayers.

Hence, we continue to recommend that the NYISO eliminating transaction fees and uplift charges on CTS transactions between PJM and NYISO. NYISO would benefit from eliminating CTS transaction fees even if PJM does not.

2015-16: Dynamically adjust operating reserve requirements to align with reliability needs. (Current Effort, High Priority)

Unused import capability into a local area can reduce the operating reserves NYISO must hold on internal resources. In many cases, it is less costly to increase local generation, reduce imports, and treat the released interface capability as reserves. We recommend NYISO modify market software to optimize reserve procurement quantities for each requirement.

This recommendation is high priority because it would let NYISO schedule and price operating reserves efficiently as it implements other high-priority recommendations. This will become more important as New York's resource mix and transmission network evolve.

²⁰¹ See *Dynamic Reserves*, presented to the Market Issues Working Group on January 25, 2024.

Status: NYISO is working toward deploying Dynamic Reserves in 2028, although we have raised concerns about elements of the proposal.²⁰² The following lists benefits of well-designed dynamic reserves and summarizes our concerns with the proposed 2028 deployment:

- Element 1 – NYCA reserve requirements will be set dynamically based on actual scheduling of the two largest contingencies, which will be lower than the current requirement to hold reserves equal to 200 percent of the largest possible single contingency. This will reduce costs and compensate resources more efficiently.
 - Concern 1 – NYISO does not propose to allocate reserve costs efficiently to large-contingency resources, which will likely shift \$15 to \$30 million per year of reserve charges to LSEs and lead to inefficient scheduling of the largest resources.
- Element 2 – NYISO will secure N-1 transmission constraints in NYC to long-term emergency limits using 10-minute reserves, replacing a procedure that would sometimes inefficiently reduce flows into NYC load pockets. The benefits of this element will grow if the Narrows and Gowanus generators retire.
- Element 3 – Zonal N-1-1 and N-1-0 constraints will be met with internal or imported reserves, replacing a requirement to meet them only with generation in the reserve zone. This will reduce production costs and compensate resource investors more efficiently.
 - Future Enhancement 1 – Most of the benefits of Element 3 will be achieved with more intrazonal reserve constraint modeling that the NYISO is not including in the initial implementation. This is tentatively planned after 2030.
 - Future Enhancement 2 – Element 3 would allow day-ahead modeling of N-1-1-0 requirements in NYC and Long Island, but this is not currently planned. Pricing and scheduling for these requirements in the day ahead market would be highly beneficial because most supplemental commitments in NYC are made for them.
 - Concern 2 – NYISO will allow non-firm and virtual imports to satisfy day-ahead reserve requirements, which will under-schedule supply and perpetuate reliance on out-of-market commitments.
 - Concern 3 – Local 30-minute reserve requirements will be based on Scheduled Load rather than forecast load, inflating day-ahead local reserve needs when day-ahead participants have incentives to over-schedule load. For example, when thunderstorms are expected near ConEd, NYISO must convert reserves to energy in real time before an actual contingency, which incentivizes over-scheduling downstate load day-ahead and causes NYISO to over-procure reserves.
 - Concern 4 – In some cases, reserves scheduled outside a reserve zone will not be fully deliverable after a contingency because a local transmission constraint limits the generator holding reserves. To limit reliability risk, NYISO plans to retain

²⁰² See *Summary of MMU Comments on NYISO's Dynamic Reserve Market Design Proposal*, Dec. 2024, [here](#).

some static local reserve requirements after implementing dynamic reserves, which will reduce the benefits of dynamic reserves.²⁰³

- Concern 5 – NYISO plans to allocate the congestion component of reserve clearing prices to LSEs statewide rather than to LSEs in the relevant reserve area. Hence, meeting local reserve needs met with imported reserves by producing more energy in the local area would be borne locally, while the costs of meeting local reserve needs with internal reserves would be allocated statewide.

Overall, shortcomings and deferred elements in the current Dynamic Reserves design eliminate most of the day-ahead market benefits where most reserve expenditures occur. The design will likely deliver more real-time benefits, but Concerns 1 and 4 also reduce these benefits. NYISO should consider design changes before moving forward.

2012-8: Operate PAR-controlled lines between New York City and Long Island to minimize production costs and create financial rights that compensate affected TOs.

Improving the operation of the PAR-controlled lines between New York City and Long Island (i.e., the 901 and 903 lines) could yield significant efficiency gains. These lines are scheduled under legacy wheeling agreements that pre-date open access transmission tariffs and NYISO markets.

In 2025, both lines were scheduled in the day-ahead market in the inefficient direction (from the high-priced area to the low-priced area) 78 percent of the time. We estimate this increased production costs by approximately \$12 million. It also increased New York State carbon dioxide emissions by 1.2 percent (433 thousand tons) and nitrous oxide emissions by 4.4 percent (498 tons).²⁰⁴

We recommend NYISO work with parties to the underlying wheeling agreements to explore changes or to identify how the agreements can be accommodated more efficiently in the markets. Because more efficient operation would financially benefit one party at another's expense, a financial settlement mechanism could compensate the party that would give up benefits under current operations. We discuss such a mechanism in Section III.I of the Appendix.

Broad Market Enhancements

2024-2: Evaluate potential reforms to enhance incentives for demand-side interconnection and participation in the NYISO market by: (a) allowing

²⁰³ See Market Issues Working Group presentation *Updates to NYC Reserve Requirements for CHPE*, dated March 16, 2026.

²⁰⁴ See Section V.E in the Appendix.

DERs to submit offer parameters comparable to generators, and (b) Coordinating with local entities regarding new load interconnections.

For more than a decade, much of the U.S. saw flat or declining electric load as energy efficiency and behind-the-meter solar offset demand growth. As these trends slow, electricity demand growth is expected to rise because of heating electrification, electric vehicle adoption, and the interconnection of new large loads such as data centers. These trends challenge centralized wholesale markets such as NYISO, which use efficient market incentives to maintain reliability in both operations and planning time horizons, enable rapid interconnection of new supply and load, and adapt to generator retirements driven by environmental policies.

These trends will require substantial wholesale market reforms in coming years. One key area is demand-side interconnection and market participation. Reforms would help soften the impact of large-scale supply and demand changes. Regulators have promoted demand-side participation and sought to facilitate new energy-intensive investments such as data centers. NYISO has improved demand-side access to the wholesale market, but important gaps remain.

First, NYISO has sought to transition capacity-selling loads from the SCR program (an emergency demand response program) to the DER participation model, but current rules impose significant burdens on DERs beyond those applied to generators that sell capacity. For example, generators can satisfy capacity obligations by offering into the day-ahead market with start-up notification time, minimum run time, and start-up cost so they can recover commitment costs if scheduled. DERs can sell capacity only if they are willing to be curtailable with little notice, without minimum duration, and without commitment cost parameters to ensure recovery of commitment costs. We recommend the NYISO consider reforms that would allow DERs to submit offer parameters that reflect their physical and economic characteristics in a manner comparable to generation resources.

Second, NYISO's planning department recognizes that many new load interconnections may not require firm service because they can be curtailable, can shift consumption away from tight-supply periods, or may rely on onsite backup generation rather than NYISO system resources.²⁰⁵ Local utilities sign interconnection agreements with new large load facilities, but it is unclear whether these agreements prevent non-firm loads from becoming firm without a new interconnection request. As a result, NYISO's reliability planning process lacks information on whether loads that are currently non-firm or curtailable (e.g., a crypto-mining operation) can later become firm (e.g., a traditional data center with firm energy requirements). Hence, it will be important for the NYISO to coordinate with local entities to: (a) improve the information that drives its planning processes and (b) ensure local entities are aware of the bulk power system

²⁰⁵ For example, see NYISO's *2024 Reliability Needs Assessment*, pages 13 and 34-37.

reliability impacts of each new load addition before it receives siting and interconnection approvals.

Enhance Planning Processes

2022-3: Improve transmission planning assumptions and metrics to better identify and fund economically efficient transmission projects. (Current Effort)

In recent years, NYISO transmission planning has been driven solely by the need to integrate expected future renewable resources under the Public Policy Transmission Process (PPTP). NYISO's Economic Planning Process focuses on long-term forecasting of the resource mix and congestion patterns (in the Outlook), which supports later PPTP project evaluation. Methodological deficiencies in benefit evaluation may cause NYISO-led public policy transmission solicitations to select projects that do not efficiently address the underlying need or are not the best among competing projects. We recommend the following enhancements to select more cost-effective projects in future solicitations:

- a) Update the Outlook methodology to better account for market incentives of renewable and storage resources;
- b) Evaluate economic and PPTN projects using a project case that reflects resource mix changes caused by including the project; and
- c) Estimate transmission project benefits based on NYISO market value.

Status: NYISO plans to address part “a” of this recommendation with enhancements to the method for placing new resources in the 2026 Outlook Report.

C. Discussion of Recommendations Made in Previous SOM Reports

During development of each State of the Market Report, we review progress on evaluating and/or implementing recommendations from prior reports. We typically remove a recommendation if NYISO has addressed its substance by changing an operating practice or filing market rule changes that the Commission has accepted (or that are largely uncontested). We may also remove a recommendation if implementation costs are materially higher than anticipated, underlying drivers have materially changed, or adoption is unlikely.

Recommendations removed since the 2024 SOM Report

We removed the following recommendations from the 2024 State of the Market Report:

2023-5: Update market processes to mitigate the risk of extreme pricing caused by inaccuracies in the Winter-Summer Ratio parameter. – NYISO recently filed its Winter Capacity Enhancements, which include provisions to address this issue when CHPE enters the market and changes the Winter-Summer Ratio for New York City.

2021-4: Improve capacity modeling and accreditation for specific types of resources. – NYISO has addressed this recommendation through several initiatives in recent years to implement marginal capacity accreditation using its resource adequacy assessment tools.

2021-1: Evaluate need for longer lead time reserve products to address increasing operational uncertainties. – We dropped this recommendation because New York State appears to be introducing intermittent resources and energy storage more slowly than prior plans, reducing the importance of the issues this recommendation would address.

2020-1: Consider enhancements to the scheduling of duct-firing capacity in the real-time market that more appropriately reflects its operational characteristics. – NYISO expects to largely address this recommendation by implementing the Duct Firing Improvements project, scheduled for deployment in 2026-Q2 (although this will likely be delayed until later in the year).

2013-1c: Evaluate locational marginal pricing of capacity (C-LMP) that minimizes the cost of satisfying planning requirements. – Recommendation 2022-4 supersedes this recommendation and likely has better prospects for implementation.

2012-1c: Grant financial capacity transfer rights between zones for market-based transmission upgrades that help satisfy planning reliability needs. – Recommendation 2022-4 supersedes this recommendation and likely has better prospects for implementation.

Analytic Appendix

2025 State of the Market Report
For the
New York ISO Markets

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I. MARKET PRICES AND OUTCOMES

The New York ISO operates a multi-settlement wholesale market system consisting of financially binding day-ahead and real-time markets for energy, operating reserves, and regulation (i.e., automatic generation control). Through these markets, the NYISO commits generating resources, dispatches generation, procures ancillary services, schedules external transactions, and sets market-clearing prices based on supply offers and demand bids. Additionally, the NYISO operates markets for transmission congestion contracts and installed capacity, which are evaluated in Sections III and I of the Appendix.

This section of the appendix summarizes the market results and performance in 2024 in the following areas:

- Wholesale market prices;
- Fuel prices and generation by fuel type;
- Fuel usage under tight gas supply conditions;
- Emissions from internal generators;
- Load levels;
- Day-ahead ancillary services prices;
- Price corrections;
- Day-ahead energy market performance; and
- Day-ahead ancillary services market performance.

A. Wholesale Market Prices

Figure A-1: Average All-In Price by Region

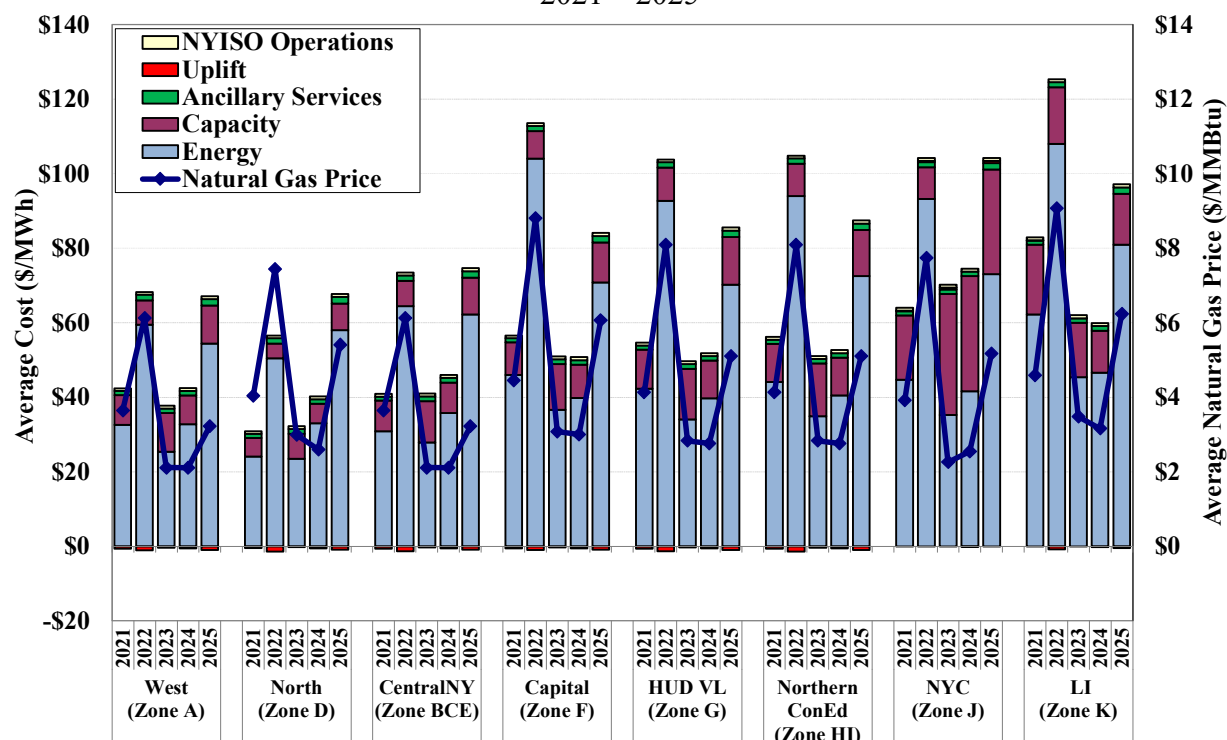
The first analysis displays the total costs of serving load from the NYISO markets as the all-in price for electricity. This value represents the sum of all wholesale market costs, including energy, uplift, capacity, ancillary services, and NYISO cost of operations. The all-in price is calculated for various locations in New York State, reflecting the substantial variability in capacity and energy prices by location. In this metric:

- **The energy component** is load-weighted average real-time energy prices.
- **The capacity component** is derived from clearing prices in the monthly spot auctions and capacity obligations in each area, allocated over energy consumption in that area.
- **The uplift component** includes both local and statewide uplift costs from Schedule 1 charges, allocated over the energy consumed in the area.
- **Ancillary services costs** are distributed evenly across all locations for purposes of this metric.

Figure A-1 shows the average all-in prices along with the average natural gas prices from 2021 to 2025 at the following eight locations: (a) the West Zone (i.e., Zone A); (b) the North Zone

(i.e., Zone D); (c) Central New York (i.e., Zones B, C, and E); (d) the Capital Zone (i.e., Zone F); (e) the Hudson Valley (i.e., Zone G); (f) Northern ConEd (i.e., Zones H and I), (g) New York City (i.e., Zone J); and (h) Long Island (i.e., Zone K). The majority of congestion in New York occurs between and within these regions.

Figure A-1: Average All-In Price by Region
2021 – 2025



Natural gas prices are based on the following gas indices (plus a transportation charge): (a) the Niagara index from December to March and Tennessee Zone 4 200L index during the rest of the year for the West Zone and Central New York; (b) the Iroquois Waddington index for the North Zone; (c) the minimum of Tennessee Zone 6 and Iroquois Zone 2 indices for the Capital Zone; (d) the average of Iroquois Zone 2 index and the TETCO M3 index for Hudson Valley and Northern ConEd; (e) the Transco Zone 6 (NY) index for New York City, and (f) the Iroquois Zone 2 index for Long Island.²⁰⁶ An incremental 6.9 percent tax rate is also reflected in the natural gas prices for New York City. An incremental 1 percent tax rate is reflected for Long Island.

Figure A-2: Day-Ahead Electricity and Natural Gas Costs

Figure A-2 shows load-weighted average natural gas costs and load-weighted average day-ahead energy prices in each month of 2025 for the seven locations shown in Figure A-2. The inset

²⁰⁶ The following transportation costs are included in the delivered prices for each region: (a) \$0.27 per MMBtu for Zones A through I, (b) \$0.20 per MMBtu for New York City, and (c) \$0.25 per MMBtu for Long Island.

table shows the annual averages of natural gas costs and LBMPs for 2024 and 2025. Although hydro and nuclear generators produce much of the electricity used by New York consumers, natural gas units usually set the energy price as the marginal unit, especially in Eastern New York.²⁰⁷

Figure A-2: Day-Ahead Electricity Prices and Natural Gas Costs
By Month, 2025

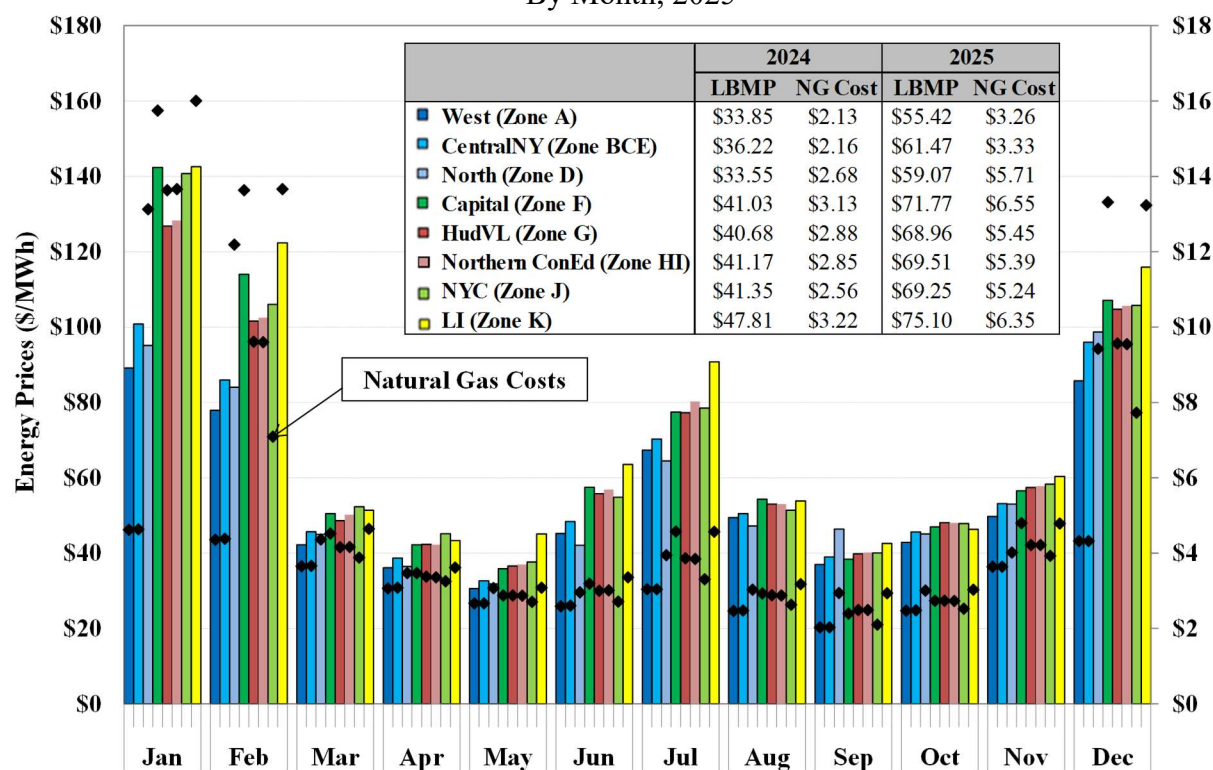


Figure A-3: Average Monthly Implied Marginal Heat Rate

The following figure summarizes the monthly average implied marginal heat rate, which highlights changes in electricity prices that are not driven by fuel price fluctuations.

The *Implied Marginal Heat Rate* is calculated as the day-ahead electricity price minus a generic unit Variable Operations and Maintenance (“VOM”) cost then divided by the fuel cost that includes the natural gas cost and greenhouse gas emission cost (i.e., RGGI Allowance Cost). For example, if the electricity price is \$40 per MWh, the VOM cost is \$3 per MWh, the natural gas price is \$3 per MMBtu, and the RGGI clearing price is \$13 per CO₂ allowance, then, the implied marginal heat rate would indicate that a generator with a 9.8 MMBtu per MWh heat rate is on the margin.²⁰⁸

²⁰⁷ The prevalence of natural gas units as the marginal resource is apparent from the strong correlation between LBMPs and natural gas prices, particularly in Eastern New York.

²⁰⁸ In this example, the implied marginal heat rate is calculated as $(\$40/\text{MWh} - \$3/\text{MWh}) / (\$3/\text{MMBtu} + \$13/\text{ton} * 0.06 \text{ ton/MMBtu emission rate})$, which equals 9.8 MMBtu per MWh.

Figure A-3: Average Monthly Implied Marginal Heat Rate
Day-Ahead Market, 2025

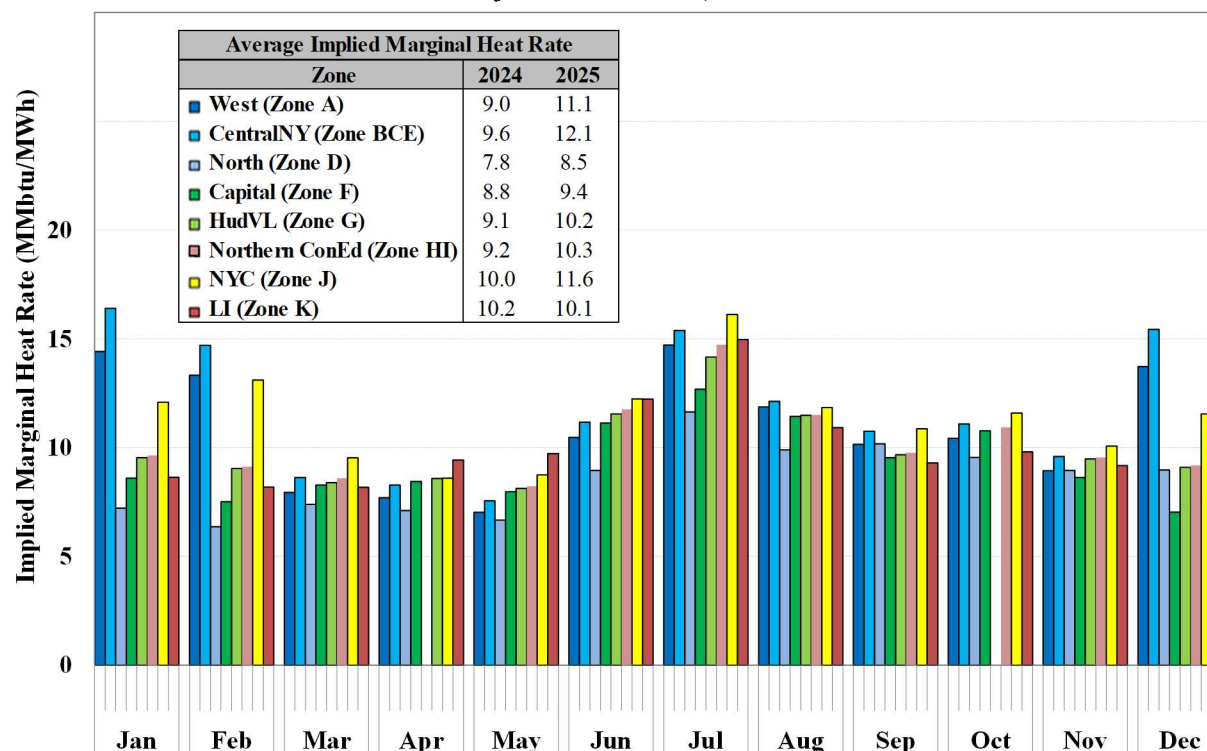


Figure A-3 shows the load-weighted average implied marginal heat rate in each month of 2024 for the eight locations shown in Figure A-1 and Figure A-2. The table in the chart shows the annual averages of the implied marginal heat rates in 2023 and in 2024 at these locations. By adjusting for variation in natural gas prices, the implied marginal heat rate shows more clearly the seasonal variation in electricity prices.

Figure A-4 – Figure A-5: Price Duration Curves and Implied Heat Rate Duration Curves

The following two analyses illustrate how prices varied across hours and locations. Figure A-4 shows eight price duration curves for 2025, one for each of the following locations: (a) the West Zone (i.e., Zone A); (b) the North Zone (i.e., Zone D); (c) Central New York (i.e., Zones B, C, and E); (d) the Capital Zone (i.e., Zone F); (e) the Hudson Valley region (i.e., Zone G); (f) Northern ConEd (i.e., Zones H and I); (g) New York City (i.e., Zone J); and (h) Long Island (i.e., Zone K). Each curve in Figure A-4 shows the number of hours (horizontal axis) when the load-weighted average real-time price in each region exceeded the corresponding price level (vertical axis). Additionally, the table in the chart shows the number of hours in 2025 at each location when the real-time price exceeded \$100, \$200, and \$500 per MWh.

The price duration curves show the distribution of prices in wholesale electricity markets, where a small number of hours exhibited very high prices that are typically associated with shortages.

Prices during shortages may rise to more than ten times the annual average price level, meaning that even a limited number of price spikes can significantly impact the average price level.²⁰⁹

Figure A-4: Real-Time Price Duration Curves by Region
2025

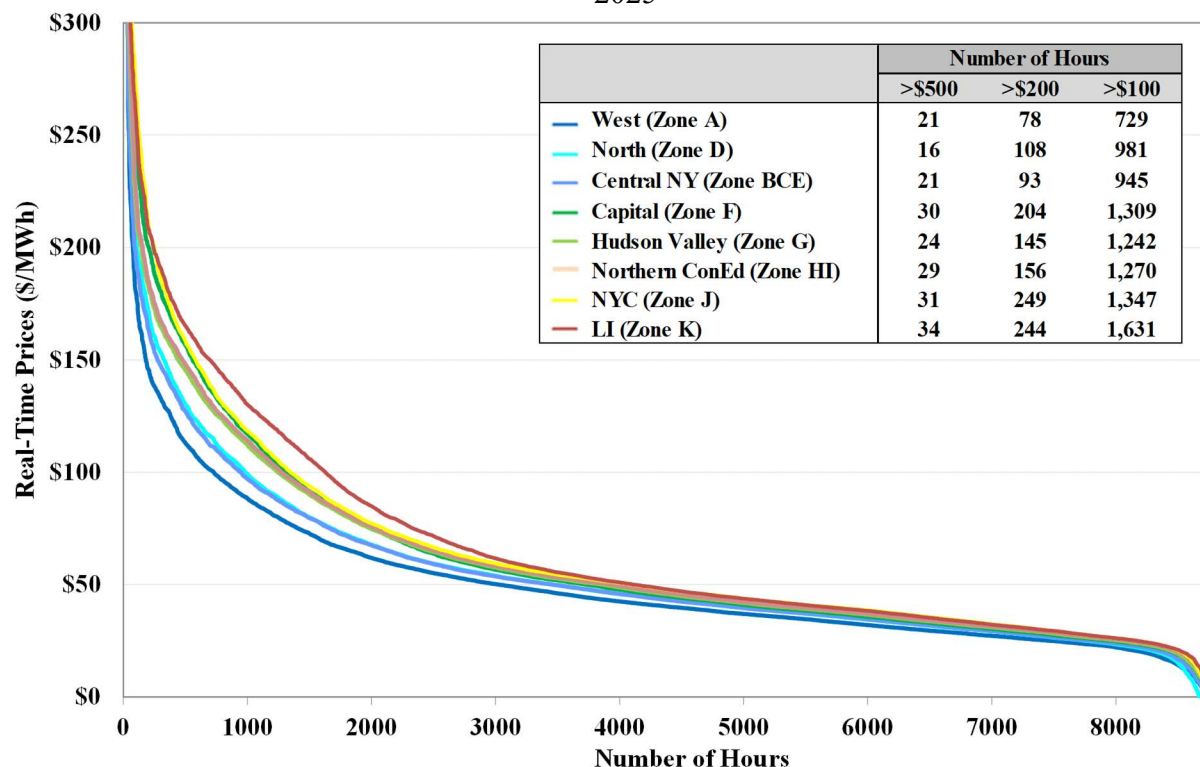
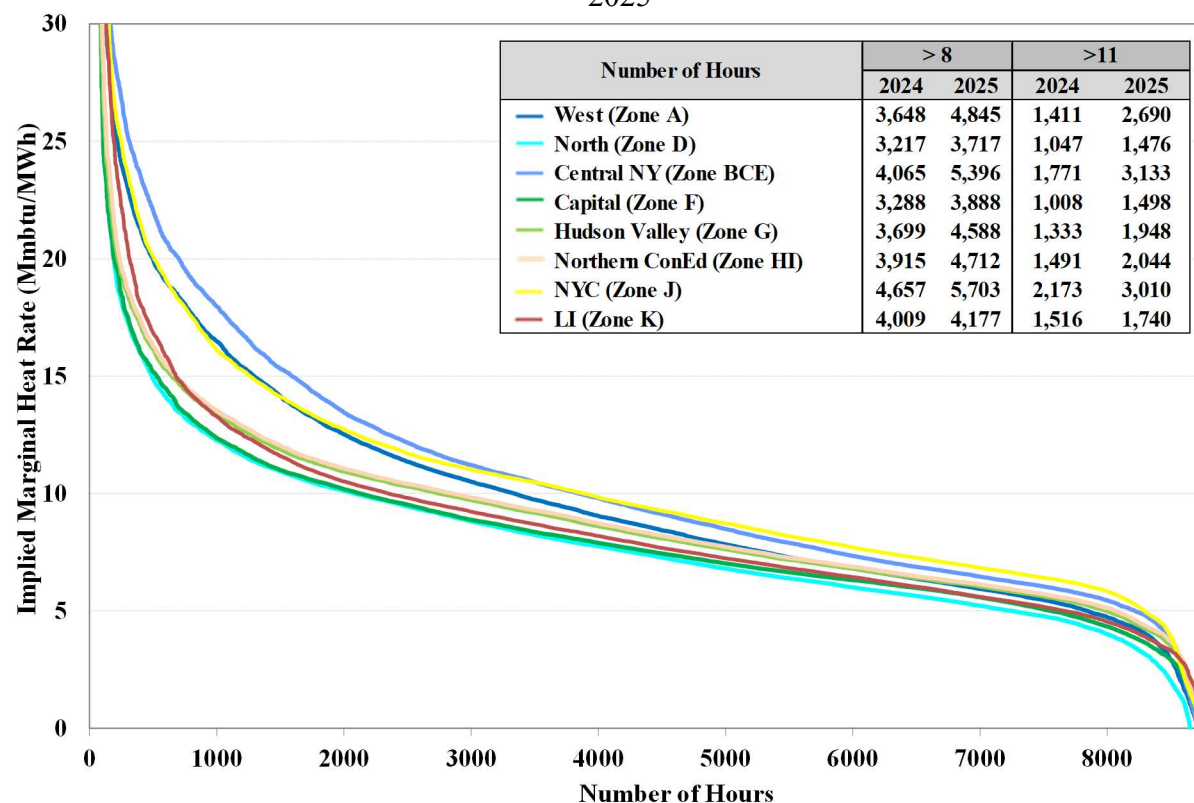


Figure A-5 shows the implied marginal heat rate duration curves for each location from the previous chart during 2025. Each curve shows the number of hours (horizontal axis) when the implied marginal heat rate for each sub-region exceeded the corresponding level (vertical axis). The calculation of the implied marginal heat rate follows the same methodology as in Figure A-3 except that this is based on real-time prices rather than day-ahead prices. The inset table compares the number of hours in each region when the implied heat rate exceeded 8 and 11 MMBtu per MWh between 2024 and 2025.

²⁰⁹ In other words, the distribution of energy prices across the year is “right skewed”, meaning that the average price is greater than the median price due to the impact of shortage pricing hours.

Figure A-5: Implied Heat Rate Duration Curves by Region
2025



B. Fuel Prices and Generation by Fuel Type

Figure A-6 to Figure A-9: Fuel Prices, RGGI Costs, and Generation by Fuel Type

Fluctuations in fossil fuel prices, especially natural gas prices, have been the primary driver of changes in wholesale electricity prices over the past several years.²¹⁰ This is because fuel costs account for most of the marginal production costs of fossil fuel generators.

Some generators in New York have dual-fuel capability, allowing them to burn either oil or natural gas. These generators usually burn the most economic fuel, which often translates to using natural gas as the default choice for most of the year. Situations may arise, however, where some generators opt to burn oil even if it is more expensive, due to specific circumstances or operational considerations.²¹¹ Since most large steam units can burn either oil or natural gas,

²¹⁰ Although much of the electricity generated in New York is from hydroelectric and nuclear generators, natural gas units are usually the marginal source of generation. Hence, natural gas prices more directly affect wholesale electricity prices.

²¹¹ For instance, natural gas may be difficult to obtain on short notice. In addition, New York City and Long Island reliability rules sometimes require that certain units burn oil to limit the exposure of the electric grid to possible disruptions in the supply of natural gas.

the effects of natural gas price spikes on electricity prices during periods of high volatility are partly mitigated by generators switching to oil.²¹²

Natural gas price patterns are normally consistent between different regions in New York, with eastern regions typically having a small premium in price compared to the western zones. However, bottlenecks on the natural gas system can lead to significant differences in delivered gas costs by area, particularly during peak winter conditions. This in turn can produce comparable differences in energy prices when network congestion occurs. The natural gas price differences generally emerge by pipeline and by zone. We track natural gas prices for the following pipelines/zones, which serve different areas in New York.

- Tennessee Zone 6 prices are representative of natural gas prices in the Capital Zone as well as in portions of New England;
- Transco Zone 6 (NY) prices are representative of natural gas prices in New York City;
- Iroquois Zone 2 prices are representative of natural gas prices in the Capital Zone and Long Island;
- TETCO M3 prices and Iroquois Zone 2 are representative of natural gas prices in various locations of the Lower Hudson Valley region; and
- Tennessee Zone 4 200L prices are representative of natural gas prices in portions of Western New York.

Figure A-6 shows average natural gas and fuel oil prices by month from 2022 to 2025. The table compares the annual average fuel prices for these four years.²¹³

Reginal Greenhouse Gas Initiative (RGGI) allowance prices have increased substantially over the past few years, contributing significantly to the costs of electricity production in 2025. In Figure A-7, the upper portion shows monthly RGGI allowance prices 2020 to 2025, while the lower portion illustrates the equivalent energy costs, expressed in \$ per MWh, for hypothetical generating units with heat rates of 7,500 Btu/kWh (once for gas emission rates and again for oil emission rates) and 10,000 Btu/kWh, respectively.²¹⁴

Figure A-8 shows the quantities of generation by fuel type in seven regions of New York in each quarter of 2025 as well as for all the NYCA.²¹⁵ The table in the chart shows annual average generation by fuel type from 2023 to 2025.

²¹² Conventional steam units that have dual-fuel capability are required to burn No. 2 oil (ULSD) in New York City, but they generally burn No. 6 residual fuel oil in other areas.

²¹³ These are index prices that do not include transportation charges or applicable local taxes.

²¹⁴ The equivalent energy cost equals RGGI price * 0.06 short ton/MMBtu * unit heat rate for a generator burning natural gas. The oil emission rate is assumed to be 0.08 short ton/MMBtu.

²¹⁵ Pumped-storage resources in pumping mode are treated as negative generation. The “Other” category includes methane, refuse, solar, and wood.

Figure A-6: Monthly Average Fuel Index Prices
2022 – 2025

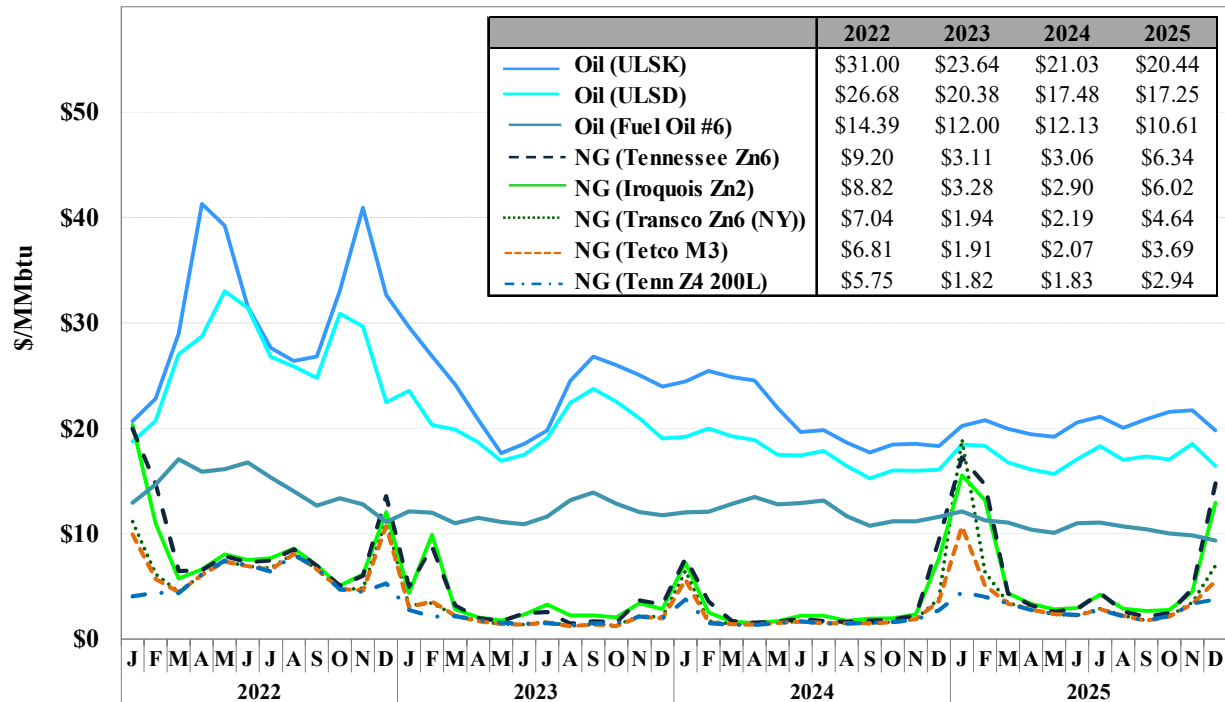


Figure A-7: RGGI Allowances Prices and Equivalent Energy Cost
By Month, 2020 - 2025

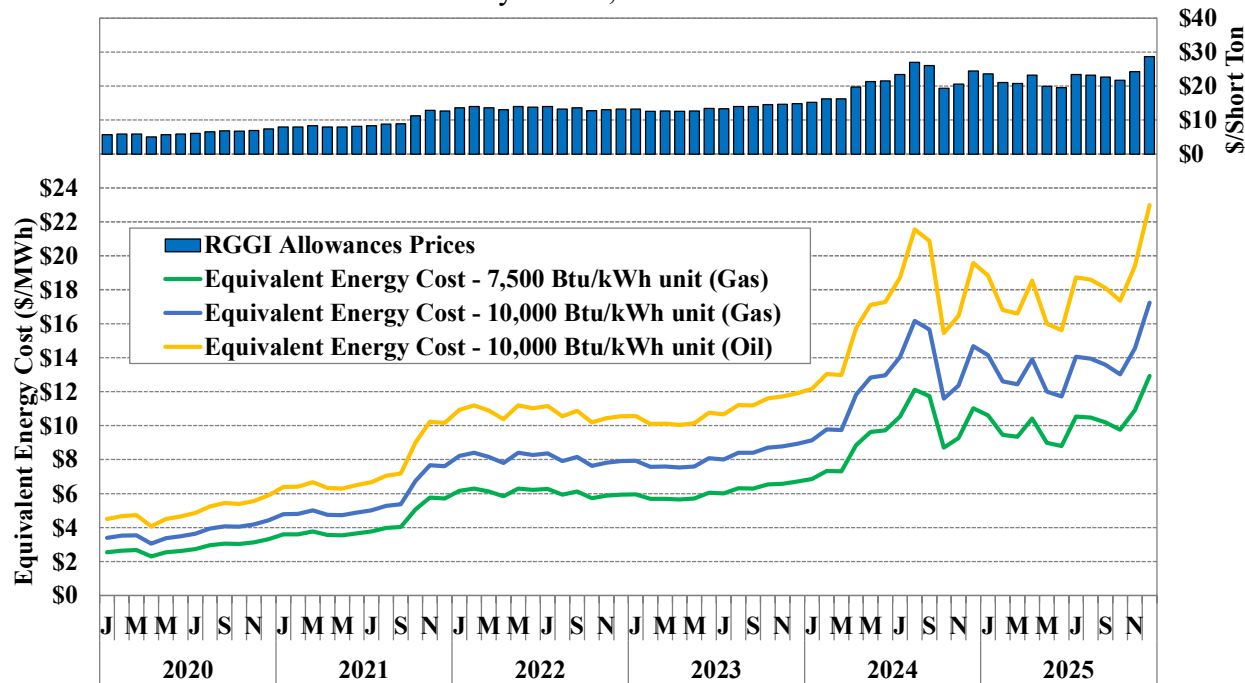


Figure A-8: Generation by Fuel Type in New York
By Quarter by Region, 2025

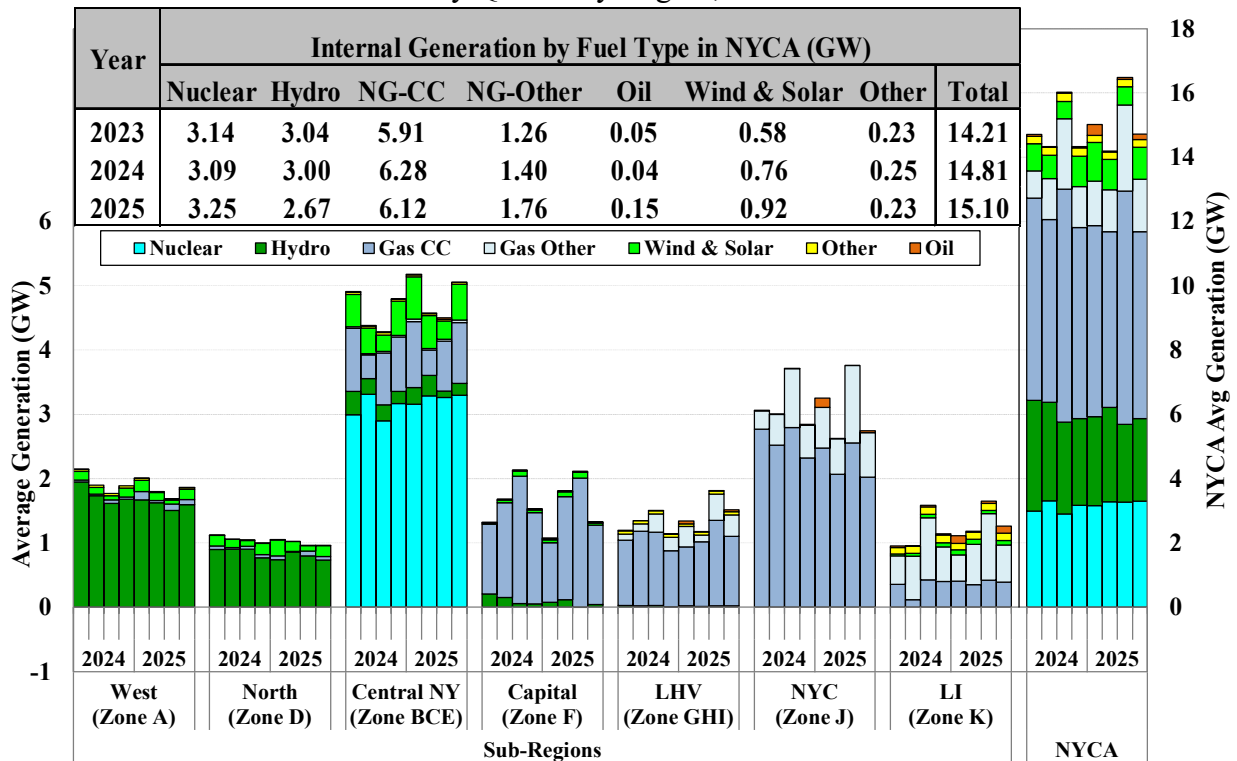
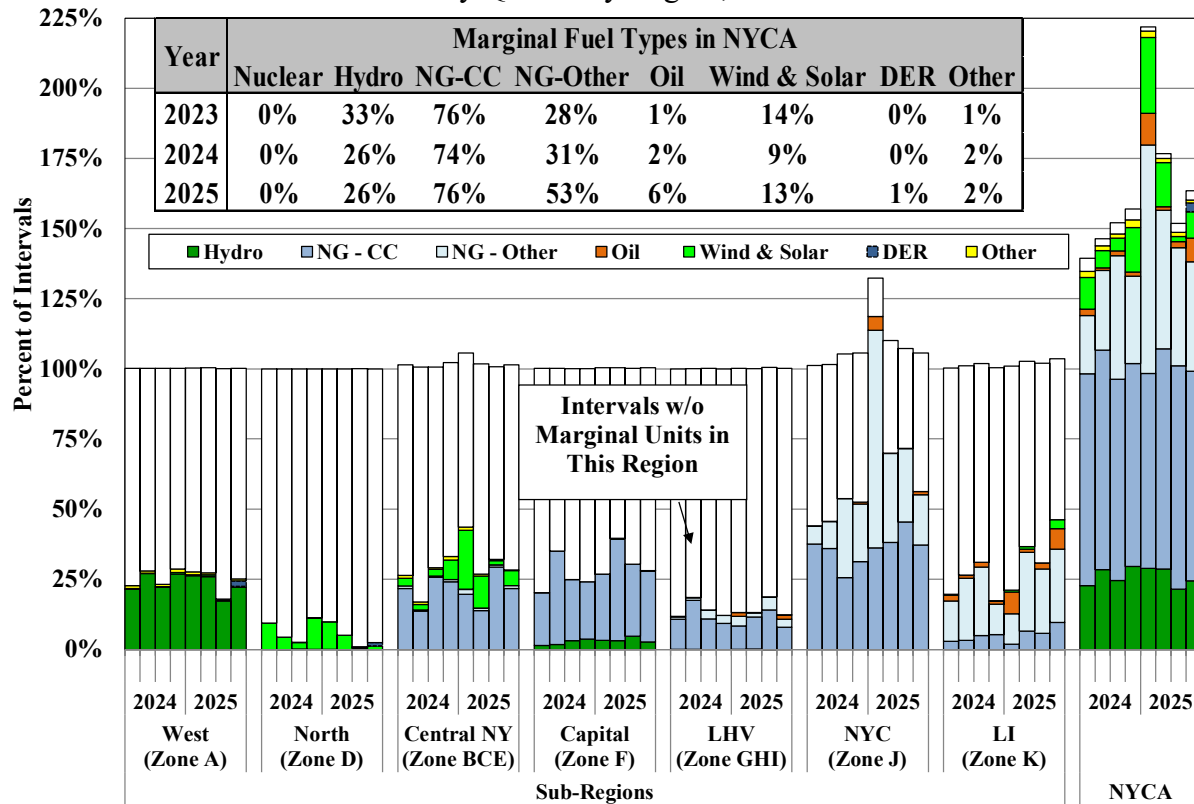


Figure A-9 summarizes how frequently each fuel type was on the margin and setting real-time energy prices in New York State and in each region of the state during 2025. The table in the chart shows annual statistics by fuel type from 2023 to 2025. More than one type of unit may be marginal in an interval, particularly when a transmission constraint is binding (different fuels may be marginal in the constrained and unconstrained areas). Hence, the total for all fuel types may be greater than 100 percent. For example, if hydro units and gas units were both on the margin in every interval, the total frequency shown in the figure would be 200 percent. When no unit is on the margin in a particular region, the LBMPs in that region are set by: (a) generators in other regions in most intervals; or (b) shortage pricing of ancillary services or transmission constraints in a small share of intervals.

The fuel type for each generator in both Figure A-8 and Figure A-9 is based on its actual fuel consumption reported to the U.S. Environmental Protection Agency (“EPA”) and the U.S. Energy Information Administration (“EIA”).

Figure A-9: Fuel Types of Marginal Units in the Real-Time Market in New York
By Quarter by Region, 2025



C. Fuel Usage Under Tight Gas Supply Conditions

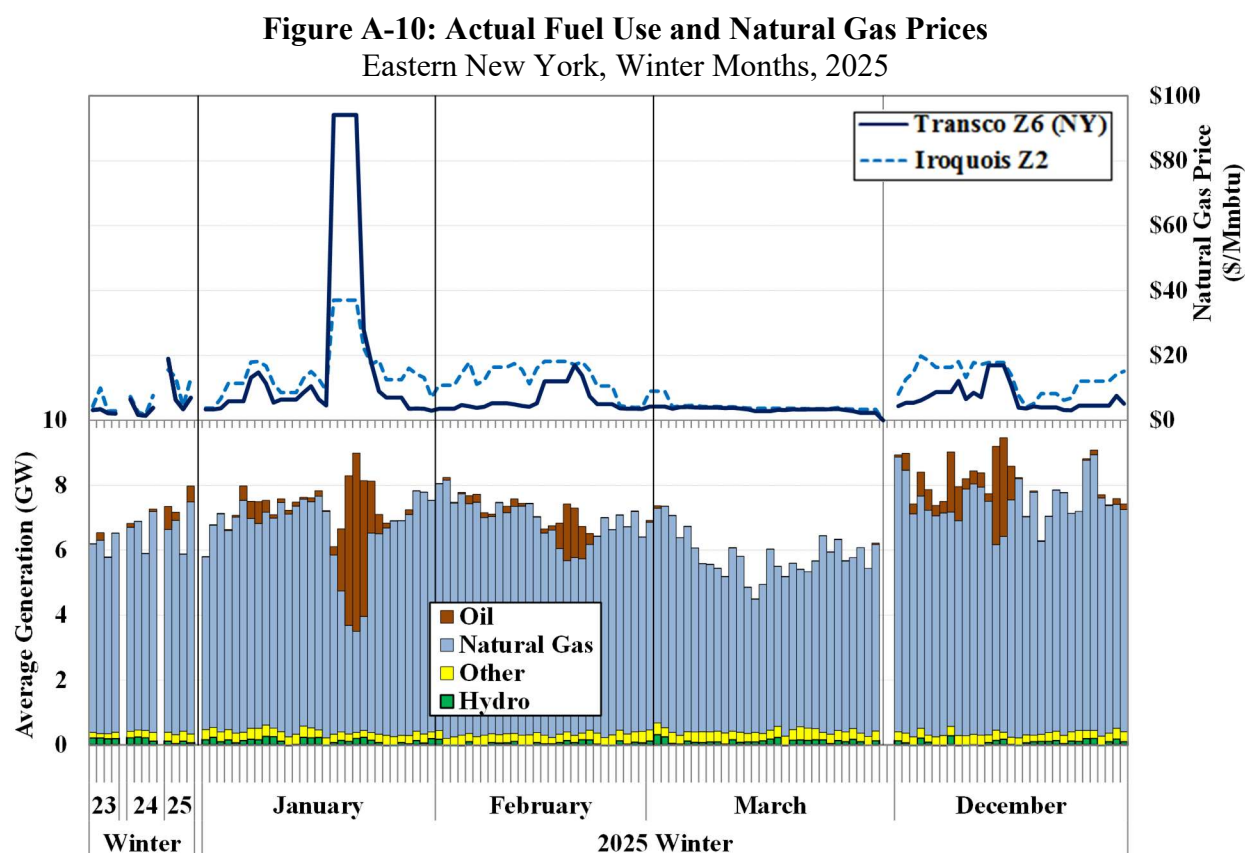
The supply of natural gas is usually tight in the winter season due to increased demand for heating. Extreme weather conditions often lead to high and volatile natural gas prices. A large share of generators in Eastern New York has dual-fuel capability, allowing them to switch to an alternative fuel when natural gas becomes expensive or unavailable. However, the increase in oil-fired generation during such periods may be limited by several factors, including:

- Not having the necessary air permits;
- Not having oil-firing equipment in serviceable condition;
- Low on-site oil inventory;
- Physical limitations and gas scheduling timeframes that may limit the flexibility of dual-fueled units to switch from one fuel to the other; and
- NOx emissions limitations.

This subsection examines actual fuel usage in the winter of 2025, focusing on the portion of the year where the supply of natural gas is likely to be tight. This has historically had a big impact on the system operations, especially in Eastern New York.

Figure A-10: Actual Fuel Use and Natural Gas Prices in the Winter

Figure A-10 summarizes the average hourly generation by fuel consumed in Eastern New York daily during the winter months of 2025 (including the months of January, February, March, and December).



The figure shows actual generation for the following fuel categories: (a) oil; (b) natural gas; (c) hydro; and (d) all other fuel types as a group. In addition, the figure shows the day-ahead natural gas price index for Iroquois Zone 2 and Transco Zone 6 (NY). The figure also compares these quantities by month for the same four-month period between 2023 and 2025. Each day in the chart represents a 24-hour gas day, which starts from 10 am on each calendar day and ends at 10 am on the next calendar day.

D. Emissions from Internal Generation

Power plants generate three main air pollutants when generating electricity: sulfur dioxide (SO₂), nitrogen oxides (NO_x), and carbon dioxide (CO₂). These emissions from electricity generation vary by type of fuel, energy technology, and power plant efficiency and have declined substantially since the inception of the NYISO markets. Policy makers have set up aggressive agenda in recent years for an ambitious clean energy transition from conventional energy resources. It is important for the NYISO markets to provide strong and clear incentives to attract new technologies and help integrate clean energy resources. This subsection examines the

emission levels of the three major pollutants from internal generation resources in the NYISO markets.

Figure A-11: Historical Emissions by Quarter in NYCA

Figure A-11 shows the historical trend of annual total emissions since 2000 in the NYISO footprint for CO₂, NO_x, and SO₂ by quarter.

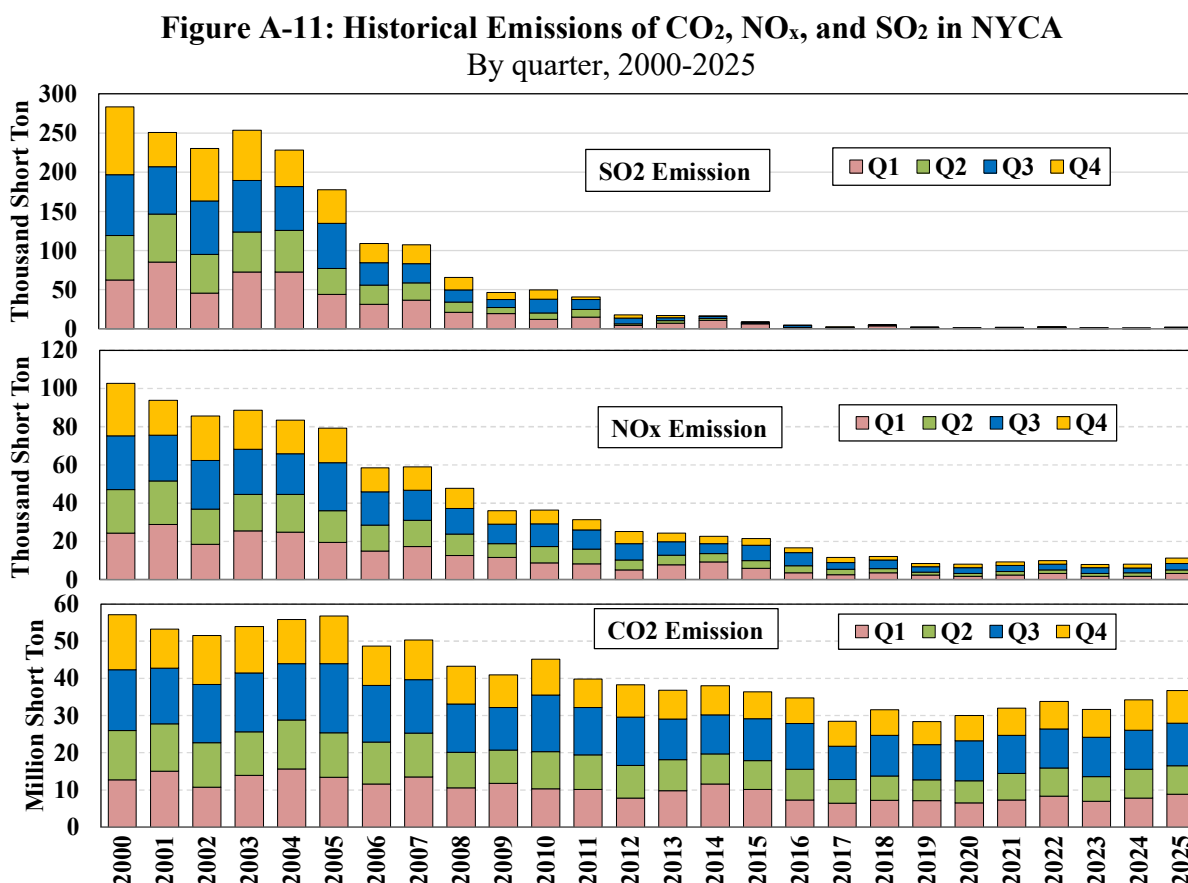


Figure A-12 - Figure A-13: Emissions by Region by Fuel Type

The following two figures show quarterly emissions across the system by generation fuel type for CO₂ and NO_x, respectively. Emission values are given for seven regions as well as the whole system for 2024 and 2025. The emission tonnage is given by aggregating the total pollution from operations on the various fossil fuel types for each month of the quarter. The inset table in each chart provides summary data on the total tonnage of emissions by fuel type.

Figure A-12: CO₂ Emissions by Region by Fuel Type
by quarter, 2024-2025

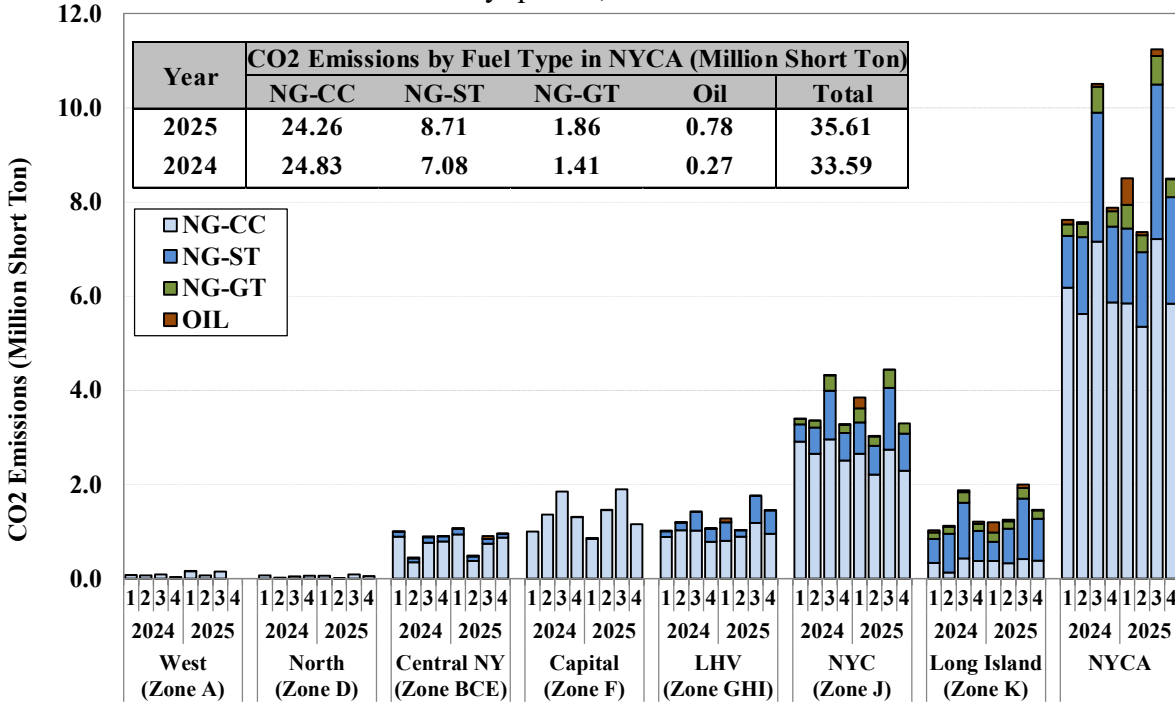
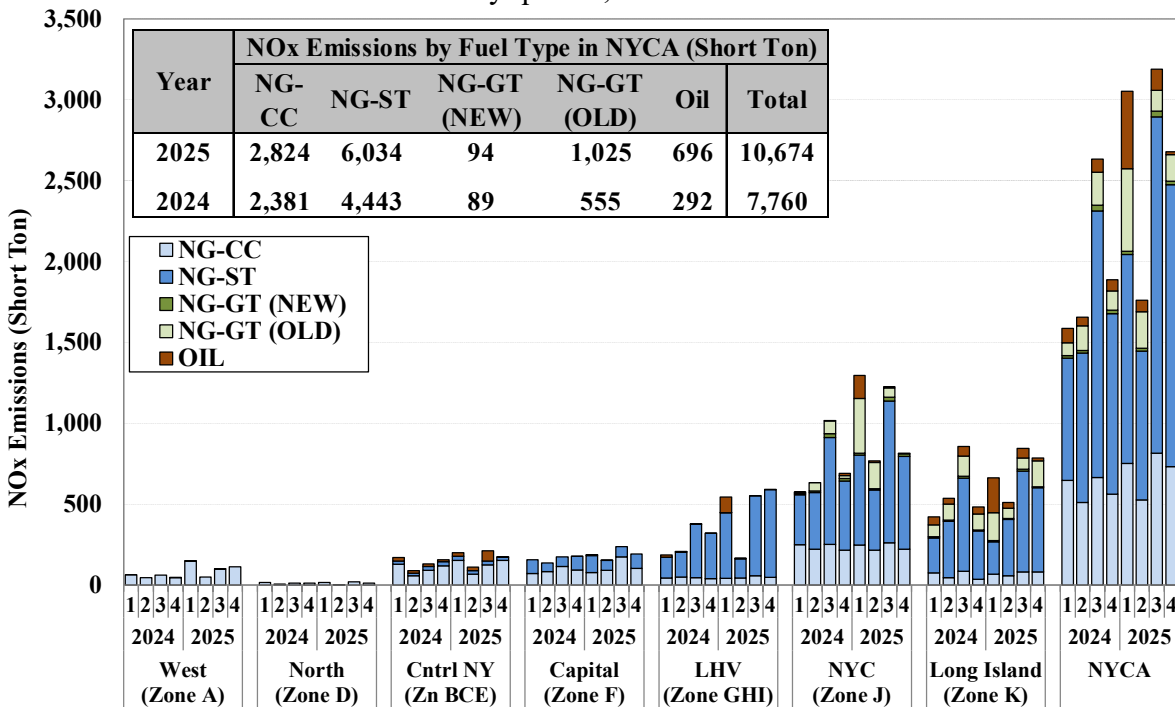


Figure A-13: NO_x Emissions by Region by Fuel Type
by quarter, 2024-2025

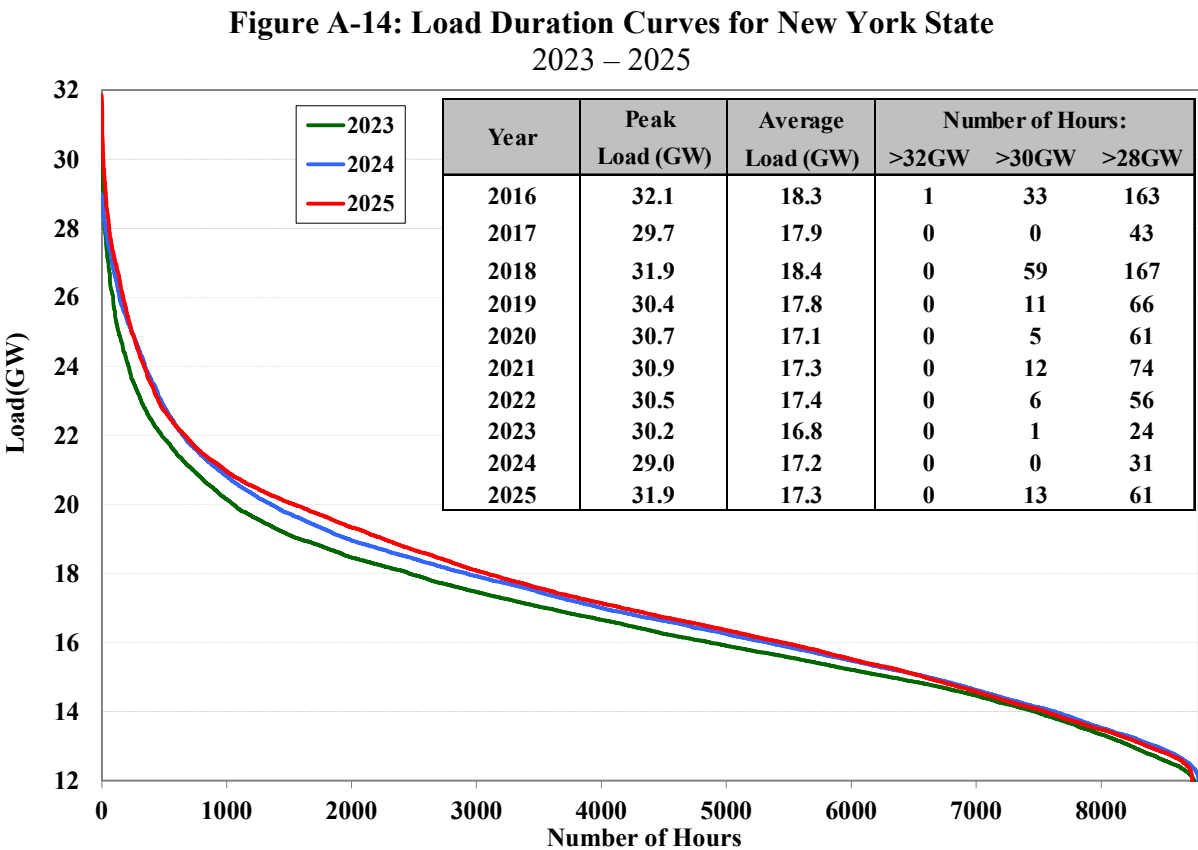


E. Load Levels

Figure A-14: Load Duration Curves for New York State

The interaction between electric supply and consumer demand is another key driver of price movements in New York. Since year-to-year changes in supply are usually small, fluctuations in electricity demand explain much of the short-term variations in electricity prices. The highest-load hours are particularly important, as they account for a disproportionately large share of both market costs to consumers and revenues to generators.

Figure A-14 presents load duration curves that illustrate demand variation over the past three years. These curves show the number of hours (horizontal axis) in which the statewide load was greater than or equal to the corresponding level (vertical axis). Additionally, the inset table provides the annual average load levels for the past ten years and the number of hours in each year when the system load exceeded 28, 30, and 32 GW.



F. Day-Ahead Ancillary Services Prices

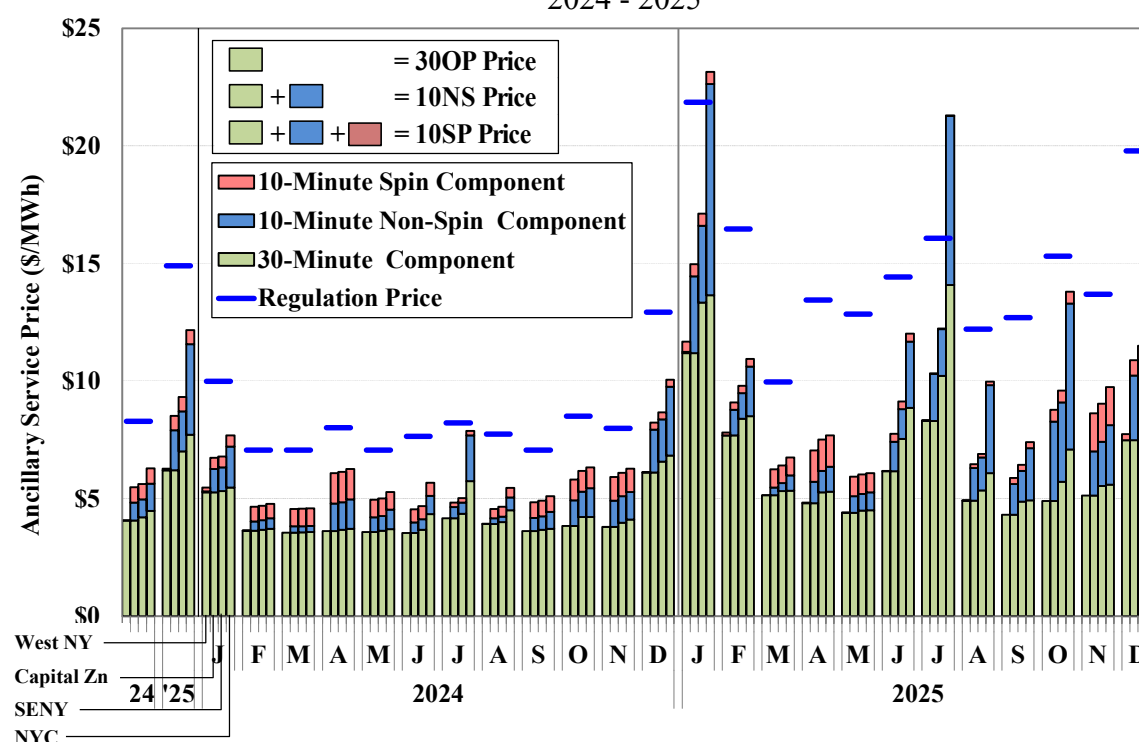
Figure A-15: Day-Ahead Ancillary Services Prices

The NYISO schedules resources to provide energy, operating reserves, and regulation service in the day-ahead and real-time markets. The NYISO co-optimizes the scheduling of these products such that the combined cost of all products is minimized. Given that available supplies must

satisfy energy demand and ancillary services requirements simultaneously, energy and ancillary services prices both reflect the costs to the system of diverting resources to provide ancillary services that would otherwise provide energy. Hence, ancillary services prices generally rise and fall with the price of energy because it influences the level of these opportunity costs.

NYISO has four market-based ancillary services products: 10-minute spinning reserves, 10-minute total reserves, 30-minute reserves, and regulation. In addition, the NYISO has locational reserve requirements that result in differences between Western, Eastern, Southeast New York and New York City reserve prices. The figure shows the average day-ahead prices for these four ancillary services products in each month of 2024 and 2025. The prices are shown separately for the following four distinct regions: (a) New York City, (b) Southeast New York (including Zones G-I and Zone K); (c) the Capital Zone (Zone F, in Eastern New York but outside Southeast New York); and (d) West New York (including Zones A-E).

Figure A-15: Day-Ahead Ancillary Services Prices
2024 - 2025



The stacked bars show three price components for each region: the 10-minute spinning component, the 10-minute non-spin component, and the 30-minute component, each representing the cost of meeting applicable reserve requirements. Take New York City as an example:

- The 30-minute component represents the cost to simultaneously meet the 30-minute reserve requirements for New York City, Southeast New York, East New York, and NYCA;
- The 10-minute non-spin component represents the cost to simultaneously meet the 10-minute total reserve requirements for New York City, East New York and NYCA (Southeast New York does not have a separate 10-minute total reserve requirement); and

- The 10-minute spinning component represents the cost to simultaneously meet the 10-minute spinning reserve requirements for East New York and NYCA (New York City and Southeast New York do not have separate 10-minute spinning reserve requirements).

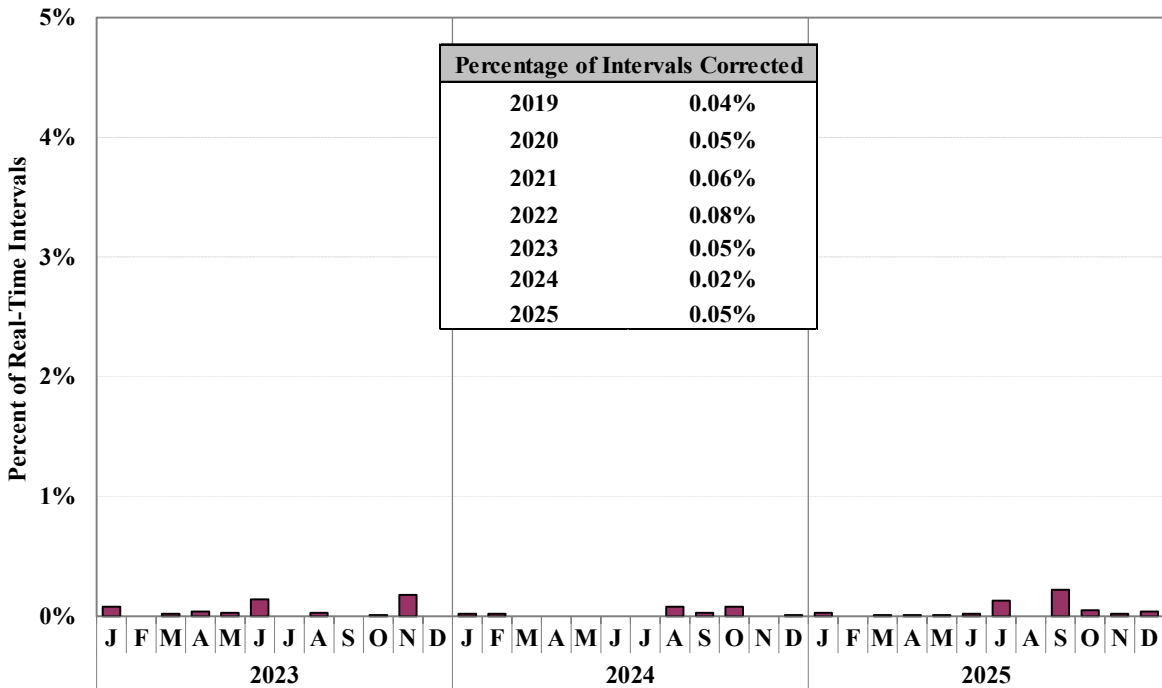
Therefore, in the figure, the 30-minute reserve price in each region equals its 30-minute component, the 10-minute non-spin reserve price equals the sum of its 30-minute component and 10-minute non-spin component, and the 10-minute spinning reserve price equals the sum of all three price components. The blue dashes give the day-ahead regulation capacity prices for the system.

G. Price Corrections

Figure A-16: Frequency of Real-Time Price Corrections

All real-time energy markets are subject to some level of price corrections to account for metering errors and other data input problems. Moreover, price corrections are required when flaws in the market operations software or operating procedures lead prices to be calculated erroneously. Accurate prices are critical for settling market transactions fairly and sending reliable real-time price signals. Less frequent corrections reduce administrative burdens and uncertainty for market participants. Hence, it is important to resolve problems that lead to price corrections quickly to maximize price certainty.

Figure A-16: Frequency of Real-Time Price Corrections
2023 - 2025



The figure summarizes the frequency of price corrections in the real-time energy market in each month from 2023 to 2025. The table in the figure indicates the change in the frequency of price corrections over the past several years. Price corrections continue to be very infrequent.

H. Day-Ahead Energy Market Performance

The day-ahead market allows participants to make forward purchases and sales of power for delivery in real-time. Participants can use the day-ahead market to hedge risks associated with the real-time market, and the system operator uses day-ahead bids and offers to improve the commitment of resources. Similarly, loads can insure against price volatility in the real-time market by purchasing in the day-ahead market. Suppliers can avoid the risk of starting-up their generators on an unprofitable day since the day-ahead auction market will only accept their offers when commitments are profitable. In addition to the value it provides individual market participants, perhaps the greatest value of the day-ahead market is that it coordinates the overall commitment of resources to satisfy the next day's needs at least cost.

In a well-functioning system with day-ahead and real-time markets, we expect that day-ahead and real-time prices will not systematically diverge from one another. If day-ahead prices were predictably higher than real-time prices, buyers would increase purchases in real-time. Alternatively, if day-ahead prices were foreseeably lower than real-time prices, buyers would increase purchases day-ahead (vice versa for sellers).

Price convergence is desirable because it promotes efficient commitment of generating resources, procurement of fuel, and scheduling of external transactions. Persistent differences between day-ahead and real-time prices can undermine incentives for suppliers to offer at marginal cost in the day-ahead market. We expect random variations from unanticipated changes in supply and demand between the two markets on an hour-to-hour basis, but persistent systematic differences between day-ahead and real-time prices would raise potential concerns.

Figure A-17 & Figure A-18: Average Day-Ahead and Real-Time Energy Prices

In general, day-ahead prices are based on the expectations of real-time market outcomes and are influenced by several uncertainties. First, demand can be difficult to forecast with precision and the availability of supply may change due to forced outages or numerous other factors. For example, the operators may commit additional generation for reliability after the day-ahead market, increasing the supply available to the real-time market. Second, special operating conditions, such as thunderstorm alerts, may alter the capability of the transmission system in ways that are difficult to arbitrage in day-ahead markets. Accordingly, day-ahead prices reflect the probability-weighted expectation of infrequent high-priced events in the real-time market.

Figure A-17 and Figure A-18 compare day-ahead and real-time energy prices in West Zone, Central Zone, North Zone, Capital Zone, and Hudson Valley, New York City, and Long Island.

Figure A-17: Average Day-Ahead and Real-Time Energy Prices in Western New York
West, Central, and North Zones – 2025

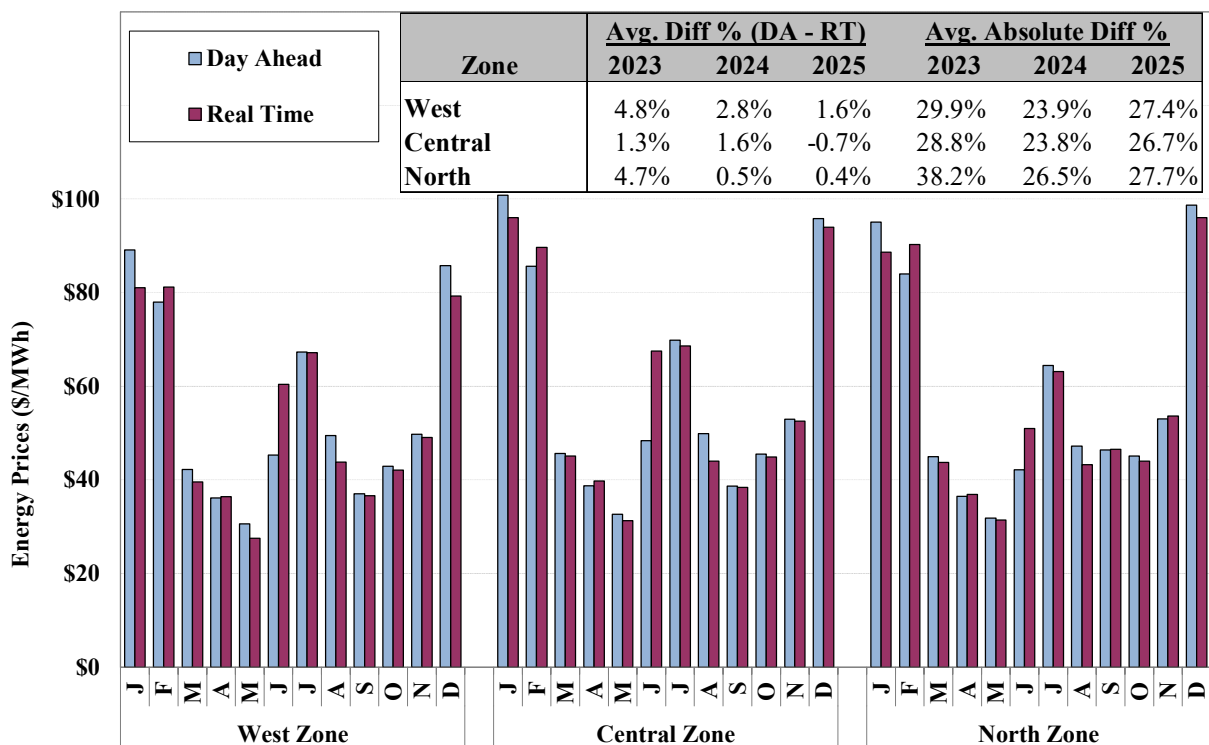
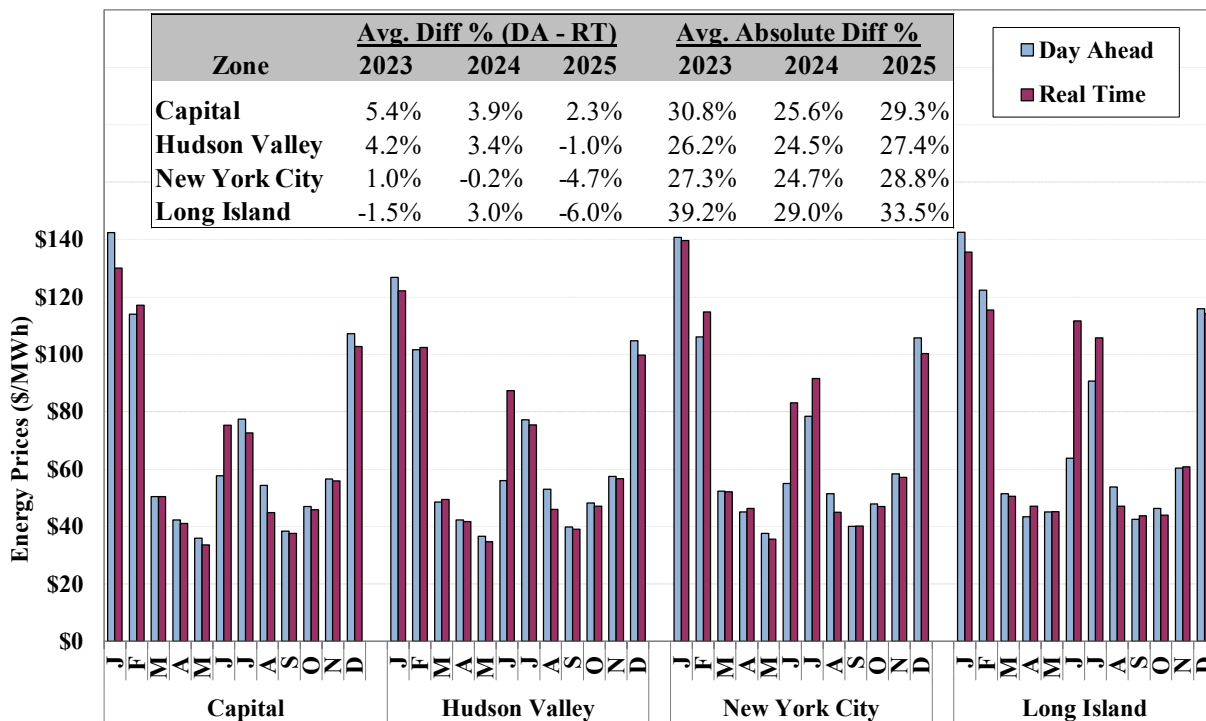


Figure A-18: Average Day-Ahead and Real-Time Energy Prices in Eastern New York
Capital, Hudson Valley, New York City, and Long Island – 2025



The figures are intended to reveal whether there are persistent systematic differences between the load-weighted average day-ahead prices and real-time prices at key locations in New York. The bars show average monthly day-ahead and real-time prices weighted on the hourly day-ahead load in each zone. The inset tables report the percentage difference between the average day-ahead price and the average real-time price, as well as the average absolute value of the difference between hourly day-ahead and real-time prices in the past three years. The latter metric measures the typical difference between the day-ahead and real-time prices in each hour, regardless of which is higher. This metric is substantially affected by real-time price volatility.

Transmission congestion can lead to a wide variation in nodal prices within a zone, while the price of each zone is a load-weighted average of the nodal prices in the zone. Hence, the pattern of intrazonal congestion may differ between the day-ahead market and the real-time market, leading to poor convergence at individual nodes even though convergence is good at the zonal level.

The pattern of intrazonal congestion may change between the day-ahead market and the real-time market for many reasons:

- Generators may change their offers after the day-ahead market. This is common during periods of fuel price volatility or when natural gas is more easily procured day-ahead.
- Generators may be committed or de-committed after the day-ahead market, changing the pattern of transmission flows.
- Constraint limits used to manage congestion may change from the day-ahead market to the real-time market.
- Transmission constraints that are sensitive to the level of demand may become more or less acute after the day-ahead market due to differences between expected load and actual load.
- Transmission forced outages, changes in the scheduled transmission maintenance, and differences in phase angle regulator settings can result in different congestion patterns.

In general, virtual trading and price-sensitive load bidding help improve convergence by facilitating arbitrage between day-ahead and real-time prices. But the NYISO is currently unable to allow market participants to submit either virtual trades or price sensitive load bids at the load pocket level or a more disaggregated level. Thus, good convergence at the zonal level may mask a significant lack of convergence within the zone.

I. Day-Ahead Reserve Market Performance

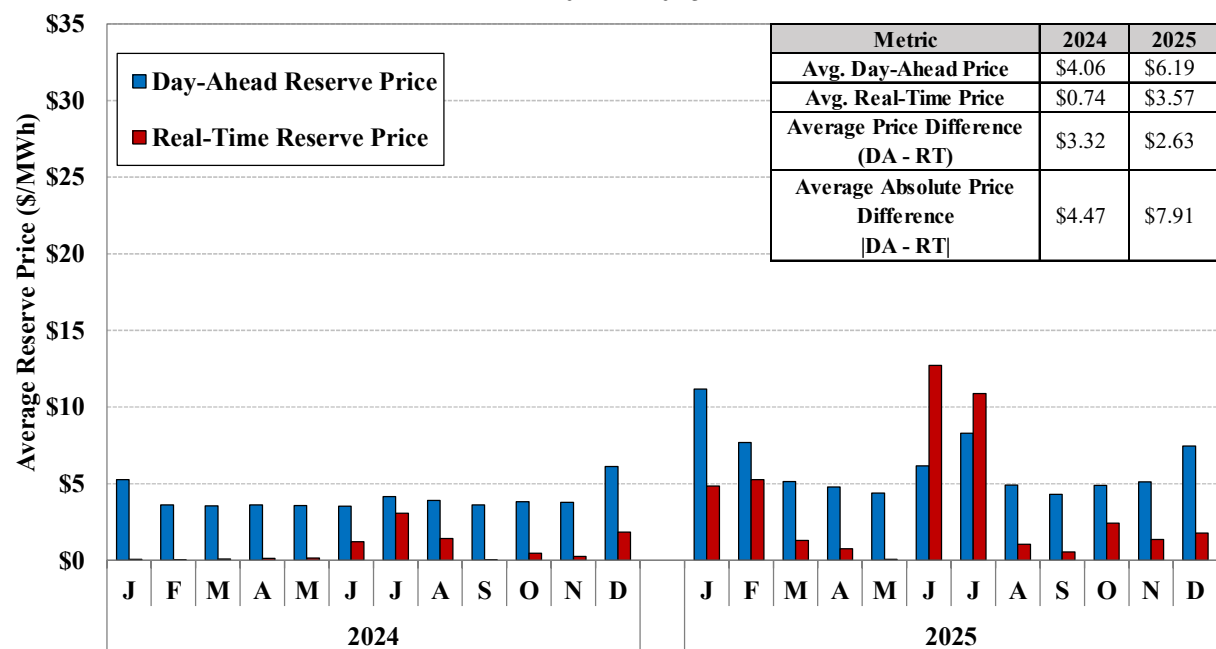
The NYISO co-optimizes the scheduling of energy, operating reserves, and regulation service such that the combined production cost of all products is minimized in the day-ahead and real-time markets. The energy and ancillary services markets place demand on the same supply resources, so prices for energy and ancillary services are highly correlated, and scarcity in the energy market is generally accompanied by a scarcity of ancillary services. As in the day-ahead energy market, a well-performing day-ahead ancillary service market will produce prices that converge well with real-time market prices.

In the market for energy, virtual trading improves convergence between day-ahead and real-time prices, which helps the ISO commit an efficient quantity of resources in the day-ahead market. In the ancillary services markets, on the other hand, only ancillary services suppliers directly participate and no virtual trading of ancillary services is allowed. Procurement of ancillary services is managed by the ISO, which obtains the same amounts of ancillary services in the day-ahead and real-time markets based on reliability criteria and without regard to price. Therefore, when systematic differences arise between day-ahead and real-time ancillary services prices, ancillary services suppliers are the only entities able to arbitrage them and improve convergence.

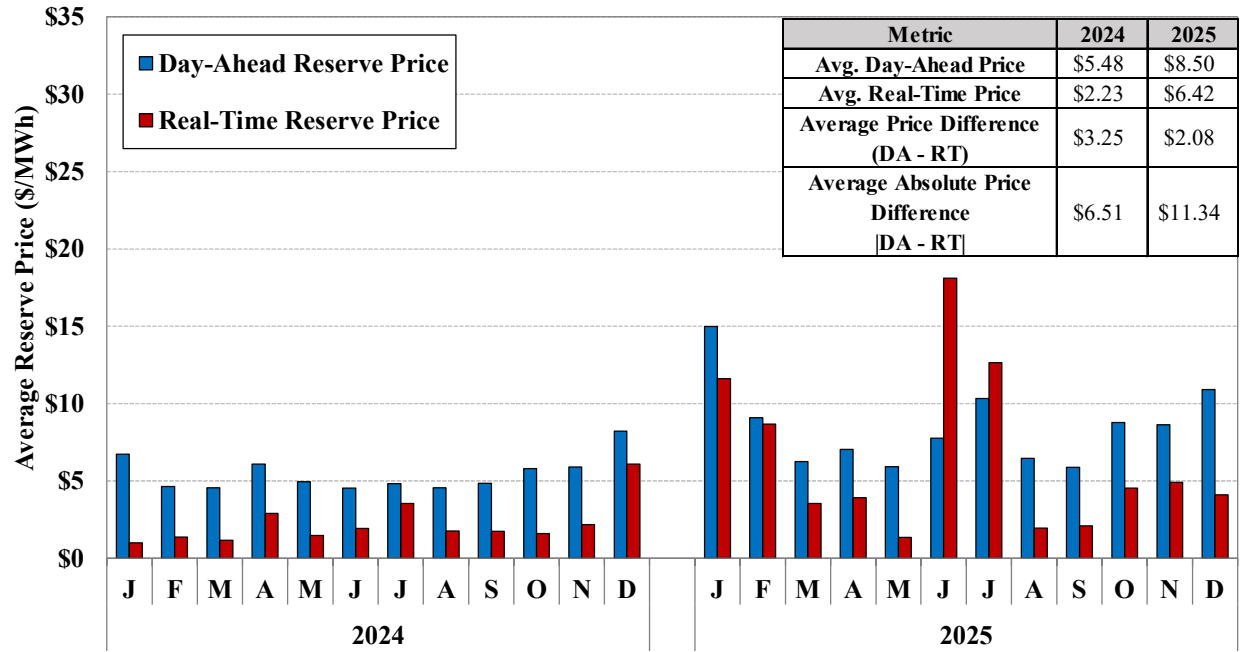
Figure A-19 to Figure A-21: Distribution of day-ahead price premiums for reserves

To evaluate the performance of the day-ahead ancillary service markets, the following three figures show the monthly day-ahead and real-time average prices for: (a) Western 30-minute reserve prices; (b) Eastern 10-minute spinning reserve prices; and (c) Eastern 10-minute non-spin reserve prices. These prices are shown for each month of the past two years. The inset table for each chart shows the annual averages for each year of: (a) the average day-ahead price; (b) the average real-time price; (c) the difference between the average day-ahead price and the average real-time price; and (d) the average absolute difference between the day-ahead price and the real-time price. Average absolute difference between the two prices provides a better metric for how consistent the convergence between day-ahead and real-time prices is than the simple average.

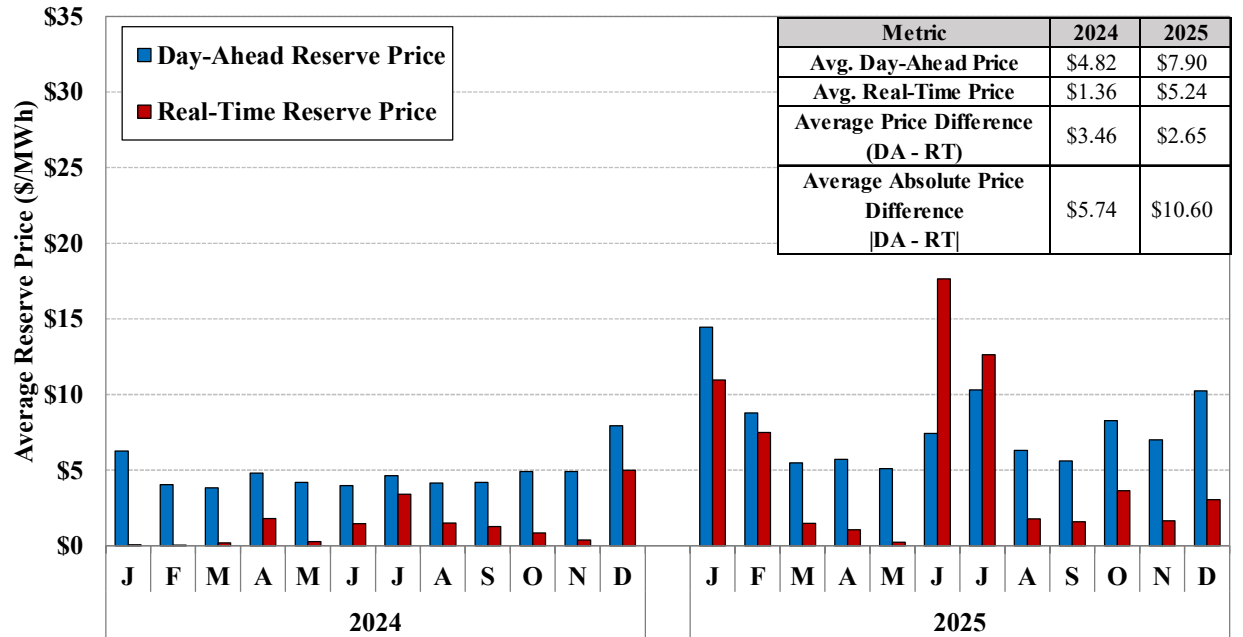
Figure A-19: Day-Ahead Premiums for 30-Minute Reserves in West New York
2024 – 2025



**Figure A-20: Day-Ahead Premiums for 10-Minute Spinning Reserves in East New York
2024 – 2025**



**Figure A-21: Day-Ahead Premiums for 10-Minute Non-Spin Reserves in East New York
2024 – 2025**



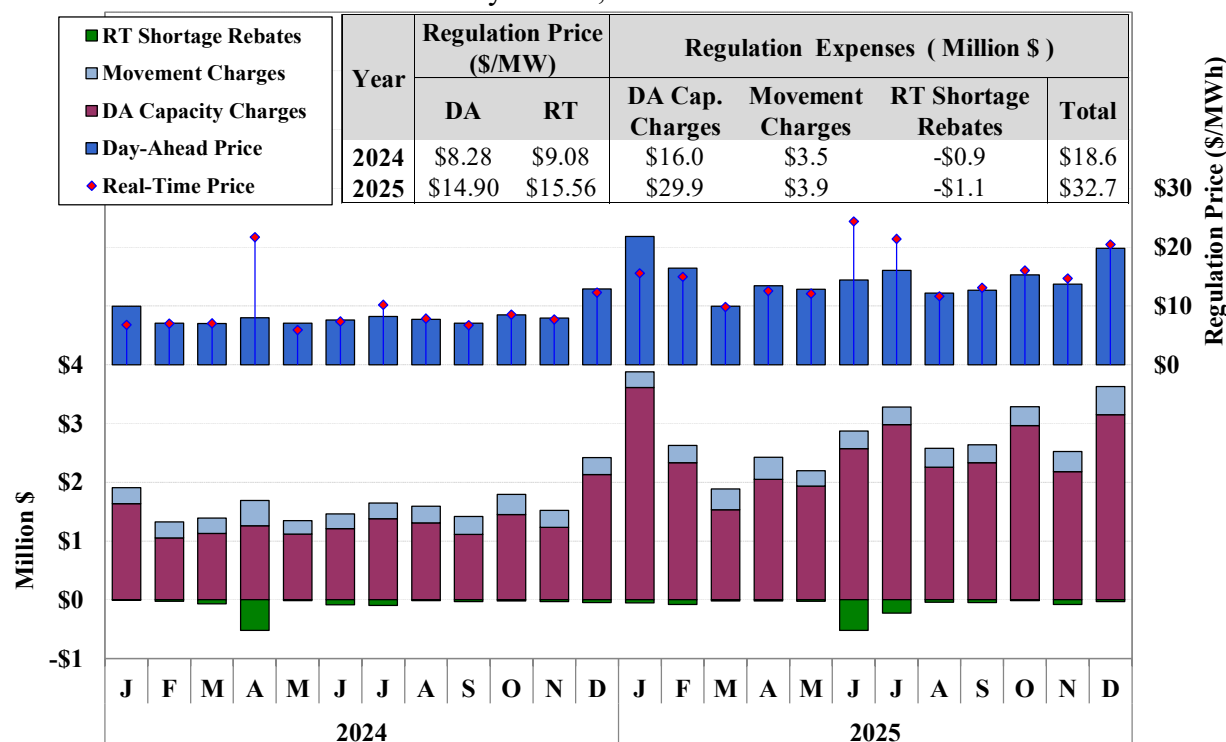
J. Regulation Market Performance

Figure A-22: Regulation Prices and Expenses

Figure A-22 shows the regulation prices and expenses in each month of the past two years. The upper portion of the figure compares the regulation prices in the day-ahead and real-time markets.²¹⁶ The lower portion of the figure summarizes regulation costs to NYISO customers, which include:

- **Day-Ahead Capacity Charge** – This equals day-ahead capacity clearing price times regulation capacity procured in the day-ahead market.
- **Real-Time Shortage Rebate** – This arises when a regulation shortage occurs in the real-time market and regulation suppliers have to buy back the shortage quantity at the real-time prices.
- **Movement Charge** – This is the compensation to regulation resources for dispatching up and down to provide regulation service. The payment amount equals the product of: (i) the real-time regulation movement price; (ii) the instructed regulation movement; and (iii) the performance factor calculated for the regulation service provider.

Figure A-22: Regulation Prices and Expenses
by Month, 2024 – 2025



²¹⁶ The day-ahead and real-time regulation prices shown in the upper portion of the chart represent the composite value of the capacity price and a movement component.

II. ANALYSIS OF ENERGY AND ANCILLARY SERVICES BIDS AND OFFERS

In this section, we analyze energy and ancillary services bid and offer patterns to assess market efficiency and ensure that market participant conduct aligns with effective competition. Specifically, this section evaluates the following areas:

- Potential physical withholding;
- Potential economic withholding;
- Market power mitigation;
- Operating reserves offers in the day-ahead market;
- Load-bidding patterns; and
- Virtual trading behavior.

Suppliers with market power can exercise it in electricity markets by withholding resources to increase the market clearing price. Physical withholding occurs when a resource is derated or not offered into the market despite being economic to produce energy (i.e., when the market clearing price exceeds the resource's marginal cost). Suppliers may also physically withhold by providing inaccurate operating characteristics, such as excessively long start-up notification times. Economic withholding occurs when a supplier raises its offer price to reduce its output below competitive levels or otherwise increase market clearing prices. Potential physical and economic withholding are evaluated in subsections A and B.

In the NYISO's market design, a competitive generator offer equals its marginal production cost. Absent market power, a supplier maximizes profits by producing output whenever its production cost is lower than the LBMP. However, a supplier with market power may profit from withholding when its losses from reduced outputs are offset by its gains from higher LBMPs. Accordingly, NYISO's market power mitigation measures cap suppliers' offers at their estimated marginal costs if their uncapped offers substantially exceed their estimated marginal cost and would materially impact LBMPs. In recent years, marginal cost estimates have become more uncertain during peak winter periods because of gas scheduling limitations and gas price volatility. As a result, the efficiency of mitigation measures depend on the accuracy of fuel cost estimates. Market power mitigation is evaluated in subsection C.

The NYISO co-optimizes the scheduling of energy and ancillary services in both the day-ahead and real-time markets. This co-optimization ensures that prices for energy and ancillary services reflect proper opportunity costs of diverting resources from energy to ancillary services. Co-optimization also reduces the potential for suppliers to exercise market power for a particular ancillary service product by allowing the market to flexibly shift resources between products, thereby increasing the competition to provide each product. Offer patterns for key operating reserve products in the day-ahead market are evaluated in subsection D.

Buyer behavior also influences energy prices. Under-scheduling load generally lowers day-ahead prices and leads to insufficient commitment for real-time needs. Alternatively, over-scheduling load tends to raise day-ahead prices above real-time prices. Thus, market participants

have incentives to schedule amounts of load consistent with real-time load. The consistency of day-ahead load scheduling with actual load is evaluated in subsection E.

Virtual trading plays an important role in overall market efficiency by improving price convergence between day-ahead and real-time markets, thereby promoting efficient commitment and scheduling of resources in the day-ahead market. When virtual trading is profitable, it generally promotes convergence between day-ahead and real-time prices and tends to improve the efficiency of resource commitment and scheduling. The efficiency of virtual trading is evaluated in subsection F.

A. Potential Physical Withholding

We evaluate potential physical withholding by analyzing day-ahead and real-time generator deratings of economic capacity as well as economic capacity that is unoffered in real-time. A derating occurs when a participant reduces the maximum output available from the plant. This can occur for a planned outage, a long-term forced outage, a short-term forced outage, or without any logged outage record. A derating can be either partial (maximum output is reduced but greater than zero) or complete (maximum output is zero). Unoffered economic capacity in real-time includes quick-start units that do not offer in real-time and online baseload units that offer less than their full capability. The figures in this section show the quantity of deratings and unoffered real-time capacity as a percent of total Dependable Maximum Net Capability (“DMNC”) from all generators in a region based on the most recent DMNC test value of each generator. *Short-term Deratings* include capacity that is derated for seven days or fewer. The remaining deratings are shown as *Long-Term Deratings*.²¹⁷

We focus particularly on short-term deratings and real-time unoffered capacity because they are more likely to reflect attempts to physically withhold than are long-term deratings, since it is less costly to withhold a resource for a short period. Taking a long-term forced outage would cause a supplier to forego the opportunity to earn profits during more hours when the supplier does not have market power. Nevertheless, the figures in this subsection evaluate long-term deratings as well, since they still may be an indication of withholding.

We focus on suppliers in Eastern New York, since this area includes roughly two-thirds of the State’s load, contains several areas with limited import capability, and is more vulnerable to the exercise of market power than is Western New York.

We also focus on economic capacity, since derated and unoffered capacity that is uneconomic does not raise prices above competitive levels and, therefore, is not an indicator of potential withholding. The figures in this subsection show the portion of derated and unoffered capacity that would have been economic based on Reference Levels and market prices.²¹⁸ This assessment determines economic commitment of baseload units based on day-ahead prices, considering start-up, minimum generation, and incremental costs. Economic dispatch of

²¹⁷ For our analyses of physical and economic withholding, we exclude unoffered capacity from hydro, solar, wind, landfill-gas and biomass generators as well as nuclear units on planned maintenance outages.

²¹⁸ This evaluation includes a modest threshold, which is described in subsection B as “Lower Threshold 1.”

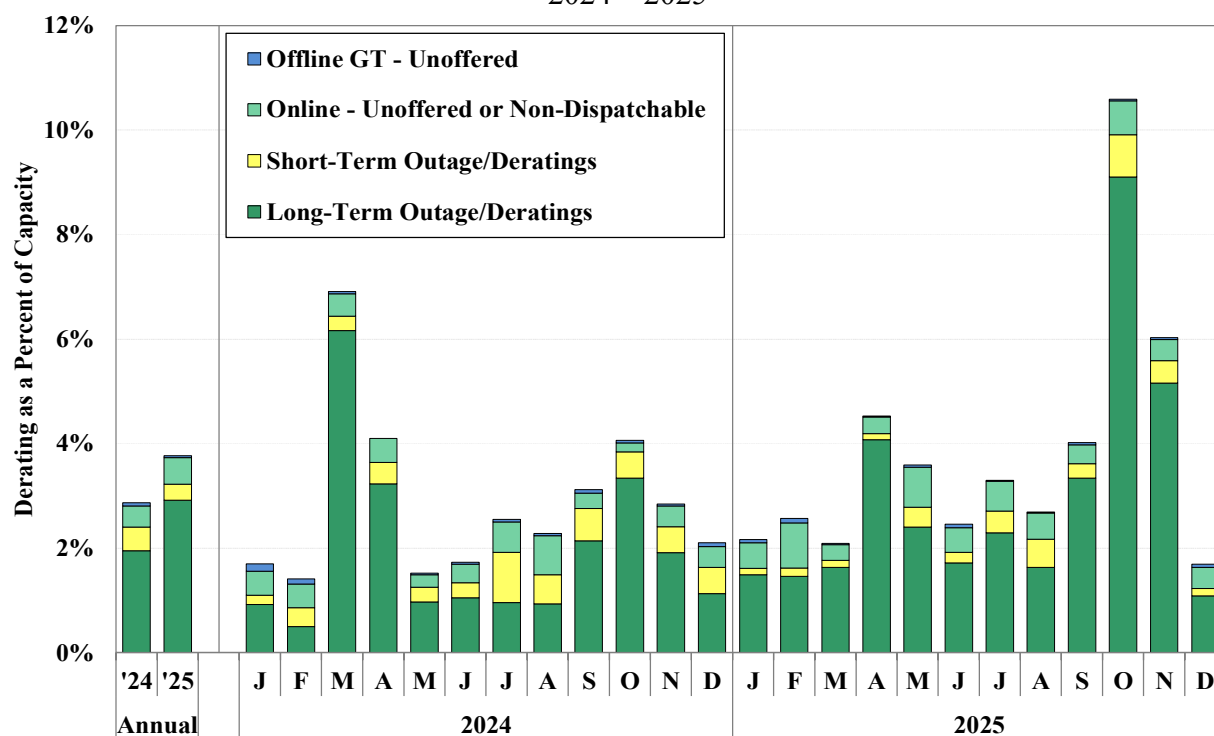
baseload units is based on RTD prices considering ramp rate limitations.²¹⁹ Quick-start units that were economic to commit must have been economic at both forecast RTC prices and settlement RTD prices.²²⁰

Figure A-23 - Figure A-26: Unoffered Economic Capacity by Month, Load Level, & Portfolio Size

Figure A-23 and Figure A-24 show the broad patterns of deratings and real-time unoffered capacity in New York State and Eastern New York in each month of 2024 and 2025.

Most wholesale electricity production comes from baseload and intermediate-load generating resources. Higher-cost resources are used to meet peak loads and constitute a very small portion of the total supply. This causes the market supply curve to be comparatively flat at low and moderate output levels and steeply sloped at high output levels. Therefore, as demand increases from low load levels, prices rise gradually until demand approaches peak levels, at which point prices can increase quickly as the costlier units are required to meet load. The shape of the market supply curve has implications for evaluating market power, namely that suppliers are more likely to have market power in broad areas under higher load conditions.

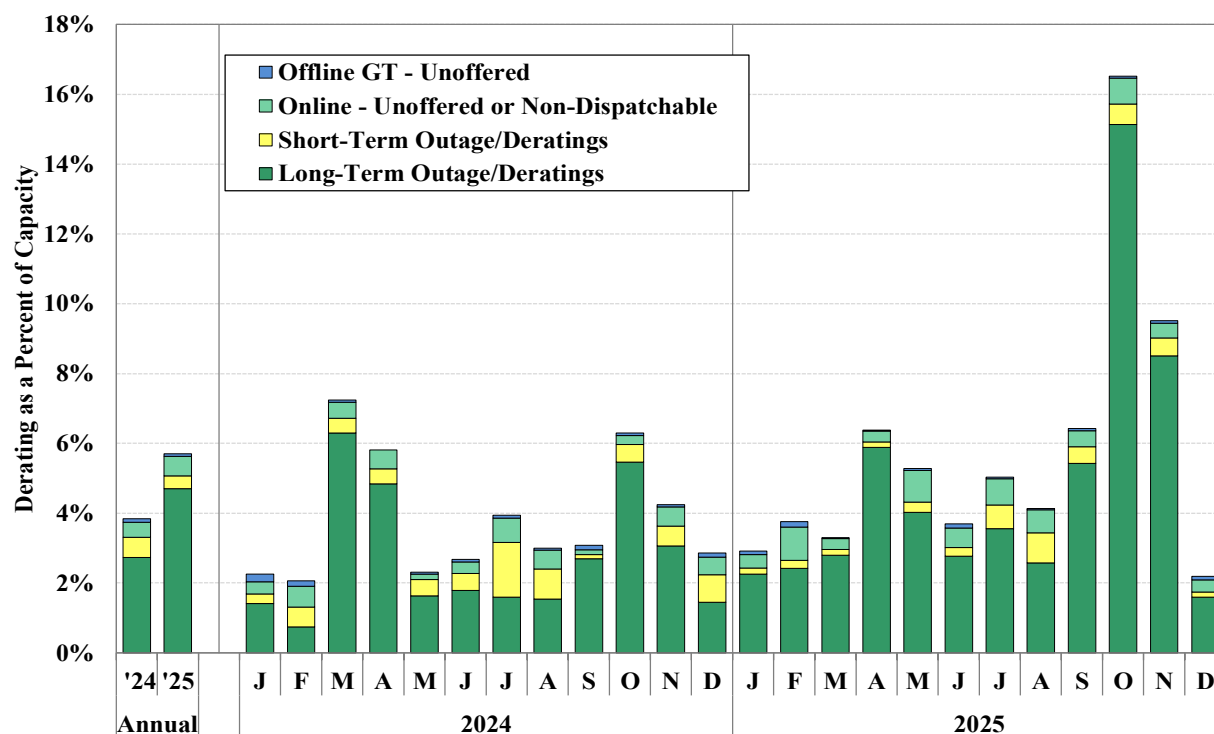
Figure A-23: Unoffered Economic Capacity by Month in NYCA
2024 – 2025



²¹⁹ If a baseload unit was committed by the DAM, optimal dispatch and potential physical withholding of incremental energy ranges was evaluated at RTD prices, even if the units DAM reference costs were above the DAM prices.

²²⁰ In this paragraph, “prices” refers to both energy and reserves prices.

**Figure A-24: Unoffered Economic Capacity by Month in East New York
2024 - 2025**



To distinguish between strategic and competitive conduct, we evaluate potential physical withholding considering market conditions and participant characteristics that would tend to create both the ability and the incentive to exercise market power. Under competitive conditions, suppliers maximize profits by increasing their offer quantities during the highest load periods to sell more power at the higher peak prices. Thus, we expect competitive suppliers to schedule maintenance outages during low-load periods, whenever possible. Nonetheless, more frequent operation of generators during high load periods increases the frequency of forced outages, which can reduce the amount of capacity offered into the market. Capacity that is on forced outage is more likely to be economic during high-load periods than during low-load periods.

As noted previously, a supplier with market power is most likely to profit from withholding in periods when the market supply curve becomes steep (e.g., high-demand periods) because that is when prices are most sensitive to withholding. Hence, we evaluate the conduct relative to load and participant size in Figure A-25 and Figure A-26 to determine whether the conduct is consistent with workable competition.

Figure A-25: Unoffered Economic Capacity by Supplier by Load Level in New York
2024 – 2025

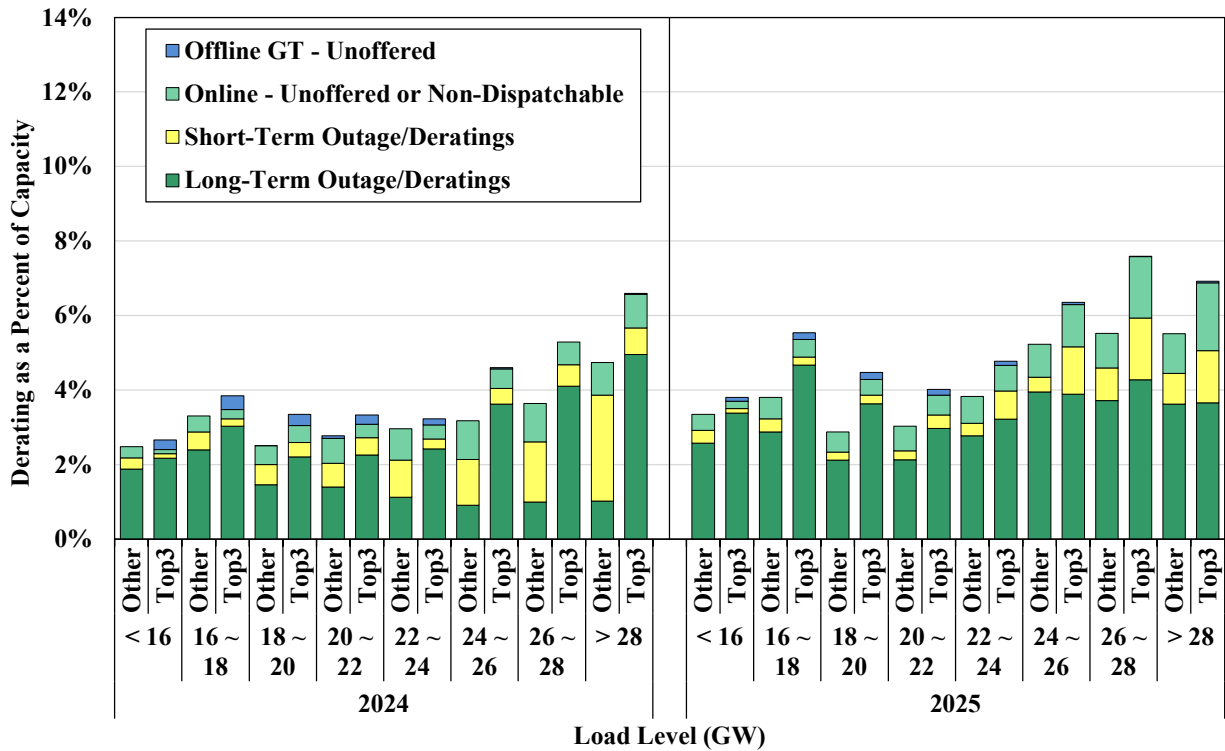
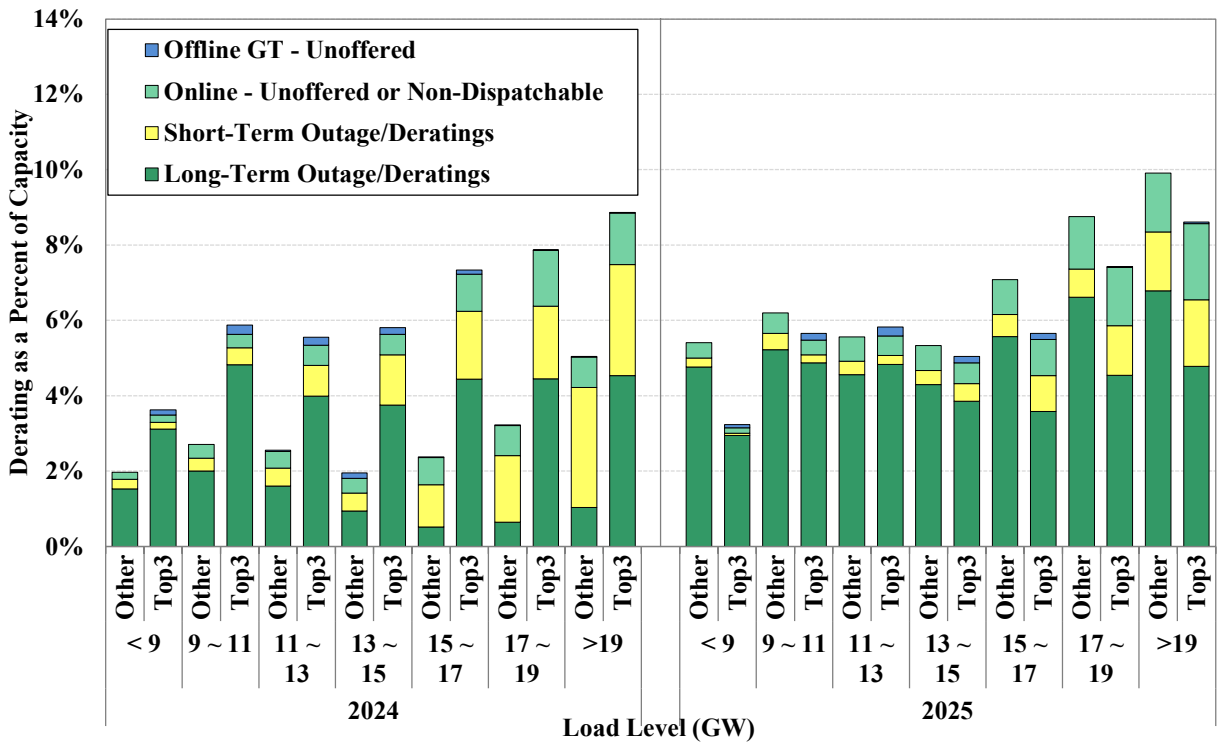


Figure A-26: Unoffered Economic Capacity by Supplier by Load Level in East New York
2024 – 2025



B. Potential Economic Withholding: Output Gap Metric

Economic withholding is an attempt by a supplier to inflate its offer price to raise LBMPs above competitive levels. In general, a supplier without market power maximizes profit by offering at marginal cost because inflated offer prices or other offer parameters prevent the unit from being dispatched when it would have been profitable. Hence, we analyze economic withholding by comparing a generator’s supply offers with its reference levels, which is an estimate of marginal cost that is used for market power mitigation.^{221, 222} An offer parameter is generally considered to be above the competitive level if it exceeds the reference level by a given threshold.

Figure A-27 to Figure A-30: Output Gap by Month, Supplier Size, and Load Level

One useful metric for identifying potential economic withholding is the “output gap.” The output gap is the amount of generation that appears to be economic at the market clearing price but is not scheduled, either due to bids that exceed the reference levels or due to other factors.²²³ We assume that the unit’s competitive offer price is equal to its reference level. To determine whether a unit is economic, we evaluate whether it would have been economic to commit based on day-ahead prices and whether its incremental energy would have been economic to produce based on real-time prices. Since gas turbines can be started in real-time, they are evaluated based on real-time prices. Like the prior analysis of potential physical withholding, we examine the broad patterns of output gap in New York State and Eastern New York, and we address the relationship of the output gap to the market demand level and participant size.

The following four figures show the output gap using three thresholds: the state-wide mitigation threshold (i.e., the level used for mitigation outside New York City), which is the lower of \$100 per MWh or 300 percent of a generator’s reference level; and two additional thresholds: Threshold 1 is 25 percent of a generator’s reference level, and Threshold 2 is 100 percent of a generator’s reference level. The two non-mitigation thresholds are used to identify abuses of market power that do not trigger the thresholds specified in the tariff for imposition of mitigation measures. However, since there is uncertainty in the estimation of the marginal costs of individual units, lower thresholds are more likely to flag behavior that is actually competitive.

Like the analysis of deratings in the prior subsection, it is useful to examine the output gap by load level and size of supplier because the incentive to economically withhold resources is positively correlated with these factors. Hence, these figures indicate how the output varies as

²²¹ The method of calculating reference levels is described in NYISO Market Services Tariff, Attachment H – NYISO Market Monitoring Plan-Market Mitigation Measures, Section 23.3.1.4. For some generators, the reference levels are based on an average of the generators’ accepted bids during competitive periods over the previous 90 days. The theory underlying this approach is that competitive conditions that prevail in most hours provide a strong incentive for suppliers to offer marginal costs. Hence, past accepted offers provide a benchmark for a generator’s marginal costs. For some generators, the reference level is based on an estimate of its fuel costs, other variable production costs, and any other applicable costs.

²²² Due to the Fuel Cost Adjustment (FCA) functionality, a generator’s reference level can be adjusted directly by a generator for a particular hour or day to account for fuel price changes. The NYISO monitors these generator-set FCA reference levels and may request documentation substantiating a generator FCA.

²²³ The output gap calculation excludes capacity that is more economic to provide ancillary services.

load increases and whether the largest three suppliers exhibit substantially different conduct than other suppliers.

Figure A-27: Output Gap by Month in New York State
2024 – 2025

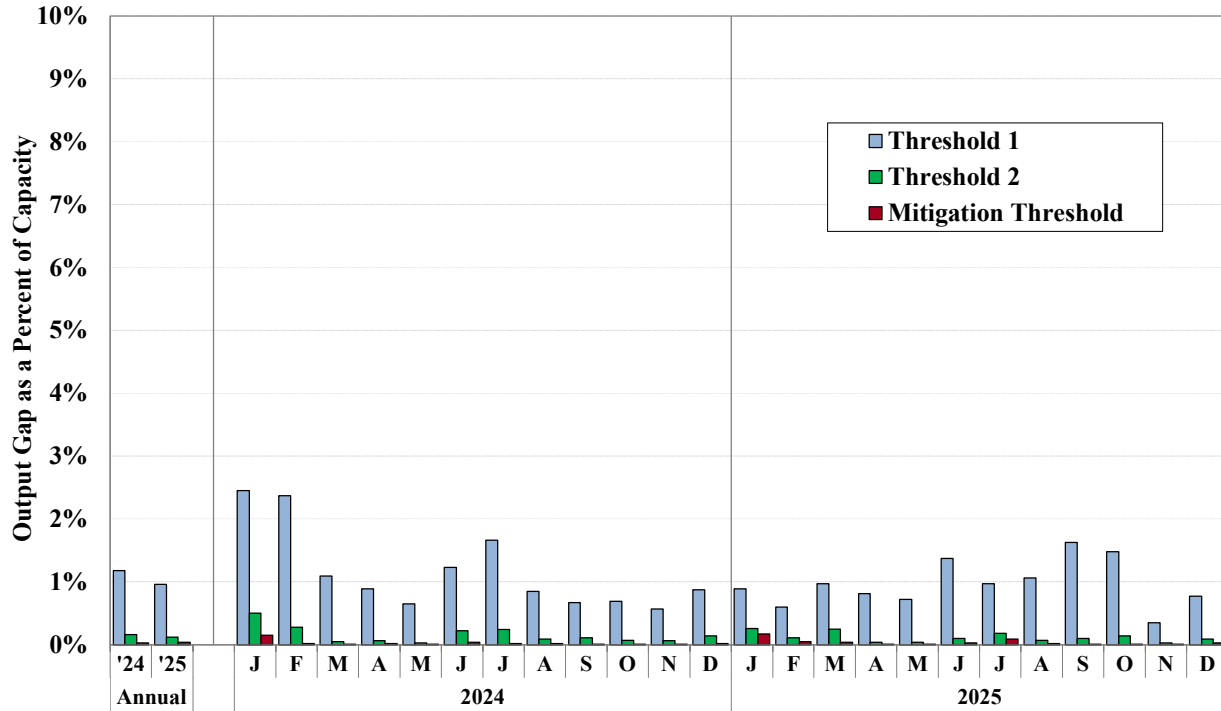
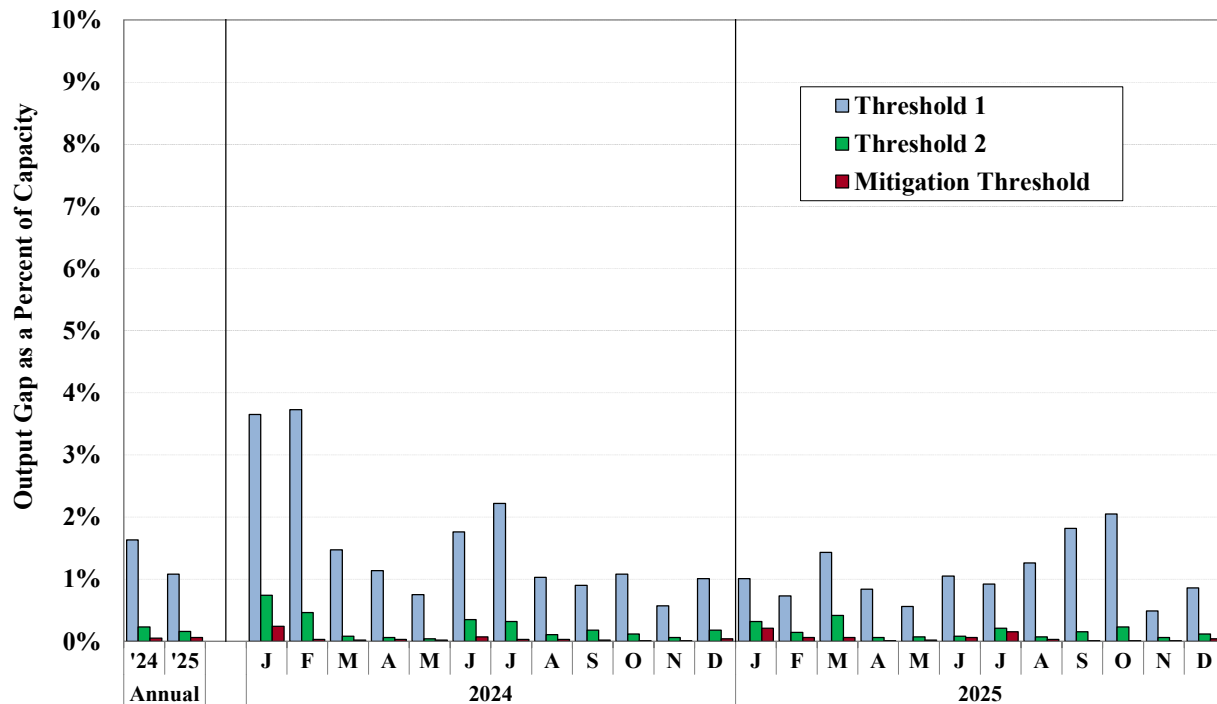
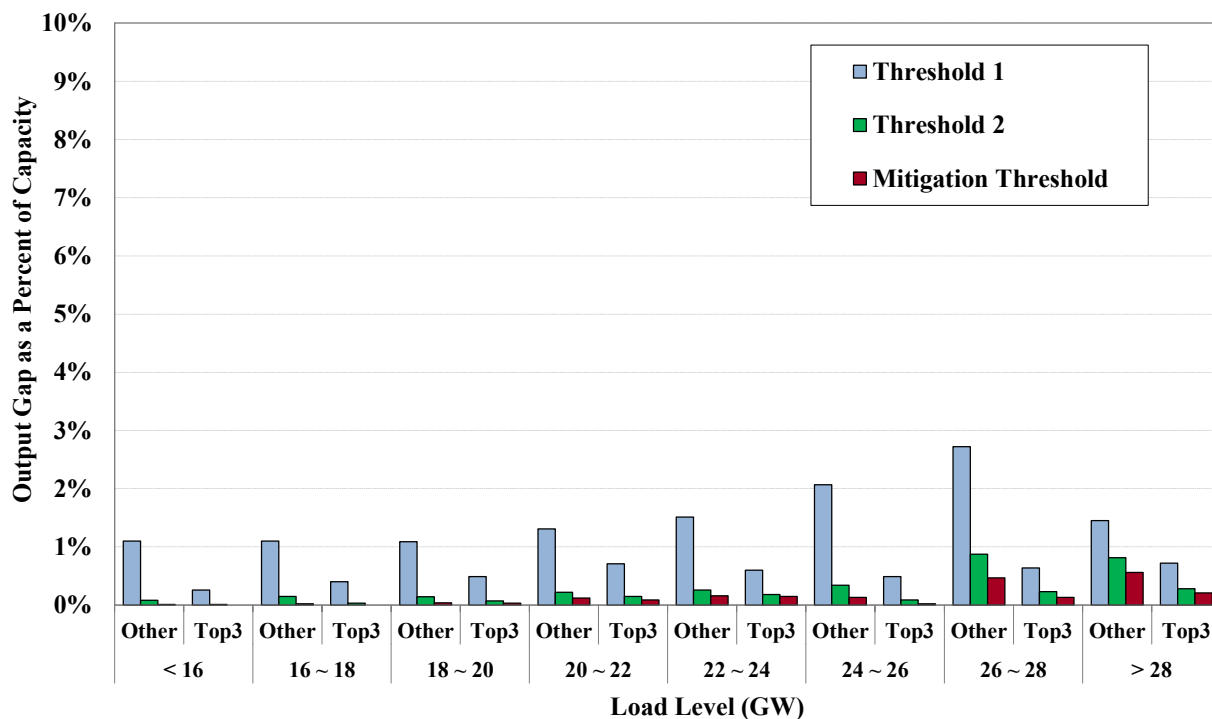


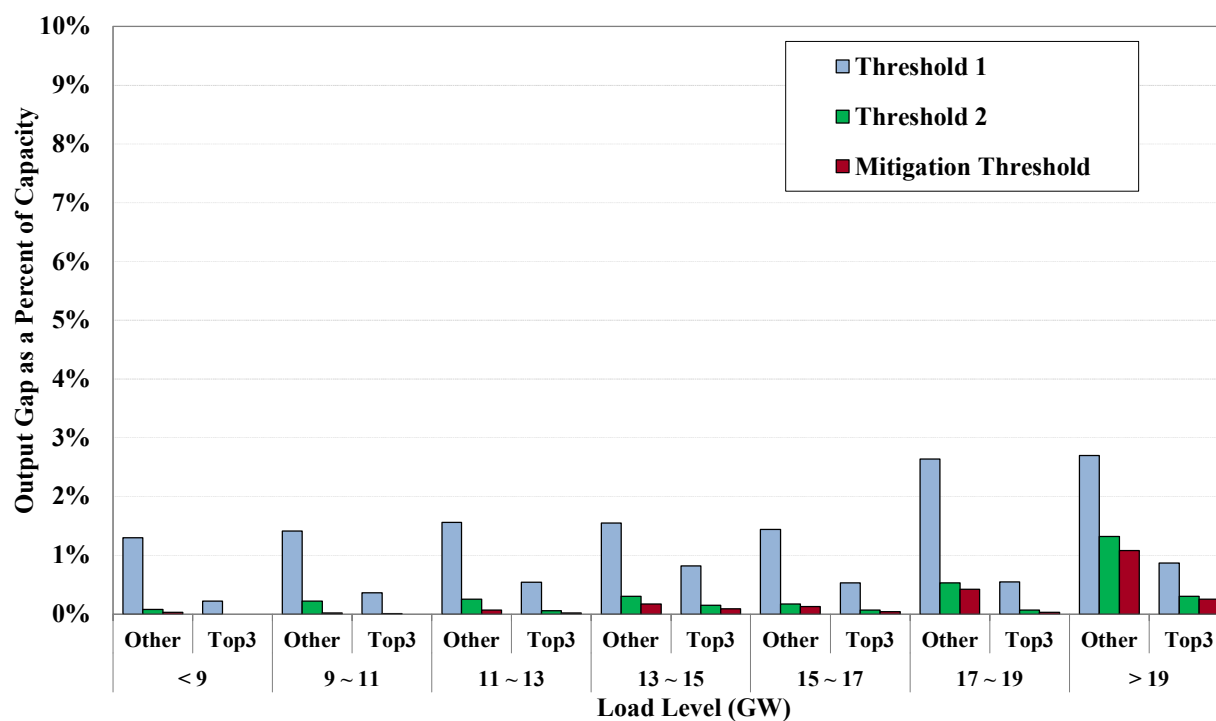
Figure A-28: Output Gap by Month in East New York
2024 - 2025



**Figure A-29: Output Gap by Supplier by Load Level in New York State
2025**



**Figure A-30: Output Gap by Supplier by Load Level in East New York
2025**



C. Day-Ahead and Real-Time Market Power Mitigation

Mitigation measures are intended to mitigate abuses of market power while minimizing interference with the market when it is workably competitive. NYISO applies conduct and impact tests that can result in mitigation of a participant's bid parameters (i.e., incremental energy offers, start-up and minimum generation offers, and physical parameters). The mitigation measures are only imposed when suppliers' conduct exceeds well-defined conduct thresholds and when the effect of that conduct on market outcomes exceeds well-defined market impact thresholds.²²⁴ This framework prevents mitigation when it is not necessary to address market power, while allowing high prices during legitimate periods of shortage.

The day-ahead and real-time market software is automated to perform conduct and impact tests and impose mitigation when appropriate. The mitigation measures are designed to allow prices to rise efficiently to reflect legitimate supply shortages while mitigating inflated offer prices that would otherwise lead to prices above competitive levels due to economic withholding.

When a transmission constraint is binding, one or more suppliers may be in a position to exercise market power due to the lack of competitive alternatives in the constrained area. For this reason, more restrictive conduct and impact thresholds are used for import-constrained load pockets in New York City. The in-city load pocket conduct and impact thresholds are determined by a formula that is based on the number of congested hours experienced over the preceding twelve-month period.²²⁵ This approach permits the in-city conduct and impact thresholds to increase as the frequency of congestion decreases, whether due to additional generation or increases in transmission capability. An in-city offer fails the conduct test if it exceeds the reference level by the threshold or more. In-city offers that fail the conduct test are tested for price impact by the market software. If their price impact exceeds the threshold, they are mitigated.

When local reliability criteria necessitate the commitment of additional generation, suppliers may be able to exercise market power due to the lack of competition in the local area. Hence, NYISO has more restrictive conduct and impact thresholds when a single supplier is pivotal for satisfying local reliability criteria outside New York City.²²⁶ The tariff limits the start-up cost offers to 10-percent of the reference level and the minimum generation cost offers of such units to conduct thresholds of the higher of \$10 per MWh or 10 percent of the reference level.²²⁷

While uncommon, a generator can be mitigated initially in the day-ahead or real-time market and unmitigated after consultation with NYISO.²²⁸ Mitigation can be reversed for several reasons:

²²⁴ See NYISO Market Services Tariff, Sections 23.3.1.2 and 23.3.2.1.

²²⁵ $\text{Threshold} = (0.02 * \text{Average Price} * 8760) / \text{Constrained Hours}$. This threshold is defined in the NYISO Market Services Tariff, Section 23.3.1.2.2.1.

²²⁶ In New York City, the start-up cost and minimum generation cost offers of units committed for local reliability are effectively subject to \$0 thresholds. See NYISO Market Services Tariff, Section 23.5.2.1.

²²⁷ See NYISO Market Services Tariff, Section 23.3.1.2.3.

²²⁸ NYISO Market Services Tariff, Section 23.3.3 lays out the requirements for consultation. This occurs after the market date, so any effect of the mitigation on LBMPs is unchanged by un-mitigation.

- A generator’s reference level is inaccurate and the supplier-initiated consultation with NYISO to increase the reference level before the generator was mitigated.
- A generator’s reference level on a particular day is lower than the consultative reference level that NYISO approved for the generator before the generator was mitigated.²²⁹
- The generator took appropriate steps to inform NYISO of a fuel price change prior to being scheduled (through an FCA or other means), but the generator was still mitigated.
- A generator’s fuel cost may change significantly by time of day, although the day-ahead market software is unable to use reference levels that vary by time of day, so it may be mitigated in a particular hour of the day-ahead market and then unmitigated once the proper reference level is reflected.

NYISO also reviews the markets for potential abuses of market power in the form of uneconomic overproduction from generation facilities. While the mitigation provisions for withholding aim to prevent a generator from underproducing in order to increase prices, mitigation provisions for uneconomic overproduction prevent generators from increasing output in order to reduce prices below competitive levels. There are several reasons why a market party operating a generator with local market power may be incentivized to over produce and reduce prices to benefits its portfolio, including:

- Create a constraint that raises prices downstream for other generators in its portfolio;
- Buy out of a day-ahead position at very low or negative LBMPs; and/or
- Benefit a financial position that profits from lower prices.

Similar to the economic and physical withholding provisions, uneconomic overproduction mitigation measures employ conduct and impact thresholds to identify such behavior.²³⁰ The NYISO’s established mitigation measures generally deter behavior that could lead to the three concerns listed above. However, we have identified a concern with the lack of financial incentives for intermittent generators to follow curtailment instructions under certain conditions. When these resources do not follow curtailment instructions, it threatens system security and may lead to inefficient market operations. Appendix Section V.B provides analysis of this issue and our recommendation to address it.

Figure A-31 & Figure A-32: Summary of Day-Ahead and Real-Time Mitigation

Figure A-31 and Figure A-32 summarize the amount of mitigation in New York that occurred in the day-ahead and the real-time markets in 2024 and 2025. These figures do not include guarantee payment mitigation that occurs in the settlement system.

²²⁹ The hierarchy of information that is used to calculate reference levels is provided in NYISO Market Services Tariff, Section 23.3.1.4. It is possible for a generator to have a bid-based or LBMP-based reference level that is less accurate than the reference level determined through consultation.

²³⁰ See tariff sections 23.3.1.3 and 23.3.2.1.1.1.

Figure A-31: Summary of Day-Ahead Mitigation
2024 – 2025

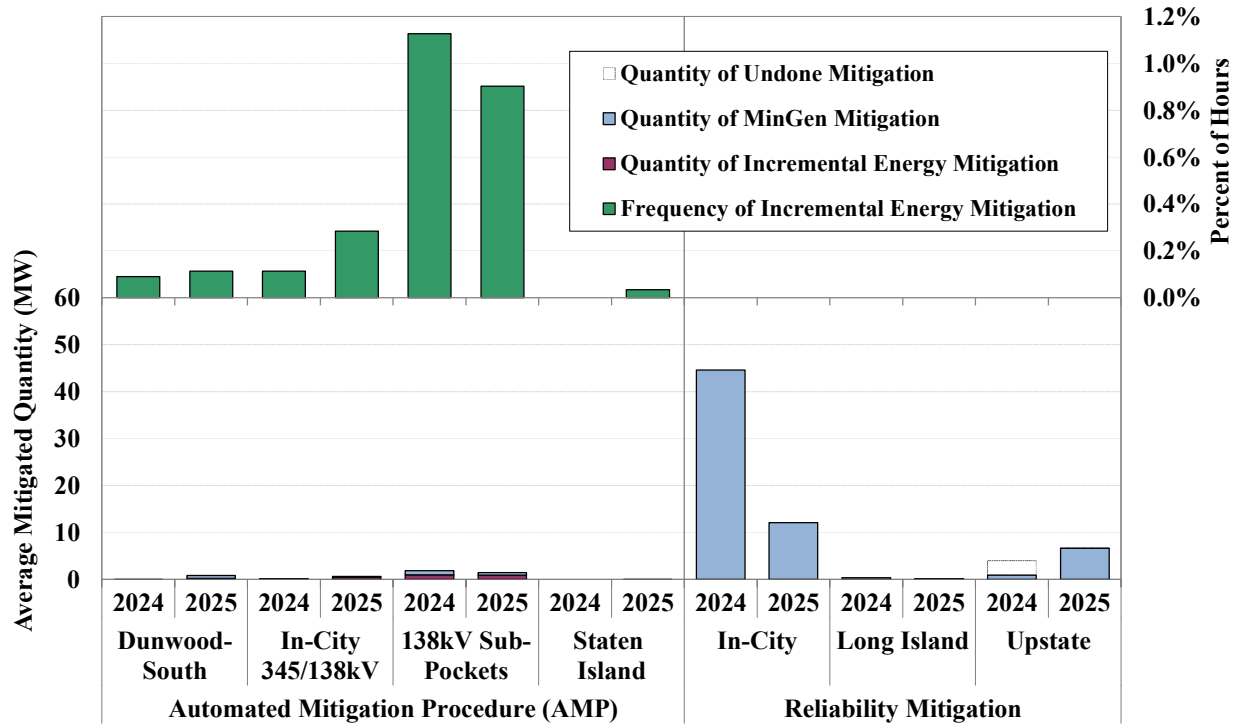
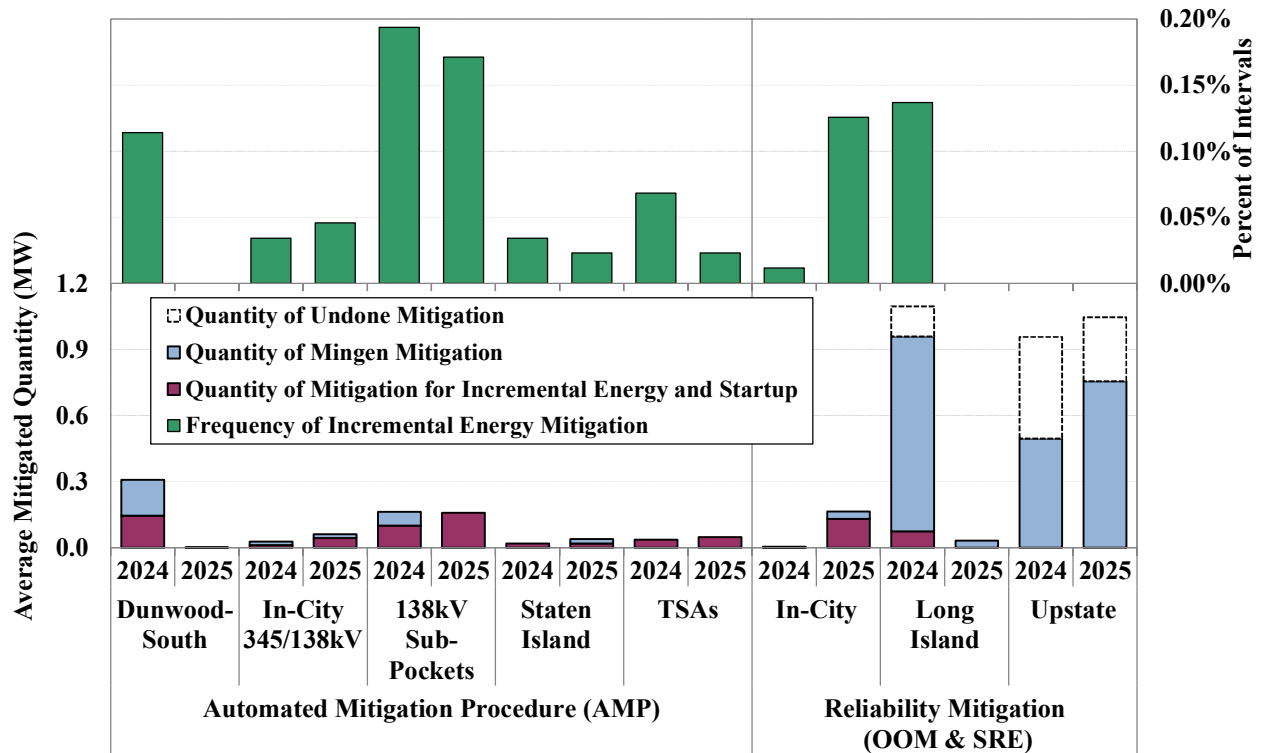


Figure A-32: Summary of Real-Time Mitigation
2024 – 2025



The bars in the upper panel of the figures indicate the percent of hours when incremental energy offer mitigation was imposed on one or more units in each category, while the bars in the lower panel indicate the average amount of capacity mitigated in hours when mitigation occurred (as well as the portion that was unmitigated). Mitigated quantities are shown separately for the flexible output ranges of units (i.e. Incremental Energy) and the non-flexible portions (i.e. MinGen).²³¹ In each figure, the left portion shows the amount of mitigation by the Automated Mitigation Procedure (“AMP”) on the economically committed units in load pockets of New York City, and the right portion shows the amount of mitigation on the units committed for reliability in New York City, Long Island, and the upstate area.

D. Operating Reserves Offers in the Day-Ahead Market

Multiple factors, including opportunity costs, demand curves, and offers, determine the prices of ancillary services. The NYISO co-optimizes the scheduling of energy and ancillary services in the day-ahead and real-time markets. Co-optimization causes the prices of energy and ancillary services to reflect the costs to the system of diverting resources to provide ancillary services that would otherwise provide energy.

The ancillary services markets use demand curves that represent the economic value placed on each class of reserves. When the reserve requirements cannot be satisfied at a cost lower than the demand curve, the system is in shortage and the reserve demand curve value is included in the reserve price and the energy price. This approach is recognized for producing efficient prices during shortages of reserves because it provides a mechanism for reflecting the value of reserves in the price of energy during shortages.

This subsection focuses on offer patterns in the day-ahead market for several key operating reserve products. In an efficient market, we expect suppliers to respond to predictable differences between day-ahead and real-time prices by raising or lowering their offer prices in the day-ahead market. However, the high volatility of real-time clearing prices is difficult to predict in the day-ahead market. High volatility of real-time prices is a source of risk for suppliers that sell reserves in the day-ahead market, since suppliers must forego real-time scarcity revenues if they have already sold reserves in the day-ahead market. Some suppliers may reduce their exposure to this risk by raising their reserves offer prices in the day-ahead market.

Figure A-33 to Figure A-34: Day-Ahead Reserve Offers That Satisfy NYCA 30-Minute Requirement and Eastern New York 10-Minute Reserve Requirement

Figure A-33 summarizes reserve offers that can satisfy NYCA 30-minute operating reserve requirement in each quarter of the past three years. These quantities include both 10-minute and 30-minute and both spinning and non-spin reserve offers, although they are not shown separately in the figure. Only offers from day-ahead committed (i.e., online) resources and available offline quick-start resources are included in this evaluation, since they directly affect the reserve prices.

²³¹ Mitigation of gas turbine capacity is shown in the Incremental Energy category whenever the incremental energy offer or the startup offer is mitigated.

The stacked bars in the Figure A-33 show the amount of reserve offers in selected price ranges for West New York (Zones A to E), East New York (Zones F to J), and NYCA (excluding Zone K). Long Island is excluded because the current rules limit its reserve contribution to the broader areas (i.e., SENY, East, NYCA). As a result, Long Island reserve offers have little impact on NYCA reserve prices.

The black bar in the figure represents the equivalent average 30-minute reserve requirements for areas outside Long Island. This is calculated as NYCA 30-minute reserve requirement *minus* 30-minute reserves scheduled on Long Island. Where the line intersects the bar provides a rough indication of reserve prices, which, however, is generally lower than actual reserve prices because opportunity costs are not reflected in the figure.

Figure A-33: Day-Ahead Reserve Offers That Satisfy NYCA 30-Minute Requirement
Committed and Available Offline Quick-Start Resources, 2023 – 2025

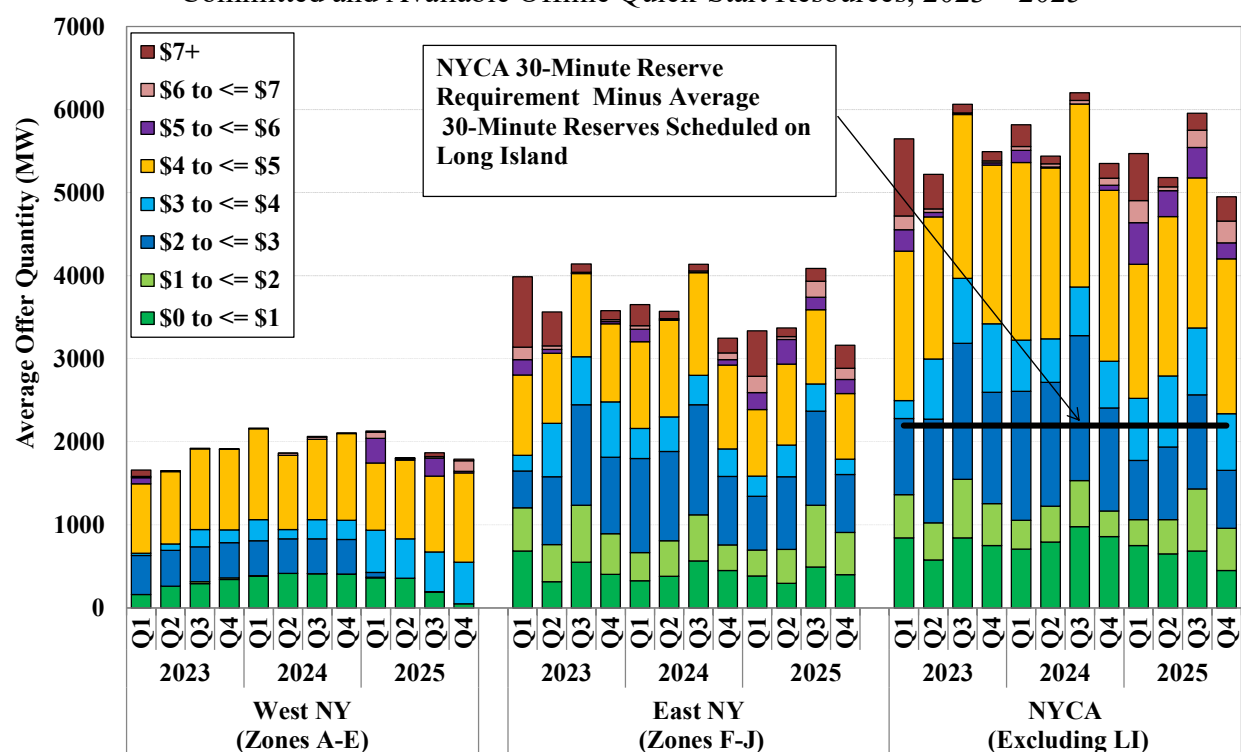
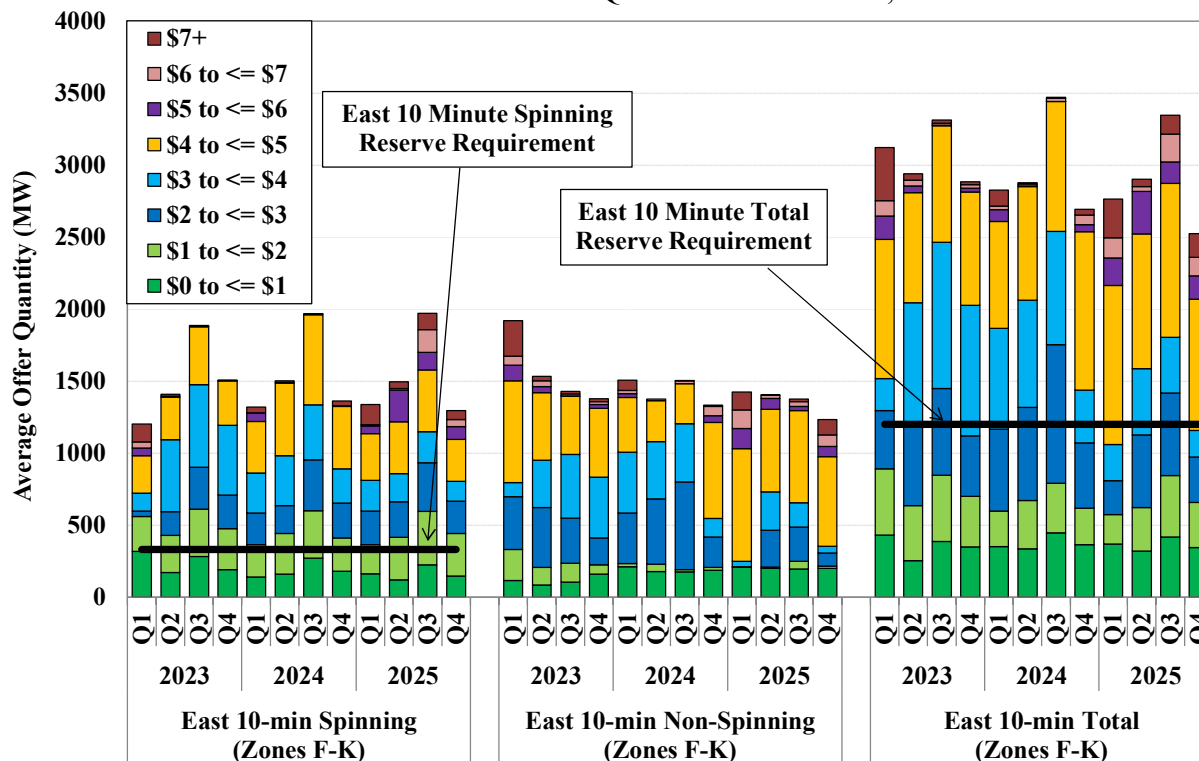


Figure A-34 summarizes offers that can satisfy the Eastern New York reserves requirement and shows generator offers for 10-minute reserves from committed resources and available offline quick-start resources. The first set of stacked bars shows the offers from generators for the 10-minute spinning requirement (set at 330 MWs and shown with a black bar) while the second set of stacked bars show the offers for 10-minute non-spinning reserves. The final stack is the sum of the first two and is shown with a black bar designating the Eastern NY total 10-minute requirement of 1200 MWs. Similar to Figure A-33, the intersection of the black bars with the stacked lines is a rough indication of reserve prices but is generally lower than actual reserve prices because opportunity costs are not reflected in the figure.

Figure A-34: Day-Ahead Reserve Offers that Satisfy ENY 10-Minute Reserve Requirement
Committed and Available Offline Quick-Start Resources, 2023 - 2025



E. Analysis of Load Bidding and Virtual Trading

In addition to screening suppliers for physical and economic withholding, it is important to evaluate how buyer behavior influences energy prices. Therefore, we evaluate whether load bidding is consistent with the principles of workable competition. Load can be scheduled in one of the following five ways:

- **Physical Bilateral Contracts** – These schedules allow participants to settle transmission charges (i.e., congestion and losses) with the NYISO between two points while privately settling the commodity sale with their counterparties. This does not represent all bilateral contracting in New York because participants have the option of entering into bilateral contracts that are settled privately (e.g., contracts for differences).
- **Day-Ahead Fixed Load** – This represents load scheduled in the day-ahead market for receipt at a specific bus, regardless of the day-ahead price. It is equivalent to a load bid with an infinite bid price.
- **Price-Capped Load Bids** – These are load bids submitted into the day-ahead market with a specific bid price, indicating the maximum amount the Load-Serving Entity (“LSE”) is willing to pay.²³²

²³²

For example, a LSE may make a price-capped bid for 500 MW at \$60 per MWh. If the day-ahead clearing price at its location is above \$60, the bid would not be accepted in the day-ahead market.

- **Virtual Load Bids** – These are bids to purchase energy in the day-ahead market with a bid price indicating the maximum amount the bidder is willing to pay. Virtual load scheduled in the day-ahead market is subsequently sold back in the real-time market. The virtual buyer earns or pays the difference between the day-ahead and real-time prices. Virtual trading is currently allowed only at the load zone level.
- **Virtual Exports** – These are external transactions in the export direction that are scheduled in the day-ahead market but are withdrawn or bid at high price levels in real time. They function similarly to virtual load bids but are placed at the external proxy buses rather than at the eleven load zones.

Each of these load categories is important because they tend to increase the amount of physical resources scheduled in the day-ahead market. Conversely, virtual supply and virtual imports tend to reduce the amount of physical resources scheduled in the day-ahead market. Virtual supply is energy offered for sale in the day-ahead market, with an offer price indicating the minimum amount the market participant is willing to accept. Virtual supply sold in the day-ahead market is subsequently purchased back in the real-time market.

Figure A-35: Day-Ahead Load Schedules versus Actual Load

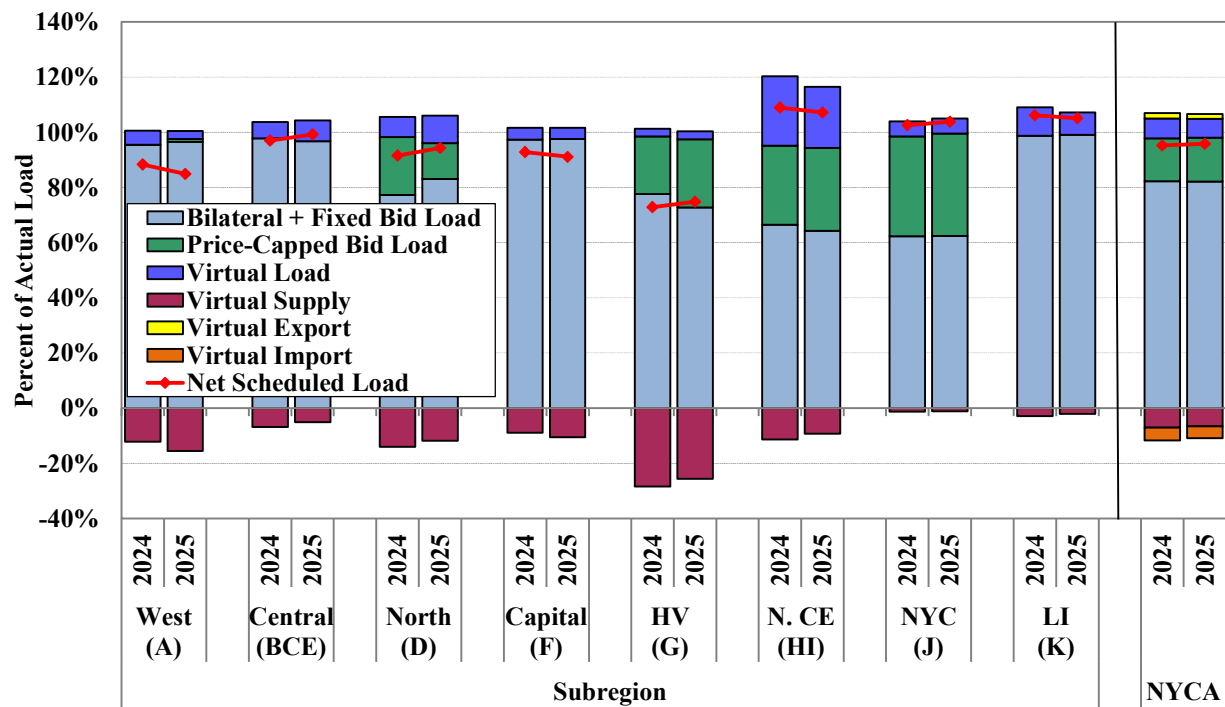
Many generating units have long lead times and substantial commitment costs. Their owners must decide whether to commit them well in advance of real-time, often before they can be certain that the unit will be economic. The day-ahead market provides suppliers with a means of being committed only when economically justified. These suppliers are more likely to sell into the day-ahead market if day-ahead prices are generally consistent with real-time prices. Thus, efficient unit commitment relies on consistency between the day-ahead and the real-time markets. The following figure evaluates the consistency between day-ahead load scheduling patterns and actual load, offering insights into the overall efficiency of the day-ahead market.

In a well-functioning market, day-ahead load schedules are expected to be generally consistent with actual load. Under-scheduling load can lead to lower day-ahead prices and insufficient commitment for real-time needs, while over-scheduling may raise day-ahead prices above real-time prices. As a result, market participants have incentives to schedule load amounts that are consistent with real-time load.

The following figure shows day-ahead load schedules and bids as a percentage of real-time load during daily peak load hours in 2024 and in 2025 at various locations in New York, based on an annual average. Since virtual load (including virtual exports) has the same effect on day-ahead prices and resource commitment as physical load, they are shown together in this analysis. Conversely, virtual supply (including virtual imports) has the same effect on day-ahead prices and resource commitment as a reduction in physical load, so it is treated as negative load for the purposes of this analysis.

For each period, physical load and virtual load are represented by bars in the positive direction, while virtual supply is represented by bars in the negative direction. Net scheduled load, indicated by the line, is the sum of scheduled physical and virtual load minus scheduled virtual supply. Virtual imports and exports are shown only for NYCA and are not displayed for individual sub-areas in New York.

Figure A-35: Day-Ahead Load Schedules versus Actual Load
By Region, 2024 – 2025



F. Virtual Trading in New York

Virtual trading plays an important role in market efficiency by improving price convergence between day-ahead and real-time markets, thereby promoting efficient commitment and scheduling of resources in the day-ahead market. Virtual trading in the day-ahead consists of purchases or sales of energy that are not associated with physical load or physical resources. Virtual bids and offers provide liquidity to the day-ahead because they constitute a substantial share of the price-sensitive supply and demand that establish efficient day-ahead prices.

Virtual transactions that are scheduled in the day-ahead market settle against real-time energy prices. Virtual demand bids are profitable when the real-time energy price is higher than the day-ahead price, while virtual supply offers are profitable when the day-ahead energy price is higher than the real-time price. If prices are lower in the day-ahead market than in the real-time market, a virtual trader may purchase energy in the day-ahead market and sell it back in the real-time market, which will tend to increase day-ahead prices and improve price convergence with the real-time market. Hence, profitable virtual transactions improve the performance of the day-ahead market. The New York ISO currently allows virtual traders to schedule transactions to arbitrage the price differences at the load zone level between day-ahead and real-time.

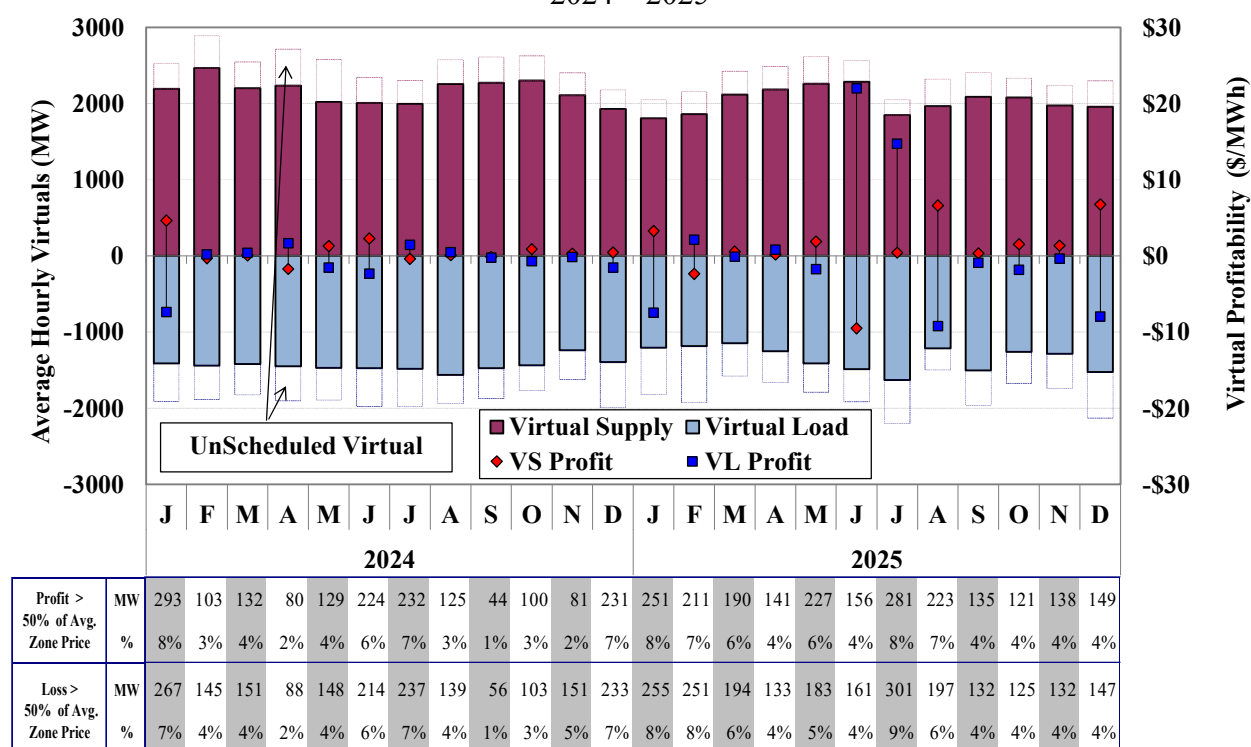
Market participants can schedule virtual-type transactions at the external proxy buses, which are referred to as Virtual Imports and Virtual Exports in this report. These types of external transactions act the same way as the virtual bids placed at the load zones (i.e., the imports and exports that are scheduled in the day-ahead market do not flow in real-time). Since the virtual

imports and exports have a similar effect on scheduling and pricing as virtual load and supply, they are evaluated as part of virtual trading in this section.

Figure A-36: Virtual Trading Volumes and Profitability

The figure summarizes recent virtual trading activity in New York by showing monthly average scheduled quantities, unscheduled quantities, and gross profitability for virtual transactions in 2024 and 2025. The amount of scheduled virtual supply in the figure includes scheduled virtual supply at the load zones and virtual imports at the external proxy buses. Likewise, the amount of scheduled virtual load in the chart includes scheduled virtual load at the load zones and scheduled virtual exports at the external proxy buses. Gross profitability is the difference between the price at which virtual traders bought and sold positions in the day-ahead market compared to the price at which these positions were covered in the real-time market.^{233,234}

Figure A-36: Virtual Trading Volumes and Profitability
2024 – 2025



The table below the figure shows a screen for relatively large profits or losses, which identifies virtual transactions with gross profits (or losses) larger than 50 percent of the average zone (or proxy bus) price. For example, an average of 231 MW of virtual transactions (or 7 percent of all virtual transactions) netted profits larger than the 50 percent of their zone (or proxy bus) prices in

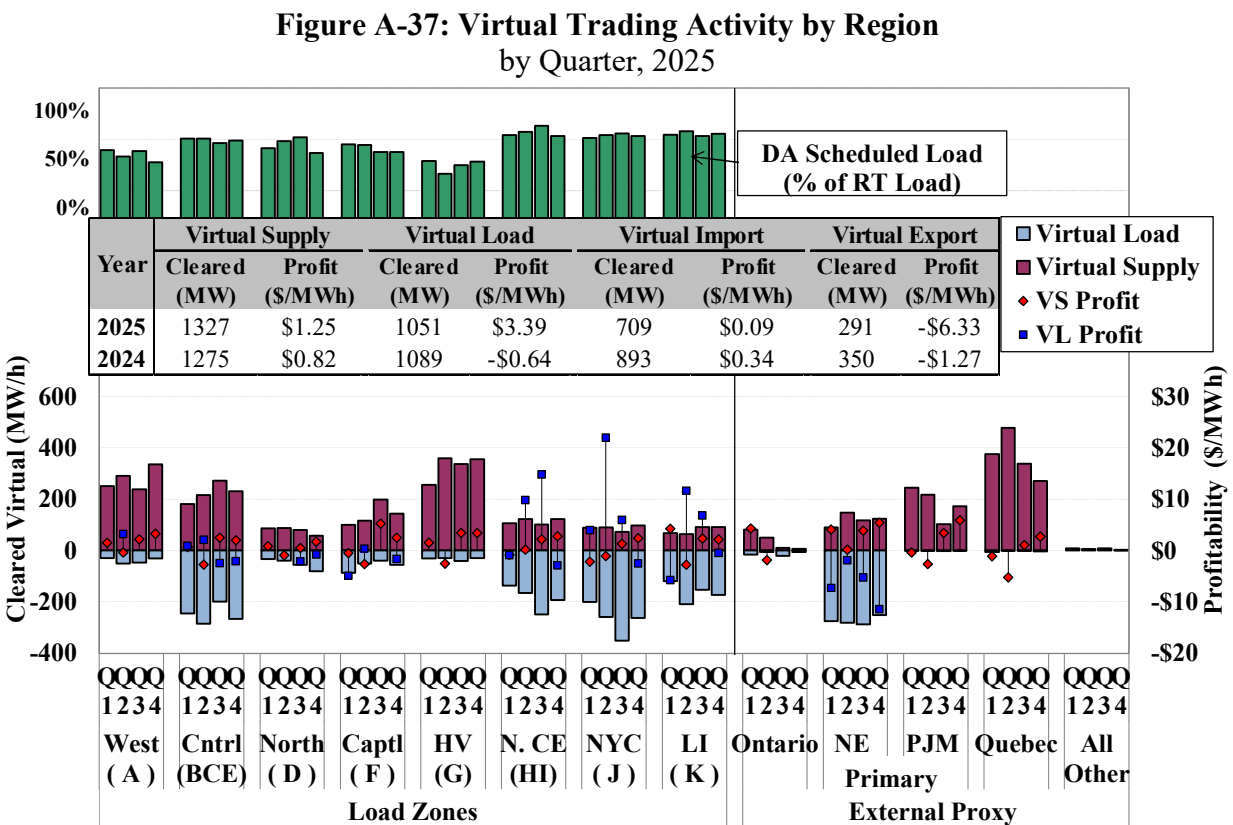
²³³ The gross profitability shown here does not account for any other related costs or charges to virtual traders.

²³⁴ The calculation of the gross profitability for virtual imports and exports does not account for the profit (or loss) related to price differences between day-ahead and real-time in the neighboring markets.

December of 2024. Large profits may be an indicator of a modeling inconsistency, while sustained losses may be an indicator of potential manipulation of the day-ahead market.

Figure A-37: Virtual Trading Activity by Region

Figure A-37 summarizes virtual trading by geographic region. The eleven zones in New York are broken into eight geographic regions based on typical congestion patterns. Zone A (the West Zone) is shown separately because of increased congestion in recent years. Zone D (the North Zone) is shown separately because generation in that zone exacerbates transmission congestion on several interfaces, particularly the Central-East interface. Zone F (the Capital Zone) is shown separately because it is constrained from Western New York by the Central-East Interface and from Southeast New York by constraints in the Hudson Valley. Zones J (New York City) and K (Long Island) are shown separately because congestion frequently leads to price separation between them and other areas. The figure also shows virtual imports and exports with neighboring control areas. The Ontario proxy bus, the primary PJM proxy bus (i.e., the Keystone proxy bus), and the primary New England proxy bus (i.e., the Sandy Pond proxy bus) are evaluated separately from all other proxy buses.



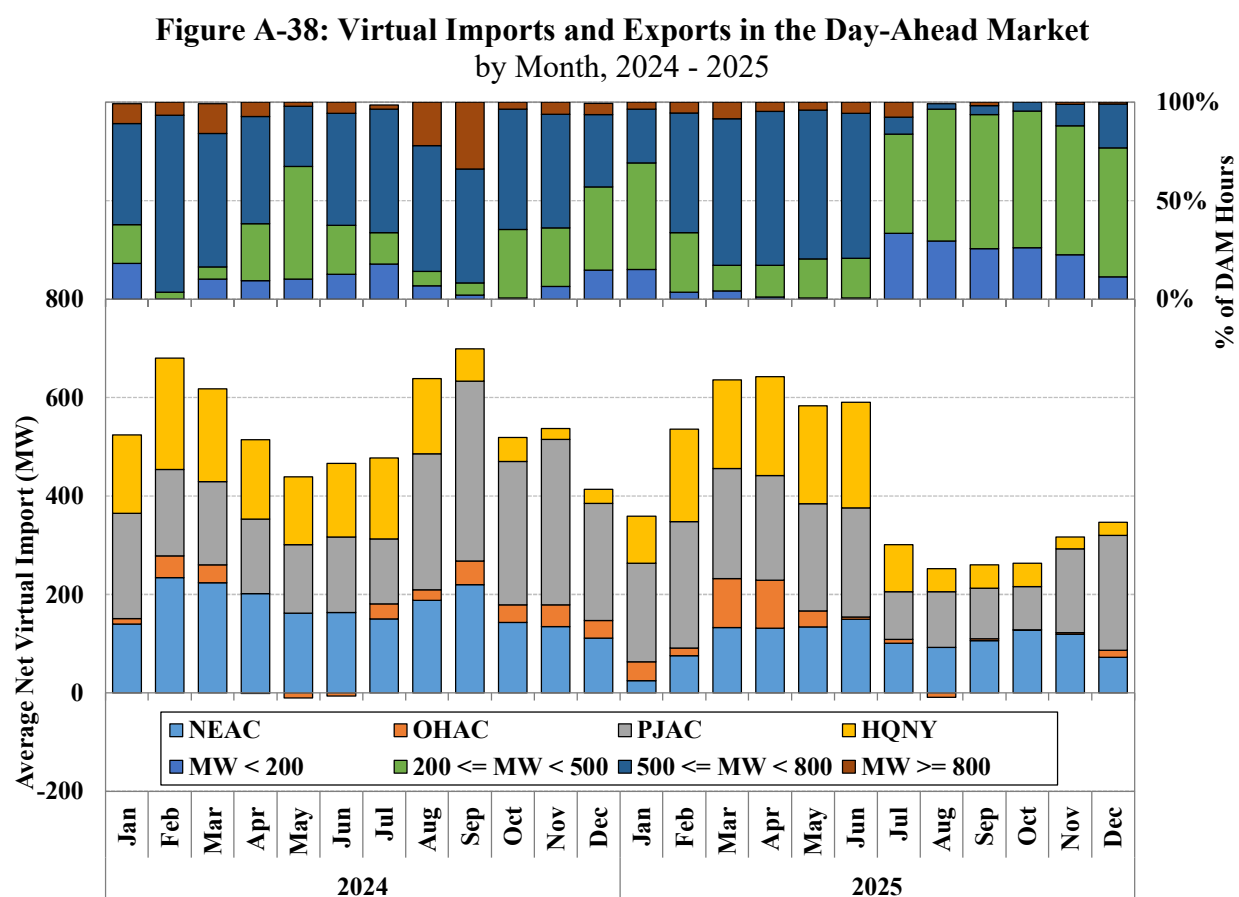
The lower portion of the figure shows average quantities of scheduled virtual supply and virtual load and their gross profitability for the seven regions and four groups of external proxy buses in each quarter of 2025. Profits or losses are not shown for a category if the average scheduled quantity is less than 50 MW. The upper portion of the figure shows the average day-ahead scheduled load (as a percentage of real-time load) at each geographic region. The table in the middle compares the overall virtual trading activity in 2024 and 2025.

Figure A-38: Virtual Imports and Exports in the Day-ahead Market

The following chart evaluates scheduled virtual imports and exports in the day-ahead market. In this analysis, virtual imports and exports are defined as external transactions that are scheduled in the day-ahead market but withdrawn from the real-market market (i.e., no real-time bids submitted). Virtual wheels are excluded from this analysis.

The bottom portion of the chart shows the hourly average quantity of net virtual imports for each month. The bars represent the average net virtual imports scheduled across the four primary interfaces between NYISO and neighboring control areas. Virtual imports and exports are rare across the Scheduled-Line interfaces, which are excluded from this analysis.

The top portion of the chart shows the percentage of hours in each month when total net virtual imports across the four primary interfaces fall into the following ranges: (a) less than 200 MW; (b) between 200 and 500 MW; (c) between 500 and 800 MW; and (d) more than 800 MW.



III. TRANSMISSION CONGESTION

Congestion arises when the transmission network is bottlenecked, limiting dispatch of the least expensive generators to satisfy system demand. When congestion occurs, the market software establishes clearing prices that vary by location to reflect the cost of meeting load at each location. These Location-Based Marginal Prices (“LBMPs”) reflect that higher-cost generation is required at locations where transmission constraints prevent the free flow of power from lower-cost resources.

The day-ahead market is a forward market that facilitates financial transactions among participants. NYISO allows market participants to schedule transactions in the day-ahead market based on the predicted transmission capacity, resulting in congestion when some purchase bids and sell offers in merit order are not scheduled to reduce flows over constrained facilities. Congestion charges are applied to purchases and sales scheduled in the day-ahead and real-time markets based on the congestion component of the LBMP. Bilateral transactions scheduled through the ISO are charged the difference between the LBMPs of the two locations (i.e., the price at the sink minus the price at the source).

Market participants can hedge congestion charges in the day-ahead market by owning Transmission Congestion Contracts (“TCCs”), which entitle the holder to payments corresponding to the congestion charges between and the source and sink locations. For example, if a participant holds 150 MW of TCCs from zone A to zone B, this participant is entitled to 150 times the difference between the congestion prices at zone B and zone A. Excepting transmission losses, a participant can perfectly hedge a bilateral contract between two points if it owns a TCC between the points.

Incremental changes in generation and load from the day-ahead market to the real-time market are subject to congestion charges or payments in the real-time market. As in the day-ahead market, charges for real-time bilateral transactions are based on the difference between the locational prices at the two locations of the bilateral contract in the real-time market. There are no TCCs for real-time congestion.

This section summarizes the following aspects of transmission congestion and locational pricing:

- Congestion Revenues and Patterns – Subsections A, B, and C evaluate congestion revenues collected by the NYISO from the day-ahead market and patterns of congestion in the day-ahead and real-time markets.
- Constraints Requiring Frequent Out-of-Market Actions – Subsection D evaluates the management of transmission constraints that are frequently resolved using out-of-market actions, including 115 kV and 69 kV networks in New York.
- Linear Constraints to Model Long Island East End TVR Requirements – Subsection E describes a modeling approach to more efficiently schedule and price resources to satisfy the Transient Voltage Recovery (“TVR”) requirements on the East End of Long Island.
- Congestion Revenue Shortfalls – Subsection F analyzes congestion shortfalls in the day-ahead and real-time markets and identify major causes of shortfalls.

- **Transmission Line Ratings** – Subsection G analyzes the potential congestion benefit of using ambient-temperature adjusted line ratings in the market model.
- **TCC Prices and Day-Ahead Market Congestion** – Subsection H reviews the consistency of TCC prices and day-ahead congestion, which determine payments to TCC holders.
- **Efficient Allocation of Congestion Rents** – Subsection I outlines the flow-based concept supporting our proposal to more efficiently allocate day-ahead congestion rents and how flow-based congestion rent allocation could enable key transmission facilities between New York City and Long Island to be used more efficiently.
- **Statistical Modeling for Thunderstorm Forecasting** – Subsection J presents a well-established statistical model for predicting thunderstorm occurrences. It outlines the selected weather variables and indices used in the modeling process, as well as the data sources for training and testing.

A. Summary of Congestion Revenue and Shortfalls

This subsection summarizes congestion revenues and shortfalls that are collected and settled through the NYISO markets. Most congestion revenues are collected in the day-ahead market and known as *day-ahead congestion revenues*. These are collected by NYISO when power is scheduled to flow across congested interfaces in the day-ahead market. Revenue collected is equal to the marginal cost of relieving the constraint (i.e., constraint shadow price) in the day-ahead market multiplied by the scheduled flow across the constraint in the day-ahead market.²³⁵

In addition to day-ahead congestion revenues, NYISO incurs two types of shortfalls that occur when there are inconsistencies between the transmission capability modeled in the TCC market, the day-ahead market, and the real-time market:

- *Balancing Congestion Shortfalls* – These arise when day-ahead scheduled flows over a constraint exceed what can flow over the constraint in the real-time market.²³⁶ To reduce flows in real time below the day-ahead schedule, the NYISO must redispatch generators by increasing generation downstream of the constraint and reducing generation upstream of the constraint. These redispatch costs (i.e., the difference between the payments for increased generation and the revenues from reduced generation in the two areas) are the balancing congestion shortfall that is recovered through uplift.
- *Day-ahead Congestion Shortfalls* – These occur when the day-ahead congestion revenues collected by NYISO are less than the payments to TCC holders. Shortfalls generally arise when the quantity of TCCs sold on a path exceeds the transfer capability of the path

²³⁵ The shadow price of a transmission constraint represents the marginal value to the system of one MW of transfer capability. For example, if 100 MW is scheduled to flow across a constrained line with a shadow price of \$50/MWh in a particular hour in the day-ahead market, the NYISO collects \$5,000 in that hour (100 MW * \$50/MWh).

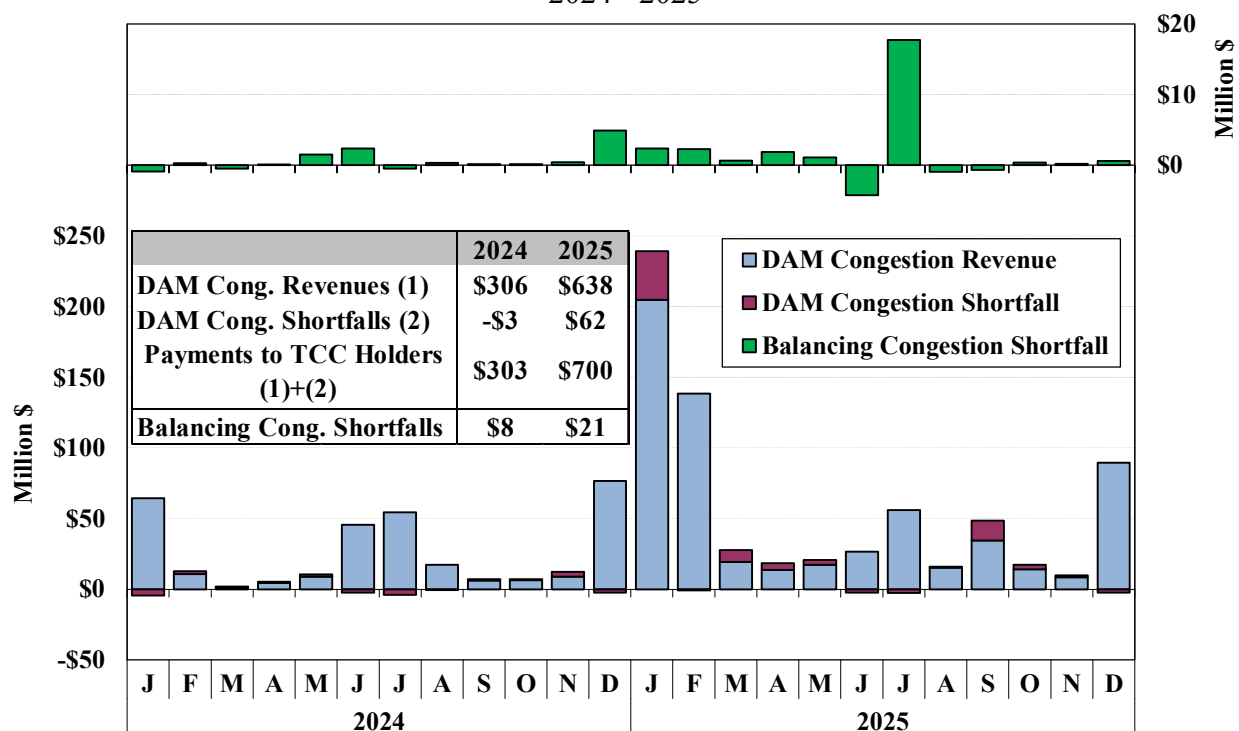
²³⁶ For example, suppose 100 MW is scheduled to flow across a particular line in the day-ahead market. If 90 MW flows across the line when it has a shadow price of \$70/MWh in an hour in the real-time market, the NYISO will have a balancing congestion shortfall of \$700 in that hour ((100 MW – 90 MW) * \$70/MWh).

modeled in the day-ahead market when it is congested.²³⁷ Day-ahead congestion shortfalls equal the difference between payments to TCC holders and day-ahead congestion revenues. These are partly offset by the revenues from selling excess TCCs.

Figure A-39: Congestion Revenue Collections and Shortfalls

Figure A-39 shows day-ahead congestion revenues and the two classes of congestion shortfalls for each month over the past two years. The upper portion of the figure shows balancing congestion shortfalls. The lower portion shows day-ahead congestion revenues collected by the NYISO and day-ahead congestion shortfalls and the sum of these two categories is equal to the total net payments to TCC holders in each month. The table in the figure reports these categories on an annual basis.

Figure A-39: Congestion Revenue Collections and Shortfalls
2024 - 2025



B. Congestion on Major Transmission Paths

Transmission lines moving power from low-cost to high-cost areas provide considerable value. In New York, eastern supply is generally more expensive than western supply, but most demand is in the East. This creates transmission bottlenecks from West to East, leading to significant

²³⁷ For example, suppose 120 MW of TCCs are sold across a particular line. If 100 MW is scheduled to flow when the constraint has a shadow price of \$50/MWh in an hour in the day-ahead market, the NYISO will have a day-ahead congestion shortfall of \$1,000 in that hour ((120 MW – 100 MW) * \$50/MWh).

congestion-related price differences between regions. This subsection examines congestion patterns in the day-ahead and real-time markets.

In the day-ahead market, the NYISO schedules generation and load based on the bids and offers submitted by market participants and the assumed transfer capability of the transmission network. When scheduling between regions reaches the limits of the transmission network, congestion price differences arise between regions in the day-ahead market.

Market participants submit bids in the day-ahead market that reflect their expectations of real-time prices and congestion, so day-ahead congestion prices are generally consistent with real-time congestion prices. To the extent that differences arise between day-ahead and real-time congestion patterns, it suggests that unexpected operating conditions may have occurred in the real-time market. Consistency between day-ahead and real-time prices is beneficial for market efficiency because it helps ensure that the resources committed each day are the most efficient ones to satisfy the system needs in real-time. Therefore, it is useful to evaluate the consistency of congestion patterns between the day-ahead and real-time markets.

Figure A-40 to Figure A-42: Day-Ahead and Real-Time Congestion by Path

Figure A-40 to Figure A-42 show the value and frequency of congestion along major transmission lines in the day-ahead and real-time market. Figure A-40 compares these quantities in 2024 and 2025 on an annual basis, while Figure A-41 and Figure A-42 show the quantities separately for each quarter of 2025. The figures measure congestion in two ways:²³⁸

- The frequency of binding constraints; and
- The value of congestion, which is equal to the marginal cost of relieving the constraint (i.e., constraint shadow cost) multiplied by the scheduled flow across the constraint.²³⁹

In the day-ahead market, the value of congestion equals the congestion revenues collected by the NYISO, which is the primary funding source for TCC payments. In the real-time market, the value of congestion does not equal the congestion revenues collected by the NYISO, since most real-time power flows settle at day-ahead prices rather than real-time prices. Nonetheless, the real-time congestion value provides the economic significance of congestion in the real-time market. The figure groups congestion along the following transmission paths:

- West Zone Lines: Transmission lines in the West Zone.
- Oswego Complex Export: Transmission lines that carry power out of the Oswego Complex generation pocket.
- West to Central: Primarily West-to-Central interface, Dysinger East interface, and transmission facilities in the Central Zone, excluding the Oswego Complex Export lines.
- North to Central: Primarily transmission facilities within and out of the North Zone.

²³⁸ Binding transmission constraints with a shadow cost lower than \$0.1/MWh are not included.

²³⁹ The shadow cost of a transmission constraint represents the marginal value to the system of one MW of transfer capability.

- Central to East: Transmission facilities from Western and Central New York to Eastern New York, including the Central-to-East interface.
- Capital to Hudson Valley: Primarily lines leading into Southeast New York.
- Hudson Valley to Dunwoodie: Lines and interfaces from Hudson Valley to Dunwoodie.
- New York City: Lines leading into and within the NYC 345 kV system, line leading into and within New York City load pockets and groups of lines into load pockets that are modeled as interface constraints.
- Long Island: Lines leading into and within Long Island.
- External Interface: Congestion related to the total transmission limits or ramp limits of external interfaces. The two primary free-flow interfaces (NYISO-PJM and NYISO-New England), are shown separately, while all other interfaces are presented in aggregate.

Figure A-40: Day-Ahead and Real-Time Congestion by Transmission Path
2024 – 2025

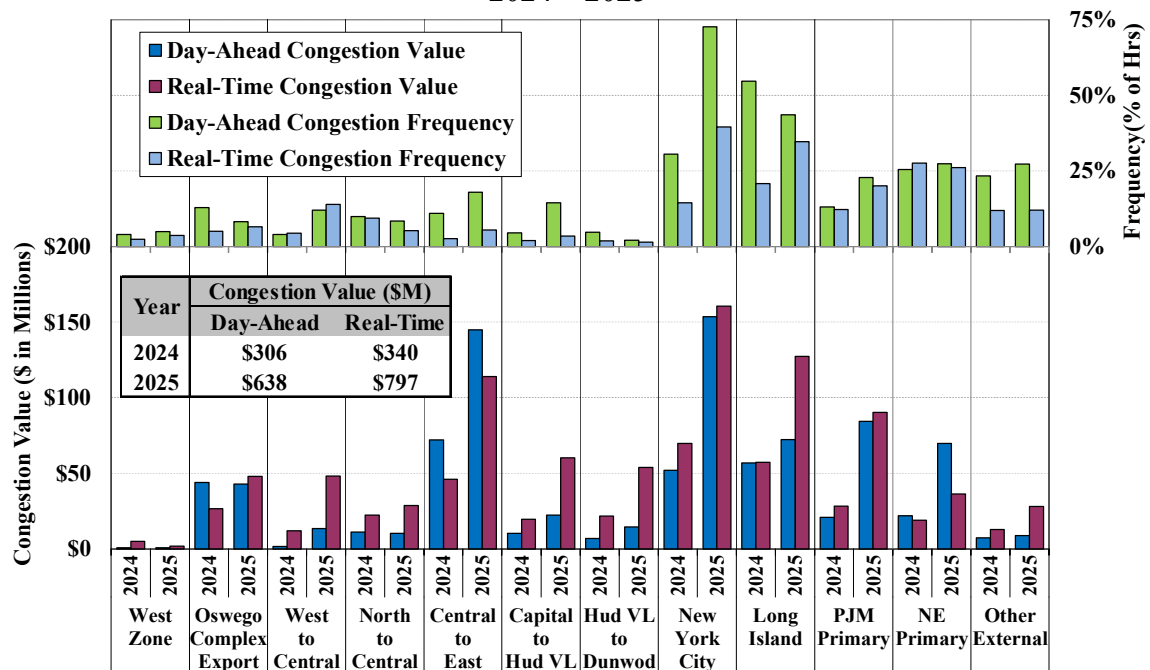


Figure A-41: Day-Ahead Congestion by Transmission Path
By Quarter, 2025

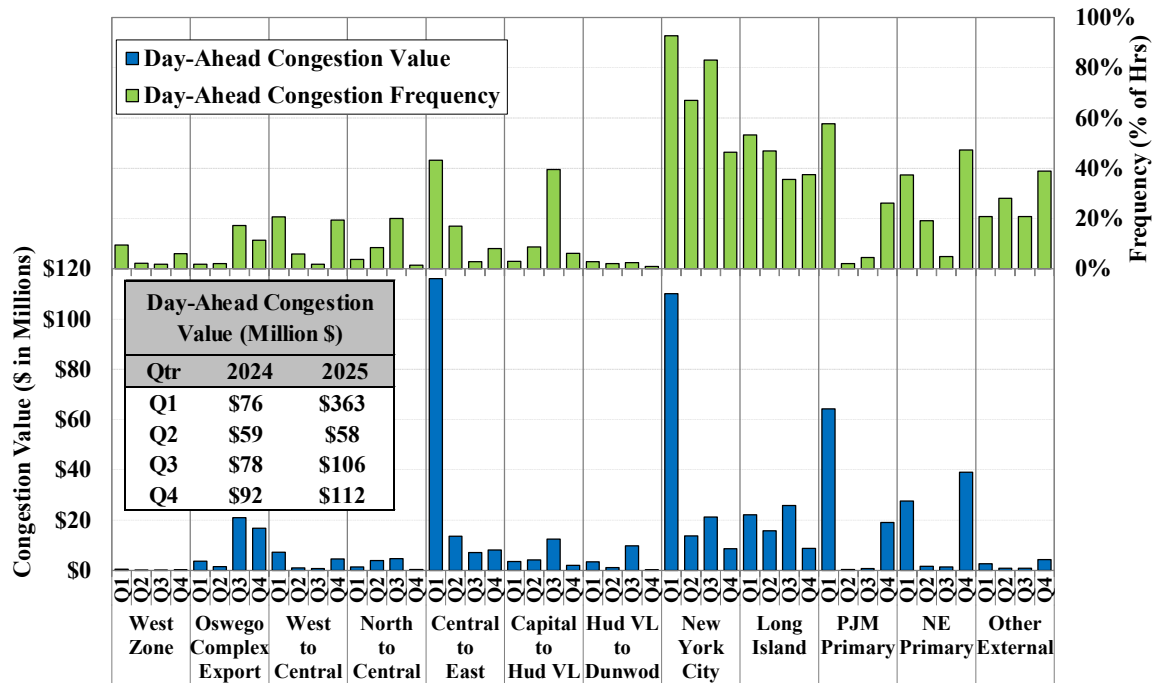
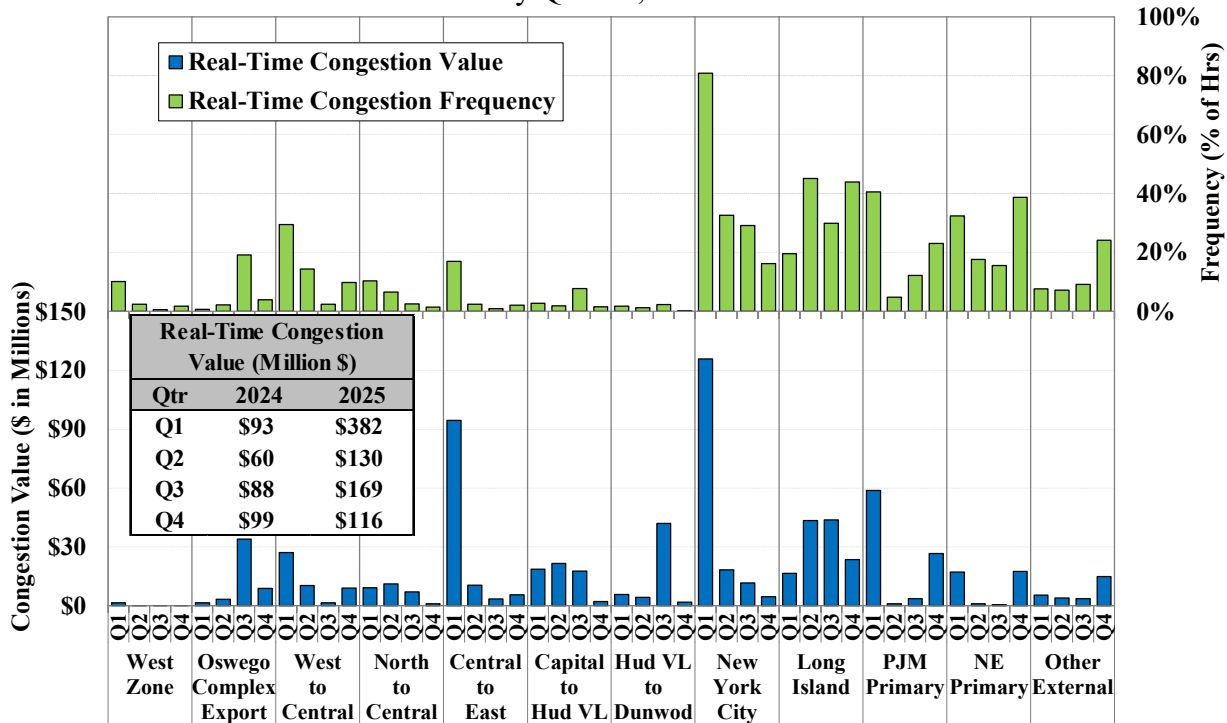


Figure A-42: Real-Time Congestion by Transmission Path
By Quarter, 2025

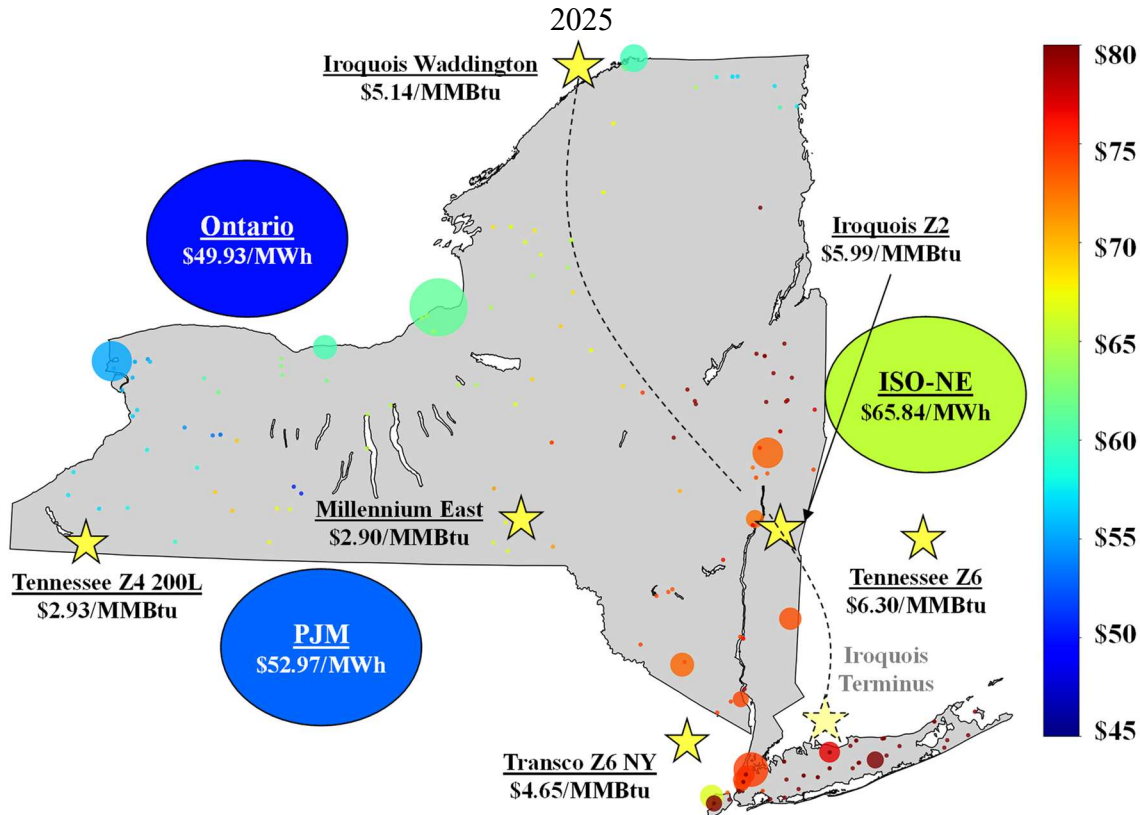


C. Real-Time Congestion Map by Generator Location

Figure A-43 to Figure A-44: Real-Time Load-Weighted Congestion Maps by Location

The previous subsection reports congestion patterns on a zonal basis or along large inter-zonal interfaces, while this subsection displays more granular information pertaining to congestion across generator nodes. Figure A-43 and Figure A-44 are two congestion maps showing such information for the entire system and New York City, respectively.

Figure A-43: NYCA Real-Time Load-Weighted Generator Congestion Map



The maps display differences in LBMPs between generator nodes across the system,²⁴⁰ illustrating transmission bottlenecks not only between broader areas but also within smaller subareas, highlighting the prevalence of intra-zonal price divergence between generation pockets and load pockets. Often, significant congestion arises from an abundance of inexpensive generation located in an export pocket driving bottlenecks on transmission lines servicing load pockets with a small number of competing generators. It also highlights where generation or transmission investment is likely to be most valuable, which can help guide investment. Each map shows details of nodal congestion in the real-time market in 2025, specifically:

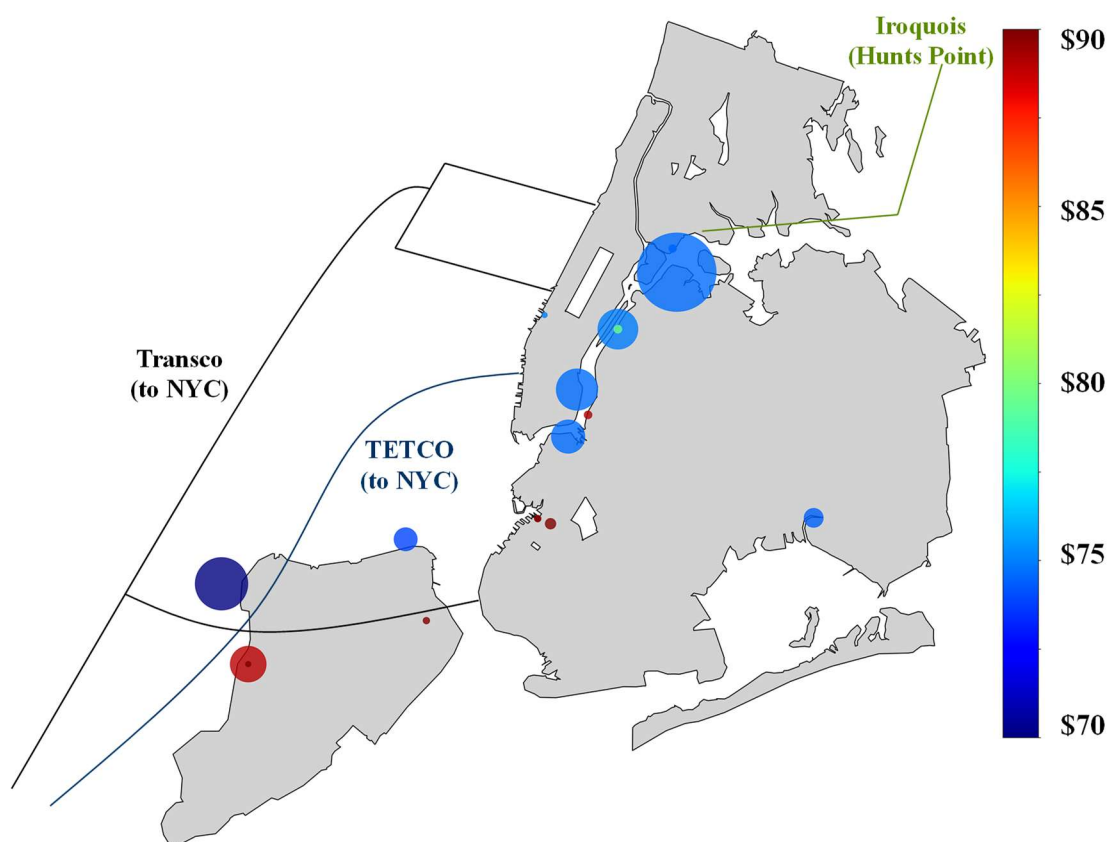
- Load-weighted hourly average real-time LBMP at each generator node within the region;

²⁴⁰

Although the differences in LBMPs include the differences in congestion and losses, the differences in losses are usually much smaller than the differences in congestion, particularly between generator nodes that are within smaller subareas.

- For the systemwide map, real-time prices on the neighboring area’s side of the external interface are load-weighted using NYCA systemwide load and presented as additional bubbles. These bubbles are not sized based on average generation levels;²⁴¹ and
- Pertinent gas market information including regional gas prices in the systemwide map and key operational points of gas delivery in the NYC map.²⁴²

Figure A-44: NYC Real-Time Load-Weighted Generator Congestion Map
2025



The generator bubbles are sized based on annual average generation MWh, however the sizing of these bubbles differs between the two maps due to the disparities in geographical sizes of the entire system versus New York City. In each case, however, a floor value is set such that generators at or below a certain annual average output all appear with the same size (i.e., the smallest sized bubble on the map), while generators with greater annual average outputs are shown with a size that is in proportion to their annual average generation. Portfolios with multiple generator PTIDs at the same station or within close proximity to each other are aggregated into one bubble and sized based on average portfolio generation. Each generator

²⁴¹ The external interface prices are sourced from the respective system operator web platforms for each region. These prices can be found for each region at PJM, ISO-NE, and IESO web platforms.

²⁴² Natural gas prices are based on the average index prices without additional adders sourced from Platts.

bubble is colored based on a heat mapping scale included to the right of each map. Prices along the color-scale are included with colder colors representing lower load-weighted real-time prices.

D. Transmission Constraints Managed with OOM Actions

Transmission constraints on the high-voltage network (including 230 and 345 kV facilities in upstate New York and most 138 kV facilities in New York City and Long Island) are generally managed through the day-ahead and real-time market systems. This provides several benefits including: (a) that the market optimization balances the costs of satisfying demand, ancillary services, and transmission security requirements, resulting in more efficient scheduling decisions; and (b) that the market optimization also produces a set of transparent clearing prices, which provide efficient signals for longer lead time decisions such as fuel procurement, generator commitment, external transaction scheduling, and investment in new and existing resources and transmission.

Although a significant portion of transmission constraints on the low-voltage (i.e., 115 kV and lower) network has been incorporated into the market systems since 2015, many remaining constraints continue to be managed through out-of-market operator actions. Typical operator actions used to resolve constraints on the low-voltage network include:

- Out of merit dispatch and supplemental commitment of generation;
- Curtailment of external transactions and limitations on external interface transfer limits;
- Use of an internal interface/constraint transfer limit that functions as a proxy for the limiting transmission facility; and
- Adjusting PAR-controlled line flows on the high voltage network.²⁴³

In this subsection, we evaluate:

- The frequency of such OOM actions used to manage transmission constraints on the low voltage network in New York (including 115 kV and 69 kV facilities) that are not incorporated in the market systems; and
- The potential pricing impact in several load pockets on Long Island.

Figure A-45 & Figure A-46: OOM-Managed Constraints on the Low Voltage Network

Figure A-45 shows the number of days in 2025 when various resources were used out of merit to manage constraints in seven areas of New York: (a) West Zone; (b) Central Zone; (c) Capital Zone; (d) North & Mohawk Valley Zones; (e) Hudson Valley Zone; (f) New York City; and (g) Long Island. In addition, the figure also reports the number of days when out-of-merit commitments were made to satisfy voltage needs or N-1-1 reserve needs in several local load pockets.

²⁴³ These constraints are sometimes managed with the use of line switching on the distribution system, but this is not included in our analysis here.

Figure A-45: OOM-Managed Constraints in New York
Summary of Resources Used to Manage Constraints, 2025

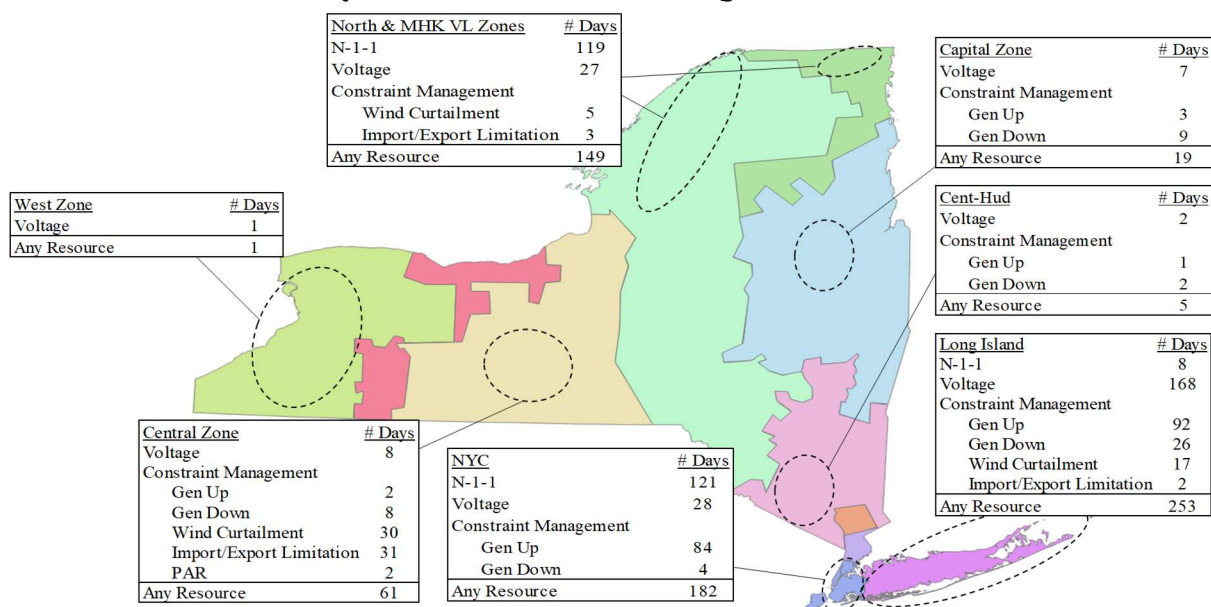


Figure A-46 focuses on the area of Long Island, showing the number of hours and days in 2025 when various resources were used to manage 69 kV (labeled as “69 kV OOM”) and TVR constraints (labeled as “TVR OOM”) in four load pockets of Long Island:

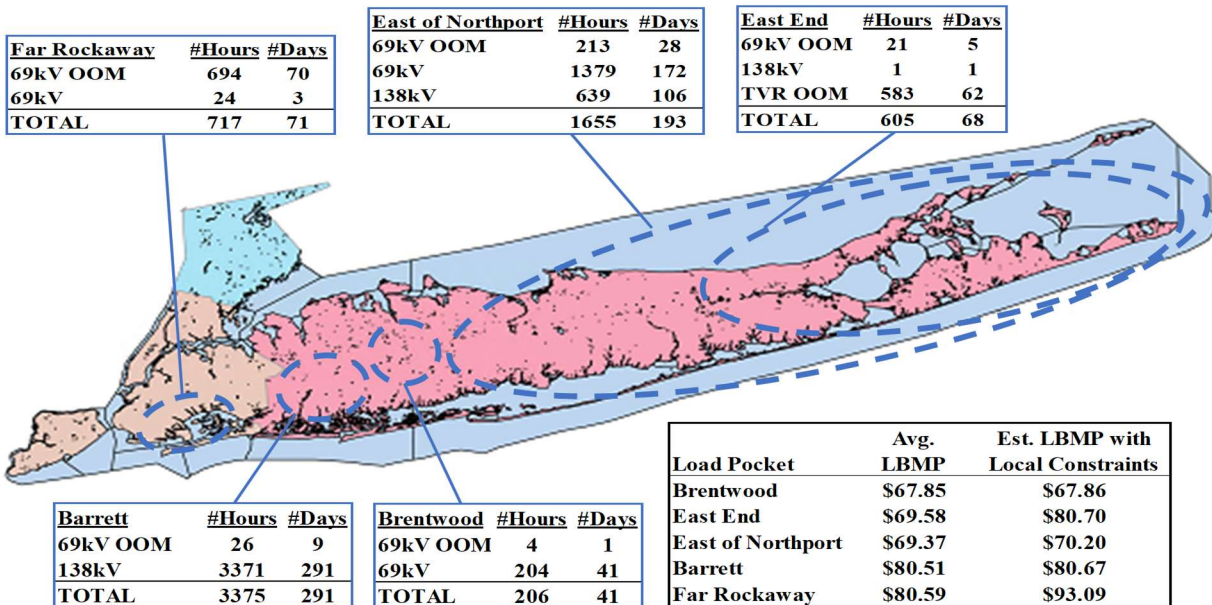
- Barrett: Mostly constraints around the Barrett bus;
- Far Rockaway: Constraints into the Far Rockaway load pocket
- Brentwood: Mostly constraints around the Brentwood bus;
- East of Northport: Mostly the Central Islip-Hauppauge and the Elwood-Deposit circuits;
- East End: Mostly the constraints around the Riverhead bus and the TVR requirement.

For a comparison, the tables also show the frequency of congestion management on the 69 kV and 138 kV constraints via the market model. Figure A-46 also shows our estimated price impacts in each Long Island load pocket that result from explicitly modeling these 69 kV and TVR constraints in the market software.²⁴⁴

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The following generator locations are chosen to represent each load pocket: (a) Barrett ST for the Barrett pocket; (b) Far Rockaway GT for the Far Rockaway pocket; (c) NYPA Brentwood GT for the Brentwood pocket; (d) Holtsville IC for the East of Northport pocket; and (e) Green Port GT for the East End pocket.

Figure A-46: Constraints on the Low Voltage Network on Long Island
Frequency of Action Used to Manage Constraints, 2025



E. Linear Constraints to Model Long Island East End TVR Requirements

Certain resources are required to be online to satisfy the Transient Voltage Recovery (“TVR”) requirement on the East End of Long Island.²⁴⁵ These resources are expensive oil peakers, which are rarely committed economically. As a result, OOM commitments are made by the local TO based on operating guidelines.²⁴⁶ These OOM commitments not only generate uplift but also depress real time prices on Long Island (see Figure A-46). Integrating TVR requirements into the market software could enable more efficient scheduling and pricing of resources on Long Island. This subsection outlines an approach to modeling TVR requirements using linear constraints for scheduling and pricing purposes.

The following three tables, excerpted from the latest *East End Operating Guideline*, detail operating options under various outage conditions and load levels.²⁴⁷ Table A-1 summarizes generator start procedures for 20 operating scenarios based on the availability of four voltage-control devices: the 9EU DVAR, the 9C DRSS DVAR, the 9C DRSS CAPs, and the Canal DRSS. Depending on their statuses, two generator equivalent support tables are utilized to guide the commitments of the five oil-fired peakers on the East End.

For instance, under Operating Scenario 1, where all four voltage-control devices are fully available, oil peakers must be committed to address the TVR need when South Fork load

²⁴⁵ Includes Global Greenport GT, East Hampton units, South Hampton IC, and Southold IC.

²⁴⁶ See *East End Operating Guideline*, available at: <https://www.psegliny.com/oasis/transmission-owner-information-being-released-to-market>.

²⁴⁷ See the *East End Operating Guideline*, released on 07/22/2022.

exceeds 173 MW. In such cases, the “*Equivalent Unit Support for South Fork Load to Resolve TVR*” values under Canal DRSS Status B serve as a reference for resource commitment.

Table A-1: East End Generation Start Procedures

Operating Scenarios	9EU DVAR	9C DRSS DVAR	9C DRSS CAPS	GEN START	ARM VOLTAGE COLLAPSE	East End Generator Equivalent Support Table for South Fork Load to Resolve TVR
1	Available	100%	Available	173	294	Canal DRSS Status B
2	Available	75%	Available	172	281	Canal DRSS Status A
3	Available	50%	Available	171	280	
4	Available	25%	Available	170	279	
5	Available	0%	Unavailable	169	278	
6	Available	100%	Unavailable	171	280	Canal DRSS Status A
7	Available	75%	Unavailable	170	279	Canal DRSS Status A
8	Available	50%	Unavailable	170	279	
9	Available	25%	Unavailable	169	278	
10	Available	0%	Unavailable	169	278	
11	Unavailable	100%	Available	155	276	Canal DRSS Status B
12	Unavailable	75%	Available	154	263	Canal DRSS Status A
13	Unavailable	50%	Available	153	262	
14	Unavailable	25%	Available	152	261	
15	Unavailable	0%	Unavailable	151	260	
16	Unavailable	100%	Unavailable	153	262	Canal DRSS Status A
17	Unavailable	75%	Unavailable	152	261	Canal DRSS Status A
18	Unavailable	50%	Unavailable	152	261	
19	Unavailable	25%	Unavailable	151	260	
20	Unavailable	0%	Unavailable	151	260	

Units	Canal DRSS Status A: ⁽¹⁾ Not Fully Available	Canal DRSS Status B: 100% Available
Greenport GT	34 MW	46 MW
East Hampton GT	24 MW	37 MW
East Hampton Diesel	10 MW	16 MW
Southold GT	20 MW	22 MW
Southampton GT ⁽²⁾	13 MW	0 MW
Total Unit Support	101 MW	121 MW

(1) There is an additional 8 MW of support for first unit dispatched due to change in limiting condition (i.e. if Greenport GT is dispatched first, the unit will provide 34+8=42 MW of support)

(2) When Canal DRSS is not fully available, Southampton GT cannot be dispatched first; while when Canal DRSS is 100% available, southampton GT does not provide support for TVR, but can be dispatched for load and/or thermal constraints.

The following linear constraints can be developed to represent Scenarios A and B. As shown in Table A-2, the dispatch levels of the five generators are labeled G1 through G5, while their commitment statuses are denoted as C1 through C5. For pricing purposes, all five blocked-on resources can be dispatched flexibly between zero and their respective UOLs.

Table A-2: East End Generator Commitment and Dispatch Parameters

Resource Name	Generator Dispatch [0, UOL]	Commitment Status {0, 1}	Bid UOL
Greenport GT	G1	C1	UOL1
East Hampton GT	G2	C2	UOL2
East Hampton Diesel	G4	C3	UOL3
Southold GT	G3	C4	UOL4
South Hampton GT	G5	C5	UOL5

The requirement under the Canal DRSS Status B scenario is straightforward to model with the following two linear constraints:

- For commitment,

$$46 * C1 + 37 * C2 + 16 * C3 + 22 * C4 + 0 * C5 \geq \text{South Fork Load} - \text{Load Trigger},$$
where C1 to C5 are binary commitment variables (0 or 1).
- For pricing,

$$\frac{46}{UOL1} G1 + \frac{37}{UOL2} G2 + \frac{16}{UOL3} G3 + \frac{22}{UOL4} G4 + \frac{0}{UOL5} G5 \geq \text{South Fork Load} - \text{Load Trigger},$$
where G1 to G5 are dispatchable from 0 to their individual UOL.

The Canal DRSS Status B scenario requires additional constraints, thus it is segmented to account for the additional 8 MW of support from the first committed resource.

Case 1: When $0 \leq \text{South Fork Load} - \text{Load Trigger} \leq 8$,

- For commitment,

$$34 * C1 + 24 * C2 + 10 * C3 + 20 * C4 + 13 * C5 \geq \text{South Fork Load} - \text{Load Trigger}$$
- For pricing,

$$\frac{34}{UOL1} G1 + \frac{24}{UOL2} G2 + \frac{10}{UOL3} G3 + \frac{20}{UOL4} G4 + \frac{13}{UOL5} G5 \geq \text{South Fork Load} - \text{Load Trigger}.$$

Case 2: When $\text{South Fork Load} - \text{Load Trigger} > 8$,

- For commitment,

$$34 * C1 + 24 * C2 + 10 * C3 + 20 * C4 + 13 * C5 \geq \text{South Fork Load} - \text{Load Trigger} - 8$$
- For pricing,

$$\frac{34}{UOL1} G1 + \frac{24}{UOL2} G2 + \frac{10}{UOL3} G3 + \frac{20}{UOL4} G4 + \frac{13}{UOL5} G5 \geq \text{South Fork Load} - \text{Load Trigger} - 8$$

Additionally, to ensure that South Hampton GT is not dispatched first, the following constraint must be enforced:

- $$C5 \leq C1 + C2 + C3 + C4$$

These linear constraints could provide a mechanism to efficiently schedule and price the TVR requirement through the market software rather than inefficient OOM actions and uplift payments.

F. Day-Ahead and Balancing Congestion Shortfalls by Path or Constraint

Congestion shortfalls generally occur because of inconsistent modeling of the transmission system between markets. Day-ahead congestion shortfalls indicate inconsistencies between the TCC and day-ahead market, while balancing congestion shortfalls indicate inconsistencies between the day-ahead market and the real-time market. These two classes of shortfalls are evaluated in this subsection.

Figure A-47 - Figure A-50: Day-ahead Congestion Revenue Shortfalls

Day-ahead congestion revenue shortfalls generally arise when the quantity of TCCs sold for a particular path exceeds the transfer capability of the path modeled in the day-ahead market during periods of congestion. Similarly, surpluses occur when the quantity of TCCs sold for a path is less than the transfer capability of the path in the day-ahead market during periods of congestion. The NYISO minimizes day-ahead congestion revenue surpluses and shortfalls by offering TCCs in the forward auction that reflect the expected transfer capability of the system. In addition, transmission owners can reduce potential day-ahead congestion revenue shortfalls by restricting the quantities of TCCs that are offered by NYISO.

NYISO determines the quantities of TCCs to offer in a TCC auction by modeling the transmission system to ensure that the TCCs sold are simultaneously feasible. NYISO uses a power flow model that includes an assumed configuration of the transmission system. The simultaneous feasibility condition requires that the TCCs awarded be feasible in a contingency constrained economic dispatch of the NYISO transmission system. If this condition is satisfied, the congestion revenues collected are expected to be sufficient to fully fund awarded TCCs. However, if transmission outages occur that were not modeled in the TCC auction or the assumptions used in the TCC auctions (e.g., assumptions related to PAR schedules and loop flows) are inconsistent with the assumptions used in the day-ahead market, the congestion revenues collected may be insufficient to meet TCC obligations.

Congestion shortfalls and surpluses resulting from differences between the TCC auctions and the day-ahead market are allocated to transmission owners as charges or credits. NYISO currently uses a two-stage process defined in the OATT for the allocations, as illustrated in Figure A-47:

- First, congestion residuals resulted from Qualifying facility changes (e.g., outages, return-to-services, and uprate/derate) are allocated to responsible transmission owners. This allocation is based on the flow impact of these change factors on the binding constraints in the day-ahead market, adhering to the cost causation principle.²⁴⁸

²⁴⁸ See OATT, Attachment N, Formula N-6 through N-14 for the calculation of these allocations.

- Second, the remaining congestion residuals, referred to as *Net Congestion Rents*, are allocated to transmission owners in a different way.²⁴⁹ These allocations are in proportion to the auction revenues from each TO's TCC holdings rather than day-ahead congestion patterns, which may not necessarily align with the cost causation principle.

Figure A-47: Illustration of Allocation of DAM Congestion Residuals

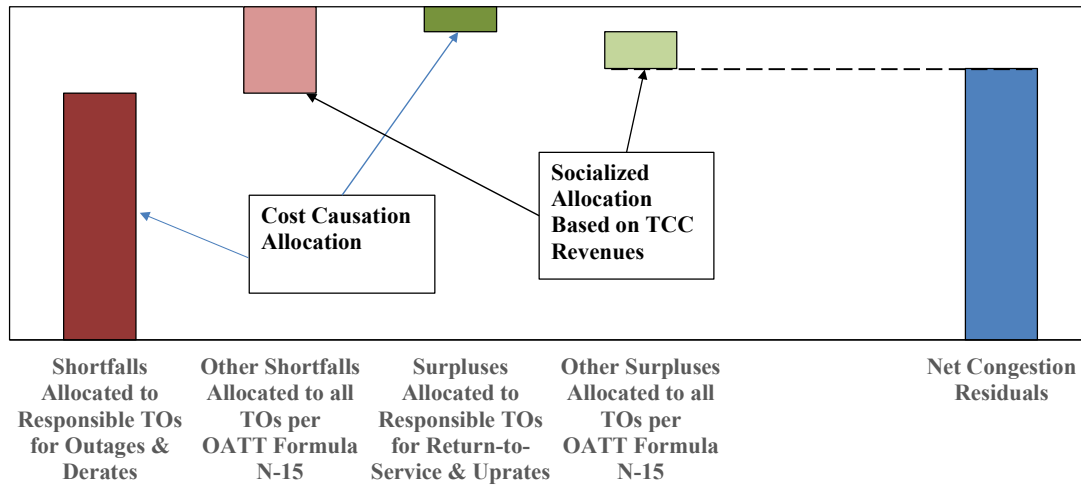


Figure A-48 shows actual allocations of day-ahead congestion residuals for each month over the past two years. The blue bars represent the portion allocated in the first stage based on a cost causation principle, while the red bars represent the portion that was allocated in the second stage based on TCC revenues using Formula N-15 in the OATT Attachment N. The positive numbers represent congestion shortfalls, while the negative numbers represent congestion surpluses. The inset table provides an annual summary of the net amount of congestion residuals allocated through these two methods for the past five years.

Figure A-49 presents our estimates of the first-stage cost-causation allocations to responsible Transmission Owners by month over the past two years. The ownership of transmission facilities is based on information provided in *Attachment A of the Outage Scheduling Manual: NYISO Facilities Requiring Coordination Notification*. Shortfalls accrued on a facility are allocated evenly among transmission owners with “O” designations, which indicate full or partial ownership of the facility. If a facility does not have a “O” designation, the “S” designation is used instead, which identifies the entity with switching authority over the facility.

²⁴⁹ See OATT, Attachment N, Formula N-15 for the calculation of these allocations.

Figure A-48: Allocation of DAM Congestion Residuals
By Month, 2024-2025

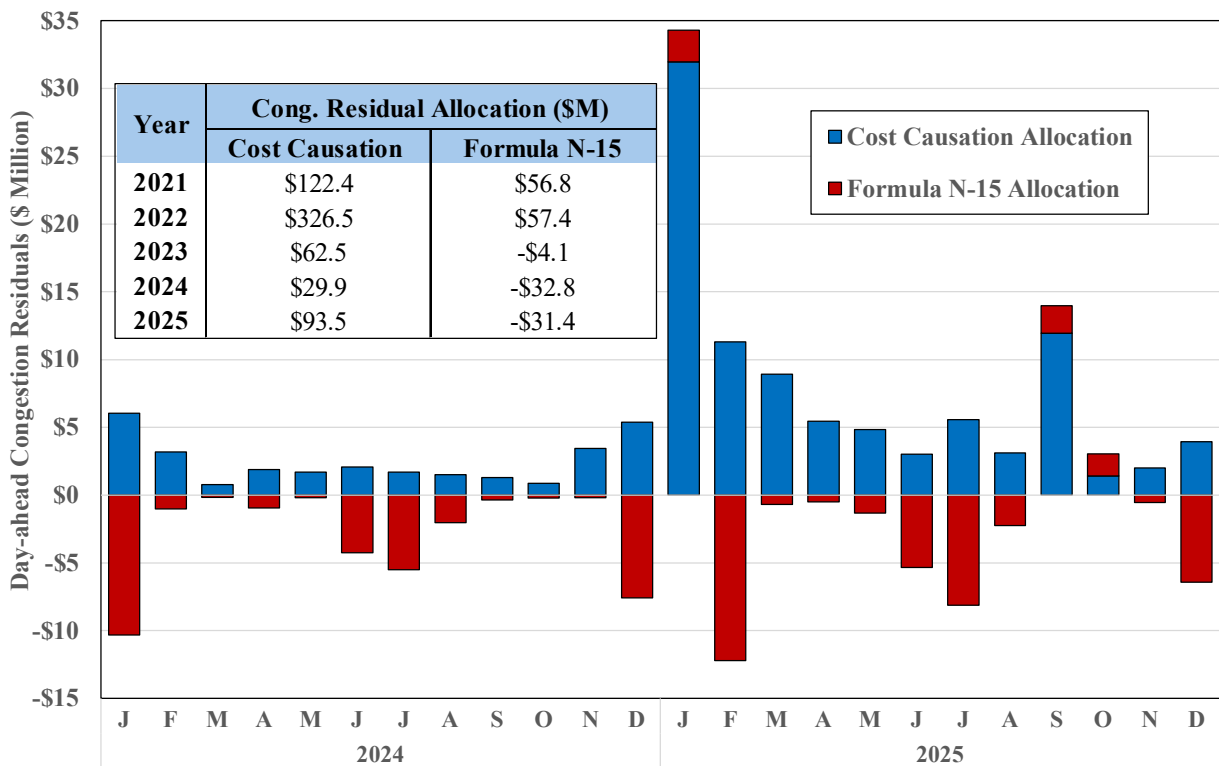


Figure A-49: First Stage Allocations of DAM Congestion Residuals to Responsible TOs
2024 - 2025

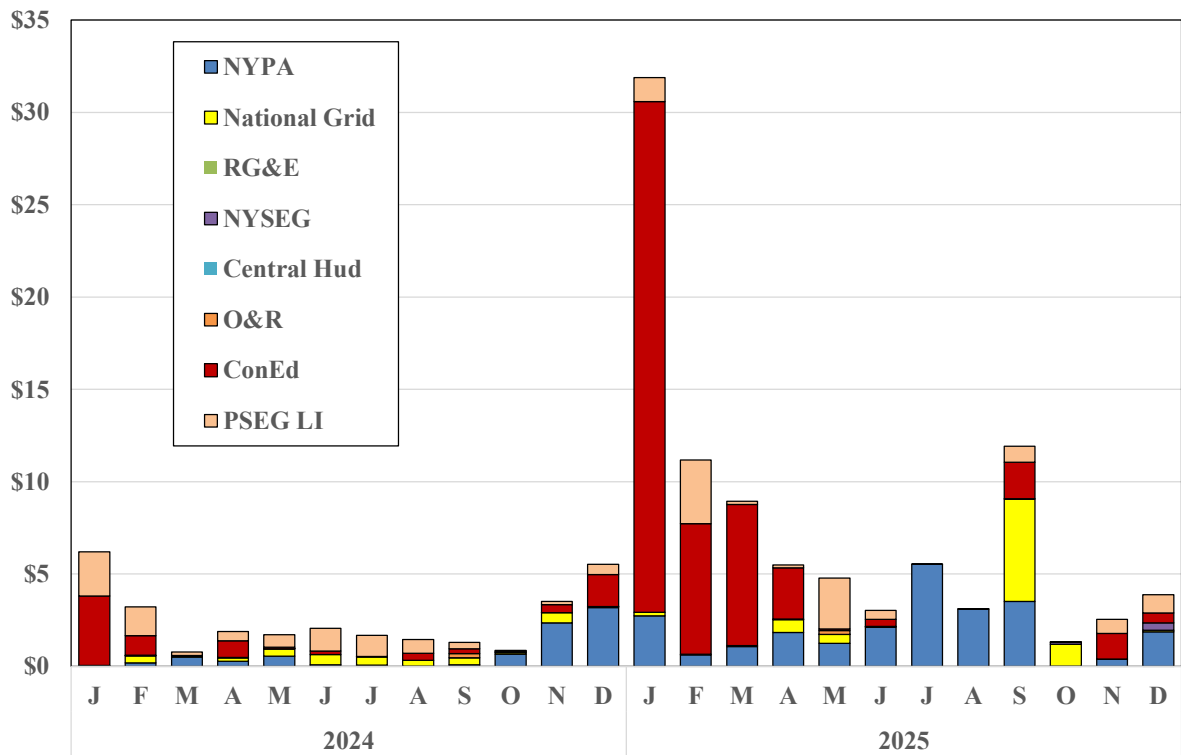


Figure A-50 presents the quantities of remaining congestion residuals, also referred to as *Net Congestion Rents*, which socialized among all transmission owners in the second stage of allocations. These quantities are categorized by the following transmission facility groups:

- North to Central: Transmission lines in the North Zone, the Moses-South Interface, EDIC-Marcy 345 line, and Marcy 765-Marcy 345 line.
- West to Central: Primarily West-to-Central interface, Dysinger East interface, and transmission facilities in the Central Zone
- Central to East: Primarily the Central-East interface.
- Capital to Hudson Valley: Lines from Capital zone to Hudson Vally zone.
- New York City Lines: Lines leading into and within New York City.
- Long Island Lines: Lines leading into and within Long Island.
- External: Related to the total transmission limits or ramp limits of the external interfaces. The primary PJM interface is shown separately from other interfaces within the group.
- All Others: All other types of constraints collectively.

The figure also shows the shortfalls resulting from some unique factors separately from other reasons for select transmission paths. For Long Island lines, the figure shows separately the shortfalls resulted from differences in assumed schedules across the two PAR controlled lines between Lake Success and Valley Stream in Long Island and Jamaica in New York City (i.e., 901/903 lines) between the TCC auction and the day-ahead market, labeled as “901/903 PARs” in the figure.

Figure A-50: Day-Ahead Net Congestion Rents
2024 – 2025

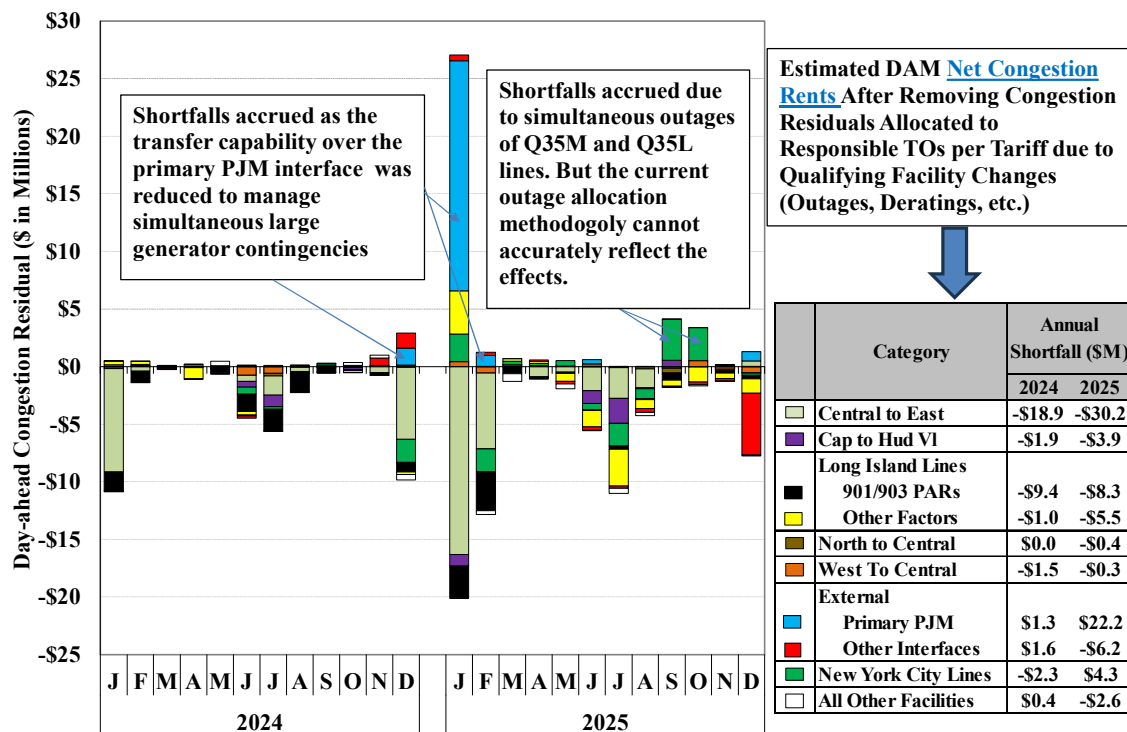


Figure A-51: Balancing Congestion Revenue Shortfalls

Balancing congestion revenue shortfalls occur when day-ahead scheduled flows across a particular line or interface exceed its real-time transfer capability, which often requires the ISO to redispatch in real time by purchasing additional generation in the import-constrained area (where real-time prices are high) and selling back energy in the export-constrained area (where real-time prices are low). The cost of this redispatch is the balancing congestion shortfall.

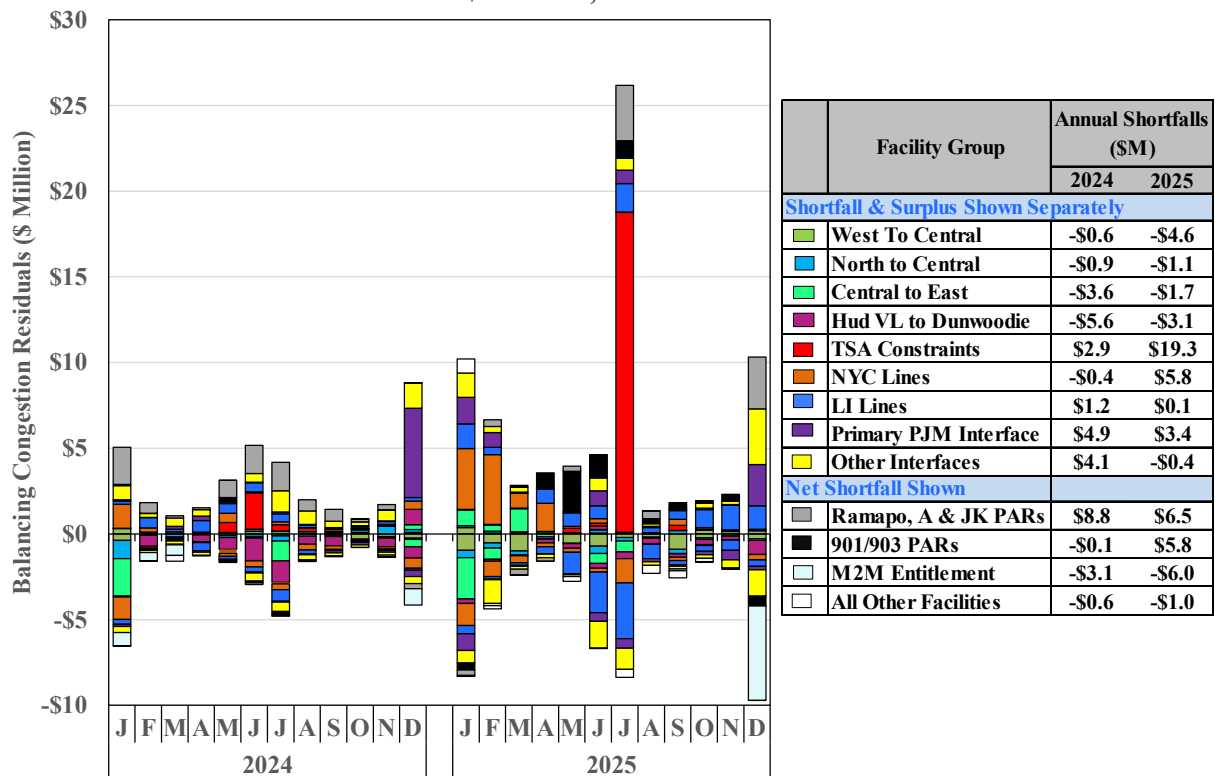
Key factors causing changes in transfer capability between day-ahead and real-time markets include:

- **Transmission Deratings and Outages** – When these occur after the day-ahead market, they reduce the transfer capability of relevant transmission interfaces or facilities. They may also change the size of the largest contingency relative to a particular transmission interface or the distribution of flows over the transmission system, thereby reducing the available transfer capability of other transmission facilities.
- **Unmodeled Constraints in the Day-ahead Market** – Reliability rules require the NYISO to reduce actual flows across certain key interfaces during TSA events. Since TSA events are not modeled in the day-ahead market, they generally result in reduced transfer capability between the day-ahead market and real-time operation. The imposition of simplified interface constraints in New York City load pockets in the real-time market that are not modeled comparably in the day-ahead market also results in reduced transfer capability between the day-ahead market and real-time operation.
- **Fast-Start Pricing** – This methodology treats physically inflexible gas turbines as flexible in the pricing logic of the real-time market model. Differences between the physical dispatch logic and the pricing logic can lead to unutilized transfer capability on interfaces that are congested in real time, leading to balancing congestion revenue shortfalls.
- **PAR Controlled Line Flows** – The flows across PAR-controlled lines are adjusted in real-time operations, which can result in flows that are very different from the day-ahead assumptions. These differences can affect the flows across multiple interfaces. This includes flow adjustments on PAR-controlled lines that result from the Coordinated Congestion Management (“M2M”) process between NYISO and PJM.
- **Unscheduled Loop Flows** – loop flows from other regions use a portion of the transmission capability across many interfaces in New York, reducing the portion of transmission capability available to the NYISO market in the direction of the loop flows. A balancing congestion revenue shortfall occurs when the loop flows assumed in the day-ahead market are lower than the actual loop flows on congested interfaces in real time.

The net cost of the redispatch in real-time due to changes from day-ahead (i.e., balancing congestion shortfalls) is collected from loads through uplift charges, most of which is allocated to load throughout the state. However, a portion associated with facilities that require special operation during TSA events is charged to Consolidated Edison whose customers benefit most directly from the additional reliability.

Figure A-51 shows balancing congestion shortfalls by transmission path or facility in each month of 2024 and 2025. Congestion shortfalls and surpluses are grouped and shown separately within each facility group. The figure also shows the shortfalls resulted from certain unique factors separately from other reasons, including PAR operations over the 901 and 903 PARs between Long Island and New York City, PAR operations over the A, JK, and Ramapo PARs under the M2M constraint management agreement between PJM and New York, and the associated M2M entitlement payment from PJM. Positive values indicate shortfalls, while negative values indicate surpluses.

Figure A-51: Balancing Congestion Shortfalls²⁵⁰
In \$Millions, 2024 – 2025



G. Transmission Line Ratings

Transmission line ratings define the maximum transfer capability of each line, impacting commitment, dispatch, congestion and prices. Accurate line ratings are essential - understated ratings can lead to inefficient market outcomes (e.g., higher production costs, and unnecessarily high congestion and energy prices), while overstated ratings may pose reliability risks.

²⁵⁰

The balancing congestion shortfalls estimated in this figure may differ from actual balancing congestion shortfalls because the figure: (a) is partly based on real-time schedules rather than metered injections and withdrawals; and (b) uses the original constraint shadow costs from the dispatch model therefore does not reflect the effect of any ex-post price corrections.

Line ratings are typically limited by thermal, voltage, or stability constraints, with thermal limits usually being the most restrictive. Thermal limits are typically influenced by ambient conditions (e.g., temperature, wind speed, and solar irradiance). For example, when ambient temperatures are cooler than the typical assumptions used for rating the facilities, additional power flows can be accommodated.

Currently, the NYISO primarily uses static seasonal line ratings for most facilities in the day-ahead and real-time markets. Some Ambient Adjusted Ratings (“AARs”) are applied in the real-time market, but static line ratings remain prevalent. This subsection examines the potential economic benefits of implementing hourly AARs in NYISO’s markets.

Figure A-52: Potential Congestion Benefit of Using Ambient-Temperature Adjusted Ratings

Figure A-52 shows our estimate of potential congestion benefit from using ambient-temperature adjusted line ratings for 2024 to 2025.

We estimate ambient-adjusted ratings based on the following assumptions:²⁵¹

- Summer line ratings are developed based on an ambient temperature of 95°F (or 35°C);
- Winter line ratings are developed based on an ambient temperature of 50°F (or 10°C); and
- For overhead lines, the relationship between the ambient-adjustment rating factor and the ambient temperature is close to linear in a wide range of normal weather conditions.

Therefore, we extrapolate the ambient adjusted ratings from the straight line that connects the summer and winter ratings and their assumed rating temperatures.²⁵² Wind speed is a critical parameter that impacts equipment thermal ratings, but its variation is not considered in this calculation.

In the figure, the bars in the bottom of the chart represent the estimated potential benefit, which equals the constraint shadow cost times the additional transfer capability from the estimated potential ambient adjustment.²⁵³ These estimates are done separately for the day-ahead and real-time markets on an hourly basis. Results are presented separately for the following facility groups: a) within the West Zone; b) West to Central; c) out of the Oswego Complex generation pocket; d) North to Central; e) Capital to Hudson Valley; and f) Hudson Valley to Dunwoodie. The bars in the top portion of the chart show the potential benefit as a percentage of total congestion values in each facility group. The inset table summarizes these quantities on an annual basis for all shown facilities combined.

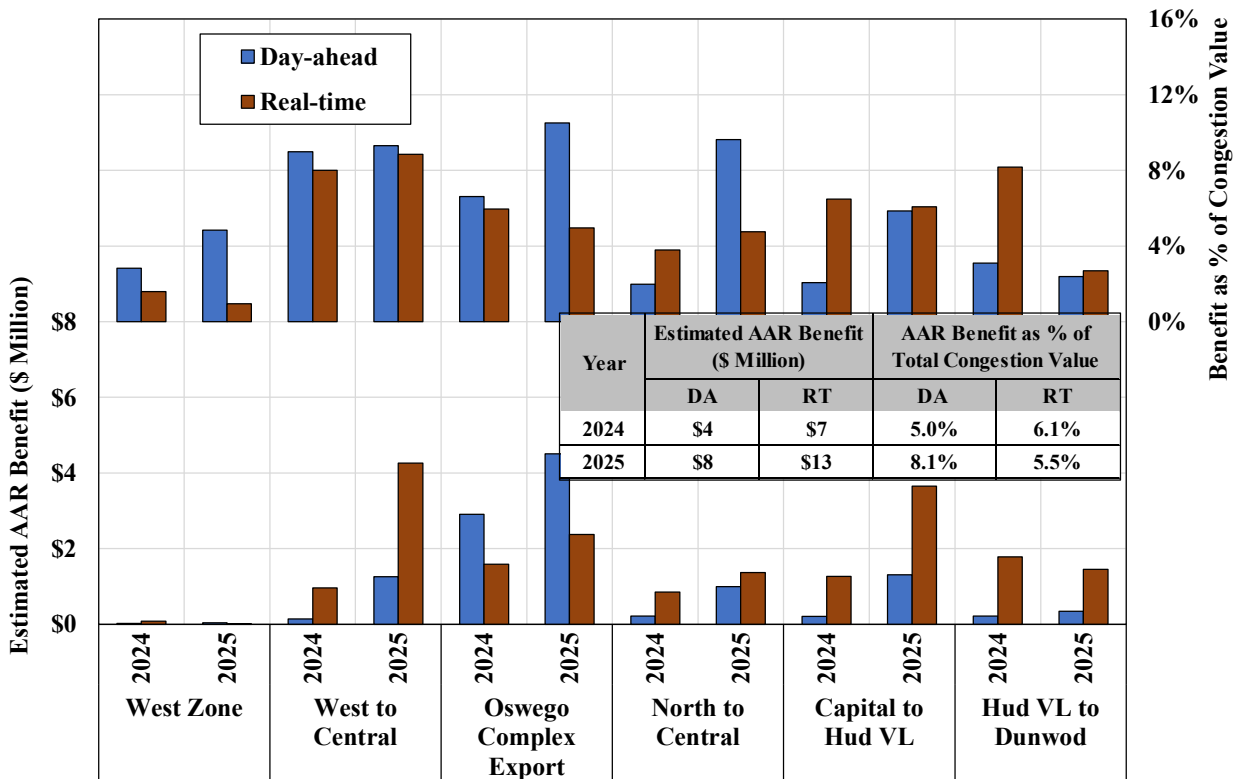
²⁵¹ See “Tie-Line Ratings Task Force Final Report on Tie-Line Ratings” by New York Power Pool, 1995.

²⁵² For example, if the line rating for a facility is 100 MW in the summer and 145 MW in the winter, then the ambient adjusted rating at 80°F is calculated as $100 + (80-95)*(145-100)/(50-95) = 115$ MW.

²⁵³ For example, if NYISO uses a rating of 120 MW for one transmission facility in the market model, the facility is binding with a shadow cost of \$100/MWh, and our estimated ambient adjusted rating is 150 MW, then the potential congestion benefit is estimated as $(150-120)*100 = \$3000$.

The Central-East interface is not included in this analysis because its rating is based on the voltage collapse limit, which is not typically affected by ambient temperature. The transmission facilities in New York City and Long Island are also excluded because most of these facilities are underground cables, whose ratings are not as sensitive to ambient air temperature as overhead lines.

Figure A-52: Potential Congestion Benefit of Using AAR Line Ratings
2024-2025



H. TCC Prices and DAM Congestion

In this subsection, we evaluate whether clearing prices in the TCC auctions were consistent with congestion prices in the day-ahead market. TCCs provide an entitlement to the holder for the day-ahead congestion between two points. In a well-functioning market, the price for the TCC should reflect a reasonable expectation of the day-ahead congestion. Perfect convergence cannot be expected because many factors affecting congestion are not known at the time of the auctions, including forced outages of generators and transmission, fuel prices, weather, etc. There are two types of TCC auctions: Centralized TCC Auctions and Reconfiguration Auctions.

- Centralized TCC Auctions** – TCCs are sold in these auctions as 6-month products for the Summer Capability Period (May to October) or the Winter Capability Period (November to April), as 1-year products for two consecutive Capability Periods, and as 2-year products for four consecutive Capability Periods. Most transmission capability is auctioned as 6-month products. The Capability Period auctions consist of a series of rounds, in which a portion of the capability is offered, resulting in multiple TCC awards

and clearing prices. Participants may offer TCCs for resale or submit bids to purchase additional TCCs in these auctions.

- *Balance-of-Period Auctions*²⁵⁴ – The NYISO conducts a Balance-of-Period Auction once every month for the remaining months in the same Capability Period for which the TCC will be effective. Participants may offer TCCs for resale or submit bids to purchase additional TCCs in the Balance-of-Period Auction. Each monthly Balance-of-Period Auction consists of only one round.

Figure A-53: TCC Cost and Profit by Auction Round and Path Type

Figure A-53 summarizes TCC cost and profit for the Winter 2024/25 and Summer 2025 Capability Periods (i.e., the 12-month period from November 2024 through October 2025). The *TCC Cost* measures what market participants paid to obtain TCC rights from the TCC auctions. For a particular path, the *TCC Cost* is equal to the purchased TCC MW multiplied by the TCC price for that path. The *TCC Profit* measures the difference between the *TCC Payment*, which is equal to the TCC MW between two points multiplied by the congestion cost difference in the day-ahead market between the two points, and the *TCC Cost*.

The figure shows the TCC costs and profits for each round of auction in the 12-month period, which includes: (a) three rounds of one-year auctions for the exact same 12-month Capability Period; (b) four rounds of six-month auctions for the Winter 2024/25 Capability Period; (c) four rounds of six-month auctions for the Summer 2025 Capability Period; and (d) twelve Balance-of-Period auctions for each month of the 12-month Capability Period.²⁵⁵ The figure includes the TCCs that were purchased and sold by Market Participants in these auctions.

For the purposes of the figure, each TCC is broken into inter-zonal and intra-zonal components, making it possible to identify portions of the transmission system that generate the most revenue in the TCC auction and that are most profitable for the buyers of TCCs. Each TCC has a Point-Of-Injection (“POI”) and a Point-Of-Withdrawal (“POW”). The POI and POW may be a generator bus, a NYCA Zone, the NYISO Reference Bus, or an external proxy bus. For purposes of this analysis, all transacted TCCs in the auctions are unbundled into the following standard components: (a) POI to POI Zone Proxy, (b) POI Zone Proxy to the POW Zone Proxy, and (c) POW Zone Proxy to POW.²⁵⁶ When a TCC is unbundled into standard components for

²⁵⁴ The Balance-of-Period Auction started with the September 2017 monthly auction, which replaced the previous Reconfiguration Auction that was conducted only for the next one-month period.

²⁵⁵ In the figure, the bars in the ‘Monthly’ category represent aggregated values for the same month from all applicable BOP auctions.

²⁵⁶ For each load zone, one node is selected as the Zone Proxy to represent the congestion level on the high-voltage network within the zone. The following Zone Proxies are used in this analysis: (a) Kintigh for Zone A; (b) Red Rochester for Zone B; (c) Carr St. for Zone C; (d) St. Lawrence for Zone D; (e) Maple Ridge WT for Zone E; (f) Athens STG for Zone F; (g) Pleasant Valley for Zone G; (h) E. Fishkill for Zone H; (i) Dunwoodie for Zone I; (j) Rainey for Zone J; and (k) Glenwood for Zone K.

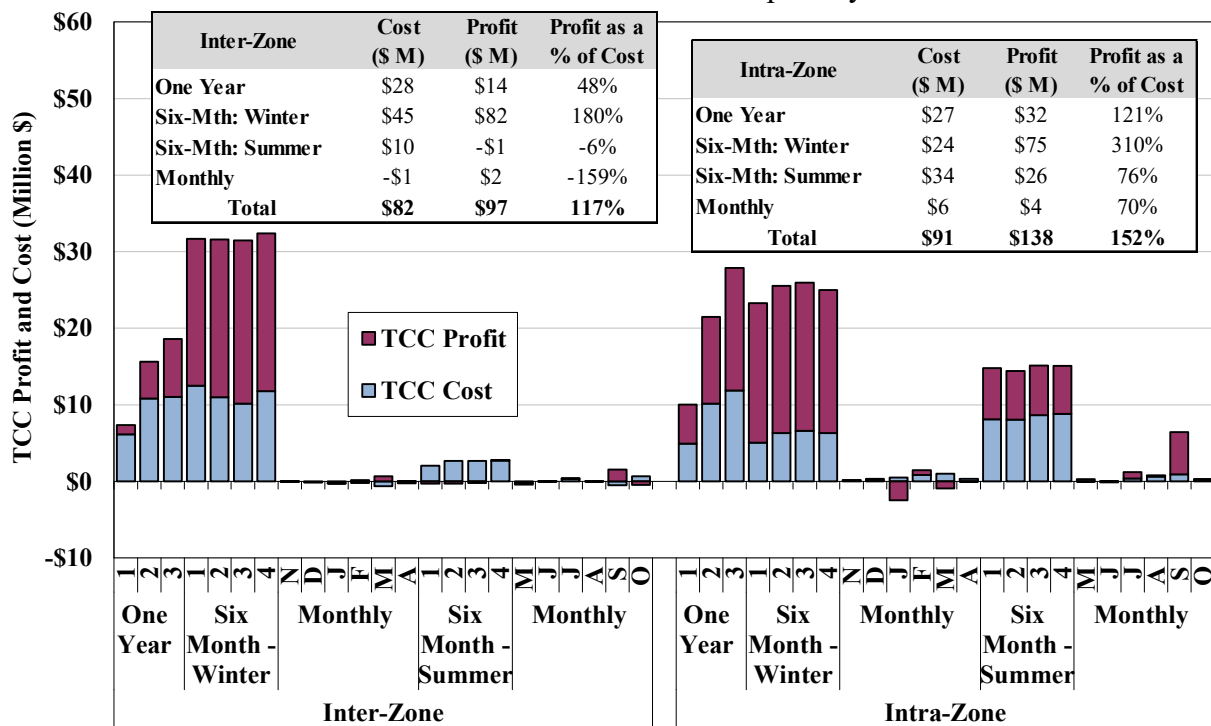
this analysis, the original TCC is replaced by up to three TCCs. The three standard components are further grouped into two categories:²⁵⁷

- **Inter-Zone TCCs**, which include all unbundled POI Zone Proxy to POW Zone Proxy TCCs and represent congestion rights on the high-voltage transmission network between load zones; and
- **Intra-Zone TCCs**, which include POI to POI Zone Proxy TCCs and POW Zone Proxy to POW TCCs and represent congestion rights from the higher-voltage network to the lower-voltage network, as well as within lower-voltage network in each load zone.

The figure shows the costs and profits separately for the intra-zone and inter-zone components of TCCs. The table in the figure summarizes the TCC cost, profit, and profitability for each type of TCC auction for the two categories of TCC paths. The profitability is measured by the total TCC profit as a percentage of total TCC cost.

Figure A-53: TCC Cost and Profit by Auction Round and Path Type

Winter 2024/25 and Summer 2025 Capability Periods



²⁵⁷

For example, a 100 MW TCC from Niagara to Arthur Kill 2 is unbundled to three components: (a) A 100 MW TCC from Niagara to Kintigh; (b) A 100 MW TCC from Kintigh to Rainey; and (c) A 100 MW TCC from Rainey to Arthur Kill 2. Components (a) and (c) belong to the intra-zone category and Component (b) belongs to inter-zone category.

Table A-3 & Table A-4: TCC Cost and Profit by Path

The following two tables compare TCC costs with TCC profits for both intra-zonal paths and inter-zonal paths during the Winter 2024/25 and Summer 2025 Capability Periods (i.e., the 12-month period from November 2024 through October 2025). Each pair of POI and POW represents all paths sourcing from the POI and sinking at the POW. Inter-zonal paths are represented by pairs with different POI and POW, while intra-zonal paths are represented by pairs with the same POI and POW. TCC costs and profits that are higher than \$2 million are highlighted with green, while TCC costs and profits that are lower than -\$2 million are highlighted with light red.

Table A-3: TCC Cost by Path
Winter 2024/25 and Summer 2025 Capability Periods

POW POI	WEST	GENESE	CENTRL	MHK VL	NORTH	CAPITL	HUD VL	MILLWD	DUNWOD	N.Y.C.	LONGIL	O H	H Q	NPX	PJM	Total
WEST	\$1	\$0	\$1	\$0	\$0	\$0	\$19	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$20
GENESE	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
CENTRL	-\$1	\$0	\$26	-\$3	\$0	\$0	\$11	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$33
MHK VL	\$0	\$1	\$1	\$12	-\$1	-\$5	\$1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$8
NORTH	\$0	\$1	\$0	-\$1	\$4	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4
CAPITL	-\$1	\$0	-\$2	-\$2	\$0	\$10	-\$1	\$0	\$0	\$0	\$0	\$0	\$0	\$2	\$0	\$6
HUD VL	-\$1	\$0	-\$6	\$0	\$0	\$1	\$4	\$0	-\$1	\$7	\$0	\$0	\$0	\$26	-\$1	\$28
MILLWD	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1	\$0	\$0	\$0	\$0	\$1
DUNWOD	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3	\$0	\$0	\$0	\$0	\$4
N.Y.C.	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	-\$1	\$18	\$1	\$0	\$0	\$0	\$0	\$17
LONGIL	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$15	\$0	\$0	\$0	\$0	\$14
O H	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1
H Q	\$0	\$0	\$0	-\$1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	-\$1
NPX	\$0	\$0	\$0	\$0	\$0	-\$1	-\$1	\$0	\$0	\$0	-\$1	\$0	\$0	\$0	\$0	-\$2
PJM	\$0	\$0	-\$2	\$0	\$0	\$0	\$32	\$0	\$0	\$0	\$0	\$0	\$0	\$11	\$0	\$41
Total	-\$1	\$1	\$17	\$3	\$2	\$4	\$65	-\$1	-\$1	\$26	\$21	\$0	\$0	\$39	-\$2	\$173

Table A-4: TCC Profit by Path
Winter 2024/25 and Summer 2025 Capability Periods

POW POI	WEST	GENESE	CENTRL	MHK VL	NORTH	CAPITL	HUD VL	MILLWD	DUNWOD	N.Y.C.	LONGIL	O H	H Q	NPX	PJM	Total
WEST	-\$1	\$0	\$0	\$0	\$0	\$0	-\$2	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	-\$4
GENESE	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
CENTRL	\$0	\$0	\$4	\$0	\$0	\$0	-\$2	\$0	\$0	\$0	\$0	\$0	\$0	\$0	-\$1	\$3
MHK VL	\$0	\$0	\$0	\$2	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2
NORTH	\$0	\$0	\$0	\$1	\$10	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$11
CAPITL	\$0	\$0	\$1	\$1	\$0	\$13	\$1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$17
HUD VL	\$0	\$0	\$1	\$0	\$0	-\$1	\$4	\$0	\$2	-\$4	\$0	\$0	\$0	\$7	-\$1	\$9
MILLWD	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1	\$0	\$0	\$0	\$0	\$2
DUNWOD	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3	\$0	\$0	\$0	\$0	\$2
N.Y.C.	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1	\$97	\$1	\$0	\$0	\$0	\$0	\$99
LONGIL	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$9	\$0	\$0	\$0	\$0	\$9
O H	\$5	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$6
H Q	\$0	\$0	\$0	-\$2	-\$1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	-\$3
NPX	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
PJM	\$0	\$0	\$6	\$0	\$0	\$0	\$57	\$0	\$0	\$0	\$0	\$0	\$0	\$16	\$0	\$81
Total	\$5	\$1	\$12	\$1	\$10	\$12	\$59	\$1	\$3	\$93	\$13	\$0	\$0	\$24	-\$1	\$234

I. Efficient Allocation of Day-ahead and Balancing Congestion Rents

This subsection discusses two proposals by the MMU related to the assignment of congestion rents from the NYISO market:

- Recommendation #2023-1 proposes NYISO allocate surpluses and shortfalls from the day-ahead market to individual facilities using a flow-based approach similar to the one

used to allocate TCC revenues. This would improve incentives to schedule transmission facilities more efficiently in the day-ahead market.

- Recommendation #2012-8 proposes NYISO create a financial right to compensate Con Edison for changes in schedule of the lines between NYC and Long Island between the DAM and RT to provide Con Edison with incentives to operate the lines efficiently (rather than according to a fixed schedule) in real time.

If Recommendation #2012-8 is implemented without #2023-1, then it would be beneficial for ConEd to receive financial rights for schedule differences between the TCC auction and the DAM as well. Note, these recommendations would not alter the TCCs possessed by any market party.

Flow-based Method for DAMCR Allocation under Recommendation #2023-1

A large share of the cost of maintaining the high voltage transmission system is recovered through the collection of DAM Congestion Revenues and the auctioning of TCCs. TCC auction revenues are allocated to each NYTO in proportion to the value of its transmission facilities in the auctions, while charges are assessed to each NYTO to the extent that outages of its equipment reduce scheduled transfers in the DAM. However, when additional transmission capability is scheduled in the DAM (above what was sold in the TCC auction), the resulting revenues are allocated in proportion to the TCC revenue allocation, regardless of which NYTO's facilities allowed the additional scheduling. Consequently, NYTOs do not recover the actual value of their transmission assets when their assets are not sold in the TCC auctions, which discourages efficient scheduling of transmission assets in the day-ahead market. We recommend NYISO revise the allocation of DAM congestion residuals based on changes in scheduled utilization of the transmission system between the TCC auctions and the day-ahead market. This would allow each NYTO to recover the value of transmission scheduled in the DAM even if the capacity was not sold in the TCC auctions.²⁵⁸

Concept for Financial Transmission Right Proposed under Recommendation #2012-8

An efficient financial right should compensate ConEd for the quantity of congestion relief provided at a price that reflects the marginal cost of relieving congestion on each flow gate in the day-ahead and real-time markets. However, if Recommendation #2023-1 is implemented, it would result in an efficient assignment of DAMCR to ConEd for the scheduling of the 901 and 903 lines, so Recommendation #2012-8 would only need to be applied to the RTM payment (but not the DAM payment) described below. ConEd should be paid:

DAM Payment =

²⁵⁸ For additional details about this approach, see MIWG presentations titled *MMU Net Congestion Rent Assignment Proposal*, dated June 3, 2025, and *Net Congestion Rent Assignment Simulation Examples*, dated July 8, 2025.

$$\sum_{l=901,903} \left([DAM MW_l - TCC MW_l] \times \sum_{c=constraint} [-DAM SF_{l,c} \times DAM SP_c] \right)$$

RTM Payment =

$$\sum_{l=901,903} \left([RTM MW_l - DAM MW_l] \times \sum_{c=constraint} [-RTM SF_{l,c} \times RTM SP_c] \right)$$

Total Payment = DAM Payment + RTM Payment, where a negative payment would result in a charge to ConEd. To illustrate, suppose there is congestion in the DAM on the interface from upstate to Long Island (Y50 Line), from upstate to NYC (Dunwoodie), and into the Valley Stream load pocket (262 Line) while the 901 Line flows are reduced below the contract amount:

- TCC MW₉₀₁ = 96 MW
- DAM MW₉₀₁ = 60 MW
- DAM SP_{Y50} = \$10/MWh
- DAM SP_{Dunwoodie} = \$5/MWh
- DAM SP₂₆₂ = \$15/MWh
- DAM SF_{901, Y50} = 100%
- DAM SF_{901, Dunwoodie} = -100%
- DAM SF_{901, 262} = 100%
- DAM Payment₉₀₁ = \$720 per hour = (60 MW – 96 MW) x {(-100% x \$10/MWh) + (100% x \$5/MWh) + (-100% x \$15/MWh)}

Since DAM payments are made for deviations from the TCC modeling assumptions, the new financial transmission right would not alter the TCCs possessed by any market party. As mentioned above, if Recommendation #2024-1 is implemented, it would result in an efficient assignment of DAMCR to ConEd for the scheduling of the 901 and 903 lines, so Recommendation #2012-8 would only need to be applied to the RTM payment (but not the DAM payment).

Revenue Adequacy of Recommendation #2012-8

Just as the LBMP compensation to generators is generally revenue adequate, the new financial transmission right would also be revenue adequate. This is illustrated by the following scenarios:

- Basecase Scenario – Provides an example of the current market rules where the NYISO receives revenues from loads that exceed payments to generators, thereby contributing to DAM congestion revenues.
- PAR Relief Scenario – Shows how a PAR-controlled line could be used to reduce congestion, allowing the owner of the line to be compensated without increasing uplift from DAMCRs.

- PAR Loading Scenario – Shows how the owner of the line would be charged if the DAM schedule increased congestion relative to the TCC schedule assumption.

These scenarios evaluate the allocation of costs and revenues from the DAM based on differences between the TCC auction outcomes and the DAM outcomes, but the allocation of costs and revenues from the RTM is similarly based on differences between the DAM and RTM outcomes. These scenarios use a simplified four node network, including: Upstate, NYC, Valley Stream, and Rest of Long Island. The four nodes are interconnected by four interfaces:

- The Dunwoodie interface from Upstate to NYC,
- The Y50 Line from Upstate to Rest of Long Island,
- The 262 Line from Rest of Long Island to Valley Stream, and
- The PAR-controlled 901 Line from Valley Stream to NYC.

For simplicity, the 901 Line contract amount that is used in the TCC auction is rounded to 100 MW.

The Base Case Scenario shows that a net of \$22,500 of DAM congestion revenue is collected from scheduling by generators and loads. The table also shows the amount of DAM congestion revenue that accrues on each constrained facility. In this example, DAMCR equals \$0 because the flows on each constrained facility are equal to the capability/assumption in the TCC model. Since the 901 Line contract moves power from a high LBMP area to a low LBMP area, it reduces congestion revenue by \$2,000, but it does not cause DAMCR because it is consistent with the TCC auction.

The PAR Relief Scenario shows that if the 901 Line flow is reduced from 100 MW to 10 MW, it reduces the generation needed in Valley Stream and increases generation in NYC, reducing overall production costs by \$1,800 as compared to the Basecase Scenario. Since LBMPs do not change in this example, payments by loads are unchanged and \$1,800 of additional congestion revenues are collected. The collection of additional congestion revenues allows the NYISO to compensate ConEd \$1,800 for the PAR adjustment, and DAMCR remains at \$0.

The PAR Relief Scenario shows that if the 901 Line flow is increased from 100 MW to 120 MW, it increases the generation needed in Valley Stream and reduces generation in NYC, increasing overall production costs by \$400 as compared to the Basecase Scenario. Since LBMPs do not change in this example, payments by loads are unchanged and \$400 less congestion revenue is collected. The collection of less congestion revenue requires the NYISO to charge ConEd \$400 for exceeding the contract amount, and DAMCR remains at \$0.

BASECASE SCENARIO

	Node	LBMP	Load	Generation	Load Revenue	Generator Payments
Gen/Load Payments	Upstate	\$25	10000	13000	\$250,000	\$325,000
	NYC	\$30	4000	1900	\$120,000	\$57,000
	Valley Stream	\$50	350	150	\$17,500	\$7,500
	Rest of Long Is.	\$35	2500	1800	\$87,500	\$63,000
	Total		16850	16850	\$475,000	\$452,500
	Net (Gen minus Load)			0		\$22,500
	Interface	Shadow Price	Interface Flow			Congestion Revenue
Transmission Revenue	Dunwoodie	\$5	2000			\$10,000
	Y50	\$10	1000			\$10,000
	262 Line	\$15	300			\$4,500
	901 Line Contract	-\$20	100			-\$2,000
	Total					\$22,500
	DAMCR (Gen minus Load minus Congestion)					\$0

PAR RELIEF SCENARIO (901 Line Flow Reduced from 100 MW to 10 MW)

	Node	LBMP	Load	Generation	Load Revenue	Generator Payments
Gen/Load Payments	Upstate	\$25	10000	13000	\$250,000	\$325,000
	NYC	\$30	4000	1990	\$120,000	\$59,700
	Valley Stream	\$50	350	60	\$17,500	\$3,000
	Rest of Long Is.	\$35	2500	1800	\$87,500	\$63,000
	Total		16850	16850	\$475,000	\$450,700
	Net (Gen minus Load)			0		\$24,300
	Interface	Shadow Price	Interface Flow			Congestion Revenue
Transmission Revenue	Dunwoodie	\$5	2000			\$10,000
	Y50	\$10	1000			\$10,000
	262 Line	\$15	300			\$4,500
	901 Line Contract	-\$20	100			-\$2,000
	901 Line Adjust	-\$20	-90			\$1,800
	Total					\$24,300
	DAMCR (Gen minus Load minus Congestion)					\$0

PAR LOADING SCENARIO (901 Line Flow Increased from 100 MW to 120 MW)

	Node	LBMP	Load	Generation	Load Revenue	Generator Payments
Gen/Load Payments	Upstate	\$25	10000	13000	\$250,000	\$325,000
	NYC	\$30	4000	1880	\$120,000	\$56,400
	Valley Stream	\$50	350	170	\$17,500	\$8,500
	Rest of Long Is.	\$35	2500	1800	\$87,500	\$63,000
	Total		16850	16850	\$475,000	\$452,900
	Net (Gen minus Load)			0		\$22,100
	Interface	Shadow Price	Interface Flow		Congestion Revenue	
Transmission Revenue	Dunwoodie	\$5	2000		\$10,000	
	Y50	\$10	1000		\$10,000	
	262 Line	\$15	300		\$4,500	
	901 Line Contract	-\$20	100		-\$2,000	
	901 Line Adjust	-\$20	20		-\$400	
	Total				\$22,100	
	DAMCR (Gen minus Load minus Congestion)					\$0

J. Statistical Modeling for Thunderstorm Forecasting

Numerous studies have examined methods for forecasting thunderstorms.^{259 260 261} Among these, logistic regression is a widely used statistical classification technique for modeling binary outcomes (0 = No, 1 = Yes). This method estimates the probability of an event occurring, with probabilities above 50 percent indicating a high likelihood of occurrence and probabilities below 50 percent indicating a low likelihood.

Logistic regression is based on the logistic (sigmoid) function, which maps real-valued inputs to probabilities between 0 and 1:

$$P(z) = \frac{1}{1 + e^{-z}}$$

Using this framework, we developed the following model to predict thunderstorm occurrence:

$$P(\text{Thunderstorm}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_6 x_6)}}$$

²⁵⁹ Sánchez, J. L., Ortega, E. G., & Marcos, J. L. (2001). Construction and assessment of a logistic regression model applied to short-term forecasting of thunderstorms in Leon (Spain). *Atmospheric research*, 56(1-4), 57-71.

²⁶⁰ Arora, K., Ray, K., Ram, S., & Mehajan, R. (2023). The role of instability indices in forecasting thunderstorm and non-thunderstorm days across six cities in India. *Climate*, 11(1), 14.

²⁶¹ Kamangir, H., Collins, W., Tissot, P., & King, S. A. (2020). A deep-learning model to predict thunderstorms within 400 km² South Texas domains. *Meteorological Applications*, 27(2), e1905.

The explanatory variables were selected based on prior meteorological studies and empirical fitting to weather conditions specific to the Con Edison service territory. These variables include:

- **Convective Available Potential Energy (CAPE):** A measure of atmospheric instability that reflects the potential strength of updrafts. Higher values indicate a greater likelihood of thunderstorm development.
- **K-Index:** Also known as George's Index, a measure of thunderstorm potential in meteorology, defined as
$$K = (T_{850} - T_{500}) + T_{d_{850}} - (T_{700} - T_{d_{700}}),$$
where T_{850} , T_{700} , and T_{500} are temperatures at pressure levels of 850 hPa, 700 hPa, and 500 hPa, and $T_{d_{850}}$, $T_{d_{700}}$ are dew points at 850 hPa and 700 hPa. Higher values indicate greater thunderstorm potential.
- **Lifted Index (LI):** An index that measures atmospheric instability by comparing the temperature of a lifted air parcel to the surrounding environment at 500 hPa. More negative values indicate stronger instability and greater potential for convection.
- **Relative Humidity at 500 hPa:** Indicates mid-level (usually chosen at 500 hPa pressure level) moisture content. Higher values support cloud formation and sustained convection.
- **V component of wind at 500 hPa:** Represents the northward component of the wind, which can influence storm organization and movement.
- **Vertical Velocity at 850 hPa:** Measures vertical air motion at 850 hPa pressure level. Negative values indicate upward motion, which is associated with convective development.

These variables are well-established indicators of thunderstorm formation in the meteorological literature and are publicly available from the ERA5 reanalysis dataset, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 provides hourly global weather data from 1940 to present (<https://cds.climate.copernicus.eu/datasets>).

The model was trained using weather data from 2023 and 2024, yielding the following coefficients:

Coefficient	Variables	Coefficient
β_0	Intercept	-4.02
β_1	CAPE	0.50
β_2	K-Index	1.68
β_3	Lifted Index (LI)	-0.99
β_4	Relative Humidity (500 hPa)	0.23
β_5	V component of wind (500 hPa)	0.53
β_6	Vertical Velocity (850 hPa)	-0.46

The estimated coefficients are consistent with both physical intuition and prior research. For example, the positive coefficients on CAPE and relative humidity indicate that greater instability and moisture increase the likelihood of thunderstorms. Similarly, the negative coefficient on vertical velocity reflects that stronger upward motion (more negative values) is associated with a higher probability of convection.

The model demonstrates strong predictive performance. On the 2025 test dataset, it achieves an AUC exceeding 0.93, indicating a high ability to distinguish between thunderstorm and non-thunderstorm hours.

IV. EXTERNAL INTERFACE SCHEDULING

New York imports a substantial amount of power from four adjacent control areas: New England, PJM, Ontario, and Quebec. Additionally, five controllable lines (the Cross Sound Cable, the 1385 Line, the Linden VFT Line, the HTP Line, and the Neptune Cable) connect Long Island and New York City directly to PJM and New England, collectively providing up to 2.2 GW of imports to downstate areas.^{262,263} Given the substantial transfer capability between New York and the adjacent regions relative to New York’s total power consumption, efficient scheduling of these interfaces is crucial.

Efficient use of transmission interfaces between regions offers two key benefits:

- First, access to external resources reduces costs of serving New York load when lower-cost external resources are available. Likewise, lower-cost internal resources gain the ability to compete to serve load in adjacent regions.
- Second, the ability to draw on neighboring systems for emergency power, reserves, and capacity helps lower the cost of meeting reliability standards in each control area.

This section evaluates transaction scheduling between New York and adjacent control areas:

- Subsection A summarizes scheduling between New York and adjacent control areas;
- Subsection B evaluates convergence of prices between New York and neighboring control areas;
- Subsection C examines the efficiency of Coordinated Transaction Scheduling (“CTS”), including an evaluation of transaction offer patterns and profitability;
- Subsection D provides a systematic evaluation of factors that lead to inconsistencies between the RTC evaluation, which schedules CTS transactions every 15 minutes, and the RTD evaluation, which determines real-time prices every five minutes that are used for settlements; and
- Subsection E examines several key factors that lead to inconsistencies between RTC and RTD in more details.

²⁶² Cross Sound Cable (“CSC”) connects Long Island to Connecticut with a transfer capability of 330 MW. Neptune Cable connects Long Island to New Jersey with a transfer capability of 660 MW. Northport-to-Norwalk line (“1385 Line”) connects Long Island to Connecticut with a transfer capability of 200 MW. Linden VFT Line connects New York City to PJM with a transfer capability of 315 MW. Hudson Transmission Project (“HTP Line”) connects New York City to New Jersey with a transfer capability of 660 MW.

²⁶³ In addition to the controllable lines connecting New York City and Long Island to adjacent control areas, there is a small controllable line between upstate New York and Quebec that is known as the “Dennison Scheduled Line” and is scheduled separately from the primary interface between New York and Quebec.

A. Summary of Scheduled Imports and Exports

Figure A-54 to Figure A-57: Average Net Imports from Ontario, PJM, Quebec, and New England

The following four figures summarize the net scheduled interchanges in real-time between New York and neighboring control areas in 2024 and 2025. The net scheduled interchange does not include unscheduled power flows (i.e., loop flows). For each interface, average scheduled net imports are shown by month for peak (i.e., 6 am to 10 pm, Monday through Friday) and off-peak hours. This is shown for the primary interfaces with Ontario and PJM in Figure A-54, the primary interfaces with Quebec and New England in Figure A-55, and the controllable lines connecting Long Island and New York City with PJM and New England in Figure A-56 and Figure A-57.

Figure A-54: Monthly Average Net Imports from Ontario and PJM
2024 – 2025

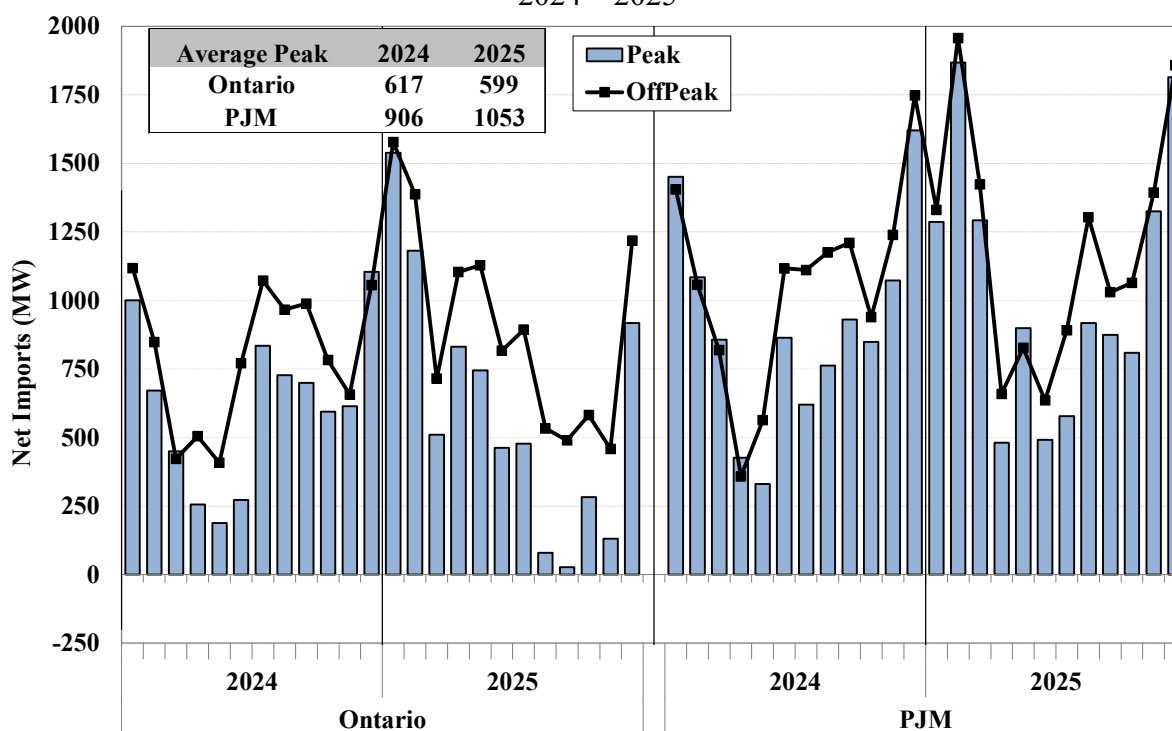


Figure A-55: Monthly Average Net Imports from Quebec and New England
2024 – 2025

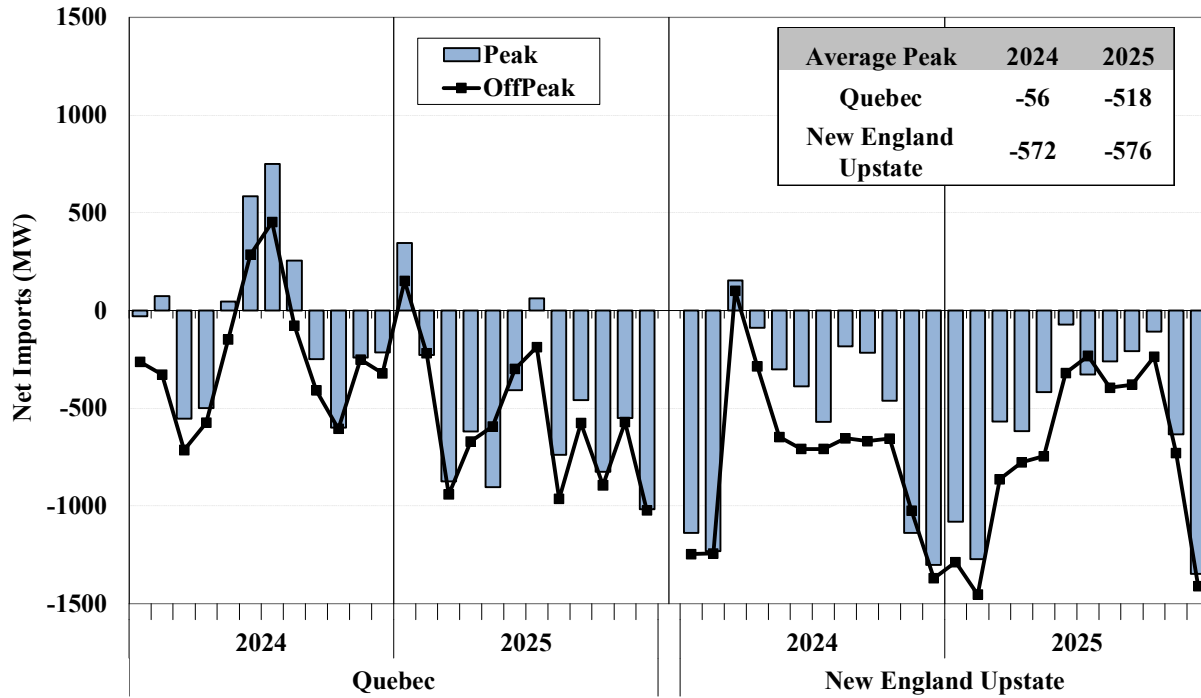


Figure A-56: Monthly Average Net Imports into New York City
2024 – 2025

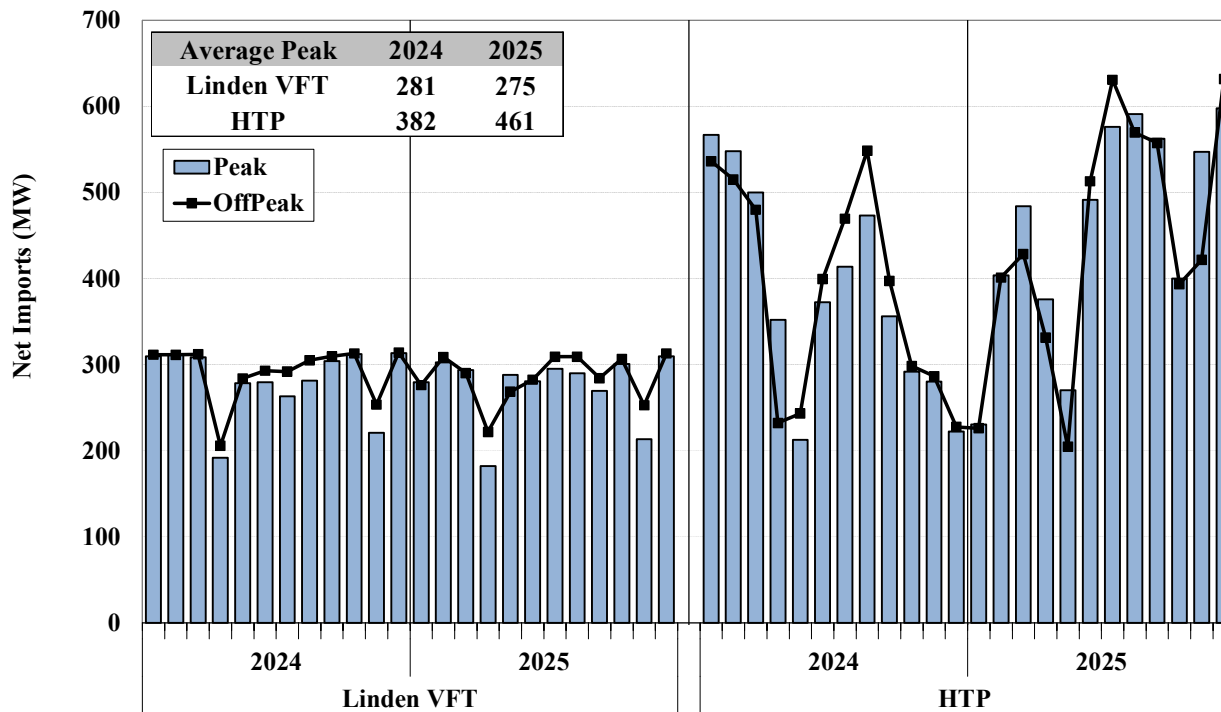
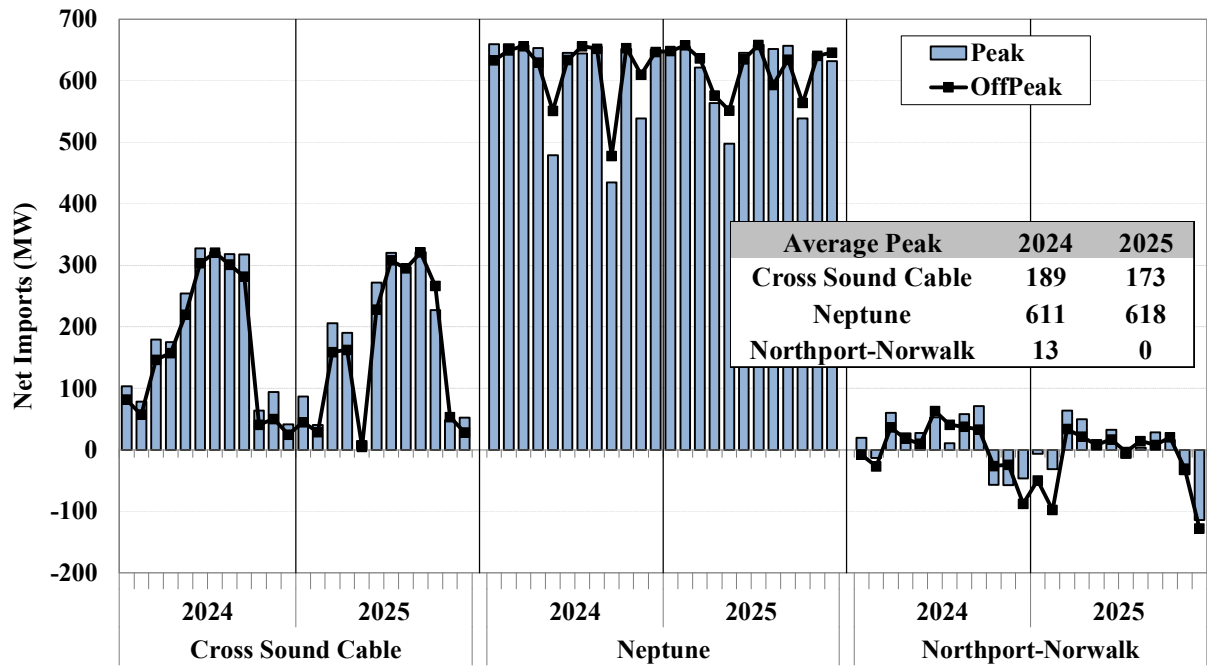


Figure A-57: Monthly Average Net Imports into Long Island
2024 – 2025



B. Price Convergence and Efficient Scheduling with Adjacent Markets

The performance of New York’s wholesale electricity markets depends not only on the efficient use of internal resources, but also on the efficient use of transmission interfaces between New York and neighboring control areas. Trading between neighboring markets tends to bring prices together as participants arbitrage price differences. When an interface is used efficiently, prices in adjacent areas should be consistent unless the interface is constrained. A lack of price convergence indicates that resources are being used inefficiently, as higher-cost resources are operating in the high-priced region that could have been supplanted by increased output from lower-cost resources in the low-priced region. Efficient scheduling is particularly important during shortages when flows between regions have the largest economic and reliability consequences. Moreover, efficient scheduling can also alleviate over-generation conditions that can lead to negative price spikes.

However, one cannot expect that trading by market participants alone will optimize the use of the interface. Several factors prevent real-time prices from being fully arbitrated.

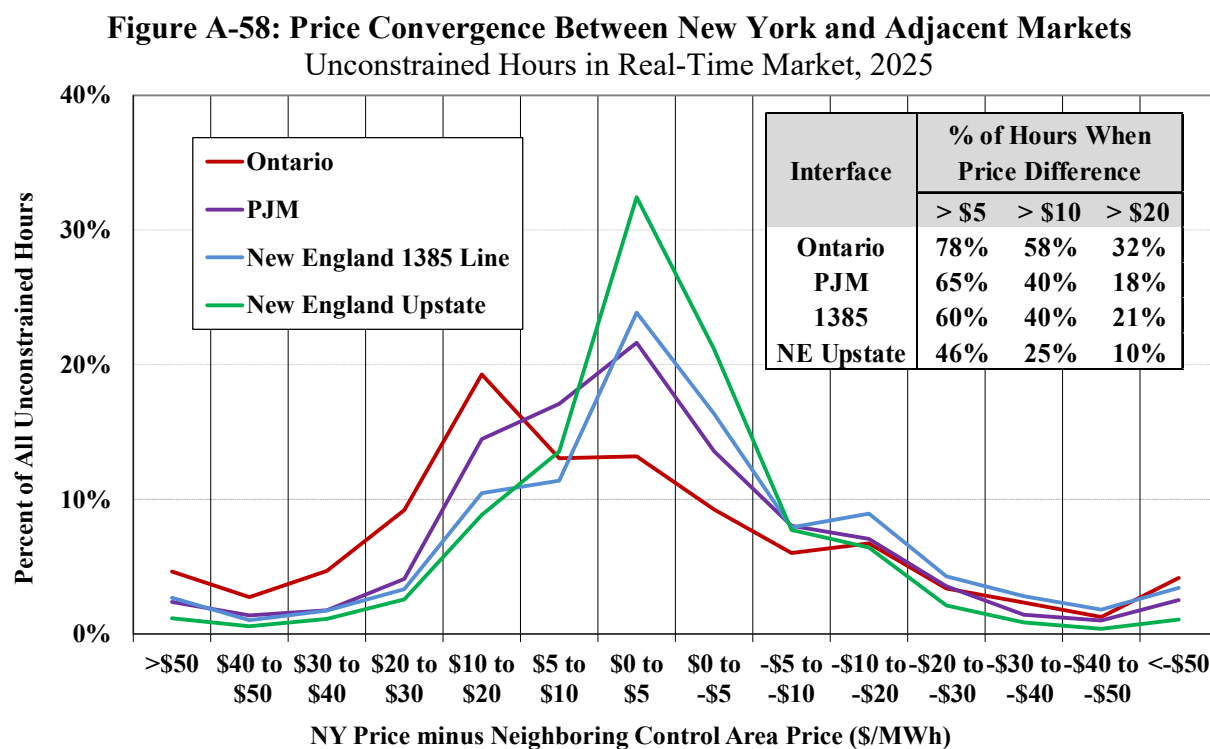
- Market participants do not operate with perfect foresight of future market conditions at the time that transaction bids must be submitted. Without explicit coordination between the markets by the ISOs, complete arbitrage will not be possible.
- Differences in scheduling procedures and timing in the markets are barriers to arbitrage.
- There are transaction costs associated with scheduling imports and exports that diminish the returns from arbitrage. Participants would not be willing to schedule additional power between regions unless they anticipate a price difference greater than these costs.

- The risks associated with curtailment and congestion reduce participants' incentives to schedule external transactions when expected price differences are small.

Figure A-58: Price Convergence Between New York and Adjacent Markets

Figure A-58 evaluates scheduling between New York and adjacent RTO markets across interfaces with open access scheduling. The Neptune Cable, the Linden VFT Line, the HTP Line, and the Cross Sound Cable are omitted because these are Designated Scheduled Lines, which have alternate systems to allocate transmission reservations. RTOs have real-time markets, which allow participants to schedule market-to-market transactions based on transparent price signals in each region. Based on the prevailing prices in each market, we can evaluate whether the interface is scheduled efficiently.

Figure A-58 summarizes price differences between New York and neighboring markets during unconstrained hours in 2025. In these hours, there were no NYISO constraints that prevented scheduling. However, in some of these hours, constraints in neighboring ISOs may have limited transaction scheduling.²⁶⁴ In the figure, the horizontal axis shows the range of price differences between New York and the adjacent control areas at the border. The lines represent the distribution of hours falling within each price difference category.



²⁶⁴

In these hours, prices in neighboring RTOs (i.e., prices at the NYISO proxy in each RTO market) reflect transmission constraints in those markets.

Table A-5: Efficiency of Inter-Market Scheduling

Table A-5 evaluates the consistency of the direction of external transaction scheduling and price differences between New York and New England, PJM, and Ontario during 2025. It evaluates transaction schedules and clearing prices between New York and the three markets across the three primary interfaces and five scheduled lines (i.e., the 1385 Line, the Cross Sound Cable, the Neptune Cable, the HTP Line, and the Linden VFT interface).

The table shows the following quantities:

- The estimated production cost savings resulting from the flows across each interface. The estimated production cost savings in each hour are based on the price difference across the interface multiplied by the scheduled power flow across the interface.²⁶⁵
- Average hourly flows between neighboring markets and New York. A positive number indicates a net import from neighboring areas to New York.
- Average price differences between markets across each interface. A positive number indicates that the average price was higher on the New York side of the interface.^{266 267}
- The share of the hours when power was scheduled in an efficient direction (i.e., from the lower-priced market to the higher-priced market).

The vast majority of power is scheduled in the day-ahead market, while relatively small balancing adjustments are typically made in the real-time market. Accordingly, this analysis is shown separately for the portion of flows scheduled in the day-ahead market and the portion resulting from balancing adjustments in the real-time market.²⁶⁸

²⁶⁵ For example, if 100 MW flows from PJM to New York across its primary interface during one hour, the price in PJM is \$50 per MWh, and the price in New York is \$60 per MWh, then the estimated production cost savings is \$1,000 (=100 * \$10). This is because each MW of flow saves \$10 by allowing a \$60 per MWh resource in New York to ramp down and be replaced by a \$50 per MWh resource in PJM. This method of calculating production cost savings tends to under-estimate the actual production cost savings when power flows from the low-priced region to the high-priced region, since if flows in the efficient direction were reduced, the cost of the marginal resource in the importing region would rise while the cost of the marginal resource in the exporting region would fall. However, this method of calculating production cost savings tends to over-estimate the actual production cost increases when power flows from the high-priced region towards the low-priced region, since if flows were reduced, the cost differential between the marginal resources in each region would converge.

²⁶⁶ Prior to May 1, 2025, the real-time Hourly Ontario Energy Price (“HOEP”) was used at the Ontario side of the interface for both the day-ahead and real-time markets. Beginning May 1, 2025, the day-ahead and real-time Intertie Prices are used for their respective markets.

²⁶⁷ Beginning March 1, 2025, both the FER (“Forecast Energy Requirement”) prices and the LMPs were used at the ISO-NE side of the interface for the day-ahead market calculations.

²⁶⁸ For example, if 100 MW is scheduled from the low-priced to the high-priced region in the day-ahead market, the day-ahead schedule would be considered *efficient direction*, and if the relative prices of the two regions was switched in the real-time market and the flow was reduced to 80 MW, the adjustment would be shown as -20 MW and the real-time schedule adjustment would be considered *efficient direction* as well.

Table A-5 evaluates the efficiency of the net scheduled interchange rather than individual transactions. Although individual transactions may be scheduled in an inefficient direction, such transactions tend to induce other firms to schedule counter-flow transactions, thereby offsetting their effects. Ultimately, it is the net scheduled interchange that determines the extent to which generation resources in one control area are used to satisfy load in another control area, and whether the external interface is used efficiently.

**Table A-5: Efficiency of Inter-Market Scheduling
Over Primary Interfaces and Scheduled Lines – 2025**

	Day-Ahead Market				Adjustment in Real-Time			
	Average Net Imports (MW)	Avg Internal Minus External Price (\$/MWh)	Percent of Hours in Efficient Direction	Estimated Production Cost Savings (Million \$)*	Average Net Imports (MW)	Avg Internal Minus External Price (\$/MWh)	Percent of Hours in Efficient Direction	Estimated Production Cost Savings (Million \$)*
Free-flowing Ties								
New England	-847	\$0.22	48%	-\$1.0	190	\$2.54	61%	\$10.9
Ontario	689	\$6.36	72%	\$59.8	64	\$3.55	55%	\$1.8
PJM	1,126	\$2.65	67%	\$26.8	-1	\$1.87	57%	\$10.1
Controllable Ties								
1385 Line	-3	-\$2.33	62%	\$3.2	-5	\$1.27	56%	\$2.7
Cross Sound Cable	168	-\$1.31	54%	\$1.6	-2	\$3.61	57%	\$1.6
Neptune	631	\$23.45	96%	\$132.7	-13	\$23.17	26%	-\$0.8
HTP	453	\$19.08	94%	\$72.0	-2	\$19.25	57%	\$0.5
Linden VFT	279	\$12.94	89%	\$34.5	2	\$15.57	65%	\$4.3

* The estimated production cost savings tend to: 1) under-estimate actual savings when power flows from the low-priced region to the high-priced region, since if flows were reduced, the cost of the marginal resource in the importing region would rise while the cost of the marginal resource in the exporting region would fall; while 2) over-estimate actual cost increases when power flows from the high-priced region to the low-priced region, since if flows were reduced, the cost differential between the marginal resources in each region would converge.

C. Evaluation of Coordinated Transaction Scheduling

Coordinated Transaction Scheduling (“CTS”) enhances efficiency by allowing two wholesale market operators to exchange price information shortly before real-time, aiding market participants in scheduling external transactions more efficiently. Compared to hourly LBMP-based scheduling, the CTS intra-hour scheduling system offers two key advantages:

- **Timeliness:** CTS schedules transactions less than 30 minutes ahead, whereas hourly LBMP-based schedules are established up to 105 minutes in advance, benefiting from more accurate system information.
- **Flexibility:** Interface flows adjust every 15 minutes instead of every 60 minutes, enabling a more efficient response to changing real-time conditions.

Figure A-59: Bidding Patterns of CTS at the Primary PJM and NE Interfaces

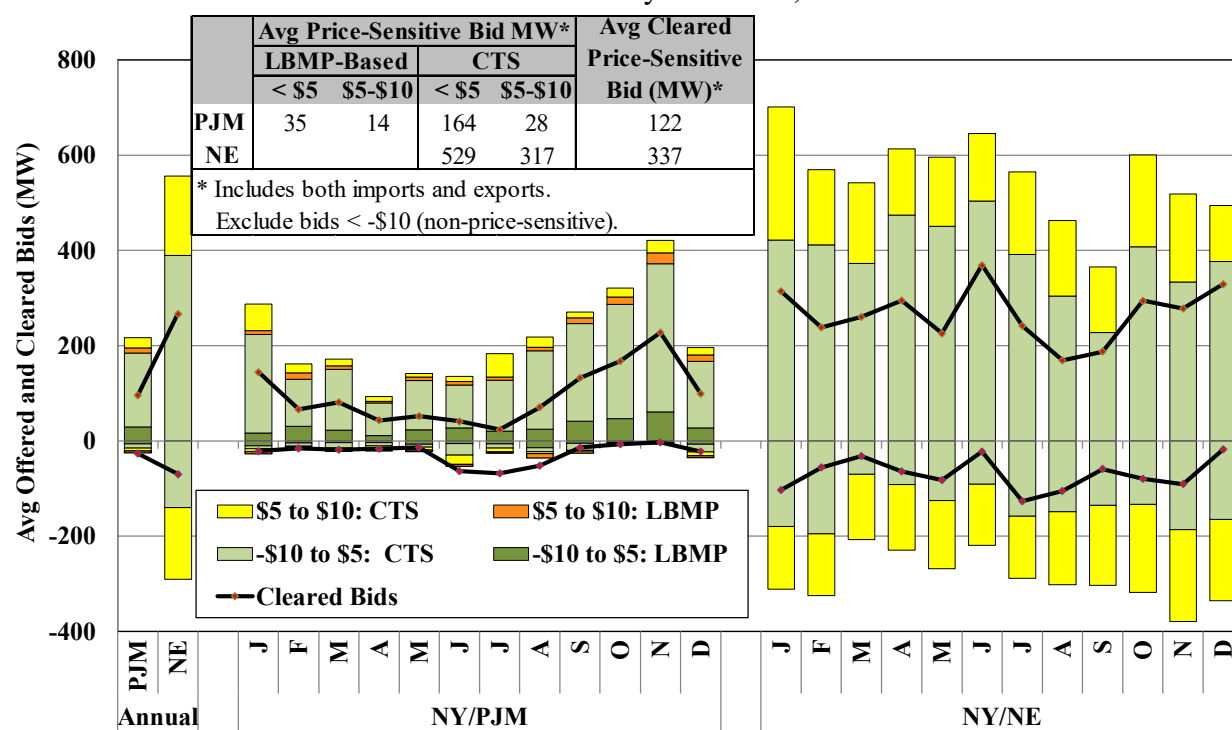
Figure A-59 shows the average quantity of CTS transactions offered and scheduled at the primary PJM and New England interfaces during peak hours (i.e., HB 7 to 22) in each month of 2025. Positive numbers indicate import offers to New York and negative numbers represent export bids to PJM or New England. Stacked bars show the average quantities of price-sensitive CTS bids for the following two price ranges: (a) between -\$10 and \$5 per MWh; and (b) between

\$5 and \$10 per MWh.²⁶⁹ Bids that are offered below -\$10 per MWh or above \$10 per MWh are considered price insensitive for this analysis. Unlike the primary New England interface where only CTS bids are allowed, traditional LBMP-based bids and CTS bids are both allowed at the PJM interface. To make a fair comparison between the two primary interfaces, LBMP-based bids at the PJM interface are converted to equivalent CTS bids and are shown in the figure as well. The equivalent CTS bids are constructed as:

- Equivalent CTS bid to import = LBMP-based import offer – PJM Forecast Price
- Equivalent CTS bid to export = PJM Forecast Price – LBMP-based export bid

The two black lines in the chart indicate the average scheduled price-sensitive imports and exports (including both CTS and LBMP-based bids) in each month. The table in the figure summarizes for the two CTS-enabled interfaces: a) the average amount of price-sensitive bids with low offer prices, which are either less than \$5 per MWh or between \$5 and \$10 per MWh; and b) the average cleared price-sensitive bids in 2025.

Figure A-59: Price-Sensitive Real-Time Transaction Bids and Offers by Month
PJM and NE Primary Interfaces, 2025



²⁶⁹ RTC evaluates whether to schedule a CTS bid to import assuming it has a cost equal to the sum of: (a) the bid price and (b) PJM's or NE's forecast marginal price at the border. Likewise, RTC evaluates whether to schedule a CTS bid to export assuming it is willing to export at a price up to: (a) PJM's or NE's forecast marginal price at the border less (b) the bid price.

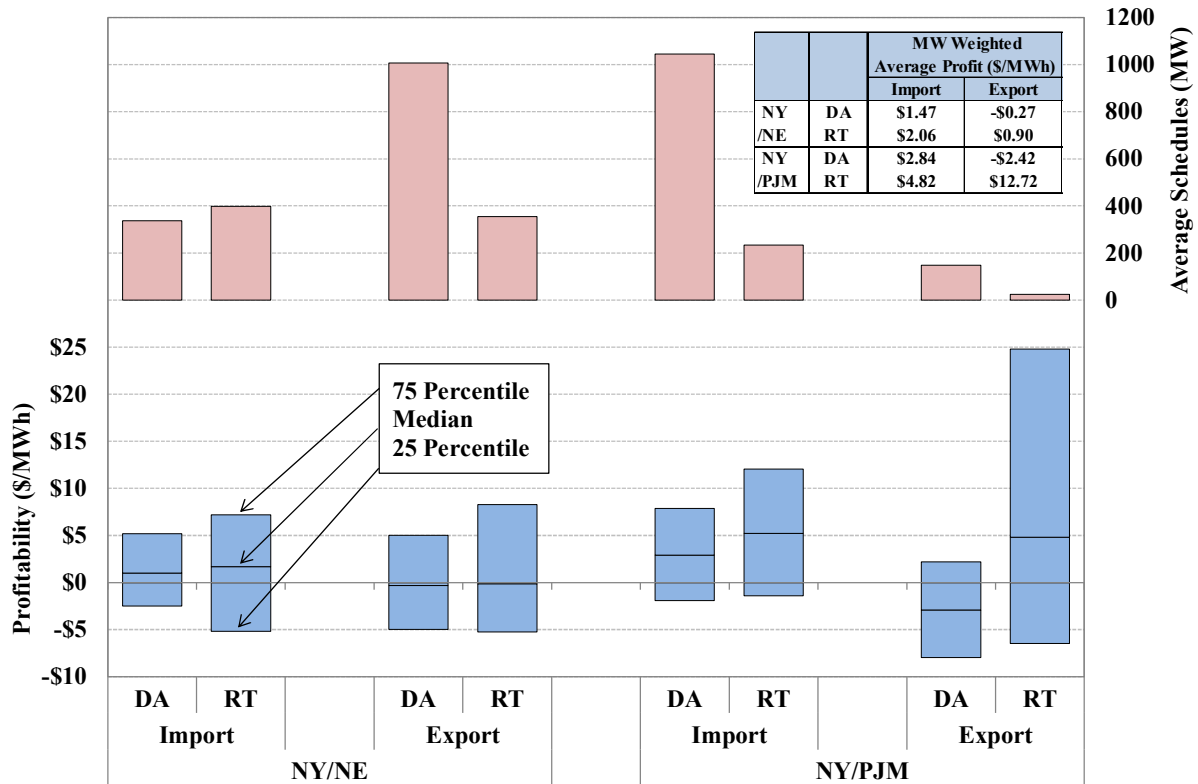
Figure A-60: Transaction Profitability at the Primary PJM and NE Interfaces

This analysis examines the profitability of scheduled transactions at the two CTS-enabled free-flow interfaces. In the bottom portion of Figure A-60, the column bars indicate the profitability spread of the middle two quartiles (i.e., the 25th to 75th percentiles) in 2025, and the line inside each bar denotes the median value of the distribution. Results are shown separately for imports and exports at the two interfaces.

Scheduled transactions are categorized in the following two groups:

- *Day-ahead* – Transactions that are scheduled in the day-ahead market and actually flow in real-time. This excludes virtual imports and exports, which have a day-ahead schedule but do not bid or offer in real-time.
- *Real-time* – Transactions that are not offered or scheduled in the day-ahead market but are scheduled in real-time (i.e., day-ahead schedules are zero, but real-time schedules are not zero).

Figure A-60: Profitability of Scheduled External Transactions
PJM and NE Primary Interfaces, 2025



The bars in the top portion of the figure show the average quantity of scheduled transactions for each category in 2025, and the inset table summarizes the annual average profit.²⁷⁰

²⁷⁰ The profit calculation for the day-ahead transactions between NYISO and ISO-NE incorporates both LMPs and FER prices from ISO-NE's day-ahead market.

Table A-6: Efficiency of Intra-Hour Scheduling Under CTS

This analysis evaluates the efficiency of the CTS-enabled intra-hour scheduling process (relative to our estimates of the scheduling outcomes that would have occurred under the hourly scheduling process) with PJM and New England.

To estimate the adjustment in the interchange schedule attributable to the intra-hour CTS scheduling process, it is first necessary to estimate the hourly interchange schedule that would have flowed if the intra-hour process were not in place. We estimate the base interchange schedule by calculating the average of the four advisory quarter-hour schedules during the hour for which RTC₁₅ determined final schedules at each hourly-scheduling interface.²⁷¹

Table A-6 examines the performance of the intra-hour scheduling process under CTS at the primary PJM and New England interfaces in 2025. The table shows the following quantities:

- % of All Intervals with Adjustment – This shows the percent of quarter-hour intervals during which interface flows were adjusted by CTS (relative to the estimated hourly schedule) in the scheduling RTC interval.
- Average Flow Adjustment – This measures the difference between the estimated hourly schedule and the final schedule. Positive numbers indicate flow adjustments in the import direction (i.e., from PJM or New England to New York) and negative numbers indicate flow adjustments in the export direction (i.e., from New York to PJM or New England).
- Production Cost Savings – This measures the market efficiency gains (and losses) that resulted from the CTS processes.
 - Projected Savings at Scheduling Time – This measures the expected production cost savings at the time when RTC determines the interchange schedule across the two primary interfaces.²⁷²
 - Net Over-Projected Savings – This estimates production cost savings that are over-projected. CTS bids are scheduled based partly on forecast prices. If forecast prices deviate from actual prices, transactions may be over-scheduled, under-scheduled,

²⁷¹ RTC₁₅ is the RTC run that posts the results by the time 15 minutes past each hour. The first interval of each RTC₁₅ is ending at 30 minutes past each hour. For each hourly-scheduling interface, each RTC₁₅ makes binding schedules for the second calendar hour in its two-and-a-half optimization period. For example, the first RTC₁₅ of each day posts market results by 0:15 am; the first interval of its two-and-a-half optimization period is ending at 0:30 am; and it makes binding transaction schedules for all hourly-scheduling interfaces for the hour beginning at 1:00 am.

²⁷² This is calculated as (final RTC schedule – estimated hourly schedule)*(RTC price at the PJM/NE proxy – PJM/NE forecast price at the NYIS proxy). An adjustment was also made to this estimate, which is described in Footnote 277.

- and/or scheduled in the inefficient direction. This estimates the portion of savings that inaccurately projected because of PJM, NYISO, and ISO-NE forecast errors.²⁷³
- Other Unrealized Savings – This measures production cost savings that are not realized once the following factors are taken into account:
 - Real-time Curtailment²⁷⁴ - Some of RTC scheduled transactions may not actually flow in real-time for various reasons (e.g., check-out failures, real-time cuts for security and reliability concerns, etc.). The reduction of flows in the efficient direction reduces market efficiency gains.
 - Interface Ramping²⁷⁵ - RTD and RTC have different assumptions regarding interface schedule ramping. In RTD, interface flows start to ramp at 5 minutes before each quarter-hour interval and reach the target level at 5 minutes after. RTC assumes that the target flow level is reached at the top of the quarter-hour interval. Therefore, an inherent difference exists between RTD flows and RTC flows at the top of each quarter-hour interval, which will lead a portion of projected savings to be unrealized in real time.
 - Price Curve Approximation – This applies only to the CTS process between New York and New England. CTSPE forecasts a 7-point piecewise linear supply curve and NYISO transfers it into a step-function curve for use in the CTS process (as shown in Figure A-62). This leads to differences between the marginal cost of interchange estimated by ISO-NE and the assumptions used by the NYISO for scheduling.
 - Actual Savings^{276,277} – This is equal to (Projected Savings – Net Over-Projected Savings - Unrealized Savings).

²⁷³ This is calculated as: a) (final RTC schedule – estimated hourly schedule)*(RTD price – RTC price) for NYISO forecast error; b) (final RTC schedule – estimated hourly schedule)*(PJM forecast price – PJM RT price) for PJM forecast error; and c) (final RTC schedule – estimated hourly schedule)*(NE forecast price – NE RT price) for NE forecast error.

²⁷⁴ This is calculated as (final RTD schedule – final RTC schedule with ramping assumption at the top of quarter-hour interval)*(RTD price at the PJM/NE proxy – PJM/NE RT price at the NY proxy).

²⁷⁵ This is calculated as (final RTC schedule with ramping assumption at the top of quarter-hour interval – final RTC schedule without ramping assumption)*(RTD price at the PJM/NE proxy – PJM/NE RT price at the NY proxy).

²⁷⁶ This is also calculated as (final RTD schedule – estimated hourly schedule)*(RTD price at the PJM/NE proxy – PJM/NE RT price at the NY proxy) + an Adjustment (as described below).

²⁷⁷ The marginal cost of production is estimated from LBMPs that result from scheduling a transaction, but the marginal cost of production varies as the interface schedule is adjusted. For example, if 100 MW is scheduled to flow from PJM or NE to NYISO, reducing the price spread between markets from \$12/MWh to \$5/MWh, our unadjusted production cost savings estimate from the transaction would be \$500/hour (= 100 MW x \$5/MWh). However, if the change in production costs was linear in this example, the true savings would be \$850/hour (= 100 MW x Average of \$5 and \$12/MWh). We make a similar adjustment

- Interface Prices – These show actual real-time prices and forecasted prices at the time of RTC scheduling.
- Price Forecast Errors – These measure the performance of price forecasting by showing the average difference and the average absolute difference between the actual and forecasted prices on both sides of the interfaces.

To examine how price forecast errors affected efficiency gains, these metrics are reported separately for intervals in which forecast errors are less than \$20 per MWh and intervals in which forecast errors exceed \$20 per MWh.

Table A-6: Efficiency of Intra-Hour Scheduling Under CTS
Primary PJM and New England Interfaces, 2025

			Average/Total During Intervals w/ Adjustment							
			CTS - NY/NE				CTS - NY/PJM			
			Both Forecast Errors <= \$20	Any Forecast Error > \$20	Total		Both Forecast Errors <= \$20	Any Forecast Error > \$20	Total	
% of All Intervals w/ Adjustment			68%	15%	83%		41%	22%	63%	
Average Flow Adjustment (MW)		Net Imports	38	52	41		7	-38	-9	
		Gross	117	155	124		87	110	95	
Production Cost Savings (\$ Million)	Projected at Scheduling Time		\$9.0	\$7.6	\$16.6		\$2.9	\$13.9	\$16.7	
	Net Over-Projection by:	NY	-\$0.4	-\$1.4	-\$1.9		-\$0.1	-\$0.3	-\$0.4	
		NE or PJM	\$0.2	-\$0.8	-\$0.6		-\$0.6	-\$12.6	-\$13.2	
	Other Unrealized Savings		-\$0.3	-\$1.6	-\$1.9		-\$0.1	-\$0.4	-\$0.5	
	Actual Savings		\$8.5	\$3.8	\$12.3		\$2.1	\$0.5	\$2.5	
Interface Prices (\$/MWh)	NY	Actual	\$48.02	\$106.20	\$58.71	\$59.58	\$43.28	\$81.37	\$56.66	\$56.18
		Forecast	\$48.92	\$108.66	\$59.90	\$60.70	\$44.09	\$81.54	\$57.24	\$56.87
	NE or PJM	Actual	\$46.94	\$106.09	\$57.81	\$62.99	\$37.48	\$77.02	\$51.37	\$48.41
		Forecast	\$47.08	\$103.50	\$57.45	\$62.95	\$39.72	\$117.35	\$66.98	\$60.61
Price Forecast Errors (\$/MWh)	NY	Fcst. - Act.	\$0.90	\$2.46	\$1.18	\$1.12	\$0.81	\$0.17	\$0.59	\$0.69
		Abs. Val.	\$3.64	\$37.26	\$9.82	\$9.58	\$3.39	\$19.68	\$9.11	\$8.21
	NE or PJM	Fcst. - Act.	\$0.15	-\$2.59	-\$0.35	-\$0.04	\$2.24	\$40.32	\$15.61	\$12.20
		Abs. Val.	\$4.13	\$31.27	\$9.12	\$9.55	\$6.29	\$75.46	\$30.58	\$26.08
			For Adjustment Intervals Only				For All Intervals			

For Adjustment Intervals Only

For All Intervals

Figure A-61 & Figure A-62: Price Forecast Errors Under CTS

The next analysis compares the performance of price forecasting by the three ISOs in the CTS process. Figure A-61 shows the cumulative distribution of forecasting errors in 2025. The price forecast error in each 15-minute period is measured as the absolute value of the difference between the forecast price and actual price.

to our estimate of marginal cost of production assuming that: a) the supply curve was linear in all three markets; b) at the NY/PJM border, a 100 MW movement in the supply curve changes the marginal cost by 7.5 percent of NY LBMP in the New York market and 2.5 percent of PJM LBMP in the PJM market; and c) at the NY/NE border, a 100 MW movement in the supply curve changes the marginal cost by 15 percent of NY LBMP in the New York market and 5 percent of NE LBMP in the NE market.

Figure A-61 shows the ISO-NE forecast error in two ways: (a) based on the piece-wise linear curve produced by its forecasting model, and (b) based on the step-function curve that the NYISO model uses to approximate the piece-wise linear curve.

Figure A-61: Distribution of Price Forecast Errors Under CTS
NE and PJM Primary Interfaces, 2025

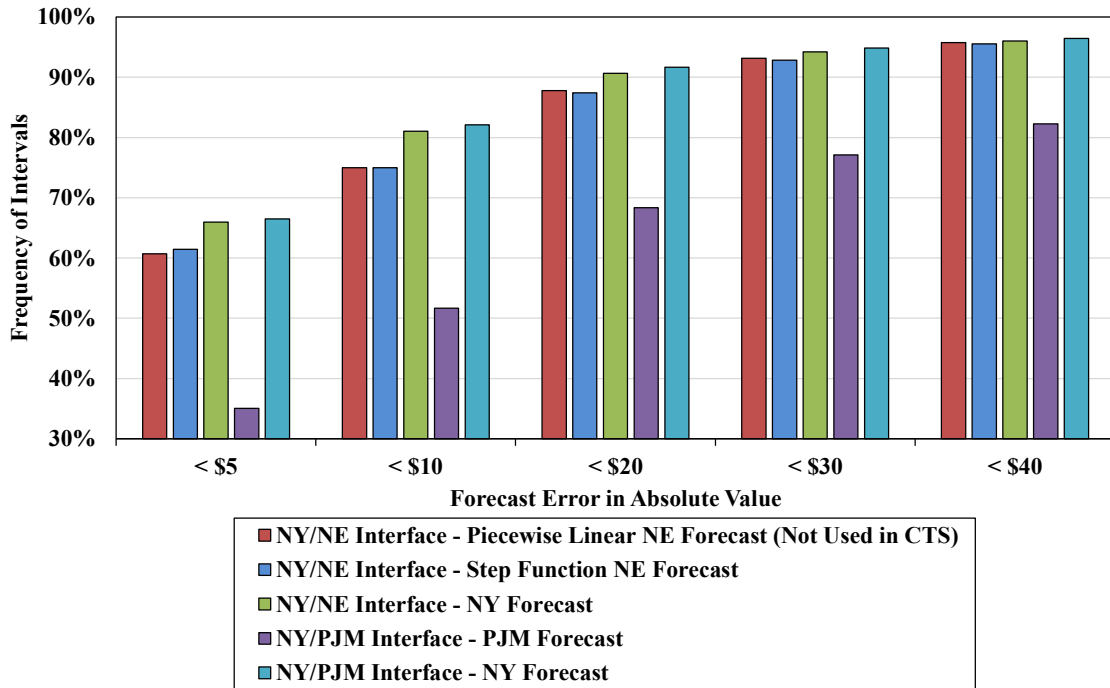


Figure A-62: Example of Supply Curve Produced by ISO-NE and Used by RTC

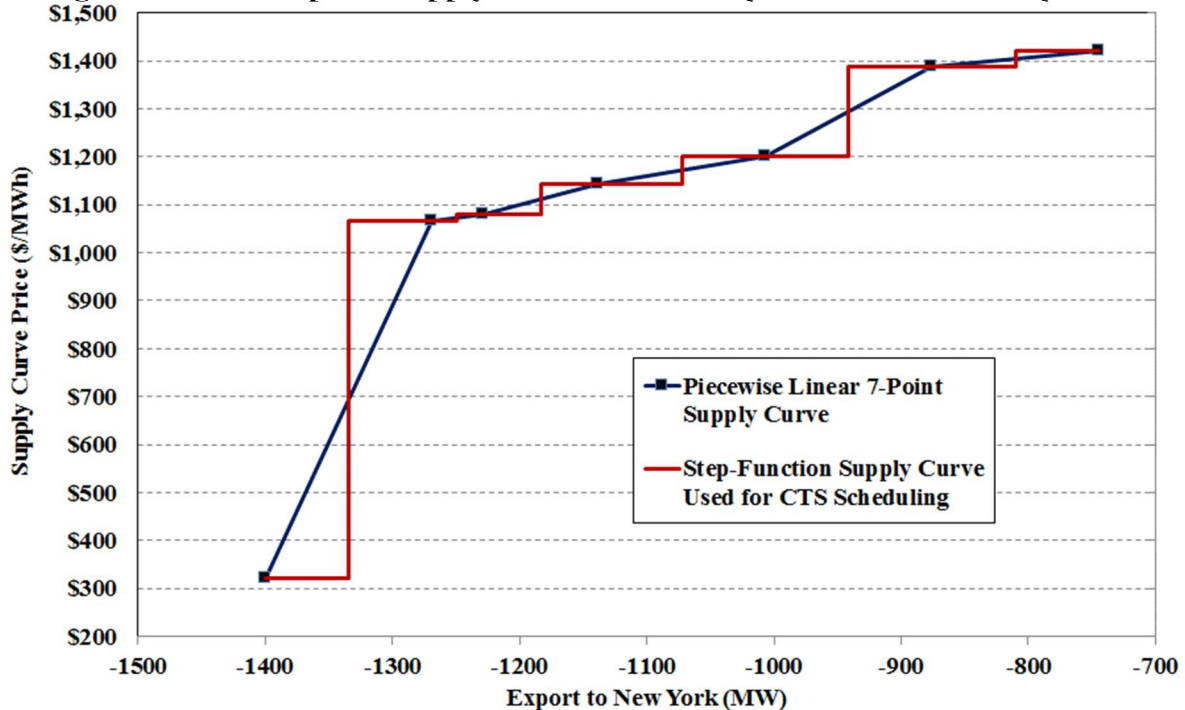


Figure A-62 illustrates this with example curves.²⁷⁸ The blue squares in the figure show the seven price/quantity pairs produced by the ISO-NE price forecast engine (CTSPE). The blue line connecting these points represents a piecewise linear supply curve at the New England border. The red step-function curve is an approximation of the piecewise linear curve and is the curve actually used in RTC for scheduling CTS transactions at the New England border.

D. Evaluation of Factors Contributing to Inconsistency between RTC and RTD

RTC schedules gas turbines and external transactions shortly in advance of the 5-minute real-time market, so its assumptions regarding factors such as the load forecast, the wind forecast, and the ramp profile of individual resources are important.

Figure A-63 to Figure A-65: Forecast Assumptions Used by RTC to Schedule CTS Transactions and Their Price Impact

Figure A-63 to Figure A-65 provide the results of our systematic evaluation of factors that lead to inconsistent results in RTC and RTD. This assesses the magnitude of the contribution of various factors using a metric that is described below. An important feature of this metric is that it distinguishes between factors that *cause* differences between RTC forecast prices and actual RTD prices (which we call “detrimental” factors) and factors that *reduce* differences between RTC forecast prices and actual RTD prices (which we call “beneficial” factors).²⁷⁹

RTC schedules resources with lead times of 15 minutes to one hour, including fast start units and external transactions. Inconsistency between RTC and RTD prices is an indication that some scheduling decisions may be inefficient. For example, suppose that RTC forecasts an LBMP of \$45/MWh and this leads RTC to forego 100 MW of CTS import offers priced at \$50/MWh, and suppose that RTD clears at \$65/MWh because actual load is higher than the load forecast in RTC and RTD satisfies the additional load with 100 MW of online generation priced at \$65/MWh. In this example, the under-forecast of load leads the NYISO to use 100 MW of \$65/MWh generation rather than \$50/MWh of CTS imports, resulting in \$1,500/hour (= 100 MW * {\$65/MWh - \$50/MWh}) of additional production costs. Thus, the inefficiency resulting from poor forecasting by RTC is correlated with: (a) the inconsistency between the MW value used in RTC versus the one used in RTD, and (b) the inconsistency between the price forecasted by RTC versus the actual price determined by RTD. Hence, we use a metric that multiplies the MW-differential between RTC and RTD with the corresponding price-differential for resources that are explicitly considered and priced by the real-time models.

For a generator, external transaction, or load i , our inconsistency metric is calculated as follows:

²⁷⁸ The two curves are forecasted supply curves used in the market on January 5, 2016.

²⁷⁹ Although RTC produces ten forecasts looking 150 minutes into the future, and RTD produces four forecasts looking one hour into the future that are in addition to the binding schedules and prices that are produced for the next five minutes, this metric is calculated comparing just the 15-minute ahead forecast of RTC (which sets the interchange schedules for the interfaces with PJM and ISO-NE that use CTS) to the 5-minute financially binding interval of RTD. Future reports will perform the analysis based on other time frames as well.

$$\text{Metric}_i = (\text{NetInjectionMW}_{i,\text{RTC}} - \text{NetInjectionMW}_{i,\text{RTD}}) * (\text{Price}_{i,\text{RTC}} - \text{Price}_{i,\text{RTD}})^{280}$$

Hence, for the load forecast in the example above, the metric is:

$$\text{Metric}_{\text{load}} = 100 \text{ MW} * (\$45/\text{MWh} - \$65/\text{MWh}) = -\$2,000/\text{hour}$$

For the high-cost generator in the example above, the metric is:

$$\text{Metric}_{\text{generator}} = -100 \text{ MW} * (\$45/\text{MWh} - \$65/\text{MWh}) = +\$2,000/\text{hour}$$

For the foregone CTS imports in the example above, the metric is:

$$\text{Metric}_{\text{import}} = 0 \text{ MW} * (\$45/\text{MWh} - \$65/\text{MWh}) = \$0/\text{hour}$$

The metric produces a negative value for the load forecast, indicating that the under-forecast of load was a “detrimental” factor that contributed to the divergence between the RTC forecast price and the actual RTD price. The metric produces a positive value for the generator that responded to the need for additional supply in RTD, indicating that the generator’s response was a “beneficial” factor that helped limit the divergence between the RTC forecast price and the actual RTD price. The metric produces a zero value for the foregone CTS imports, recognizing that the divergence was not caused by the CTS imports not being scheduled, but rather that their not being scheduled was the result of poor forecasting.

For PAR-controlled line i , our inconsistency metric is calculated across binding constraints c :

$$\text{Metric}_i = (\text{FlowMW}_{i,\text{RTC}} - \text{FlowMW}_{i,\text{RTD}}) * \sum_c \{(\text{ShadowPrice}_{c,\text{RTC}} * \text{ShiftFactor}_{i,c,\text{RTC}} - \text{ShadowPrice}_{c,\text{RTD}} * \text{ShiftFactor}_{i,c,\text{RTD}})\}$$

Hence, for a PAR-controlled line that is capable of relieving congestion on a binding constraint, if the flow on the PAR-controlled line is higher in RTD than in RTC and the shadow price of the constraint is higher in RTD than in RTC, the metric will produce a positive value, indicating that the PAR-controlled line had a beneficial inconsistency (i.e., it helped reduce the divergence between RTC and RTD congestion prices). However, if the flow on the PAR-controlled line decreases in RTD while the shadow price is increasing, the metric will produce a negative value, indicating that the PAR-controlled line had a detrimental inconsistency (i.e., it contributed to the divergence between RTC and RTD congestion prices). This calculation is performed for both “optimized” PARs and “non-optimized” PARs.²⁸¹

For transmission constraints that are modeled, it is also important to quantify inconsistencies that lead to divergence between RTC and RTD. To the extent that such inconsistencies result from reductions in available transfer capability that increase congestion, the metric will produce a negative (i.e., detrimental) result. On the other hand, if inconsistencies result from an increase in

²⁸⁰ Note, that this metric is summed across energy, operating reserves, and regulation for each resource.

²⁸¹ A PAR is called “non-optimized” if the RTC and RTD models treat the flow as a fixed value in the optimization engine, while a PAR is called “optimized” if the optimization engines of the RTC and RTD models treat the flow as a flexible within some range.

transfer capability that helps ameliorate an increase in congestion, the metric will produce a positive (i.e., beneficial) result. For each limiting facility/contingency pair c , the calculation utilizes the shift factors and schedules for resources and other inputs i :

$$\text{Metric_BindingTx}_c = \text{ShadowPrice}_{c,\text{RTC}} * \sum_i \{ \text{ShiftFactor}_{i,c,\text{RTC}} * (\text{MW}_{i,\text{RTC}} - \text{MW}_{i,\text{RTD}}) \} \\ - \text{ShadowPrice}_{c,\text{RTD}} * \sum_i \{ \text{ShiftFactor}_{i,c,\text{RTD}} * (\text{MW}_{i,\text{RTC}} - \text{MW}_{i,\text{RTD}}) \}$$

Once the metric is calculated for each optimized PAR and each binding constraint, the transmission system is divided into regions and if a particular region has optimized PARs and/or binding constraints with positive and negative values, the following adjustments are used. If the sum across all values is positive, then each positive value is multiplied by the ratio of: $\{(\text{TotalGrossPositive} + \text{TotalGrossNegative})/\text{TotalGrossPositive}\}$ and each negative value is discarded. If the sum across all values is negative, then each negative value is multiplied by the ratio of: $\{(\text{TotalGrossPositive} + \text{TotalGrossNegative})/\text{TotalGrossNegative}\}$ and each positive value is discarded. This is done because when transfer capability on one facility in a particular region is reduced, the optimization engine often increases utilization of parallel circuits, so the adjustments above are helpful in discerning whether the net effect was beneficial or detrimental.

Example 1

The following two-node example illustrates how the metrics would be calculated if a transmission line tripped after the RTC run, causing a divergence between RTC and RTD prices. Suppose, RTC forecasts:

- Load_A = 100 MW and Load_B = 200 MW;
- Three transmission lines (Lines 1, 2, and 3) with equal impedance connect A to B and the lowest rated line (Line 1) has 50 MW of capability, so the shift factor of node A on Line 1 is 0.333 (assuming node B is the reference bus);
- Gen_A produces 250 MW at a cost of \$20/MWh and Gen_B produces 50 MW at a cost of \$30/MWh; and
- Thus, in RTC, Price_A = \$20/MWh, Price_B = \$30/MWh, Flow_{AB1} on Line 1 = 50 MW, so the ShadowPrice_{AB1} = \$30/MWh.

Suppose that before RTD runs, Line 2 trips, reducing flows from Node A to Node B and requiring output from a \$45/MWh generator at Node B. This will lead to the following changes:

- Only two transmission lines (Lines 1 and 3) with equal impedance connect A to B, so the shift factor of node A on Line 1 is 0.5 (assuming node B is the reference bus);
- Gen_A produces 200 MW at a cost of \$20/MWh, Gen_B produces 50 MW at a cost of \$30/MWh, and Gen_{B2} produces 50 MW at a cost of \$45/MWh; and
- Thus, in RTD, Price_A = \$20/MWh, Price_B = \$45/MWh, Flow_{AB1} on Line 1 = 50 MW, so the ShadowPrice_{AB1} = \$50/MWh.

In this example, the metric would be calculated as follows for each input:

- Metric_Load_A = \$0 = (-100MW - -100MW) * (\$20/MWh - \$20/MWh)
- Metric_Load_B = \$0 = (-200MW - -200MW) * (\$30/MWh - \$45/MWh)

- $\text{Metric_Gen}_A = \$0 = (250\text{MW} - 200\text{MW}) * (\$20/\text{MWh} - \$20/\text{MWh})$
- $\text{Metric_Gen}_B = \$0 = (50\text{MW} - 50\text{MW}) * (\$30/\text{MWh} - \$45/\text{MWh})$
- $\text{Metric_Gen}_{B2} = \$750/\text{hour} = (0\text{MW} - 50\text{MW}) * (\$30/\text{MWh} - \$45/\text{MWh})$
- $\text{Metric_BindingTx} = -\$750/\text{hour} = \$30/\text{MWh} * 0.333 * (250\text{MW} - 200\text{MW}) - \$50/\text{MWh} * 0.5 * (250\text{MW} - 200\text{MW})$
- Metric_BindingTx exhibits a negative value, indicating a detrimental factor because the divergence between RTC prices and RTD prices was caused by a reduction in transfer capability from Node A to Node B. Metric_Gen_{B2} exhibits a positive value, indicating a beneficial factor because the divergence between RTC prices and RTD prices was limited by the response of additional generation at Node B. All of the other factors have a zero value because they neither contributed to convergence or divergence between RTC and RTD prices.

Example 2

The following two-node example illustrates how the metrics would be calculated if a generator tripped after the RTC run, causing a divergence between RTC and RTD prices. Suppose RTC forecasts:

- $\text{Load}_A = 100 \text{ MW}$ and $\text{Load}_B = 200 \text{ MW}$;
- Three transmission lines (Lines 1, 2, and 3) with equal impedance connect A to B and the lowest rated line (Line 1) has 50 MW of capability, so the shift factor of node A on Line 1 is 0.333 (assuming node B is the reference bus);
- Gen_A produces 200 MW at a cost of $\$20/\text{MWh}$ and Gen_B produces 100 MW at a cost of $\$20/\text{MWh}$; and
- Thus, in RTC, $\text{Price}_A = \$20/\text{MWh}$, $\text{Price}_B = \$20/\text{MWh}$, Flow_{AB1} on Line 1 = 33.33 MW, so the $\text{ShadowPrice}_{AB1} = \$0/\text{MWh}$.

Suppose that before RTD runs, Gen_B trips, increasing flows from Node A to Node B from 100 MW to 150 MW, requiring 50 MW of additional production from Gen_A and requiring 50 MW of production from a $\$45/\text{MWh}$ generator at Node B. This will lead to the following changes:

- Gen_A produces 250 MW at a cost of $\$20/\text{MWh}$ and Gen_{B2} produces 50 MW at a cost of $\$45/\text{MWh}$; and
- Thus, in RTD, $\text{Price}_A = \$20/\text{MWh}$, $\text{Price}_B = \$45/\text{MWh}$, Flow_{AB1} on Line 1 = 50 MW, so the $\text{ShadowPrice}_{AB1} = \$75/\text{MWh}$.

In this example, the metric would be calculated as follows for each input:

- $\text{Metric_Load}_A = \$0 = (-100\text{MW} - -100\text{MW}) * (\$20/\text{MWh} - \$20/\text{MWh})$
- $\text{Metric_Load}_B = \$0 = (-200\text{MW} - -200\text{MW}) * (\$20/\text{MWh} - \$45/\text{MWh})$
- $\text{Metric_Gen}_A = \$0 = (200\text{MW} - 250\text{MW}) * (\$20/\text{MWh} - \$20/\text{MWh})$
- $\text{Metric_Gen}_B = -\$2,500/\text{hour} = (100\text{MW} - 0\text{MW}) * (\$20/\text{MWh} - \$45/\text{MWh})$
- $\text{Metric_Gen}_{B2} = \$1,250/\text{hour} = (0\text{MW} - 50\text{MW}) * (\$20/\text{MWh} - \$45/\text{MWh})$

- $\text{Metric_BindingTx} = \$1,250/\text{hour} = \$0/\text{MWh} * 0.333 * (200\text{MW} - 250\text{MW}) - \$75/\text{MWh} * 0.333 * (200\text{MW} - 250\text{MW})$
- Metric_BindingTx exhibits a positive value, indicating a beneficial factor because excess transfer capability was utilized to reduce the divergence between RTC prices and RTD prices that was caused by the generator trip at Node B. Metric_Gen_{B2} exhibits a positive value, indicating a beneficial factor because the divergence between RTC prices and RTD prices was limited by the response of additional generation at Node B. All of the other factors have a zero value because they neither contributed to convergence or divergence between RTC and RTD prices.

Categories of Factors Affecting RTC/RTD Price Divergence

RTC and RTD forecasts are based on numerous inputs. We summarize inputs that change between RTC and RTD in the following ten categories for the purposes of this analysis:

- Load Forecast Error – Combines the forecast of the load forecasting model with any upward or downward adjustment by the operator.
- Wind Forecast Error – Uses the blended value that is a weighted average of the wind forecasting model and the current telemetered value.
- External Transaction Curtailments and Checkout Failures
- Generator Forced Outages and Derates
- Generator Not Following Schedule – Includes situations where a generator's RTD schedule is affected by a ramp-constraint and where the ramp-constraint was tighter as a result of the generator not following its schedule in a previous interval.
- Generator on OOM Dispatch
- Generator Dispatch In Merit
- NY/NJ PARs and Other Non-Optimized PARs – Includes the A, J, K, and 5018 PAR-controlled lines.
- Transmission Utilization – Includes contributions from binding constraints and optimized PARs. This category is organized into the following regional transmission corridors:
 - West Zone
 - West Zone to Central NY
 - North Zone to Central NY
 - Central East
 - UPNY-SENY & UPNY-ConEd
 - New York City
 - Long Island
- Schedule Timing and Ramp Profiling – This includes differences that result from inconsistent timing and treatment of ramp between RTC and RTD for load forecast,

external interchange, self-scheduled generation, and dispatchable generation. This is illustrated for external interchange in Figure A-68.

Figure A-63 summarizes the RTC/RTD divergence metric results for detrimental factors in 2025, while Figure A-64 provides the summary for beneficial factors. Figure A-65 summarizes the beneficial and detrimental metric results for Transmission Utilization.

Figure A-63: Detrimental Factors Causing Divergence between RTC and RTD
2025

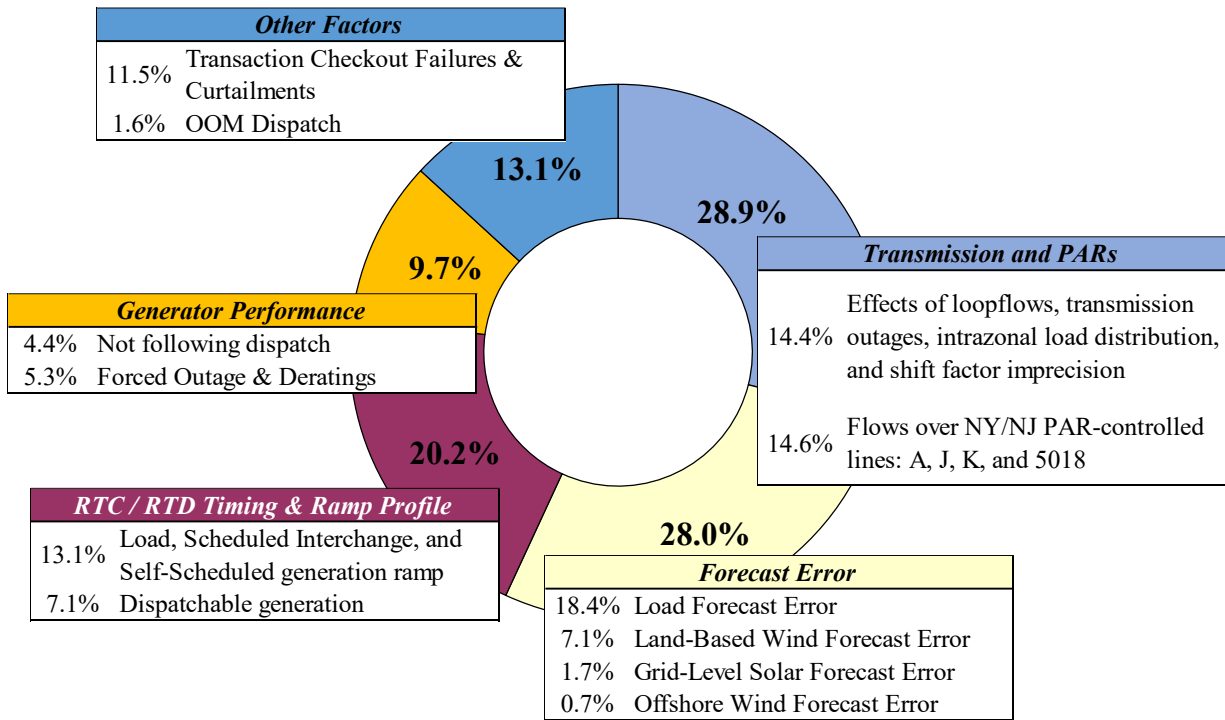


Figure A-64: Beneficial Factors Reducing Divergence between RTC and RTD
2025

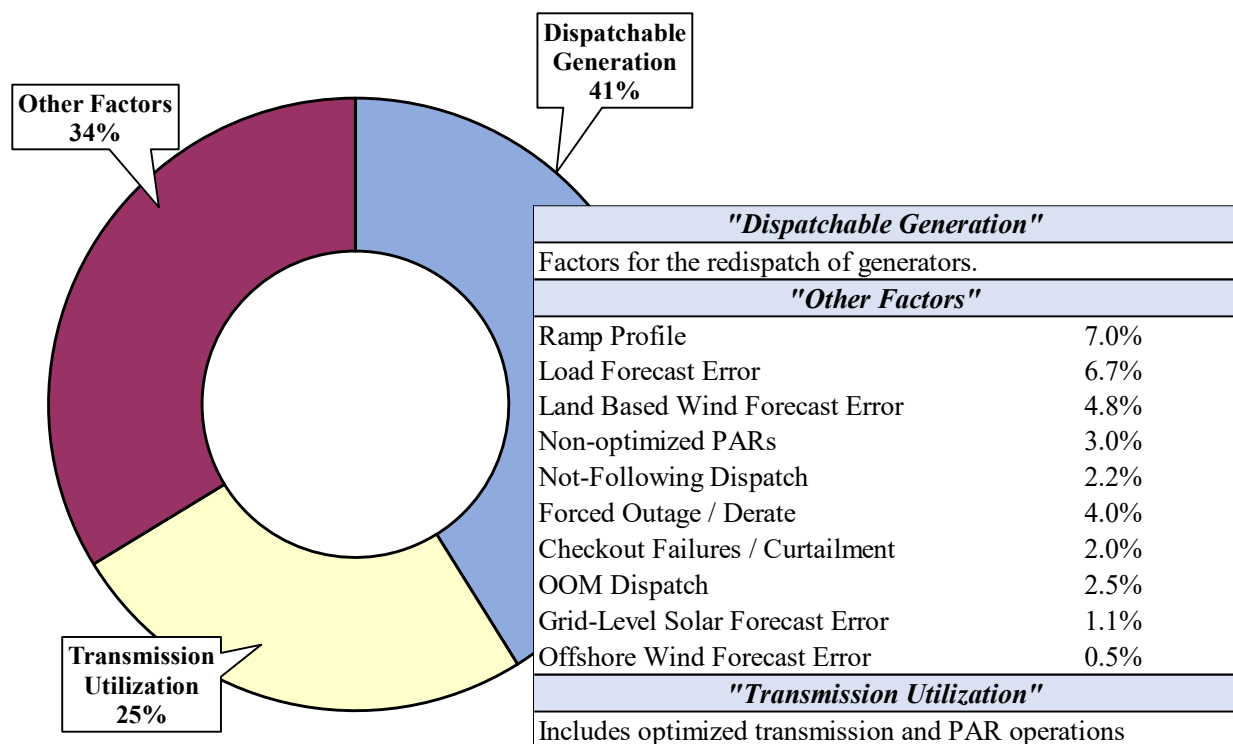
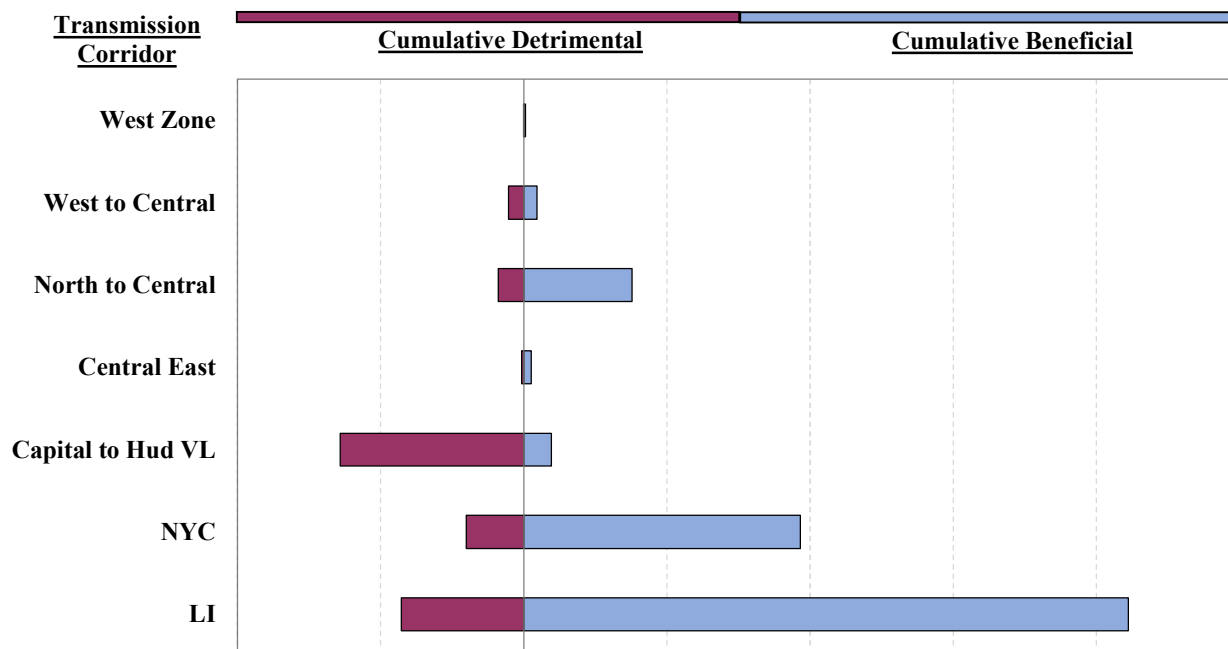


Figure A-65: Effects of Network Modeling on Divergence between RTC and RTD
By Region, 2025



E. Patterns of Key Factors Driving Price Differences between RTC and RTD

The following analyses focus on several key factors contributing to inconsistency between RTC and RTD, which (a) evaluate the magnitude and patterns of forecast errors of these factors and (b) examine how these affect the accuracy of RTC's price forecasting.

Figure A-66 & Figure A-67: Differences in Prices vs Differences in Assumptions of Net Interchanges between RTC and RTD

Figure A-66 shows a histogram of the differences in 2025 between (a) the net interchange assumed by RTC; and (b) the actual net interchange reflected in RTD at the quarter-hour intervals (i.e., at :00, :15, :30, :45).

For each histogram tranche, the figure summarizes the accuracy of the RTC price forecast by showing:

- The average of the RTD LBMP minus the RTC LBMP;
- The median of the RTD LBMP minus the RTC LBMP; and
- The mean absolute difference between the RTD and RTC LBMPs.

LBMPs are shown at the NYISO Reference Bus at the quarter-hour intervals for RTC and RTD.

Figure A-66: Histogram of Differences Between RTC and RTD Prices and Net Interchange 2025

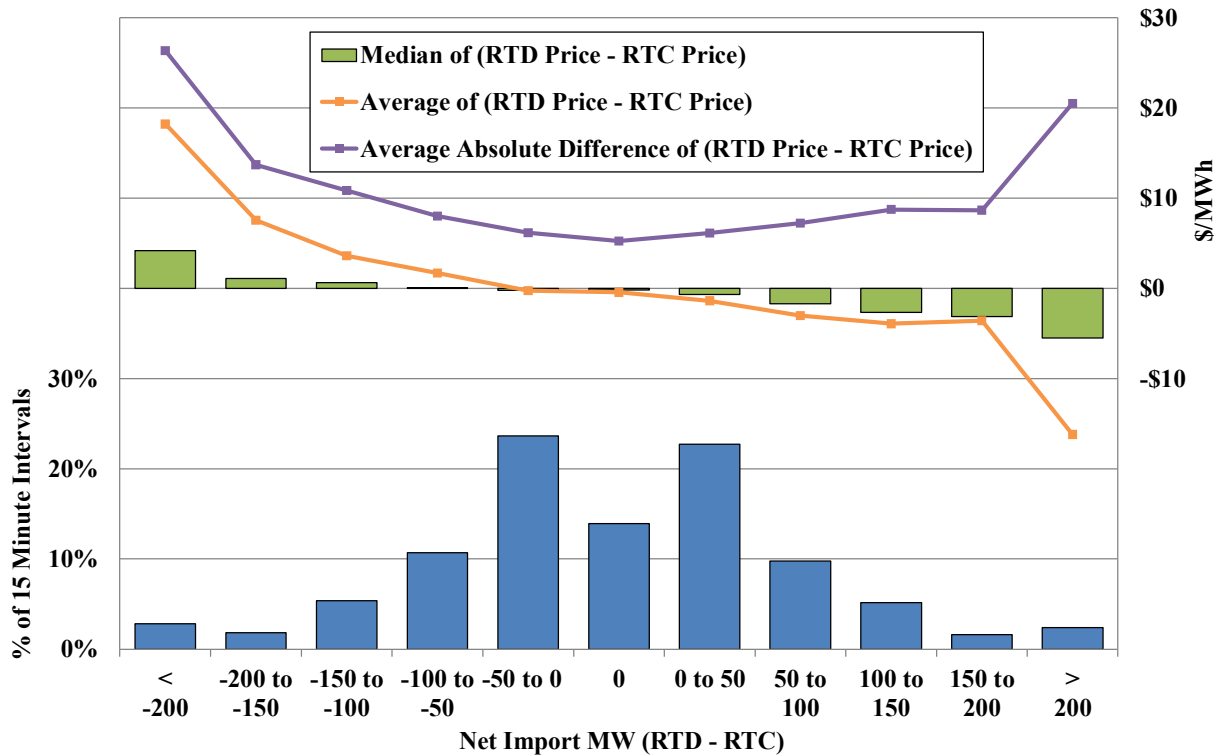


Figure A-67 shows pricing and scheduling differences by time of day. The stacked bars in the lower portion of the figure show the frequency, direction, and magnitude of differences between RTC and RTD net import levels for 100+ MW tranches. The upper portion of the figure summarizes the accuracy of the RTC price forecast by showing the average RTD LBMP minus the average RTC LBMP and the mean absolute difference between the RTD and RTC LBMPs.

Figure A-67: Differences Between RTC and RTD Prices and Net Interchange Schedules by Time of Day, 2025

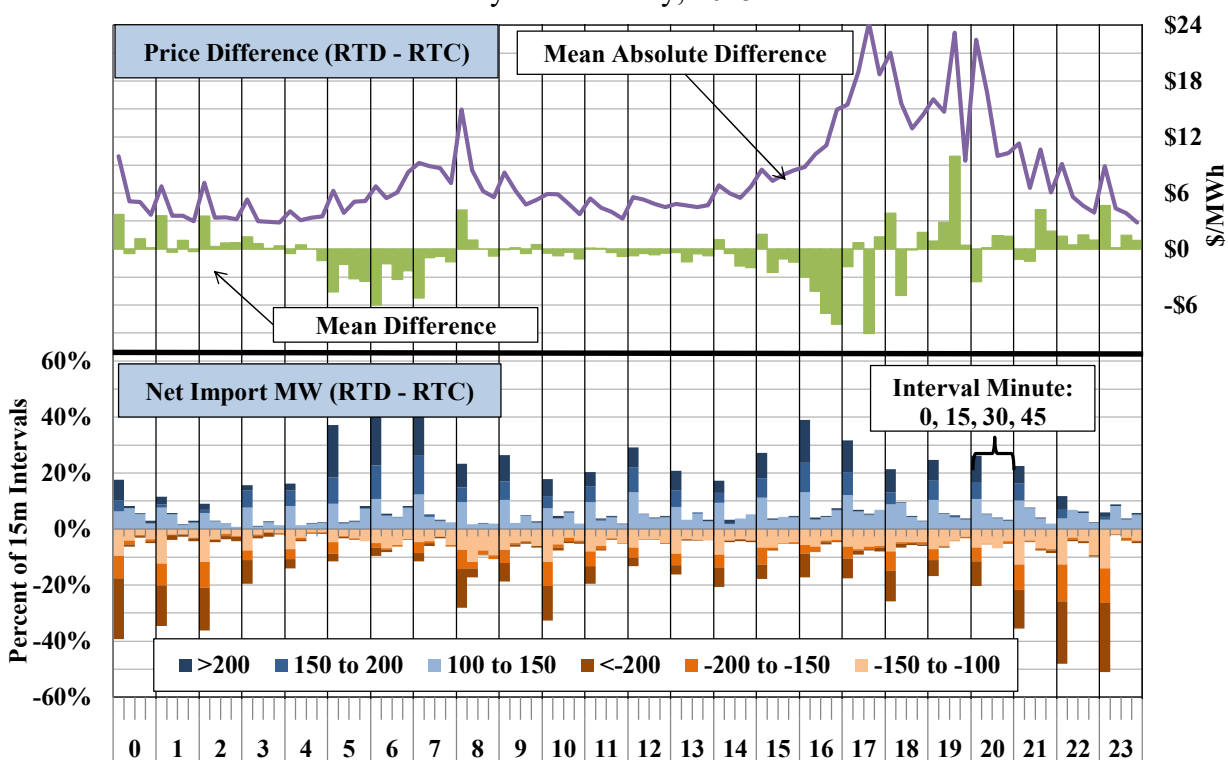


Figure A-68: Illustration of the ramp profiles that are assumed by RTC and RTD

The differences in net interchange schedules between RTC and RTD result from factors such as transaction checkout failures, curtailments by operators, and different ramp assumptions used in RTC and RTD. Figure A-68 provides an illustration of the ramp profiles that are assumed by RTC and RTD. The different ramp profiles lead to inconsistencies between RTC and RTD in the level of net imports, which contribute to differences between the RTC price forecast and actual 5-minute RTD clearing prices. Although inconsistent ramp profile assumptions are not the only source of inconsistent RTC and RTD prices, they illustrate how inconsistent modeling assumptions can lead to inconsistent pricing outcomes.

In RTD, the assumed level of net imports is based on the scheduled interchange at the end of each 5-minute period. Transactions are assumed to move over a 10-minute period from one scheduling period to the next for both hourly and 15-minute interfaces. The 10-minute period goes from five minutes before the top-of-the-hour or quarter-hour to five minutes after. On the other hand, RTC schedules transactions as if they reach their schedule at the top-of-the-hour or quarter-hour, which is five minutes earlier than RTD. Green arrows are used to show intervals

when RTD imports exceed the assumption used in RTC. Red arrows are used to shown intervals when imports assumed in RTC exceed the RTD imports.

Figure A-68: Illustration of External Transaction Ramp Profiles in RTC and RTD

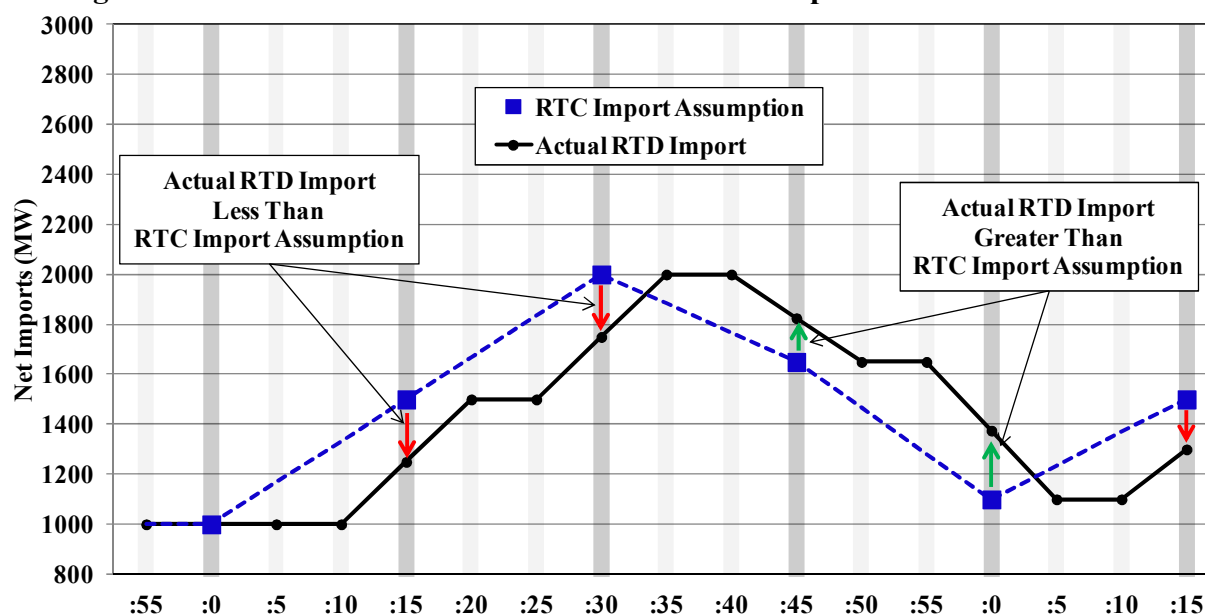


Figure A-69 & Figure A-70: Differences in Prices vs Differences in Load Forecasts between RTC and RTD

Figure A-69 shows a histogram of the differences in systemwide load forecasts (including load biases by operators) between RTC and RTD at the quarter-hour intervals (i.e., at :00, :15, :30, :45) for 2025. For each tranche of the histogram, the figure summarizes the accuracy of the RTC price by showing:

- The average of the RTC LBMP minus the RTD LBMP;
- The median of the RTC LBMP minus the RTD LBMP; and
- The mean absolute difference between the RTD and RTC LBMPs.

LBMPs are shown as zonal-load-weighted prices at the quarter-hour intervals for both RTC and RTD.

Figure A-70 shows these pricing and load forecasting differences by time of day. The stacked bars in the lower portion of the figure show the frequency, direction, and magnitude of differences between RTC and RTD load forecast levels in tranches. The upper portion of the figure summarizes the accuracy of the RTC price forecast by showing:

- The average RTC LBMP minus the average RTD LBMP; and
- The mean absolute difference between the RTD and RTC LBMPs.

Figure A-69: Histogram of Differences Between RTC and RTD Prices and Load Forecasts 2025

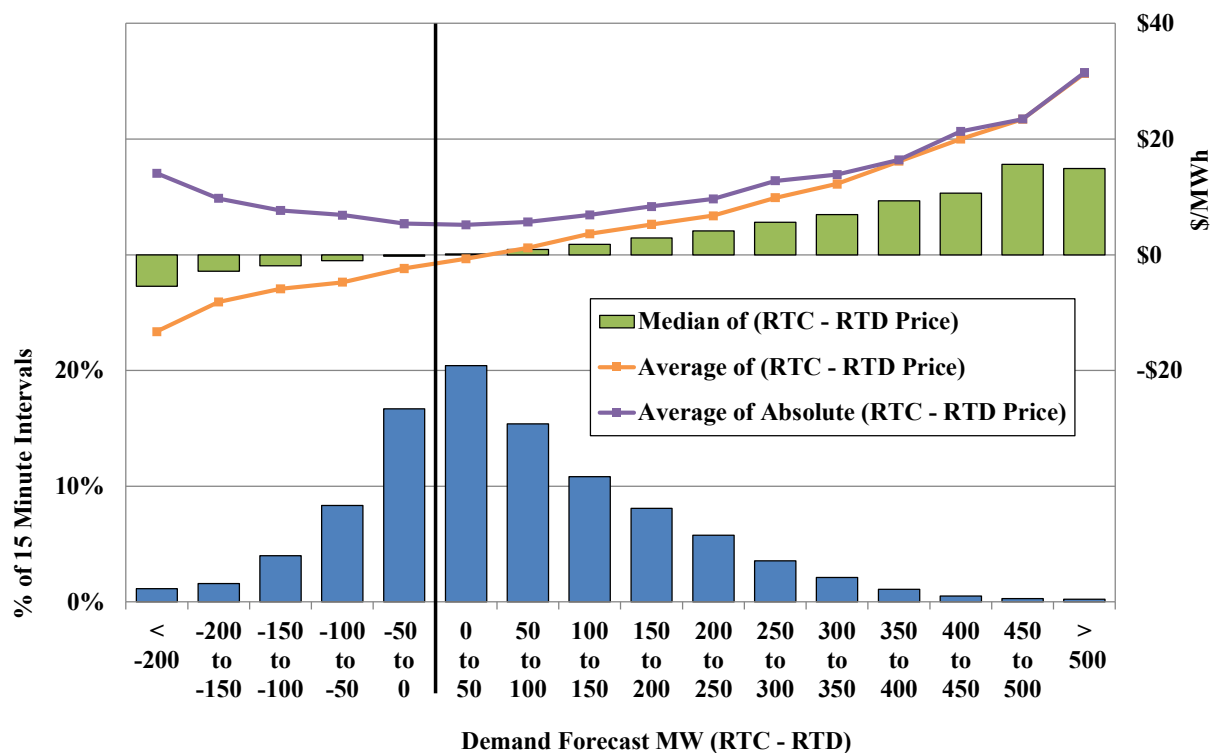


Figure A-70: Differences Between RTC and RTD Prices and Load Forecasts by Time of Day, 2024

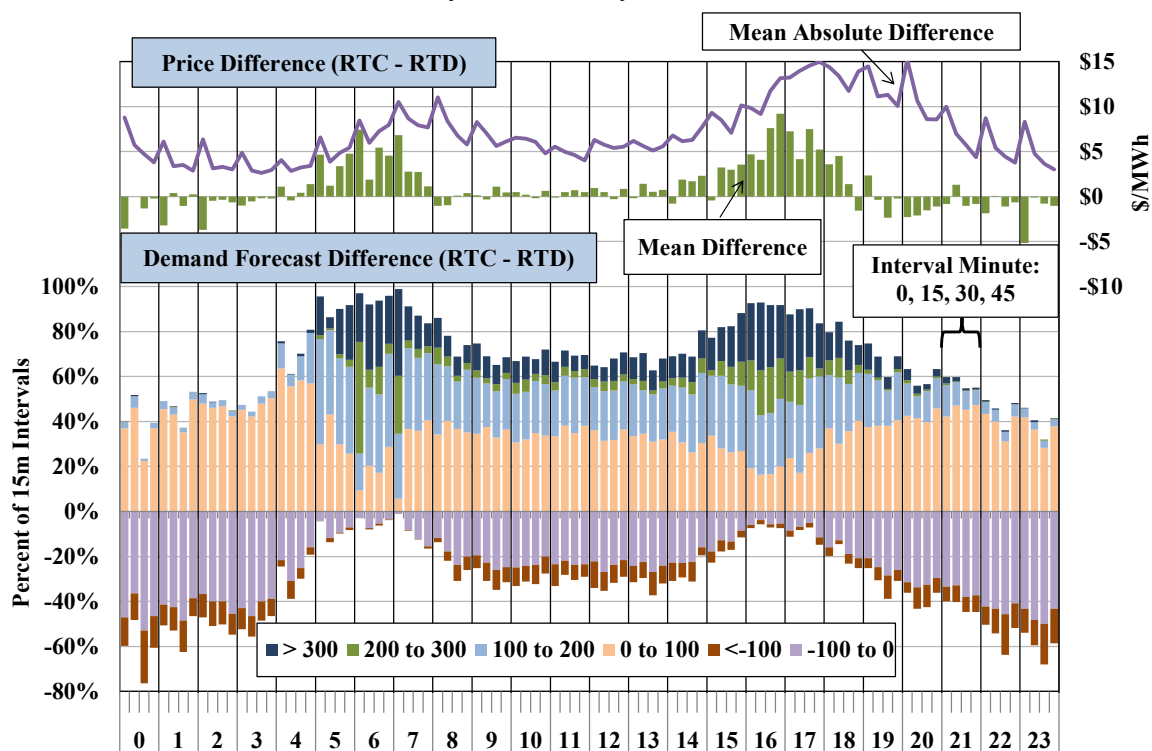
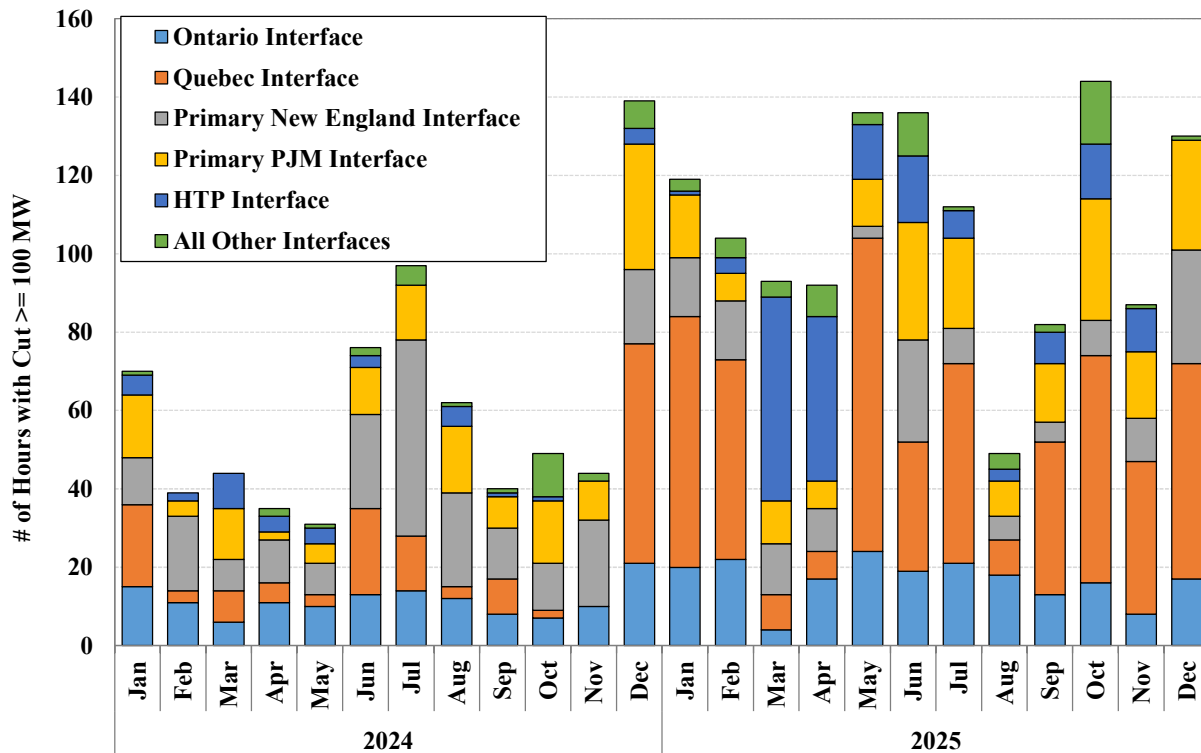


Figure A-71 & Figure A-72: Curtailments on RTC/RTD Divergence

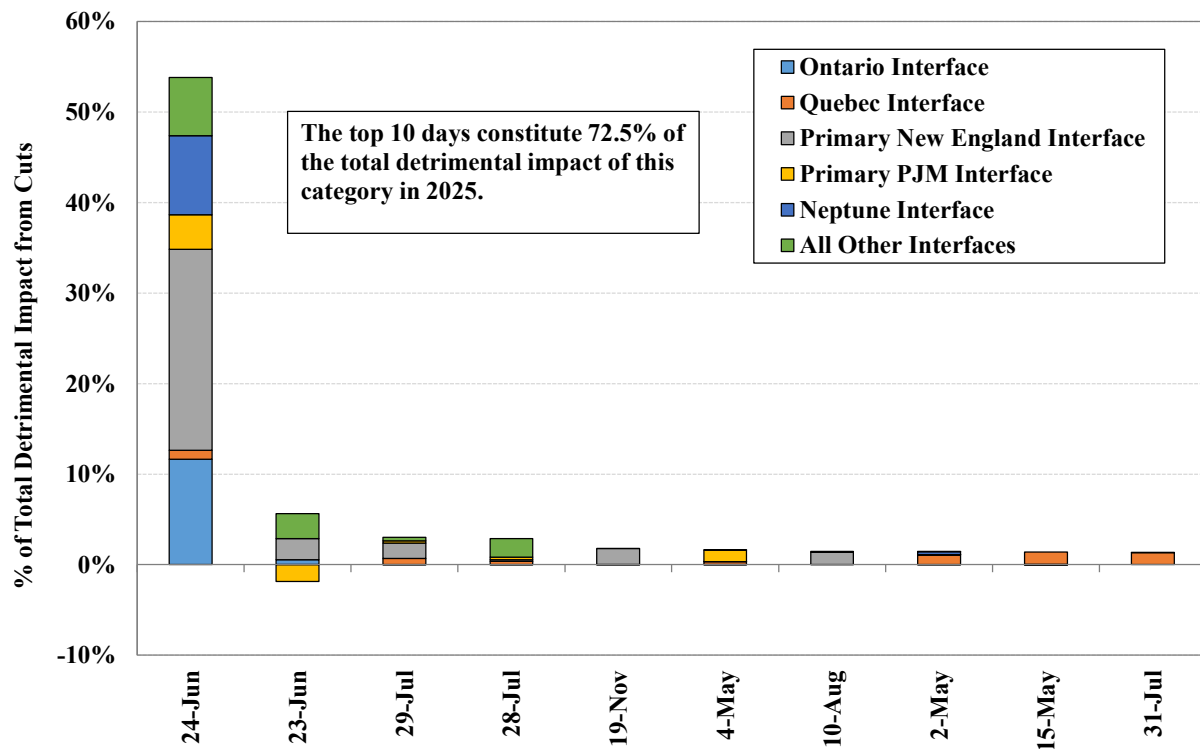
Figure A-71 compares the frequency of external transaction curtailments by month in 2024 and 2025. This is shown separately for the Ontario interface, the Quebec interface, the primary New England interface, the Neptune interface, and the primary PJM interface. All other interfaces are grouped together. For one particular interface, one hour is counted towards the curtailment frequency if the quantity of net curtailments in either import direction or export direction was more than 100MW in any intervals within the hour.

Figure A-72 shows the 10 days in 2025 which contributed the most significant impact to the category of detrimental curtailments causing divergence between RTD and RTC. Of this category, the bulk of the impact (nearly 80 percent) is contained within the top 10 days.

Figure A-71: Number of Hours with External Transaction Curtailments by Interface
By Month, 2024-2025



**Figure A-72: Top 10 Days in Detrimental Curtailment Category
2025**



V. MARKET OPERATIONS

The objective of the wholesale market is to coordinate resources efficiently to satisfy demand while maintaining reliability. The day-ahead market should commit the lowest-cost resources to meet expected conditions on the following day, and the real-time market should deploy the available resources efficiently. Clearing prices should be consistent with the costs of deploying resources to satisfy demand while maintaining reliability. Under shortage conditions, the real-time market should provide incentives for resources to help the NYISO maintain reliability and set clearing prices that reflect the shortage of resources.

The operation of the real-time market plays a critical role in the efficiency of the market outcomes because changes in operations can have large effects on wholesale market outcomes and costs. Efficient real-time price signals are beneficial because they encourage competitive conduct by suppliers, participation by demand response, and investment in new resources and transmission where they are most valuable.

In this section, we evaluate the following aspects of wholesale market operations in 2025:

- *Real-Time Price-Setting by Gas Turbines with Multi-Hour Minimum Run Times* – This subsection evaluates the consistency of pricing with gas turbine commitment and dispatch decisions in the real-time market, focusing on a subset of gas turbines that offer multi-hour minimum run times.
- *Dispatch Performance of Intermittent Generators when Curtailed* – This evaluates the performance of intermittent generators when operators curtail them for system security.
- *Performance of Operating Reserve Providers* – This subsection analyzes: a) the performance of gas turbines in responding to a signal to start-up in the real-time market; and b) how the expected performance of operating reserve providers affects the cost of congestion management in New York City.
- *M2M Coordination* – This subsection evaluates the operation of PAR-controlled lines under market-to-market coordination (“M2M”) between PJM and the NYISO.
- *Operation of Controllable Lines* – This subsection evaluates the efficiency of real-time flows across controllable lines more generally.
- *Regulation Movement-to-Capacity Ratio* – This subsection evaluates the actual movement-to-capacity for individual regulation providers versus the single common multiplier used in the regulation scheduling process.
- *Pricing Under Shortage Conditions* – We evaluate two types of shortage conditions: (a) shortages of operating reserves and regulation, and (b) transmission shortages.
- *Supplemental Commitment for Reliability* – These are necessary when the market does not provide incentives for suppliers to satisfy certain reliability requirements. However, supplemental commitments raise concerns because they indicate the market does not provide sufficient incentives, they dampen market signals, and they lead to uplift charges.
- *BPCG Uplift Charges* – This subsection evaluates BPCG uplift charges resulted primarily from supplemental commitment and out-of-merit dispatch.

A. Real-Time Price-Setting by Gas Turbines with Multi-Hour Minimum Run Times

The ISO schedules resources to provide energy and ancillary services in real time using two models.

First, the Real Time Dispatch model (“RTD”) typically executes every five minutes, deploying resources that are flexible enough to adjust their output on a five-minute basis. RTD also starts 10-minute units when it is economic to do so.²⁸² RTD models dispatch across roughly a one-hour time horizon, better enabling it to determine when a gas turbine will be economic to start or when a generator should begin ramping in anticipation of a constraint in a future interval.

Second, the Real Time Commitment model (“RTC”) executes every 15 minutes and evaluates conditions across a two-and-a-half-hour time horizon. RTC is primarily responsible for scheduling resources that are not flexible enough to be dispatched by RTD. RTC starts up and shuts down 10-minute and 30-minute units when it is economic to do so.²⁸³ RTC also schedules bids and offers to export, import, and wheel-through power to and from other control areas.

The real-time scheduling process ignores minimum run time offers and assumes a default one-hour minimum run time for all fast-start units. Nonetheless, fast-start units that submit bids with multi-hour minimum run times are excluded from setting prices. Therefore, the real-time costs of these units are not properly reflected in the LBMPs. This subsection evaluates the potential market impact from this discrepancy between scheduling and pricing in the real-time markets.

Figure A-73: Real-Time Prices during Commitment of GTs with Multi-Hour MRT

Figure A-73 evaluates prices during commitments of gas turbines offering multi-hour minimum run times in the real-time market in the past two years. The evaluation focuses on economic commitments made by RTC, RTD, or RTD-CAM,²⁸⁴ excluding self-schedules and out-of-market commitments made by operators.

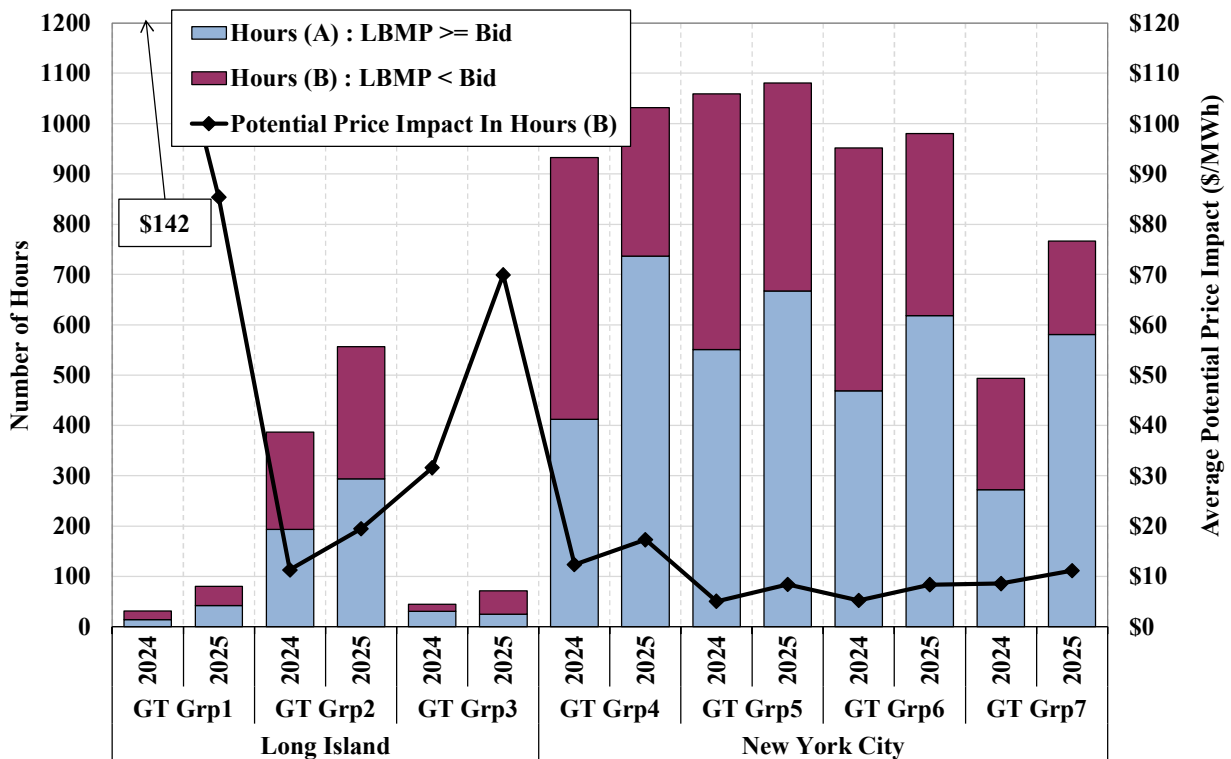
The bars in the figure show the total number of hours when GTs were economically committed each year. The blue bars indicate the number of hours when LBMPs exceeded GT costs (i.e., incremental cost + amortized startup cost), while the red bars represent the number of hours when LBMPs were below GT costs. The black line shows our estimate of potential price impact if these GTs were allowed to set prices. GTs are combined into seven groups in New York City and Long Island based on their electric connection to the grid.

²⁸² 10-minute units can start quickly enough to provide 10-minute non-synchronous reserves.

²⁸³ 30-minute units can start quickly enough to provide 30-minute non-synchronous reserves, but not quickly enough to provide 10-minute reserves.

²⁸⁴ The Real-Time Dispatch – Corrective Action Mode (RTD-CAM) is version of RTD that NYISO operators can run on-demand to address abnormal or unexpected system conditions.

**Figure A-73: Prices During Commitments of GTs Offering Multi-Hour Min Run Times
2024-2025**



B. Performance Evaluation of Intermittent Power Resources

Table A-7 displays the performance of wind-and-solar powered Intermittent Power Resources (“IPRs”) in 2025 when receiving an Output Limit (i.e., being curtailed). Each such resource is placed into a performance range based on how much it reduced output during economic curtailments in 2025. The performance grouping for the year is based on the average performance of an IPR during each discrete economic curtailment event that :

$$\text{Performance} = 1 - \left(\frac{\text{Overgeneration MWh}}{\text{Instructed Curtailment MWh}} \right)$$

Where:

$$\text{Overgeneration MWh} = \text{Max}(0, \text{Actual Output MWh} - ([\text{Economic Basepoint MWh}] + (0.03 * \text{UOL})))$$

Additional information for each performance group of IPRs documents the number of unique resources per bucket, total output for the year, the percentage of intervals where resources were curtailed (economically or manually), and the percentage of curtailment intervals where the curtailment instruction was issued manually. Manual curtailments of IPRs are identified through a combination of approaches:

- Review of NYISO operator logs; and

- Sharp reductions in actual output that persist for several intervals and do not coincide with forecasted drops in wind output.

Table A-7: Performance Evaluation of IPRs²⁸⁵
2025

Performance Group	Installed Capacity (MW)	Group Composite Performance	No. of Units	Total Generation (GWh)	Percent of Intervals Curtailed	Percent of Curtailment Intervals Communicated Manually
0%-10%	0		0			
10%-20%	50	12.7%	≤ 3	130	0.4%	33%
20%-30%	0		0			
30%-40%	0		0			
40%-50%	20	43.5%	≤ 3	40	0.0%	
50%-60%	250	56.4%	≤ 3	500	0.2%	1%
60%-70%	400	68.5%	5	920	1.0%	70%
70%-80%	530	76.1%	11	920	0.7%	73%
80%-90%	870	85.5%	11	2,210	2.3%	48%
90%-100%	1,430	94.6%	12	3,380	2.1%	49%

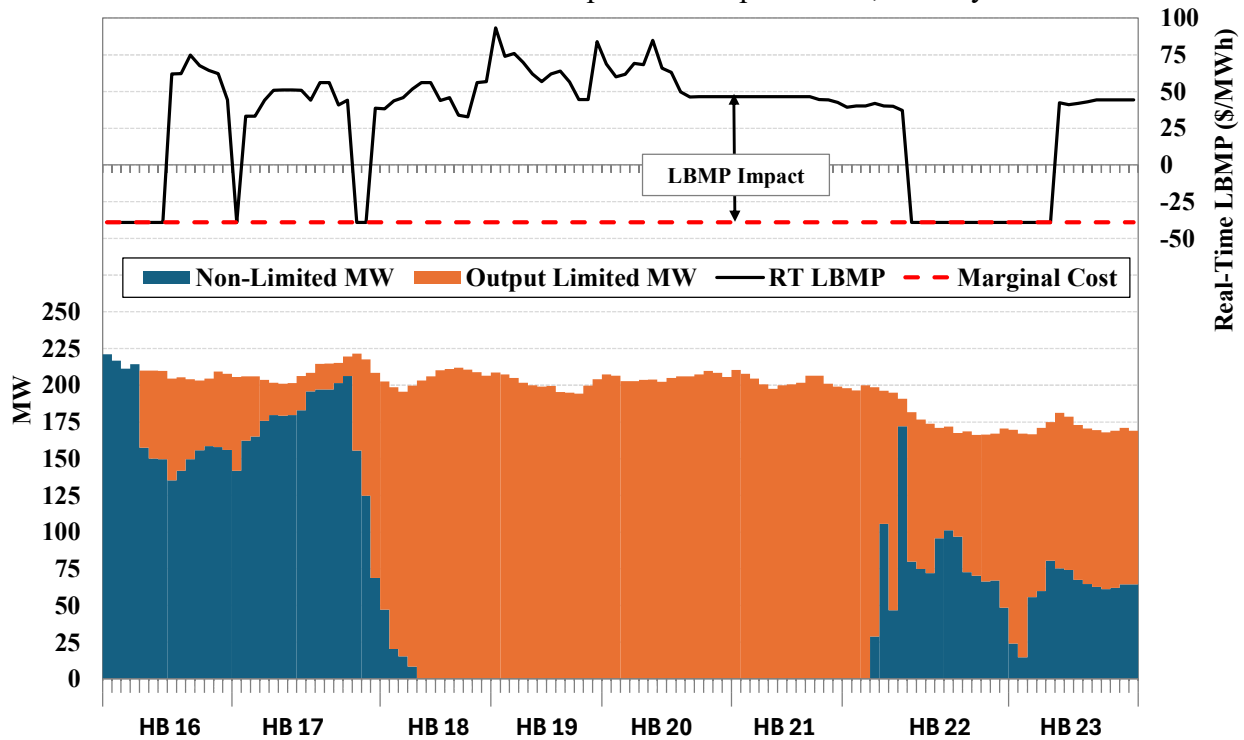
While most intermittent generators comply well with curtailment instructions and most curtailment instructions are given to the high performing resources, a small number of generators perform significantly worse than average. Figure A-74 shows the performance of one wind resource that did not respond appropriately during a January 2024 event. Since this unit did not respond to Output Limits, the operators were forced to issue manual curtailment instructions to other nearby wind units to maintain transmission security. The primary axis shows total generation from the non-compliant unit broken into two categories:

- The actual output from the unit that would have been produced even if it had followed its Output Limit (blue columns), and
- The actual output from the unit that would have been curtailed if the resource had followed its Output Limit (orange columns).

The amount of curtailment megawatts that would have applied to the unit were calculated based on the amount of output curtailed manually by operators from other IPRs and adjusted based on the shift factors from all relevant units on the active constraint. The secondary axis shows the real-time nodal LBMP (black line) at the non-responding IPR along with an estimate of its marginal cost (red dashed line). Whenever these two lines diverge, it indicates that the magnitude of the manual curtailments issued by the NYISO caused the constraint to not bind in the real-time market.

²⁸⁵ The “***” connotation in the table marks information redacted to preserve confidentiality.

Figure A-74: Failure to Follow Curtailment Instructions
Event where IPR Did Not Respond to Output Limits, January 2024



The following summarizes the overgeneration penalties for intermittent generators that do not obey curtailment instructions.

Explanation of Overgeneration Charge Shortcoming

When the real-time LBMP is negative, as is usually the case during an interval with an Output Limit, the NYISO balancing settlement is determined based on the following simplified formula:²⁸⁶

$$(E_{RT} - E_{DA}) * LBMP_{RT} + P$$

Where:

E_{RT} = Real-time Actual Output in MW from the resource

E_{DA} = Day-ahead scheduled Output in MW from the resource

$LBMP_{RT}$ = Real-time LBMP

P = Overgeneration Charge which is 0 if the Actual Output is less than or equal to the Baseline plus 3% of UOL.²⁸⁷

²⁸⁶ See B.2 of the Accounting and Billing [Manual](#).

²⁸⁷ The overgeneration charge is based on the maximum of the day-ahead and real-time regulation capacity price for the impacted intervals.

However, the resource will also receive compensation based on state Renewable Energy Certificates (“RECs”) and federal production tax credits (“PTC”) for the actual output that it produces. This means that the true balancing settlement to the resource, including both the NYISO settlement and the production credits is:

$$(E_{RT} - E_{DA}) * LBMP_{RT} + (E_{RT} * CREDIT) + P$$

Where CREDIT is the sum of the value per MWh of the applicable PTC and RECs to the resource.

When the IPR fails to follow dispatch, its actual output exceeds the economic basepoint (“E_{BP}”) to which the model instructs it, i.e., $E_{RT} > E_{BP}$. The change in settlements to the resource from all sources can be described as:

$$\{(E_{RT} - E_{DA}) * LBMP_{RT} + (E_{RT} * CREDIT) + P\} - \{(E_{BP} - E_{DA}) * LBMP_{RT} + (E_{BP} * CREDIT)\}$$

Which is equal to:

$$(E_{RT} - E_{BP}) * (LBMP_{RT} + CREDIT) + P$$

This equation will yield a positive value if $(CREDIT + LBMP_{RT}) > P$. Therefore, if the resource’s LBMP is set at a price above its short-run marginal cost ($= -1 * CREDIT$) by an amount greater than the value of the overgeneration charge, i.e., the maximum of the day-ahead and real-time regulation capacity charge, it stands to benefit from ignoring an Output Limit instruction. If the LBMP is similar to the short-run marginal cost, the IPR has a weak disincentive to over-generate by more than 3% of its UOL. If the LBMP is much higher than its short-run marginal cost (as occurred in the event summarized in Figure A-74), then the IPR will profit significantly by not complying with curtailment instructions.

C. Performance of Operating Reserve Providers

Wholesale markets should provide efficient incentives for resources to help the ISO maintain reliability by compensating resources consistent with the value they provide. This subsection evaluates: a) the performance of GTs in responding to start-up instructions in the real-time market; and b) how the expected performance of operating reserve providers affects the cost of congestion management in New York City.

Table A-8: Average GT Performance after a Start-Up Instruction

Table A-8 summarizes the performance of offline GTs in responding to start-up instructions that result from NYISO audits and economic commitments (including commitments by RTC, RTD, and RTD-CAM).²⁸⁸ The table’s rows categorize performance into 10-percent increments from 0 to 100 percent. A unit’s performance for a given start is measured based on its output level at its expected full output time (i.e., at 10 or 30 minutes after receiving a start-up instruction), expressed as a percentage of its Upper Operating Limit (“UOL”). For example, if a 40 MW 10-minute GT produces 30 MW at the 10-minute mark after receiving a start-up instruction, its

²⁸⁸ This evaluation does not include OOM start-ups by either NYISO or TO as we do not have reliable data for the instructed starting times.

performance is 75 percent, which falls into the 70-to-80-percent category. The performance category represents a unit’s average performance across all economic starts and NYISO audits in 2025. For each performance category, the table shows:

- Number of Units;
- Total Number of Associated Unit-Starts;
- Average Performance On Time: measured at the unit’s expected full output time; and
- Average Performance 10 Minute Later.

Performance metrics are also broken down for two different operating conditions:

- **RPUs + Unforeseen Economic Starts & Audits:** These include Reserve Pickup (“RPU”) events, random NYISO audits, and economic starts that are *NOT* anticipated in the look-ahead advisory evaluations.
- **Remaining Economic Starts and Audits:** These include re-tests conducted within days after an initial audit failure and economic starts that are anticipated in the look-ahead advisory evaluations.

Table A-8: Average GT Performance After a Start-up Instruction
2025

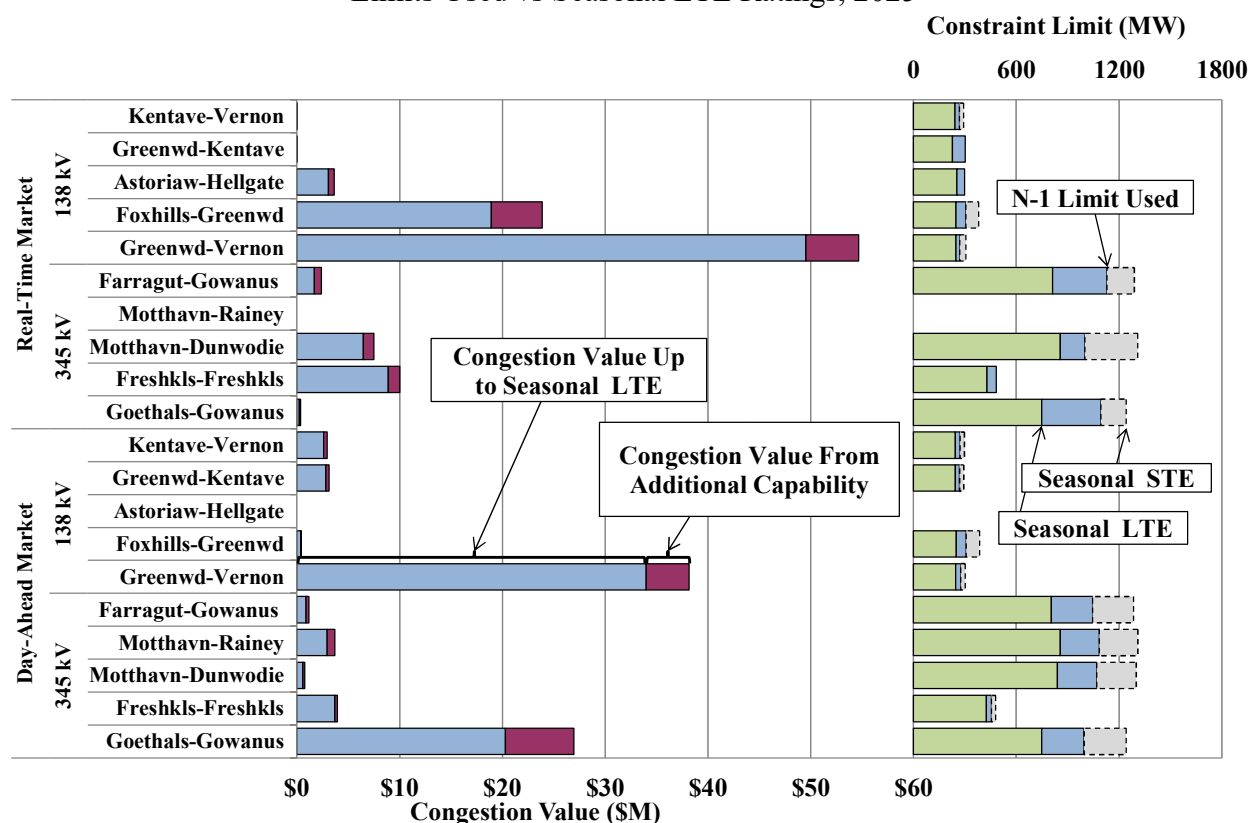
10/30-Minute GT Start Performance - 2025						
Performance Category	No. of Units	Total No. of Starts Evaluated	RPUs + Unforeseen Economic Starts and Audits		Remaining Economic Starts and Audits	
			Performance On Time	Performance 10 Min Later	Performance On Time	Performance 10 Min Later
0% - 10%	1	1			0.0%	0.0%
10% - 20%	0	0				
20% - 30%	0	0				
30% - 40%	1	10	19.0%	23.0%	57.8%	72.3%
40% - 50%	1	15	52.2%	54.9%	33.3%	40.0%
50% - 60%	3	56	58.4%	87.2%	57.2%	71.7%
60% - 70%	3	41	67.7%	78.1%	67.7%	80.8%
70% - 80%	6	273	72.9%	79.9%	78.8%	85.9%
80% - 90%	34	4103	86.8%	93.8%	87.1%	94.3%
90% - 100%	52	5436	93.8%	96.8%	94.3%	96.4%

Figure A-75: Use of Operating Reserves to Manage Congestion in New York City

The NYISO sometimes operates a facility above its Long-Term Emergency (“LTE”) rating if post-contingency actions (e.g., deployment of operating reserves) would be available to quickly reduce flows to LTE. The use of post-contingency actions is important because it allows NYISO to increase flows into load centers and reduce congestion costs. However, the service provided by these actions is not properly compensated.

Figure A-75 shows such select N-1 constraints in New York City. The left panel in the figure summarizes their day-ahead and real-time congestion values in 2025. The blue bars represent the congestion values measured up to the seasonal LTE ratings of the facilities.²⁸⁹ The red bars represent the congestion values measured for the additional transfer capability above LTE.²⁹⁰ The bars in the right panel show the average seasonal LTE and STE ratings for these facilities, compared to the average N-1 constraint limits used in the market software.

Figure A-75: Use of Operating Reserves to Manage N-1 Constraints in New York City
Limits Used vs Seasonal LTE Ratings, 2025



D. Market-to-Market Coordination with PJM

Coordinated congestion management between NYISO and PJM (“M2M”) commenced in January 2013. This process allows each RTO to relieve congestion more efficiently on its constraints with re-dispatch from the other RTO’s resources when it is less costly for them to do so.²⁹¹ M2M includes two types of coordination:

²⁸⁹ Congestion value up to seasonal LTE = constraint shadow cost × seasonal LTE rating summed across all market hours / intervals.

²⁹⁰ Congestion value for additional capability above LTE = constraint shadow cost × (modeled constraint limit - seasonal LTE rating) summed across all market hours / intervals.

²⁹¹ The terms of M2M coordination are set forth in NYISO OATT Section 35.23, which is Attachment CC Schedule D.

- Re-dispatch Coordination – If one of the pre-defined flowgates becomes congested in the monitoring RTO, the non-monitoring RTO will re-dispatch its generation to help manage congestion when economic.
- PAR Coordination – If certain pre-defined flowgates become congested in one or both RTOs, three sets of PAR-controlled lines between New York and New Jersey can be adjusted to reduce overall congestion.²⁹²

Ramapo PARs have been used for the M2M process since its inception, while ABC and JK PARs were incorporated into this process later in May 2017 following the expiration of the ConEd-PSEG Wheel agreement. The NYISO and PJM have an established process for identifying constraints that will be on the list of pre-defined flow gates for Re-dispatch Coordination and PAR Coordination.²⁹³

Figure A-76: NY-NJ PAR Operation under M2M with PJM

The use of Re-dispatch Coordination has been infrequent since the inception of M2M, while the use of PAR Coordination had far more significant impacts on the market. Hence, the following analysis focuses on the operation of NY-NJ PARs in 2025.

Figure A-76 evaluates operations of these NY-NJ PARs under M2M with PJM during periods of noticeable congestion differential between NY and PJM. For each PAR group in the figure, the evaluation is done for the following periods:

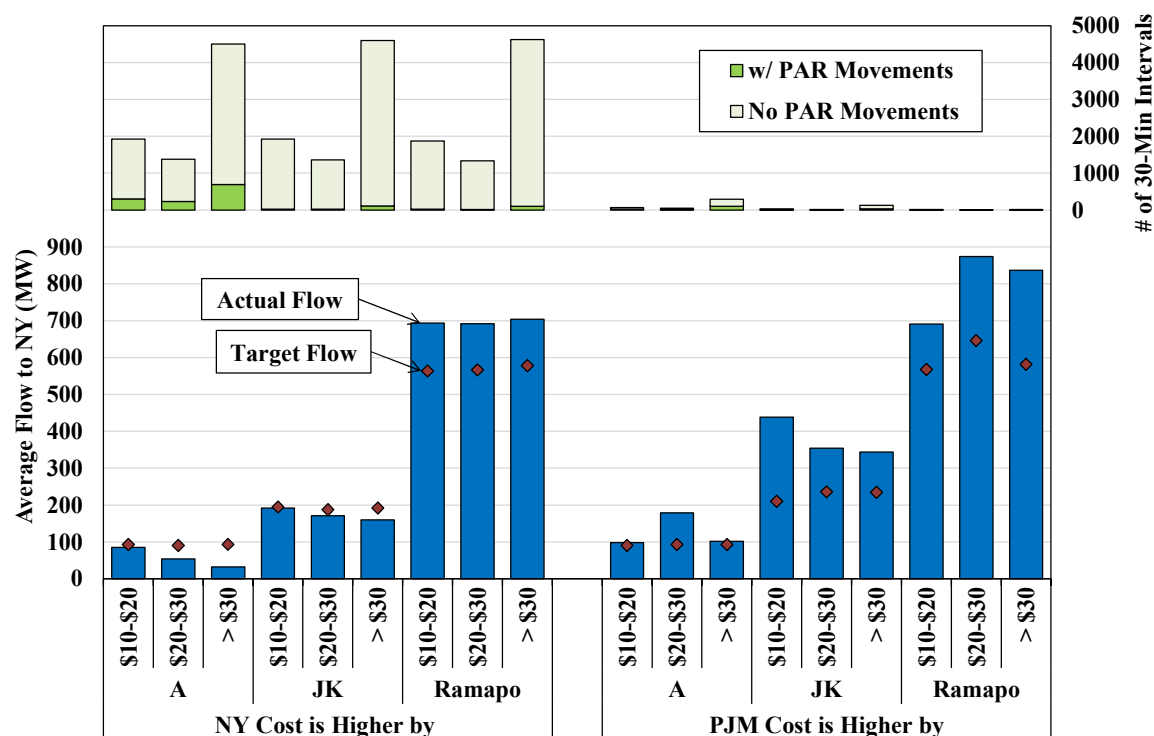
- When NY costs on relevant M2M constraints exceed PJM costs by: a) \$10/MWh to \$20/MWh; b) \$20/MWh to \$30/MWh; or c) more than \$30/MWh.
- When PJM costs on relevant M2M constraints exceed NY costs by: a) \$10/MWh to \$20/MWh; b) \$20/MWh to \$30/MWh; or c) more than \$30/MWh;

The market cost is measured as the constraint shadow price multiplied by the PAR shift factor, summed over relevant M2M constraints in each 5-minute market interval and then averaged over each half-hour period. In the figure, the top portion shows two stacked bars for each evaluation group, representing the total number of 30-minute intervals with and without any PAR tap movements; while the bottom portion shows average actual PAR flows (blue bar), compared with their average M2M targets (red diamond).

²⁹² These include two Ramapo PARs that control the 5018 line, three Waldwick PARs that control the J and K lines, and one PAR that controls the A line.

²⁹³ The list of pre-defined flowgates, *Coordinated Flowgates and Entitlements*, is posted [here](#) in the subgroup “Notices” under “General Information”.

**Figure A-76: NY-NJ PAR Operation under M2M with PJM
2025**



E. Operation of Controllable Lines

The majority of transmission lines that make up the bulk power system are not controllable, and thus, must be secured by redispatching generation in order to maintain flows below applicable limits. However, there are still a significant number of controllable transmission lines that source and/or sink in New York. This includes HVDC transmission lines, PAR-controlled lines, and VFT-controlled lines. Controllable transmission lines allow power flows to be channeled along paths that lower the overall cost of satisfying the system's needs. Hence, they can provide greater benefits than conventional AC transmission lines.

Controllable transmission lines that source and/or sink in NYCA are scheduled in three ways. First, some controllable transmission lines are scheduled as external interfaces using external transaction scheduling procedures.²⁹⁴ Such lines are analyzed in Section V.E of the Appendix, which evaluates external transaction scheduling. Second, "optimized" PAR-controlled lines are optimized in the sense that they are normally adjusted by the local TO to reduce generation redispatch (i.e., to minimize production costs) in the day-ahead and real-time markets. Third, "non-optimized" PAR-controlled lines are scheduled according to various operating procedures that are not primarily focused on reducing production costs in the day-ahead and real-time markets. This sub-section evaluates the use of non-optimized PAR-controlled lines.

²⁹⁴

This includes the Cross Sound Cable (an HVDC line), the Neptune Cable (an HVDC line), the HVDC line connecting NYCA to Quebec, the Dennison Scheduled Line (partly VFT-controlled), the 1385 Scheduled Line (PAR-controlled), and the Linden VFT Scheduled Line.

Table A-9 & Figure A-77: Scheduling of Non-Optimized PAR-Controlled Lines

PARs are commonly used to control line flows on the bulk power system. Through control of tap positions, power flows on a PAR-controlled line can be changed to facilitate power transfer between regions or to manage congestion within and between control areas. This subsection evaluates efficiency of PAR operations during 2025.

Table A-9 evaluates the consistency of the direction of power flows on non-optimized PAR-controlled lines and LBMP differences across these lines during 2025. The evaluation is done for the following eight PAR-controlled lines:

- One between IESO and NYISO: St. Lawrence – Moses PAR (L34 line).
- One between ISO-NE and NYISO: Sand Bar – Plattsburgh PAR (PV20 line).
- Four between PJM and NYISO: Two Waldwick PAR-controlled lines (J & K lines), one Branchburg-Ramapo PAR-controlled line (5018 line), and one Linden-Goethals PAR (A line). These are discussed in sub-section D.
- Two between Long Island and New York City: Lake Success-Jamaica PAR (903 line) and Valley Stream-Jamaica PAR (901 line), which are usually scheduled to support a wheel of up to 300 MW from upstate New York through Long Island to New York City.

For each group of PAR-controlled lines, Table A-9 shows:

- Average hourly net flows into NYCA or New York City;
- Average price at the interconnection point in the NYCA or NYC minus the average price at the interconnection point in the adjacent area (the external control area or Long Island);
- The share of the hours when power was scheduled in the efficient direction (i.e., from the lower-price market to the higher-price market); and
- The estimated production cost savings that result from the flows across each line. The estimated production cost savings in each hour is based on the price difference across the line multiplied by the scheduled power flow across the line.²⁹⁵

This analysis is shown separately for the portion of flows scheduled in the day-ahead market versus the portion that is from balancing adjustments in the real-time market.²⁹⁶ The vast

²⁹⁵ For example, if 100 MW flows from Lake Success to Jamaica in one hour, the price at Lake Success is \$50 per MWh and the price at Jamaica is \$60 per MWh, then the estimated production cost savings is \$1,000 (=100 * \$10) because each MW of flow saves \$10 by allowing a \$60 per MWh resource in New York City to ramp down and a \$50 per MWh resource in Long Island to ramp up. This method tends to under-estimate the actual production cost savings when power flows from the low-priced region to the high-priced region, since if flows in the efficient direction were reduced, the cost of the marginal resource in the importing region would rise while the cost of the marginal resource in the exporting region would fall. However, this method of calculating production cost savings tends to over-estimate the actual production cost increases when power flows from high-priced region towards the low-priced region, since if flows were reduced, the cost differential between the marginal resources in each region would converge.

²⁹⁶ For example, if 100 MW is scheduled from the low-priced region to the high-priced region in the day-ahead market, the day-ahead schedule is considered *efficient direction*, and if the relative prices of the two

majority of power is scheduled in the day-ahead market, while small balancing adjustments are typically made in the real-time market.

Table A-9: Efficiency of Scheduling on Non-Optimized PAR Controlled Lines
2025

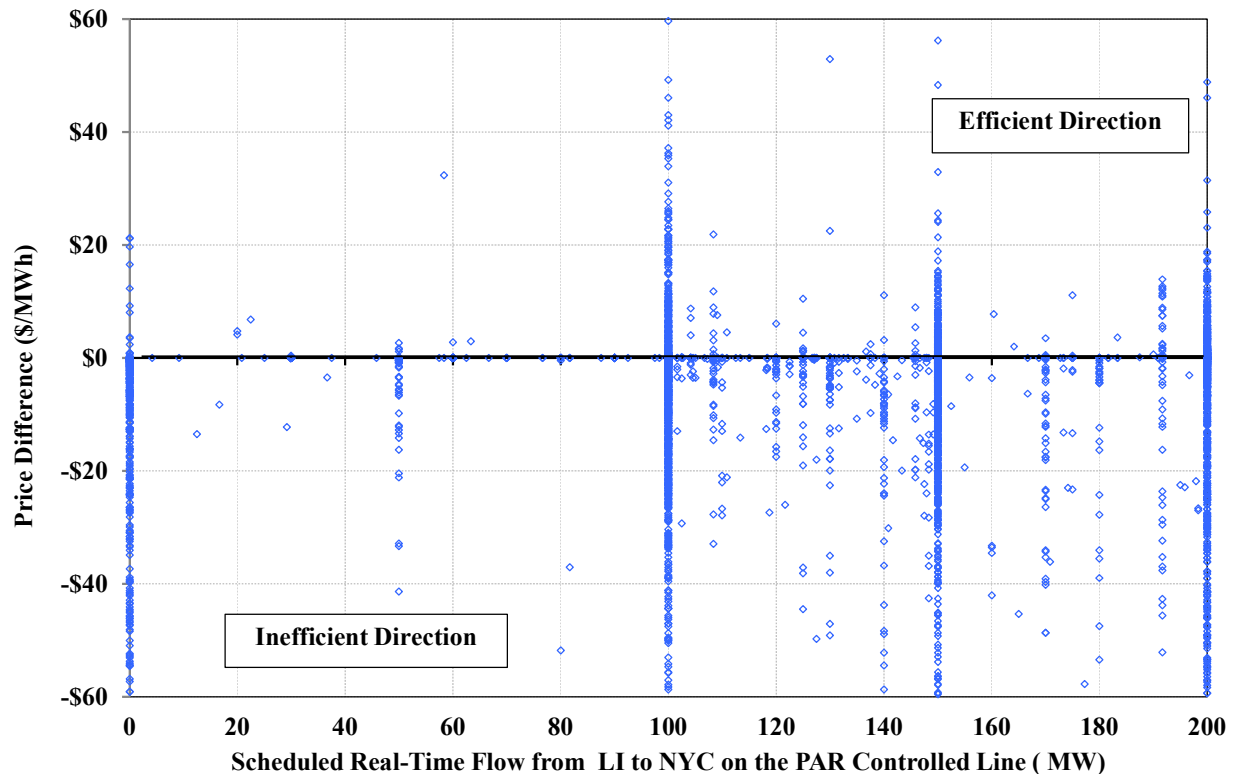
	Day-Ahead Market Schedule				Adjustment in Real-Time			
	Avg Flow (MW)	Avg NYCA Price minus Avg Outside Price (\$/MWh)	Percent of Hours in Efficient Direction	Estimated Production Cost Savings (Million \$)	Avg Flow (MW)	Avg NYCA Price minus Avg Outside Price (\$/MWh)	Percent of Hours in Efficient Direction	Estimated Production Cost Savings (Million \$)
Ontario to NYCA St. Lawrence	-4	\$9.67	51%	\$0.3	30	\$8.83	59%	\$5.4
New England to NYCA Sand Bar	-74	-\$15.18	70%	\$11	-2	-\$9.00	56%	\$0.6
PJM to NYCA Waldwick	164	\$16.02	89%	\$30	16	\$15.13	52%	-\$3.8
Ramapo	502	\$20.53	95%	\$106	122	\$20.42	72%	\$22
Goethals	78	\$14.80	90%	\$13	0.2	\$16.49	51%	-\$8.4
Long Island to NYC Lake Success	102	-\$4.44	22%	-\$4	9	-\$6.10	42%	-\$0.3
Valley Stream	82	-\$8.28	23%	-\$6	5	-\$14.92	34%	-\$1.4

* The estimated production cost savings tend to: 1) under-estimate actual savings when power flows from the low-priced region to the high-priced region, since if flows were reduced, the cost of the marginal resource in the importing region would rise while the cost of the marginal resource in the exporting region would fall; while 2) over-estimate actual cost increases when power flows from the high-priced region to the low-priced region, since if flows were reduced, the cost differential between the marginal resources in each region would converge.

Figure A-77 provides additional detail on the efficiency of scheduling for one of the lines in the table. The figure is a scatter plot of power flows versus price differences across the Lake Success-Jamaica line. The figure shows hourly price differences in the real-time market on the vertical axis versus power flows scheduled in the real-time market on the horizontal axis. Points above the \$0-dollar line in the figure are characterized as scheduled in the efficient direction. Power scheduled in an efficient direction flows from the lower-priced market to the higher-priced market. Similarly, points below the \$0-dollar line are characterized as scheduled in the inefficient direction, corresponding to power flowing from the higher-priced market to the lower-priced market. Good market performance would be indicated by a large share of hours scheduled in an efficient direction.

regions is switched in the real-time market and the flow was reduced to 80 MW, the adjustment is shown as -20 MW and the real-time schedule adjustment is considered *efficient direction* as well.

Figure A-77: Efficiency of Scheduling on PAR Controlled Lines
Lake Success-Jamaica Line – 2025



F. Regulation Movement-to-Capacity Ratio

Regulation sellers submit a two-part offer indicating two separate costs of providing regulation services. One is the capacity offer indicating the cost associated with setting aside capacity for regulation. The other is the movement offer that indicates additional cost associated with moving the resource up and down every six seconds when deployed to provide regulation. Under the current market rules, a composite offer is calculated equal to (*capacity offer*) plus (*movement offer*) times (*movement multiplier*) for each regulation provider that estimates its overall cost of providing regulation and is used in the market software for scheduling and pricing.

Resources are currently scheduled assuming a uniform Regulation Movement Multiplier of 13 per MW of capability,²⁹⁷ but they are deployed based on individual ramping capability and are compensated according to actual movement. This inconsistency between assumed costs and actual costs incurred can lead to inefficiency in the resource scheduling and pricing. Accordingly, this subsection focuses on actual regulation movement versus assumed common multiplier.

²⁹⁷ The uniform Regulation Movement Multiplier was changed from 8, which had been in place since September 2021, to 13 in May 2025.

Figure A-78 & Figure A-79: Regulation Movement-to-Capacity Ratio

Figure A-78 shows a distribution of average actual movement-to-capacity ratio of all scheduled regulation suppliers in 2025. The blue bars show the average scheduled regulation capacity in each movement-to-capacity ratio. The solid blue line represents the capacity weighted average actual movement-to-capacity ratio, compared to the multiplier of 13 that is currently used for all resources when formulating the composite regulation offer.

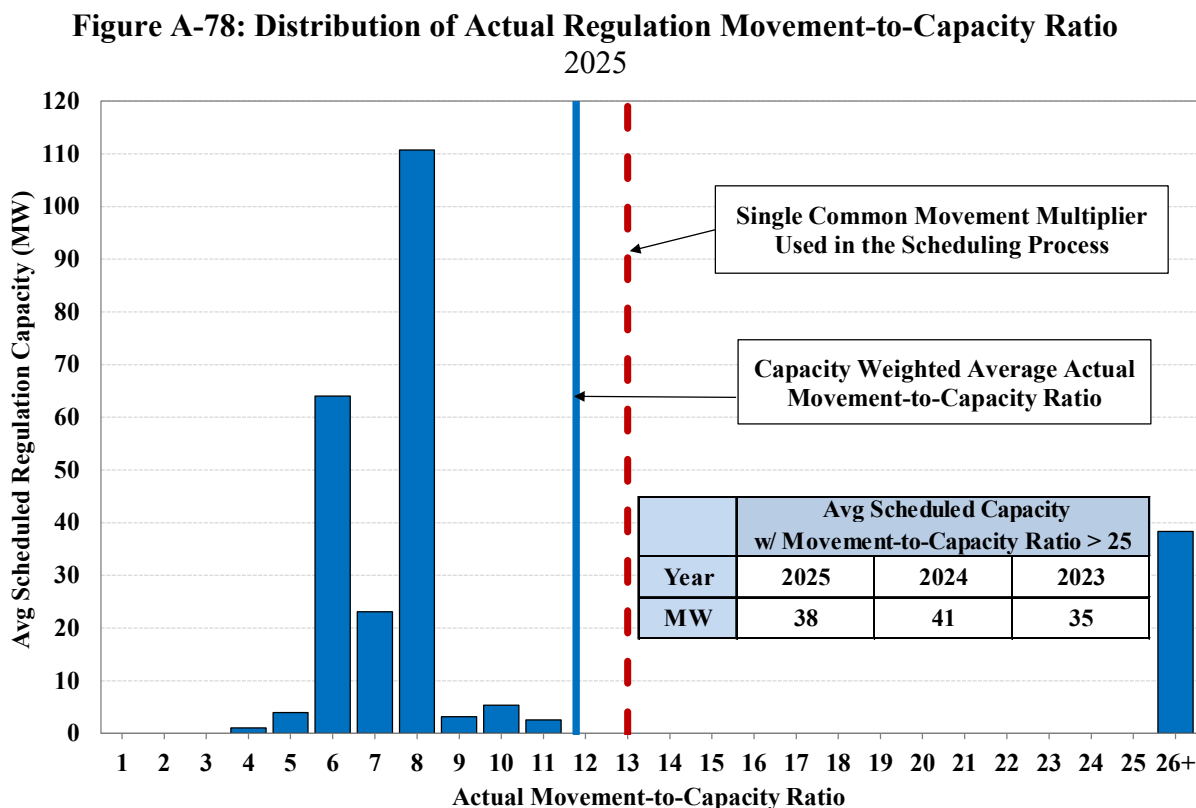
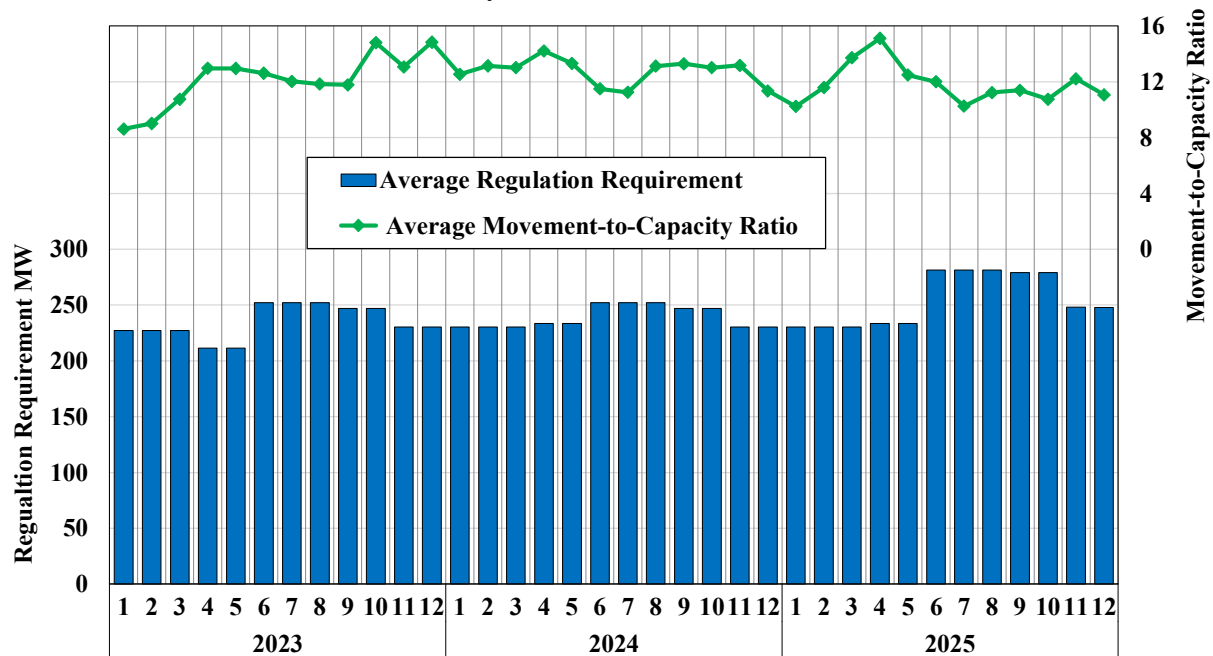


Figure A-79 tracks the variation of regulation movement-to-capacity ratio in recent years, summarizing the following quantities by month:

- Average regulation requirement – The regulation requirement varies by hour by season. This is the hourly average regulation requirement for each month.
- Average actual regulation movement-to-capacity ratio – This is calculated as total regulation movement MW from all resources divided by total scheduled regulation capacity in each month.

Figure A-79: Regulation Requirement and Movement-to-Capacity Ratio
By Month, 2023-2025



G. Market Operations under Shortage Conditions

Prices that occur under shortage conditions (i.e., when resources are insufficient to meet the system’s energy, operating reserves, and regulation requirements while satisfying transmission security constraints) are an important component to efficient price signals.

In the long run, prices should signal to market participants where and when new investment in generation, transmission, and demand response would be most valuable. In the short run, prices should provide incentives for market participants to commit sufficient resources in the day-ahead market to satisfy anticipated system conditions and should incentivize suppliers and demand response resources perform reliably, particularly during real-time shortages. However, shortage pricing should occur only during legitimate shortage conditions and not as a result of anticompetitive behavior or inefficient market operations.

The importance of setting efficient real-time price signals during shortages is well-recognized. NYISO’s market design includes three provisions that facilitate shortage pricing. First, NYISO uses operating reserves and regulation demand curves to set real-time clearing prices during operating reserves and regulation shortages. Second, NYISO uses a transmission demand curve to set real-time clearing prices during transmission shortages. Third, NYISO allows demand response resources to set clearing prices when an operating reserve shortage is avoided by the deployment of demand response.

We evaluate market operations and resulting prices under the first type of shortage conditions in this subsection. Transmission shortages are evaluated in Subsection H, and demand response deployments are evaluated in Section VIII.

Figure A-80: Real-Time Prices During Physical Ancillary Services Shortages

NYISO’s approach to efficient pricing during operating reserves and regulation shortages is to use ancillary services demand curves. The real-time dispatch model (“RTD”) co-optimizes energy and ancillary services procurement, efficiently allocating resources every five minutes to meet both energy demand and ancillary service requirements.

When RTD cannot simultaneously satisfy energy demand and ancillary services requirements with available resources, the ancillary services demand curves rationalize pricing of energy and ancillary services during shortage periods by causing prices to reflect the value of foregone ancillary services. The demand curves also set limits on the costs that can be incurred to maintain operating reserves and regulation.

Figure A-80 summarizes physical ancillary services shortages and their effects on real-time prices in 2024 and 2025 for the following categories:²⁹⁸

- 30-minute NYCA – The ISO is required to hold 2,620 MW of 30-minute reserves in the state and has a demand curve value of \$40/MW if the shortage is up to 200 MW, \$100/MW if the shortage is between 200 and 325 MW, \$175/MW if the shortage is between 325 and 380 MW, \$225/MW if the shortage is between 380 and 435 MW, \$300/MW if the shortage is between 435 and 490 MW, \$375/MW if the shortage is between 490 and 545 MW, \$500/MW if the shortage is between 545 and 600 MW, \$625/MW if the shortage is between 600 and 655 MW, and \$750/MWh if the shortage is more than 655 MW.
- 10-minute NYCA – The ISO is required to hold 1,310 MW of 10-minute operating reserves in the state and has a demand curve value of \$750/MW.
- 10-Spin NYCA – The ISO is required to hold 655 MW of 10-minute spinning reserves in the state and has a demand curve value of \$775/MW.
- 10-minute East – The ISO is required to hold 1200 MW of 10-minute operating reserves in Eastern New York and has a demand curve value of \$775/MW.
- 30-minute SENY – The ISO is required to hold at least 1300 MW of 30-minute operating reserves in Southeast New York for all hours and has a demand curve value of \$500/MW. Additional 30-minute operating reserves are required for a subset of hours and has a demand curve value of \$40/MW in the incremental range.
- Regulation – The ISO is required to hold 150 to 300 MW of regulation capability in the state and has a demand curve value of \$25/MW if the shortage is less than 25 MW, \$525/MW if the shortage is between 25 and 80 MW, and \$775/MW if the shortage is more than 80 MW.

The top portion of the figure shows the frequency of physical shortages. The bottom portion shows the average shadow price during physical shortage intervals and the applicable demand curve level for each requirement. The inset table shows the average shadow prices during physical shortages multiplied by the frequency of those shortages, indicating the overall price

²⁹⁸ See *NYISO Ancillary Services Manual* for more details.

impact by product and the total impact by region. The table also shows the cumulative effect of all ancillary services shortages on average real-time energy clearing prices in:

- Western New York – This is based on the sum of shadow prices of the NYCA reserve requirements as well as the effects of positive and negative regulation spikes; and
- Eastern New York (outside New York City) – This equals the Western New York effect plus the sum of shadow prices of eastern reserve requirements.

Figure A-80: Real-Time Prices During Ancillary Services Shortages
2024 – 2025

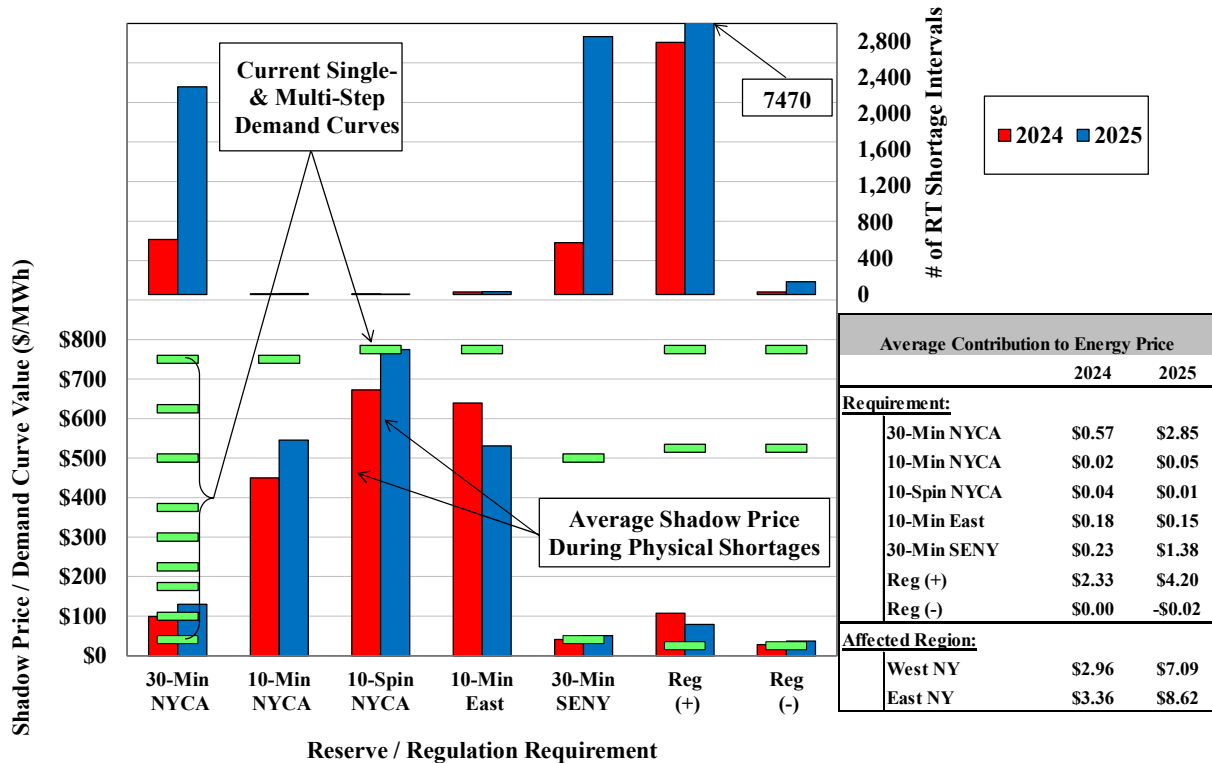


Figure A-81 & Table A-10: Reserves Shortages in New York City

NYISO currently models two reserves requirements in NYC:

- 10-minute Reserves Requirement – The ISO is required to hold 500 MW of 10-minute operating reserves in New York City and has a demand curve value of \$25/MWh; and
- 30-minute Reserves Requirement – The ISO is required to hold 1,000 MW of 30-minute operating reserves in New York City and has a demand curve value of \$25/MWh.

Table A-10 shows the real-time market performance during reserves shortages in New York City for each month in 2025. The table shows the following quantities:

- # Intervals – This is the total number of real-time intervals in each month when either 10-minute reserves or 30-minute reserves or both were short in New York City.

- Average Shortage MW – This is the average quantity of reserve shortages over all shortage intervals in each month. In each interval, the shortage quantity is equal to the higher amount of 10-minute and 30-minute shortages.
- # Intervals with ‘toNYC’ Congestion – This is the total number of real-time shortage intervals that coincided with congestion on transmission paths into New York City.

Table A-10: Real-Time Reserve Shortages in New York City
2025

	RT Reserve Shortages in NYC in 2025		
Month	# Intervals	Avg. Shortage MW	#Intervals w/ toNYC Congestion
Jan	2689	97	0
Feb	1000	90	0
Mar	14	39	0
Apr	4	42	0
May	13	20	0
Jun	668	221	0
Jul	924	92	0
Aug	373	30	0
Sep	49	11	1
Oct	756	63	90
Nov	118	28	0
Dec	1263	124	0
Total	7871	102	91

Figure A-81 illustrates a sample real-time shortage event on July 15, 2025 when New York City was short of reserves (either 10-minute or 30-minute or both) primarily in the afternoon hours. For each interval from the beginning of hour 8 to the end of hour 21, the figure shows:

- The amount of reserve shortages (red bar); and
- Net imports from upstate areas (blue bar).²⁹⁹

When net imports to New York City drop significantly because New York City generators increase output, it creates a reserve import capability that can be used during a contingency. Therefore, when reserve import capability is available into the city, less reserve capacity needs to be held on generators in New York City to maintain reliability.

²⁹⁹ This is calculated as (NYC load) minus (NYC gen) minus (HTP imports) minus (VFT imports) minus (flows on the 901/903 lines into NYC) minus (flows on the A line into NYC).

Figure A-81: Real-Time Reserve Shortages in New York City
Sample Event on July 15, 2025

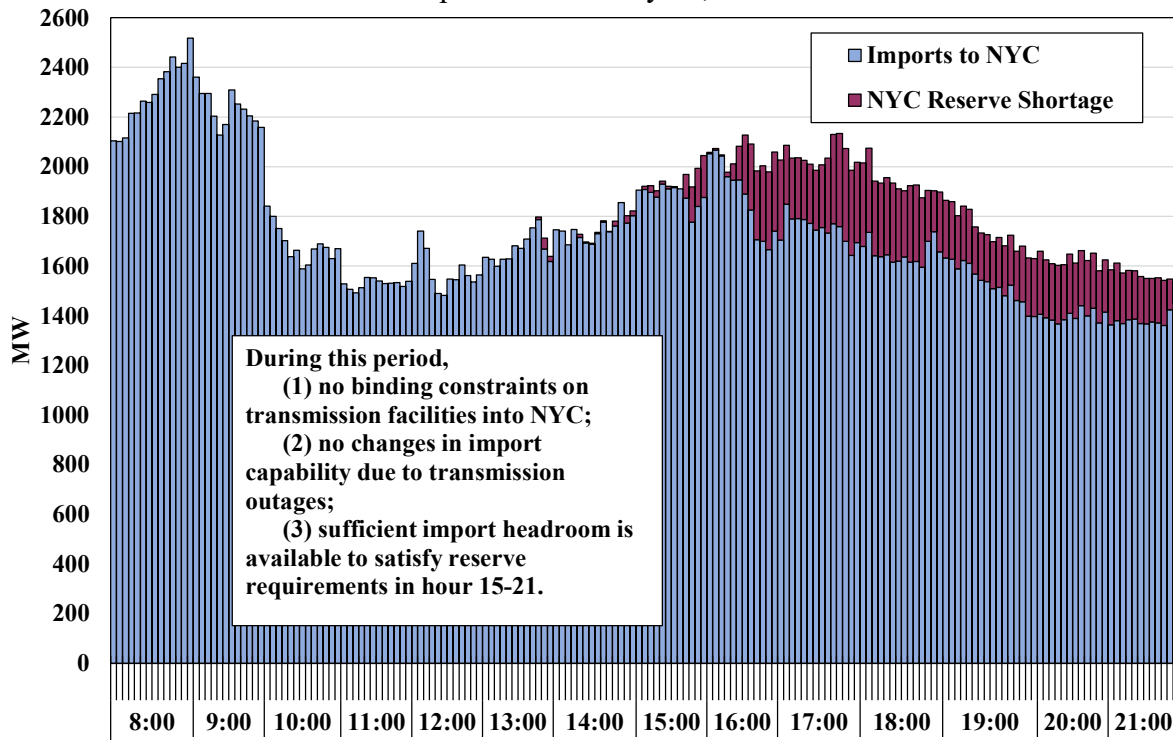


Figure A-82: Comparison of Shortage Pricing in NYISO and Neighboring Markets

In recent years, shortage pricing values in the neighboring PJM and ISO-NE regions have increased dramatically relative to NYISO. While ISO-NE has reserve shortage pricing incentives of \$1,000/MWh during a 30-minute reserve shortage, ISO-NE implemented Pay-for-Performance in its capacity market in 2018, which currently provides an additional real-time performance incentive of \$9,337/MWh. PJM Capacity Performance rules provide real-time performance incentives of approximately \$2,200/MWh,³⁰⁰ in addition to reserve shortage prices that reach \$1,275/MWh during a 10-minute reserve shortage.

These stronger incentives should encourage generators to invest in making their units more reliable and available during tight operating conditions. However, when there is an imbalance between the market incentives provided in two adjacent regions, it can lead market participants to schedule interchange from the area with weaker incentives to the area with stronger incentives even when the area with weaker incentives is in a less-reliable state. In some cases, this could lead the operators of the control area with weaker incentives to maintain reliability through out-of-market actions (e.g., purchases of emergency energy). This may be necessary to maintain reliability in the short term, but it tends to undermine incentives for investment in the long term.

³⁰⁰

This is calculated as the PJM RTO Net Cone in ICAP terms (\$/MW-yr) divided by 30 PAH (Performance Assessment Hours) for the 2025/2026 delivery year.

Figure A-82 compares incentives for NYISO resources during real-time shortage events to those in neighboring markets. These include maximum 30-minute and 10-minute Non-Spin operating reserve demand curve values as well as Pay-for-Performance penalty rates. A resource may face a total incentive that is the sum of each of these sources when multiple reserve product shortages and/or pay-for-performance scarcity conditions are in effect simultaneously. Values shown for NYISO reflect the revised operating reserve demand curves approved by FERC in 2021, which increased some shadow prices. NYISO ‘locational’ prices are shown for the regions at the border of each neighboring ISO to indicate the comparative incentives faced by NYISO suppliers when shortage pricing in the neighboring area is in effect.³⁰¹

Figure A-82: Shortage Pricing in NYISO vs. Neighboring Markets

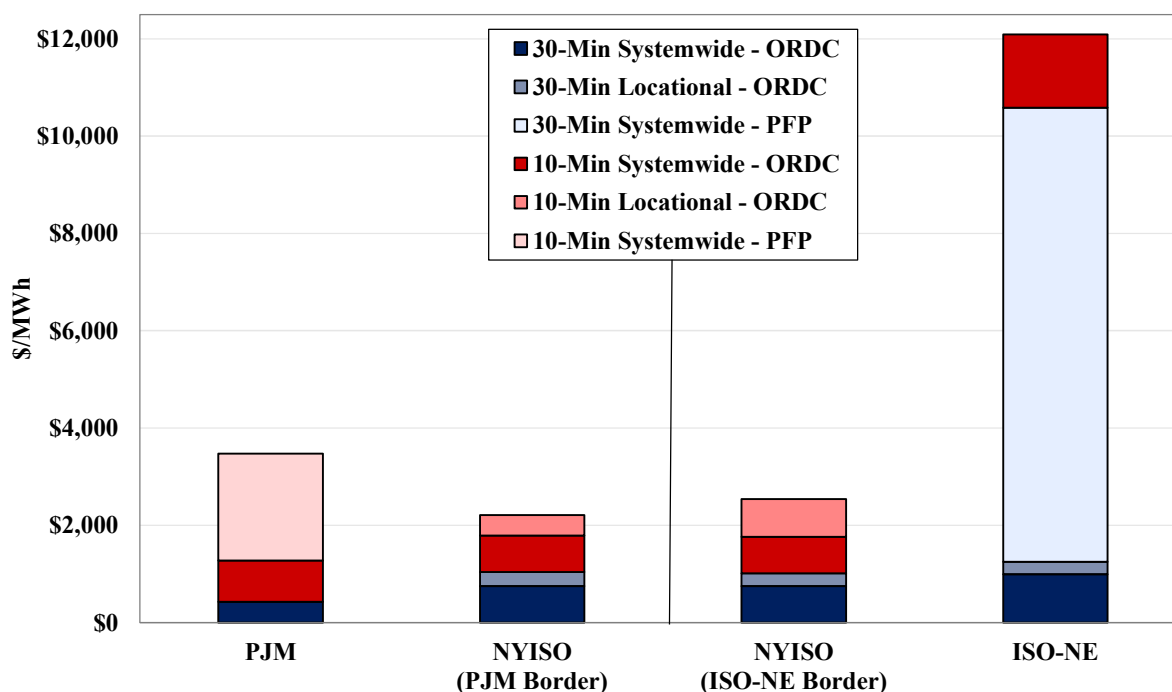


Figure A-83 - Figure A-85: NYISO Operating Reserves Demand Curve vs. MMU EVOLL Curve

The Value of Lost Load (“VOLL”) is a well-recognized metric that quantifies the economic impact on consumers during electricity service interruptions. Essentially, VOLL captures the economic value of reliable service and is commonly determined by assessing outage costs. Outage costs are most accurately estimated through survey-based studies, as they leverage real customer experiences to generate more accurate data on outage costs. Survey methodologies underpin the major benchmark studies of outage costs within US jurisdictions including key

³⁰¹ Locational prices for ISO-NE refer to Connecticut. Locational prices for NYISO (PJM Border) assign 54 percent weight to East 30-minute, SENY 30-minute, and East 10-minute shadow prices. Locational prices for NYISO (ISO-NE Border) include the full value of East 30-minute and East 10-minute shadow prices and a 45 percent weight to the SENY 30-minute shadow price. PJM ORDC prices reflect a \$1,275/MWh price cap for 10-minute non-spinning reserve shortages, divided between \$850 (10 minute) and \$425 (30 minute) prices. A shortage of only 30-minute or 10-minute reserves would result in a price of \$850/MWh.

meta studies that have established versatile outage cost estimators. The most widely referenced meta studies were conducted by Sullivan, et al. from the Berkeley National Laboratory. The initial study was conducted in 2009 (“2009 Berkeley Study”) and was subsequently updated in 2015 (“2015 Berkeley Study”). These studies utilize an econometric model to evaluate the impact of various parameters on outage costs across different customer categories. The coefficients derived from this model can then be utilized to estimate outage costs tailored to specific regions, timeframes, and customer segments. Drawing from these research findings, a VOLL estimate of \$30,000 per MWh is considered appropriate for evaluating outage costs within the NYISO market.

The Operating Reserve Demand Curve (“ORDC”) represents the marginal reliability value of maintaining certain amount of reserves to avoid shedding load. The marginal reliability value of reserves at any reserve shortage level can be estimated as the Expected Value of Lost Load (“EVOLL”) = VOLL × conditional probability of losing load at that shortage level. The slope of the ORDC is influenced by the rate at which the probability of losing load increases as operating reserves decreases, which is estimated from the likelihoods of random contingencies and conditions that could arise during a shortage in the NYISO market.

To account for these unpredictable factors, we employed a Monte Carlo simulation to estimate the conditional probability of losing load for any given level of reserves. This simulation considered random forced generation outages, wind forecast errors, load forecast errors, and import curtailments by neighboring areas.

Generation Forced Outages and Deratings

We utilize a stochastic Markov Process to model random forced outages and deratings for generation resources, including conventional thermal generators and large hydro generators. This modeling approach excludes small run-of-river hydro units and intermittent renewable resources. For each resource, a stochastic Markov Process is developed, where a state space is defined to represent different levels of deratings and a transition matrix is established to capture the transition rates between these capability states. The Markov Process has the following property:

Let T_{ij} be the time the Markov Process spends in state i before entering into a different state j . The time T_{ij} is exponentially distributed with transition rate a_{ij} , and the transition probability from state i to state j over a time interval Δt is:

$$P_{ij}(\Delta t) = \Pr(T_{ij} \leq \Delta t) = 1 - e^{-a_{ij}\Delta t}$$

During the Monte Carlo simulation, this probability is compared to a random number between zero and one to simulate forced outages and deratings for each resource. For this analysis, we utilize the transition rate matrices developed for the annual IRM/LCR study conducted for the NYISO capacity market. Additionally, we model all existing resources as being online but their available capability is adjusted using the following formula to reflect average participation during summer peak conditions:

$$\text{Modeled Capacity} = \text{ICAP} * \text{Participation Factor}$$

For each resource, its Participation Factor is calculated as the ratio of the actual total online capacity to the total ICAP during the afternoon peak hours (from HB15 to HB 20) in July and August. It is important to note that this metric assumes resources are fully contributing to meet energy, ancillary services, headroom, and ramp capability needs. This approach differs from a traditional capacity factor, which measures the energy output as a ratio of generation capability.

Wind Forecast Errors

Intermittent resources are represented in our simulation as forecast uncertainties. For the purpose of this analysis, we only consider land-based wind resources, given the limited capacity currently available from in-service solar, offshore wind, and battery storage resources within the NYISO market. However, as the penetration of these resources grows in the coming years, our methodology can be expanded to include them.

To quantify forecast errors, we computed aggregate forecast discrepancies from select historical periods across various forecast windows (e.g., 15 minutes, 30 minutes, or 60 minutes, etc.). The errors equal the difference between actual wind outputs in time t and the forecasted outputs at different time intervals preceding t (e.g., 15 minutes prior to t). We then modeled these actual error distributions using standardized normal distributions, with mean and standard deviations derived from the observed data.

During the Monte Carlo simulation, a distinct random number between zero and one is generated for each iteration, which serves as the probability distribution for wind forecast errors. The simulated wind forecast error is determined by the corresponding inverse of the normal cumulative distribution. We model both over-forecasts and under-forecasts in our analysis.

Net Load Forecast Errors

Net load (= load – BTM solar) forecast uncertainties are considered in our simulation. Similar to simulating wind forecast uncertainties, we represent net load forecast errors with standardized normal distribution curves, with mean and standard deviations calculated from select historical periods across different forecast windows. The Monte Carlo simulation utilizes random numbers between zero and one and their corresponding inverse of these normal cumulative distributions to model net load forecast uncertainties. Both over-forecasts and under-forecasts of net load are simulated in this analysis.

Import Curtailments from Neighboring Areas

Neighboring control areas often curtail their exports to New York after being scheduled by RTC due to various reasons, such as unforeseen reliability issues, bid mismatches, checkout failures, or transmission delivery bottlenecks. These close-to-real-time curtailments introduce unexpected supply losses to the NYISO market, which our simulation accounts for.

We calculated the aggregate import curtailments across all interfaces between New York and neighboring control areas using data from select historical periods. Our simulation incorporates the observed frequency of curtailments, while the magnitude of these curtailments is estimated using a standardized exponential distribution. The mean of this distribution is derived from the observed data. In the Monte Carlo simulation, random numbers between zero and one are

generated for each iteration. These numbers are then used with the inverse of the exponential cumulative distribution to model the quantity of import curtailments.

These four random factors described above are then summed together to calculate the net supply loss in each iteration of the Monte Carlo simulation:

$$\text{Net Supply Loss} = \text{Forced-Out Generation Capacity} + \text{Wind Over-forecast} + \text{Net Load Under-forecast} + \text{Import Curtailment}$$

The conditional probability of lost load at any point (x MW) on the ORDC curve is then calculated as:

$$\Pr\{\text{Load Shed} | x \text{ MW of reserves available}\} = \frac{\text{The number of iterations yielding net supply loss} > x \text{ MW}}{\text{The total number of iterations}}$$

The EVOLL at x MW on the ORDC curve equals:

$$\text{EVOLL} = \text{VOLL} * \Pr\{\text{Load Shed} | x \text{ MW of reserves available}\}$$

Figure A-83 shows our estimated EVOLL curves for NYCA-wide operating reserves for an outage recovery period of 15 minutes, 30 minutes, one hour, and two hours, respectively. These EVOLL curves are compared to existing ORDCs in the NYISO market.

We compare the EVOLL curves with the stacked ORDCs because they represent the cumulative value of reserves available for deployment within each respective outage recovery period. For example, only 10-minute reserves can be deployed within the 15-minute outage recovery period, while both 10-minute and 30-minute reserves can be deployed within the 30-minute outage recovery period. Consequently, the 15-minute EVOLL curve indicates the economic value of 10-minute reserves, compared to the combined 10-Spin ORDC and 10-Minute ORDC. This comparison is shown separately in Figure A-84. The analysis reveals that approximately half of the current ORDC curve significantly undervalues the marginal reliability of 10-minute reserves. Moreover, the MMU EVOLL curve extends beyond the existing 1310 MW of 10-minute reserve requirements. This extended portion could serve as the pricing basis for additional reserves in the form of 10-minute reserves that are procured to address uncertainties associated with intermittent resource availability.³⁰²

Likewise, the 30-minute EVOLL curve represents the economic value of both 10-minute and 30-minute reserves, compared to the combined 10-Spin ORDC, 10-Minute ORDC, 30-Minute ORDC. We derive an EVOLL-based 30-minute ORDC as the difference between the 30-minute EVOLL and the 15-minute EVOLL. Figure A-85 shows this MMU economic 30-minute ORDC, compared to the current 30-minute ORDC.

³⁰² See *Balancing Intermittency: Percentiles and Shortage Pricing Curves*, ICAPWG/MIWG, March 4, 2024.

Figure A-83: NYISO ORDCs vs. MMU EVOLL Curves for up to 2 Hour Reserves

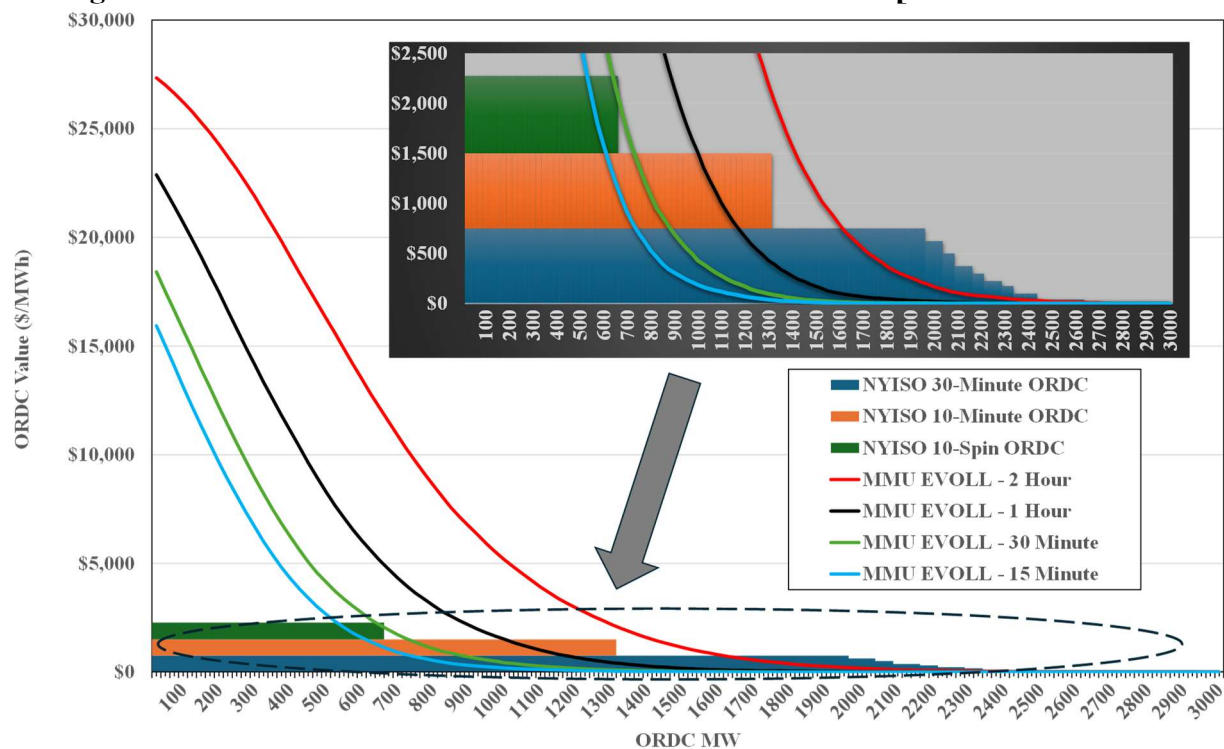


Figure A-84: NYISO ORDC vs. MMU EVOLL Curve for 10-Minute Reserves

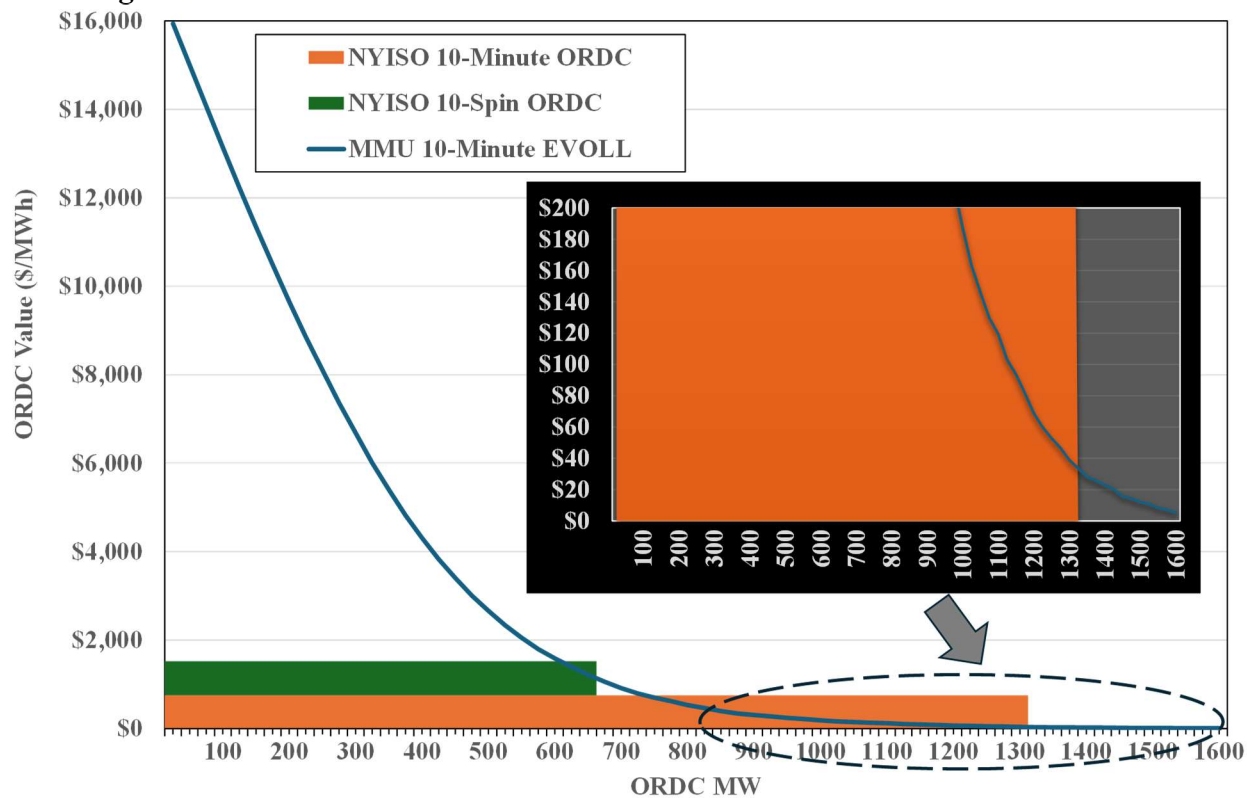
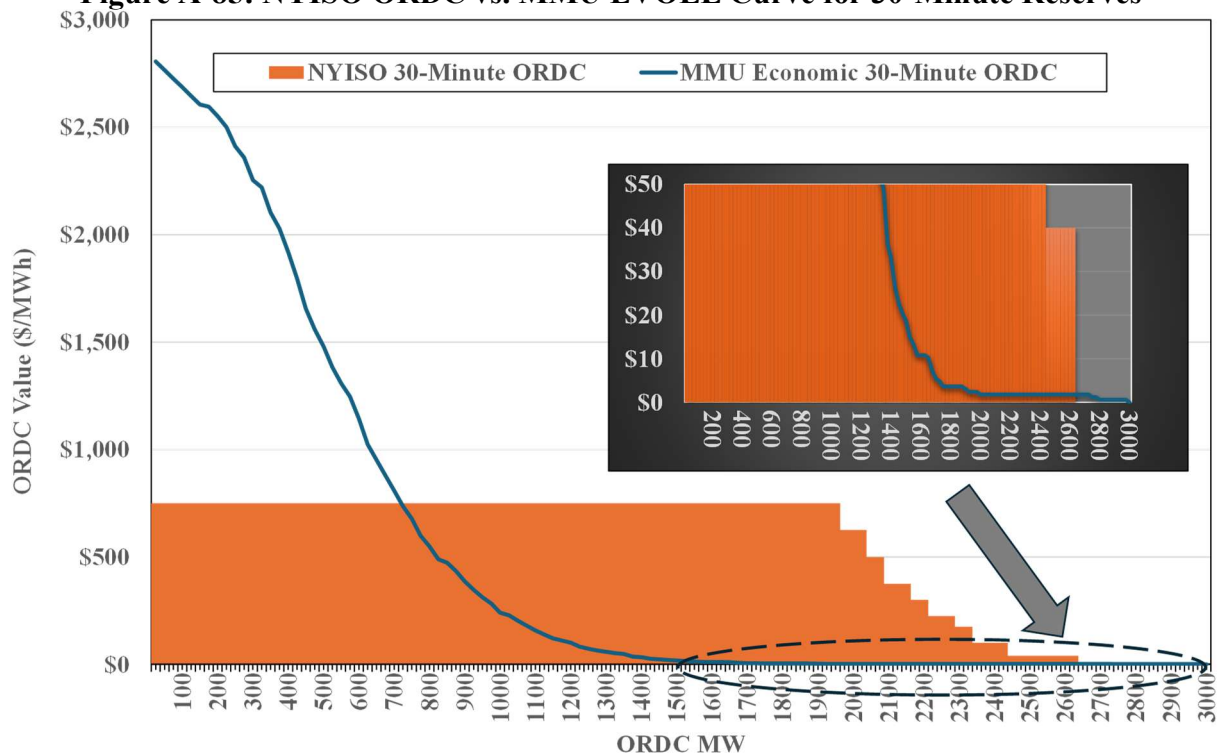


Figure A-85: NYISO ORDC vs. MMU EVOLL Curve for 30-Minute Reserves

H. Offline GT Pricing and Transmission Shortages

Transmission shortages occur when power flows exceed the limit of a transmission constraint. While such transmission shortages may require the ISO to shed firm load to maintain system security, they often persist for many hours without necessitating load shedding or causing equipment damage. During transmission shortages, it is important for wholesale markets to set efficient prices that accurately reflect the acuteness of operating conditions. Efficient pricing provides incentives for generation and demand response resources to respond in ways that support reliability.

The real-time dispatch model (“RTD”) manages transmission constraints by redispatching available capacity, including online units that ramp in five minutes and offline quick-start gas turbines that can start and synchronize within 10 minutes. If the available physical capacity is insufficient to resolve a transmission constraint, NYISO uses a Graduated Transmission Demand Curve (“GTDC”) to set prices under shortage conditions. NYISO first implemented the GTDC

approach on February 12, 2016,³⁰³ and made two subsequent enhancements to improve market efficiency during transmission shortages.^{304,305}

Additionally, a condition similar to a shortage can occur when an offline quick-start gas turbine is counted towards resolving a transmission constraint but is not given a startup instruction.³⁰⁶ In such cases, the marginal cost of the resources actually dispatched to relieve the constraint is lower than the shadow price set by the offline gas turbine (which is not actually started). The Commission has recognized that it is not efficient for such units to set the clearing price because they: (a) do not reflect the marginal cost of supply that is available to relieve the constraint in that time interval, and (b) do not reflect the marginal value of the constraint that may be violated when it does not generate as assumed in RTD.³⁰⁷ This category of shortage is evaluated in this section.

Figure A-86: RT Congestion Management with GTDCs

Offline GTs have been used far more frequently to manage congestion on Long Island’s transmission facilities compared to other regions in recent years. Accordingly, Figure A-86 focuses on analyzing the price effects of offline GTs on transmission constraints specific to Long Island in 2025, grouped based on CRMs used: (a) the 345 kV transmission circuits from upstate to Long Island with a CRM of 50 MW; (b) the 138 kV transmission constraints on Long Island with a CRM of 20 MW; and (c) the 69 kV transmission constraints on Long Island with a CRM of 10 MW.

The scatter plots show transmission constraint shadow prices on the y-axis and transmission violations on the x-axis. For a given constraint shadow price, the blue diamond represents the transmission violation as recognized by RTD, while the red diamond represents the violation after excluding the relief provided by offline GTs.

³⁰³ See Section V.F in the Appendix of our *2016 State of Market Report* for a detailed description of the initial implementation of the GTDC.

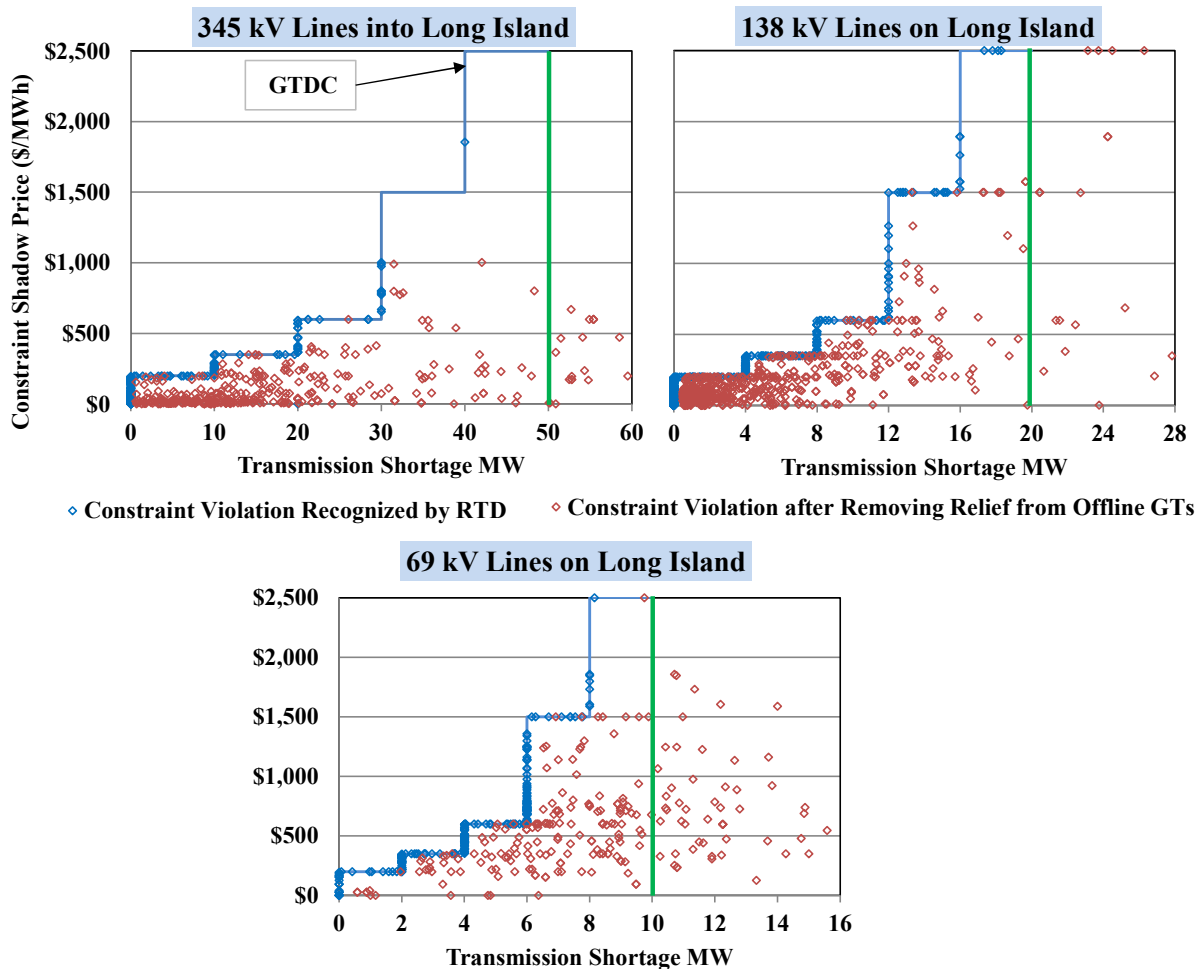
³⁰⁴ The first enhancement was made on June 20, 2017. Key changes include: 1) modifying the second step of the GTDC from \$2350 to \$1175/MWh; and 2) removing the “feasibility screen” and applying the GTDC to all constraints with a non-zero Constraint Reliability Margin (“CRM”). A CRM is a reduction in actual physical limit used in the market software, largely to account for loop flows and other un-modeled factors. These changes are discussed in detail in Commission Docket ER17-1453-000.

³⁰⁵ The second enhancement was made on November 14, 2023. Key changes include: 1) replacing the three-step GTDC curve with a six-step curve with distinct shortage values at \$200, \$350, \$600, \$1500, \$2500, and \$4000, respectively; and 2) replacing the static 20 MW GTDC curve with a CRM-dependent curve for each transmission facility. Each of the first five steps of the GTDC curve equals to 20 percent of the assigned CRM value. These changes are discussed in detail in Commission Docket ER23-1863-000.

³⁰⁶ Offline quick-start gas turbine is usually the most expensive available capacity due to their commitment costs, so offline gas turbines are usually not counted towards resolving the constraint unless all available online generation has already been scheduled. If a gas turbine is scheduled by RTD but does not satisfy the start-up requirement (i.e., economic for at least three intervals and scheduled at the full output level for all five intervals), it will not be instructed to start-up after RTD completes execution.

³⁰⁷ In Docket RM17-3-000, see the Commission’s NOPR on Fast Start Pricing, dated December 15, 2016, and comments of Potomac Economics, dated March 1, 2017.

Figure A-86: Transmission Constraint Shadow Prices and Violations
 With and Without Relief from Offline GTs, 2025



I. Supplemental Commitment for Reliability

When the wholesale market does not meet all forecasted load and reliability requirements, the NYISO (or an individual Transmission Owner) commits additional resources to ensure that sufficient resources will be available in real-time. Supplemental commitments increase the amount of supply available in real-time, leading to distorted real-time market prices, which tend to undermine market incentives for meeting reliability requirements and generate expenses that are uplifted to the market. Hence, it is important for supplemental commitments to be as limited as possible.

In this subsection, we examine supplemental commitment for reliability and focus particularly on New York City where most reliability commitments occur. In the next subsection, we summarize uplift charges that result from guarantee payments received by generators, which are primarily caused by supplemental commitments for local reliability.

Figure A-87: Supplemental Commitment for Reliability in New York

Supplemental commitment occurs when a generator is not committed by the economic pass of the day-ahead market but is needed for reliability. Supplemental commitment primarily occurs in the following three ways:

- Day-Ahead Reliability Units (“DARU”) Commitment, which typically occurs at the request of local Transmission Owner prior to the economic commitment in SCUC;
- Day-Ahead Local Reliability (“LRR”) Commitment, which takes place during the economic commitment pass in SCUC to secure reliability in New York City; and
- Supplemental Resource Evaluation (“SRE”) Commitment, which occurs after the day-ahead market closes.

Generators committed for reliability are generally not economic at prevailing market prices, but they affect the market by: (a) reducing prices below levels that would otherwise result from a purely economic dispatch; and (b) increasing non-local reliability uplift, since lower LBMPs can result in additional guarantee payments to economically committed generators whose revenues are insufficient to cover their as-bid costs. Hence, it is important to commit these units as efficiently as possible.

To the extent LRR constraints in SCUC accurately reflect the reliability requirements in New York City, the local Transmission Owner should not need to rely on DARU or SRE commitments. LRR commitments are generally more efficient than DARU and SRE commitments because they are selected within the SCUC optimization rather than outside the economic evaluation. However, to commit units efficiently, SCUC must have accurate assumptions regarding the needs in each local reliability area.

Figure A-87 shows quarterly quantities of total capacity (stacked bars) and minimum generation (markers) committed for reliability by commitment type and region in 2024 and 2025. Four types of commitments are shown in the figure: DARU, LRR, SRE, and Forecast Pass. The first three categories are primarily for local reliability needs. The Forecast Pass represents additional commitments in the forecast pass of SCUC after the economic pass, which ensures that sufficient physical resources are committed in the day-ahead market to meet forecasted load.

The figure shows these supplemental commitments separately for four regions: (a) West Upstate (Zones A-E); (b) East Upstate (Zones F-I); (c) New York City (Zone J); and (d) Long Island (Zone K). The inset table summarizes annual totals for each region for both years.

Figure A-87: Supplemental Commitment for Reliability in New York
By Category and Region, 2024 – 2025

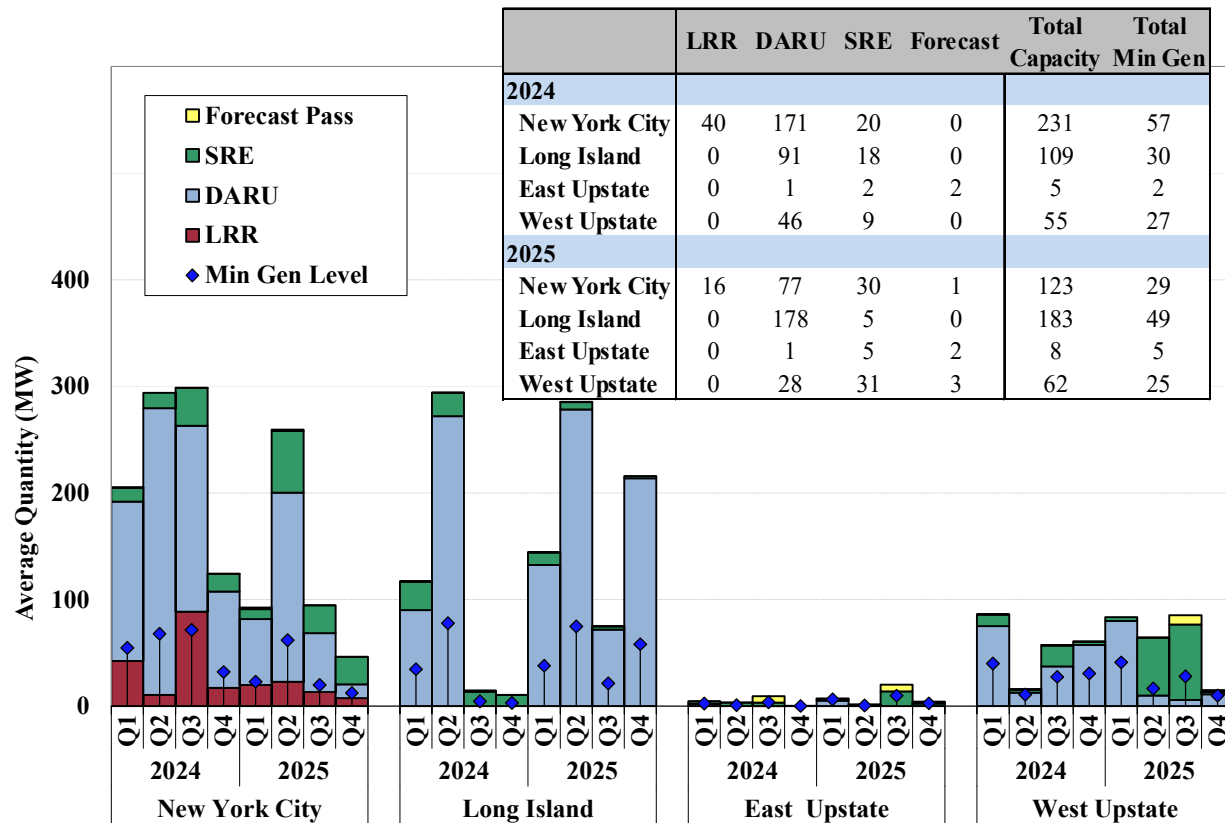


Figure A-88 & Figure A-89: Forecast-Pass Commitment in New York

In the day-ahead market, when the Bid Load Pass does not commit enough physical resources to meet forecast load and reserves requirements, the subsequent Forecast Pass will commit additional physical resources, as indicated by the yellow bars in Figure A-87. However, this need is not currently priced in the market software. As a result, units committed through the Forecast Pass often depress market clearing prices and receive BPCG uplift. Although the amount of FCT-committed capacity was modest on the vast majority of days, reflecting the underlying needs through market signals would still be beneficial.

Figure A-88 examines Forecast Pass commitments. The x-axis shows all days in 2025 when Forecast Pass commitments occurred. The solid blue bars show, for each day, the total MWh committed in the Forecast Pass, including capacity from slow-start units and non-blocked quick-start units. The hollow bars show the available offline capacity from non-blocked quick-start units during the hours when FCT commitments occurred. This available capacity is currently not treated the same as blocked quick-start units in the FCT pass for satisfying load and reserve requirements. If these units were recognized as quick-start by the market software, most FCT commitments would not have been needed.

The inset table summarizes annual totals for the past three years, including: (i) the number of days with FCT commitments; (ii) the total MWh committed in the FCT pass; and (iii) the available offline capacity from non-blocked quick-start units during FCT commitment hours.

Figure A-89 compares the FCT commitment with forecast physical energy needs in the day-ahead market in 2025, summarizing the following quantities on a daily basis:

- *Forecast Required Energy for Dispatch* – This summarizes the difference between NYISO forecasted load and scheduled physical energy in the economic pass, in total MWh for each day; and
- *Forecast-Pass Committed Capacity* – Summarizes additional capacity committed in the forecast pass to meet NYISO forecast load on each day. The reported quantity includes capacity from internal slow-start resources and non-blocked quick-start units in the hours where it is not online in the economic pass but is online in the forecast pass.

Figure A-88: Forecast-Pass Commitment
2025

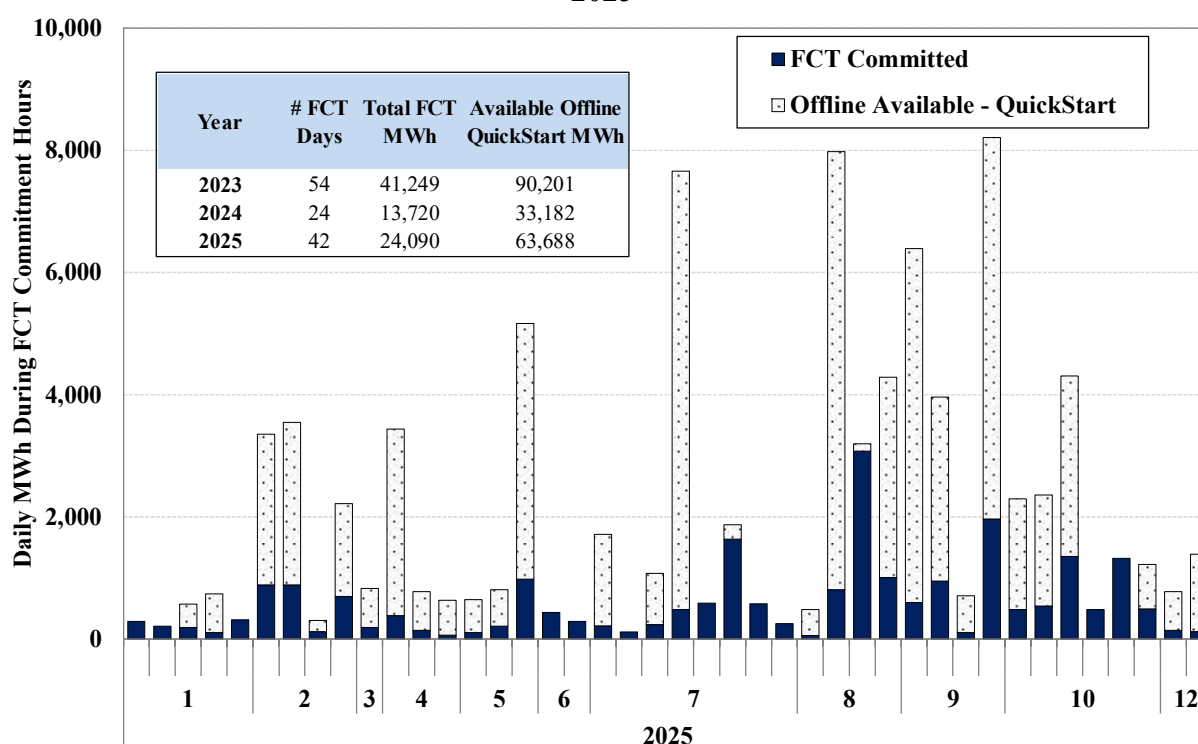


Figure A-89: FCT Commitment and DAM Forecast Physical Energy Needs
By Day, 2025

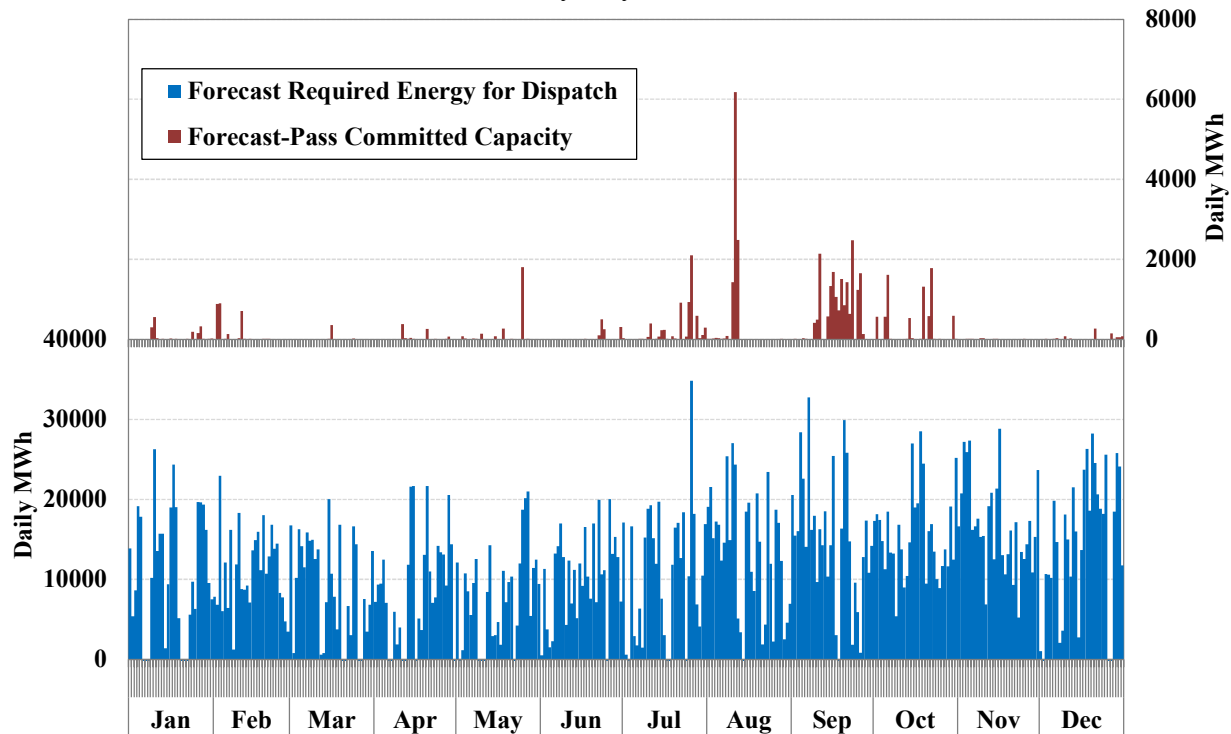


Figure A-90: SRE Commitments and SCR Deployments for Systemwide Capacity

The following chart presents our evaluation of the efficiency of SRE commitments and SCR deployments for systemwide capacity needs on each applicable day in 2024 and 2025.

Several categories of supply and demand changes occurring after the day-ahead market can reduce the capacity margin and lead the NYISO to commit SRE resources. The main categories include:

- **Reduction in Expected Imports:** This category represents expected reductions of in scheduled net imports, primarily from virtual external transactions. Additional reductions may arise from physical transactions that fail to clear the day-ahead checkout process or are expected to be reduced because of real-time system conditions.
- **Increases in Load Forecast:** This category shows the reduction in supply margin due to upward adjustments in load forecasts.
- **Generator Derates and Outages:** This category represents the reduction in generating capacity caused by resource outages and deratings.

When the total loss in supply exceeds day-ahead scheduled supply margin, NYISO initiates an SRE commitment to secure additional resources. These SREs reflect reserve needs that are not fully represented in the day-ahead market.

The top portion of the chart presents the quantity associated with the first category, reductions in expected imports. The bottom portion of the figure shows the quantity of SRE commitments

made by NYISO, compared to the forecasted capacity deficiency. The black bar indicates the projected capacity deficiency at the time the SRE commitments were made. A positive value indicates a capacity deficiency, while a negative number indicates a surplus.

The portion of SRE capacity that falls below the black bar is categorized as “Needed”, while the portion above the black bar is categorized as “Headroom”. Similarly, on high-load days when additional SCR resources were activated to meet systemwide capacity needs, the portion of estimated SCR deployments below the black bar is categorized as “Needed”, and the remaining portion is categorized as “Headroom”.

Figure A-90: Key Drivers to SRE Commitments for Systemwide Capacity
2024 – 2025

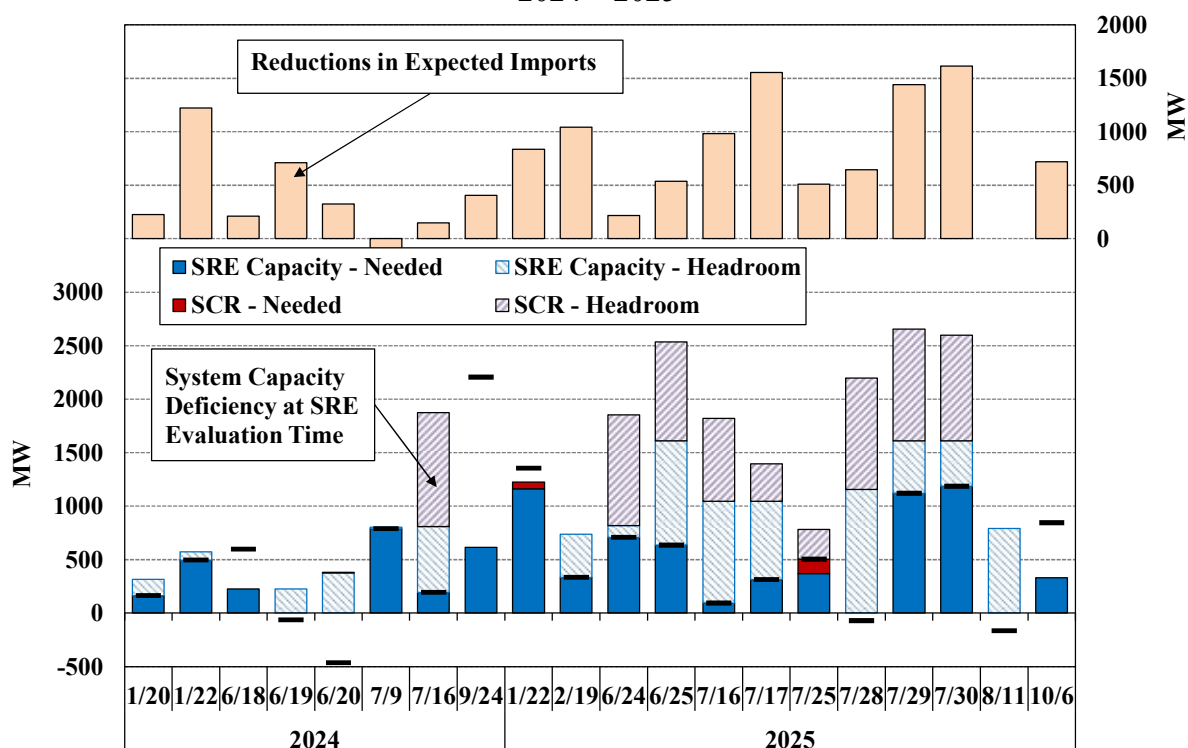


Figure A-91 & Figure A-92 : Supplemental Commitment for Reliability in New York City and North Country Load Pockets

Figure A-91 examines the necessity of reliability commitments in New York City, which accounted for the majority of the reliability commitments in NYCA during 2025. The figure shows the reliability commitment quantities in stacked bars for each month of 2025 in four distinct categories:

- **Economic MWh:** This category represents the total MWh of the initial DARU commitments that eventually qualify as economic capacity within the scheduling software (because they are still committed if the DARU and LRR requirements are removed from the SCUC run).
- **Verified – Needed MWh:** This category represents the total MWh of the initial DARU commitments and applicable LRR and SRE commitments that do not qualify as

Economic but are verified by the MMU’s assessment as necessary for maintaining reliability (including known thermal and voltage requirements) in the applicable load pockets.

- Our assessment relies on information available in the day-ahead and real-time markets, including factors such as load forecast, resource availability, and transmission network conditions.
- **Verified – Headroom MWh:** This category represents the total MWh that are associated with **Verified** commitments but exceed the amount of **Needed** MWh.
 - For example, if a 100 MW unit is verified for a reliability need of 50 MWh over two hours but has a minimum run time commitment of five hours, the headroom MWh would be 450 MWh ($= 5 \times 100 - 50$).
- **Unverified MWh:** This category represents the remaining DARU and SRE commitments that do not fit into the other three categories.

Figure A-91: Evaluation of DARU/LRR/SRE Commitments in New York City
By Month, 2025

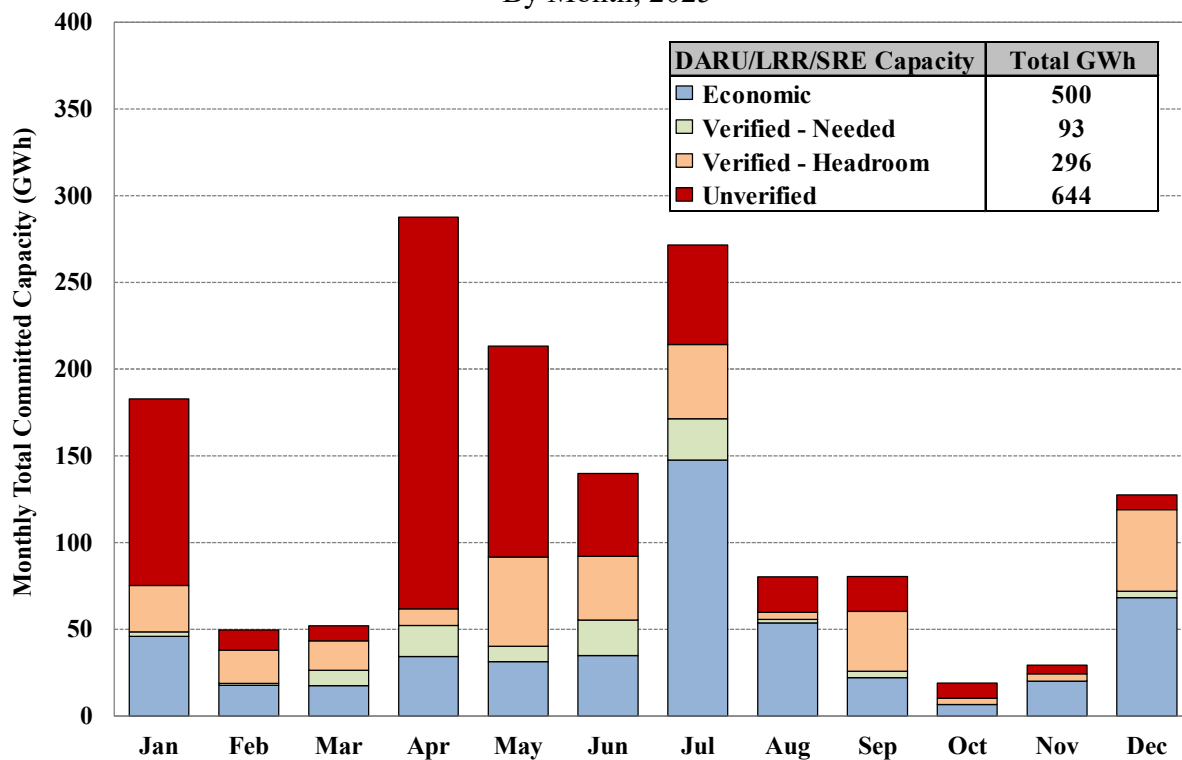
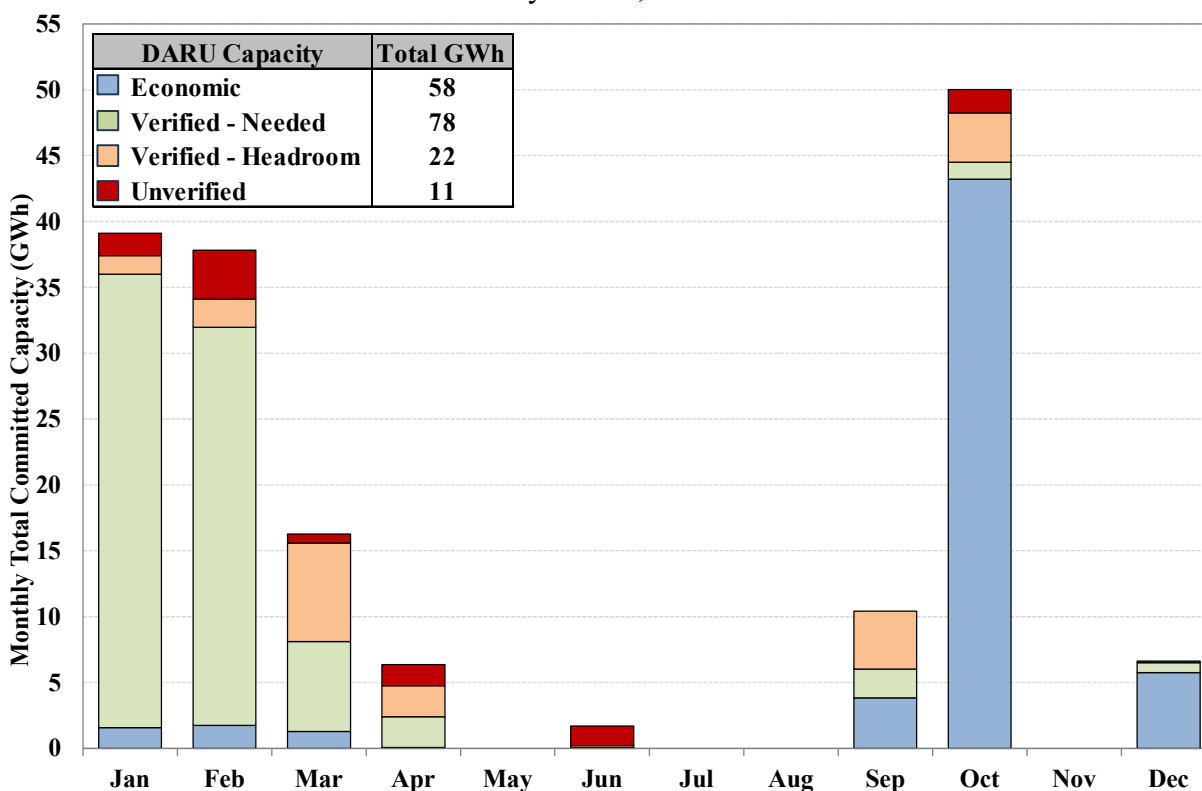


Figure A-92 examines the necessity of day-ahead reliability commitments of fossil fuel generators in the North Zone for North Country Reliability across 2025. The figure shows the reliability commitment quantities in the same four categories as used in the N.Y.C. analysis of Figure A-91. Reliability commitments in the region have been abnormally high in recent years due to ongoing maintenance work related to the various Smart Path transmission upgrades.

Figure A-92: Evaluation of DARU Commitment for North Country Reliability
By Month, 2025



J. Uplift Costs from Guarantee Payments

Uplift charges from guarantee payments accrue from the operation of individual generators for local and non-local reliability reasons in the day-ahead and real-time markets. The following figures summarize the three categories of non-local reliability uplift, which are allocated to all Load Serving Entities (“LSEs”), and the four categories of local reliability, which are allocated to local Transmission Owners.

The three categories of non-local reliability uplift are:

- **Day-Ahead Market** – This category primarily includes guarantee payments to generators that are economically committed in the day-ahead market. These generators receive payments when day-ahead clearing prices are insufficient to cover their total as-bid costs, including start-up, minimum generation, and incremental costs. In addition, when a DARU unit is committed by the NYISO for statewide reliability, the resulting guarantee payments are uplifted statewide. However, these account for a small portion of total DARU capacity.
- **Real-Time Market** – Guarantee payments are made primarily to gas turbines economically committed by RTC and RTD that do not receive sufficient revenue to cover start-up and other running costs over their run time. Guarantee payments in this category are also made for: a) SRE commitments and out-of-merit dispatch for bulk power system

reliability; b) imports scheduled with an offer price greater than the real-time LBMP; and c) demand response resources (i.e., EDRP/SCRs) deployed for system reliability.

- Day-Ahead Margin Assurance Payment – Guarantee payments made to cover losses in margin for generators dispatched by RTD below their day-ahead schedules. When a unit is dispatched or committed for local reliability, any day-ahead margin assurance payments it receives are allocated as local reliability uplift. However, the majority of day-ahead margin assurance payments are allocated as non-local reliability uplift.

The four categories of local reliability uplift are:

- Day-Ahead Market – Guarantee payments are made to generators committed in the SCUC due to the Local Reliability Rule (“LRR”) or as Day-Ahead Reliability Units (“DARU”) for local reliability needs at the request of local Transmission Owners. Although uplift associated with these payments is allocated to the local area, these commitments tend to decrease day-ahead prices. As a result of lower prices, additional (non-local reliability) uplift may be paid to generators that are economically committed in the Bid Load pass.
- Real-Time Market – Guarantee payments are made to generators committed and redispatched for local reliability reasons after the day-ahead market. While this can occur for a variety of reasons, the majority of this uplift is related to Supplemental Resource Evaluation (“SRE”) commitments.
- Minimum Oil Burn Compensation Program – Guarantee payments made to generators that cover the spread between oil and gas prices when generators burn fuel oil to help maintain reliability in New York City due to potential natural gas supply disruptions.
- Day-Ahead Margin Assurance Payment – Guarantee payments made to cover losses in margin for generators dispatched out-of-merit for local reliability reasons below their day-ahead schedules.

Figure A-93 - Figure A-94: Uplift Costs from Guarantee Payments

Figure A-93 shows the seven categories of uplift costs associated with guarantee payments on a monthly basis for 2024 and 2025. The uplift costs associated with EDRP/SCR resources are shown separately from other real-time statewide uplift costs. The inset table summarizes the total annual uplift costs for each category for both years.

Figure A-94 shows the seven categories of uplift charges on a quarterly basis in 2024 and 2025 across four regions in New York: (a) West Upstate (Zones A-E); (b) East Upstate (Zones F-I); (c) New York City (Zone J); and (d) Long Island (Zone K). In the figure, uplift costs paid to import transactions from neighboring control areas and to EDRP/SCR resources are shown separately from those associated with generation resources in the four regions. The inset table summarizes the total annual uplift costs for each region for both years.

It is also noted that both figures are based on information available at the time of reporting and do not reflect any subsequent billing adjustments. Accordingly, the values shown may differ from final settlement amounts.

Figure A-93: Uplift Costs from Guarantee Payments by Month
2024 – 2025

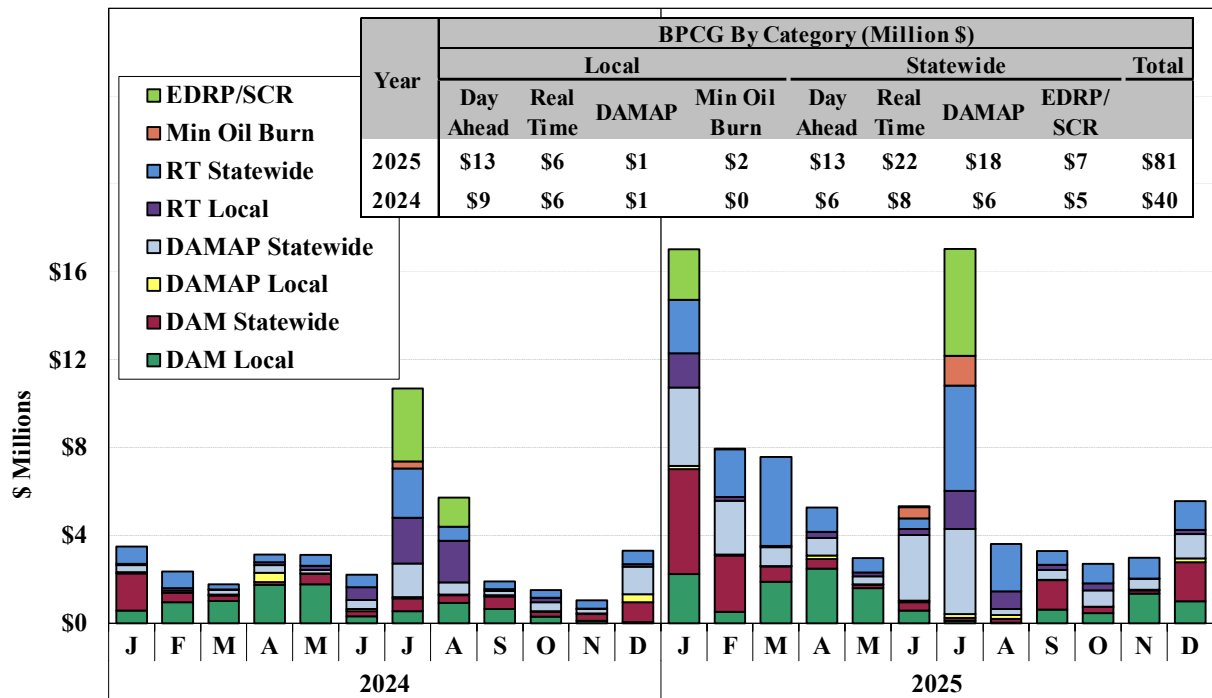
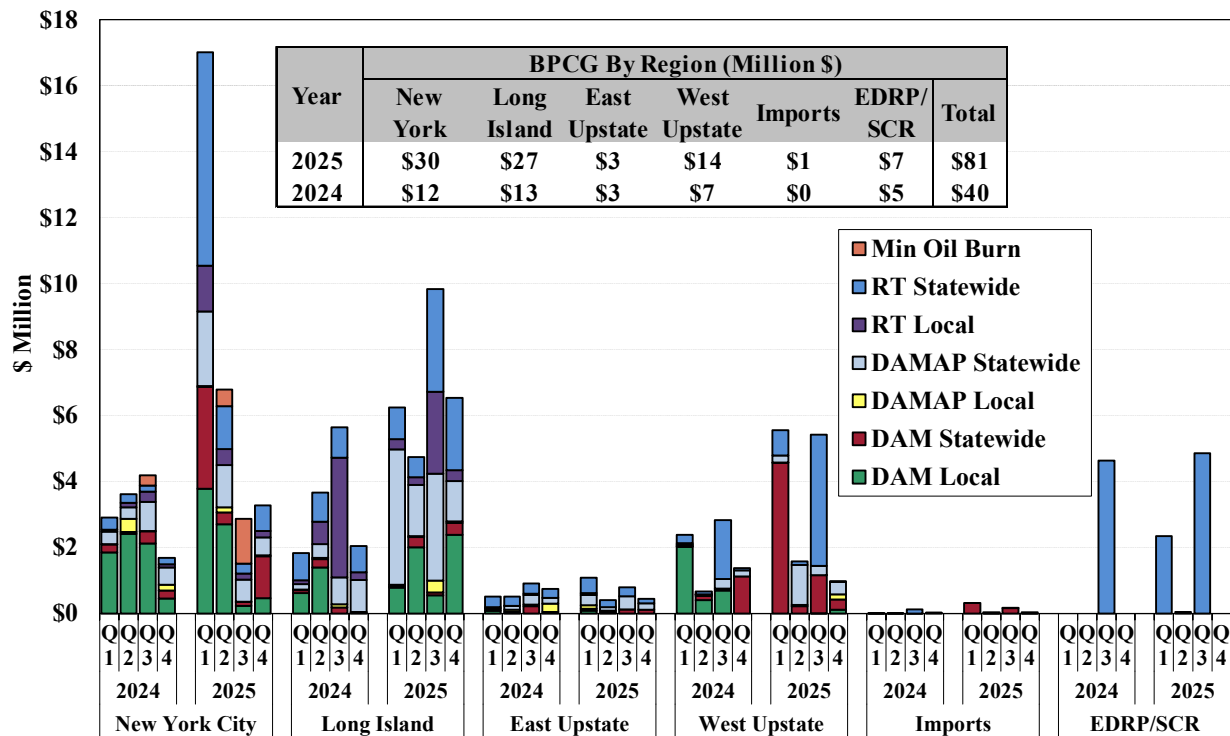


Figure A-94: Uplift Costs from Guarantee Payments by Region
2024 – 2025



VI. CAPACITY MARKET

This section evaluates the performance of the capacity market, which is designed to ensure that sufficient resources are available to satisfy New York’s planning reserve margin requirements. The capacity market supplements the incentives provided by the energy and ancillary services markets. In combination, these three sources of revenue provide incentives for new investment, retirement decisions, and participation by demand response.

The New York State Reliability Council (“NYSRC”) determines the Installed Reserve Margin (“IRM”) for NYCA, which is the amount of planning reserves necessary to meet the reliability standards for New York State. NYISO uses the IRM in conjunction with the annual peak load forecast to calculate the Installed Capacity (“ICAP”) requirement for NYCA.³⁰⁸ NYISO also determines the Minimum Locational Installed Capacity Requirements (“LCRs”) for New York City, the G-J Locality, and Long Island, which it uses in conjunction with the locational annual peak load forecast to calculate the locational ICAP requirement.³⁰⁹

Since NYISO operates an Unforced Capacity (“UCAP”) market, the ICAP requirements are translated into UCAP requirements, using location-wide availability rates known as Derating Factors. The obligations to satisfy the UCAP requirements are allocated to the LSEs in proportion to their annual coincident peak load in each area. LSEs can satisfy their UCAP requirements by purchasing capacity through bilateral contracts, by self-scheduling their own capacity, or by participating in UCAP market auctions run by NYISO.

NYISO conducts three UCAP auctions: a forward strip auction where UCAP is transacted in six-month blocks for the upcoming capability period, a monthly forward auction where UCAP is transacted for the remaining months of the capability period, and a monthly spot auction. The two forward markets are voluntary, but all requirements must be satisfied at the conclusion of the spot market immediately prior to each month. Market participants that have purchased more than their obligation prior to the spot auction sell the excess into the spot auction. Demand curves are used to determine the clearing prices and quantities purchased in each locality in each monthly UCAP spot auction.³¹⁰ The amount of UCAP purchased is determined by the intersection of UCAP supply offers in the spot auction and the demand curve (adjusted for capacity sales through bilateral contracts and forward auctions). Hence, the spot auction purchases more capacity than is necessary to satisfy the UCAP requirement when more capacity is available.

The demand curve for a Locality is defined as a straight line through the following two points:

³⁰⁸ The ICAP requirement = $(1 + \text{IRM}) \times \text{Forecasted Peak Load}$. The IRM was set at 24.4 percent in the most recent Capability Year (i.e., the period from May 2025 to April 2024). NYSRC’s annual IRM reports may be found at “http://www.nysrc.org/NYSRC_NYCA_ICR_Reports.html”.

³⁰⁹ The locational ICAP requirement = $\text{LCR} \times \text{Forecasted Peak Load}$ for the location. These are set in the annual Locational Minimum Installed Capacity Requirements Study, which may be found [here](#).

³¹⁰ The capacity demand curves are not used in the forward strip auction and the forward monthly auction. The clearing prices in these two forward auctions are determined based on participants’ offers and bids.

- Net CONE at Level of Excess – The demand curve price equals the levelized cost of a new peaking unit (net of estimated energy and ancillary services revenue) when the quantity of UCAP procured exceeds the UCAP requirement by a small margin known as the “Level of Excess”.³¹¹
- \$0 at Zero Crossing Point – The demand curve price equals \$0 when the quantity of UCAP procured exceeds the UCAP requirement by 12 percent for NYCA, 15 percent for the G-J Locality, and 18 percent for both New York City and Long Island.

Every four years, NYISO establishes the capacity demand curves through a study that includes a review of the selection, costs, and revenues of the peaking technology. Each year, NYISO further adjusts the demand curve to account for changes in Net CONE of a new peaking unit.

This report evaluates a period when there were four capacity market Localities: G-J Locality (Zones G to J), New York City (Zone J), Long Island (Zone K), and NYCA (Zones A to K). New York City, Long Island and the G-J Locality are each nested within the NYCA Locality. New York City is additionally nested within the G-J Locality. The clearing price in a nested Locality cannot be lower than the clearing price in the surrounding Locality.

This appendix includes information on the following aspects of the capacity market:

- Trends in internal installed capacity, capacity exports, and imports from neighboring control areas (sub-sections A and B);
- Equivalent Forced Outage Rates (“EFORDs”) and Derating Factors (sub-section C);
- Capacity supply and quantities purchased each month as well as clearing prices in monthly spot auctions (sub-section D);
- Additional analysis supporting recommendations to improve the capacity market in Section IX of the Main Body of this report:
 - Analysis of the marginal reliability impact (MRI) and Cost of Reliability Improvement (CRI) at locations within capacity regions (sub-section E)
 - Detailed proposal for establishing more granular capacity zones (sub-section F),
 - Proposed approach for adjusting the capacity payments of generators that have a positive or negative impact on capacity transfer capability (sub-section G).

A. Installed Capacity of Generators in NYCA

Figure A-95 - Figure A-96: Installed Summer Capacity and Forecasted Peak Demand

The following figures show the amount of installed capacity in specific regions by fuel and technology type and how they have changed over time. Capacity has shifted away from coal and nuclear toward natural gas and renewable resources. Since the retirement of the Indian Point nuclear units in 2020 and 2021, Eastern New York has become largely dependent on fossil-fueled capacity with virtually all renewable, hydro, and nuclear resources in upstate regions.

³¹¹ The demand curves have maximum price levels which apply when UCAP procured falls substantially below the UCAP requirement. The demand curves can be found on NYISO’s webpage, available [here](#).

The bottom panel of Figure A-95 shows the total installed summer capacity of generation (by prime mover) and the forecasted and actual summer peak demands for the New York Control Area for the years 2016 through 2025.^{312, 313} The top panel of Figure A-95 shows the amount of capacity that entered or exited the market during each year.³¹⁴

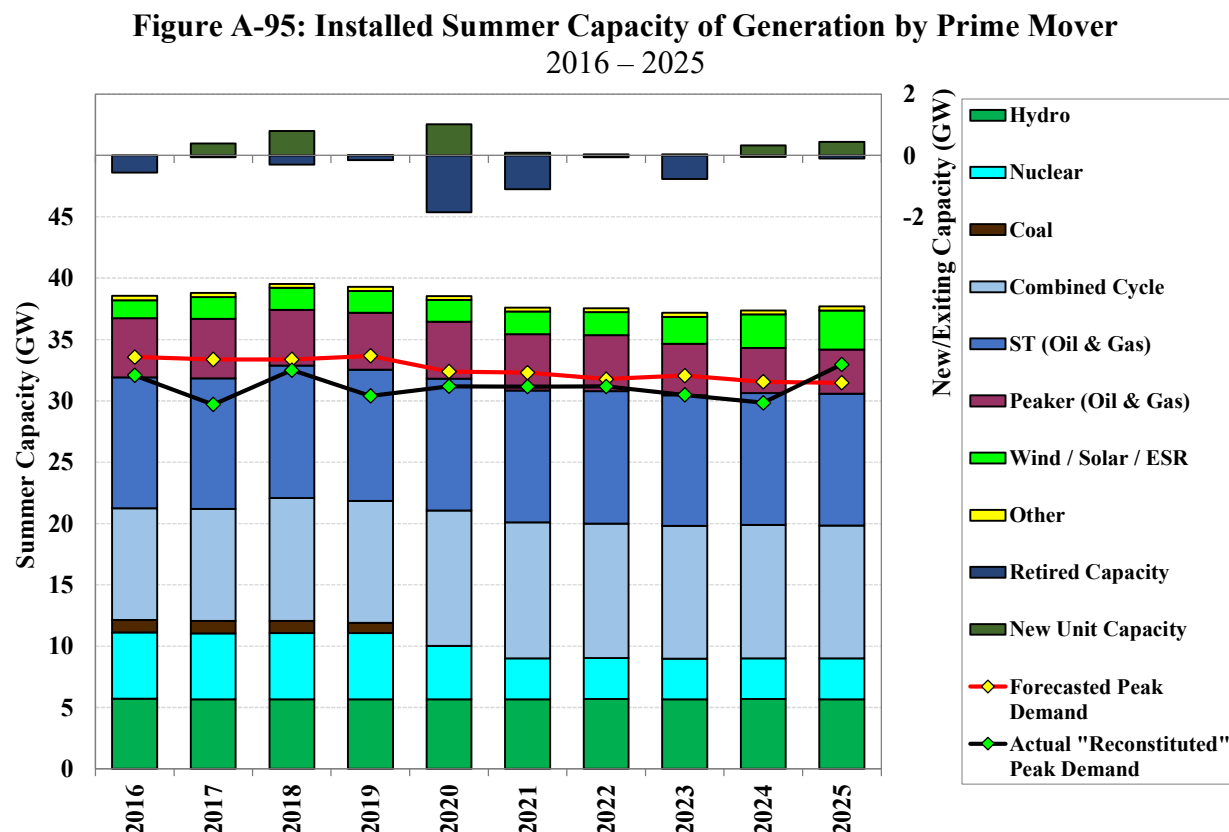


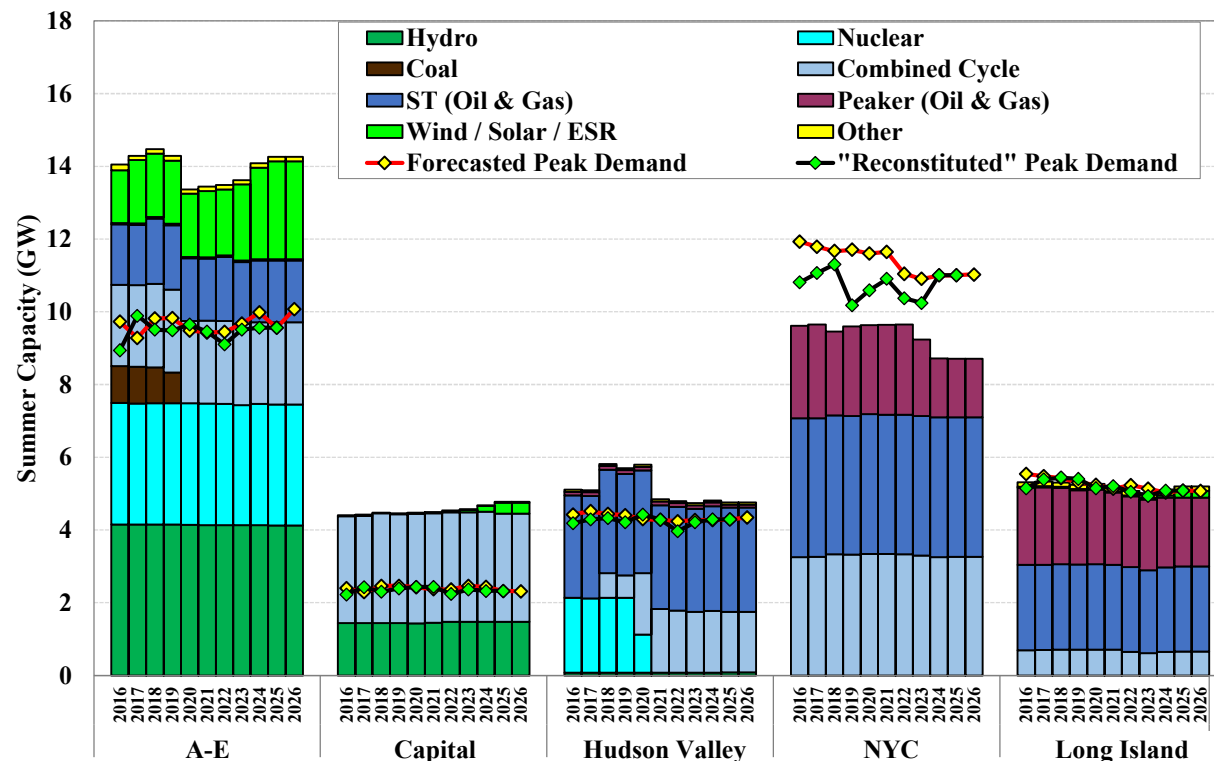
Figure A-96 shows a regional distribution of generation resources and the forecasted and actual non-coincident peak demand levels for each region over the same timeframe. The installed capacity shown for each year is based on the summer rating of resources that are operational at the beginning of the Summer Capability Period of that year (i.e., capacity online by May 1 of each given year).

³¹² Forecasted peak demand shown is based on the forecasted NYCA coincident peak demand from the Gold Book of each year.

³¹³ Reconstituted peak demand values in Figure A-95 and Figure A-96 include demand reductions from NYISO and utility-based programs.

³¹⁴ Both the annual capacity and capacity from new additions from wind resources are given for units with both ERIS and CRIS rights. ERIS-only wind units do not appear in this chart as capacity resources.

Figure A-96: Installed Summer Capacity of Generation by Region and by Prime Mover
2016 – 2026

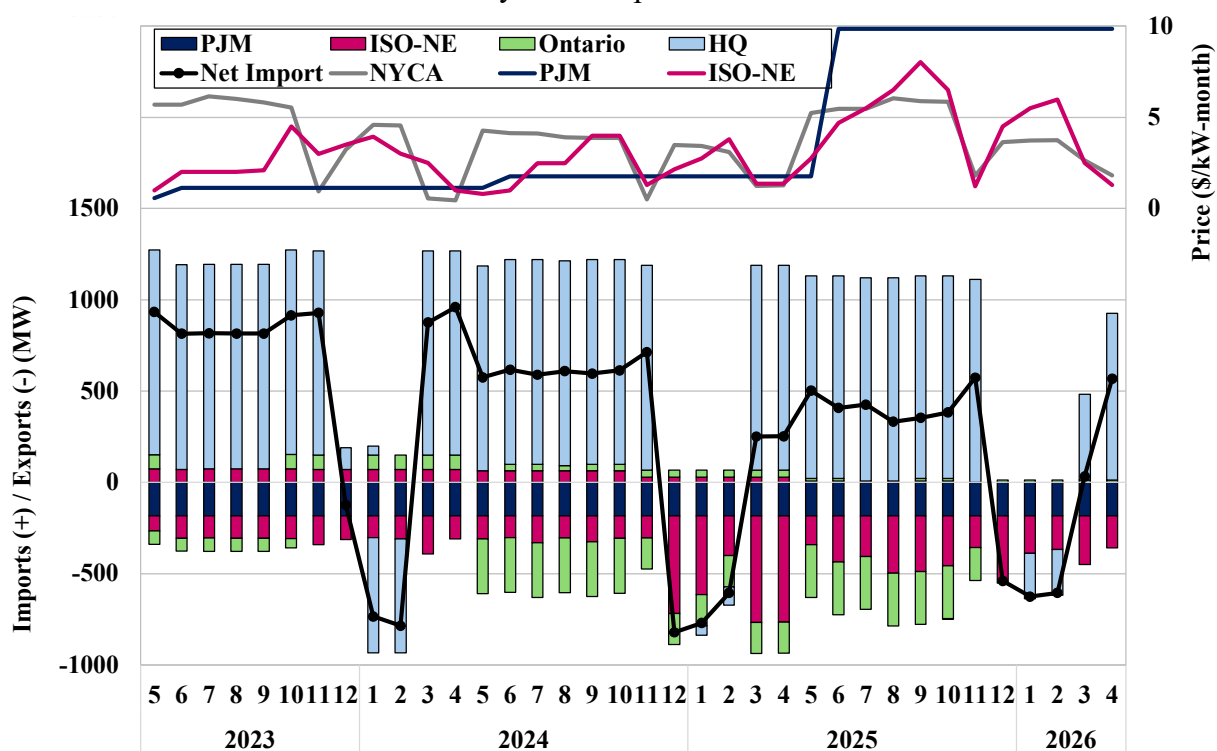


B. Capacity Imports and Exports

Figure A-97: NYISO Capacity Imports and Exports by Interface

NYISO procures a portion of its installed capacity from neighboring regions, and some internal capacity is sold to neighboring regions. Figure A-97 shows the monthly average of net capacity imports (UCAP) into the NYISO from neighboring control areas from May 2023 through April 2026 along with capacity prices in the New York Control Area and its neighboring control areas, including Hydro Quebec (“HQ”), Ontario (“OH”), PJM, and ISO-NE. The capacity imported from each region is shown by the positive value stacked bars, while the capacity exported from NYCA is shown as negative value bars. The capacity prices shown in the figure are: (a) the NYCA spot auction price for NYISO; (b) the RTO price in the Third Incremental Auction for PJM; and (c) the NY AC Ties price in the Monthly Reconfiguration Auction for ISO-NE.

Figure A-97: NYISO Capacity Imports and Exports by Interface
May 2023 –April 2026



C. Derating Factors and Equivalent Forced Outage Rates

The UCAP of a resource is equal to its installed capacity adjusted to reflect its expected availability, as measured by its Equivalent Forced Outage Rate on demand (“EFORD”). A generator with a high frequency of forced outages over the preceding two years’ Capability Periods (i.e. a unit with a high EFORD) would not be able to sell as much UCAP as a reliable unit (i.e. a unit with a low EFORD) with the same installed capacity. For example, a unit with 100 MW of tested capacity and an EFORD of 7 percent would be able to sell 93 MW of UCAP.³¹⁵ This gives suppliers an incentive to perform reliably.

The locality-specific derating factors are used to translate ICAP requirements into UCAP requirements for each capacity zone. The NYISO computes the derating factor for each capability period based on the weighted-average EFORD of the capacity resources that are electrically located within the zone. For each Locality, a derating factor is calculated from the two most recent like-Capability Period average EFORD values of resources in the Locality in accordance with Section 4.5 of the NYISO’s Installed Capacity Manual.³¹⁶

³¹⁵ The variables and methodology used to calculate EFORD for a resource can be found [here](#).

³¹⁶ The Derating Factor used in each six-month capability period for each Locality may be found [here](#).

Table A-11: Historic Derating Factors by Locality

Table A-11 shows the derating factors the NYISO calculated for each capacity zone from Summer 2021 onwards. Derating factors tend to be highest in regions with the most intermittent capacity and most volatile year-over-year in regions with older generation fleets.

Table A-11: Derating Factors by Locality
Summer 2021 – Winter 2025/26

Locality	Summer 2025	Summer 2024	Summer 2023	Summer 2022	Summer 2021	Winter 2025/26	Winter 2024/25	Winter 2023/24	Winter 2022/23	Winter 2021/22
G-I	10.27%	13.49%	11.82%	7.63%	5.45%	14.90%	12.49%	10.62%	10.39%	8.41%
LI	9.02%	8.66%	7.29%	6.27%	4.91%	9.47%	9.36%	10.66%	10.31%	7.21%
NYC	5.18%	4.62%	1.64%	3.26%	2.69%	6.80%	4.13%	4.12%	3.41%	2.48%
A-F	17.51%	18.03%	14.48%	14.20%	13.27%	17.94%	18.68%	13.16%	10.70%	11.36%
NYCA	13.00%	13.21%	10.14%	9.78%	8.77%	14.05%	13.46%	10.39%	8.91%	8.40%

Table A-12 to Table A-13: Unavailable Capacity to RTC & RTD from Various Technologies in Summer Capability Periods

The NYISO tariff describes a DMNC testing process to determine the ICAP ratings for traditional generators such as nuclear units, combined cycles, steam turbines, and peaking facilities. The process is similar for each of these unit types, but it takes into consideration certain technology-specific characteristics in fine tuning testing obligations.³¹⁷ One such technology-specific obligation that exists is for “internal combustion, combustion units, and combined cycles” to temperature-adjust their DMNC test results based on an output factor curve that is dependent on one variable, ambient air temperatures, and a seasonal peak temperature rating determined by the previous Transmission District peak conditions across the most recent four like-Capability Periods. Functionally, this tends to cause the ICAP ratings for these unit types during the summer Capability Periods to be lower than the value at which they test since tests are often done at cooler temperatures than the seasonal peak.

NYISO filed and were approved for tariff revisions that reduced and are expected to further reduce DMNC values in the 2025/26 and future Capability Years. Table A-12 summarizes the status of current enacted, pending, and unresolved issues in this area:

Rule changes that impacted 2025/26 Capability Year results included the sunseting of the Capacity Limited Resource (“CLR”) program. This shifted the DA offer obligation of most capacity resources from the Emergency Upper Operating Limit (“UOLe”) to the Normal Upper Operating Limit (“UOLn”) strengthening performance obligations on seldom-used capacity ranges. Of the installed capacity previously offered as emergency-only (i.e., above the UOLn and below UOLe), approximately 210 MW ceased being sold into the capacity market during Summer 2025.

³¹⁷ See Section 4.2 of the ICAP Manual.

**Table A-12: Excessive Qualification of ICAP Progress
2025**

Category	MMU Recommendation	NYISO Adaptation	Implementation Year
Emergency Capacity	Require emergency capacity to be accredited based on the marginal capacity value of the calculated EFORD of that operating range.	Sunset of the Capacity Limited Resource program.	2025/26
Relative Humidity	Require combustion turbines equipped with certain inlet cooling systems to adjust to make a relative humidity adjustment to DMNC test results.	Comparable to MMU's recommended approach.	2026/27
Barometric Pressure	Require combustion turbines to adjust DMNC test results to a barometric pressure consistent with expectations at peak conditions.	N/A	N/A
Water Temperature	Require once-through water cooled generators to adjust DMNC test results to expected water temperatures at peak summer conditions.	Require water-cooled generators to conduct DMNC tests during July or August between the hours of 10 AM and 10 PM.	2026/27
Tidal Conditions	Require coastal once-through cooled water dependent generators to adjust DMNC test results to water levels consistent with average tidal depth.	N/A	N/A
Cogeneration Steam Demand	Require cogeneration units to reduce ICAP sales based on actual host steam demand during peak load hours comparable to BTM:NG resources.	Strict enforcement of of NYISO Market Services Tariff 5.12.1.5.1	2025/26

Beginning with the Summer 2026 Capability Period, fossil fuel resources with certain evaporative cooling equipment installed will be required to adjust to temperature and humidity conditions applicable to peak summer conditions. Additionally, certain water-cooled generators must perform DMNC tests during a restricted window which may better capture the effects on generator output from warmer water temperatures. These changes should improve capacity accreditation for the impacted generators and we will report on the effects of this adjustment in future reports. Additional improvements are necessary to accurately account for expected ambient-adjusted capability of resources impacted by ambient water temperature conditions and barometric pressure conditions (which are discussed below).

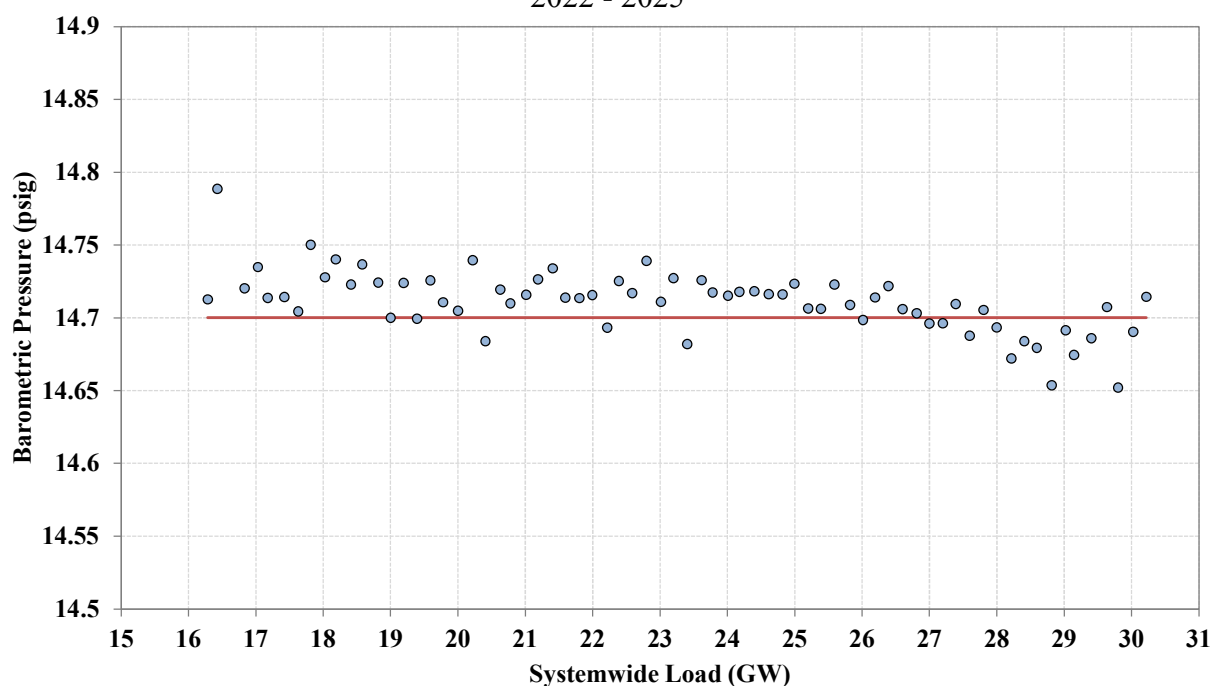
Other Ambient – Barometric Pressure on Combustion Turbines

Barometric pressure affects the power output and efficiency of certain generation technologies, particularly CTs in simple cycle and combined cycle configurations. This is because air density falls as barometric pressure drops leading to lower mass flow of air through the CT.³¹⁸

Therefore, barometric pressure and the CT maximum output are correlated. Barometric pressure conditions are inversely correlated with NYISO load values on the warmest summer days.

Figure A-98 shows the relationship between barometric pressure values at the generator locations for all generators with at least one CT in the NYISO versus the systemwide hourly demand from each hour of Summers 2022-2025. Each point represents the average barometric pressure across all generators during hours where the real-time system demand was within defined 200 MW buckets. For example, the first point shows the average barometric pressure when systemwide demand was between 16.3 GW and 16.5 GW was 14.71 psig (y-axis) with an average system load in that range of 16.31 GW (x-axis). The red line denotes the average barometric pressure at sea level, which is a standard benchmark for air pressure also known as “1 atmosphere”.

**Figure A-98: Barometric Pressure vs Systemwide Load at Generator Locations
2022 - 2025**



This analysis shows that average barometric pressure does not vary much under most load conditions.³¹⁹ Under summer peak load conditions when demand surpassed roughly 27 to 28

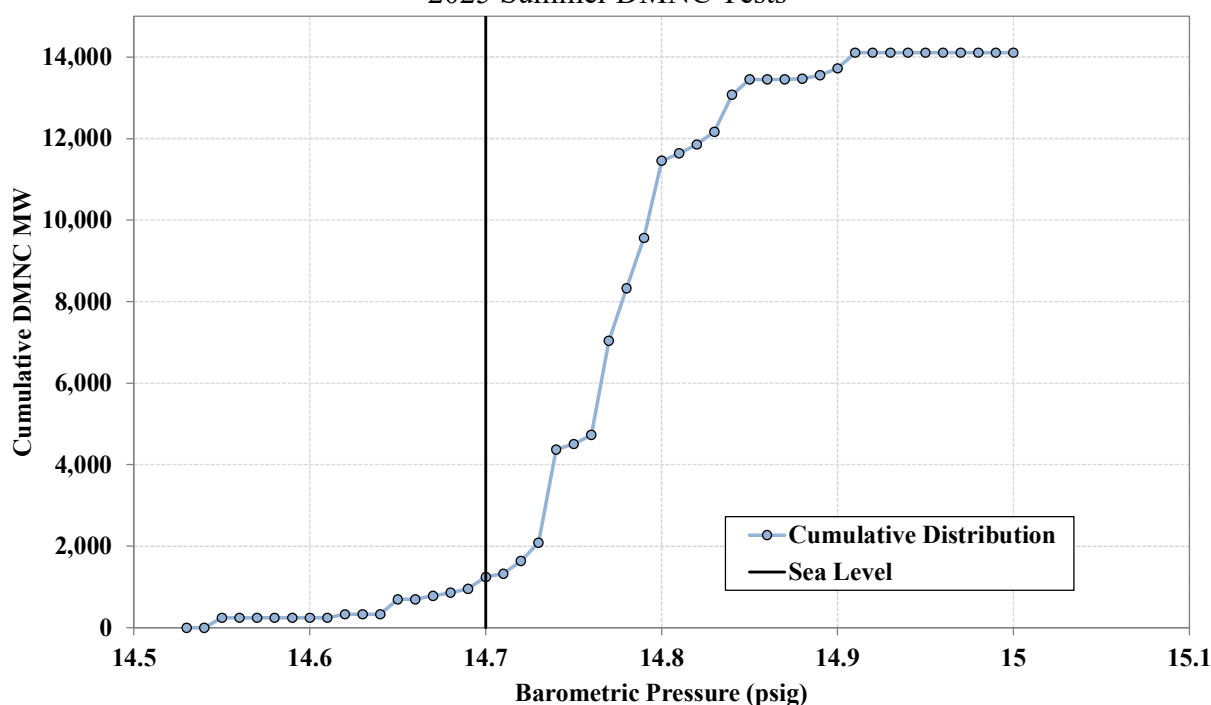
³¹⁸ See: https://www.governova.com/content/dam/gepower-new/global/en_US/downloads/gas-new-site/resources/reference/ger-3567h-ge-gas-turbine-performance-characteristics.pdf

³¹⁹ Warmer air is less dense which means barometric pressure predictably drops as temperatures rise. See <https://www.noaa.gov/jetstream/atmosphere/air-pressure#:~:text=This%20change%20in%20pressure%20is%20caused%20by,are%20farther%20apart%20than%20in%20cooler%20air.>

GW, the air was less dense than average. Therefore, CT capability at peak conditions will, on average, be lower at peak conditions than throughout the rest of the summer due to predictably lower pressure conditions.

The next figure evaluates the average barometric pressure at each generating station in NYISO with at least one CT during its DMNC test in the 2025 Summer Capability Period. The combined DMNC MW submitted from these tests is given by the blue line, which shows the amount of capacity that was tested at barometric pressures at or below the value given on the x-axis. The average barometric pressure corresponding to sea level (14.7 psig) is shown with the black vertical line as a point of reference.

Figure A-99: Cumulative Capacity by Barometric Pressure during DMNC Tests
2025 Summer DMNC Tests



Roughly 91 percent of all capacity from those generators that are impacted by barometric pressure conditions came from DMNC tests that occurred at pressure conditions that exceeded sea level conditions. The seasonal DMNC rating is meant to quantify the capability of the CT at peak load conditions, i.e., when air temperatures are highest. However, the DMNC testing procedures allow testing during any hours of June through mid-September for combined cycle units and CTs. Most generators test at optimal barometric conditions (i.e., cool, sunny days when barometric pressures are above the pressure at sea level). Hence, total installed capacity is likely overstated since no barometric pressure adjustment is included in the requirements of ambient conditions dependent resources.

Table A-13 summarizes the MMU's estimates of overvalued ICAP ratings from barometric pressure-impacted generators based on: (a) barometric pressure at sea level and (b) actual

conditions during DMNC tests from each of the past three summers.³²⁰ The relationship between station max output and barometric pressure used in this analysis was 10.34 percent per 1 psig change in pressure.³²¹ Since barometric pressure predictably falls with hotter air temperatures and most DMNC tests occur at milder conditions, the total impact of barometric pressure on functionally unavailable capacity is estimated at 101.4 MW from the 2025 Summer DMNC tests, with roughly 70 percent of that value coming from combined cycle stations.

Table A-13: Overestimate of ICAP due to Barometric Pressure by Unit Type
Based on Adjusting DMNC to Pressure at Sea Level, 2023-2025

Type	2023	2024	2025
CC	53.5	46.1	69.7
Peaker	7.8	10.7	31.7
Total	61.3	56.9	101.4

Although NYISO has filed tariff changes to improve the accuracy of DMNC values, additional improvements are necessary to accurately account for expected ambient-adjusted capability of resources impacted by ambient water temperature (shown in Table A-12) and barometric pressure conditions (shown in Table A-13).

Water temperatures tend to rise steadily over the summer. Temperatures are predictably lower in early-July than in late-July and mid-August. Insofar as the peak conditions occur further into the summer months, tests conducted in early-July will likely underestimate the water temperature impact on these resources. Additionally, there is more than 9 GW of once through cooled fossil-fired steam turbine capacity in southeast NY which is impacted by tidal conditions. Ambient-adjusted capability is highest at high tide and can fall as the tides drop. Any DMNC test conducted at above-average tidal conditions is likely to minimize or mask the effects of tides.

Barometric pressure affects the capability of CTs in a predictable way with higher pressures allowing for increased CT output. Hot air tends to be lower pressure than cooler air. Since most generators with one or more CTs conduct DMNC tests on mild, high-pressure days in the early or late summer, actual capability tends to be lower at peak conditions. Therefore, the following changes would result in more accurate DMNC values:

- Ambient water & tidal dependent resources adjust DMNC test results to peak temperature and average tide conditions using an approach similar to what the NYISO has chosen for ambient air temperature and humidity conditions dependent resources; and
- Generators that are impacted by barometric pressure make an adjustment from DMNC test conditions to an ISO-determined pressure level typical of peak load conditions or to another benchmark such as 14.7 psig (i.e., barometric pressure at sea level).

³²⁰ We pull weather conditions for each generator location based on data at the ZIP Code level.

³²¹ See Figure 10 of: https://www.gevernova.com/content/dam/gepower-new/global/en_US/downloads/gas-new-site/resources/reference/ger-3567h-ge-gas-turbine-performance-characteristics.pdf

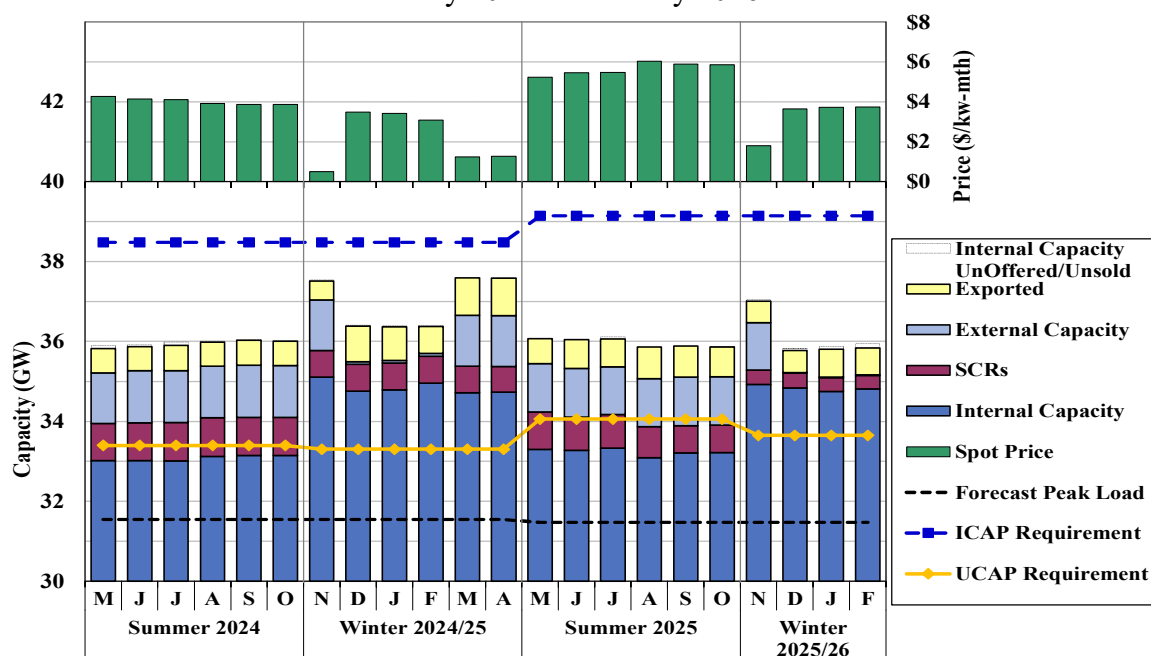
D. Capacity Market Results

Figure A-100 – Figure A-103: Capacity Sales and Prices

Figure A-100 shows capacity market results in the NYCA for the past four six-month Capability Periods. In the lower portion of each figure, the bars show the quantities of internal capacity sales, which include sales related to Unforced Deliverability Rights (“UDRs”) and sales from SCRs.³²² The hollow portion of each bar represents the In-State capacity in each region not sold (including capacity not offered) in New York or in any adjacent market. The line indicates the capacity requirement for each Capability Period for NYCA. Additionally, the figure shows sales from external capacity resources into NYCA and exports of internal capacity to other control areas. The upper portion of the figure shows clearing prices in the monthly spot auctions for NYCA (i.e., the Rest of State).

The capacity sales and requirements in the figure are shown in the UCAP terms, which reflect the amount of resources available to sell capacity. The changes in the UCAP requirements are affected by changes in the forecasted peak load, the minimum capacity requirement, and the Derating Factors. To better illustrate these changes over the period examined, the figure also shows the forecasted peak load and the ICAP requirements.

Figure A-100: UCAP Sales and Prices in NYCA
May 2024 to February 2026



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Special Case Resources (“SCRs”) are Demand Side Resources whose Load is capable of being interrupted upon demand, and/or Demand Side Resources that have a Local Generator, which is not visible to the ISO’s Market Information System and is rated 100 kW or higher, that can be operated to reduce Load from the NYS Transmission System and/or the distribution system at the direction of the NYISO.

Figure A-101 to Figure A-103 show capacity market results in New York City, Long Island, and the G-J Locality for the past four six-month Capability Periods. These charts display the same quantities as Figure A-100 does for the NYCA region and compare the spot prices in each Locality to the Rest-Of-State prices.

Figure A-101: UCAP Sales and Prices in New York City

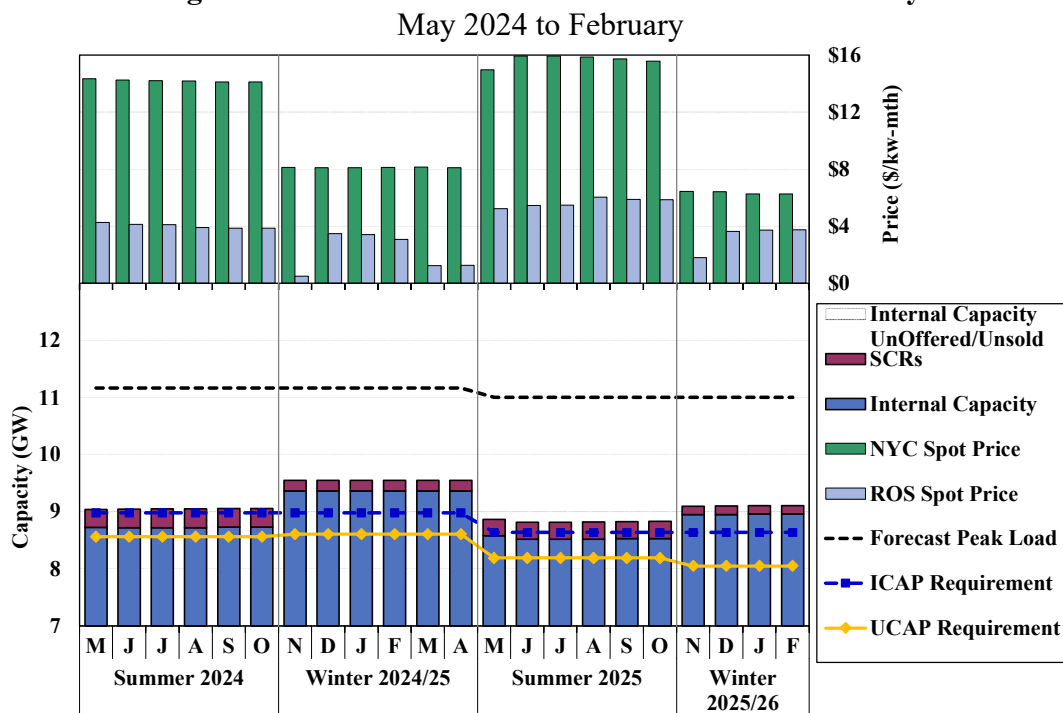


Figure A-102: UCAP Sales and Prices in Long Island

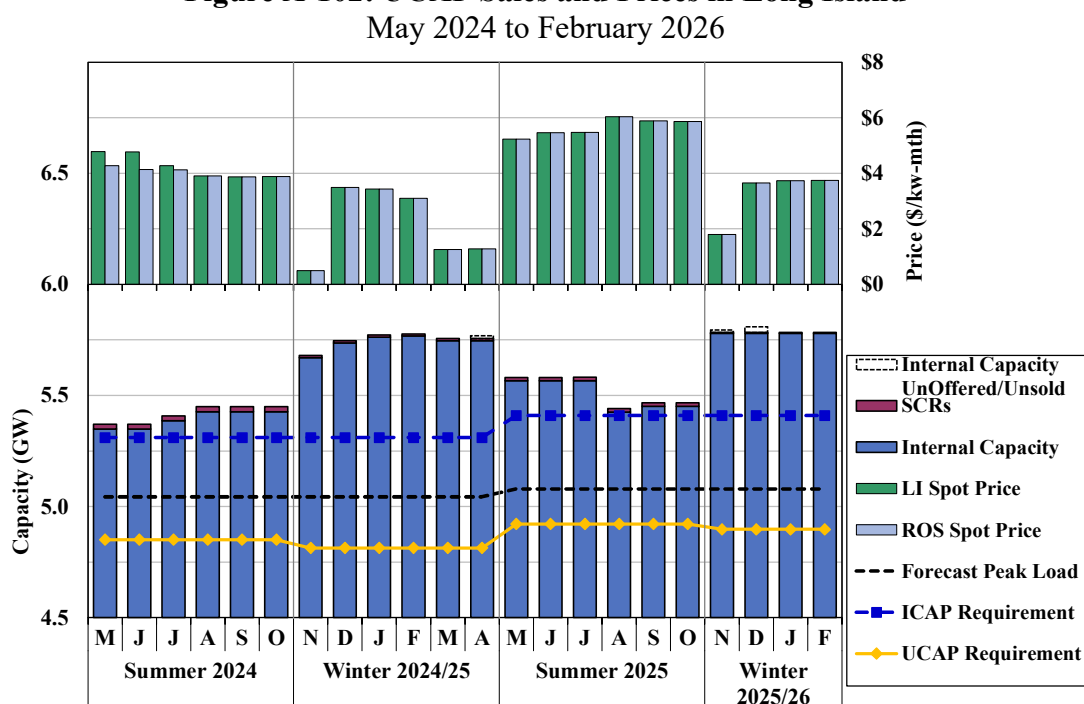
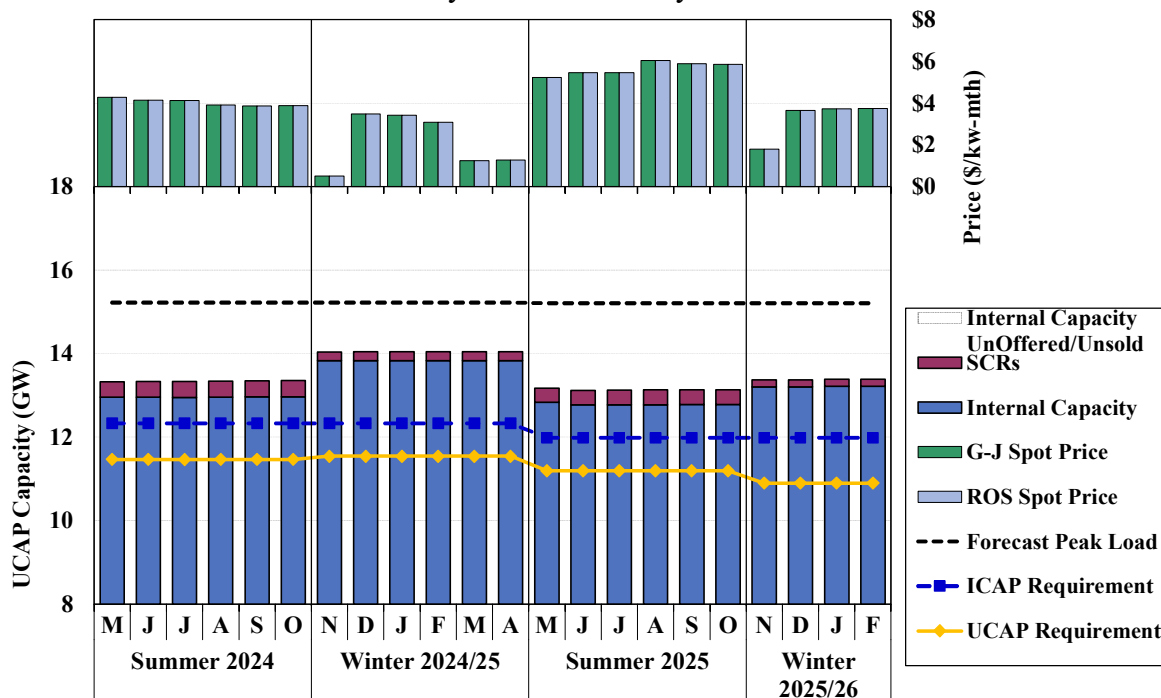


Figure A-103: UCAP Sales and Prices in the G-J Locality
May 2024 to February 2026

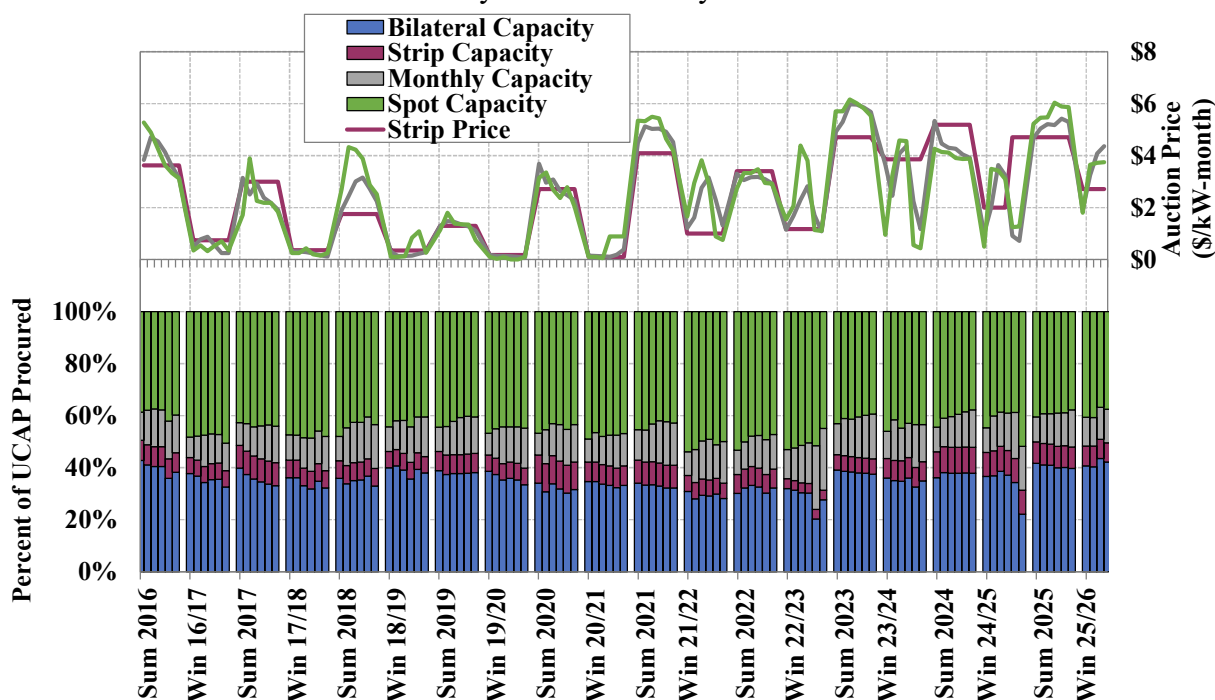


In addition to the changes that affect the NYCA capacity requirements (e.g., forecasted peak load and the Derating Factors), requirements in the local capacity zones can also be affected by changes in the Local Capacity Requirement that are unrelated to load changes.

Figure A-104: Capacity Procurement by Type and Auction Price Differentials

Figure A-104 describes the breakdown of capacity procured by mechanism (bilateral markets, strip auctions, monthly auctions and spot auctions) and the resulting prices for various auctions over the last ten Capability Years. Bilateral prices are not reported to the NYISO and are not included in this figure. The stacked columns correspond to the left vertical axis and indicate the percentage of total capacity procured via the four procurement methods for each month in a given Capability Period. The top panel of the chart (corresponding to the left vertical axis) shows the monthly prices for each of the spot, monthly and strip auctions since the Summer 2016 capability period on a dollar-per-kilowatt-month basis.

Figure A-104: Auction Procurement and Price Differentials in NYCA
May 2016 – February 2026



E. Cost of Reliability Improvement from Additional Capacity

An efficient capacity market would signal for capacity to locate where it is most cost-effective to improve system reliability. In this subsection, we discuss a framework for measuring capacity prices relative to this objective and evaluate the effectiveness of NYISO market at meeting it.

Since the inception of NYISO, the installed capacity requirements have been primarily based on resource adequacy criteria, which require sufficient capacity to maintain the likelihood of a load shedding event in the NYCA below the prescribed level (i.e., 1 day in 10 years). Hence, the capacity price in a particular location should depend on how much capacity at that location would reduce the likelihood of load shedding in NYCA. Since implementing downward-sloping capacity demand curves in 2004, NYISO has used the cost of new entry as the basis for placing the demand curve sufficiently high to allow a hypothetical new entrant to recover its capital costs over an assumed project life. Hence, capacity markets should provide price signals that reflect: the reliability impact and the cost of procuring additional capacity in each location.

The Cost of Reliability Improvement (“CRI”), which is defined as the cost of additional capacity to a zone that would improve LOLE by 0.001, characterizes the value of additional capacity in a zone and captures the two key factors that should be considered while determining capacity prices. Under an efficient market design, the CRI should be the same in every zone under long term equilibrium conditions. This will reduce the overall cost of maintaining reliability and direct investment to the most valuable locations. To achieve these efficient locational capacity prices, the market should procure amounts of capacity in each area that minimize the cost of satisfying the resource adequacy standard.

NYISO’s methodology for determining the LCRs (“Optimized LCRs Method”) seeks to minimize the total procurement cost of capacity under long term equilibrium while conforming to: (a) an LOLE of less than 0.1 days per year, (b) the NYSRC-determined IRM, and (c) transmission security limits (“TSL”) for individual Localities. The “Optimized LCRs Method” minimizes procurement costs (i.e., capacity clearing price times quantity) rather than investment costs (i.e., the marginal cost of supply in the capacity market). Minimizing procurement costs is inefficient because it does not necessarily select the lowest cost supply to satisfy reliability. Minimizing investment costs is efficient because it selects the lowest cost resources just as the energy and ancillary services markets select the lowest cost resources to satisfy load and ancillary services requirements.

Table A-14: Cost of Reliability Improvement

Table A-14 shows the CRI in each zone based on the long-term equilibrium that is modeled in the demand curve reset process. Under these conditions, each locality has a modest excess (known as its “Excess Level”) so that the system is more reliable than the 0.1 LOLE minimum criteria.³²³ An Excess Level is assumed so that the demand curve in each area is set sufficiently high to ensure the system never exceeds the 0.1 LOLE criteria. This modest excess results in an LOLE of 0.070 in the 2026/27 Capability Year. The table shows the following for each area:

- *Net CONE of Demand Curve Unit* – Based on the Net CONE curves filed by NYISO for the 2026/2027 Capability Year, converted to UCAP using the 2026/27 CAF values.
- *NYCA LOLE at Excess Level in Demand Curve Reset* – This is a single value for NYCA that is found by setting the capacity margin in each area to the Excess Level from the last demand curve reset.
- *LOLE from 100 MW UCAP Addition* – The estimated LOLE from placing 100 MW of additional UCAP in the area.³²⁴
- *Marginal Reliability Impact (“MRI”)* – The estimated reliability benefit (reduction in LOLE) from placing 100 MW of additional UCAP in the area. This is calculated as the difference between the NYCA LOLE at Excess Level and the LOLE from adding 100 MW of UCAP to the area.

³²³ The demand curve reset process is required by tariff to assume that the average level of excess in each capacity region is equal to the size of the demand curve unit in that region. The last demand curve reset assumed proxy units of approximately 200 MW (ICAP) in each area. For the MARS results discussed in this section, the base case was set to the “Level of Excess” in each area.

³²⁴ These values were obtained by starting with the system at Level of Excess with an LOLE of 0.070 and calculating the change in LOLE from a 100-MW perfect capacity addition in each area.

- *Cost of Reliability Improvement (“CRI”)* – The annual levelized investment cost for a 0.001 improvement in LOLE from placing capacity in the area.^{325, 326} This is calculated based on the ratio of the *Net CONE of Demand Curve Unit* to the *MRI* for each area.

Table A-14: Cost of Reliability Improvement
2026/27 Capability Year

Locality/Zone	Net CONE of Demand Curve Unit \$/kW-yr UCAP	NYCA LOLE at Excess Level	LOLE with 100 MW UCAP Addition	Marginal Reliability Impact $\Delta LOLE$ per 100MW/MS	Cost of Reliability Improvement \$ per 0.001 $\Delta LOLE$
NYCA					
A	\$92		0.063	0.0068	\$1.3
B	\$92		0.063	0.0068	\$1.3
C	\$92		0.063	0.0068	\$1.3
D	\$92		0.063	0.0068	\$1.3
E	\$92		0.063	0.0068	\$1.4
F	\$92		0.063	0.0068	\$1.3
G-J Locality					
G	\$88	0.070	0.063	0.0068	\$1.3
H	\$88		0.063	0.0068	\$1.3
I	\$88		0.063	0.0068	\$1.3
NYC					
J	\$232		0.063	0.0068	\$3.4
Long Island					
K	\$107		0.063	0.0066	\$1.6
Other Areas					
Staten Island (J3)	\$232		0.063	0.0068	\$3.4

Figure A-105 and Figure A-106 - Cost and CRI Curves in LCR Optimizer

Figure A-105 and Figure A-106 illustrate how the current design of the LCR Optimizer contributes to volatility and inefficient outcomes. Both figures compare the marginal cost of capacity for the 2022/23 capability year based on two formulations:

- *Investment cost minimization* – This uses the Net CONE curves to represent marginal investment cost.³²⁷ These are shown in the top panel of each figure, which are monotonic upward-sloping marginal cost curves. These investment costs include the categories of costs that could be saved by procuring capacity more efficiently.

³²⁵ For example, for Zone F: $\$92/\text{kW-year} \times 1000\text{kW}/\text{MW} \div (0.0068 \text{ LOLE change}/100\text{MW}) \times 0.001 \text{ LOLE change} = \1.3 million .

³²⁶ Note, this value expresses the marginal rate at which LOLE changes from adding capacity when at the Excess Level. However, the actual cost of improving the LOLE by 0.001 might be somewhat higher since the impact of additional capacity tends to fall as more capacity is added at a particular location.

³²⁷ This is because the first-order conditions of the investment cost minimizing optimization problem include the Net CONE functions in each location.

- *LCR Optimizer formulation* – The marginal cost function at each location is derived assuming NYISO minimizes overall consumer costs.³²⁸ In the bottom panel of each figure, these non-monotonic marginal cost curves are shown to be discontinuous with irregular *downward* steps because the marginal consumer cost is strongly affected by slight changes in the steepness of Net CONE curve steps.³²⁹ For example, Figure A-111 shows that if the LCR in Zone J rises from 80.6 to 86.6 percent, the Net CONE rises just 4.4 percent, while the corresponding marginal consumer cost curve falls by 24.0 percent.

For each locality in each formulation, Figure A-105 shows the marginal cost of capacity per kW-year, while Figure A-106 shows the CRI curve. Each CRI curve equals the marginal cost curve from Figure A-105 divided by the marginal reliability impact of capacity in the locality.^{330 331} Thus, the CRI curve is the marginal cost of capacity per unit of LOLE improvement. The red diamonds indicate simulated LCRs determined using the LCR Optimizer cost function.

In Figure A-106, the bottom panel illustrates how the LCR Optimizer seeks a solution that equalizes CRI values across localities while satisfying the LOLE criterion, IRM, and TSLs.³³² However, because the Optimizer calculates the marginal cost of capacity based on consumer costs, it relies on CRI curves that are not monotonic and may produce similar values for multiple different LCRs. Ultimately, this raises the following concerns:

- Because the Optimizer computes each locality’s CRI in a way that produces the same value at multiple different LCRs, changes in model assumptions may lead to unpredictable and volatile changes in LCRs.
- The Optimizer does not produce efficient LCRs. In this example, the Zone K LCR is inefficiently low and the G-J Locality LCR is inefficiently high.

³²⁸ Marginal cost functions are derived from first order conditions of the consumer cost minimization problem.

³²⁹ Monotonicity is an important because it allows a solver to find the unique cost-minimizing solution more quickly, while non-monotonic cost functions make the problem non-convex and more difficult to solve.

³³⁰ We simulated MRI curves based on MARS-derived LOLE values for various combinations of LCRs. Because all possible combinations of LCRs cannot be feasibly tested, the MRI curves are approximate.

³³¹ This chart assumes a fixed IRM at the 2022/23 level of 119.6 percent. With a constant IRM, the CRI in locality X is equal to: (marginal cost of adding capacity in locality X – marginal cost of adding capacity in NYCA) / (MRI of locality X – MRI of NYCA).

³³² The simulated LCRs indicated by the red diamonds in Figure A-106 differ slightly from the actual 2022/23 LCRs because MRI was simulated using a limited number of MARS data points.

Figure A-105: Optimizer Cost Curves vs. Net CONE Curves
Example based on 2022/2023 Capability Year

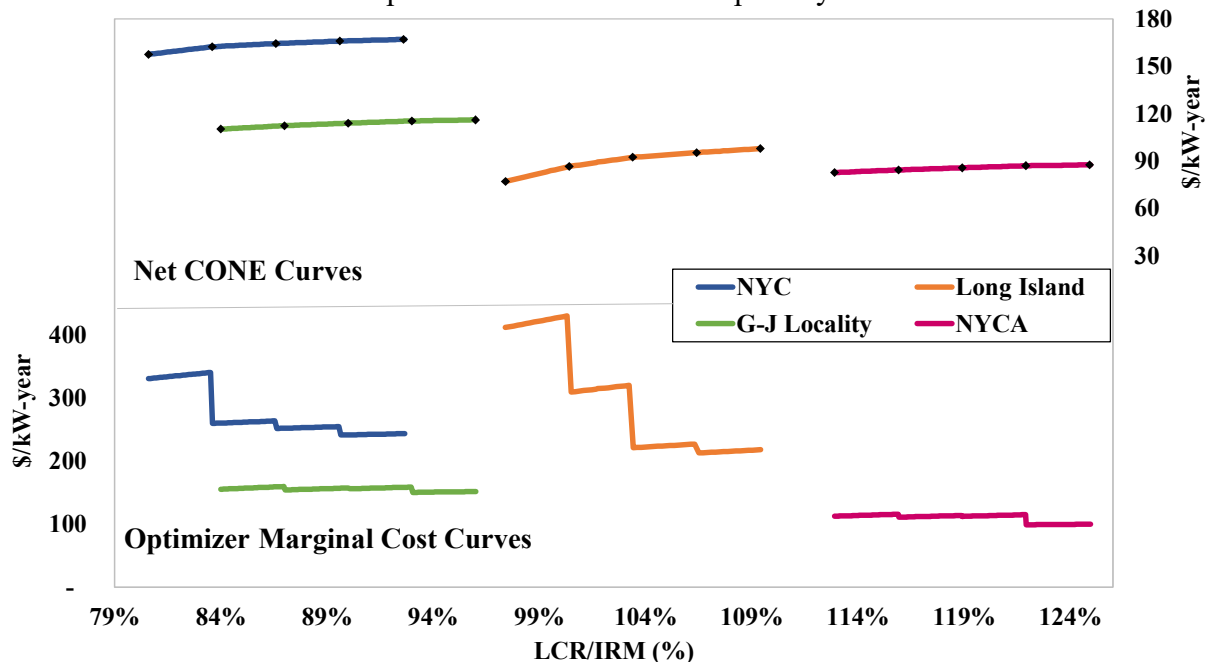
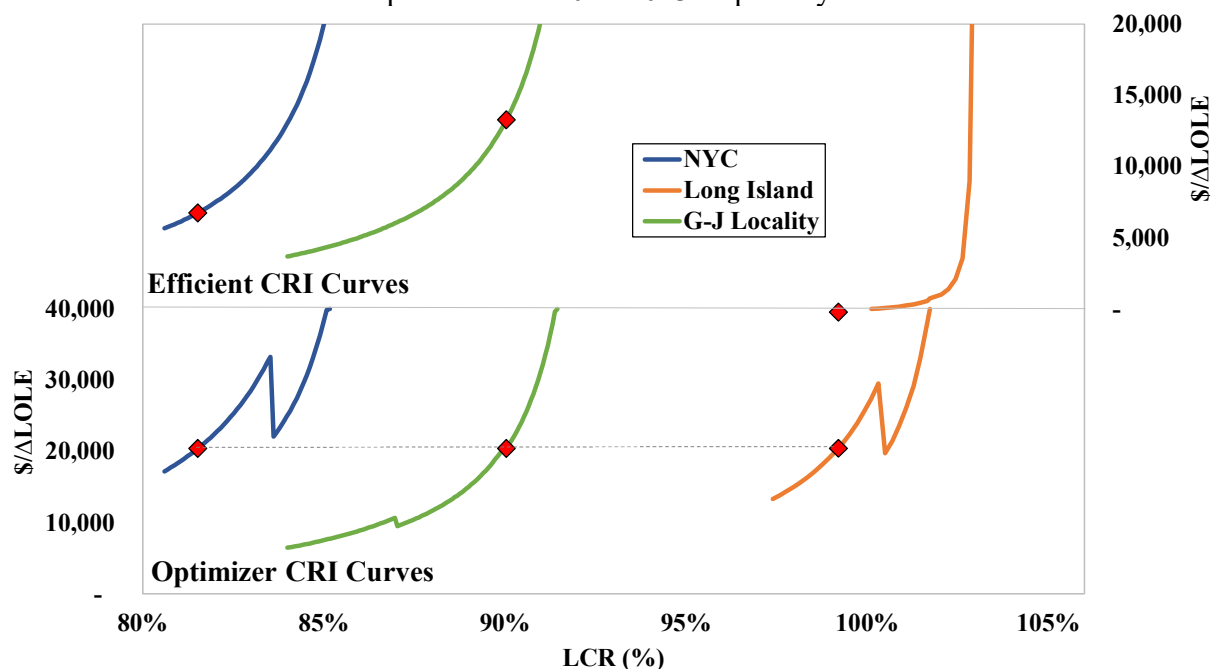


Figure A-106: Optimizer CRI Curves vs. Efficient CRI Curves
Example based on 2022/2023 Capability Year



By contrast, the top panel shows how calculating the marginal cost of capacity based on investment costs produces uniformly upward sloping CRI curves, which allow the optimal solution to be found more quickly and reliably.

F. Details of Granular Capacity Zones Recommendation

Section IX.B discusses shortcomings of NYISO’s current use of four capacity market pricing regions and introduces our proposals to improve locational pricing signals (Recs. 2022-4 and 2025-3). This subsection elaborates on our proposal to implement more granular capacity zones and a dynamic process for updating the zones (2022-4).

Overview of Granular Capacity Zones Proposal

We recommend that NYISO establish a dynamic process to update capacity zones used to set prices (Recommendation 2022-4).³³³ This would expand the number of capacity zones and replace the existing zone creation process, while keeping the structure of the capacity market largely intact. Its primary effect would be to: (1) discount capacity payments to export-constrained areas that are overpriced, and (2) allow for reliability needs to be efficiently reflected in prices as they emerge. We discuss this proposed process for establishing capacity zones and requirements in this subsection:

6. Represent all major capacity deliverability bottlenecks in the resource adequacy model;
7. Designate capacity zones as import or export-constrained capacity zones based on the configuration of binding transmission constraints in the resource adequacy model;
8. Determine ICAP requirements for all import and export zones;
9. Establish import and export demand curves for use in the Spot Auction;
10. Apply a financial Capacity Constraint Pricing Credit or Charge to capacity payments of resources that positively or negatively impact aggregate deliverability between regions.

1. Represent all major deliverability bottlenecks in the resource adequacy model

NYISO’s resource adequacy model GE-MARS is a probabilistic simulation of load shedding risk that accounts for transmission limits between regions. It is used in the IRM and LCR studies to determine the ICAP Requirements in the capacity market. The representation of the NYCA region in the IRM and LCR studies includes areas based on the eleven historic load zones (zones A-K) with transmission limits between them.³³⁴ In reality, there are also internal bottlenecks within the load zones that limit deliverability of capacity. For example, recent deliverability studies indicate that binding export constraints exist in Staten Island and eastern and central Long Island, with deliverability headroom tightening in other areas. New intra-zonal constraints

³³³ In this subsection, a “capacity zone” refers to a pricing zone with a capacity market demand curve (such the NYCA and G-J Locality zones), a “region” refers to a part of a capacity zone that may have a distinct price (such as the Rest of State and GHI regions within the NYCA and G-J Locality), and an “area” refers to a part of the system that is separated from other areas by transmission bottlenecks. Areas are represented as “bubbles” in the GE-MARS topology and include (but are not limited to) the 11 historic load zones (A-K).

³³⁴ Transmission limits between Staten Island and New York City are modeled indirectly by a dynamic limit on PJM imports to Zone J via the “J3” area.

may arise over time and pricing capacity in these areas efficiently requires that they be represented in the resource adequacy model so the bottled capacity can be quantified.

Hence, the annual process used by NYISO and NYSRC to update the transmission topology for the IRM study could identify intra-zonal capacity bottlenecks based on power flow simulation and represent those constraints in GE MARS. New constraints could be represented by modeling additional areas in MARS with transmission interfaces between adjacent areas. Not all constraints detected this way will lead to binding constraints in MARS since the probabilistic outcomes of MARS will differ from a deterministic power flow assessment.³³⁵ It may be necessary to establish a threshold for representation of a new area in MARS so that only bottlenecks that affect a significant amount of capacity are represented.

2. Designate capacity zones as import or export-constrained based on configuration of binding constraints in the resource adequacy model

After the previous step identifies individual capacity zones and transmission interfaces, individual zones can be classified as either import or export zones:

- An import zone consists of one or more areas whose ability to import capacity is constrained during all or some hours of reliability risk. Import zones would function like NYISO’s existing capacity zones and could be nested within other import zones. For example, a constraint on the UPNY-CONED interface between zones H and G could lead to an import zone within the existing G-J locality consisting of zones H-J.
- An export zone is an area that has surplus capacity facing export bottlenecks to a “parent” region. When exports from an area to its parent region are constrained in MARS, an export zone should be created. Each export zone would be nested inside of an import zone. The process for compensating capacity in these zones is discussed further below.

This process would largely eliminate the need to assign mandatory SDUs to new projects seeking CRIS. Instead, all resources in a bottled region receive lower capacity prices reflecting the value of capacity in that region. Informational studies could be regularly conducted by NYISO to inform developers of potential new zones likely to emerge in the coming years based on the locations of projects in the interconnection queue. Developers entering bottled regions could elect to fund transmission upgrades and earn financial rights allowing them to benefit from the capacity value of the upgrades.

3. Determine ICAP Requirements for all import and export zones

³³⁵ A difference in the value of capacity between zones can be observed by calculating the Marginal Reliability Impact (MRI) of each zone when the system is modeled at the target reliability criteria. A difference in MRI between zones indicates a binding transmission constraint between those zones.

NYISO would continue to use the LCR Optimizer to establish LCRs for each import zone while satisfying the minimum TSL-based floors.³³⁶ This method implicitly accounts for both the cost and the marginal reliability benefit of procuring capacity in each region as the amount procured changes. As a result, the optimized LCRs will maximize procurement in lower-cost regions until transmission constraints begin to limit the effectiveness of capacity there.³³⁷

Under this process, the ICAP Requirements of import zones would represent the targeted minimum amount of capacity to be maintained in that zone. For export zones, the ICAP Requirement would represent the maximum amount of capacity that would be fully deliverable to the parent zone.³³⁸ The requirements of export zones would be set such that any additional capacity will cause the export constraints to bind during critical hours in MARS.³³⁹ The requirements of export zones would be included in the requirements of the parent import zone.

4. Establish import and export demand curves for use in the Spot Auction

Currently, the capacity market’s spot auction is cleared using demand curves that are designed to encourage new entry when capacity in a zone approaches that zone’s requirement. Under our proposal, this process would remain largely unchanged for import zones. Each import zone would clear based on its own demand curve, and each supplier would receive the highest clearing price among import zones to which it belongs.

In the current framework, certain key demand curve parameters including the net cost of new entry (Net CONE) and demand curve length are determined in the Demand Curve Reset (DCR) process every four years. This process may not anticipate every import zone and determine parameters for it. Hence, it will be necessary to: (1) determine Net CONE values for a set of locations and use the Net CONE value of the “parent” zone for any new import zone that is created before the next DCR, and (2) establish a process to automatically determine demand curve lengths for new zones based on the marginal reliability impact (MRI) of surplus capacity.

³³⁶ The NYCA IRM is currently determined prior to the LCRs by the NY State Reliability Council (NYSRC) using a different process from the LCR Optimizer. If NYISO modifies the objective function of the LCR Optimizer to minimize investment costs (rather than consumer costs), it would be more efficient to determine the IRM and LCRs simultaneously using the LCR Optimizer, but this is not necessary for Recommendation #2022-4 to be implemented.

³³⁷ This assumes that NYISO modifies the objective function of the LCR Optimizer to minimize investment costs (rather than consumer costs) before implementing Granular Capacity Zones. NYISO has signaled that it intends to do so.

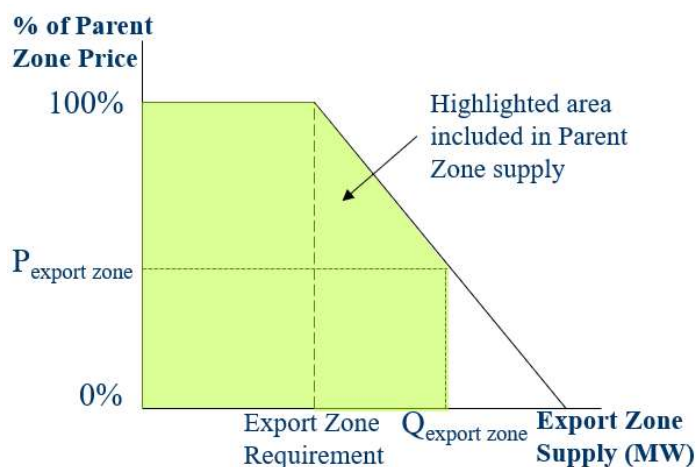
³³⁸ To determine the amount of fully deliverable capacity from the exporting zone, the MRI of the exporting zone for a MARS case at the Level of Excess would be compared to the MRI of the downstream portion of the parent zone. If the MRI of the exporting area was lower by at least a modest (~5 percent) amount, the exporting zone would be modeled as a distinct capacity pricing location, but the export-constrained zone would not need its own Net CONE estimate.

³³⁹ This implies export zone capacity should have an MRI very close to that of the parent zone when capacity is equal to the requirement, and a declining MRI relative to the parent zone if additional capacity were added.

For export zones, we recommend creating export zone demand curves whose purpose is to discount payments to resources in bottlenecked areas. Each export zone demand curve would determine the percentage of the parent zone's price to be paid to resources in the export zone, as a function of the export zone's capacity surplus. Hence, export zone demand curves would not require a separate Net CONE estimate. Capacity in the export zone that is fully deliverable during critical hours should be counted towards the requirement of the parent zone when clearing the auction, and partially deliverable capacity should be counted on a discounted basis.

Figure A-107 illustrates this concept. When capacity in the export zone is less than or equal to the export zone requirement, payments are equal to the parent zone price because all of the export zone's capacity is deliverable. Surplus beyond the export zone's requirement causes the price paid to fall as a percentage of the parent zone price. The export zone demand curve has a slope because export zone capacity may be bottlenecked in some, but not all hours when surpluses are modest. The area under the export zone demand curve (which counts all capacity up to the export zone requirement and discounts any surplus) is then counted as supply towards meeting the import zone's requirement.

Figure A-107: Export Zone Demand Curve



5. Apply a Capacity Constraint Pricing (CCP) component to capacity payments of resources that positively or negatively affect transmission limits between zones.

Each resource in a capacity zone is currently paid the same capacity price even though not all resources within a zone contribute equally to loading of constraints affecting that zone. Hence, as part of Recommendation 2022-4, we propose a financial Capacity Constraint Pricing Credit or Charge that modifies the capacity payments of resources that increase or decrease the total amount of capacity deliverable over a binding constraint. These variable effects of different resources on the constraint are reflected in their generation shift factors (GSFs) when NYISO performs a transfer limit analysis. For example, generation added in an export-constrained area at a bus with a very low or negative GSF on the constrained facility may increase the total deliverable capacity by displacing other resources with higher GSFs on the constraint.³⁴⁰

Efficient prices reward investment at locations that improve deliverability and discourage investment at locations that diminish deliverability. We propose NYISO apply a CCP credit or charge to reward or penalize resources that modify a zone's import or export limit. We propose the following process:

³⁴⁰ In situations where the GSFs do not accurately approximate generators' impact on the relevant constraint, such as for voltage-based transfer limits, other methods may be used.

1. Calculate a set of generator CCPs for each interface between capacity zones (e.g., between two nested Import Zones or between and Import and Export zone). The CCP Factor is the amount by which an additional MW of output at a generator's location would change the total deliverable capacity, either positively or negatively. The CCP Factors are specific to each interface between zones.
2. Calculate the price difference across each interface between nested capacity zones. This is the difference in capacity price between the capacity zones connected by the interface.
3. Each generator earns a total capacity payment equal to: (a) its UCAP MW times the sum of the zonal Capacity Price plus (b) the generator's unique CCP Credit/Charge.

The following section of this Appendix provides an example of the calculation of CCP Factors and generator payments. The CCP Credit/Charge would produce substantial benefits by providing much more accurate locational incentives in each capacity zone. This is key because generators in any fixed capacity zone will have different effects on key constraints. It will also mitigate issues that arise when new capacity zones are not created to reflect key deliverability constraints.

6. Conclusions Regarding the Granular Capacity Zones Proposal

The current zonal structure of the capacity market does not capture important distinctions in the value of capacity by location and will become increasingly disconnected from the needs of the system over time. As a result, the capacity market will send inefficient signals for investment and retirements and the flawed deliverability process will continue to be a major barrier to new investment in certain locations. In this subsection, we have proposed a process to define new capacity zones that will better signal where additional capacity is and is not valuable. In particular, this proposal will:

- Avoid over-compensating resources in bottled areas and facilitate retirement of non-deliverable capacity;
- Reduce capacity costs because LCRs will not rise to compensate for bottled capacity; and
- Attract and retain capacity in locations where it is more valuable to the system

G. Mathematical Example of Capacity Constraint Pricing (CCP)

The previous section of this report discusses our proposal to create a more granular set of capacity pricing zones. As part of this proposal, we recommend applying a financial Capacity Constraint Pricing (CCP) Credit or Charge to capacity payments of resources that positively or negatively impact aggregate deliverability between zones. We propose the following process to determine generator payments:

- Calculate a set of generator Capacity Constraint Pricing Factors (CCP Factors) for each interface between nested capacity zones (e.g. between two nested Import Zones or between and Import and Export zone). The CCP Factor is the amount by which an additional MW of output at a generator's location would cause the total amount of capacity deliverable over the interface to change. Each generator would be assigned a

CCP Factor for each interface between the generator’s zone and an adjacent zone. The CCP Factor can be positive, negative, or zero, indicating that the generator improves, harms or does not affect the interface limit.

- Calculate the zonal price difference for each interface between nested capacity zones. This is the difference in capacity prices between the zones connected by the interface.
- Each generator earns a total capacity payment equal to its UCAP MW times the sum of the zonal Capacity Price and generator’s unique CCP Credit/Charge. The CCP Credit/Charge is calculated as the sum of the zonal price difference times the generator’s CCP Factor for each constraint.

This subsection provides an example of how CCP Factors would be calculated and how they would affect resources’ total capacity-related compensation.

Capacity Constraint Pricing Factors with One Transmission Constraint

Table A-15 through Table A-17 provide an illustrative example in which flows on the interface between an export zone and an import zone are limited by one constrained facility. The example includes five generators (units A-E), each of which have a different generator shift factor (GSF) on the most constrained facility that limits transfers over the interface (Line 1). The generators are classified as belonging to the import zone or export zone based on the direction of their GSF on Line 1. Load is assumed to be only in the import zone.

To estimate the interface limit from the export zone to the import zone, the output of all five generators is adjusted until the maximum amount of combined output in the export zone is reached, subject to constraints: (1) total generation equals total load, (2) flows on Line 1 cannot exceed its limit of 45 MW, and (3) each generator’s output cannot exceed its maximum capacity. Flows on Line 1 are calculated as the product of each generator’s scaled output level and its GSF (plus the load times the load shift factor). The maximum export zone output of 232.5 MW is reached by maximizing output from generators with lower GSFs on Line 1 (Units A, B and E) and reducing output from generators with higher GSFs on Line 1 (units C and D).

The export zone has 7.5 percent more capacity than its export limit (row (i)). We assume a 20 percent export zone demand curve length and a \$10 per kW-month price in the import zone. Based on the export demand curve proposal discussed earlier this results in a capacity price of \$6.2 per kW-month (62 percent of the import zone price) in the export zone. The discounted price in the export zone reflects the reduced value of capacity there due to the presence of a binding transmission constraint.

Finally, each unit’s CCP Factor (row (n)) is calculated by increasing the maximum capacity of that unit by 1 MW and then recalculating the interface limit (row (g)) by adjusting all units to maximize output while maintaining the load balance and Line 1 limit. Units C and D each have a CCP Factor of zero, because additional capacity at these locations would not change the maximum amount of output that can occur in the export zone without violating the limit of Line 1 (however, additional capacity at unit C would increase the capacity surplus in the export zone and lower its price). Units A, B and E have positive CCP Factors because these units have low GSFs on Line 1, so additional capacity at these locations would allow a larger total amount of

output in the export zone. For example, an additional MW at Unit A would cause the interface limit of 232.5 MW to increase by 0.5 MW.

Table A-15: Line and Unit Characteristics in One Line CCP Factor Example

Line Limits

Line 1 Limit (MW)	45
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Unit Characteristics and Output Levels

Unit		A	B	C	D	E	Load	Total
Net Gen Capacity (MW)	(a)	100.0	100.0	50.0	100.0	100.0	-400.0	50.0
GSF	(b)	0.10	0.12	0.30	-0.10	-0.20	-0.10	
Zone	(c)	Export	Export	Export	Import	Import	Import	
Output Scalar	(d)	1.00	1.00	0.65	0.67	1.00	1.00	
Scaled Net Gen (MW)	(e) = (a) * (d)	100.0	100.0	32.5	67.5	100.0	-400.0	0.0
Impact on Line 1 (MW)	(f) = (b) * (e)	10.00	12.00	9.75	-6.75	-20.00	40.00	45.00

Table A-16: Calculation of Capacity Prices in One Line CCP Factor Example

Calculation of Zonal Prices

Export Zone Limit (MW)	(g) = Sum of (e) in export zone	232.50
Export Zone Supply (MW)	(h) = Sum of (a) in export zone	250.00
Export Zone Surplus	(i) = (h) / (g) - 1	7.5%
Export Demand Curve Length	(j)	20%
Import Zone Price (\$/kW-mo)	(k)	10.00
Export Zone Price (\$/kW-mo)	(l) = (k) * (1 - (i) / (j))	6.24
Zonal Price Difference	(m) = (k) - (l)	3.76

Table A-17: CCP Factors in One Line CCP Factor Example

CCP Factors and Credit/Charge		A	B	C	D	E
CCP Factor	(n)	0.50	0.45	0.00	0.00	0.25
Zonal Capacity Price	(o) = (k) or (l)	6.24	6.24	6.24	10.00	10.00
CCP Credit	(p) = (m) * (n)	1.88	1.69	0.00	0.00	0.94
Total Payment	(q) = (o) + (p)	8.12	7.93	6.24	10.00	10.94

CCP Factors with Two Transmission Constraints

Table A-18 through Table A-20 provide an illustrative example in which flows on the interface between an export zone and an import zone are limited by more than one constrained facility. It uses the same assumptions as the example with one line shown in Table A-15, but includes a second line (Line 2) with a limit of 135 MW. Each generator's GSF on Line 2 is not necessarily the same as its GSF on Line 1. To determine the interface limit from the export zone to the import zone, each generator's output level is adjusted to maximize output in the export zone

while maintaining the load balance and respecting the limits of *both* Line 1 and Line 2. In this example, the optimal export limit of 230.2 MW occurs when both lines are constrained.

Compared to the example with one line, fewer units have positive CCP Factors. This is because additional capacity at a generator's location will not improve the interface limit unless the generator has a low GSF on *both* constraints. For example, raising output at the location of Unit A would allow for less loading on Line 1 (which resulted in a positive CCP Factor in the example with one line), but cause additional loading on Line 2, so it will not allow more total generation in the export zone. Unit B has a GSF below the 'marginal' GSF on both constraints, so it has a positive GSF and received a CCP Credit which increases its capacity payment.

Table A-18: Line and Unit Characteristics in Two Line CCP Factor Example

Line Limits

Line 1 Limit (MW)	45
Line 2 Limit (MW)	135

Unit Characteristics and Output Levels

Unit		A	B	C	D	E	Load	Total
Net Gen Capacity (MW)	(a)	100.0	100.0	50.0	100.0	100.0	-400.0	50.0
GSF on Line 1	(b)	0.10	0.12	0.30	-0.10	-0.20	-0.10	
GSF on Line 2	(c)	0.32	0.25	0.30	-0.30	-0.30	-0.30	
Zone	(d)	Export	Export	Export	Import	Import	Import	
Output Scalar	(e)	0.95	1.00	0.70	0.70	1.00	1.00	
Scaled Net Gen (MW)	(f) = (a) * (e)	95.3	100.0	34.8	69.8	100.0	-400.0	0.0
Impact on Line 1 (MW)	(g) = (b) * (f)	9.5	12.0	10.5	-7.0	-20.0	40.0	45.0
Impact on Line 2 (MW)	(h) = (c) * (f)	30.5	25.0	10.5	-21.0	-30.0	120.0	135.0

Table A-19: Calculation of Capacity Prices in Two Line CCP Factor Example

Calculation of Zonal Prices

Export Zone Limit (MW)	(i) = Sum of (f) in export zone	230.2
Export Zone Supply (MW)	(j) = Sum of (a) in export zone	250.0
Export Zone Surplus	(k) = (j) / (i) - 1	8.6%
Export Demand Curve Length	(l)	20%
Import Zone Price (\$/kW-mo)	(m)	10.00
Export Zone Price (\$/kW-mo)	(n) = (m) * (1 - (k) / (l))	5.69
Constraint Shadow Price	(o) = (m) - (n)	4.31

Table A-20: CCP Factors in Two Line CCP Factor Example

CCP Factors and Credit/Charge		A	B	C	D	E
CCP Factor	(p)	0.00	0.11	0.00	0.00	0.02
Zonal Capacity Price	(q) = (m) or (n)	5.69	5.69	5.69	10.00	10.00
CCP Credit	(r) = (o) * (p)	0.00	0.47	0.00	0.00	0.09
Total Payment	(s) = (q) + (r)	5.69	6.16	5.69	10.00	10.09

This proposal has two primary advantages. First, it differentiates payments of resources based on their ability to improve deliverability across constrained transmission interfaces. This will improve the efficiency of capacity market signals for motivating resources to enter at the most valuable locations and avoid less-valuable locations. Second, the use of CCP Factors reduces the risk that a resource will receive inefficient capacity payments due to being grouped into a capacity zone to which it does not fully belong. For example, a resource that is located in an export zone but has a very low GSF on the constraint that determines that zone’s export interface limit would be compensated under this proposal at a price similar to resources in the import zone. A resource could incur a CCP Charge (a negative adjustment to its capacity payment) if it is defined as belonging to an import zone but contributed to increased loading on a constrained interface into that zone.

The CCP Charge/Credit for generators that decrease/increase transfer capability that affects resource adequacy assessment is a particular type of Financial Capacity Transfer Right (“FCTR”). FCTRs can also be used to compensate merchant transmission investors that construct facilities that increase transfer capability that improves resource adequacy.

VII. NET REVENUE ANALYSIS

Revenues from the energy, ancillary services, and capacity markets provide the signals for investment in new generation and the retirement of existing generation. The decision to build or retire a generation unit depends on the expected net revenues the unit will receive. Net revenue is defined as the total revenue (including energy, ancillary services, and capacity revenues) that a generator would earn in the New York markets less its variable production costs.

If there is not sufficient net revenue in the short-run from these markets to justify entry of a new generator, then one or more of the following conditions exist:

- New capacity is not needed because sufficient generation is already available;
- Load conditions are below expectations due to mild weather or reduced demand, leading to lower energy prices than expected; and
- Market rules or conduct are causing revenues to be reduced inefficiently.

Alternatively, if prices provide excessive revenues in the short-run, this would indicate a shortage of capacity, unusually high load conditions, or market rules or conduct resulting in inflated prices. Therefore, the evaluation of the net revenues produced from the NYISO's markets is one of our principal means for assessing whether the markets are designed to provide efficient long-run economic signals.

In this section, we estimate the net revenues the markets would have provided to: new and existing gas-fired units (subsection A), existing nuclear plants (subsection B), new utility-scale solar PV, land-based wind, and offshore wind units (subsection C), and new battery storage (subsection D). Net revenues vary substantially by location, so we estimate the net revenues that each unit would have received at a number of locations across New York.

A. Gas-Fired and Dual Fuel Units Net Revenues

We estimate the net revenues from the market for four types of hypothetical gas-fired units:

- A new frame-type H-Class simple-cycle combustion turbine ("New CT") unit
- An existing Steam Turbine ("ST") unit
- An existing 10-minute Gas Turbine ("GT-10") unit,
- An existing 30-minute Gas Turbine ("GT-30") unit, and
- An existing small combined cycle (CC) unit of 1990s/early 2000s vintage.

We estimate the historical net energy and ancillary services revenues for gas-fired units in Long Island, the 345kV portion of New York City, the Hudson Valley Zone, and the Rest of State region. For energy and ancillary services revenues for units in the Capital Zone and Central Zone, energy prices are based on average zonal LBMPs. For Long Island, results are shown for the Caithness CC1 generator bus, which is representative of most areas of Long Island. For New York City, results are shown for the Ravenswood GT3/4 generator bus, which is representative

of most areas of the 345kV system in New York City.³⁴¹ For the Hudson Valley zone, results are shown for the average of LBMPs at the Roseton 1 and Bowline 1 generator buses, since these are representative of areas in the zone that are downstream of the UPNY-SENY interface. We also estimate historical capacity revenues based on spot capacity prices.

Table A-21 and Table A-22: Assumptions for Net Revenues of Fossil Fuel Units

Our net revenue estimates for gas-fired units are based on the following assumptions:

- All units are scheduled based on day-ahead prices, considering commitment costs, minimum run times, minimum generation levels, and other physical limits:
- ST units may sell energy, 10-minute spinning reserves, and 30-minute reserves; while CTs may sell energy and 10-minute or 30-minute non-spinning reserves. Small CC units are assumed not to sell reserves.
- CTs (including older gas turbines) are committed in real-time based on RTC prices.³⁴² CTs settle with the ISO according to real-time market prices and the deviation from their day-ahead schedule. To the extent that these combustion turbines are committed uneconomically by RTC, they may receive DAMAP and/or Real-Time BPCG payments. Consistent with the NYISO tariffs, DAMAP payments are calculated hourly, while Real-Time BPCG payments are calculated over the operating day.
- Online units are dispatched in real-time consistent with the hourly real-time LBMP and settle with the ISO on the deviation from their day-ahead schedule. However, for the ST unit, a limitation on its ramp capability is assumed to keep the unit within a certain margin of the day-ahead schedule. The margin is assumed to be 25 percent of UOL.
- We estimate the portion of revenue that would come from operating on oil for generators with dual-fuel capability. During hourly OFOs, GT and CC units are assumed to offer on oil and Dual-fuel STs are assumed to offer Min Gen on the least expensive fuel and to offer incremental energy on residual oil in the DAM.
- Regional Greenhouse Gas Initiative (“RGGI”) compliance costs are included.
- Fuel costs include a 6.9 percent natural gas excise tax for New York City units, a one percent gas excise tax for Long Island units, and transportation and other charges on top of the day-ahead index price. Intraday gas purchases are assumed to be at a premium due to gas market illiquidity and balancing charges, while intraday gas sales are assumed to be at a discount for these reasons.
- All peaking units incur a \$2.00/MWh cost when committed to provide operating reserves. This assumption is reflective of historical reserve market offers and is intended to represent costs incurred to make a generator available, secure fuel, and/or compensate for performance risks when providing reserves.

³⁴¹ Prices at locations on the 345 kV network in New York city often differ from those on the lower-voltage 138 kV network, which typically experiences more localized congestion.

³⁴² We assume a Frame unit is committed for an hour if the average LBMP in RTC at its node is greater than the applicable start-up and incremental energy cost of the unit for the full 2.5-hour RTC look-ahead period.

- Reserve providers on Long Island are paid the SENY reserve price, but NYISO limits the amount of reserves that are scheduled on Long Island due to export constraints to the rest of the region. Hence, we prorate reserve revenues of GTs on Long Island based on analysis of GT capacity that remains uncommitted when reserve commitment based on price would be economic. This results in an approximately 60-70 percent reduction of reserve revenues.

The tables below summarize the unit parameters and gas prices used in our analysis:

Table A-21: Gas-fired Unit Parameters for Net Revenue Estimates³⁴³

Characteristics	ST	GT-10	GT-30 (Older)	GT-30 (Newer)	Small CC	New CT (7HA.03)	New CT (7HA.02)
Summer Capacity (MW)	360	32	16	40	80	400	365
Winter Capacity (MW)	360	40	20	40	85	417	388
Heat Rate (Btu/kWh)	10,000	15,000	17,000	10,600	8,800	9,000	9,220
Min Run Time (hrs)	24	1	1	1	6	1	1
Variable O&M (2024\$/MWh)	\$10.7	\$5.4	\$6.6	\$7.2	\$5.9	\$3.0	\$2.6
Startup Cost (2024\$)	\$7,185	\$1,436	\$621	\$0	\$13,000	\$23,100	\$23,000
Startup Cost (MMBTU)	3500	50	60	2400	1100	376	240
EFORd	5.1%	10.5%	19.7%	19.7%	10.0%	4.1%	4.1%

Table A-22: Gas and Oil Price Indices and Other Charges by Region³⁴⁴

Region	Gas Price Index	Transportation & Other Charges (\$/MMBTU)			Intraday Premium/ Discount
		Natural Gas	Diesel/ ULSD	Residual Oil	
West	April - November:				
	Tennessee Zn 4 - 200 Leg	\$0.27	\$2.00	\$1.50	10%
	December - March:				
	Niagara				
Capital	Iroquois Zn 2	\$0.27	\$2.00	\$1.50	10%
Hudson Valley	Iroquois Zn2	\$0.27	\$1.50	\$1.00	10%
New York City	Transco Zn6	\$0.20	\$1.50	\$1.00	20%
Long Island	Iroquois Zn 2	\$0.25	\$1.50	\$1.00	30%

Figure A-108 to Figure A-111: Net Revenues Estimates for Fossil Fuel Units

The following four figures summarize our net revenue and run hour estimates for dual-fuel units in various locations across New York. They also indicate the levelized Cost of New Entry (CONE for new units or going-forward cost (GFC) for existing units. Net revenues and CONE values are shown per kW-year of Summer Installed Capability. Net revenues from the sale of energy in the day-ahead market are shown separately for hours when the unit would operate on gas and hours when it would operate on fuel oil:

³⁴³ The parameters for the new CT are based on the most recent NYISO Demand Curve Reset study. We use the 7HA.02 technology for Long Island and the 7HA.03 technology for all other areas.

³⁴⁴ The analysis assumes that the units in New York City region would switch from Transco Zn6 to Iroquois Zn2 when the Transco Zn6 pipeline is congested.

Figure A-108: Net Revenue & Cost for New Frame CT Units
2023-2025

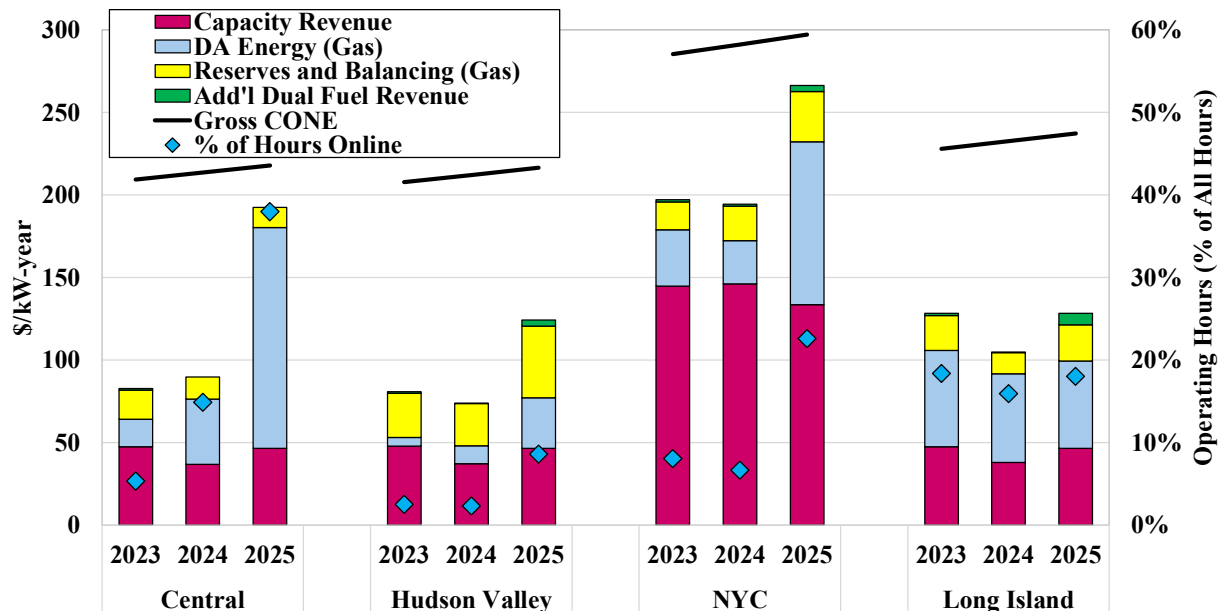


Figure A-109: Net Revenue & Cost for Existing Fossil Units in Rest of State
2023-2025

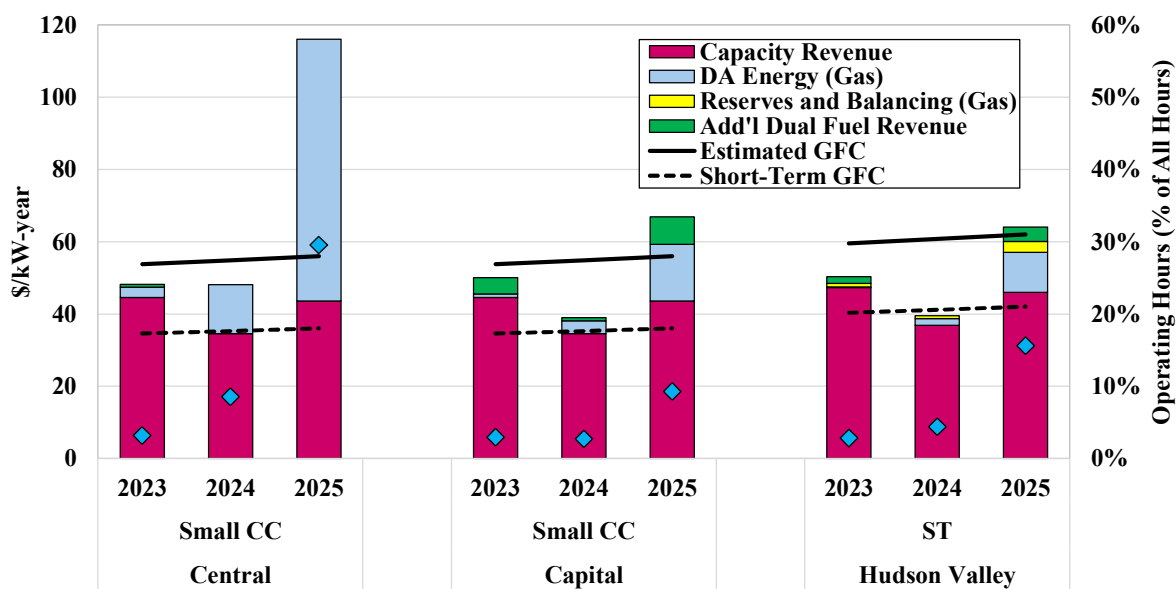


Figure A-110: Net Revenue & Cost for Existing Fossil Units in New York City
2023-2025

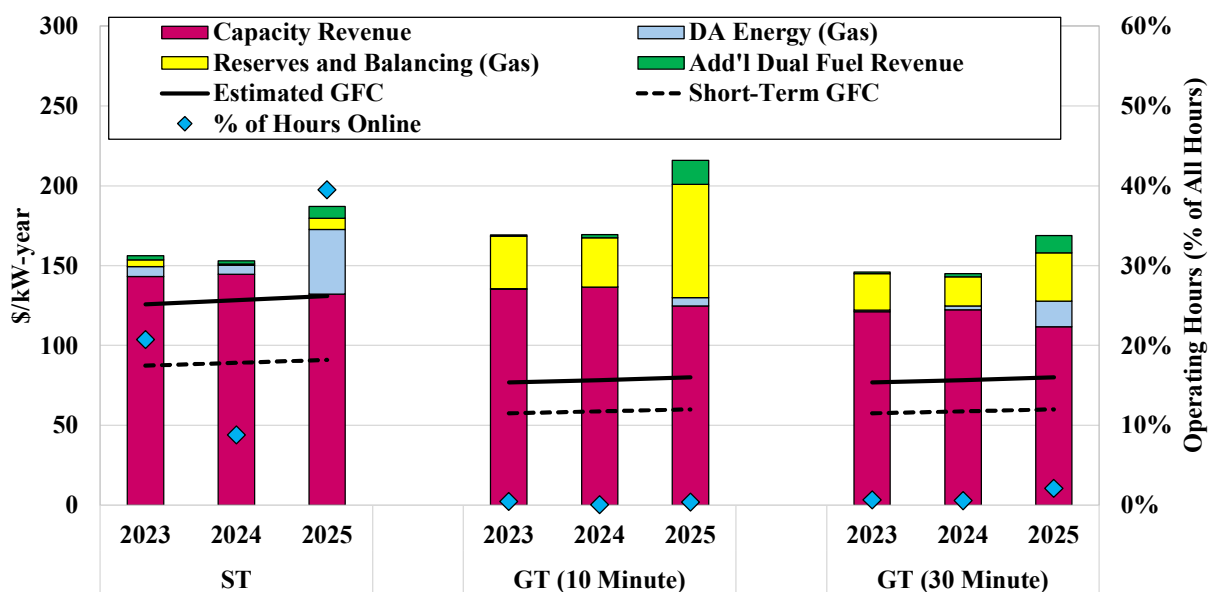
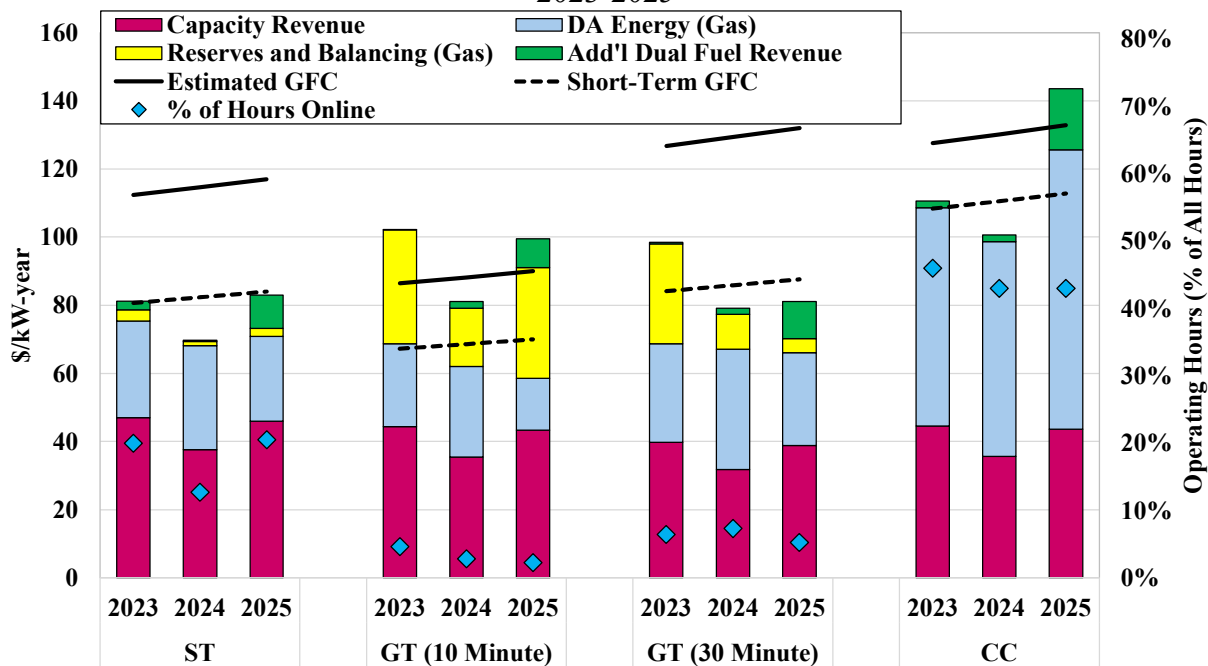


Figure A-111: Net Revenue & Cost for Existing Fossil Units in Long Island
2023-2025



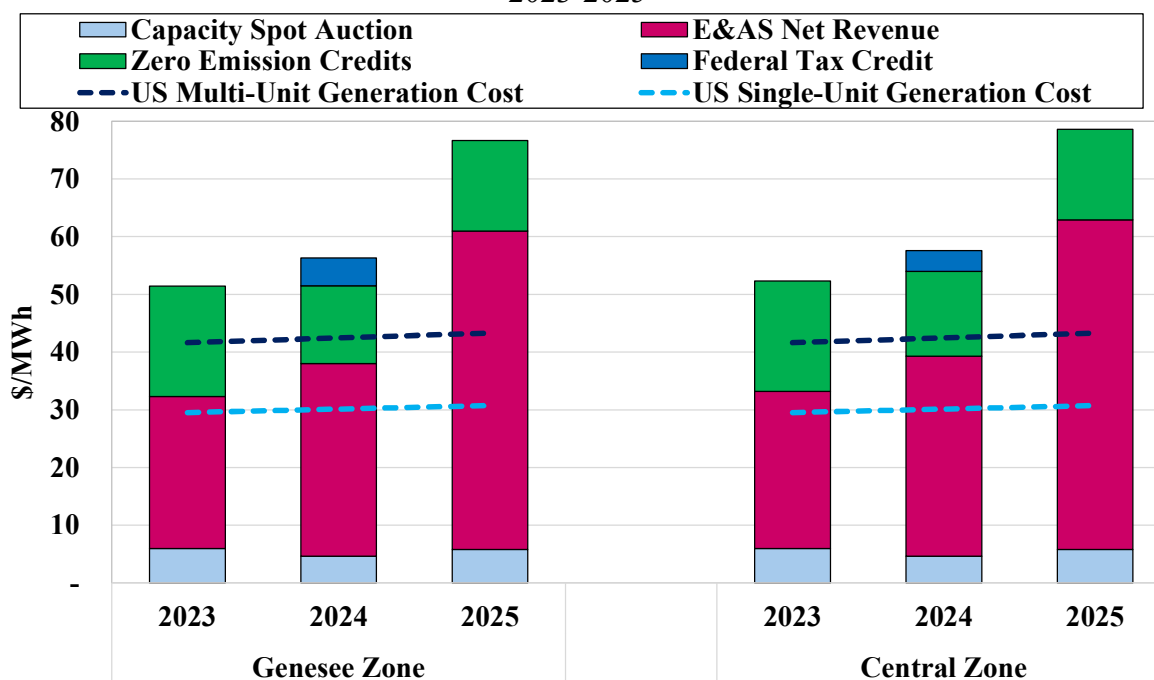
B. Nuclear Unit Net Revenues

We estimate the net revenues the markets provide to the nuclear plants in the Genesee and Central Zones. The estimates are based on LBMPs at the Ginna bus (for Genesee), and the Fitzpatrick and Nine Mile Unit 1 buses (for Central Zone).

Figure A-112: Net Revenues for Nuclear Plants

Figure A-112 shows the net revenues and the US-average operating costs for the nuclear units from 2023 to 2025. Estimated net revenues assume that nuclear plants are scheduled day-ahead and only sell energy and capacity. Nuclear units are assumed to earn energy revenues throughout the year except during periods of forced outages and outages related to refueling. We assumed an EFORD of two percent and a capacity factor of 67 percent during March and April to account for reduced output during refueling.³⁴⁵

**Figure A-112: Net Revenue of Existing Nuclear Units
2023-2025**



The costs of generation (including O&M, fuel, and capex) for nuclear plants are highly plant-specific and vary significantly based on several factors that include number of units at the plant, technology, age, and location. Our assumptions for operating costs for single-unit and larger nuclear plants are based on observed average costs of nuclear plants in the US in 2023.³⁴⁶ The nuclear units located in upstate zones are eligible for additional revenue in the form of Zero

³⁴⁵ The refueling cycle for nuclear plants is typically 18-24 months. We assume a reduced capacity factor in March and April every year to enable a year over year comparison of net revenues.

³⁴⁶ The average cost of operation of nuclear plants in the US are based on NEI/EUCG reports and presentations. See [here](#).

Emission Credits (“ZECs”).³⁴⁷ The ZEC price is fixed in two-year tranches by the NYPSC and was \$21.38 per MWh for the period April 2021 to March 2023, \$18.27 per MWh for the period April 2023 to March 2025, and \$14.76 per MWh for the period April 2025 to March 2027. Beginning in 2024, nuclear facilities qualified for federal incentives under section 45U of the Inflation Reduction Act, with a maximum rate of \$15 per MWh reduced by plant revenues in excess of \$25 per MWh (\$2022). State ZEC payments are contractually reduced by the amount of these federal incentive payments.

C. Renewable Units Net Revenues

We estimate the net revenues the markets would have provided to utility-scale solar PV in the Central and Capital zones, land-based wind in the Central and North zones, and offshore wind plants interconnecting in Long Island and New York City. For each of these technologies, we estimated the revenues from the NYISO markets and the state and federal incentive programs.

Table A-23 and Figure A-113: Costs, Performance Parameters, and Net Revenues of Renewable Units

Table A-23 shows cost estimates for solar PV, land-based wind and offshore wind units we used for a unit that commence operations in 2025. We estimate costs using publicly available sources including NREL’s Annual Technology baseline (ATB) and Lazard’s Levelized Cost of Energy. The table also shows the capacity factor and capacity value assumptions we used for calculating net revenues for these renewable units.

Assuming the operating and cost parameters shown in Table A-23, Figure A-113 shows the net revenues and the estimated CONE for each of the units during years 2023-2025. The CONE and net revenues of a unit in a given year correspond to those of a representative unit that commences operation in the same year.

Our methodology for estimating net revenues and the CONE for utility-scale solar PV and land-based wind units assumes net E&AS revenues are calculated using real time energy prices. Energy production is estimated using technology and location-specific hourly capacity factors. The capacity factors are based on location-specific resource availability and technology performance data.³⁴⁸

³⁴⁷ See State of New York PSC’s “Order adopting a clean energy standard”, issued on August 1, 2016 at page 130. The price of ZECs is determined by 1) starting with the U.S. government’s estimate of the social cost of carbon; 2) subtracting fixed baseline portion of this cost already captured in current wholesale power prices through the forecast RGGI prices embedded in the CARIS phase 1 report; and 3) converting the value from \$/ton to \$/MWh, using a measure of the New York system’s carbon emissions per MWh. These prices are subject to reduction by any increase in forecasted Zone A forward capacity and energy prices above a threshold of \$39/MWh. ZEC prices are fixed in advance for two year tranches and published by the NYDPS in Case 15-E-0302.

³⁴⁸ Assumed yearly capacity factors for solar PV, land-based wind, and offshore wind units are sourced from the 2024 NREL ATB and operational data from NYISO resources.

Table A-23: Cost and Performance Parameters of Renewable Units

Parameter	Utility-Scale Solar PV	Onshore Wind	Offshore Wind
Investment Cost (2024) (2025\$/kW AC basis)	<i>Upstate NY</i> : \$1,850	<i>Upstate NY</i> : \$2,440	<i>Long Island/NYC</i> : \$6,550 / \$7,000
Fixed O&M (2025\$/kW-yr)	\$15	\$44	\$76
Federal Incentives	ITC	PTC	ITC
Project Life	30 years	20 years	25 years
Debt Term	20 years		
Depreciation Schedule	5-years MACRS		
Average Annual Capacity Factor	20.0%	35.0%	45.0%
Unforced Capacity Percentage	12.2%	16.8%	35.8%
Renewable Energy Credits (Nominal \$/MWh)	REC payments estimated using average Index REC strike prices reported by NYSERDA for resources that entered service during 2023-2025 (land based renewables) and active contracts (offshore wind): Solar PV: \$73.6/MWh Land Based Wind: \$70/MWh Offshore Wind: \$155/MWh (NYC), \$146/MWh (Long Island)		

The capacity revenues for solar PV, land-based wind, and offshore wind units are calculated using prices from the spot capacity market. Capacity values are based on the latest Capacity Accreditation Factors (CAFs) beginning in 2024.³⁴⁹

We estimated the value of Renewable Energy Credits (“RECs”) and Offshore RECs (“ORECs”) using Index REC strike prices published by NYSERDA. Under this contract structure, the monthly value of RECs is equal to a fixed strike price minus the zonal monthly average day-ahead LBMP and a reference capacity payment.³⁵⁰

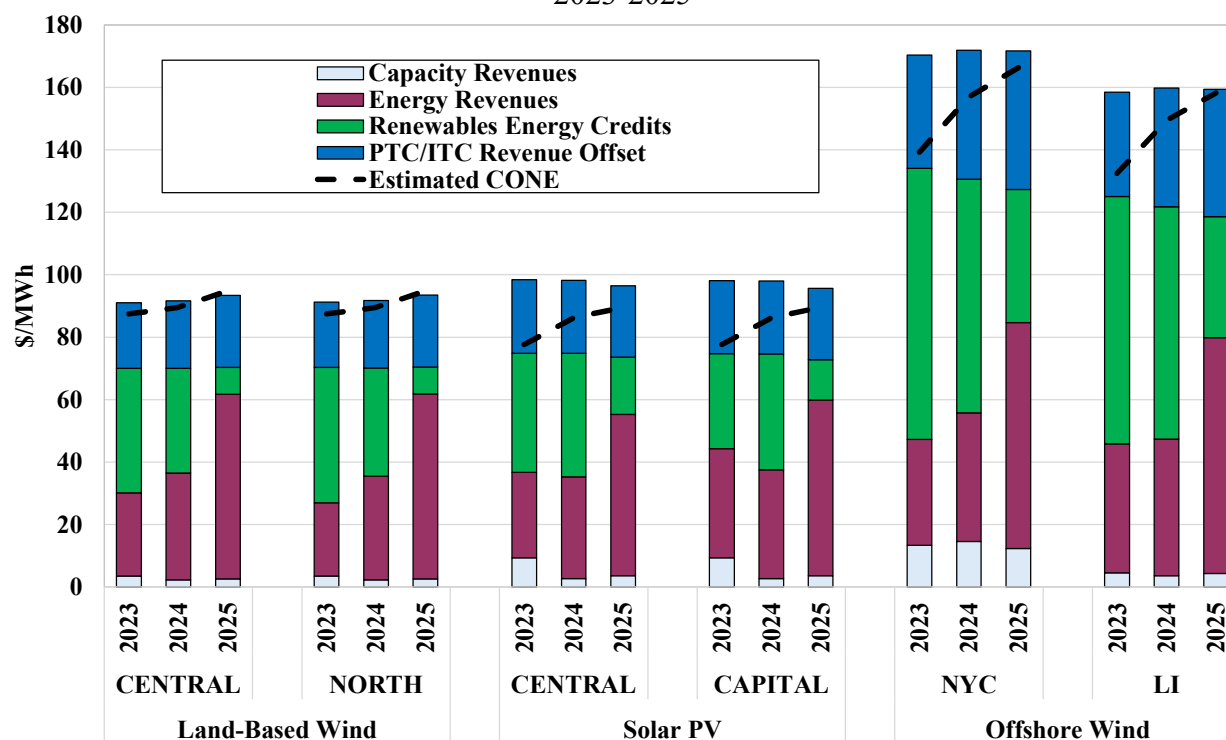
Solar PV, offshore wind, and land-based wind plants are eligible for the federal Investment Tax Credit (“ITC”) or the Production Tax Credit (“PTC”). The ITC reduces the federal income tax of the investors by a portion of a unit’s eligible investment costs depending on the resource type, and is realized in the first year of the project’s commercial operation. The PTC is a per-kWh tax credit for the electricity produced by a wind facility over a period of 10 years. Land-based wind projects that entered service in the years studied were eligible for a 2.6 cent per kWh PTC, while solar and offshore wind projects were eligible for a 30 percent ITC. We levelized the value of these tax credits over the project life using the after-tax WACC.³⁵¹

³⁴⁹ Capacity values before 2024 are defined in NYISO’s ICAP Manual. Beginning in the 2024/25 capability year, capacity values in all months are determined by the new Capacity Accreditation process (see [here](#)).

³⁵⁰ For more information on the Index REC contract structure, see [here](#). Data on NYSERDA-contracted projects can be found at [data.ny.gov](#) [here](#).

³⁵¹ In addition to these federal programs, renewable power projects may qualify for several other state or local-level incentives (e.g., property tax exemptions) in New York. However, our analysis does not consider any

**Figure A-113: Net Revenues of Solar, Land-based Wind and Offshore Wind Units
2023-2025**



Renewable generators are assumed to incur a lower cost of capital due to the availability of revenues from sale of renewable energy credits, which carry a lower risk relative to NYISO market revenues. Accordingly, we assumed a weighted average cost of capital reflecting a mix of merchant and regulated risk based on publicly available information about the cost of financing for renewable projects.³⁵² Our estimated CONE for renewable generators assumes a 2 percent annual escalation of revenues after Year 1.

D. Energy Storage Revenues

We estimate the revenues the markets would have provided to energy storage resources in the NYC and Long Island zones. For each of these zones, we estimate the revenues from the NYISO markets and from state and federal incentive programs.

other renewables-specific revenue streams or cost offsets beyond the revenues from sale of Renewable Energy Credits and the PTC or the ITC.

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See Norton Rose Fulbright Cost of Capital: 2026 Outlook, available [here](#). We estimated cost of capital in each year assuming a pre-tax cost of debt equal to the Secured Overnight Financing Rate (SOFR) plus indicated lender spreads, a debt to equity ratio targeting a debt service coverage ratio (DSCR) that reflects a combination of merchant and contractual revenues, and a cost of equity based on the NYISO's 2024 Demand Curve Reset. For 2025, we calculate an ATWACC of 8.0 percent for wind, 7.8 percent for solar, and 9.1 percent for merchant storage.

Figure A-114: Costs, Performance Parameters, and Net Revenues of Energy Storage Units

The assumed operating characteristics are as follows:

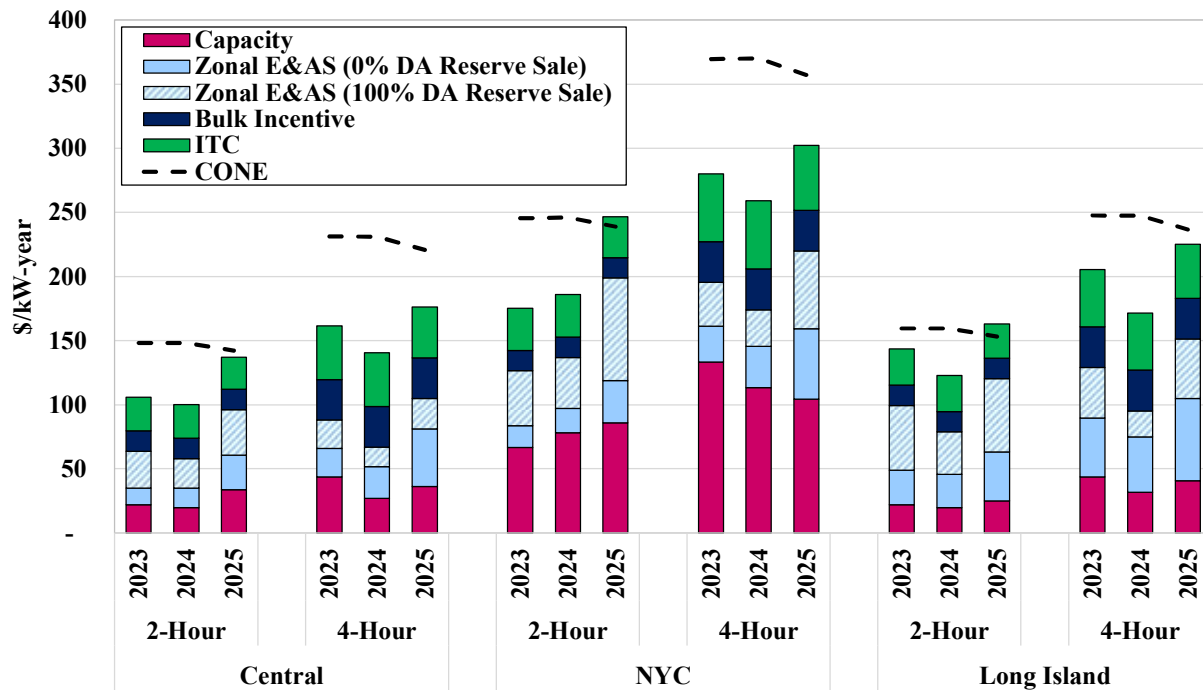
- We studied a grid-scale battery storage unit with a power rating of one MW and four hours or two hours of storage capacity. We assume a roundtrip efficiency of 85 percent.
- We model storage revenues assuming that the battery offers a portion of its capacity as 10-minute spin reserves in the day-ahead market, then self-schedules to charge or discharge in the real time market to take advantage of energy arbitrage and real-time reserve opportunities. We assume that the battery operator lacks perfect foresight of real-time market prices. Instead, we develop threshold prices at which to charge or discharge using an algorithm that considers the day-ahead forecast, RTC forecast, and backward-looking prices from the week prior to each operating day. Figure A-114 assumes that 100 percent of the battery's capacity is offered as day-ahead reserves, which was the highest-revenue strategy in the period 2023-2025.
- Capacity credit for a four-hour storage resource is based on the final capacity accreditation factors (CAFs) for the 2024/25 and 2025/26 capability years and previous default values (90 percent for four-hour, 45 percent for two-hour) in prior years. In 2025/26, the CAF for a four-hour battery was 78.5 percent in New York City and 87.1 percent in Long Island and the CAF for a two-hour battery was 64.9 percent in New York City and 52.7 percent in Long Island.
- Cost assumptions are based on NYISO's 2024 Demand Curve Reset study and publicly available sources including Lazard's Levelized Cost of Storage. Assumed capital costs for a four-hour battery in 2025 are \$2,420 per kW in NYC, \$1,790 per kW in Long Island, and \$1,700 per kW upstate. Assumed capital costs for a two-hour battery in 2025 are \$1,635 per kW in NYC, \$1,170 per kW in Long Island, and \$1,100 per kW upstate. We assume a 20 year project life and merchant cost of capital with after-tax WACC of 9.1 percent in 2025.³⁵³
- Bulk storage resources are assumed to be eligible for the NYSERDA Bulk Storage Incentive program, at a rate of \$75 per kWh of installed storage capacity.³⁵⁴ We levelize this benefit over the course of the project's life using a merchant cost of capital.
- Standalone battery storage entering service beginning in 2023 are eligible for the 30 percent federal Investment Tax Credit. We levelized the value of the ITC over the project's life.

Figure A-114 shows the net revenues and the estimated CONE for each of the units during years 2023-2025. The CONE and net revenues of a unit in a given year correspond to those of a representative unit that commences operation in the same year.

³⁵³ See description of our methodology for estimating cost of capital in Section C.

³⁵⁴ See [here](#). Bulk projects in Con Edison service territory are eligible to compete for contracted payments from the utility instead of this bulk incentive; for this analysis we assume the incentives to resources in Zone J under this approach are comparable to the bulk storage incentive available elsewhere.

Figure A-114: Net Revenues and CONE of Energy Storage Units³⁵⁵
2023-2025



355 Capacity revenues are shown for each calendar year.

VIII. DEMAND RESPONSE PROGRAMS

Demand response contributes to reliable system operations, long-term resource adequacy, lower production costs, decreased price volatility, and reduced supplier market power. Even modest reductions in consumption by end users during high-price periods can significantly reduce the costs of committing and dispatching generation to satisfy the needs of the system. These benefits underscore the value of designing wholesale markets that provide transparent economic signals and market processes that facilitate demand response.

NYISO currently operates two demand response programs that allow it to curtail end users in real-time for reliability reasons. These programs include:

- Emergency Demand Response Program (“EDRP”) – These resources are paid the higher of \$500/MWh or the real-time clearing price. There are no consequences for enrolled EDRP resources that fail to curtail.³⁵⁶
- Installed Capacity/Special Case Resource (“ICAP/SCR”) Program – These resources are paid the higher of their strike price (which can be up to \$500/MWh) or the real-time clearing price. These resources sell capacity in the capacity market in exchange for the obligation to respond when deployed.³⁵⁷

NYISO also operates the Targeted Demand Response Program (“TDRP”) to curtail EDRP and SCR resources when called by the local Transmission Owner for reliability reasons at the sub-load pocket level in New York City only.³⁵⁸

NYISO underwent a shift in participation programs allowing demand response to participate in the day-ahead and real-time energy and ancillary services markets. Prior to November 2025, certain qualified demand response resources could participate in the ancillary services markets via the Demand Side Ancillary Services Program (“DSASP”).³⁵⁹ The Day-Ahead Demand Response Program (“DADRP”) existed to facilitate participation in the day-ahead energy market but never attracted significant participation.³⁶⁰ Both programs were phased out on October 31,

³⁵⁶ Resources participate in EDRP through Curtailment Service Providers (“CSPs”), which serve as the interface between the NYISO and resources.

³⁵⁷ SCRs participate through Responsible Interface Parties (“RIPs”). Resources are obligated to curtail when called upon by NYISO to do so with two or more hours in-day notice, provided that the resource is informed on the previous day of the possibility of such a call.

³⁵⁸ EDRP resources are paid the higher of \$500/MWh or the real-time clearing price. SCRs are paid the higher of their strike price (which is capped at \$500/MWh) or the real-time clearing price. Response from these resources is voluntary.

³⁵⁹ This program allowed Demand Side Resources to offer their load curtailment capability to provide regulation and operating reserves in both day-ahead and real-time markets. DSASP resources that are dispatched for energy in real-time were not paid for that energy, DSASP resources were eligible for DAMAP.

³⁶⁰ This program allowed curtailable loads to offer into the day-ahead market (subject to a floor price) like any supply resource. If the offer cleared in the day-ahead market, the resource was paid the day-ahead clearing

2025 to facilitate the transition for participation of demand response resources in the day-ahead and real-time energy and ancillary services markets via the Distributed Energy Resource (“DER”) and Aggregation Participation Model:

- **DER and Aggregation Participation Model** – The NYISO implemented its DER and Aggregation participation model on April 16, 2024, allowing individual large consumers and aggregations of small consumers to participate more directly in the market. This model enables resources to better reflect duration limitations in their offers, payments, and obligations.

Participation of demand response resources in the DER program increased in 2025. The number of participating resources increased from zero in early 2025 to six in early-2026. All of these participants are former DSASP resources that transitioned to DER with the sunseting of the old participation programs.

Despite the advantages of the DER program, significant barriers to participation in the wholesale market by loads remain and registration has been slow. The most significant barriers stem from the dispatchability requirements of the program.³⁶¹ Unlike generators, DER resources cannot reflect reasonable operating constraints into their energy offers, including Start Up Notification Time (“SUNT”), Minimum Run Time (“MRT”), and Minimum Down Time (“MDT”). The inability to reflect these operating constraints discourages some loads from being available to curtail. To attract more beneficial participation, it will be important to develop programs to provide efficient incentives to demand response resources and facilitate their participation in the real-time market.

NYISO has been working on a series of market design projects that are intended to facilitate more active participation by consumers. These projects include:

- **Meter Service Entity (“MSE”)** for DER – The MSE rules went into effect in May 2020, which authorize third party metering that provides greater flexibility to consumers and retail load serving entities for demand side participation.
- **Dual Participation (“DP”)** – The DP rules went into effect in May 2020, which allow resources that provide wholesale market services to also provide retail market services.

This section evaluates the existing demand-side programs in 2025 in the following subsections: (a) participation of demand-side resources and price-sensitive loads and (b) evaluation of the DER program.

A. Demand-Side Participation and Price-Responsive Load

The EDRP and SCR programs enable NYISO to deploy reliability demand response resources when the NYISO and/or a TO forecast a reliability issue. The DER program allows demand

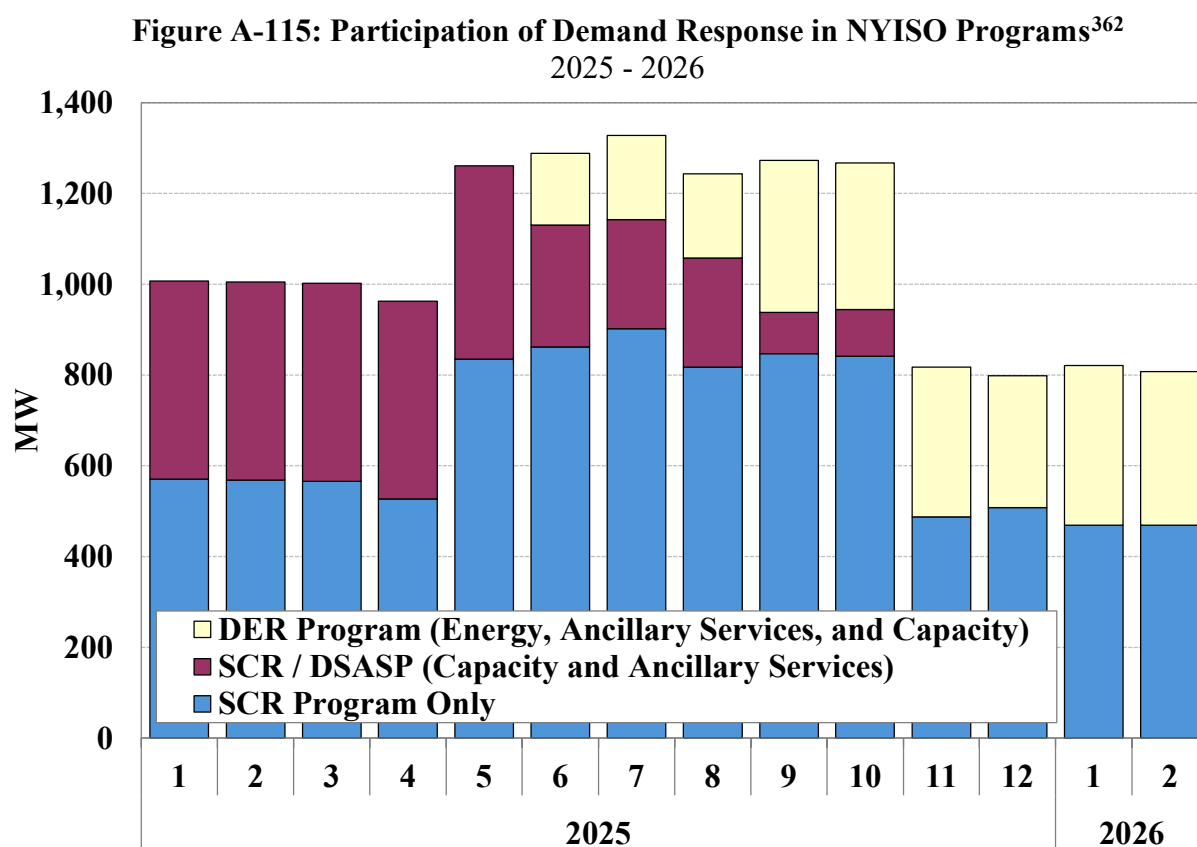
price and must curtail its load in real-time accordingly. Failure to curtail resulted in penalties being assessed in accordance with applicable rules.

³⁶¹ Aggregations are considered dispatch-only and do not receive unit commitments from the NYISO. See §2.2 of the [Aggregations Manual](#).

response resources to be economically dispatched for energy and reserves in a comparable manner to generators. Before its sunset in October 2025, the DSASP program allowed certain qualified loads to sell operating reserves in the NYISO ancillary services market. Historically, most DSASP resources jointly participated in the SCR program. Beginning in June 2025 and expanding across the year, most of the capacity associated with the DSASP program transitioned to participation in the DER program.

Figure A-115: Registration in NYISO Demand Response Reliability Programs

Figure A-115 summarizes registration of demand response resources in the SCR, DSASP, and DER programs in each month of 2025 through early-2026. Capacity values are listed in terms of ICAP sold in the NYISO installed capacity market. Since EDRP resources and SCRs in New York City participate in the TDRP program on a voluntary basis, TDRP resources are not shown separately.



Nearly all of the DSASP resources participated in the SCR program. With the NYISO's announcement of the DSASP program sunset by October 31, 2025, many of the DSASP resources registered to participate in the DER program.

³⁶² This figure is excerpted from *NYISO 2024 Annual Report on Demand Response Programs*, February 28, 2024, available at: www.nyiso.com/demand-response under "DR05 - NYISO Semi-Annual Demand Response Report - 2025."

Over the past decade, SCRs have accounted for nearly all of the total enrolled MWs in the reliability-based programs because capacity market revenues account for most of the revenues available to emergency demand response resources. The Expanding Capacity Eligibility market rules became effective in May 2021 and began to discount the capacity payments to SCRs by the same amount as 4-hour duration limited resources. In May 2021, the Duration Adjustment Factor was 90 percent.³⁶³ After the Capacity Accreditation rules went into effect, the Capacity Accreditation Factor of SCRs ranged between roughly 64 and 79 percent (depending on the Capacity Region) in the 2024/25 Capability Year and between 68 and 77 percent in the 2025/26 Capability Year.

Demand Response and Scarcity Pricing

In an efficient market, clearing prices should reflect the cost of deploying resources to satisfy demand and maintain reliability, particularly under scarcity conditions. Ordinarily, to be involved with setting prices in the real-time market, resources must be dispatchable by the real-time market model on a five-minute basis. EDRP and SCR resources must be called in advance based on projections of operating conditions; they are not dispatchable by the real-time model. Hence, there is no guarantee that these resources will be “in-merit” relative to the real-time clearing price, and their deployment can lower prices. Prices can be very low after EDRP and SCR resources are curtailed, if adequate resources are available to the system in real-time. NYISO currently has two market rules that improve the efficiency of real-time prices when demand response resources are deployed.

First, to minimize the price-effects of “out-of-merit” demand response resources, NYISO has the TDRP currently available in New York City. This program enables the local Transmission Owner in New York City to call EDRP and SCR resources in blocks smaller than an entire zone. This prevents the local Transmission Owner from calling all the EDRP and SCR resources in New York City to address local issues and avoids deploying substantial quantities of demand response that provide no reliability benefit but unnecessarily depress real-time prices and increase uplift.

Second, NYISO has special scarcity pricing rules for periods when demand response resources are deployed. Generally, when a shortage of 30-minute reserves is prevented by the deployment of demand response in certain regions (e.g., state-wide, Eastern New York, or Southeastern New York), real-time energy prices will be set to \$500/MWh or higher within the region. This rule helps reflect the cost of maintaining adequate reserve levels in real-time clearing prices and improves the efficiency of real-time prices during scarcity conditions. Prior to June 22, 2016, the real-time LBMPs during EDRP/SCR activations were set in an *ex-post* fashion, which tended to cause inconsistencies between resource schedules and pricing outcomes and result in potential uplift costs. NYISO implemented a Comprehensive Scarcity Pricing on June 22, 2016, to address this issue. Under this enhanced rule, the 30-minute reserve requirement in the applicable region is increased to reflect the expected EDRP/SCR deployment in the pricing logic, setting the LBMPs in the applicable region at a proper level in an *ex-ante* fashion.

³⁶³ See Section 4.1.1 of ICAP Manual for more details.

Table A-24 – Figure A-116: Reliability Demand Response Deployments by NYISO

Table A-24 summarizes the reliability demand response events in 2025. The table lists for each event the program type (i.e., TDRP or SCR/EDRP), the start and end times, required zones, and obligated ICAP MWs. The table also indicates whether the scarcity pricing rule was triggered during the event and affected LBMPs in how many intervals.

Table A-24: Reliability Program Demand Response Activations
2025

DR Program	Event Date	Start Time	End Time	Event Zone	Obligated ICAP MW	# of RT Market Intervals w/ Scarcity Pricing Triggered
SCR/EDRP	1/21/2025	16:00	22:00	A-K	1026.1	3
SCR/EDRP	1/22/2025	16:00	22:00	A-K	1026.1	0
TDRP	2/20/2025	12:00	23:59	J3,J4	56.8	N/A
TDRP	2/21/2025	0:00	23:59	J3,J4	56.8	N/A
TDRP	6/19/2025	15:00	20:00	J3	53	N/A
TDRP	6/20/2025	14:00	20:00	J3	53	N/A
SCR/EDRP	6/23/2025	14:00	21:00	A-K	1192.8	57
SCR/EDRP	6/24/2025	14:00	21:00	A-K	1192.8	95
SCR/EDRP	6/25/2025	14:00	21:00	A-K	1192.8	39
TDRP	6/29/2025	13:00	23:00	J8	106.1	N/A
TDRP	6/30/2025	13:00	23:00	J8	106.1	N/A
TDRP	7/1/2025	13:00	23:00	J8	107.9	N/A
TDRP	7/8/2025	15:00	22:00	J8	107.9	N/A
SCR/EDRP	7/16/2025	15:00	22:00	A-K	1204.6	0
TDRP	7/17/2025	9:00	15:00	J3	54.2	N/A
SCR/EDRP	7/17/2025	15:00	22:00	G-K	549.9	0
TDRP	7/17/2025	22:00	23:00	J3	54.2	N/A
TDRP	7/19/2025	15:00	23:00	J8	107.9	N/A
SCR/EDRP	7/25/2025	16:00	22:00	J,K	475.9	7
SCR/EDRP	7/28/2025	15:00	22:00	A-K	1204.6	47
SCR/EDRP	7/29/2025	15:00	22:00	A-K	1204.6	48
SCR/EDRP	7/30/2025	15:00	22:00	A-K	1204.6	15

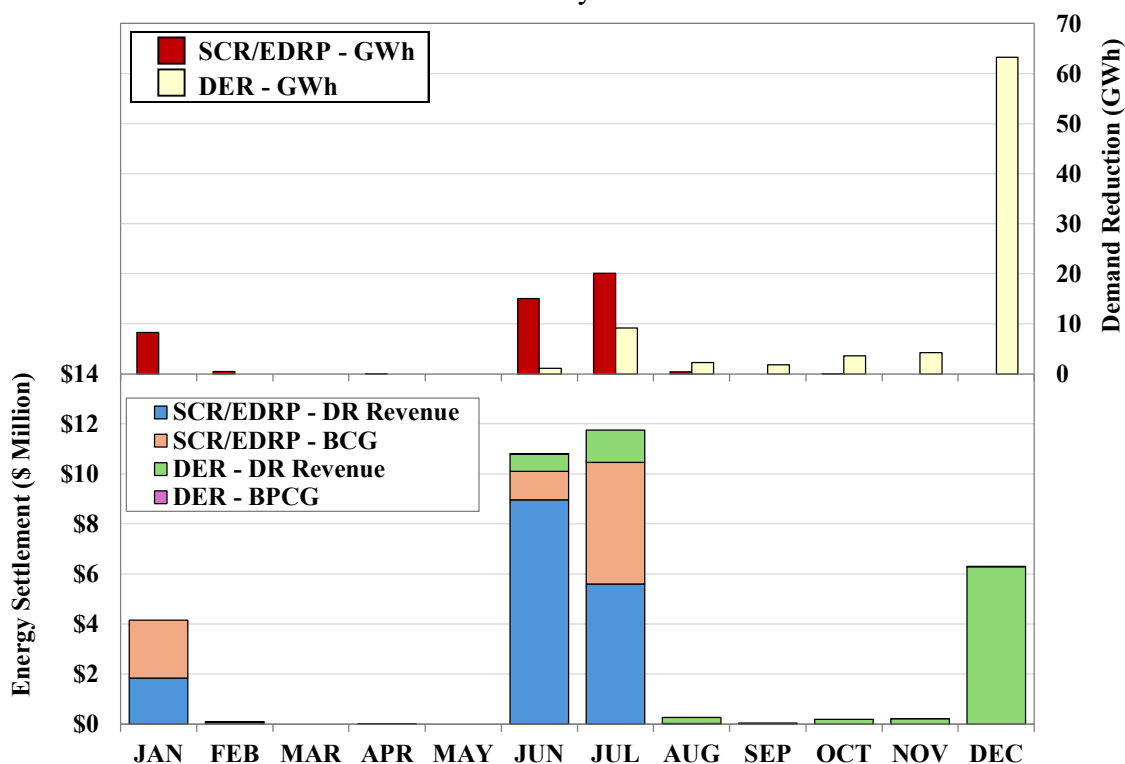
Notes: Obligated MW is defined as the Installed Capacity Equivalent of the UCAP sold by SCRs in a Load Zone during the calendar month in which the event occurred. The numbers reported here use ACL as baseline.

Figure A-116: SCR and DER Energy Deployments and Revenues

Demand reduction from demand response resources is treated in the NYISO markets as equivalent to energy generation. Consequently, the demand reduction energy provided by DR settles in the NYISO markets based on energy prices. Settlement rules differ between activations in the SCR, EDRP, and DER programs. EDRP resources settle at the greater of \$500 per MWh or RTD Zonal LBMP during deployments. SCR resources settle at RTD Zonal LBMP with bid

cost guarantee (“BCG”) eligibility based on the resource minimum payment nomination. DER settle at RTD nodal LBMP subject to eligibility against the monthly Net Benefits Threshold (“NBT”). Figure A-116 summarizes the energy settlements and BCG for demand response resources participating in SCR, EDRP, and DER across each month of 2025. It also displays the total demand reduction energy supplied by these resources per month in 2025.

Figure A-116: SCR and DER Energy Deployments and Revenues
Monthly 2025



B. Evaluation of DER Program

NYISO implemented its DER and Aggregation participation model on April 16, 2024, allowing individual large consumers and aggregations of small consumers to participate more directly in the market. Demand response resource participating in this program no longer register as SCR resources, enabling more efficient scheduling of demand reduction and ancillary services directly in the market. Despite the advantages, registration in this program has been slow.³⁶⁴ Only with the sunseting of the DSASP program in 2025 did any demand response resources make the

³⁶⁴ All resources participating in the DER program as of early 2026 are demand response resources. Prior to January 2026, only cryptocurrency mining loads had joined the program. These resources are better able to navigate the difficulties of the program due to their fast responsiveness, low cost of curtailment, and ability to cycle often.

switch to DER. The first resource to register did so in June 2025, with four other resources active by the end of 2025. Registration has been slow due to several factors:

- All DERs are considered “dispatch-only” and, therefore, not committed. Physical parameters that a generator would normally include in its offers are not considered when dispatching DERs, excluding most resources that cannot respond in five minutes or less to basepoints from participating in the program.
- Assessing compliance with energy and reserve offer obligations is difficult due to the rules for the Economic Customer Baseline Load (“ECBL”) calculation. The formulation of ECBL makes it difficult to harmonize with energy and reserve offers includes:
 - ECBL is calculated at the 5-minute interval level while energy and reserve offers are hourly.
 - ECBL proxy load calculation only factors in demand reductions that occur at prices that exceed the NBT. Many ramping intervals during dispatch occur at lower prices than NBT which make proxy load artificially low.
 - The in-day adjustment calculation for ECBL does not prevent the ECBL from potentially exceeding the installed load capacity of the DR, especially if there are proxy load issues that impact the in-day adjustment intervals.

Figure A-117 summarizes daily load reductions, uneconomic loads, and “energy” scheduled from DR participating in the DER program from November 2025 through January 2026. Uneconomic load consumption (i.e., consumption at LBMPs exceeding the DER’s estimated value of consumption) is bifurcated into: (a) loads that pay retail rates and (b) loads that pay wholesale rates. The top portion of the chart displays the monthly NBT, the opportunity cost of bitcoin mining, and the capacity-weighted median incremental energy offer from participating cryptocurrency mining loads. chart displays the monthly NBT, the opportunity cost of bitcoin mining, and the capacity-weighted median incremental energy offer from participating cryptocurrency mining loads.³⁶⁵

³⁶⁵

The opportunity cost of bitcoin mining was estimated using the S19k XP antminer as the default miner. The daily value is calculated as equal to [Bitcoin Network Hash Price] * [Hash per miner] / [Power Requirement] * 1,000,000 / 24 * Efficiency. The assumptions for the default miner are: Power Requirement: 3,010 W; Hash per Miner: 0.14 PH, Efficiency: 0.95. Bitcoin Network Hash Price sourced from: <https://www.coinglass.com/pro/i/hashprice-index>.

Figure A-117: Inefficient Curtailment and Loads from Order 745 Requirements
Nov 2025 – Jan 2026

